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Quantum Electrodynamics

Course of Theoretical Physics
Volume IV

New English translation from the fourth corrected Russian edition

Contents

Introduction	1
1 Uncertainty Relations in the Relativistic Domain	1
I Photon	5
2 Quantization of the Free Electromagnetic Field	5
3 Photons	9
4 Gauge invariance	12
5 The electromagnetic field in quantum theory	13
6 Angular momentum and parity of the photon	15
7 Spherical waves of photons	17
8 Polarization of the photon	22
9 The two-photon system	27
II Bosons	31
10 Wave equation for particles with spin 0	31
11 Particles and antiparticles	35
12 Strictly neutral particles	38
13 The transformations C, P, T	41
14 Wave equation for spin-1 particles	47
15 Wave equation for particles with higher integer spins	50
16 Helicity states of a particle ¹	52
III Fermions	59
17 Four-dimensional spinors	59
18 Relation of spinors to 4-vectors	61
19 Inversion of spinors	65
20 The Dirac equation in spinor representation	69
21 Symmetric form of the Dirac equation	71
22 Algebra of Dirac matrices	76
23 Plane waves	79
24 Spherical waves	82
25 The connection between spin and statistics	85

26	Charge conjugation and time reversal of spinors	88
27	Internal symmetry of particles and antiparticles	93
28	Bilinear forms	95
29	Polarisation density matrix	99
30	Two-component fermions	103
31	Wave equation for a spin-3/2 particle	107
IV	A Particle in an External Field	109
32	The Dirac equation for an electron in an external field	109
33	Expansion in powers of $1/c$	113
34	Fine structure of hydrogen atom levels	116
35	Motion in a centrally symmetric field	118
36	Motion in a Coulomb field	122
37	Scattering in a centrally symmetric field	129
38	Scattering in the ultra-relativistic case	131
39	System of wave functions of the continuous spectrum for scattering in a Coulomb field	132
40	Electron in the field of a plane electromagnetic wave	136
41	Spin motion in an external field	139
42	Scattering of neutrons in an electric field	144
V	Radiation	147
43	The operator of electromagnetic interaction	147
44	Emission and absorption	149
45	Dipole radiation	151
46	Electric multipole radiation	153
47	Magnetic multipole radiation	158
48	Angular distribution and polarisation of radiation	160
49	Radiation of atoms. Electric type ²	168
50	Radiation of atoms. Magnetic type	172
51	Radiation of atoms. Zeeman and Stark effects	175
52	Radiation of atoms. Hydrogen atom	178
53	Radiation of diatomic molecules. Electronic spectra	183
54	Radiation of diatomic molecules. Vibrational and rotational spectra	189
55	Radiation of nuclei	190
56	Photoeffect. Non-relativistic case	193
57	Photoeffect. Relativistic case	197
58	Photodisintegration of the deuteron	201
VI	Scattering of Light	205

²In §§ 49, 51, 53–55 we use ordinary units.

59	Scattering tensor	205
60	Scattering by freely orientable systems	214
61	Scattering on molecules	221
62	Natural width of spectral lines	224
63	Resonance fluorescence	227
VII	The Scattering Matrix	231
64	Scattering amplitude	231
65	Reactions with polarised particles	236
66	Kinematic invariants	239
67	Physical regions	242
68	Partial-wave expansion	245
69	Symmetry of helicity amplitudes	248
70	Invariant amplitudes	253
71	Unitarity condition	257
VIII	Invariant Perturbation Theory	263
72	Chronological product	263
73	Feynman diagrams for electron scattering	266
74	Feynman diagrams for photon scattering	271
75	Electron propagator	274
76	Photon propagator	277
77	General rules of diagram technique	281
78	Crossing invariance	287
79	Virtual particles	288
IX	Interaction of Electrons	293
80	Scattering of an electron in an external field	293
81	Scattering of electrons and positrons on an electron	297
82	Ionization losses of fast particles	306
83	Breit equation	312
84	Positronium	318
85	Interaction of atoms at large distances	322
X	Interaction of Electrons with Photons	327
86	Scattering of a photon by an electron	327
87	Scattering of a photon by an electron. Polarisation effects	331
88	Two-photon annihilation of an electron-positron pair	339
89	Annihilation of positronium	342
90	Magnetobremssstrahlung	347
91	Pair production by a photon in a magnetic field	354
92	Bremssstrahlung of an electron on a nucleus. Non-relativistic case	357

93	Bremsstrahlung of an electron on a nucleus. Relativistic case	368
94	Pair production by a photon in the field of a nucleus	377
95	Exact theory of pair production in the ultrarelativistic case	380
96	Exact theory of bremsstrahlung in the ultrarelativistic case	385
97	Bremsstrahlung of an electron on an electron in the ultrarelativistic case	391
98	Emission of soft photons in collisions	395
99	The method of equivalent photons	401
100	Pair production in particle collisions	407
101	Emission of a photon by an electron in the field of an intense electro- magnetic wave	411
XI	Exact Propagators and Vertex Parts	417
102	Field operators in the Heisenberg representation	417
103	The exact photon propagator	419
104	The photon self-energy function	424
105	The exact electron propagator	427
106	The vertex operator	430
107	Dyson equations	433
108	Ward identity	434
109	Electron propagator in an external field	436
110	Physical conditions of renormalisation	441
111	Analytic properties of the photon propagator	446
112	Regularisation of Feynman integrals	448
XII	Radiative Corrections	453
XIII	Asymptotic Formulae of Quantum Electrodynamics	455
XIV	Electrodynamics of Hadrons	457

Introduction

§ 1. Uncertainty Relations in the Relativistic Domain

The quantum theory presented in Vol. III of this course is non-relativistic in its very substance, and it fails for phenomena in which motion occurs at speeds no longer small next to the speed of light. One might at first imagine that a relativistic theory could be reached by generalizing the formalism of non-relativistic quantum mechanics in a more or less straightforward manner. A closer look shows, however, that a logically complete relativistic theory cannot be built without invoking fresh physical principles.

We begin by recalling several physical notions on which non-relativistic quantum mechanics rests (Vol. III, § 1). A fundamental part was played there, as we saw, by the concept of measurement — the process in which a quantum system interacts with a “classical object” (an “apparatus”) and thereby comes to possess definite values of particular dynamical variables (coordinates, velocities, and so forth). We saw, too, that quantum mechanics places severe limits on the extent to which different dynamical variables of an electron³ can exist together. Thus the uncertainties Δq and Δp with which coordinate and momentum can coexist obey the relation $\Delta q \Delta p \sim \hbar$;⁴ the more precisely either of these quantities is measured, the less precisely the other can be known at the same moment.

What matters, however, is that any single dynamical variable of the electron, on its own, admitted measurement of unlimited precision, and within a time interval that could be taken as short as desired. On this circumstance the whole of non-relativistic quantum mechanics depends in a fundamental way. It alone makes it possible to introduce the wave function, the central object of the theory’s formalism. The physical content of the wave function $\psi(q)$ is, after all, that its squared modulus gives the probability that a measurement performed at a given instant will yield one or another value of the electron’s coordinate. Evidently the very notion of such a probability presupposes that a coordinate measurement of unlimited precision and speed can be realized in principle; failing that, the notion would be empty of content and would forfeit its physical meaning.

The existence of a limiting speed (the speed of light c) leads to new limitations, of a fundamental kind, on the possibilities of measuring various physical quantities (L. D. Landau, R. Peierls, 1930).

In Vol. III, § 44 the relation

$$(v' - v) \Delta p \Delta t \sim \hbar \tag{1.1}$$

was obtained; it relates the uncertainty Δp of a momentum measurement on the electron to the length Δt of the measuring process itself, v and v' denoting the electron’s velocity before and after that process. It follows that momentum can be measured accurately within a short time (small Δp together with small Δt) only if the measurement itself

³As in Vol. III, § 1, we say “electron” for brevity, understanding by it any quantum system.

⁴Ordinary units are used in this section.

alters the velocity strongly enough. In the non-relativistic theory this expresses the fact that a momentum measurement repeated after a short interval will not reproduce its result, but it places no restriction of principle on how accurate a single measurement of the momentum may be, the difference $v' - v$ being capable of unlimited increase.

The presence of a limiting speed, however, changes the state of affairs radically. The difference $v' - v$, like the velocities themselves, can now not exceed c (more precisely, $2c$). Replacing $v' - v$ in (1.1) by c , we obtain the relation

$$\Delta p \Delta t \sim \hbar/c, \quad (1.2)$$

which determines the best accuracy of momentum measurement attainable in principle for a given duration Δt of the measurement. Thus in the relativistic theory an arbitrarily accurate and arbitrarily rapid measurement of the momentum turns out to be impossible in principle. An exact measurement of the momentum ($\Delta p \rightarrow 0$) is possible only in the limit of an infinitely long duration of the measurement.

One has reason to expect that the measurability of the electron's coordinate as such is likewise affected. Within the mathematical formalism of the theory this appears as a conflict between an exact coordinate measurement and the statement that a free particle's energy is positive. As we shall see, the full set of eigenfunctions of a free particle's relativistic wave equation contains, besides the solutions whose time dependence is the "proper" one, solutions of "negative frequency" as well. These will in general appear also in the expansion of a wave packet describing an electron confined to a small region of space.

As will be shown, the "negative-frequency" wave functions are tied to the existence of antiparticles — the positrons. Their presence in the expansion of the wave packet reflects the creation of electron–positron pairs, unavoidable in the general case, during a measurement of the electron's coordinates. Since the emergence of these new particles cannot be controlled by the measuring process itself, coordinate measurement for the electron is drained of meaning.

In the rest frame of the electron the minimum error in a measurement of its coordinates is

$$\Delta q \sim \hbar/mc. \quad (1.3)$$

This value (the only one that dimensional considerations would permit) answers to a momentum uncertainty $\Delta p \sim mc$, which corresponds in turn to the minimum threshold energy needed to create a pair.

In a reference frame where the electron moves with energy ε , (1.3) is replaced by

$$\Delta q \sim c\hbar/\varepsilon. \quad (1.4)$$

In the extreme ultra-relativistic limit, in particular, energy and momentum are related by $\varepsilon \approx cp$, so that

$$\Delta q \sim \hbar/p, \quad (1.5)$$

the error Δq then coinciding with the particle's de Broglie wavelength.⁵

⁵What is meant here are measurements any outcome of which permits an inference about the state of

For photons the ultra-relativistic case always holds, so that expression (1.5) is valid. This means that it makes sense to speak of the coordinates of a photon only in those cases where the characteristic dimensions are large compared with the wavelength. But this is nothing other than the “classical” limiting case, corresponding to geometrical optics, in which one may speak of the propagation of light along definite trajectories — rays. In the quantum case, however, when the wavelength cannot be regarded as small, the concept of the coordinates of a photon becomes vacuous. We shall see later (see § 4) that in the mathematical formalism of the theory the non-measurability of the photon’s coordinates already shows itself in the impossibility of forming from its wave function a quantity that could play the part of a probability density satisfying the requirements which relativistic invariance demands.

On the basis of all that has been said, it is natural to suppose that the future theory will renounce altogether the consideration of the time development of the processes of interaction of particles. It will show that in these processes no exactly definable characteristics exist (even within the limits of the usual quantum-mechanical accuracy), so that the description of a process in time will turn out to be just as illusory as classical trajectories proved to be in non-relativistic quantum mechanics. The only observable quantities will be the characteristics (momenta, polarizations) of free particles — the initial particles that enter into the interaction and the final particles that have arisen as a result of the process (*L. D. Landau, R. Peierls, 1930*).

The characteristic formulation of a problem in relativistic quantum theory consists in the determination of the probability amplitudes of transitions connecting given initial and final states (that is, at $t \rightarrow \mp\infty$) of a system of particles. The totality of the transition amplitudes between all possible states constitutes the *scattering matrix*, or *S-matrix*. This matrix will be the carrier of all the information about the processes of interaction of particles that possesses an observable physical meaning (*W. Heisenberg, 1938*).

At the present time a complete, logically closed relativistic quantum theory does not yet exist. We shall see that the existing theory introduces new physical aspects into the manner of describing the states of particles, which acquires certain features of field theory (see § 10). The theory is constructed, however, to a considerable degree on the model of, and with the help of the concepts of, ordinary quantum mechanics. This way of building the theory has led to success in the domain of quantum electrodynamics. The absence of complete logical closure shows itself in this theory in the existence of divergent expressions when its mathematical apparatus is applied directly; but for the removal of these divergences there exist entirely unambiguous procedures. Nevertheless these procedures retain to a considerable degree the character of semi-empirical recipes, and our confidence in the correctness of the results obtained in this way rests, in the end, on their excellent agreement with experiment, and not on the internal consistency and logical coherence of the fundamental principles of the theory.

the electron; we thus set aside coordinate measurements by collision, in which over the time of observation the result occurs with a probability short of unity. In such a case a deflection of the particle does allow the electron’s position to be inferred, but from the absence of a deflection nothing at all can be concluded.

Photon

§2. Quantization of the Free Electromagnetic Field

In order to treat the electromagnetic field as a quantum object, it is convenient to start from a classical description in which the field is characterized by a discrete, albeit infinite, set of variables; such a description will allow one to apply the standard formalism of quantum mechanics directly. A description by means of potentials specified at every point of space is, in essence, a description in terms of a continuous set of variables.

Let $\mathbf{A}(\mathbf{r}, t)$ be the vector potential of the free electromagnetic field, satisfying the “transversality condition”

$$\nabla \cdot \mathbf{A} = 0. \quad (2.1)$$

The scalar potential then vanishes, $\Phi = 0$, and the fields $\mathbf{E}$ and $\mathbf{H}$ are

$$\mathbf{E} = -\dot{\mathbf{A}}, \quad \mathbf{H} = \nabla \times \mathbf{A}. \quad (2.2)$$

Maxwell’s equations reduce to the wave equation for $\mathbf{A}$:

$$\Delta \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0. \quad (2.3)$$

As is well known (see Vol. II, § 52), in classical electrodynamics the passage to a description in terms of a discrete set of variables is effected by considering the field inside a large but finite volume V .¹ Let us recall, omitting the details of the calculation, how this is done.

A field in a finite volume can be expanded in travelling plane waves, so that its potential takes the form of a series

$$\mathbf{A} = \sum_{\mathbf{k}} (\mathbf{a}_{\mathbf{k}} e^{i\mathbf{k}\mathbf{r}} + \mathbf{a}_{\mathbf{k}}^* e^{-i\mathbf{k}\mathbf{r}}), \quad (2.4)$$

where the coefficients $\mathbf{a}_{\mathbf{k}}$ depend on time according to

$$\mathbf{a}_{\mathbf{k}} \sim e^{-i\omega t}, \quad \omega = |\mathbf{k}|. \quad (2.5)$$

By virtue of condition (2.1) the complex vectors $\mathbf{a}_{\mathbf{k}}$ are orthogonal to the corresponding wave vectors: $\mathbf{a}_{\mathbf{k}} \mathbf{k} = 0$.

The summation in (2.4) runs over an infinite discrete set of values of the wave vector (its three components k_x, k_y, k_z). The transition to an integration over a continuous distribution is made with the aid of the expression

$$\frac{d^3 k}{(2\pi)^3}$$

¹To avoid burdening formulae with superfluous factors, we set $V = 1$.

for the number of allowed values of $\mathbf{k}$ falling within a volume element $d^3k = dk_x dk_y dk_z$ of $\mathbf{k}$ -space.

Once the vectors $\mathbf{a}_\mathbf{k}$ are given, the field inside the volume in question is completely determined. One may therefore view these quantities as a discrete set of classical “field variables”. In order to find the passage to a quantum theory, however, a further transformation of these quantities is needed, one that brings the field equations into a form analogous to the canonical (Hamiltonian) equations of classical mechanics. The canonical variables of the field are introduced by means of

$$\mathbf{Q}_\mathbf{k} = \frac{1}{\sqrt{4\pi}}(\mathbf{a}_\mathbf{k} + \mathbf{a}_\mathbf{k}^*), \quad \mathbf{P}_\mathbf{k} = -\frac{i\omega}{\sqrt{4\pi}}(\mathbf{a}_\mathbf{k} - \mathbf{a}_\mathbf{k}^*) = \dot{\mathbf{Q}}_\mathbf{k} \quad (2.6)$$

(they are evidently real). The vector potential is expressed through the canonical variables as

$$\mathbf{A} = \sqrt{4\pi} \sum_{\mathbf{k}} \left(\mathbf{Q}_\mathbf{k} \cos \mathbf{k}\mathbf{r} - \frac{1}{\omega} \mathbf{P}_\mathbf{k} \sin \mathbf{k}\mathbf{r} \right). \quad (2.7)$$

To find the Hamiltonian H one must evaluate the total field energy

$$\frac{1}{8\pi} \int (\mathbf{E}^2 + \mathbf{H}^2) d^3x,$$

expressed in terms of the quantities $\mathbf{Q}_\mathbf{k}$, $\mathbf{P}_\mathbf{k}$. Substituting the expansion (2.7) for $\mathbf{A}$, computing $\mathbf{E}$ and $\mathbf{H}$ by means of (2.2), and integrating, one obtains

$$H = \frac{1}{2} \sum_{\mathbf{k}} (\mathbf{P}_\mathbf{k}^2 + \omega^2 \mathbf{Q}_\mathbf{k}^2).$$

Each of the vectors $\mathbf{P}_\mathbf{k}$ and $\mathbf{Q}_\mathbf{k}$ is perpendicular to the wave vector $\mathbf{k}$ and therefore has two independent components. The direction of these vectors fixes the polarization direction of the corresponding wave. Denoting the two components of $\mathbf{Q}_\mathbf{k}$, $\mathbf{P}_\mathbf{k}$ (in the plane perpendicular to $\mathbf{k}$) by $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ ($\alpha = 1, 2$), we rewrite the Hamiltonian as

$$H = \sum_{\mathbf{k}\alpha} \frac{1}{2} (P_{\mathbf{k}\alpha}^2 + \omega^2 Q_{\mathbf{k}\alpha}^2). \quad (2.8)$$

The Hamiltonian has thus broken up into a sum of independent terms, each involving just one pair $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$. Every term answers to a travelling wave of a given wave vector and polarization and reproduces the structure of a single harmonic-oscillator Hamiltonian. One accordingly speaks of this decomposition as the expansion of the field in *oscillators*.

We now turn to the quantization of the free electromagnetic field. The method of classical description just set out makes the route to quantum theory obvious. The canonical variables — the generalized coordinates $Q_{\mathbf{k}\alpha}$ and generalized momenta $P_{\mathbf{k}\alpha}$ — are now to be treated as operators obeying the commutation rule

$$\hat{P}_{\mathbf{k}\alpha} \hat{Q}_{\mathbf{k}\alpha} - \hat{Q}_{\mathbf{k}\alpha} \hat{P}_{\mathbf{k}\alpha} = -i \quad (2.9)$$

(operators with different $\mathbf{k}\alpha$ all commute with one another). Together with them the potential $\mathbf{A}$ and, by (2.2), the field strengths $\mathbf{E}$ and $\mathbf{H}$ also become (Hermitian) operators.

A consistent determination of the field Hamiltonian requires evaluation of the integral

$$\hat{H} = \frac{1}{8\pi} \int (\hat{\mathbf{E}}^2 + \hat{\mathbf{H}}^2) d^3x, \quad (2.10)$$

in which $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ are expressed through the operators $\hat{P}_{\mathbf{k}\alpha}$, $\hat{Q}_{\mathbf{k}\alpha}$. In practice, however, the non-commutativity of the latter does not make itself felt here, since the products $Q_{\mathbf{k}\alpha}P_{\mathbf{k}\alpha}$ enter with the factor $\cos \mathbf{k}\mathbf{r} \cdot \sin \mathbf{k}\mathbf{r}$, which vanishes upon integration over the full volume. Consequently the Hamiltonian takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}_{\mathbf{k}\alpha}^2 + \omega^2 \hat{Q}_{\mathbf{k}\alpha}^2), \quad (2.11)$$

exactly reproducing the classical Hamiltonian, as one would naturally expect.

Finding the eigenvalues of this Hamiltonian requires no special computation, since the problem reduces to the familiar one of the energy levels of linear oscillators (see Vol. III, § 23). We can therefore write down at once the energy levels of the field:

$$E = \sum_{\mathbf{k}\alpha} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right) \omega, \quad (2.12)$$

where $N_{\mathbf{k}\alpha}$ are non-negative integers.

Discussion of this formula will be taken up in the next section; here we write out the matrix elements of the quantities $Q_{\mathbf{k}\alpha}$, which can be obtained directly from the known formulae for the matrix elements of the oscillator coordinate (see Vol. III, § 23). The non-vanishing matrix elements are

$$\langle N_{\mathbf{k}\alpha} | Q_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{\frac{N_{\mathbf{k}\alpha}}{2\omega}}. \quad (2.13)$$

The matrix elements of $P_{\mathbf{k}\alpha} = \dot{Q}_{\mathbf{k}\alpha}$ differ from those of $Q_{\mathbf{k}\alpha}$ only by a factor $\pm i\omega$.

For subsequent calculations it will, however, be more convenient to work not with $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ themselves but with their linear combinations $\omega Q_{\mathbf{k}\alpha} \pm iP_{\mathbf{k}\alpha}$, which have matrix elements only for the transitions $N_{\mathbf{k}\alpha} \rightarrow N_{\mathbf{k}\alpha} \pm 1$. Accordingly we introduce the operators

$$\hat{c}_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}} (\omega \hat{Q}_{\mathbf{k}\alpha} + i \hat{P}_{\mathbf{k}\alpha}), \quad \hat{c}_{\mathbf{k}\alpha}^+ = \frac{1}{\sqrt{2\omega}} (\omega \hat{Q}_{\mathbf{k}\alpha} - i \hat{P}_{\mathbf{k}\alpha}) \quad (2.14)$$

(the classical quantities $c_{\mathbf{k}\alpha}$, $c_{\mathbf{k}\alpha}^*$ coincide, up to a factor $\sqrt{2\pi/\omega}$, with the coefficients $a_{\mathbf{k}\alpha}$, $a_{\mathbf{k}\alpha}^*$ in the expansion (2.4)).

The matrix elements of these operators are

$$\langle N_{\mathbf{k}\alpha} - 1 | \hat{c}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} \rangle = \langle N_{\mathbf{k}\alpha} | \hat{c}_{\mathbf{k}\alpha}^+ | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{N_{\mathbf{k}\alpha}}. \quad (2.15)$$

The commutation rule between $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$ follows from the definition (2.14) and the rule (2.9):

$$\hat{c}_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha}^+ - \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha} = 1. \quad (2.16)$$

Returning to the vector potential, we adopt an expansion of the same type as (2.4), but with the coefficients now promoted to operators:

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{A}_{\mathbf{k}\alpha}^*), \quad (2.17)$$

where

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} e^{i\mathbf{k}\mathbf{r}}. \quad (2.18)$$

Here $\mathbf{e}^{(\alpha)}$ denotes a unit vector specifying the polarization direction of the oscillator; the vectors $\mathbf{e}^{(\alpha)}$ are perpendicular to the wave vector $\mathbf{k}$, and for each $\mathbf{k}$ there are two independent polarizations.

Analogously, for the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ we write

$$\hat{\mathbf{E}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{E}_{\mathbf{k}\alpha}^*), \quad \hat{\mathbf{H}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{H}_{\mathbf{k}\alpha}^*), \quad (2.19)$$

with

$$\mathbf{E}_{\mathbf{k}\alpha} = i\omega \mathbf{A}_{\mathbf{k}\alpha}, \quad \mathbf{H}_{\mathbf{k}\alpha} = \mathbf{n} \times \mathbf{E}_{\mathbf{k}\alpha} \quad (\mathbf{n} = \mathbf{k}/\omega). \quad (2.20)$$

The vectors $\mathbf{A}_{\mathbf{k}\alpha}$ are mutually orthogonal in the sense that

$$\int \mathbf{A}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}'\alpha'}^* d^3x = \frac{2\pi}{\omega} \delta_{\alpha\alpha'} \delta_{\mathbf{k}\mathbf{k}'}. \quad (2.21)$$

To see this, note that when $\mathbf{A}_{\mathbf{k}\alpha}$ and $\mathbf{A}_{\mathbf{k}'\alpha'}^*$ carry different wave vectors their product involves $e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}}$, which integrates to zero over the volume; when only the polarizations differ, the orthogonality $\mathbf{e}^{(\alpha)} \mathbf{e}^{(\alpha')*} = 0$ of the two independent polarization directions provides the required vanishing. Analogous relations hold for the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$. Their normalization is conveniently written as

$$\frac{1}{4\pi} \int (\mathbf{E}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}'\alpha'}^* + \mathbf{H}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}'\alpha'}^*) d^3x = \omega \delta_{\mathbf{k}\mathbf{k}'} \delta_{\alpha\alpha'}. \quad (2.22)$$

Substituting the operators (2.19) into (2.10) and integrating with the help of (2.22), we get the field Hamiltonian expressed through the operators $\hat{c}$, $\hat{c}^+$:

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} \omega (\hat{c}_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha}^+ + \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha}). \quad (2.23)$$

In the representation under discussion (with the matrix elements of $\hat{c}$, $\hat{c}^+$ given by (2.15)) this operator is diagonal, and its eigenvalues coincide, naturally, with (2.12).

In classical theory the field momentum is defined by the integral

$$\mathbf{P} = \frac{1}{4\pi} \int \mathbf{E} \times \mathbf{H} d^3x.$$

On passing to the quantum theory by replacing $\mathbf{E}$ and $\mathbf{H}$ with the operators (2.19), one readily finds

$$\hat{\mathbf{P}} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}_{\mathbf{k}\alpha}^2 + \omega^2 \hat{Q}_{\mathbf{k}\alpha}^2) \mathbf{n} \quad (2.24)$$

— which mirrors the familiar classical link between energy and momentum of plane waves. This operator has eigenvalues

$$\mathbf{P} = \sum_{\mathbf{k}\alpha} \mathbf{k} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right). \quad (2.25)$$

The operator representation realized by the matrix elements (2.15) is called the “occupation-number representation”: it amounts to characterizing the state of the system (the field) through the quantum numbers $N_{\mathbf{k}\alpha}$ (*occupation numbers*). Within this representation the field operators (2.19) (and hence the Hamiltonian (2.11)) act on the system’s wave function, written as a function of the numbers $N_{\mathbf{k}\alpha}$; we denote it $\Phi(N_{\mathbf{k}\alpha}, t)$. The field operators (2.19) do not depend on time explicitly. This corresponds to the Schrödinger representation of operators, familiar from non-relativistic quantum mechanics. What does depend on time is the state of the system, $\Phi(N_{\mathbf{k}\alpha}, t)$, its time dependence being governed by the Schrödinger equation

$$i \frac{\partial \Phi}{\partial t} = \hat{H} \Phi.$$

This description of the field is relativistically invariant in essence, since it rests on the invariant Maxwell equations. Yet the invariance is not exhibited explicitly — above all because the spatial coordinates and the time enter the description in a highly asymmetric way.

Within a relativistic framework it is desirable to give the description a more overtly invariant appearance. For this purpose one uses the so-called Heisenberg representation, wherein the explicit dependence on time is shifted onto the operators (see Vol. III, § 13). Time and spatial coordinates then appear symmetrically in the field-operator expressions, while the system state Φ depends only on the occupation numbers.

For the operator $\hat{\mathbf{A}}$ the passage to the Heisenberg representation amounts to replacing, in each term of the sum in (2.17), (2.18), the factor $e^{i\mathbf{k}\mathbf{r}}$ by $e^{i(\mathbf{k}\mathbf{r}-\omega t)}$; that is, one is to understand by $\mathbf{A}_{\mathbf{k}\alpha}$ the time-dependent functions

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} e^{-i(\omega t - \mathbf{k}\mathbf{r})}. \quad (2.26)$$

This is easily verified by noting that the matrix element of a Heisenberg operator for the transition $i \rightarrow f$ must contain the factor $\exp\{-i(E_i - E_f)t\}$, where E_i and E_f are the energies of the initial and final states (see Vol. III, § 13). For a transition in which $N_{\mathbf{k}}$ decreases or increases by one, this factor reduces to $e^{-i\omega t}$ or $e^{i\omega t}$, respectively. The indicated replacement satisfies this requirement.

Henceforth (in treating both the electromagnetic field and the fields of particles) we shall always understand the Heisenberg representation of operators.

§3. Photons

Let us now turn to a discussion of the quantization formulae derived above.

To begin with, formula (2.12) for the energy of the field exposes the following difficulty. At the lowest energy level, all oscillator quantum numbers $N_{\mathbf{k}\alpha}$ vanish (this

state is known as the *electromagnetic field vacuum*). Yet even in this state every oscillator retains a nonzero “zero-point energy” $\omega/2$. Summing over the entire infinite set of oscillators then produces an infinite result. One is thus confronted with one of the “divergences” stemming from the absence of full logical self-consistency in the existing theory.

As long as one is concerned only with the eigenvalues of the field energy, this difficulty can be eliminated by simply subtracting away the zero-point oscillation energy, i.e. by writing for the energy and momentum of the field²

$$E = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha} \omega, \quad \mathbf{P} = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha} \mathbf{k}. \quad (3.1)$$

These formulae make it possible to introduce the concept fundamental to all of quantum electrodynamics: that of *light quanta*, or *photons*³. Namely, the free electromagnetic field can be regarded as a collection of particles, each carrying energy $\omega (= \hbar\omega)$ and momentum $\mathbf{k} (= \hbar\omega/c)$. The relationship between a photon’s energy and momentum is precisely what relativistic mechanics requires for particles of zero rest mass that propagate at the speed of light. The occupation numbers $N_{\mathbf{k}\alpha}$ acquire the meaning of the number of photons with given momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. The polarization property of a photon is analogous to the notion of spin for other particles (the specific features of the photon in this regard will be examined below, in § 6).

It is easy to see that the mathematical formalism developed in the preceding section is fully consistent with the picture of the electromagnetic field as an assembly of photons; it is nothing other than the apparatus of so-called second quantization as applied to a system of photons⁴. In this method (see Vol. III, § 64) the occupation numbers serve as the independent variables, and the operators act on functions of those numbers. The central role is played by the “annihilation” and “creation” operators for particles, which respectively decrease or increase an occupation number by one. Exactly such operators are $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$: the operator $\hat{c}_{\mathbf{k}\alpha}$ destroys a photon in the state $\mathbf{k}\alpha$, while $\hat{c}_{\mathbf{k}\alpha}^+$ creates one in that state.

The commutation rule (2.16) corresponds to particles obeying Bose statistics. Photons are therefore bosons, as one might have anticipated: the number of photons occupying any given state can be arbitrary (we shall return to the significance of this fact in § 5).

The plane waves $\mathbf{A}_{\mathbf{k}\alpha}$ (2.26), which appear in the operator $\hat{\mathbf{A}}$ (2.17) as coefficients multiplying the photon annihilation operators, can be interpreted as the wave functions of photons possessing definite momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. This interpretation corresponds to expanding the ψ -operator as a series in the stationary-state wave functions of the particle within the non-relativistic second-quantization framework (unlike the

²This subtraction can be carried out in a formally consistent manner by agreeing to interpret the operator products in (2.10) as “normal-ordered”, i.e. arranged so that the operators $\hat{c}^+$ always stand to the left of the operators $\hat{c}$. Formula (2.23) then takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} (\omega \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha}).$$

³The photon concept was first put forward by *Einstein* (A. Einstein, 1905).

⁴The method of second quantization as applied to radiation theory was developed by *Dirac* (P. A. M. Dirac, 1927).

non-relativistic case, however, the expansion (2.17) involves both annihilation and creation operators; the significance of this distinction will become clear later, see § 12).

The wave function (2.26) is normalized by the condition

$$\int \frac{1}{4\pi} (|\mathbf{E}_{\mathbf{k}\alpha}|^2 + |\mathbf{H}_{\mathbf{k}\alpha}|^2) d^3x = \omega \quad (3.2)$$

This amounts to normalizing to one photon in a volume $V = 1$. Indeed, the integral on the left-hand side of the equation represents the quantum-mechanical expectation value of the photon's energy in a state described by the given wave function⁵. On the right-hand side of (3.2) stands the energy of a single photon.

The role of the “Schrödinger equation” for the photon is played by Maxwell's equations. In the present case (for the potential $\mathbf{A}(\mathbf{r}, t)$ satisfying condition (2.1)) this is the wave equation

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta \mathbf{A} = 0.$$

In the general case of arbitrary stationary states, the photon “wave functions” are complex solutions of this equation whose time dependence enters through the factor $e^{-i\omega t}$.

Speaking of the photon wave function, let us emphasize once more that it cannot by any means be treated as a probability amplitude for the spatial localization of a photon — in contrast to the basic meaning of the wave function in non-relativistic quantum mechanics. This is linked to the fact that (as noted in § 1) the notion of photon coordinates has no physical meaning at all. We shall revisit the mathematical side of this issue at the end of the next section.

The Fourier-transform components of $\mathbf{A}(\mathbf{r}, t)$ with respect to the coordinates constitute the photon wave function in the momentum representation; we denote it $\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k})e^{-i\omega t}$. Thus, for a state with a definite momentum $\mathbf{k}$ and polarization $\mathbf{e}^{(\alpha)}$ the momentum-representation wave function is given simply by the coefficient of the exponential factor in (2.26):

$$\mathbf{A}_{\mathbf{k}\alpha}(\mathbf{k}', \alpha') = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} \delta_{\mathbf{k}'\mathbf{k}} \delta_{\alpha\alpha'}. \quad (3.3)$$

In keeping with the measurability of the momentum of a free particle, the momentum-representation wave function carries deeper physical content than its coordinate-space counterpart: it provides a way to compute the probabilities $w_{\mathbf{k}\alpha}$ of the various values of momentum and polarization for a photon in a given state. According to the general rules of quantum mechanics, $w_{\mathbf{k}\alpha}$ is given by the squared modulus of the expansion coefficients of $\mathbf{A}(\mathbf{k}')$ in the wave functions of states with definite $\mathbf{k}$ and $\mathbf{e}^{(\alpha)}$:

$$w_{\mathbf{k}\alpha} \propto \left| \sum_{\mathbf{k}'\alpha'} \mathbf{A}_{\mathbf{k}\alpha}^*(\mathbf{k}', \alpha') \mathbf{A}(\mathbf{k}') \right|^2$$

⁵Note that the coefficient $1/(4\pi)$ in the integral (3.2) is twice the usual coefficient $1/(8\pi)$ in (2.10). This difference is ultimately connected with the complex character of the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$, as opposed to the Hermitian field operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$.

(the proportionality coefficient depends on the choice of normalization for the functions). Substituting (3.3), we obtain

$$w_{\mathbf{k}\alpha} \propto |\mathbf{e}^{(\alpha)} \mathbf{A}(\mathbf{k})|^2. \quad (3.4)$$

Summing over the two polarizations, we find the probability that a photon has momentum $\mathbf{k}$:

$$w_{\mathbf{k}} \propto |\mathbf{A}(\mathbf{k})|^2. \quad (3.5)$$

§4. Gauge invariance

As is well known, the choice of field potentials in classical electrodynamics is not unique: the components of the 4-potential A_μ may be subjected to an arbitrary *gauge* (or *gradient*) transformation of the form

$$A_\mu \rightarrow A_\mu + \partial_\mu \chi, \quad (4.1)$$

where χ is an arbitrary function of the coordinates and time (see Vol. II, § 18). In the case of a plane wave, restricting oneself to transformations that preserve the functional form of the potential (its proportionality to $\exp(-ik_\mu x^\mu)$), the non-uniqueness amounts to the freedom of adding to the wave amplitude an arbitrary 4-vector proportional to k^μ .

The ambiguity of the potential persists, of course, in the quantum theory as well — as applied to the field operators or to the photon wave functions. Without prejudging the choice of potentials, one should write in place of (2.17) an analogous expansion for the operator 4-potential

$$\hat{A}^\mu = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} A_{\mathbf{k}\alpha}^\mu + \hat{c}_{\mathbf{k}\alpha}^+ A_{\mathbf{k}\alpha}^{\mu*}), \quad (4.2)$$

where the wave functions $A_{\mathbf{k}\alpha}^\mu$ are 4-vectors of the form

$$A_{\mathbf{k}}^\mu = \sqrt{4\pi} \frac{e^\mu}{\sqrt{2\omega}} e^{-ik_\nu x^\nu}, \quad e_\mu e^{\mu*} = -1,$$

or, in a compact notation that suppresses the four-dimensional vector indices:

$$A_{\mathbf{k}} = \sqrt{4\pi} \frac{e}{\sqrt{2\omega}} e^{-ikx}, \quad ee^* = -1. \quad (4.3)$$

Here the 4-momentum is $k_\mu = (\omega, \mathbf{k})$ (so that $kx = \omega t - \mathbf{k}\mathbf{r}$), and e is the unit polarization 4-vector⁶.

If one restricts oneself to gauge transformations that do not alter how the function (4.3) depends on the coordinates and time, they amount to the replacement

$$e_\mu \rightarrow e_\mu + \chi k_\mu, \quad (4.4)$$

⁶Expression (4.3) does not have an entirely relativistically covariant (4-vector) form, which is related to the non-invariant character of our normalization to a finite volume $V = 1$. This is, however, of no fundamental significance and is fully compensated by the convenience of such a normalization scheme. We shall see later that it ensures simple and automatic derivation of real physical quantities in the proper invariant form.

where $\chi = \chi(k^\mu)$ is an arbitrary function. The transversality of the polarization means that one can always choose a gauge in which the 4-vector e takes the form

$$e^\mu = (0, \mathbf{e}), \quad \mathbf{ek} = 0 \quad (4.5)$$

(we shall call such a gauge *three-dimensionally transverse*). In an invariant four-dimensional notation this requirement is written as the condition of *four-dimensional transversality*

$$ek = 0. \quad (4.6)$$

Note that this condition (as well as the normalization condition $ee^* = -1$) is not violated by the transformation (4.4), by virtue of $k^2 = 0$. On the other hand, the vanishing of the 4-momentum squared of a particle signifies that its mass is zero. A direct link between gauge invariance and the vanishing of the photon mass is thereby revealed (further aspects of this connection will be discussed in § 14).

No measurable physical quantity should change under a gauge transformation of the wave functions of the photons taking part in a given process. This requirement of *gauge invariance* plays an even greater role in quantum electrodynamics than in the classical theory. As we shall see from numerous examples, it serves here, alongside the requirement of relativistic invariance, as a powerful heuristic principle.

In turn, the gauge invariance of the theory is closely bound up with the law of conservation of electric charge; we shall dwell on this aspect in § 43.

We already remarked in the preceding section that the coordinate-space wave function of a photon cannot be interpreted as a probability amplitude for its spatial localization. From a mathematical standpoint this circumstance manifests itself in the impossibility of constructing, out of the wave function, a quantity whose formal properties alone would qualify it to serve as a probability density. Such a quantity would have to be an essentially positive bilinear combination of the wave function A_μ and its complex conjugate. Moreover, it would need to satisfy definite requirements of relativistic covariance — namely, to be the time component of a 4-vector (the point being that the continuity equation expressing conservation of particle number takes the four-dimensional form of a vanishing 4-divergence of the current 4-vector; the time component of this current is, in the present context, the probability density for the localization of the particle; see Vol. II, § 29). On the other hand, gauge invariance demands that A_μ enter the current only through the antisymmetric tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -i(k_\mu A_\nu - k_\nu A_\mu)$. The current 4-vector would therefore have to be assembled bilinearly from $F_{\mu\nu}$ and $F_{\mu\nu}^*$ (together with components of the 4-vector k_μ). But no such 4-vector can be constructed at all: every expression satisfying the stated conditions (for instance, $k^\lambda F_{\mu\nu}^* F_{\lambda}{}^\nu$) vanishes by the transversality condition ($k^\lambda F_{\nu\lambda} = 0$), quite apart from the fact that it would not be essentially positive (since it contains odd powers of the components k_μ).

§ 5. The electromagnetic field in quantum theory

Treating the field as a collection of photons is the sole description that fully captures the physical meaning of the electromagnetic field within quantum theory. It replaces the classical characterization through field strengths. The field strengths themselves enter the mathematical apparatus of the photon picture as operators of second quantization.

It is well known that a quantum system behaves classically when the quantum numbers characterizing its stationary states become large. For the free electromagnetic field (within a given volume) this requires that the oscillator quantum numbers — that is, the photon occupation numbers $N_{\mathbf{k}\alpha}$ — be large. The fact that photons obey Bose statistics is of profound importance in this context. In the mathematical formalism, the link between Bose statistics and classical field behavior is encoded in the commutation rules for $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^\dagger$. When $N_{\mathbf{k}\alpha}$ is large, so that the matrix elements of these operators are also large, the unity appearing on the right-hand side of the commutation relation (2.16) may be dropped, giving

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}^\dagger \simeq \hat{c}_{\mathbf{k}\alpha}^\dagger\hat{c}_{\mathbf{k}\alpha},$$

i.e. the operators go over into mutually commuting classical quantities $c_{\mathbf{k}\alpha}$, $c_{\mathbf{k}\alpha}^*$ that determine the classical field strengths.

The quasi-classicality condition for the field requires, however, a further refinement. The point is that if all the numbers $N_{\mathbf{k}\alpha}$ are large, then upon summing over every state $\mathbf{k}\alpha$ the field energy is in any case infinite, rendering the condition meaningless.

A physically sensible formulation of the quasi-classicality criterion rests on examining values of the field averaged over some small time interval Δt . If one writes the classical electric field $\bar{\mathbf{E}}$ (or the magnetic field $\bar{\mathbf{H}}$) as a Fourier integral over time, then upon time-averaging over an interval Δt only those Fourier components whose frequencies satisfy $\omega\Delta t \lesssim 1$ contribute appreciably to the mean value of $\bar{\mathbf{E}}$; in the opposite limit the oscillating factor $e^{-i\omega t}$ nearly averages to zero. Consequently, in establishing the quasi-classicality condition for the time-averaged field, one need consider only those quantum oscillators whose frequencies satisfy $\omega \lesssim 1/\Delta t$. It suffices to require that the quantum numbers of these oscillators be large. The number of oscillators with frequencies between zero and $\omega \sim 1/\Delta t$ (referred to the volume $V = 1$) is of order⁷

$$\left(\frac{\omega}{c}\right)^3 \sim \frac{1}{(c\Delta t)^3}. \quad (5.1)$$

The total energy of the field per unit volume is of order $\bar{\mathbf{E}}^2$. Dividing this quantity by the number of oscillators and by a characteristic energy of one photon ($\sim \hbar\omega$), one obtains an estimate for the photon occupation numbers

$$N_{\mathbf{k}} \sim \frac{\bar{\mathbf{E}}^2 c^3}{\hbar\omega^4}.$$

Demanding that this number be large, we arrive at the inequality

$$|\mathbf{E}| \gg \frac{\sqrt{\hbar c}}{(c\Delta t)^2}. \quad (5.2)$$

This is the sought-after condition permitting a classical treatment of the time-averaged (over intervals Δt) field. We see that the field must be sufficiently strong — all the stronger, the shorter the averaging interval Δt . For time-varying fields this interval must not, of course, exceed the characteristic periods over which the field changes appreciably.

⁷In this section we use ordinary units.

Consequently, sufficiently weak time-varying fields can under no circumstances be quasi-classical. Only for static (time-independent) fields may one set $\Delta t \rightarrow \infty$, so that the right-hand side of the inequality (5.2) vanishes. This means that a static field is always classical.

It has already been noted that the classical expressions for the electromagnetic field in the form of a superposition of plane waves must be treated as operator-valued in the quantum theory. The physical meaning of these operators is, however, rather limited. Indeed, a physically meaningful field operator ought to yield vanishing field values in the photon-vacuum state. Yet the expectation value of the squared-field operator $\hat{\mathbf{E}}^2$ in the ground state, which coincides up to a factor with the zero-point energy of the field, turns out to be infinite (by “expectation value” we mean the quantum-mechanical average, i.e. the corresponding diagonal matrix element of the operator). This divergence cannot be circumvented even by any formal subtraction procedure (as can be done for the field energy), since in the present case we would need to carry out the subtraction through some sensible modification of the operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$ themselves (rather than their squares), which proves to be impossible.

§ 6. Angular momentum and parity of the photon

As with any particle, the photon can possess a definite angular momentum. In order to elucidate the properties of this quantity for the photon, let us begin by recalling how the angular momentum of a particle is connected with its wave function in the formalism of quantum mechanics.

The total angular momentum $\mathbf{j}$ of a particle consists of the orbital part $\mathbf{l}$ and the intrinsic part — the spin $\mathbf{s}$. A particle of spin s is described by a symmetric spinor of rank $2s$, i.e. a set of $2s + 1$ components that transform into one another under coordinate-system rotations according to a specific law. The orbital angular momentum is instead linked to how the wave functions depend on coordinates: states with orbital angular momentum l have wave-function components that are expressed (linearly) in terms of spherical harmonics of order l .

The ability to distinguish consistently between spin and orbital angular momentum therefore demands that the “spin” and “coordinate” properties of the wave functions be independent: the coordinate dependence of each spinor component (at a given instant of time) should not be constrained by any supplementary conditions. Passing to the momentum representation, coordinate dependence becomes dependence on $\mathbf{k}$. In the three-dimensionally transverse gauge the photon wave function is the vector $\mathbf{A}(\mathbf{k})$. Because a vector is equivalent to a rank-2 spinor, one could formally ascribe spin 1 to the photon. This vector wave function, however, must satisfy the transversality requirement $\mathbf{k}\mathbf{A}(\mathbf{k}) = 0$, an extra constraint on the way $\mathbf{A}(\mathbf{k})$ depends on momentum. It follows that the dependence on $\mathbf{k}$ can no longer be chosen freely for every vector component at once, making it impossible to disentangle orbital angular momentum and spin.

We note also that for the photon one cannot define spin as the angular momentum of the particle at rest, since for a photon traveling at the speed of light no rest frame exists at all.

Thus, for a photon one can speak only of its total angular momentum. It is clear

from the outset that the total angular momentum can assume only integer values. This is already evident from the fact that among the quantities characterizing the photon there are no spinors of odd rank.

As for any particle, the state of a photon is further characterized by its parity, associated with the behavior of the wave function under spatial inversion (see Vol. III, § 30). In the momentum representation, a reversal of the sign of the coordinates corresponds to reversing the sign of every component of $\mathbf{k}$. The action of the inversion operator $\hat{P}$ on a scalar function $\varphi(\mathbf{k})$ amounts simply to this sign change: $\hat{P}\varphi(\mathbf{k}) = \varphi(-\mathbf{k})$. When acting on the vector function $\mathbf{A}(\mathbf{k})$, one must additionally take into account that reversing the axes also reverses the sign of every component of the vector; hence⁸

$$\hat{P}\mathbf{A}(\mathbf{k}) = -\mathbf{A}(-\mathbf{k}). \quad (6.1)$$

Although the decomposition of the photon's angular momentum into orbital and spin parts lacks physical meaning, it is nevertheless convenient to introduce "spin" s and "orbital" angular momentum l in a formal sense as auxiliary concepts describing the transformation properties of the wave function with respect to rotations: the value $s = 1$ reflects the vectorial character of the wave function, while l is the order of the spherical harmonics entering it. Here we have in mind the wave functions of states with definite values of the photon's angular momentum, which for a free particle take the form of spherical waves. The number l determines, in particular, the parity of the photon state, equal to

$$P = (-1)^{l+1} \quad (6.2)$$

In the same formal sense one can write the angular-momentum operator $\hat{\mathbf{j}}$ as the sum $\hat{\mathbf{s}} + \hat{\mathbf{l}}$. As is well known, the angular-momentum operator is associated with infinitesimally small rotations of the coordinate system; here, with the action of such a rotation on a vector field. Within the decomposition $\hat{\mathbf{s}} + \hat{\mathbf{l}}$, the operator $\hat{\mathbf{s}}$ acts on the vector index and intermixes the various components of the vector. The operator $\hat{\mathbf{l}}$ acts on these same components viewed as functions of momentum (or of coordinates). Let us count the number of states (at a given energy) that are possible for a given value j of the photon's angular momentum (setting aside the trivial $(2j + 1)$ -fold degeneracy with respect to the direction of the angular momentum).

For independent $\mathbf{l}$ and $\mathbf{s}$ this counting is carried out simply by enumerating the ways in which, according to the rules of the vector model, the angular momenta $\hat{\mathbf{l}}$ and $\mathbf{s}$ can be composed so as to yield the required value of j . For a particle with spin $s = 1$ we would find in this way (for a given nonzero value of j) three states with the following values of l and parity:

$$l = j, \quad P = (-1)^{l+1} = (-1)^{j+1}; \quad l = j \pm 1, \quad P = (-1)^{l+1} = (-1)^j.$$

⁸We adopt the convention of defining the parity of a state by the action of the inversion operator on the polar vector $\mathbf{A}$ (or, equivalently, the corresponding electric vector $\mathbf{E} = i\omega\mathbf{A}$). This differs in sign from the action on the axial vector $\mathbf{H} = i[\mathbf{k}\mathbf{A}]$, since inversion does not reverse the direction of such a vector:

$$\hat{P}\mathbf{H}(\mathbf{k}) = \mathbf{H}(-\mathbf{k}).$$

If $j = 0$, there is only a single state (with $l = 1$) having parity $P = +1$.

This counting, however, has not accounted for the transversality condition on $\mathbf{A}$; all three components were regarded as independent. From the result obtained one must therefore subtract the number of states associated with a longitudinal vector. A longitudinal vector can be expressed as $\mathbf{k}\varphi(\mathbf{k})$, and it is apparent that, owing to its transformation behavior (with respect to rotations), its three components reduce to a single scalar φ ⁹. The extra state that is excluded by transversality would therefore belong to a particle whose wave function is a scalar (a spinor of rank zero), that is, a particle with “spin 0”¹⁰. For this state the angular momentum j equals the order of the spherical harmonics appearing in φ . Its parity, viewed as a photon state, follows from how the inversion operator acts on the vector function $\mathbf{k}\varphi$:

$$\hat{P}(\mathbf{k}\varphi) = -(-\mathbf{k})\varphi(-\mathbf{k}) = (-1)^j \mathbf{k}\varphi(\mathbf{k}),$$

i.e. it equals $(-1)^j$. Thus, from the above count of states with parity $(-1)^j$ (two for $j \neq 0$ and one for $j = 0$) one must subtract one.

We arrive finally at the result that for nonzero photon angular momentum j there exist one even-parity and one odd-parity state. For $j = 0$ we obtain no states whatsoever. This means that the photon can never have zero angular momentum, so that j takes only the values $1, 2, 3, \dots$. The impossibility of $j = 0$ is, moreover, obvious: the wave function of a state with vanishing angular momentum would have to be spherically symmetric, which is manifestly impossible for a transverse wave.

There is a standard nomenclature for the various photon states. When a photon carries angular momentum j with parity $(-1)^j$ it is called an *electric 2^j -pole* (or Ej) photon; when its parity is $(-1)^{j+1}$ it is called a *magnetic 2^j -pole* (or Mj) photon. In particular, the electric dipole photon has odd parity and $j = 1$; the electric quadrupole photon has even parity and $j = 2$; the magnetic dipole photon has even parity and $j = 1$ ¹¹.

§7. Spherical waves of photons

Having established the admissible values of the photon's angular momentum, our task is now to find the wave functions that correspond to them¹².

Let us first consider a formal problem: to identify the vector functions that serve as eigenfunctions of the operators $\hat{\mathbf{j}}^2$ and $\hat{j}_z$; in doing so we do not stipulate in advance which particular functions enter the photon wave functions of interest, nor do we impose the transversality constraint.

⁹Indeed, when one speaks of the transformation character of a quantity under rotation, one means a transformation at a given point, i.e. at a given $\mathbf{k}$. Under such a transformation $\mathbf{k}\varphi(\mathbf{k})$ does not change at all, i.e. it behaves as a scalar.

¹⁰Let us emphasize once more that what is meant here is not the state of any real particle. The counting carried out here is purely formal and amounts, mathematically, to sorting the complete collection of quantities that mix under rotations into irreducible representations of the rotation group.

¹¹This nomenclature mirrors the usage in the classical theory of radiation: as will be shown later (§§ 46–47), the production of electric- and magnetic-type photons is controlled by the respective electric and magnetic multipole moments of the system of charges.

¹²This problem was first studied by Heitler (W. Heitler, 1936). The form of the solution presented here is due to V. B. Berestetskii (1947).

The functions will be sought in the momentum representation, where the coordinate operator takes the form $\hat{\mathbf{r}} = i\partial/\partial\mathbf{k}$ (see III, (15.12)). For the orbital angular momentum operator one then obtains

$$\hat{\mathbf{l}} = [\hat{\mathbf{r}}\mathbf{k}] = -i \left[\mathbf{k} \frac{\partial}{\partial\mathbf{k}} \right],$$

i.e. it differs from the angular momentum operator in coordinate space only by the replacement of $\mathbf{r}$ with $\mathbf{k}$. Accordingly, the formal solution of the problem is identical in both representations.

Let us denote the desired eigenfunctions by $\mathbf{Y}_{jm}$ and call them *spherical vectors*. They must satisfy the equations

$$\hat{\mathbf{j}}^2 = j(j+1)\mathbf{Y}_{jm}, \quad \hat{j}_z \mathbf{Y}_{jm} = m\mathbf{Y}_{jm} \quad (7.1)$$

(the z axis being a fixed direction in space). We shall demonstrate that any function of the form $\mathbf{a}Y_{jm}$ possesses this property, where $\mathbf{a}$ is a vector constructed from the unit vector $\mathbf{n} = \mathbf{k}/\omega$ and Y_{jm} are the ordinary (scalar) spherical harmonics. The latter will everywhere be defined in accordance with III, § 28:

$$Y_{lm}(\mathbf{n}) = (-1)^{\frac{m+|m|}{2}} i^l \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}} P_l^{|m|}(\cos\theta) e^{im\varphi} \quad (7.2)$$

(θ, φ being the spherical angles specifying the direction of $\mathbf{n}$)¹³.

To show this, recall the commutation rule III, (29.4):

$$\{\hat{l}_i, a_k\}_- = ie_{ikl}a_l.$$

The right-hand side can be written as $(-\hat{s}_i a_k)$, where $\hat{\mathbf{s}}$ is the spin-1 operator (its action on a vector function is precisely $\hat{s}_i a_k = -ie_{ikl}a_l$; see III, § 57, problem 2). Consequently,

$$\hat{l}_i a_k - a_k \hat{l}_i = -\hat{s}_i a_k.$$

Making use of this relation, we find

$$\hat{j}_i a_k = (\hat{l}_i + \hat{s}_i) a_k = a_k \hat{l}_i.$$

It follows that

$$\hat{\mathbf{j}}^2(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{\mathbf{l}}^2 Y_{jm}, \quad \hat{j}_z(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}_z Y_{jm}.$$

Since the spherical harmonic Y_{jm} is an eigenfunction of $\hat{l}^2$ and $\hat{l}_z$ with eigenvalues $j(j+1)$ and m respectively, we arrive at equations (7.1).

¹³For future reference, note the value of these functions at $\theta = 0$ ($\mathbf{n}$ along the z axis):

$$Y_{lm}(\mathbf{n}_z) = i^l \sqrt{\frac{2l+1}{4\pi}} \delta_{m0}. \quad (7.2a)$$

Three essentially distinct types of spherical vectors are obtained by choosing the vector $\mathbf{a}$ to be one of the following¹⁴:

$$\frac{\nabla_{\mathbf{n}}}{\sqrt{j(j+1)}}, \quad \frac{[\mathbf{n}\nabla_{\mathbf{n}}]}{\sqrt{j(j+1)}}, \quad \mathbf{n}. \quad (7.3)$$

The spherical vectors are thus defined as follows:

$$\begin{aligned} \mathbf{Y}_{jm}^{(e)} &= \frac{1}{\sqrt{j(j+1)}} \nabla_{\mathbf{n}} Y_{jm}, & P &= (-1)^j; \\ \mathbf{Y}_{jm}^{(m)} &= [\mathbf{n}\mathbf{Y}_{jm}^{(e)}], & P &= (-1)^{j+1}; \\ \mathbf{Y}_{jm}^{(l)} &= \mathbf{n}Y_{jm}, & P &= (-1)^j. \end{aligned} \quad (7.4)$$

Next to each vector we have indicated its parity P . The spherical vectors of the three types are mutually orthogonal; $\mathbf{Y}_{jm}^{(l)}$ is longitudinal, while $\mathbf{Y}_{jm}^{(e)}$ and $\mathbf{Y}_{jm}^{(m)}$ are transverse with respect to $\mathbf{n}$.

The spherical vectors can be expressed through scalar spherical harmonics. In this expansion, $\mathbf{Y}_{jm}^{(m)}$ involves harmonics of a single order $l = j$ only, whereas $\mathbf{Y}_{jm}^{(e)}$ and $\mathbf{Y}_{jm}^{(l)}$ involve harmonics of orders $l = j \pm 1$. This is self-evident: one need only compare the parities listed in (7.4) with the parity $(-1)^{l+1}$ of a vector field, expressed via the order of the spherical harmonics it contains.

The spherical vectors within each type are mutually orthogonal and normalized according to

$$\int \mathbf{Y}_{jm} \mathbf{Y}_{j'm'}^* d\Omega = \delta_{jj'} \delta_{mm'}. \quad (7.5)$$

For the vectors $\mathbf{Y}_{jm}^{(l)}$ this follows immediately from the normalization of the spherical harmonics Y_{jm} . For $\mathbf{Y}_{jm}^{(e)}$ the normalization integral becomes

$$\frac{1}{j(j+1)} \int \nabla_{\mathbf{n}} Y_{jm} \nabla_{\mathbf{n}} Y_{j'm'}^* d\Omega = -\frac{1}{j(j+1)} \int Y_{j'm'}^* \Delta_{\mathbf{n}} Y_{jm} d\Omega,$$

and since $\Delta_{\mathbf{n}} Y_{jm} = -j(j+1)Y_{jm}$, we recover (7.5). The normalization of the vectors $\mathbf{Y}_{jm}^{(m)}$ reduces to an identical integral.

We remark that the spherical vectors (7.4) could also have been obtained without the direct verification of equations (7.1) carried out above — simply on the basis of general transformation-property arguments. Such reasoning led us in the preceding section to conclude that a vector function of the form $\mathbf{n}\varphi$ corresponds to a value of j equal to

¹⁴The operator $\nabla_{\mathbf{n}} \equiv |\mathbf{k}| \nabla_{\mathbf{k}}$ acts on functions that depend only on the direction of $\mathbf{n}$. In spherical coordinates it has just two components:

$$\nabla_{\mathbf{n}} = \left(\frac{\partial}{\partial \theta}, \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} \right).$$

The operator denoted below by $\Delta_{\mathbf{n}}$ is the angular part of the Laplacian:

$$\Delta_{\mathbf{n}} = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}.$$

the order of the spherical harmonics appearing in φ ; setting $\varphi = Y_{jm}$ further fixes the projection m . In this way one immediately arrives at the spherical vectors $\mathbf{Y}_{jm}^{(1)}$. The transformation-property arguments of § 6, however, remain valid if the factor $\mathbf{n}$ in the product $\mathbf{n}\varphi$ is replaced by $\nabla_{\mathbf{n}}$ or by $[\mathbf{n}\nabla_{\mathbf{n}}]$, yielding in this manner the spherical vectors of the other two types.

Let us return to the photon wave functions. For an electric-type photon (Ej) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^j$. Among the spherical vectors, $\mathbf{Y}_{jm}^{(e)}$ and $\mathbf{Y}_{jm}^{(l)}$ possess this parity; of the two, however, only the former satisfies the transversality condition. For a magnetic-type photon (Mj) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^{j+1}$; only the spherical vector $\mathbf{Y}_{jm}^{(m)}$ has this parity. The wave functions of a photon with definite angular momentum j , projection m , and energy ω are therefore

$$\mathbf{A}_{\omega jm}(\mathbf{k}) = \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) \mathbf{Y}_{jm}(\mathbf{n}), \quad (7.6)$$

where one should write $\mathbf{Y}_{jm}^{(e)}$ or $\mathbf{Y}_{jm}^{(m)}$ for $\mathbf{Y}_{jm}$ according as the photon is of electric or magnetic type; the specified energy is incorporated through the factor $\delta(|\mathbf{k}| - \omega)$. The functions (7.6) are normalized by the condition

$$\frac{1}{(2\pi)^4} \int \omega \omega' \mathbf{A}_{\omega' j' m'}^*(\mathbf{k}) \mathbf{A}_{\omega jm}(\mathbf{k}) d^3 k = \omega \delta(\omega' - \omega) \delta_{jj'} \delta_{mm'}. \quad (7.7)$$

For the coordinate-representation wave functions, condition (7.7) is equivalent to¹⁵

$$\frac{1}{4\pi} \int \left\{ \mathbf{E}_{\omega' j' m'}^*(\mathbf{r}) \mathbf{E}_{\omega jm}(\mathbf{r}) + \mathbf{H}_{\omega' j' m'}^*(\mathbf{r}) \mathbf{H}_{\omega jm}(\mathbf{r}) \right\} d^3 x = \omega \delta(\omega' - \omega) \delta_{jj'} \delta_{mm'}. \quad (7.8)$$

Indeed, when written in terms of potentials, the integral on the left-hand side takes the form

$$\frac{1}{2\pi} \int \mathbf{A}_{\omega' j' m'}^*(\mathbf{r}) \mathbf{A}_{\omega jm}(\mathbf{r}) \omega' \omega d^3 x.$$

One must substitute here the Fourier expansions

$$\begin{aligned} \mathbf{A}_{\omega jm}(\mathbf{r}) &= \int \mathbf{A}_{\omega jm}(\mathbf{k}) e^{i\mathbf{k}\mathbf{r}} \frac{d^3 k}{(2\pi)^3}, \\ \mathbf{A}_{\omega' j' m'}^*(\mathbf{r}) &= \int \mathbf{A}_{\omega' j' m'}^*(\mathbf{k}') e^{-i\mathbf{k}'\mathbf{r}} \frac{d^3 k'}{(2\pi)^3}. \end{aligned} \quad (7.9)$$

After that, integration over $d^3 x$ produces the delta function $(2\pi)^3 \delta(\mathbf{k}' - \mathbf{k})$, which is removed by the integration over $d^3 k'$, bringing the integral to the form (7.7).

Up to this point we have assumed the transverse gauge for the potentials, in which the scalar potential $\Phi = 0$. In various applications, however, other gauge choices for a spherical wave may prove more convenient. An admissible gauge transformation of the potentials in the momentum representation consists of the substitution

$$\mathbf{A} \rightarrow \mathbf{A} + \mathbf{n}f(\mathbf{k}), \quad \Phi \rightarrow \Phi + f(\mathbf{k}),$$

¹⁵This condition is of the same type as (2.22). The appearance of the factor $\delta(\omega' - \omega)$ on the right-hand side is due to the fact that the field (a spherical wave) is being considered in all of infinite space rather than in a finite volume $V = 1$.

with $f(\mathbf{k})$ an arbitrary function. We choose it here so that the new potentials are still expressed through the same spherical harmonics and continue to have definite parity. For an electric-type photon these requirements restrict the potentials to the following form:

$$\begin{aligned} \mathbf{A}_{\omega jm}^{(e)}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) (\mathbf{Y}_{jm}^{(e)} + C \mathbf{n} Y_{jm}), \\ \Phi_{\omega jm}^{(e)}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) C Y_{jm}, \end{aligned} \quad (7.10)$$

where C is an arbitrary constant. For a magnetic-type photon, on the other hand, such an addition to $\mathbf{A}^{(m)}(\mathbf{k})$ would destroy its definite parity, so under the same conditions the choice (7.6) is uniquely fixed.

The probability that a photon of given angular momentum and parity is detected traveling in the direction $\mathbf{n}$ within the solid-angle element do equals, by (3.5) and (7.6),

$$w(\mathbf{n})do = |\mathbf{Y}_{jm}^{(e)}|^2 do. \quad (7.11)$$

We have written the expression for an E -type photon. Since, however, $|\mathbf{Y}_{jm}^{(m)}|^2 = |\mathbf{Y}_{jm}^{(e)}|^2$, the probability distributions $w(\mathbf{n})$ for photons of both types are identical.

The squared modulus $|\mathbf{Y}_{jm}^{(e)}|^2$ is independent of the azimuthal angle φ (the factors $e^{\pm im\varphi}$ in the spherical harmonics cancel). The probability distribution $w(\mathbf{n})$ is therefore axially symmetric about the z axis. Furthermore, since each spherical vector has definite parity, the squared moduli are even under inversion, i.e. under the replacement $\theta \rightarrow \pi - \theta$; this means that the function $w(\theta)$, when expanded in Legendre polynomials, contains only polynomials of even order. The determination of the expansion coefficients reduces to computing integrals of products of three spherical harmonics followed by summation over components. Both steps are carried out with the aid of formulas derived in III, §§ 107–108, and lead to the following result:

$$\begin{aligned} w(\theta) &= (-1)^{m+1} \frac{(2j+1)^2}{4\pi} \sum_{n=0}^{\infty} (4n+1) \begin{pmatrix} j & j & 2n \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} j & j & 2n \\ m & -m & 0 \end{pmatrix} \times \\ &\quad \times \left\{ \begin{matrix} j & j & 1 \\ j & j & 2n \end{matrix} \right\} P_{2n}(\cos \theta). \end{aligned} \quad (7.12)$$

Finally, we present expressions for the components of the spherical vectors as expansions in spherical harmonics. We employ the “spherical components” of a vector $\mathbf{f}$, defined in accordance with III, § 107; the components f_λ of the vector $\mathbf{f}$ are

$$f_0 = if_z, \quad f_{+1} = -\frac{i}{\sqrt{2}}(f_x + if_y), \quad f_{-1} = \frac{i}{\sqrt{2}}(f_x - if_y). \quad (7.13)$$

Introducing “circular basis vectors”:

$$\mathbf{e}^{(0)} = i\mathbf{e}^{(z)}, \quad \mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}(\mathbf{e}^{(x)} + i\mathbf{e}^{(y)}), \quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}(\mathbf{e}^{(x)} - i\mathbf{e}^{(y)}) \quad (7.14)$$

($\mathbf{e}^{(x,y,z)}$ being the unit vectors along the x, y, z axes), we have

$$\mathbf{f} = \sum_{\lambda} (-1)^{1-\lambda} f_{-\lambda} \mathbf{e}^{(\lambda)}, \quad f_{\lambda} = (-1)^{1-\lambda} \mathbf{f} \mathbf{e}^{(-\lambda)*} = \mathbf{f} \mathbf{e}^{(\lambda)}. \quad (7.15)$$

The spherical components of the spherical vectors are expressed through $3j$ -symbols and spherical harmonics by the following formulas:

$$\begin{aligned} (-1)^{j+m+\lambda+1} (\mathbf{Y}_{jm}^{(e)})_{\lambda} &= -\sqrt{j} \begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\ &\quad + \sqrt{j+1} \begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}, \\ (-1)^{j+m+\lambda+1} (\mathbf{Y}_{jm}^{(m)})_{\lambda} &= -\sqrt{2j+1} \begin{pmatrix} j & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j,m+\lambda}, \\ (-1)^{j+m+\lambda+1} (\mathbf{Y}_{jm}^{(l)})_{\lambda} &= \sqrt{j+1} \begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\ &\quad + \sqrt{j} \begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}. \end{aligned} \quad (7.16)$$

These formulas are derived as follows. Each of the three spherical vectors has the form $\mathbf{Y}_{jm} = \mathbf{a} Y_{jm}$, where $\mathbf{a}$ is one of the three vectors (7.3). Therefore

$$\mathbf{Y}_{jm} = \sum_{lm'} \langle lm' | \mathbf{a} | jm \rangle Y_{lm'},$$

so that one must determine the matrix elements of the vectors $\mathbf{a}$ between eigenstates of the orbital angular momentum. From formula (107.6) of III it follows that

$$\langle lm' | a_{\lambda} | jm \rangle = i(-1)^{j_{\max}-m'} \begin{pmatrix} l & 1 & j \\ -m' & \lambda & m \end{pmatrix} \langle l || a || j \rangle,$$

where $j_{\max}$ is the larger of the numbers l and j . It therefore suffices to know the nonvanishing reduced matrix elements $\langle l || a || j \rangle$. The relevant formulas are:

$$\begin{aligned} \langle l-1 || n || l \rangle &= \langle l || n || l-1 \rangle^* = i\sqrt{l}, \\ \langle l || \nabla_{\mathbf{n}} || l-1 \rangle &= i(l-1)\sqrt{l}, \\ \langle l-1 || \nabla_{\mathbf{n}} || l \rangle &= i(l+1)\sqrt{l}, \\ \langle l || [\mathbf{n} \nabla_{\mathbf{n}}] || l \rangle &= i\sqrt{l(l+1)(2l+1)}. \end{aligned} \quad (7.17)$$

§8. Polarization of the photon

The polarization vector $\mathbf{e}$ plays for the photon the role of the “spin part” of the wave function (with the reservations expressed in § 6 regarding the notion of photon spin).

The various cases arising for the polarization of a photon are in no way distinct from the possible types of polarization of a classical electromagnetic wave (see II, § 48).

An arbitrary polarization $\mathbf{e}$ can be represented as a superposition of two mutually orthogonal polarizations $\mathbf{e}^{(1)}$ and $\mathbf{e}^{(2)}$ ($\mathbf{e}^{(1)}\mathbf{e}^{(2)*} = 0$), chosen in some definite manner. In the decomposition

$$\mathbf{e} = e_1\mathbf{e}^{(1)} + e_2\mathbf{e}^{(2)} \quad (8.1)$$

the squared moduli of the coefficients e_1 and e_2 give the probability that the photon has polarization $\mathbf{e}^{(1)}$ or $\mathbf{e}^{(2)}$.

As these two basis polarizations one may choose two mutually perpendicular linear polarizations. Alternatively, any polarization can be decomposed into two circular ones with opposite senses of rotation. We shall denote the right and left circular polarization vectors by $\mathbf{e}^{(+1)}$ and $\mathbf{e}^{(-1)}$ respectively; in a coordinate system $\xi\eta\zeta$ with the ζ axis along the photon's direction $\mathbf{n} = \mathbf{k}/\omega$,

$$\mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}(\mathbf{e}^{(\xi)} + i\mathbf{e}^{(\eta)}), \quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}(\mathbf{e}^{(\xi)} - i\mathbf{e}^{(\eta)}). \quad (8.2)$$

The existence of two distinct polarizations of the photon (at a given momentum) means, in other words, that every eigenvalue of the momentum is doubly degenerate. This circumstance is intimately connected with the fact that the photon mass equals zero.

For a freely moving particle of nonvanishing mass a rest frame always exists. It is clear that the intrinsic symmetry properties of the particle as such manifest themselves precisely in this frame. The symmetry to be considered is with respect to all possible rotations about the center, that is, with respect to the full spherical symmetry group. The quantity characterizing the symmetry properties of the particle under this group is its spin s , which determines the multiplicity of the degeneracy (the number $2s + 1$ of distinct wave functions that transform into one another). In particular, a particle whose wave function is a vector (three components) has spin 1.

For a particle of zero mass, however, no rest frame exists — in every reference frame it travels at the speed of light. For such a particle there is always a preferred spatial direction — the direction of the momentum vector $\mathbf{k}$ (the ζ axis). In this situation the full three-dimensional rotation symmetry is evidently absent, and one may speak only of axial symmetry about the distinguished axis.

Under axial symmetry only the *helicity* of the particle is conserved — the projection of angular momentum onto the ζ axis; we denote it by λ ¹⁶. If one further demands symmetry under reflections in planes passing through the ζ axis, then states differing in the sign of λ are mutually degenerate; for $\lambda \neq 0$ there is therefore a twofold degeneracy¹⁷. The state of a photon with a definite momentum belongs to precisely this type of twofold-degenerate state. It is described by a “spin” wave function that is a vector $\mathbf{e}$ in the $\xi\eta$ plane; the two components of this vector transform into each other under all rotations about the ζ axis and under reflections in planes containing that axis.

The various polarization cases of a photon are in a definite correspondence with the possible values of its helicity. This correspondence can be established using formulas III,

¹⁶In contrast to the projection m of the angular momentum onto a fixed direction (the z axis) in space, discussed in the preceding section.

¹⁷We note that the electronic terms of a diatomic molecule are classified in exactly the same way (see III, § 78).

(57.9), which relate the components of a vector wave function to the components of the equivalent rank-2 spinor¹⁸. The projections $\lambda = +1$ and -1 correspond to vectors $\mathbf{e}$ that have only the component $e_\xi - ie_\eta$ or $e_\xi + ie_\eta$ nonzero, i.e. $\mathbf{e} = \mathbf{e}^{(+1)}$ or $\mathbf{e} = \mathbf{e}^{(-1)}$ respectively. In other words, $\lambda = +1$ and -1 correspond to the right and left circular polarizations of the photon (in § 16 the same result will be derived by directly computing the eigenfunctions of the spin-projection operator). Thus the projection of the photon's angular momentum onto the direction of its motion can take only the two values (± 1) ; the value 0 is not possible.

A photon state with definite momentum and polarization is a pure state (in the sense explained in III, § 14); it is described by a wave function and constitutes a complete quantum-mechanical specification of the particle (photon) state. There also exist “mixed” photon states corresponding to a less complete description, carried out not by a wave function but only by a density matrix.

Let us examine a photon state that is mixed in its polarization but possesses a definite momentum $\mathbf{k}$. Such a state (called a state of *partial polarization*) admits a “coordinate” wave function.

The polarization density matrix of a photon is a second-rank tensor $\rho_{\alpha\beta}$ defined in the plane orthogonal to $\mathbf{n}$ (the $\xi\eta$ plane; the indices α, β assume only two values). This tensor is Hermitian:

$$\rho_{\alpha\beta} = \rho_{\beta\alpha}^*, \quad (8.3)$$

and normalized by the condition

$$\rho_{\alpha\alpha} = \rho_{11} + \rho_{22} = 1. \quad (8.4)$$

By virtue of (8.3) the diagonal components ρ_{11} and ρ_{22} are real, and one determines the other through condition (8.4). The component ρ_{12} is complex, while $\rho_{21} = \rho_{12}^*$. Altogether, then, the density matrix is characterized by three real parameters.

Once the polarization density matrix is known, one can find the probability that the photon has any given definite polarization $\mathbf{e}$. This probability is obtained by “projecting” the tensor $\rho_{\alpha\beta}$ onto the direction of the vector $\mathbf{e}$, i.e. it equals the quantity

$$\rho_{\alpha\beta} e_\alpha^* e_\beta. \quad (8.5)$$

Thus the components ρ_{11} and ρ_{22} are the probabilities of linear polarizations along the ξ and η axes. Projection onto the vectors (8.2) yields the probabilities of the two circular polarizations:

$$\frac{1}{2} [1 \pm i(\rho_{12} - \rho_{21})]. \quad (8.6)$$

In both form and substance the properties of the tensor $\rho_{\alpha\beta}$ coincide with those of the tensor $J_{\alpha\beta}$ that describes partially polarized light in the classical theory (see II, § 50). Let us recall some of these properties here.

For a pure state of definite polarization $\mathbf{e}$, the tensor $\rho_{\alpha\beta}$ reduces to products of the components of the vector $\mathbf{e}$:

$$\rho_{\alpha\beta} = e_\alpha e_\beta^*. \quad (8.7)$$

¹⁸Recall that the wave-function components, viewed as probability amplitudes for the various values of the angular-momentum projection of the particle (which is what is meant here), correspond to the contravariant components of the spinor.

In this case the determinant $|\rho_{\alpha\beta}| = 0$. In the opposite case of an unpolarized photon, all polarization directions are equally probable, i.e.

$$\rho_{\alpha\beta} = \delta_{\alpha\beta}/2, \quad (8.8)$$

and then $|\rho_{\alpha\beta}| = 1/4$.

For the general case of partial polarization it is convenient to introduce three real Stokes parameters ξ_1, ξ_2, ξ_3 ¹⁹; the density matrix is then expressible as

$$\rho_{\alpha\beta} = \frac{1}{2} \begin{pmatrix} 1 + \xi_3 & \xi_1 - i\xi_2 \\ \xi_1 + i\xi_2 & 1 - \xi_3 \end{pmatrix}. \quad (8.9)$$

All three parameters range between -1 and $+1$. In the unpolarized state $\xi_1 = \xi_2 = \xi_3 = 0$; for a completely polarized photon $\xi_1^2 + \xi_2^2 + \xi_3^2 = 1$.

The parameter ξ_3 characterizes linear polarization along the ξ or η axis; the probability of linear polarization along these axes is $(1 + \xi_3)/2$ and $(1 - \xi_3)/2$ respectively. The values $\xi_3 = +1$ or -1 therefore correspond to complete polarization in these directions.

The parameter ξ_1 characterizes linear polarization along directions making an angle $\varphi = \pi/4$ or $\varphi = -\pi/4$ with the ξ axis. The probability of linear polarization in these directions is $(1 + \xi_1)/2$ or $(1 - \xi_1)/2$; this is easily verified by projecting the tensor $\rho_{\alpha\beta}$ onto the directions $\mathbf{e} = (1, \pm 1)/\sqrt{2}$.

Lastly, the parameter ξ_2 represents the degree of circular polarization; by (8.6), the probability of right or left circular polarization is $(1 + \xi_2)/2$ or $(1 - \xi_2)/2$ respectively. Because these two polarizations answer to helicities $\lambda = \pm 1$, it is evident that in general ξ_2 coincides with the mean helicity of the photon. Note further that for a pure state with polarization $\mathbf{e}$,

$$\xi_2 = i[\mathbf{e}\mathbf{e}^*]\mathbf{n}. \quad (8.10)$$

Recall (see II, § 50) that the quantities ξ_2 and $\sqrt{\xi_1^2 + \xi_3^2}$ are invariant under Lorentz transformations.

In what follows we shall also need to know how the Stokes parameters transform under time reversal. One readily verifies that they are invariant under this operation. Since the result clearly does not depend on whether the polarization state is pure or mixed, it is enough to check invariance for a pure state. In quantum mechanics, time reversal replaces the wave function with its complex conjugate (see III, § 18). For a plane-polarized wave the corresponding substitution is²⁰

$$\mathbf{k} \rightarrow -\mathbf{k}, \quad \mathbf{e} \rightarrow -\mathbf{e}^*. \quad (8.11)$$

Under this transformation the symmetric part of the density matrix, $(1/2)(e_\alpha e_\beta^* + e_\beta e_\alpha^*)$, and hence the parameters ξ_1 and ξ_3 remain unchanged. The invariance of ξ_2 under the

¹⁹The notation for the parameters should not be confused with that for the ξ axis!

²⁰The extra sign reversal of $\mathbf{e}$ is due to the fact that time reversal changes the sign of the electromagnetic vector potential. The scalar potential is unaffected; consequently, for the 4-vector e time reversal takes the form

$$(e_0, \mathbf{e}) \rightarrow (e_0^*, -\mathbf{e}^*). \quad (8.11a)$$

same transformation is apparent from (8.10); it is also obvious already from the physical meaning of ξ_2 as the mean helicity. Indeed, helicity is the projection of the angular momentum $\mathbf{j}$ onto the direction $\mathbf{n}$, that is, the product $\mathbf{j}\mathbf{n}$; time reversal reverses the sign of both these vectors.

For the calculations that follow, the photon density matrix must be cast in four-dimensional form, that is, expressed as a certain 4-tensor $\rho_{\mu\nu}$. When the photon is polarized and described by a 4-vector e_μ , a natural definition of this tensor is

$$\rho_{\mu\nu} = e_\mu e_\nu^*. \quad (8.12)$$

In the three-dimensionally transverse gauge, $e = (0, \mathbf{e})$, if one of the spatial coordinate axes is chosen along $\mathbf{n}$, the nonvanishing components of this 4-tensor coincide with (8.7).

For an unpolarized photon the three-dimensionally transverse gauge yields a tensor $\rho_{\mu\nu}$ with components

$$\rho_{ik} = \frac{1}{2}(\delta_{ik} - n_i n_k), \quad \rho_{0i} = \rho_{i0} = \rho_{00} = 0 \quad (8.13)$$

(if one of the axes coincides with the direction of $\mathbf{n}$, we return to (8.8)). Using the tensor $\rho_{\mu\nu}$ directly in this three-dimensional form is, however, inconvenient. We may instead perform a gauge transformation; for the density matrix this is a transformation of the type

$$\rho_{\mu\nu} \rightarrow \rho_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \quad (8.14)$$

where χ_μ are arbitrary functions. Setting

$$\chi_0 = -1/(4\omega), \quad \chi_i = k_i/(4\omega^2)$$

we obtain in place of (8.13) the simple four-dimensional expression

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \quad (8.15)$$

The four-dimensional representation of the density matrix of a partially polarized photon is readily obtained by first rewriting the two-dimensional tensor (8.9) in three-dimensional form. In the three-dimensionally transverse gauge: $e^{(1)} = (0, \mathbf{e}^{(1)})$, $e^{(2)} = (0, \mathbf{e}^{(2)})$. The four-dimensional photon density matrix is therefore

$$\begin{aligned} \rho_{\mu\nu} = & (1/2)(e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + (\xi_1/2)(e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ & - (i\xi_2/2)(e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + (\xi_3/2)(e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}). \end{aligned} \quad (8.17)$$

The convenience of any particular choice of the 4-vectors $e^{(1)}$, $e^{(2)}$ depends on the specific conditions of the problem at hand.

It should be borne in mind that the conditions (8.16) do not fix the choice of $e^{(1)}$ and $e^{(2)}$ uniquely. If any 4-vector e_μ satisfies these conditions, so does any 4-vector of the form $e_\mu + \chi k_\mu$ (since $k^2 = 0$). This ambiguity is tied to the gauge freedom of the density matrix.

The first term in (8.17) corresponds to the unpolarized state. It could therefore be replaced, by (8.15), with $-g_{\mu\nu}/2$. Such a replacement is again equivalent to a gauge

transformation. When working with 4-tensors of the form (8.17), decomposed in two independent 4-vectors, it is useful to adopt the following formal device. Writing (8.17) as

$$\rho_{\mu\nu} = \sum_{a,b=1}^2 \rho^{(ab)} e_{\mu}^{(a)} e_{\nu}^{(b)},$$

we represent the coefficients $\rho^{(ab)}$ as a 2×2 matrix

$$\rho = \begin{pmatrix} \rho^{(11)} & \rho^{(12)} \\ \rho^{(21)} & \rho^{(22)} \end{pmatrix}.$$

Any Hermitian 2×2 matrix can be expanded over four linearly independent 2×2 matrices — the Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ and the identity matrix 1:

$$\rho = (1/2)(1 + \boldsymbol{\xi} \boldsymbol{\sigma}), \quad \boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3), \quad (8.18)$$

as is easily verified by direct comparison with (8.17) using the well-known explicit forms of the Pauli matrices (18.5) (grouping the three quantities ξ_1, ξ_2, ξ_3 into a “vector” $\boldsymbol{\xi}$ is of course purely formal and serves only to shorten the notation).

The requisite generalization is achieved by replacing these 3-vectors by spacelike real 4-vectors $e^{(1)}, e^{(2)}$, orthogonal to each other and to the photon 4-momentum k :

$$e^{(1)2} = e^{(2)2} = -1, \quad e^{(1)} e^{(2)} = 0, \quad e^{(1)} k = e^{(2)} k = 0. \quad (8.16)$$

Problem

Write the photon density matrix in the representation whose “coordinate axes” are the circular basis vectors (8.2).

Solution. The components of the tensor $\rho'_{\alpha\beta}$ in the new axes ($\alpha, \beta = \pm 1$) are obtained by projecting the tensor (8.9) onto the basis vectors (8.2):

$$\rho'_{11} = \rho_{\alpha\beta} e_{\alpha}^{(+1)*} e_{\beta}^{(+1)}, \quad \rho'_{1,-1} = \rho_{\alpha\beta} e_{\alpha}^{(+1)*} e_{\beta}^{(-1)}, \quad \dots,$$

$$\rho' = \frac{1}{2} \begin{pmatrix} 1 + \xi_2 & -\xi_3 + i\xi_1 \\ -\xi_3 - i\xi_1 & 1 - \xi_2 \end{pmatrix}.$$

§9. The two-photon system

Arguments analogous to those developed in §6 make it possible to count the number of allowed states in the more involved case of a system of two photons as well (*L. Landau*, 1948).

Let us work in the center-of-inertia frame of the two photons, where $\mathbf{k}_1 = -\mathbf{k}_2 \equiv \mathbf{k}$ ²¹. In the momentum representation the wave function of the pair can be cast as a three-dimensional rank-2 tensor $A_{ik}(\mathbf{n})$ built bilinearly from the components of the individual

²¹ A center-of-inertia frame can be found in every case except when the two photons propagate in the same direction. For such a pair the total momentum $\mathbf{k}_1 + \mathbf{k}_2$ and total energy $\omega_1 + \omega_2$ are connected by exactly the same relation as for one photon, so no frame with $\mathbf{k}_1 + \mathbf{k}_2 = 0$ exists.

photon vector wave functions; each of its two indices refers to one photon ($\mathbf{n}$ denotes the unit vector in the direction of $\mathbf{k}$). The transversality requirement for each photon is encoded in the orthogonality of A_{ik} to $\mathbf{n}$:

$$A_{il}n_l = 0, \quad A_{lk}n_l = 0. \quad (9.1)$$

Interchanging the two photons amounts to transposing the indices of the tensor A_{ik} together with a simultaneous reversal of the sign of $\mathbf{n}$. Because photons obey Bose statistics,

$$A_{ik}(-\mathbf{n}) = A_{ki}(\mathbf{n}). \quad (9.2)$$

The tensor A_{ik} is, in general, not symmetric in its indices. We decompose it into its symmetric part (s_{ik}) and antisymmetric part (a_{ik}): $A_{ik} = s_{ik} + a_{ik}$. Each of these parts must individually satisfy the relation (9.2) (as well as the orthogonality conditions (9.1)). This gives

$$s_{ik}(-\mathbf{n}) = s_{ik}(\mathbf{n}), \quad (9.3)$$

$$a_{ik}(-\mathbf{n}) = -a_{ik}(\mathbf{n}). \quad (9.4)$$

Spatial inversion does not by itself reverse the sign of the components of a second-rank tensor, but it does reverse the sign of $\mathbf{n}$. From (9.3) it is therefore apparent that the wave function s_{ik} is symmetric under inversion, i.e. it corresponds to even-parity states of the photon system; the wave function a_{ik} , on the other hand, corresponds to odd-parity states.

An antisymmetric second-rank tensor is equivalent (dual) to an axial vector $\mathbf{a}$ whose components are expressed via the tensor components as $a_i = \frac{1}{2}e_{ikl}a_{kl}$, where e_{ikl} is the antisymmetric unit tensor (see II, § 6). The orthogonality of the tensor a_{kl} to the vector $\mathbf{n}$ implies that the vectors $\mathbf{a}$ and $\mathbf{n}$ are parallel²². One may therefore write $\mathbf{a} = \mathbf{n}\varphi(\mathbf{n})$, where φ is a scalar; by (9.4) we must have $\mathbf{a}(-\mathbf{n}) = -\mathbf{a}(\mathbf{n})$, and hence

$$\varphi(-\mathbf{n}) = \varphi(\mathbf{n}).$$

This relation means that the scalar φ can be built linearly only from spherical harmonics of even order L (including zero).

We see that the antisymmetric tensor a_{ik} , by its transformation properties (under rotations), is equivalent to a single scalar (cf. the footnote on p. 34). Assigning to the latter a “spin” of 0, we find that the angular momentum of the state is $J = L$. Thus the tensor a_{ik} corresponds to odd-parity states of the photon system with even total angular momentum J .

Let us turn to the symmetric tensor s_{ik} . Since it is even under reversal of the sign of $\mathbf{n}$, it describes even-parity states of the photon system. It follows likewise that all components of s_{ik} are expressed through spherical harmonics of even order L (including $L = 0$). An arbitrary symmetric second-rank tensor can be decomposed, as is well known, into a scalar (s_{ii}) and a traceless symmetric tensor (s'_{ik}) with $s'_{ii} = 0$.

²²We have: $a_{ik} = e_{ikl}a_l$, and the orthogonality condition yields

$$a_{ik}n_k = e_{ikl}a_ln_k = [\mathbf{na}]_i = 0.$$

The scalar s_{ii} is assigned “spin” 0, so the angular momentum of the corresponding states is $J = L$, which is even. The tensor s'_{ik} corresponds to “spin” 2 (see III, § 57). Composing this “spin” with the even “orbital angular momentum” L according to the addition rule for angular momenta, we find that for a given even $J \neq 0$ there are three states (with $L = J \pm 2, J$), while for odd $J \neq 1$ there are two states (with $L = J \pm 1$). The exceptions are $J = 0$ with a single state ($L = 2$) and $J = 1$ with a single state ($L = 2$).

In this counting, however, we have not yet accounted for the orthogonality of the tensor s_{ik} to the vector $\mathbf{n}$. One must therefore subtract from the number of states obtained those corresponding to a symmetric second-rank tensor that is “parallel” to the vector $\mathbf{n}$. Such a tensor (denote it s''_{ik}) can be written as

$$s''_{ik} = n_i b_k + n_k b_i,$$

where $\mathbf{b}$ is some vector. By (9.3) this vector must satisfy $\mathbf{b}(-\mathbf{n}) = -\mathbf{b}(\mathbf{n})$. The tensor s''_{ik} responsible for the “extra” states is thus equivalent to an odd-parity vector. This vector must consequently be expressed through spherical harmonics of odd orders L only. Noting further that a vector corresponds to “spin” 1, we find that for each even $J \neq 0$ there are two states (with $L = J \pm 1$), and for each odd J a single state (with $L = J$); the special case $J = 0$ has one state (with $L = 1$).

Collecting all the results obtained, we arrive at the following table giving the number of possible even- and odd-parity states of a system of two photons (with vanishing total momentum) for the various values of the total angular momentum J :

J	Even	Odd
0	1	1
1	—	—
$2k$	2	1
$2k + 1$	1	—

(9.5)

(k being a positive integer other than zero). We observe that for odd J no odd-parity states exist, while the value $J = 1$ is altogether excluded²³.

The wave function of the two-photon system A_{ik} determines the correlation between their polarizations. The probability that both photons simultaneously possess definite polarizations $\mathbf{e}_1$ and $\mathbf{e}_2$ is proportional to

$$A_{ik} e_{1i}^* e_{2k}^*.$$

In other words, if the polarization $\mathbf{e}_1$ of one photon is specified, then the polarization of the second photon, $\mathbf{e}_2$, is proportional to

$$e_{2k} \propto A_{ik} e_{1i}^*. \quad (9.6)$$

In odd-parity states of the system, A_{ik} coincides with the antisymmetric tensor a_{ik} . In that case

$$\mathbf{e}_2 \mathbf{e}_1^* \propto a_{ik} e_{1i}^* e_{1k}^* = 0,$$

²³For an alternative derivation of these results, see problem 1 to § 69.

i.e. the polarizations of the two photons are mutually orthogonal. For linear polarization this signifies that their directions are perpendicular, while for circular polarization it means opposite senses of rotation.

The even-parity state with $J = 0$ is described by a symmetric tensor reducing to a scalar,

$$s_{ik} = \text{const} \cdot (\delta_{ik} - n_i n_k).$$

From (9.6) we then obtain $\mathbf{e}_1 = \mathbf{e}_2^*$. For linear polarization this means that their directions are parallel, whereas for circular polarization it means opposite senses of rotation. The latter fact is self-evident: when $J = 0$ the sum of the angular-momentum projections of the two photons onto any common direction $\mathbf{k}$ must in all cases vanish (the projections onto the opposite directions $\mathbf{k}_1$ and $\mathbf{k}_2$, i.e. the helicities, are accordingly equal).

Bosons

§ 10. Wave equation for particles with spin 0

Chapter I demonstrated the construction of a quantum description for the free electromagnetic field, proceeding from the known classical-limit properties of the field and relying on the apparatus of ordinary quantum mechanics. The scheme of field description as a photon system obtained in this way retains many features that carry over to the relativistic quantum-theoretical treatment of particles as well.

The electromagnetic field is a system with an infinite number of degrees of freedom. For it there is no conservation law for the number of particles (photons), and among its possible states are those with any number of particles¹. However, the same property must, generally speaking, also hold in relativistic theory for systems of any particles. The conservation of particle number in non-relativistic theory is tied to the conservation of mass: the sum of the masses (rest masses) of the particles does not change during their interaction; conservation of the total mass in a system of electrons, say, implies the constancy of their number as well. In relativistic mechanics, however, no law of mass conservation exists; only the total energy of the system (which includes the rest energies of the particles) must be conserved. For this reason the particle number need no longer be conserved, and consequently every relativistic particle theory must be a theory of systems possessing an infinite number of degrees of freedom. In other words, such a particle theory acquires the character of a field theory.

The adequate mathematical apparatus for treating systems whose particle number varies is that of second quantization (see III, §§ 64, 65). In the quantum treatment of the electromagnetic field the 4-potential $\hat{A}$ serves as the second-quantization operator. It is built from the (coordinate-space) wave functions of the individual particles (photons) together with their creation and annihilation operators. A completely analogous role in the description of a system of particles belongs to the operator of the quantized wave function. Its construction requires, first and foremost, knowledge of the wave-function form for a single free particle and of the equation governing this function.

It should be stressed that the notion of free-particle fields is auxiliary in character. Real particles interact, and the task of the theory lies in studying these interactions. But every interaction reduces to a collision, before and after which the system can be regarded as a collection of free particles. In § 1 it was noted that these are the only directly measurable objects. We therefore use the fields of free particles as a means for describing initial and final states.

We begin the relativistic description of free particles with the case of spin-0 particles. The mathematical simplicity of this case makes it possible to bring out most clearly the basic ideas and characteristic features of such a description.

¹In practice, of course, the photon number changes only through various interaction processes.

The state of a free particle (without spin) can be completely specified by giving its momentum $\mathbf{p}$ alone. The energy ε of the particle² satisfies $\varepsilon^2 = \mathbf{p}^2 + m^2$ (where m is the particle mass), or in four-dimensional form:

$$p^2 = m^2. \quad (10.1)$$

It is well known that the conservation of momentum and energy is tied to the homogeneity of space and time, that is, to symmetry with respect to any parallel translation of the 4-coordinate system. Within the quantum description, this symmetry requirement implies that under such a coordinate transformation the wave function of a particle possessing a definite 4-momentum may only acquire a phase factor (whose modulus equals unity). This requirement is met solely by an exponential with an exponent that depends linearly on the 4-coordinates. Put differently, the wave function of a free particle in a state of definite 4-momentum $p^\mu = (\varepsilon, \mathbf{p})$ must have the form of a plane wave:

$$\text{const} \cdot e^{-ipx}, \quad px = \varepsilon t - \mathbf{p}\mathbf{r} \quad (10.2)$$

(the choice of sign in the exponent is in itself a convention within relativistic theory; it is made so as to agree with the non-relativistic case).

The wave equation must possess the functions (10.2) as particular solutions for every 4-vector p obeying the condition (10.1). It must be linear—this being a manifestation of the superposition principle: every linear combination of functions (10.2) likewise describes a possible particle state and hence must also satisfy the equation. Lastly, the equation should have the lowest attainable order; raising the order would bring in extraneous solutions.

Spin is the angular momentum that a particle possesses in the frame where it is at rest. When the particle's spin equals s , the wave function in the rest frame takes the form of a three-dimensional spinor of rank $2s$. For describing the particle in an arbitrary reference frame one must cast the wave function in terms of four-dimensional objects.

A particle of spin 0 is described in its rest frame by a three-dimensional scalar. This scalar may, however, have a different four-dimensional “ancestry”: it can be a four-scalar ψ , but it can equally well be the temporal component of a timelike 4-vector ψ_μ whose only component that does not vanish in the rest frame is ψ_0 ³.

For a free particle the only operator that can enter the wave equation is the 4-momentum operator $\hat{p}$. Its components are the differentiation operators with respect to the coordinates and time:

$$\hat{p}^\mu = i\partial^\mu = \left(i\frac{\partial}{\partial t}, -i\nabla\right). \quad (10.3)$$

The wave equation must represent a differential relation between the quantities ψ and ψ_μ , effected by means of the operator $\hat{p}$. This relation must, of course, be expressed through relativistically invariant equations. Such equations are

$$m\psi_\mu = \hat{p}_\mu\psi, \quad \hat{p}^\mu\psi_\mu = m\psi, \quad (10.4)$$

²We denote the energy of an individual particle by ε , as distinct from the energy E of the system of particles.

³Or, by the same token, the temporal component of a higher-rank 4-tensor; that possibility, however, would give rise to equations of higher order.

where m is a dimensional constant characterizing the particle⁴.

Substituting ψ_μ from the first equation into the second, we obtain

$$(\hat{p}^2 - m^2)\psi = 0 \quad (10.5)$$

(*O. Klein, V. A. Fock, 1926; W. Gordon, 1927*). Written out explicitly, this equation reads

$$-\partial_\mu \partial^\mu \psi \equiv \left(-\frac{\partial^2}{\partial t^2} + \Delta \right) \psi = m^2 \psi. \quad (10.6)$$

If one inserts a plane wave (10.2) for ψ , the result is $p^2 = m^2$, showing that m is indeed the particle mass. It is worth noting that the structure of equation (10.5) could, of course, have been anticipated, since $\hat{p}^2$ is the sole scalar operator one can form from $\hat{p}$ (and for this reason every component of the wave function of a particle of arbitrary spin obeys the same equation—something we shall encounter many times below).

Thus a spin-0 particle is described essentially by a single (four-dimensional) scalar ψ obeying the second-order equation (10.5). In the first-order equations (10.4) the role of the wave function is played by the aggregate of the quantities ψ and ψ_μ , the 4-vector ψ_μ reducing to the 4-gradient of the scalar ψ . In the rest frame the particle's wave function does not depend on the (spatial) coordinates, so that the spatial components of the 4-vector ψ_μ vanish, as they should.

For carrying out second quantization it proves useful to write the particle's energy and momentum as spatial integrals of certain bilinear (in ψ and ψ^*) combinations playing the role of spatial densities of these quantities. This amounts to finding the energy-momentum tensor $T_{\mu\nu}$ that corresponds to equation (10.5). With the aid of this tensor, the energy-momentum conservation law takes the form

$$\partial_\mu T_\nu^\mu = 0. \quad (10.7)$$

Following the general rules of field theory (see II, § 32), let us write down the variational principle whose consequence would be equation (10.5). Such a principle must consist in requiring the minimality of the “action integral”

$$S = \int L d^4x \quad (10.8)$$

with respect to a certain real 4-scalar L —the Lagrangian density of the field⁵. From the scalar ψ (and the operator ∂^μ) one can construct a real scalar bilinear expression of the form

$$L = \partial_\mu \psi^* \cdot \partial^\mu \psi - m^2 \psi^* \psi, \quad (10.9)$$

where m is a dimensional constant. Treating ψ and ψ^* as independent variables describing the field (“generalized coordinates” of the field q), it is easy to verify that the Lagrange

⁴The constants m are introduced in (10.4) in such a way that ψ_μ and ψ have the same dimensionality. It would be pointless to introduce different constants m_1 and m_2 in these two equations, since they could always be made equal by redefining ψ or ψ_μ .

⁵The corresponding second-quantized operator $\hat{L}$ is called the field *Lagrangian*. For brevity of terminology we shall use this term both for the “quantized” and the “unquantized” Lagrangian density.

equations

$$\frac{\partial}{\partial x^\mu} \frac{\partial L}{\partial q_{,\mu}} = \frac{\partial L}{\partial q} \quad (10.10)$$

$(q_{,\mu} \equiv \partial_\mu q)$ do indeed coincide with the equations (10.5) for ψ and ψ^* , with m being the particle mass. Note also that the expression (10.9) is written with an overall sign such that the square of the time derivative, $|\partial\psi/\partial t|^2$, enters L with a plus sign; in the opposite case the action could not have a minimum (cf. II, § 27). The choice of the overall numerical coefficient of L is a matter of convention (and is reflected only in the normalization coefficient in ψ).

The energy-momentum tensor is now computed by the formula

$$T_\mu^\nu = \sum q_{,\mu} \frac{\partial L}{\partial q_{,\nu}} - L \delta_\mu^\nu \quad (10.11)$$

(summation over all q). Substituting (10.9), we obtain

$$T_{\mu\nu} = \partial_\mu \psi^* \cdot \partial_\nu \psi + \partial_\nu \psi^* \cdot \partial_\mu \psi - L g_{\mu\nu} \quad (10.12)$$

(these quantities are, as they should be, real, which is ensured by the reality of L). In particular,

$$T_{00} = 2 \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial t} - L = \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial t} + \nabla \psi^* \cdot \nabla \psi + m^2 \psi^* \psi, \quad (10.13)$$

$$T_{i0} = \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial x^i} + \frac{\partial \psi^*}{\partial x^i} \frac{\partial \psi}{\partial t}. \quad (10.14)$$

The 4-momentum of the field is given by the integral

$$P_\mu = \int T_{\mu 0} d^3x, \quad (10.15)$$

i.e. T_{00} and T_{i0} play the role of the energy and momentum densities. We note that the quantity T_{00} is essentially positive.

Formula (10.13) can be used to normalize the wave function. A plane wave normalized to one particle in a volume $V = 1$ takes the form

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}} e^{-ipx}. \quad (10.16)$$

For this function one has $T_{00} = \varepsilon$, whence the total energy within the volume $V = 1$ coincides with the energy of one particle.

The angular momentum, whose conservation follows from the isotropy of space, likewise admits a representation as a spatial integral; we shall, however, have no occasion to use such a representation later on.

Finally, besides the conservation laws linked directly with space-time symmetry, equations (10.4) admit one more conservation law. It is easy to verify that, by virtue of (10.4) (and the analogous equations for ψ^*), the following equation holds:

$$\partial_\mu j^\mu = 0, \quad (10.17)$$

where

$$j_\mu = m(\psi^* \psi_\mu + \psi_\mu^* \psi) = i[\psi^* \partial_\mu \psi - (\partial_\mu \psi^*) \psi]. \quad (10.18)$$

It is apparent from this that j^μ plays the role of the 4-vector current density. Equation (10.17) is then the continuity equation expressing the conservation of the quantity

$$Q = \int j_0 d^3x, \quad (10.19)$$

where

$$j_0 = j^0 = i \left(\psi^* \frac{\partial \psi}{\partial t} - \frac{\partial \psi^*}{\partial t} \psi \right). \quad (10.20)$$

Let us call attention to the fact that j_0 is not a positive-definite quantity. This circumstance alone shows that in the general case it certainly cannot be interpreted as the probability density of the spatial localization of the particle. The meaning of the conservation law expressed by equation (10.17) will be clarified in the next section.

§ 11. Particles and antiparticles

In keeping with the general procedure of second quantization, one must expand an arbitrary wave function over the eigenfunctions spanning the full set of possible free-particle states, for example over the plane waves ψ_p :

$$\psi = \sum_{\mathbf{p}} a_{\mathbf{p}} \psi_p, \quad \psi^* = \sum_{\mathbf{p}} a_{\mathbf{p}}^* \psi_p^*.$$

After that, the coefficients $a_{\mathbf{p}}$, $a_{\mathbf{p}}^*$ would have to be promoted to operators $\hat{a}_{\mathbf{p}}$, $\hat{a}_{\mathbf{p}}^+$ for annihilating and creating particles in the corresponding states⁶.

At this point, however, we immediately encounter the following fundamentally new circumstance (compared with the non-relativistic theory). In a plane wave that solves equation (10.5), the energy ε must satisfy (for a given momentum $\mathbf{p}$) only the condition $\varepsilon^2 = \mathbf{p}^2 + m^2$, so that it can take two values: $\pm\sqrt{\mathbf{p}^2 + m^2}$. Physically meaningful values of the free-particle energy can, however, be only positive ones. Yet simply discarding the negative values is inadmissible: the general solution of the wave equation is formed only by a superposition of all its independent particular solutions. This points to the need for a certain re-interpretation of the expansion coefficients of ψ and ψ^* upon second quantization.

Let us write this expansion in the form

$$\psi = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a_{\mathbf{p}}^{(+)} e^{i(\mathbf{p}\mathbf{r} - \varepsilon t)} + \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a_{\mathbf{p}}^{(-)} e^{i(\mathbf{p}\mathbf{r} + \varepsilon t)}, \quad (11.1)$$

where the first sum contains plane waves normalized according to (10.16) with positive “frequencies,” and the second with negative ones; everywhere ε stands for the positive

⁶We label the ψ -functions by the 4-momentum index p ; functions with “negative frequency” will henceforth be denoted ψ_{-p} . The operators $\hat{a}$, $\hat{a}^+$, on the other hand, carry the three-momentum index $\mathbf{p}$, which completely specifies the state of a real particle.

quantity: $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$. In performing second quantization, the coefficients $a_{\mathbf{p}}^{(+)}$ in the first sum are replaced in the usual way by particle annihilation operators $\hat{a}_{\mathbf{p}}$. In the second sum we observe that upon subsequent evaluation of matrix elements the time dependence of its terms will correspond not to annihilation but to creation of particles; the factor $e^{i\varepsilon t} = (e^{-i\varepsilon t})^*$ corresponds to one extra particle with energy ε in the final state (cf. the end of § 2). Accordingly, the coefficients $a_{\mathbf{p}}^{(-)}$ are replaced by creation operators $\hat{b}_{-\mathbf{p}}^+$ for certain other particles. Changing the summation variable in the second sum of (11.1) from $\mathbf{p}$ to $-\mathbf{p}$ (so that the exponential factor takes the form $e^{-i(\mathbf{p}\mathbf{r} - \varepsilon t)}$), we arrive at the ψ -operators

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}} e^{-i\mathbf{p}\mathbf{x}} + \hat{b}_{\mathbf{p}}^+ e^{i\mathbf{p}\mathbf{x}}), \quad \hat{\psi}^+ = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}}^+ e^{i\mathbf{p}\mathbf{x}} + \hat{b}_{\mathbf{p}} e^{-i\mathbf{p}\mathbf{x}}). \quad (11.2)$$

As a result, every operator $\hat{a}_{\mathbf{p}}$ or $\hat{b}_{\mathbf{p}}$ appears multiplied by a function carrying the “correct” time factor ($\sim e^{-i\varepsilon t}$), whereas each $\hat{a}_{\mathbf{p}}^+$ or $\hat{b}_{\mathbf{p}}^+$ is accompanied by the complex conjugate of that function. This permits the identification, following the general rules, of $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ as annihilation operators and of $\hat{a}_{\mathbf{p}}^+$ and $\hat{b}_{\mathbf{p}}^+$ as creation operators for particles possessing momenta $\mathbf{p}$ and energies ε .

We thus arrive at the picture of particles of two kinds, appearing together and on an equal footing. They are referred to as *particles* and *antiparticles* (the reason for this name will become clear below). One kind corresponds, within the second-quantization formalism, to the operators $\hat{a}_{\mathbf{p}}$, $\hat{a}_{\mathbf{p}}^+$, and the other to $\hat{b}_{\mathbf{p}}$, $\hat{b}_{\mathbf{p}}^+$. Both species of particles, whose operators enter the same ψ -operator, thereby possess identical masses.

These conclusions can also be reached by starting directly from the demands of relativistic invariance.

From a mathematical viewpoint, Lorentz transformations are rotations of the four-dimensional coordinate system that change the orientation of the time axis (combined with purely spatial rotations, which leave the time axis unchanged, they constitute the transformation group known as the *Lorentz group*⁷). All these transformations share the property that they do not carry the t axis out of the corresponding half of the light cone, which expresses the physical principle of the existence of a limiting signal-propagation speed.

From a purely mathematical standpoint, however, a simultaneous reversal of the sign of all four coordinates is likewise a rotation (*four-dimensional inversion*): the determinant of this transformation equals +1, just as for every other rotational transformation. Under it the time axis passes from one half of the light cone to the other. Although this circumstance signifies the physical unrealizability of such a transformation (as a change of reference frame), mathematically the distinction reduces merely to the fact that (because of the pseudo-Euclidean character of the metric) such a rotation cannot be carried out without admitting a complex transformation of the coordinates in the process.

It is natural to expect that this distinction should be immaterial when one speaks of four-dimensional invariance. Then every expression that is invariant under Lorentz

⁷Note that the totality of all three-dimensional (spatial) rotations constitutes by itself a group, entering the Lorentz group as a subgroup. The totality of Lorentz transformations does not by itself form a group: the result of successive Lorentz transformations may reduce to a purely spatial rotation.

transformations should also be invariant under 4-inversion. The precise formulation of this requirement as applied to the scalar ψ -operator will be given in § 13. But let us immediately note that in any case it entails the simultaneous presence in the ψ -operators of terms with both signs in front of ε in the exponents, since the replacement $t \rightarrow -t$ reverses precisely this sign.

Returning to expressions (11.2), let us now determine the commutation relations among the operators $\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+$ (and $\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+$). In the photon case this was done (for $\hat{c}_{\mathbf{p}}, \hat{c}_{\mathbf{p}}^+$) by drawing on the oscillator analogy, that is, ultimately from the classical-limit behavior of the electromagnetic field. Here no comparable analogy exists. In order to decide whether the operators obey Bose or Fermi commutation rules, one can appeal only to the structure of the Hamiltonian built from these operators.

The Hamiltonian is obtained (see III, § 64) by substituting $\hat{\psi}$ and $\hat{\psi}^+$ for ψ and ψ^* in the integral $\int T_{00} d^3x$ ⁸. In this way we find

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} + \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+). \quad (11.3)$$

It is easy to see that a sensible result for the eigenvalues of this Hamiltonian is obtained only if the operators satisfy Bose commutation rules:

$$\{\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+\}_- = \{\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+\}_- \quad (11.4)$$

(all other pairs of operators commute; in particular, all particle operators $\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+$ commute with all antiparticle operators $\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+$). Indeed, in that case

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} + \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ + 1).$$

The eigenvalues of the products $\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+$ are non-negative integers $N_{\mathbf{p}}$ and $\bar{N}_{\mathbf{p}}$ —the numbers of particles and antiparticles. The infinite additive constant $\sum \varepsilon$ (“vacuum energy”) can once again simply be dropped:

$$E = \sum_{\mathbf{p}} \varepsilon(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}}) \quad (11.5)$$

(cf. formula (3.1) and the remark following it). This expression is manifestly positive and accords with the picture of two species of actually existing particles. In an analogous fashion, for the total momentum of the particle system we obtain

$$\mathbf{P} = \sum_{\mathbf{p}} \mathbf{p}(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}}). \quad (11.6)$$

Had we adopted, in place of (11.4), the Fermi commutation relations (anticommutators instead of commutators), we would have obtained

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ + 1)$$

⁸In non-relativistic theory one customarily places the conjugate operator $\hat{\psi}^+$ to the left of ψ . Here, however, the ordering is immaterial, since interchanging $\hat{\psi}^+$ and $\hat{\psi}$ would merely interchange the equivalent operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$. It is, however, necessary, once a particular ordering has been chosen, always to adhere to the same convention.

and, instead of formula (11.5), the physically meaningless expression $\sum \varepsilon(N_{\mathbf{p}} - \bar{N}_{\mathbf{p}})$. This expression is not positive-definite and therefore cannot represent the energy of a system of free particles.

Thus spin-0 particles are bosons.

Next, consider the integral Q (10.19). Replacing the functions ψ and ψ^* in j^0 by the operators $\hat{\psi}$ and $\hat{\psi}^+$ and carrying out the integration, we obtain

$$\hat{Q} = \sum_{\mathbf{p}} (\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+) = \sum_{\mathbf{p}} (\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}}^+ \hat{b}_{\mathbf{p}} - 1). \quad (11.7)$$

The eigenvalues of this operator (apart from the inessential additive constant $\sum 1$) are

$$Q = \sum_{\mathbf{p}} (N_{\mathbf{p}} - \bar{N}_{\mathbf{p}}), \quad (11.8)$$

i.e. they equal the difference between the total numbers of particles and antiparticles.

As long as we deal with free particles, abstracting from all interaction among them, the content of the conservation law for Q (as well as the conservation of total energy and momentum (11.5, 11.6)) remains, of course, largely formal: what is actually conserved is not merely this sum, but each of the individual numbers $N_{\mathbf{p}}$, $\bar{N}_{\mathbf{p}}$ separately. Whether the quantity Q will remain conserved as a result of the interaction depends on the nature of that interaction. If Q is conserved (i.e. if the operator $\hat{Q}$ commutes with the interaction Hamiltonian), then expression (11.8) shows what restriction this imposes on the possible changes in particle number: only pairs “particle + antiparticle” can appear or disappear.

If a particle carries an electric charge, its antiparticle must bear a charge of the opposite sign: were both to carry identical charges, the creation or annihilation of a pair would violate the strict law of nature—conservation of the total electric charge. We shall see below (§ 32) how this opposition of charges arises automatically within the theory (in the interaction of particles with the electromagnetic field).

The quantity Q is sometimes called the *charge* of the field of the given particles. For electrically charged particles, Q determines, in particular, the total electric charge of the system (in units of the elementary charge e). Let us stress, however, that particles and antiparticles may well be electrically neutral.

We see in this way how the character of the relativistic energy–momentum relation (the two-valuedness of the root of the equation $\varepsilon^2 = \mathbf{p}^2 + m^2$), combined with the requirements of relativistic invariance, leads in quantum theory to the emergence of a new classification principle for particles—the possibility of existence of pairs of distinct particles (“particle + antiparticle”), standing in the correspondence described above. This remarkable prediction was first made (for spin-1/2 particles) by *Dirac* in 1930, even before the actual discovery of the first antiparticle—the positron⁹.

§ 12. Strictly neutral particles

When carrying out the second quantization of the ψ -function (11.1), the coefficients $a_{\mathbf{p}}^{(+)}$ and $a_{\mathbf{p}}^{(-)}$ were treated as operators pertaining to different particles. This is, however,

⁹The notion of antiparticles was extended to bosons by *Weisskopf* and *Pauli* (V. Weisskopf, W. Pauli, 1934).

not obligatory: as a special case, the annihilation and creation operators appearing in $\hat{\psi}$ may refer to one and the same type of particle (as was the case for photons—cf. (2.17)). Denoting these operators $\hat{c}_{\mathbf{p}}$ and $\hat{c}_{\mathbf{p}}^+$ in this case, we write the ψ -operator as

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{c}_{\mathbf{p}} e^{-ipx} + \hat{c}_{\mathbf{p}}^+ e^{ipx}). \quad (12.1)$$

The field described by such an operator corresponds to a system of identical particles that can be said to “coincide with their own antiparticles.”

The operator (12.1) is Hermitian ($\hat{\psi}^+ = \hat{\psi}$); in this sense such a field has half as many “degrees of freedom” as a complex field, for which the operators $\hat{\psi}$ and $\hat{\psi}^+$ do not coincide.

In connection with this, the Lagrangian of the field, when expressed through the Hermitian operator $\hat{\psi}$, must contain an extra factor of 1/2 (compared with (10.9))¹⁰

$$\hat{L} = (1/2)(\partial_{\mu}\hat{\psi} \cdot \partial^{\mu}\hat{\psi} - m^2\hat{\psi}^2). \quad (12.2)$$

The corresponding energy–momentum tensor is

$$\hat{T}_{\mu\nu} = \partial_{\mu}\hat{\psi} \cdot \partial_{\nu}\hat{\psi} - \hat{L}g_{\mu\nu}, \quad (12.3)$$

so that the energy-density operator reads

$$\hat{T}_{00} = \left(\frac{\partial\hat{\psi}}{\partial t}\right)^2 - \hat{L} = \frac{1}{2} \left[\left(\frac{\partial\hat{\psi}}{\partial t}\right)^2 + (\nabla\hat{\psi})^2 + m^2\hat{\psi}^2 \right]. \quad (12.4)$$

Inserting (12.1) into the integral $\int \hat{T}_{00} d^3x$, we obtain the field Hamiltonian

$$\hat{H} = \frac{1}{2} \sum_{\mathbf{p}} \varepsilon (\hat{c}_{\mathbf{p}}^+ \hat{c}_{\mathbf{p}} + \hat{c}_{\mathbf{p}} \hat{c}_{\mathbf{p}}^+). \quad (12.5)$$

This again reveals the necessity of Bose quantization:

$$\{\hat{c}_{\mathbf{p}}, \hat{c}_{\mathbf{p}}^+\}_{-} = 1, \quad (12.6)$$

and the energy eigenvalues (once more dropping the additive constant) are

$$E = \sum_{\mathbf{p}} \varepsilon_{\mathbf{p}} N_{\mathbf{p}}. \quad (12.7)$$

Fermi quantization, on the other hand, would yield a meaningless result—a value of E independent of $N_{\mathbf{p}}$.

The “charge” Q of the field under consideration vanishes. This is already clear from the fact that the charge Q must change sign when particles are replaced by antiparticles,

¹⁰This is analogous to the extra factor 1/2 in the operator (2.10) for the electromagnetic field energy density (expressed via the Hermitian operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$), as compared to the photon energy density (3.2) expressed through the photon’s complex wave function; cf. the remark on p. 26.

and here the two coincide. Correspondingly, no 4-vector current density exists either. Indeed, the expression

$$\hat{j}_\mu = i[\hat{\psi}^+ \partial_\mu \hat{\psi} - (\partial_\mu \hat{\psi}^+) \hat{\psi}] \quad (12.8)$$

for the operator of the conserved 4-current $\hat{j}$ vanishes when $\hat{\psi} = \hat{\psi}^+$ (and the vector $\hat{\psi} \partial_\mu \hat{\psi}$ is not by itself conserved). This in turn signifies the absence of any special conservation law that would restrict the possible changes of the particle number. It is evident that such particles are, in any case, electrically neutral.

Particles of this kind are called *strictly neutral*, as distinct from electrically neutral particles that possess an antiparticle. While the latter can annihilate (converting into photons) only in pairs, strictly neutral particles can annihilate singly.

The structure of the ψ -operator (12.1) is the same as that of the operators (2.17)–(2.20) of the electromagnetic field. In this sense one can say that photons themselves are strictly neutral particles. In the case of the electromagnetic field, the Hermiticity of the operators was linked to the reality of the field strengths as measurable (in the classical limit) physical quantities. For the ψ -operators of particles, however, no such link exists, since they do not correspond to any directly measurable quantities at all.

The absence of a conserved 4-vector current is a general property of strictly neutral particles and is not connected with zero spin (it holds for photons as well). Physically it expresses the absence of the corresponding prohibitions on changes in particle number. From a formal standpoint there is a direct connection between the absence of a conserved current and the reality of the field—the Hermiticity of the operator $\hat{\psi}$.

The Lagrangian of a complex field

$$\hat{L} = \partial_\mu \hat{\psi}^+ \cdot \partial^\mu \hat{\psi} - m^2 \hat{\psi}^+ \hat{\psi} \quad (12.9)$$

is invariant under multiplication of the ψ -operator by an arbitrary phase factor, i.e. under transformations of the form

$$\hat{\psi} \rightarrow e^{i\alpha} \hat{\psi}, \quad \hat{\psi}^+ \rightarrow e^{-i\alpha} \hat{\psi}^+ \quad (12.10)$$

(these are termed *gauge* transformations). In particular, the Lagrangian remains unchanged under an infinitesimally small gauge transformation

$$\hat{\psi} \rightarrow \hat{\psi} + i\delta\alpha \cdot \hat{\psi}, \quad \hat{\psi}^+ \rightarrow \hat{\psi}^+ - i\delta\alpha \cdot \hat{\psi}^+. \quad (12.11)$$

Under an infinitesimally small variation of the “generalized coordinates” q , the Lagrangian undergoes the change

$$\begin{aligned} \delta\hat{L} &= \sum \left(\frac{\partial\hat{L}}{\partial q} \delta q + \frac{\partial\hat{L}}{\partial q_{,\mu}} \delta q_{,\mu} \right) = \\ &= \sum \left(\frac{\partial\hat{L}}{\partial q} - \frac{\partial}{\partial x^\mu} \frac{\partial\hat{L}}{\partial q_{,\mu}} \right) \delta q + \sum \frac{\partial}{\partial x^\mu} \left(\frac{\partial\hat{L}}{\partial q_{,\mu}} \delta q \right) \end{aligned}$$

(summation over all q). The first term vanishes by virtue of the “equations of motion” (Lagrange’s equations). Taking $\hat{\psi}$ and $\hat{\psi}^+$ as the “coordinates” q and setting

$$\delta\hat{\psi} = i\delta\alpha \cdot \hat{\psi}, \quad \delta\hat{\psi}^+ = -i\delta\alpha \cdot \hat{\psi}^+,$$

we get

$$\delta \hat{L} = i\delta\alpha \frac{\partial}{\partial x^\mu} \left(\hat{\psi} \frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}} - \hat{\psi}^+ \frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}^+} \right).$$

It is apparent that the condition of invariance of the Lagrangian ($\delta \hat{L} = 0$) is equivalent to the continuity equation ($\partial_\mu \hat{j}^\mu = 0$) for the 4-vector

$$\hat{j}^\mu = i \left(\hat{\psi}^+ \frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}^+} - \hat{\psi} \frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}} \right). \quad (12.12)$$

It is readily verified that for the Lagrangian (12.9) this formula yields the current (12.8).

We see, therefore, that in the mathematical framework of the theory the existence of a conserved current is tied to gauge invariance of the Lagrangian (W. Pauli, 1941). The Lagrangian (12.2) of a strictly neutral field, however, lacks this symmetry.

§ 13. The transformations C, P, T

Unlike four-dimensional inversion, a three-dimensional (spatial) inversion cannot be reduced to any rotation of the 4-coordinate system: the determinant of this transformation equals not +1 but -1. The symmetry properties of particles with respect to inversion (P -transformation) are therefore not dictated by the requirements of relativistic invariance¹¹.

Applied to a scalar wave function, the inversion operation amounts to the transformation

$$\hat{P}\psi(t, \mathbf{r}) = \pm\psi(t, -\mathbf{r}), \quad (13.1)$$

where the sign “+” or “-” on the right-hand side corresponds, respectively, to a true scalar or a pseudoscalar.

It is apparent from this that one must distinguish two aspects of the wave function's behavior under inversion. One of them is related to the dependence of the wave function on the coordinates. In nonrelativistic quantum mechanics only this aspect was considered—it leads to the notion of the parity of a state (which we shall henceforth call the *orbital parity*), characterizing the symmetry properties of the particle's motion. If a state possesses a definite orbital parity (+1 or -1), this means that

$$\psi(t, -\mathbf{r}) = \pm\psi(t, \mathbf{r}).$$

The second aspect is the behavior of the wave function at a given point (which one may conveniently take to be the coordinate origin) when the coordinate axes are inverted. This behavior gives rise to the concept of the *intrinsic parity of the particle*. For a spin-0 particle the two signs in definition (13.1) correspond to intrinsic-parity values of +1 and -1. The overall parity of a particle system equals the product of their individual intrinsic parities and the orbital parity associated with the relative motion.

The “intrinsic” symmetry properties of different particles manifest themselves, of course, only in the processes of their mutual transformations. The analogue of intrinsic

¹¹The Lorentz group supplemented by spatial inversion is called the *extended Lorentz group* (as opposed to the original group, which does not contain P and which in this context is termed *proper*). The extended group includes all transformations that do not carry the t axis out of the corresponding half of the light cone.

parity in nonrelativistic quantum mechanics is the parity of a bound state of a composite system (for instance, a nucleus). From the standpoint of relativistic theory, which draws no fundamental distinction between composite and elementary particles, this kind of intrinsic parity is no different from the intrinsic parity of the particles that figure as elementary ones in nonrelativistic theory. In the nonrelativistic domain, where the latter behave as immutable objects, their intrinsic symmetry properties are unobservable, and consequently any consideration of them would be devoid of physical content.

Within the second-quantization formalism, the intrinsic parity is expressed through the behavior of the ψ -operators under inversion. For scalar and pseudoscalar fields the transformation laws read

$$P : \hat{\psi}(t, \mathbf{r}) \rightarrow \pm \hat{\psi}(t, -\mathbf{r}). \quad (13.2)$$

The very meaning of the action of inversion on the ψ -operator must, however, be formulated as a definite transformation of the particle annihilation and creation operators—one such that the change (13.2) results from it. It is easy to see that the required transformation is

$$P : \hat{a}_{\mathbf{p}} \rightarrow \pm \hat{a}_{-\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \rightarrow \pm \hat{b}_{-\mathbf{p}} \quad (13.3)$$

(and likewise for the conjugate operators). Indeed, carrying out this substitution in the operator

$$\hat{\psi}(t, \mathbf{r}) = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}} e^{-i\omega t + i\mathbf{p}\mathbf{r}} + \hat{b}_{\mathbf{p}}^{\dagger} e^{i\omega t - i\mathbf{p}\mathbf{r}} \right) \quad (13.4)$$

and then relabeling the summation variable ($\mathbf{p} \rightarrow -\mathbf{p}$), we bring it to the form $\pm \hat{\psi}(t, -\mathbf{r})$. Thus, denoting by $\hat{\psi}^P(t, \mathbf{r})$ the operator in which transformation (13.3) has been performed, one may write

$$\hat{\psi}^P(t, \mathbf{r}) = \pm \hat{\psi}(t, -\mathbf{r}). \quad (13.5)$$

Note that transformation (13.3) has a perfectly natural form: inversion reverses the sign of the polar vector $\mathbf{p}$, so that particles with momentum $\mathbf{p}$ are replaced by particles with momentum $-\mathbf{p}$.

In (13.3) the operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ both transform with either the upper or the lower signs. In the second-quantization formalism this expresses the equality of the intrinsic parities of particle and antiparticle (with spin 0). This equality is in itself already obvious from the fact that particles and antiparticles (of spin 0) are described by the very same (scalar or pseudoscalar) wave functions.

In relativistic theory there also arises a symmetry with respect to a transformation that has no counterpart in nonrelativistic theory; it is called *charge conjugation* (C -transformation). If all operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ are mutually interchanged:

$$C : \hat{a}_{\mathbf{p}} \rightarrow \hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \rightarrow \hat{a}_{\mathbf{p}} \quad (13.6)$$

(i.e. particles and antiparticles are interchanged), then $\hat{\psi}$ goes over into the “charge-conjugate” operator $\hat{\psi}^C$, with

$$\hat{\psi}^C(t, \mathbf{r}) = \hat{\psi}^{\dagger}(t, \mathbf{r}) \quad (13.7)$$

This equality expresses the symmetry with which the notions of particles and antiparticles enter the theory.

Let us note that the definition of the charge-conjugation transformation contains a certain inessential formal arbitrariness. The content of the transformation is unaltered if one introduces into definition (13.6) an arbitrary phase factor:

$$\hat{a}_{\mathbf{p}} \rightarrow e^{i\alpha} \hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \rightarrow e^{-i\alpha} \hat{a}_{\mathbf{p}}.$$

Then one would have

$$\hat{\psi} \rightarrow e^{i\alpha} \hat{\psi}^+, \quad \hat{\psi}^+ \rightarrow e^{-i\alpha} \hat{\psi},$$

and a twofold application of this transformation would still yield the identity ($\hat{\psi} \rightarrow \hat{\psi}$). All such definitions are, however, equivalent to one another. Since the properties of the ψ -operators are unaffected by multiplication by a phase factor (cf. the end of the preceding section), one may simply relabel $\hat{\psi}$ as $\hat{\psi} e^{i\alpha/2}$ and then return to the definition of charge conjugation in the form (13.6), (13.7).

Because charge conjugation replaces a particle by a non-identical antiparticle, it does not in the general case give rise to any new characteristic of a particle or a system of particles as such.

Systems containing equal numbers of particles and antiparticles constitute an exception. The operator $\hat{C}$ carries such a system into itself, so that eigenstates exist with eigenvalues $C = \pm 1$ (which follow from $\hat{C}^2 = 1$). When describing charge symmetry one may treat particle and antiparticle as two distinct “charge states” of a single particle, distinguished by the charge quantum number $Q = \pm 1$. The system’s wave function then factorizes into an orbital part and a “charge” part, and must be symmetric with respect to the simultaneous interchange of all variables (spatial and charge) belonging to any pair of particles. The symmetry character of the “charge” factor fixes the charge parity of the system (see the Problem)¹².

The concept of charge parity, which arises naturally for “strictly neutral” systems, should apply also to strictly neutral “elementary” particles. In the second-quantization formalism this concept is described by the relation

$$\hat{\psi}^C = \pm \hat{\psi}; \quad (13.8)$$

the signs “+” and “−” correspond to charge-even and charge-odd particles.

In § 11 it was pointed out that relativistic invariance must also entail invariance under 4-inversion. For the operator of a scalar (in the sense of 4-rotations) field this means that under such a transformation one must have

$$\hat{\psi}(t, \mathbf{r}) = \hat{\psi}(-t, -\mathbf{r})$$

always with the same sign “+” on the right-hand side. In terms of the operators $\hat{a}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}$, the conversion of $\hat{\psi}(t, \mathbf{r})$ into $\hat{\psi}(-t, -\mathbf{r})$ is accomplished by interchanging the coefficients of e^{-ipx} and e^{ipx} in (13.4), i.e. by the replacement

$$\hat{a}_{\mathbf{p}} \rightarrow \hat{b}_{\mathbf{p}}^+, \quad \hat{b}_{\mathbf{p}} \rightarrow \hat{a}_{\mathbf{p}}^+ \quad (13.9)$$

¹²In these arguments we have in mind particles of spin 0. The method of analysis described here generalizes directly to other cases—see, for example, the Problem in § 27.

Since it replaces the a -operators by b -operators, this transformation incorporates the mutual interchange of particles and antiparticles. We see that in relativistic theory there naturally arises a requirement of invariance under a transformation in which spatial inversion (P) and time reversal (T) are accompanied by charge conjugation (C); this statement is known as the *CPT theorem*¹³.

In this connection it is, however, appropriate to stress that although the arguments presented here and in §§ 11, 12 appear to be a natural development of the concepts of ordinary quantum mechanics and classical relativity, the results obtained in this way go beyond that framework both in form (ψ -operators that simultaneously contain creation and annihilation operators for particles) and in substance (particles and antiparticles). These results cannot therefore be regarded as a matter of pure logical necessity. They embody new physical principles whose correctness can be judged only by experiment.

If we denote by $\hat{\psi}^{CPT}(t, \mathbf{r})$ the operator (13.4) in which the transformation (13.9) has been carried out, one may write

$$\hat{\psi}^{CPT}(t, \mathbf{r}) = \hat{\psi}(-t, -\mathbf{r}). \quad (13.10)$$

Having thus formulated 4-inversion as the transformation (13.9), we thereby establish for the ψ -operator also the formulation of the time-reversal transformation: together with the CP transformation (called the *combined inversion*) it must yield (13.9). Taking into account the definitions (13.3) and (13.6), we therefore find

$$T : \hat{a}_{\mathbf{p}} \rightarrow \pm \hat{a}_{-\mathbf{p}}^+, \quad \hat{b}_{\mathbf{p}} \rightarrow \pm \hat{b}_{-\mathbf{p}}^+ \quad (13.11)$$

(the signs “ $\pm$ ” match those in (13.3)). The meaning of this transformation is perfectly natural: time reversal not only converts motion with momentum $\mathbf{p}$ into motion with momentum $-\mathbf{p}$, but also interchanges the initial and final states in matrix elements; the annihilation operators for particles with momenta $\mathbf{p}$ are therefore replaced by creation operators for particles with momenta $-\mathbf{p}$. Performing the substitution (13.11) in (13.4) and relabeling the summation variable ($\mathbf{p} \rightarrow -\mathbf{p}$), we find that¹⁴

$$\hat{\psi}^T(t, \mathbf{r}) = \pm \hat{\psi}^+(-t, \mathbf{r}). \quad (13.12)$$

This result mirrors the standard time-reversal prescription of quantum mechanics: when a state has wave function $\psi(t, \mathbf{r})$, the “temporally reversed” state is represented by $\psi^*(-t, \mathbf{r})$; complex conjugation is needed in order to recover the “proper” character of the time dependence, which the sign flip of t would otherwise spoil (*E. P. Wigner*, 1932).

Since the transformation T (and with it CPT) interchange initial and final states, the concepts of eigenstates and eigenvalues have no meaning for them. They therefore do not lead to new characteristics of particles as such. The consequences to which they lead when applied to scattering processes will be discussed in §§ 69, 71.

Let us examine how the operator 4-vector of the current $\hat{j}^\mu$ (12.8) transforms under C , P , and T . Transformation (13.2) together with the replacement $(\partial_0, \partial_i) \rightarrow (\partial_0, -\partial_i)$

¹³It was formulated by *Lüders* (G. Lüders, 1954) and *Pauli* (W. Pauli, 1955).

¹⁴If one were to define the operation T independently of the other transformations, the same arbitrariness in the choice of a phase factor would arise as for the operation C . The requirement of CPT symmetry, however, leaves the phase-factor choice free in only one of the two transformations, C or T .

gives

$$P : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \rightarrow (\hat{j}^0, -\hat{\mathbf{j}})_{t,-\mathbf{r}}, \quad (13.13)$$

as it must for a true 4-vector. Transformation (13.7) would simply give

$$C : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \rightarrow (-\hat{j}^0, -\hat{\mathbf{j}})_{t,\mathbf{r}}, \quad (13.14)$$

were the operators $\hat{\psi}$ and $\hat{\psi}^+$ commutative. The noncommutativity of these operators arises, however, solely from the noncommutativity of $\hat{a}_{\mathbf{p}}$ and $\hat{a}_{\mathbf{p}}^+$ (or of $\hat{b}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}^+$) at the same $\mathbf{p}$; but by virtue of the commutation rules (11.4) the interchange of these operators merely introduces terms independent of the occupation numbers, i.e. of the field state. Discarding these terms as inessential (just as in (11.5),(11.6)), we recover rule (13.14), which has a natural meaning: by replacing particles with antiparticles, charge conjugation reverses the sign of every component of the 4-current.

Since the time-reversal operation is linked with the transposition of initial and final states, its application to a product of operators reverses the order of the factors. Thus,

$$(\hat{\psi}^+ \partial_\mu \hat{\psi})^T = (\partial_\mu \hat{\psi})^T (\hat{\psi}^+)^T.$$

In the present case, however, this is immaterial: by virtue of the commutativity of the ψ -operators (in the sense indicated above), reverting to the original order of the factors does not affect the result. Noting also that under time reversal $(\partial_0, \partial_i) \rightarrow (-\partial_0, \partial_i)$, we find the transformation rule for the current:

$$T : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \rightarrow (\hat{j}^0, -\hat{\mathbf{j}})_{-t,\mathbf{r}}. \quad (13.15)$$

The three-dimensional vector $\mathbf{j}$ changes sign in accordance with the classical meaning of this quantity.

Finally, under the CPT transformation we have

$$CPT : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \rightarrow (-\hat{j}^0, -\hat{\mathbf{j}})_{-t,-\mathbf{r}}, \quad (13.16)$$

in accord with the meaning of this operation as a 4-inversion. Let us stress in this connection that, since 4-inversion reduces to a rotation of the 4-coordinate system, with respect to it there is no distinction at all between two types (true and pseudo) of 4-tensors of any rank.

Up to now we have been considering free particles. But the quantum numbers of parity acquire real significance only when one examines interacting particles, where definite selection rules are associated with them, permitting or forbidding particular processes. Such significance, however, can attach only to conserved quantities—eigenvalues of operators that commute with the Hamiltonian of the interacting particles.

Relativistic invariance guarantees that the CPT -transformation operator commutes with the Hamiltonian in every case. As for the individual transformations C and P (together with T), experiment demonstrates that electromagnetic and strong interactions respect these symmetries, so that the associated parity quantum numbers are conserved within those interactions. In weak processes, however, these conservation laws break down ¹⁵.

¹⁵The suggestion that parity conservation might fail in weak interactions was first put forward by Lee and Yang (*T. D. Lee, C. N. Yang*, 1956). Earlier still, the broader idea that P - and T -invariance of physical laws is not mandatory was advanced by Dirac (1949).

Anticipating later developments, we point out that the operator governing the coupling of charged particles to the electromagnetic field is formed as the product of the operator 4-vectors $\hat{A}$ and $\hat{j}$. Because charge conjugation flips the sign of $\hat{j}$, electromagnetic-interaction invariance under this transformation requires that $\hat{A}$ likewise change sign. Put differently, photons are charge-odd particles.

The stated behavior of the operators $\hat{A}$ is in agreement with the properties of the 4-potential in classical theory. Indeed, from the transformations

$$C : (\hat{A}_0, \hat{\mathbf{A}}) \rightarrow (-\hat{A}_0, -\hat{\mathbf{A}})_{t, \mathbf{r}},$$

$$P : (\hat{A}_0, \hat{\mathbf{A}}) \rightarrow (\hat{A}_0, -\hat{\mathbf{A}})_{t, -\mathbf{r}},$$

$$CPT : (\hat{A}_0, \hat{\mathbf{A}}) \rightarrow (-\hat{A}_0, -\hat{\mathbf{A}})_{-t, -\mathbf{r}},$$

it follows that

$$T : (\hat{A}_0, \hat{\mathbf{A}}) \rightarrow (\hat{A}_0, -\hat{\mathbf{A}})_{-t, \mathbf{r}},$$

which corresponds to the classical rule for transforming the electromagnetic-field potentials under time reversal.

The requirement of *CPT* invariance does not impose any restrictions on the properties of particles in themselves. It does, however, lead to a definite connection between the properties of particles and antiparticles. This includes, first and foremost, the equality of their masses—a fact already clear from the connection, set out in § 11, between 4-inversion and the very origin of the particle–antiparticle concept.

Furthermore, *CPT* invariance implies that the proportionality coefficients relating the electric and magnetic moment vectors to the spin vector differ for particle and antiparticle only in sign. Indeed, the magnetic moment changes sign under *C* and *T* transformations and (being an axial vector) is *P*-invariant. Consequently, the *CPT* transformation, while converting a particle into its antiparticle, does not at the same time change the sign of the magnetic moment; the spin vector, however, does change sign. The same applies to the electric moment, which remains unchanged under time reversal and changes sign under *C*-transformation and (by the properties of a polar vector) under spatial inversion.

The requirements of *P*- or *T*-invariance (if they are obeyed), on the other hand, constrain the properties of each individual particle: they forbid the particle from possessing an electric dipole moment. Indeed, the only vector that can be constructed for an elementary particle at rest from its ψ -operators is the vector of its spin operator. This vector is *P*-even and *T*-odd; it can therefore determine only the magnetic, but not the electric moment. Let us emphasize that for this prohibition it suffices to require merely one of the two—*P*- or *T*-invariance.

Problem

Determine the charge parity and spatial parity of a system consisting of a spin-0 particle and its antiparticle, with orbital angular momentum l of the relative motion.

Solution. Permutation of the particle coordinates is equivalent to inversion (about the center of inertia) and therefore multiplies the orbital function by $(-1)^l$; permutation of

the charge variables is equivalent to charge conjugation and multiplies the “charge” factor in the wave function by the sought quantity C . From the condition $C(-1)^l = 1$ we have

$$C = (-1)^l.$$

The spatial parity P of the system is the product of the orbital parity with the intrinsic parities of the two particles. Because particle and antiparticle share the same intrinsic parity, here P simply equals the orbital parity: $P = (-1)^l$.

§ 14. Wave equation for spin-1 particles

In its rest frame a particle of spin 1 is characterized by a wave function with three components—a three-dimensional vector (one therefore often speaks of a *vector* particle). Viewed four-dimensionally, these three components can be either the spatial part of a (spacelike) 4-vector ψ^μ , or the mixed components of an antisymmetric 4-tensor of rank two, $\psi^{\mu\nu}$, for which the time (ψ^0) and purely spatial (ψ^{ik}) components vanish in the rest frame¹⁶.

The wave equation—a differential relation between the quantities ψ^μ and $\psi^{\mu\nu}$ —is established by the relations that we write as

$$i\psi_{\mu\nu} = \hat{p}_\mu\psi_\nu - \hat{p}_\nu\psi_\mu, \quad (14.1)$$

$$im^2\psi_\mu = \hat{p}^\nu\psi_{\mu\nu}, \quad (14.2)$$

where $\hat{p} = i\partial$ (A. Proca, 1936). Applying the operator $\hat{p}^\mu$ to both sides of equation (14.2), we obtain (on account of the antisymmetry of $\psi_{\mu\nu}$)

$$\hat{p}^\mu\psi_\mu = 0. \quad (14.3)$$

From (14.1, 14.2) one can eliminate $\psi_{\mu\nu}$ by inserting the first equation into the second. With the aid of (14.3) this yields

$$(\hat{p}^2 - m^2)\psi_\mu = 0, \quad (14.4)$$

from which it is again apparent (cf. § 10) that m is the particle mass. A free spin-1 particle can therefore be described entirely by a single 4-vector ψ^μ whose components satisfy the second-order equation (14.4) together with the supplementary condition (14.3), which removes from ψ^μ the part belonging to spin 0.

In the rest frame, where ψ_μ is independent of the spatial coordinates, we find that $\hat{p}^0\psi_0 = 0$. Since at the same time $\hat{p}^0\psi_0 = m\psi_0$, it follows that $\psi_0 = 0$ in the rest frame, as required. Together with ψ_0 the quantities ψ_{ik} also vanish.

A spin-1 particle can possess different intrinsic parities, depending on whether ψ^μ is a true vector or a pseudovector. In the first case

$$\hat{P}\psi^\mu = (\psi^0, -\psi^i),$$

¹⁶Looking ahead, let us mention that the set consisting of the 4-vector ψ_μ and the 4-tensor $\psi^{\nu\mu}$ corresponds to the set of four-dimensional second-rank spinors $\xi^{\alpha\beta}$, $\eta_{\dot{\alpha}\dot{\beta}}$, $\xi^{\alpha\dot{\beta}}$, where $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ are symmetric spinors that transform into each other under inversion (see § 19).

and in the second

$$\hat{P}\psi^\mu = (-\psi^0, \psi^i).$$

Equations (14.1),(14.2) can be derived from a variational principle with the Lagrangian

$$L = (1/2)\psi_{\mu\nu}\psi^{\mu\nu*} - (1/2)\psi^{\mu\nu*}(\partial_\mu\psi_\nu - \partial_\nu\psi_\mu) - \\ - (1/2)\psi^{\mu\nu}(\partial_\mu\psi_\nu^* - \partial_\nu\psi_\mu^*) + m^2\psi_\mu\psi^{\mu*}. \quad (14.5)$$

The independent generalized coordinates here are $\psi_\mu, \psi_\mu^*, \psi_{\mu\nu}, \psi_{\mu\nu}^*$ ¹⁷.

For obtaining the energy-momentum tensor, formula (10.11) is not entirely convenient in the present case, since it would lead to a non-symmetric tensor requiring additional symmetrization. One may instead use the formula

$$\frac{1}{2}T_{\mu\nu}\sqrt{-g} = -\frac{\partial}{\partial x^\lambda}\frac{\partial\sqrt{-g}L}{\partial g_{,\lambda}^{\mu\nu}} + \frac{\partial\sqrt{-g}L}{\partial g^{\mu\nu}}, \quad (14.6)$$

in which it is assumed that L is written in a form valid for arbitrary curvilinear coordinates (see II, § 94). If L involves only the components of the metric tensor $g_{\mu\nu}$ itself (but not their coordinate derivatives), the formula simplifies to

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}}\frac{\partial\sqrt{-g}L}{\partial g^{\mu\nu}} = 2\frac{\partial L}{\partial g^{\mu\nu}} - g_{\mu\nu}L$$

(recall that $d\ln g = -g_{\mu\nu}dg^{\mu\nu}$).

Since the differentiation in formula (14.6) is not performed with respect to ψ_μ or $\psi_{\mu\nu}$, when applying it there is no need to treat these quantities as independent; one may directly use relation (14.1) and rewrite the Lagrangian (14.5) as

$$L = (-1/2)\psi_{\mu\nu}\psi_{\lambda\rho}^*g^{\mu\lambda}g^{\nu\rho} + m^2\psi_\mu\psi_\nu^*g^{\mu\nu}. \quad (14.7)$$

Then

$$T_{\mu\nu} = -\psi_{\mu\lambda}\psi_\nu^{\lambda*} - \psi_{\mu\lambda}^*\psi_\nu^\lambda + m^2\left(\psi_\mu^*\psi_\nu + \psi_\nu^*\psi_\mu\right) + \\ + g_{\mu\nu}\left((1/2)\psi_{\lambda\rho}\psi^{\lambda\rho*} - m^2\psi_\lambda^*\psi^\lambda\right). \quad (14.8)$$

In particular, the energy density is given by a manifestly positive expression

$$T_{00} = (1/2)\psi_{ik}\psi_{ik}^* + \psi_{0i}\psi_{0i}^* + m^2(\psi_0\psi_0^* + \psi_i\psi_i^*). \quad (14.9)$$

The conserved 4-vector of the current density reads

$$j^\mu = i(\psi^{\mu\nu*}\psi_\nu - \psi^{\mu\nu}\psi_\nu^*). \quad (14.10)$$

¹⁷Had we carried out the variation with respect to ψ_μ alone (assuming from the outset that $\psi_{\mu\nu}$ is expressed through ψ_μ via (14.1)), equation (14.3) would have to be imposed as an auxiliary condition unrelated to the variational principle.

It can be obtained from formula (12.12) by differentiating the Lagrangian (14.5) with respect to the derivatives $\partial_\mu \psi_\nu$. In particular,

$$j^0 = i \left(\psi^{0k*} \psi_k - \psi^{0k} \psi_k^* \right) \quad (14.11)$$

and is not a positive-definite quantity.

A plane wave normalized to one particle in volume $V = 1$:

$$\psi_\mu = \frac{1}{\sqrt{2\varepsilon}} u_\mu e^{-ipx}, \quad u_\mu u^{\mu*} = -1, \quad (14.12)$$

where u_μ is a unit polarization 4-vector obeying (by virtue of (14.3)) the four-dimensional transversality condition

$$u_\mu p^\mu = 0. \quad (14.13)$$

Indeed, substituting the function (14.12) into (14.9) and (14.11) gives

$$T_{00} = -2\varepsilon^2 \psi_\mu \psi^{\mu*} = \varepsilon, \quad j^0 = 1.$$

In contrast to the photon, a vector particle of nonzero mass possesses three independent polarization directions. For the corresponding amplitudes see (16.21).

For partially polarized vector particles the polarization density matrix is introduced in such a way that for a pure state it becomes simply the product

$$\rho_{\mu\nu} = u_\mu u_\nu^*$$

(analogously to expression (8.7) for photons). According to (14.12), (14.13) it satisfies the conditions

$$p^\mu \rho_{\mu\nu} = 0, \quad \rho_\mu^\mu = -1. \quad (14.14)$$

For unpolarized particles the matrix $\rho_{\mu\nu}$ must have the form $a g_{\mu\nu} + b p_\mu p_\nu$. Fixing the coefficients a and b from (14.14), one finds

$$\rho_{\mu\nu} = -\frac{1}{3} \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right). \quad (14.15)$$

Quantization of the vector-particle field proceeds in the same way as for the scalar case, and there is no need to repeat the whole argument anew. The ψ -operators of the vector field take the form

$$\begin{aligned} \hat{\psi}_\mu &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{b}_{\mathbf{p}\alpha}^+ u_\mu^{(\alpha)*} e^{ipx} \right), \\ \hat{\psi}_\mu^+ &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\alpha}^+ u_\mu^{(\alpha)*} e^{ipx} + \hat{b}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} \right), \end{aligned} \quad (14.16)$$

where the index α labels the three independent polarizations.

The positive definiteness of expression (14.9) for T_{00} and the indefiniteness of j^0 (14.11) lead, just as in the scalar case, to the requirement of Bose quantization.

There exists a close connection between the properties of the strictly neutral vector field and the electromagnetic field. A neutral vector field is described by a Hermitian ψ -operator:

$$\hat{\psi}_\mu = \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{c}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{c}_{\mathbf{p}\alpha}^+ u_\mu^{(\alpha)*} e^{ipx} \right). \quad (14.17)$$

The Lagrangian of this field is

$$\hat{L} = \frac{1}{4} \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} - \frac{1}{2} \hat{\psi}^{\mu\nu} (\partial_\mu \hat{\psi}_\nu - \partial_\nu \hat{\psi}_\mu) + \frac{1}{2} m^2 \hat{\psi}_\mu \hat{\psi}^\mu. \quad (14.18)$$

The electromagnetic field corresponds to the case $m = 0$. The 4-vector ψ^μ then becomes the 4-potential A^μ , and the 4-tensor $\psi^{\mu\nu}$ becomes the field tensor $F^{\mu\nu}$, related to the potential through definition (14.1). Equation (14.2) turns into $\partial^\nu \psi_{\mu\nu} = 0$, which corresponds to the second pair of Maxwell's equations. Condition (14.3) no longer follows from it and thus ceases to be obligatory. In the absence of the supplementary condition there is no need to treat $\hat{\psi}_\mu$ and $\hat{\psi}_{\mu\nu}$ as independent “coordinates” in the Lagrangian, and (14.18) reduces to

$$\hat{L} = -(1/4) \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} \quad (14.19)$$

in agreement with the familiar classical expression for the electromagnetic-field Lagrangian. This Lagrangian, together with the tensor $\hat{\psi}_{\mu\nu}$, is invariant under an arbitrary gauge transformation of the “potentials” $\hat{\psi}_\mu$. The connection between this circumstance and the vanishing mass is clearly visible: the Lagrangian (14.18) lacks this property on account of the term $m^2 \hat{\psi}_\mu \hat{\psi}^\mu$.

§ 15. Wave equation for particles with higher integer spins

Since the wave equations (14.3, 14.4) follow directly from specifying the mass and spin of the particle, the practical use of the Lagrangian comes down not so much to deriving these equations as to constructing the expressions for the energy, momentum, and charge of the field.

For this purpose, as already noted, one may use expression (14.7) in place of (14.5), and the former can be further transformed. Using (14.1) and rewriting the Lagrangian (14.7) as

$$\begin{aligned} L &= -(\partial_\mu \psi_\nu^*) (\partial^\mu \psi^\nu) + (\partial_\nu \psi_\mu^*) (\partial^\mu \psi^\nu) + m^2 \psi_\mu \psi^{\mu*} = \\ &= -(\partial_\mu \psi_\nu^*) (\partial^\mu \psi^\nu) + m^2 \psi_\mu^* \psi^\mu + \partial_\nu (\psi_\mu^* \partial^\mu \psi^\nu) - \psi_\mu^* \partial^\mu \partial_\nu \psi^\nu. \end{aligned} \quad (15.1)$$

By virtue of (14.3) the last term vanishes, while the penultimate one is a total derivative. Dropping it, we arrive at the Lagrangian

$$L' = -(\partial_\mu \psi_\nu^*) (\partial^\mu \psi^\nu) + m^2 \psi_\mu^* \psi^\mu.$$

It has the same structure as the Lagrangian (10.9) for a spin-0 particle, differing only in that the scalar ψ is replaced by the 4-vector ψ_μ and the overall sign is reversed. The sign

change is due to the spacelike character of ψ_μ , so that $\psi_\mu \psi^{\mu*} < 0$, whereas for a scalar particle $\psi \psi^* > 0$.

If one constructs the energy–momentum 4-tensor and the current 4-vector with the help of Lagrangian (15.1), one obtains expressions of the same form as expressions (10.12) and (10.18) for the scalar field:

$$T_{\mu\nu} = -\partial_\mu \psi^{\lambda*} \cdot \partial_\nu \psi_\lambda - \partial_\nu \psi^{\lambda*} \cdot \partial_\mu \psi_\lambda - L' g_{\mu\nu}, \quad (15.2)$$

$$j_\mu = -i [\psi_\lambda^* \partial_\mu \psi^\lambda - (\partial_\mu \psi_\lambda^*) \psi^\lambda]. \quad (15.3)$$

Their difference from (14.8) and (14.10) likewise reduces to total derivatives. But the local values of these quantities lack (as was already stressed) any deep physical significance. Only the volume integrals P_μ (10.15) and Q (10.19) matter, and they coincide for either choice of $T_{\mu\nu}$ and j_μ .

This method of description extends straightforwardly to particles of arbitrary (integer) spin. The wave function of a particle with spin s is an irreducible 4-tensor of rank s , that is, a tensor symmetric in all of its indices and vanishing upon contraction over any pair of them:

$$\psi_{\dots\mu\dots\nu\dots} = \psi_{\dots\nu\dots\mu\dots}, \quad \psi_{\dots\mu\dots}{}^\mu{}_{\dots} = 0. \quad (15.4)$$

This tensor must obey the supplementary four-dimensional transversality condition

$$\hat{p}^\mu \psi_{\dots\mu\dots} = 0, \quad (15.5)$$

and each of its components must satisfy the second-order equation

$$(\hat{p}^2 - m^2) \psi_{\dots} = 0. \quad (15.6)$$

In the rest frame, condition (15.5) causes every component of the 4-tensor that carries a 0 among its indices to vanish. In other words, the wave function in the rest frame (i.e. in the nonrelativistic limit) reduces, as expected, to an irreducible 3-tensor of rank s with $2s + 1$ independent components.

The Lagrangian, the energy–momentum tensor, and the current vector for particles of spin s differ from (15.1)–(15.3) solely by the replacement of ψ_λ with $\psi_{\lambda\mu\dots}$.

The normalized plane wave is

$$\psi^{\mu\nu\dots} = \frac{1}{\sqrt{2\varepsilon}} u^{\mu\nu\dots} e^{-ipx}, \quad u_{\mu\nu\dots}^* u^{\mu\nu\dots} = -1, \quad (15.7)$$

with the wave amplitude satisfying the conditions

$$u^{\dots\mu\dots} p_\mu = 0. \quad (15.8)$$

There are $2s + 1$ independent polarization states.

Field quantization proceeds as an obvious generalization of the spin-0 or spin-1 cases.

The scheme presented above is entirely sufficient for the stated purpose—describing the field of free particles. The situation changes, however, if one poses the problem of describing the interaction of particles with the electromagnetic field. Such an interaction

would need to be incorporated into a Lagrangian from which all equations could be obtained without having to impose supplementary conditions separately. In practice, however, it turns out that this approach to describing the interaction works only for electrons—particles of spin $1/2$ (see § 32). For other spins the problem would therefore be of purely methodological interest.

Let us note that for all (integer and half-integer) spins $s > 1$ it proves impossible to formulate a variational principle using only a single function (tensorial or spinorial) whose rank matches the given spin. For this purpose it becomes necessary to introduce, as auxiliary quantities, tensorial or spinorial objects of lower rank as well. The Lagrangian is then chosen so that these auxiliary quantities automatically vanish by virtue of the free-particle field equations that follow from the variational principle¹⁸.

§ 16. Helicity states of a particle¹⁹

Within relativistic theory, the orbital angular momentum $\mathbf{l}$ and the spin $\mathbf{s}$ of a particle in motion lack individual conservation laws—only their sum, the total angular momentum $\mathbf{j} = \mathbf{l} + \mathbf{s}$, is conserved. Consequently, the spin projection onto a given direction (the z axis) is also nonconserved, rendering it unsuitable for labeling the polarization (spin) states of a particle in motion.

What is conserved, however, is the projection of the spin on the direction of the momentum: since $\mathbf{l} = [\mathbf{r}\mathbf{p}]$, the product $\mathbf{s}\mathbf{n}$ coincides with the conserved product $\mathbf{j}\mathbf{n}$ ($\mathbf{n} = \mathbf{p}/|\mathbf{p}|$). This quantity is called the *helicity*²⁰ (we have already considered it for the photon in § 8). Its eigenvalues will be denoted by λ ($\lambda = -s, \dots, +s$), and states of the particle with definite values of λ will be called helicity states.

Let $\psi_{\mathbf{p}\lambda}$ be the wave function (a plane wave) describing a state of the particle with definite $\mathbf{p}$ and λ , and let $u^{(\lambda)}(\mathbf{p})$ be its amplitude; for brevity we suppress the component indices of this function (for integer spin these are 4-tensor indices).

We saw in the preceding sections that a relativistic description of particles with nonzero (integer) spin requires introducing a wave function whose number of components exceeds $2s + 1$. The number of independent components, however, remains equal to $2s + 1$; the “extra” components are eliminated by imposing supplementary conditions, by virtue of which these components vanish in the rest frame (in the next chapter we shall see the same for half-integer s).

According to the transformation rules for angular momentum (see II, § 14), the helicity is invariant under Lorentz transformations that leave unchanged the direction of $\mathbf{p}$ onto which the angular momentum is projected. Therefore the number λ retains its meaning as a quantum number under such transformations, and to study the symmetry properties of helicity states one may use a reference frame in which $|\mathbf{p}| \ll m$ (in the limit, the rest frame). Then $\psi_{\mathbf{p}\lambda}$ reduces to a nonrelativistic $(2s + 1)$ -component wave function. Let us denote its amplitude by $w^{(\lambda)}(\mathbf{n})$, indicating as the argument the direction $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ along which the angular momentum is quantized. The amplitude $w^{(\lambda)}$ is an eigenfunction

¹⁸See Fierz M., Pauli W.//Proc. Roy. Soc. — 1939. — V. A 173. — P. 211. In that work the program indicated here was carried out for particles of spin $3/2$ and 2 .

¹⁹The content of this section applies to particles of any (integer or half-integer) spin.

²⁰In the Russian literature the term is *spiralnost'*.

of the operator $\mathbf{n}\hat{\mathbf{s}}$:

$$\mathbf{n}\hat{\mathbf{s}} w^{(\lambda)}(\mathbf{n}) = \lambda w^{(\lambda)}(\mathbf{n}). \quad (16.1)$$

In the spinor representation $w^{(\lambda)}$ is a contravariant symmetric spinor of rank $2s$; by the correspondence formulas (57.2) (see III) its components can also be labeled by the values of the spin projection σ on a fixed axis z ²¹.

In the momentum representation the wave functions of the states under consideration essentially coincide with the amplitudes $u^{(\lambda)}(\mathbf{p})$. Specifically:

$$\psi_{\mathbf{p}\lambda}(\mathbf{k}) = u^{(\lambda)}(\mathbf{k})\delta^{(2)}(\mathbf{v} - \mathbf{n}) = u^{(\lambda)}(\mathbf{p})\delta^{(2)}(\mathbf{v} - \mathbf{n}), \quad (16.2)$$

where the momentum as an independent variable is denoted $\mathbf{k}$, in contrast to its eigenvalue $\mathbf{p}$, and $\mathbf{v} = \mathbf{k}/|\mathbf{k}|$, in contrast to $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ ²². In the nonrelativistic limit

$$\psi_{\mathbf{n}\lambda}(\mathbf{v}) = w_{\sigma}^{(\lambda)}(\mathbf{v})\delta^{(2)}(\mathbf{v} - \mathbf{n}) = w^{(\lambda)}(\mathbf{n})\delta^{(2)}(\mathbf{v} - \mathbf{n}). \quad (16.3)$$

In more detail this expression should be written as

$$\psi_{\mathbf{n}\lambda}(\mathbf{v}, \sigma) = w_{\sigma}^{(\lambda)}(\mathbf{v})\delta^{(2)}(\mathbf{v} - \mathbf{n}),$$

where the discrete independent variable σ is also displayed explicitly.

The helicity operator $\hat{\mathbf{s}}\mathbf{n}$ is commutative with $\hat{j}_z$ and $\hat{\mathbf{j}}^2$. This follows from the fact that the angular-momentum operator corresponds to an infinitesimal coordinate rotation, while the dot product of any two vectors remains unchanged under arbitrary rotations. Consequently, stationary states exist in which the particle has simultaneously well-defined angular momentum j , its projection $j_z = m$, and helicity λ . We shall refer to such states as *spherical helicity states*.

We now find the wave functions of these states in the momentum representation. The construction proceeds by direct analogy with the expressions derived in III, § 103 for the wave functions of a symmetric top. Those results were obtained from the transformation formulas for wave functions under finite rotations (see III, § 58). Since that derivation relies exclusively on rotational-symmetry properties, it applies to momentum-representation functions just as it does to coordinate-space functions.

Along with the space-fixed coordinate system xyz (with respect to which the functions $\psi_{jm\lambda}$ are written), let us also introduce a “body-fixed” frame $\xi\eta\zeta$ with the ζ axis along the direction of $\mathbf{v}$. Without repeating the corresponding reasoning (cf. the derivation of formula (103.8) (see III)), we write

$$\psi_{jm\lambda}(\mathbf{k}) = \psi_{j\lambda}^{(0)} D_{\lambda m}^{(j)}(\mathbf{v}),$$

²¹The reasoning given here (together with the listing of allowed values of λ) pertains to particles whose mass differs from zero. For a particle of zero mass no rest frame exists, and the helicity admits only the two values $\lambda = \pm s$. This restriction traces back to the circumstance noted already in § 8: states of such a particle fall into representations of the axial-symmetry group, which permits at most a twofold degeneracy of levels (from the viewpoint of the wave equation this signifies that upon passing to the limit $m \rightarrow 0$ the system of equations for a spin- s particle separates into independent equations describing massless particles of spins $s, s-1, \dots$). In particular, for the photon $\lambda = \pm 1$, and the three-dimensional vectors $\mathbf{e}^{(\pm 1)}$ (8.2) assume the role of the corresponding amplitudes $w^{(\lambda)}$.

²²Here the delta function $\delta^{(2)}$ is defined so that $\int \delta^{(2)}(\mathbf{v} - \mathbf{n}) d\mathbf{o}_{\mathbf{v}} = 1$. In (16.2) (and in the analogous case below, see (16.4)) the delta function ensuring the given value of the energy has been omitted.

where $\psi_{j\lambda}^{(0)}$ is the wave function in the “body-fixed” frame describing a state with a definite value of the ζ -projection of the angular momentum: $j_\zeta = \lambda$; in the momentum representation this function evidently coincides with the amplitude $u^{(\lambda)}$. The normalized (see below) wave function is

$$\psi_{jm\lambda}(\mathbf{k}) = \sqrt{\frac{2j+1}{4\pi}} D_{\lambda m}^{(j)}(\mathbf{v}) u^{(\lambda)}(\mathbf{k}). \quad (16.4)$$

Here, however, a question arises regarding the choice of phases, connected with the following ambiguity. The rotation of the system xyz relative to $\xi\eta\zeta$ is specified by three Euler angles α, β, γ ; the direction of $\mathbf{v}$, on which alone the wave function of the particle can depend, is determined only by two spherical angles $\alpha \equiv \varphi, \beta \equiv \theta$. One must therefore adopt some convention for the angle γ . We shall set $\gamma = 0$, i.e. define $D_{\lambda m}^{(j)}(\mathbf{v})$ as

$$D_{\lambda m}^{(j)}(\mathbf{v}) = D_{\lambda m}^{(j)}(\varphi, \theta, 0) = e^{im\varphi} d_{\lambda m}^{(j)}(\theta). \quad (16.5)$$

By virtue of (58.21) (see III) the functions (16.5) satisfy the orthogonality and normalization conditions:

$$\int D_{\lambda_1 m_1}^{(j_1)*}(\mathbf{v}) D_{\lambda_2 m_2}^{(j_2)}(\mathbf{v}) \frac{d\mathbf{v}}{4\pi} = \frac{1}{2j+1} \delta_{j_1 j_2} \delta_{m_1 m_2} \quad (16.6)$$

($d\mathbf{v} = \sin\theta d\theta d\varphi$). Orthogonality in the index λ is guaranteed by the amplitude factor $u^{(\lambda)}$. The functions $\psi_{jm\lambda}$ are therefore mutually orthogonal, as required, across all three labels $jm\lambda$, and the prefactor adopted in (16.4) ensures their normalization according to

$$\int |\psi_{jm\lambda}|^2 d\mathbf{v} = 1. \quad (16.7)$$

Here it is assumed that the amplitudes $u^{(\lambda)}$ are normalized to unity: $u^{(\lambda)} u^{(\lambda)*} = 1$.

Let us consider the behavior of helicity-state wave functions under coordinate inversion. The product of the polar vector $\mathbf{v}$ and the axial vector $\mathbf{j}$ is a pseudoscalar. It is therefore clear that upon inversion a state of helicity λ goes over into a state with $-\lambda$; one must, however, determine the phase factors in these transformations.

Under inversion $\mathbf{v} \rightarrow -\mathbf{v}$. The vector $\mathbf{v}$ is specified by two angles φ, θ , and the transformation $\mathbf{v} \rightarrow -\mathbf{v}$ is effected by the replacement $\varphi \rightarrow \varphi + \pi, \theta \rightarrow \pi - \theta$. This determines the axis ζ , but the position of the axes ξ and η , which also depends on the third Euler angle γ , remains unspecified; the transformation of θ and φ alone does not allow one to distinguish between an inversion and a rotation of the axis ζ in this sense. In terms of all three Euler angles the inversion is the transformation

$$\alpha \equiv \varphi \rightarrow \varphi + \pi, \quad \beta \equiv \theta \rightarrow \pi - \theta, \quad \gamma \rightarrow \pi - \gamma. \quad (16.8)$$

Therefore, if $D_{\lambda m}^{(j)}(\mathbf{v})$ is defined according to (16.5) (i.e. with $\gamma = 0$), and the substitution $\mathbf{v} \rightarrow -\mathbf{v}$ is understood as the result of an inversion, then

$$D_{\lambda m}^{(j)}(-\mathbf{v}) = D_{\lambda m}^{(j)}(\varphi + \pi, \pi - \theta, \pi). \quad (16.9)$$

Using formulas (58.9), (58.16), (58.18) (see III) we therefore find

$$D_{\lambda m}^{(j)}(-\mathbf{v}) = e^{i\lambda\pi} d_{\lambda m}^{(j)}(\pi - \theta) e^{im(\varphi + \pi)} =$$

$$= (-1)^{j-\lambda} e^{im\varphi} d_{-\lambda m}^{(j)}(\theta) = (-1)^{j-\lambda} D_{-\lambda m}^{(j)}(\mathbf{v}),$$

or

$$D_{\lambda m}^{(j)}(-\mathbf{v}) = (-1)^{-\lambda} D_{-\lambda m}^{(j)}(\mathbf{v}) \quad (16.10)$$

($j - \lambda$ is an integer).

An analogous formula for the spinor $w^{(\lambda)}$ can be obtained by noting that its components $w_{\sigma}^{(\lambda)}$ coincide, up to a factor, with the functions

$$w_{\sigma}^{(\lambda)}(\mathbf{v}) \propto D_{\lambda \sigma}^{(s)}(\mathbf{v})^*. \quad (16.11)$$

Indeed, applying the transformation formula (58.7) (see III) to the spin eigenfunctions and setting the ζ -projection of the spin to the definite value λ (i.e. replacing $\psi_{jm'}$ on the right-hand side of (58.7) (see III) by $\delta_{m'\lambda}$), we find that $D_{\lambda \sigma}^{(s)}(\mathbf{v})$ are spin wave functions corresponding to definite values of the z - and ζ -projections (σ and λ). The totality of these functions ($\sigma = -s, \dots, +s$) forms (by the correspondence formulas (57.6) (see III)) a covariant spinor of rank $2s$. The components of the contravariant spinor (57.2) (see III), however—corresponding to the components $w_{\sigma}^{(\lambda)}$ —transform as the complex conjugates of the components of a covariant spinor of the same rank. From (16.10), (16.11) we have

$$w^{(\lambda)}(-\mathbf{v}) = (-1)^{s-\lambda} w^{(-\lambda)}(\mathbf{v}) \quad (16.12)$$

($s - \lambda$ is an integer). The inversion operation applied to $w^{(\lambda)}$, however, consists not only in the replacement $\mathbf{v} \rightarrow -\mathbf{v}$, but also in multiplication by an overall phase factor (the “intrinsic parity” of the particle), which we denote by η :

$$\hat{P}w^{(\lambda)}(\mathbf{v}) = \eta w^{(\lambda)}(-\mathbf{v}) = \eta(-1)^{s-\lambda} w^{(-\lambda)}(\mathbf{v}). \quad (16.13)$$

For the relativistic amplitude $u^{(\lambda)}(\mathbf{k})$ this transformation takes the form

$$\hat{P}u^{(\lambda)}(\mathbf{k}) = \eta \beta u^{(\lambda)}(-\mathbf{k}) = \eta(-1)^{s-\lambda} u^{(-\lambda)}(\mathbf{k}), \quad (16.14)$$

where β is a certain matrix that acts as the identity on the components of $u^{(\lambda)}$ surviving in the limit $|\mathbf{p}| \rightarrow 0$. It is important that this matrix is independent of the quantum numbers of the state, so that the difference between (16.13) and (16.14) is in this sense inessential²⁴.

Applying (16.14) to (16.2), we obtain the transformation law for the wave functions of the states $|\mathbf{n}\lambda\rangle$:

$$\hat{P}\psi_{\mathbf{n}\lambda}(\mathbf{v}) = \eta(-1)^{s-\lambda} \psi_{-\mathbf{n}-\lambda}(\mathbf{v}). \quad (16.15)$$

For the spherical helicity states, making use of (16.10) and (16.12), we obtain the transformation law:

$$\hat{P}\psi_{jm\lambda}(\mathbf{v}) = \eta(-1)^{j-s} \psi_{jm-\lambda}(\mathbf{v}). \quad (16.16)$$

²⁴Thus, for $s = 1$ the amplitudes $u^{(\lambda)}$ are 4-vectors (16.22); in this case β is the full identity matrix with respect to 4-vector indices: $\beta_{\mu\nu} = \delta_{\mu\nu}$. For $s = 1/2$, $u^{(\lambda)}$ is a bispinor; the phase factor is then $\eta = i$, and β is the Dirac matrix γ^0 (see (21.10)).

The states ψ_{jm0} transform, according to (16.16), into themselves, i.e. they possess a definite parity. If, however, $\lambda \neq 0$, then only superpositions of states with opposite helicities have a definite parity:

$$\psi_{jm|\lambda|}^{(\pm)} = \frac{1}{\sqrt{2}} (\psi_{jm\lambda} \pm \psi_{jm-\lambda}). \quad (16.17)$$

Under inversion they transform into themselves according to

$$\hat{P}\psi_{jm|\lambda|}^{(\pm)}(\mathbf{v}) = \pm\eta(-1)^{j-s}\psi_{jm|\lambda|}^{(\pm)}(\mathbf{v}). \quad (16.18)$$

Let us draw attention to the fact that in this section we have classified the states of a free particle with given angular momentum by operating solely with conserved quantities, without appealing to the concept of orbital angular momentum (which was used, for instance, in §§ 6, 7 for the classification of photon states).

As an illustration consider the case of spin 1. In the rest frame the amplitudes $u^{(\lambda)}$ (4-vectors) reduce to three-dimensional vectors $\mathbf{e}^{(\lambda)}$, which play here the role of the amplitudes $w^{(\lambda)}$. The action of the spin-1 operator on a vector function $\mathbf{e}$ is given by

$$(\hat{S}_i \mathbf{e})_k = -ie_{ikl}e_l \quad (16.19)$$

(see III, § 57, Problem 2). Equation (16.1) therefore assumes the form

$$i[\mathbf{n}\mathbf{e}^{(\lambda)}] = \lambda\mathbf{e}^{(\lambda)}. \quad (16.20)$$

Its solutions (in a coordinate frame $\xi\eta\zeta$ with the ζ axis along $\mathbf{n}$) coincide with the circular unit vectors (7.14)²⁶:

$$\mathbf{e}^{(0)} = i(0, 0, 1), \quad \mathbf{e}^{(\pm 1)} = \mp \frac{i}{\sqrt{2}}(1, \pm i, 0). \quad (16.21)$$

In a frame where the particle has momentum $\mathbf{p}$, the helicity-state amplitudes are the 4-vectors

$$u^{(0)\mu} = \left(\frac{|\mathbf{p}|}{m}, \frac{\varepsilon}{m} \mathbf{e}^{(0)} \right), \quad u^{(\pm 1)\mu} = (0, \mathbf{e}^{(\pm 1)}). \quad (16.22)$$

If $\mathbf{e}$ is a polar vector, then $\eta = -1$. In that case the functions (16.17) (which for $s = 1$ are three-dimensional vectors) have the following parities:

$$\begin{aligned} \psi_{jm|\lambda|}^{(+)} : \quad P &= (-1)^j, \\ \psi_{jm|\lambda|}^{(-)} : \quad P &= (-1)^{j+1}, \\ \psi_{jm0} : \quad P &= (-1)^j. \end{aligned}$$

Comparing with the definition of vector spherical harmonics (7.4), we see that these functions are identical (up to phase factors) with $\mathbf{Y}_{jm}^{(e)}$, $\mathbf{Y}_{jm}^{(m)}$, $\mathbf{Y}_{jm}^{(l)}$ respectively.

²⁶The choice of phase factors is fixed by the requirement that the matrix elements of the spin operators computed with the eigenfunctions (16.21) conform to the general definitions in III, §§ 27, 107.

Determining the phase factors (say, by comparing the values at $\theta = 0$), we arrive at the equalities

$$\begin{aligned} \mathbf{Y}_{jm}^{(e)} &= i^{j-1} \sqrt{\frac{2j+1}{8\pi}} \left(\mathbf{e}^{(1)} D_{1m}^{(j)}(\nu) + \mathbf{e}^{(-1)} D_{-1m}^{(j)}(\nu) \right), \\ \mathbf{Y}_{jm}^{(m)} &= i^{j-1} \sqrt{\frac{2j+1}{8\pi}} \left(\mathbf{e}^{(1)'} D_{1m}^{(j)}(\nu) + \mathbf{e}^{(-1)'} D_{-1m}^{(j)}(\nu) \right), \\ \mathbf{Y}_{jm}^{(l)} &= i^{j-1} \sqrt{\frac{2j+1}{4\pi}} \mathbf{e}^{(0)} D_{0m}^{(j)} \end{aligned} \quad (16.23)$$

(j is an integer!); $\mathbf{e}^{(\lambda)'} = [\mathbf{n}\mathbf{e}^{(\lambda)}]$ are circular unit vectors in axes $\xi'\eta'\zeta$ that are rotated relative to $\xi\eta\zeta$ by 90° about the ζ axis.

The last of formulas (16.23) is equivalent to the expression (58.23) (see III) for $d_{0m}^{(j)}(\theta)$. From the first (or the second) formula one can obtain a simple expression for the functions $d_{\pm 1m}^{(j)}$. We have

$$i^{j-1} \sqrt{\frac{2j+1}{8\pi}} D_{\pm 1m}^{(j)} = \mathbf{Y}_{jm}^{(e)} \mathbf{e}^{(\pm 1)*} = \frac{1}{\sqrt{j(j+1)}} \mathbf{e}^{(\pm 1)*} \nabla Y_{jm}.$$

We expand the scalar product on the right-hand side in the frame $\xi\eta\zeta$, where

$$\left(\frac{\partial}{\partial \xi}, \frac{\partial}{\partial \eta} \right) \rightarrow \left(\frac{\partial}{\partial \theta}, \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} \right).$$

Recalling definition (7.2) of the function Y_{jm} and definition (16.5), we obtain as the final result

$$d_{\pm 1m}^{(j)}(\theta) = (-1)^{m+1} \sqrt{\frac{(j-m)!}{(j+m)!j(j+1)}} \left(\pm \frac{\partial}{\partial \theta} + \frac{m}{\sin \theta} \right) P_j^m(\cos \theta), \quad m \geq 0. \quad (16.24)$$

Fermions

§ 17. Four-dimensional spinors

In nonrelativistic theory a particle of arbitrary spin s is described by a $(2s + 1)$ -component quantity—a symmetric spinor of rank $2s$. From the mathematical standpoint these are quantities that furnish the irreducible representations of the spatial rotation group.

In relativistic theory this group appears only as a subgroup of the broader group of four-dimensional rotations—the Lorentz group. In connection with this there arises the need to construct a theory of four-dimensional spinors (4-spinors)—quantities that carry the irreducible representations of the Lorentz group; the exposition of this theory occupies §§ 17–19. In §§ 17, 18 only the proper Lorentz group is considered, which does not include spatial inversion; the latter will be treated in § 19.

The construction of the 4-spinor theory proceeds along the same lines as the three-dimensional spinor theory (*B. L. van der Waerden*, 1929; *G. E. Uhlenbeck*, *O. Laporte*, 1931).

A spinor ξ^α is a two-component quantity ($\alpha = 1, 2$); regarded as components of the wave function of a spin-1/2 particle, ξ^1 and ξ^2 correspond to eigenvalues of the z -projection of the spin equal to $+1/2$ and $-1/2$ respectively. Under any (proper) Lorentz transformation the two quantities ξ^1, ξ^2 transform through each other:

$$\xi^{1'} = \alpha\xi^1 + \beta\xi^2, \quad \xi^{2'} = \gamma\xi^1 + \delta\xi^2. \quad (17.1)$$

The coefficients $\alpha, \beta, \gamma, \delta$ are definite functions of the rotation angles of the 4-coordinate system, subject to the condition

$$\alpha\delta - \beta\gamma = 1, \quad (17.2)$$

i.e. the determinant of the binary transformation (17.1) equals 1, just as the determinants of coordinate transformations in the Lorentz group do.

By virtue of conditions (17.2) the bilinear form $\xi^1\Xi^2 - \xi^2\Xi^1$ (where ξ^α and Ξ^α are two spinors) is invariant under the transformation (17.1) (it corresponds to a spin-0 particle “composed” of two spin-1/2 particles). For a natural way of writing such invariant expressions, alongside the “contravariant” components of the spinor ξ^α one also introduces “covariant” components ξ_α . The passage from one set to the other is carried out with the aid of the “spinor metric” $g_{\alpha\beta}$ ¹:

$$\xi_\alpha = g_{\alpha\beta}\xi^\beta, \quad (17.3)$$

where

$$g = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (17.4)$$

¹ Spinor indices will be denoted by the first letters of the Greek alphabet: $\alpha, \beta, \gamma, \dots$.

so that

$$\xi_1 = \xi^2, \quad \xi_2 = -\xi^1. \quad (17.5)$$

The invariant $\xi^1 \Xi^2 - \xi^2 \Xi^1$ is then written as a scalar product $\xi^\alpha \Xi_\alpha$. Note that $\xi^\alpha \Xi_\alpha = -\xi_\alpha \Xi^\alpha$.

Up to this point the properties listed above have formally coincided with those of three-dimensional spinors. A difference arises, however, when one turns to complex-conjugate spinors.

In nonrelativistic theory the sum

$$\psi^1 \psi^{1*} + \psi^2 \psi^{2*}, \quad (17.6)$$

which governs the probability density of finding particles at a given point in space, had to be a scalar, and for this purpose the components $\psi^{\alpha*}$ had to transform as covariant spinor components; put differently, the transformation (17.1) had to be unitary ($\alpha = \delta^*$, $\beta = -\gamma^*$). In the relativistic case, however, the particle density does not behave as a scalar; it forms the time component of a 4-vector. Owing to this, the requirement just stated ceases to hold, and no further restrictions on the transformation coefficients arise (apart from (17.2)). Four complex quantities $\alpha, \beta, \gamma, \delta$ constrained only by (17.2) amount to $8 - 2 = 6$ real parameters—in agreement with the number of angles specifying a 4-coordinate rotation (rotations in six coordinate planes).

Consequently, complex-conjugate binary transformations prove to be essentially different from one another, and in relativistic theory two varieties of spinors arise. To distinguish between them one adopts special notation: the indices of spinors whose transformation law is the complex conjugate of (17.1) are written as digits with dots placed above them (*dotted indices*). Thus, by definition,

$$\eta^{\dot{\alpha}} \sim \xi^{\alpha*}, \quad (17.7)$$

where the sign $\sim$ stands for “transforms as.” In other words, the transformation formulas for a “dotted” spinor are:

$$\eta^{\dot{1}'} = \alpha^* \eta^{\dot{1}} + \beta^* \eta^{\dot{2}}, \quad \eta^{\dot{2}'} = \gamma^* \eta^{\dot{1}} + \delta^* \eta^{\dot{2}}. \quad (17.8)$$

The operations of lowering and raising dotted indices are performed in the same way as for undotted ones:

$$\eta_{\dot{1}} = \eta^{\dot{2}}, \quad \eta_{\dot{2}} = -\eta^{\dot{1}}. \quad (17.9)$$

With respect to spatial rotations the behavior of 4-spinors coincides with that of 3-spinors. For the latter, as we know, $\psi_\alpha^* \sim \psi^\alpha$. By virtue of definition (17.7) the 4-spinor $\eta_{\dot{\alpha}}$ therefore behaves under rotations as a contravariant 3-spinor ψ^α . The eigenvalues of the spin projection $1/2$ and $-1/2$ accordingly correspond to the covariant components $\eta_{\dot{1}}$ and $\eta_{\dot{2}}$.

Spinors of higher rank are defined as sets of quantities that transform like products of components of several first-rank spinors. The indices of a higher-rank spinor may include both dotted and undotted ones. For instance, three types of second-rank spinors exist:

$$\xi^{\alpha\beta} \sim \xi^\alpha \Xi^\beta, \quad \zeta^{\alpha\dot{\beta}} \sim \xi^\alpha \eta^{\dot{\beta}}, \quad \eta^{\dot{\alpha}\dot{\beta}} \sim \eta^{\dot{\alpha}} H^{\dot{\beta}}.$$

Merely stating the total rank of a spinor is therefore insufficient for an unambiguous definition of the concept; when necessary we shall specify the rank as a pair of numbers (k, l) —the number of undotted and the number of dotted indices.

Since the transformations (17.1) and (17.8) are algebraically independent, there is no need to fix the ordering of dotted and undotted indices (in this sense, for example, the spinors $\zeta^{\alpha\dot{\beta}}$ and $\zeta^{\dot{\beta}\alpha}$ are one and the same).

For an expression involving spinors to have an invariant character, it must carry the same number of undotted and dotted indices on both sides; otherwise the expression will certainly be violated upon passing from one reference frame to another. One should keep in mind here that complex conjugation entails the interchange of dotted and undotted indices. Therefore the relation $\eta^{\dot{\alpha}\dot{\beta}} = (\xi^{\alpha\beta})^*$ between two spinors does have an invariant character.

Contraction of spinors or their products can be performed only over pairs of indices of the same kind—two dotted or two undotted. Summation over a pair of indices of different kinds is not an invariant operation. For this reason, from the spinor

$$\zeta^{\alpha_1 \alpha_2 \dots \alpha_k \dot{\beta}_1 \dot{\beta}_2 \dots \dot{\beta}_l}, \quad (17.10)$$

which is symmetric in all k undotted and in all l dotted indices, it is impossible to form a spinor of lower rank (recall that contracting over a pair of indices in which the spinor is symmetric gives zero). This means that from the quantities (17.10) one cannot assemble any smaller set of their linear combinations that would transform among themselves under every group transformation. Stated differently, symmetric 4-spinors realize the irreducible representations of the proper Lorentz group. Every such representation is labeled by a pair of numbers (k, l) .

Since each spinorial index runs over two values, there are $k + 1$ essentially distinct sets of numbers $\alpha_1 \alpha_2 \dots \alpha_k$ in (17.10) (containing 0, 1, 2, $\dots$, k ones and $k, k - 1, \dots$, 0 twos) and $l + 1$ sets of numbers $\dot{\beta}_1 \dot{\beta}_2 \dots \dot{\beta}_l$. Altogether, therefore, a symmetric spinor of rank (k, l) has $(k + 1)(l + 1)$ independent components; this is the dimensionality of the irreducible representation it carries.

§ 18. Relation of spinors to 4-vectors

A spinor $\zeta^{\alpha\dot{\beta}}$ carrying one dotted and one undotted index possesses $2 \cdot 2 = 4$ independent components—precisely the number of components belonging to a 4-vector. It is therefore evident that both objects furnish one and the same irreducible representation of the proper Lorentz group, and a definite correspondence must hold between their respective components.

To establish this correspondence we turn first of all to the analogous correspondence in the three-dimensional case, bearing in mind that with respect to purely spatial rotations the behavior of 3- and 4-spinors must be identical.

For a three-dimensional spinor $\psi^{\alpha\beta}$ the correspondence formulas hold (see III, § 57),

which we write here in the form

$$\begin{aligned} a_x &= \frac{1}{2}(\psi^{22} - \psi^{11}) = \frac{1}{2}(\psi_1^2 + \psi_2^1), \\ a_y &= -\frac{i}{2}(\psi^{22} + \psi^{11}) = \frac{i}{2}(\psi_2^1 - \psi_1^2), \\ a_z &= \frac{1}{2}(\psi^{12} + \psi^{21}) = \frac{1}{2}(\psi_1^1 - \psi_2^2), \end{aligned} \quad (\text{III.1})$$

where a_x, a_y, a_z are the components of a certain three-dimensional vector $\mathbf{a}$. In passing to the four-dimensional case one must replace the components ψ_β^α by $\zeta^{\alpha\beta}$, and interpret a_x, a_y, a_z as the contravariant components a^1, a^2, a^3 of a 4-vector. As for the expression for the fourth component, a^0 , its form is already clear from the circumstance noted in § 17: the quantity (17.6) must transform as a^0 . Hence $a^0 \sim \zeta^{11} + \zeta^{22}$; the proportionality coefficient is fixed by requiring that the scalar $\zeta_{\alpha\beta}\zeta^{\alpha\beta}$ coincide with $2a_\mu a^\mu \equiv 2a^2$.

We thus arrive at the following correspondence formulas:

$$\begin{aligned} a^1 &= \frac{1}{2}(\zeta^{12} + \zeta^{21}), & a^2 &= \frac{i}{2}(\zeta^{12} - \zeta^{21}), \\ a^3 &= \frac{1}{2}(\zeta^{11} - \zeta^{22}), & a^0 &= \frac{1}{2}(\zeta^{11} + \zeta^{22}). \end{aligned} \quad (18.1)$$

The inverse formulas are:

$$\begin{aligned} \zeta^{11} &= \zeta_{22} = a^3 + a^0, & \zeta^{22} &= \zeta_{11} = a^0 - a^3, \\ \zeta^{12} &= -\zeta_{21} = a^1 - ia^2, & \zeta^{21} &= -\zeta_{12} = a^1 + ia^2. \end{aligned} \quad (18.2)$$

Here

$$\zeta_{\alpha\beta}\zeta^{\alpha\beta} = 2a^2. \quad (18.3)$$

We note also that

$$\zeta_{\alpha\beta}\zeta^{\gamma\beta} = \delta_\alpha^\gamma a^2. \quad (18.4)$$

To see why this last relation holds, observe that the quantity $\zeta_{\alpha\beta}\zeta_\gamma^{\beta}$, viewed as a second-rank spinor in the indices α and γ , is antisymmetric and hence must be proportional to the metric spinor.

The link between $\zeta^{\alpha\beta}$ and a 4-vector is actually one instance of a broader principle: any symmetric spinor of rank (k, k) can be placed in one-to-one correspondence with an irreducible symmetric 4-tensor of rank k (irreducibility meaning that contraction over every index pair gives zero).

The relationship linking a spinor to a 4-vector can be written compactly with the aid of the two-by-two Pauli matrices²:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (18.5)$$

Denoting symbolically by ζ the matrix of the quantities $\zeta^{\alpha\beta}$ with upper indices (the first being undotted), formulas (18.2) take the form

$$\zeta = \mathbf{a}\boldsymbol{\sigma} + a^0 \quad (18.6)$$

²For brevity of notation, operators (matrices) that act on spin variables will be denoted by letters without carets.

(in the second term one understands, of course, the product of a^0 with the unit matrix). The inverse relations are:

$$\mathbf{a} = \frac{1}{2} \text{Sp}(\zeta \boldsymbol{\sigma}), \quad a^0 = \frac{1}{2} \text{Sp} \zeta. \quad (18.7)$$

With the help of formulas (18.6), (18.7) one can establish the connection between the transformation laws for the 4-vector and the spinor, thereby expressing the spinor transformation law through the parameters of rotations of the 4-coordinate system.

Let us write the transformation of the spinor ξ^α in the form

$$\xi^{\alpha'} = (B\xi)^\alpha, \quad B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad (18.8)$$

where B is a two-by-two matrix assembled from the coefficients of the binary transformation. The transformation of the dotted spinor is then:

$$\eta^{\dot{\beta}'} = (B^* \eta)^{\dot{\beta}} = (\eta B^+)^{\dot{\beta}}, \quad (18.9)$$

and the transformation of the second-rank spinor $\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha \eta^{\dot{\beta}}$ can be written symbolically as $\zeta' = B\zeta B^{+1}$. For an infinitesimally small transformation $B = 1 + \lambda$, where λ is a small matrix, and to first-order accuracy

$$\zeta' = \zeta + (\lambda\zeta + \zeta\lambda^+). \quad (18.10)$$

Consider first the Lorentz transformation to a reference frame moving with an infinitesimally small velocity $\delta\mathbf{V}$ (without altering the orientation of the spatial axes). Under this transformation the 4-vector $a^\mu = (a^0, \mathbf{a})$ changes according to

$$\mathbf{a}' = \mathbf{a} - a^0 \delta\mathbf{V}, \quad a^{0'} = a^0 - \mathbf{a} \delta\mathbf{V}. \quad (18.11)$$

Let us now make use of formulas (18.7). The transformation of a^0 can be expressed, on the one hand, as

$$a^{0'} = a^0 - \mathbf{a} \delta\mathbf{V} = a^0 - \frac{1}{2} \text{Sp}(\zeta \boldsymbol{\sigma} \delta\mathbf{V}),$$

and on the other hand as

$$a^{0'} = \frac{1}{2} \text{Sp} \zeta' = a^0 + \frac{1}{2} \text{Sp} \zeta (\lambda + \lambda^+).$$

These expressions must coincide identically (i.e. for arbitrary ζ). From this we obtain the equality:

$$\lambda + \lambda^+ = -\boldsymbol{\sigma} \delta\mathbf{V}.$$

By the same method, examining the transformation of $\mathbf{a}$, we find

$$\boldsymbol{\sigma} \lambda + \lambda^+ \boldsymbol{\sigma} = -\delta\mathbf{V}.$$

These equalities, regarded as equations for λ , yield the solution:

$$\lambda = \lambda^+ = -\frac{1}{2} \boldsymbol{\sigma} \delta\mathbf{V}.$$

Thus the infinitesimally small Lorentz transformation of the spinor ξ^α is effected by the matrix

$$B = 1 - \frac{1}{2}(\boldsymbol{\sigma}\mathbf{n})\delta V, \quad (18.12)$$

where $\mathbf{n}$ is a unit vector along the velocity $\delta\mathbf{V}$. From this it is straightforward to obtain the transformation for a finite velocity $\mathbf{V}$. To do so, recall that the Lorentz transformation amounts (geometrically) to a rotation of the 4-coordinate system in the $t\mathbf{n}$ plane through an angle φ related to the velocity $\mathbf{V}$ by $\text{th } \varphi = V^3$. An infinitesimally small transformation corresponds to the angle $\delta\varphi = \delta V$, while a rotation through a finite angle φ is achieved by repeating the rotation through $\delta\varphi$ a total of $\varphi/\delta\varphi$ times. Raising the operator (18.12) to the power $\varphi/\delta\varphi$ and passing to the limit $\delta\varphi \rightarrow 0$, we obtain

$$B = e^{-\frac{\varphi}{2}\mathbf{n}\boldsymbol{\sigma}}. \quad (18.13)$$

The mathematical meaning of this operator becomes transparent once one observes that, by the properties of the Pauli matrices, all even powers of $\mathbf{n}\boldsymbol{\sigma}$ equal 1 while all odd powers equal $\mathbf{n}\boldsymbol{\sigma}$. Noting that ch decomposes into even powers and sh into odd powers of its argument, we arrive at the final result

$$B = \text{ch } \frac{\varphi}{2} - \mathbf{n}\boldsymbol{\sigma} \text{sh } \frac{\varphi}{2}, \quad \text{th } \varphi = V. \quad (18.14)$$

We note that the Lorentz-transformation matrices B turn out to be Hermitian: $B = B^+$.

Consider now an infinitesimally small rotation of the spatial coordinate system. Under such a rotation the three-dimensional vector $\mathbf{a}$ transforms according to

$$\mathbf{a}' = \mathbf{a} - [\delta\boldsymbol{\theta} \mathbf{a}], \quad (18.15)$$

where $\delta\boldsymbol{\theta}$ is the vector of the infinitesimally small rotation angle. The corresponding spinor transformation could be found in an analogous manner. There is no need for this, however, since under spatial rotations the behavior of 4-spinors coincides with that of 3-spinors, and for the latter the transformation is known beforehand from the general connection between the spin operator and the operator of an infinitesimally small rotation:

$$B = 1 + \frac{i}{2}\boldsymbol{\sigma} \delta\boldsymbol{\theta}. \quad (18.16)$$

The passage to a rotation through a finite angle θ is carried out in the same way as the passage from (18.12) to (18.14):

$$B = \exp\left(\frac{i\theta}{2}\mathbf{n}\boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} + i\mathbf{n}\boldsymbol{\sigma} \sin \frac{\theta}{2}, \quad (18.17)$$

where $\mathbf{n}$ is the unit vector along the rotation axis. This matrix is unitary ($B^+ = B^{-1}$), as must be the case for a spatial rotation.

³We recall that in planes containing the time axis the metric is pseudo-Euclidean.

§ 19. Inversion of spinors

When developing the three-dimensional spinor theory (in Vol. III) we did not consider how spinors behave with respect to the spatial inversion operation, inasmuch as in the nonrelativistic framework this would not have produced any new physical consequences. It is worth pausing on this matter now, however, for a clearer understanding of the inversion properties of 4-spinors that will be discussed below.

The inversion operation does not alter the sign of an axial vector, and the spin vector is precisely of this type. Consequently the value of its projection s_z remains unchanged as well. It follows that under inversion each component of the spinor ψ^α can transform only into itself, i.e. one must have

$$\psi^\alpha \rightarrow P\psi^\alpha, \quad (19.1)$$

where P is a constant coefficient. Applying inversion twice returns us to the original coordinate system. For spinors, however, the return to the initial configuration can be understood in two distinct senses: as a rotation of the system by 0° or by 360° . For spinors these two interpretations are not equivalent, since ψ^α reverses sign under a 360° rotation. Two alternative definitions of inversion therefore arise: in the first,

$$P^2 = 1, \quad P = \pm 1, \quad (19.2)$$

and in the second,

$$P^2 = -1, \quad P = \pm i. \quad (19.3)$$

A crucial requirement is that inversion be defined in the same manner for every spinor. It would be inconsistent for distinct spinors to follow different inversion rules (one obeying (19.2), another (19.3)), for then one could not always build a scalar (or pseudoscalar) out of an arbitrary pair of spinors: if ψ^α were governed by (19.2) while φ^α were governed by (19.3), the product $\psi^\alpha \varphi_\alpha$ would pick up a factor $\pm i$ upon inversion instead of staying fixed (or simply flipping sign).

It should also be stressed that (under either definition of inversion) the assignment of one or another parity P to a spinor carries no absolute significance, because spinors change sign under a rotation through 2π , which can always be performed simultaneously with the inversion. What does possess an absolute character, however, is the “relative parity” of two spinors, defined as the parity of the scalar $\psi^\alpha \varphi_\alpha$ formed from them; since a 2π rotation reverses the sign of all spinors simultaneously, the associated ambiguity does not affect the parity of the scalar in question.

Let us turn now to four-dimensional spinors.

We note first that since inversion changes the sign of only three (x, y, z) out of the four (t, x, y, z) coordinates, it commutes with spatial rotations but does not commute with transformations that rotate the t axis. If $\widehat{L}$ denotes the Lorentz transformation to a frame moving with velocity $\mathbf{V}$, then $\widehat{P}\widehat{L} = \widehat{L}'\widehat{P}$, where $\widehat{L}'$ is the transformation to a frame moving with velocity $-\mathbf{V}$.

It follows from this that under inversion the components of a 4-spinor ξ^α cannot transform through themselves alone. If the inversion of the spinor ξ^α were still given by the transformation (19.1) (i.e. were represented by a matrix proportional to the identity),

it would commute with all Lorentz transformations whatsoever, which certainly must not be the case (since the operations $\widehat{L}$ and $\widehat{L}'$ applied to ξ^α manifestly do not coincide).

Inversion must therefore map the components of the spinor ξ^α into other quantities. These can only be the components of some other spinor $\eta_{\dot{\alpha}}$ whose transformation properties differ from those of ξ^α . Since inversion leaves the z -projection of the spin unchanged (as already noted above), the components ξ^1 and ξ^2 can pass under inversion only into η_1 and η_2 , which correspond to the same values $s_z = 1/2$ and $s_z = -1/2$. Taking inversion to be an operation that yields 1 when applied twice, one can specify its action by the formulas

$$\xi^\alpha \rightarrow \eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow \xi^\alpha. \quad (19.4)$$

For the covariant components ξ_α and the contravariant $\eta^{\dot{\alpha}}$ these transformations carry the opposite sign:

$$\xi_\alpha \rightarrow -\eta^{\dot{\alpha}}, \quad \eta^{\dot{\alpha}} \rightarrow -\xi_\alpha, \quad (19.4a)$$

since lowering and raising one and the same index are performed with opposite signs, see (17.5) and (17.9)⁴.

A certain distinction between the two definitions of inversion lies in the fact that under the second definition complex-conjugate spinors transform identically: if $\Xi_\alpha = \eta_{\dot{\alpha}}^*$, $H^{\dot{\alpha}} = \xi^{\alpha*}$, then from (19.5) one obtains $\Xi_\alpha \rightarrow -iH^{\dot{\alpha}}$, $H^{\dot{\alpha}} \rightarrow -i\Xi_\alpha$, i.e. the same rule as for $\xi_\alpha, \eta^{\dot{\alpha}}$. Under definition (19.4), by contrast, one would obtain the transformation $\Xi_\alpha \rightarrow H^{\dot{\alpha}}$, $H^{\dot{\alpha}} \rightarrow \Xi_\alpha$, which is the reverse in sign of the transformation of the spinors $\xi_\alpha, \eta^{\dot{\alpha}}$. We shall return to possible physical aspects of this distinction in § 27.

In what follows we shall everywhere adopt definition (19.5) for definiteness.

Under the rotation subgroup the spinors ξ^α and $\eta_{\dot{\alpha}}$ transform, as we know, in the same way. Forming from their components the combinations

$$\xi^\alpha \pm \eta_{\dot{\alpha}}, \quad (19.6)$$

one would obtain quantities transforming under inversion according to (19.1) with $P = \pm i$. These combinations, however, do not behave as spinors with respect to all transformations of the Lorentz group.

Incorporating inversion into the symmetry group therefore demands that one treat a pair of spinors $(\xi^\alpha, \eta_{\dot{\alpha}})$ jointly; such a pair goes by the name of a first-rank *bispinor*. The four components of a bispinor realize one of the irreducible representations that belong to the extended Lorentz group.

Given two bispinors $(\xi^\alpha, \eta_{\dot{\alpha}})$ and $(\Xi^\alpha, H_{\dot{\alpha}})$, their scalar product admits two distinct constructions. The expression

$$\xi^\alpha \Xi_\alpha + \eta_{\dot{\alpha}} H^{\dot{\alpha}} \quad (19.7)$$

⁴If instead inversion is understood in the sense that $P^2 = -1$, its action is specified by the formulas

$$\xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha \quad (19.5)$$

or, equivalently,

$$\xi_\alpha \rightarrow -i\eta^{\dot{\alpha}}, \quad \eta^{\dot{\alpha}} \rightarrow -i\xi_\alpha. \quad (19.5a)$$

is altogether unchanged by inversion, i.e. it constitutes a true scalar. The quantity

$$\xi^\alpha \Xi_\alpha - \eta_{\dot{\alpha}} H^{\dot{\alpha}} \quad (19.8)$$

is likewise invariant under rotations of the 4-coordinate system but reverses sign under inversion; in other words, it is a pseudoscalar.

A second-rank spinor $\zeta^{\alpha\dot{\beta}}$ can also be defined in two ways. Specifying its transformation law as

$$\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha H^{\dot{\beta}} + \Xi^\alpha \eta^{\dot{\beta}}, \quad (19.9)$$

one obtains quantities that transform under inversion according to

$$\zeta^{\alpha\dot{\beta}} \rightarrow \zeta_{\dot{\alpha}\beta}. \quad (19.10)$$

The 4-vector a^μ to which such a spinor is equivalent undergoes, by virtue of (18.1), the mapping $(a^0, \mathbf{a}) \rightarrow (a^0, -\mathbf{a})$, so it constitutes a genuine 4-vector (while the spatial part $\mathbf{a}$ is polar).

One may, however, also define $\zeta^{\alpha\dot{\beta}}$ according to

$$\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha H^{\dot{\beta}} - \Xi^\alpha \eta^{\dot{\beta}}. \quad (19.11)$$

Then⁵

$$\zeta^{\alpha\dot{\beta}} \rightarrow -\zeta_{\dot{\alpha}\beta}. \quad (19.12)$$

The 4-vector corresponding to such a spinor undergoes the inversion transformation $(a^0, \mathbf{a}) \rightarrow (-a^0, \mathbf{a})$, i.e. it is a 4-pseudovector (while the three-dimensional vector $\mathbf{a}$ is axial).

Symmetric second-rank spinors whose indices are of the same type obey the transformation laws:

$$\xi^{\alpha\beta} \sim \xi^\alpha \Xi^\beta + \xi^\beta \Xi^\alpha, \quad \eta_{\dot{\alpha}\dot{\beta}} \sim \eta_{\dot{\alpha}} H_{\dot{\beta}} + \eta_{\dot{\beta}} H_{\dot{\alpha}}. \quad (19.13)$$

Under inversion they pass into one another:

$$\xi^{\alpha\beta} \rightarrow -\eta_{\dot{\alpha}\dot{\beta}}. \quad (19.14)$$

The pair $(\xi^{\alpha\beta}, \eta_{\dot{\alpha}\dot{\beta}})$ constitutes a second-rank bispinor. Its number of independent components equals $3 + 3 = 6$. An antisymmetric second-rank 4-tensor $a^{\mu\nu}$ possesses the same number of independent components. A definite correspondence must therefore exist between them (both furnish equivalent irreducible representations of the extended Lorentz group).

Since the spinors $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ transform independently under the proper Lorentz group, the components of the 4-tensor $a^{\mu\nu}$ can likewise be partitioned into two groups of quantities that mix only among themselves under all rotations of the 4-coordinate system. This decomposition is carried out as follows.

⁵We emphasize that the transformation laws (19.10) and (19.12), which differ by a sign on the right-hand side, are by no means equivalent, since both sides of each involve the components of one and the same spinor (cf. the footnote on p. 92).

We introduce a polar three-dimensional vector $\mathbf{p}$ and an axial three-dimensional vector $\mathbf{a}$, related to the components of the 4-tensor $a^{\mu\nu}$ by

$$a^{\mu\nu} = \begin{pmatrix} 0 & p_x & p_y & p_z \\ -p_x & 0 & -a_z & a_y \\ -p_y & a_z & 0 & -a_x \\ -p_z & -a_y & a_x & 0 \end{pmatrix} \equiv (\mathbf{p}, \mathbf{a}) \quad (19.15)$$

$(\mathbf{p}, \mathbf{a})$ is a shorthand notation that we shall employ for listing the components of such a tensor). Here $a^{\mu\nu} = (-\mathbf{p}, \mathbf{a})$, and of the two quantities

$$\mathbf{a}^2 - \mathbf{p}^2 = \frac{1}{2} a_{\mu\nu} a^{\mu\nu}, \quad \mathbf{a}\mathbf{p} = \frac{1}{8} e_{\mu\nu\rho\sigma} a^{\mu\nu} a^{\rho\sigma}$$

the first is a scalar and the second a pseudoscalar; with respect to the proper Lorentz group both are equally invariant. Equally invariant alongside them are the squared moduli of the three-dimensional vectors $\mathbf{f}^\pm = \mathbf{p} \pm i\mathbf{a}$. This means that every rotation in 4-space amounts, for the vectors $\mathbf{f}^\pm$, to a “rotation” in three-dimensional space—generally speaking through complex angles (the six rotation angles in 4-space correspond to three complex “rotation angles” of the three-dimensional system). The operation of spatial inversion, on the other hand, reverses the sign of $\mathbf{p}$ (but not of $\mathbf{a}$) and thereby interchanges the vectors $\mathbf{f}^+$ and $-\mathbf{f}^-$. The components of these vectors form precisely the two sought-after groups of quantities built from the components of the tensor $a^{\mu\nu}$.

At the same time the correspondence between the components of the 4-tensor $a^{\mu\nu}$ and those of the spinors $\xi^{\alpha\beta}, \eta_{\dot{\alpha}\dot{\beta}}$ becomes evident. Since spatial rotations enter the Lorentz group as a subgroup, the relations linking the spinor components to the components of the three-dimensional vectors must be the same as for three-dimensional spinors:

$$\begin{aligned} f^+_x &= \frac{1}{2}(\xi^{22} - \xi^{11}), & f^+_y &= \frac{i}{2}(\xi^{22} + \xi^{11}), & f^+_z &= \xi^{12}; \\ f^-_x &= \frac{1}{2}(\eta_{2\dot{2}} - \eta_{1\dot{1}}), & f^-_y &= \frac{i}{2}(\eta_{2\dot{2}} + \eta_{1\dot{1}}), & f^-_z &= \eta_{1\dot{2}}. \end{aligned} \quad (19.16)$$

Problem Derive the general correspondence that connects spinors of even rank to 4-tensors.

Solution. Every spinor whose total rank $k + l$ is even realizes a single-valued (though in general reducible) representation of the extended Lorentz group; it is consequently equivalent to a 4-tensor that realizes the same representation⁶.

A spinor of rank (k, k) transforms under inversion according to

$$\zeta^{\alpha\beta\dots\dot{\gamma}\dot{\delta}\dots} \rightarrow \pm \zeta_{\dot{\alpha}\dot{\beta}\dots\gamma\delta\dots}$$

Such a spinor is equivalent to a symmetric irreducible 4-tensor of rank k —a true tensor or a pseudotensor depending on the sign in (1).

Spinors of ranks (k, l) and (l, k) making up a bispinor transform under inversion according to

$$\underbrace{\zeta^{\alpha\beta\dots}}_k \underbrace{\dot{\gamma}\dot{\delta}\dots}_l \rightarrow (-1)^{\frac{k-l}{2}} \chi_{\dot{\alpha}\dot{\beta}\dots\gamma\delta\dots} \quad (2)$$

⁶Spinors whose rank is odd realize, by contrast, double-valued representations: a spatial rotation through a full turn of 360° reverses the sign of these spinors, so that each group element is associated with a pair of matrices differing in sign.

When $l = k + 2$ the bispinor is equivalent to an irreducible 4-tensor $a_{[\mu\nu]\rho\sigma\dots}$ of rank $k + 2$, antisymmetric in the indices $[\mu\nu]$ and symmetric in all the rest. The irreducibility of this tensor means that it vanishes upon contraction over any pair of indices and upon dualization with respect to any triple of indices (i.e. $e^{\lambda\mu\nu\sigma}a_{[\mu\nu]\rho\sigma\dots} = 0$); the latter condition signifies that the tensor vanishes upon antisymmetrization over any index triple.

More broadly, when $l = k + 2n$ the bispinor corresponds to an irreducible 4-tensor of rank $k + 2n$ that is antisymmetric in n index pairs $[\lambda\mu][\nu\rho] \dots$ and symmetric in the remaining k indices. Tensors possessing antisymmetry of yet higher degree (in triples, quadruples, and so on of indices) are absent from this classification for a transparent reason: a completely antisymmetric tensor of third rank is dual to a pseudovector, one of fourth rank collapses to a scalar (being proportional to the Levi-Civita pseudotensor $e^{\lambda\mu\nu\rho}$), and antisymmetry in more than four indices cannot arise in a four-dimensional space at all.

§ 20. The Dirac equation in spinor representation

In its own rest frame a particle of spin $1/2$ is described by a two-component wave function—a 3-spinor. From the “four-dimensional” standpoint this quantity may originate as either an undotted or a dotted 4-spinor. To describe the particle in an arbitrary frame of reference one needs both kinds of 4-spinor; let us label them ξ^α and $\eta_{\dot{\alpha}}$ ⁷.

In the case of a free particle the sole operator entering the wave equation can be (as was noted already in § 10) the 4-momentum $\hat{p}_\mu = i\partial_\mu$. Expressed through spinor indices, this 4-vector corresponds to an operator spinor $\hat{p}_{\alpha\dot{\beta}}$ with components

$$\begin{aligned}\hat{p}^{11} &= \hat{p}_{22} = \hat{p}_0 + \hat{p}_z, & \hat{p}^{22} &= \hat{p}_{11} = \hat{p}_0 - \hat{p}_z, \\ \hat{p}^{12} &= -\hat{p}_{21} = \hat{p}_x - i\hat{p}_y, & \hat{p}^{21} &= -\hat{p}_{12} = \hat{p}_x + i\hat{p}_y.\end{aligned}\quad (20.1)$$

The wave equation is a linear differential relation linking the components of the spinors through the operator $\hat{p}_{\alpha\dot{\beta}}$. The demand of relativistic invariance singles out the following system of equations:

$$\hat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} = m\xi^\alpha, \quad \hat{p}_{\dot{\beta}\alpha}\xi^\alpha = m\eta_{\dot{\beta}}, \quad (20.2)$$

where m is a dimensional constant. Introducing different constants m_1 and m_2 into these two equations (or reversing the sign in front of m) would be pointless, since by a suitable redefinition of ξ^α or $\eta_{\dot{\alpha}}$ the equations could always be brought back to the form given above.

Eliminating one of the two spinors from equations (20.2) by substituting $\eta_{\dot{\beta}}$ from the second equation into the first, we obtain:

$$\hat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} = \frac{1}{m}\hat{p}^{\alpha\dot{\beta}}\hat{p}_{\gamma\dot{\beta}}\xi^\gamma = m\xi^\alpha.$$

⁷A three-dimensional first-rank spinor can also “descend” from 4-spinors of higher odd rank that become antisymmetric in one or several index pairs in the rest frame. Such variants, however, would lead to equations of higher order (cf. the footnote on p. 52).

But by (18.4) $\widehat{p}^{\alpha\beta}\widehat{p}_{\gamma\beta} = \widehat{p}^2\delta_\gamma^\alpha$, so that we arrive at

$$(\widehat{p}^2 - m^2)\xi^\gamma = 0, \quad (20.3)$$

from which it is clear that m is the mass of the particle.

We draw attention to the fact that the necessity of introducing a mass into the wave equation requires the simultaneous treatment of a pair of spinors (ξ^α and $\eta_{\dot{\alpha}}$): from only one of them it is impossible to compose a relativistically invariant equation that would contain any dimensional parameter. By the same token the wave equation turns out to be automatically invariant with respect to spatial inversion, provided one defines the transformation of the wave function as

$$P : \quad \xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha. \quad (20.4)$$

One readily checks that this substitution (accompanied by the replacement $\widehat{p}^{\dot{\alpha}\beta} \rightarrow \widehat{p}_{\alpha\dot{\beta}}$, obvious from (20.1)) causes the two equations (20.2) to interchange. A pair of spinors that transform into one another under inversion make up a four-component object—a bispinor.

The relativistic wave equation embodied in the system (20.2) bears the name of the *Dirac equation* (it was derived by *Dirac* in 1928). To pursue the further study and application of this equation we shall now examine the different representations in which it can be written.

With the aid of formula (18.6) we rewrite equations (20.2) as

$$(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma})\eta = m\xi, \quad (\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma})\xi = m\eta. \quad (20.5)$$

Here the symbols ξ and η stand for two-component quantities—spinors

$$\xi = \begin{pmatrix} \xi^1 \\ \xi^2 \end{pmatrix}, \quad \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \quad (20.6)$$

(the first carrying upper, the second lower indices), and when the matrices σ are multiplied by any two-component quantity f the ordinary rule of matrix multiplication is always understood here and below:

$$(\sigma f)_\alpha = \sigma_{\alpha\beta} f_\beta. \quad (20.7)$$

For ease of subsequent reference we write out the Pauli matrices once more:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (20.8)$$

and recall their principal properties:

$$\sigma_i\sigma_k + \sigma_k\sigma_i = 2\delta_{ik}, \quad \sigma_i\sigma_k = ie_{ikl}\sigma_l + \delta_{ik} \quad (20.9)$$

(see III, § 55).

Let us also write down the wave equation obeyed by the complex-conjugate wave function, which is built from the spinors

$$\xi^* = (\xi^{1*}, \xi^{2*}), \quad \eta^* = (\eta_1^*, \eta_2^*). \quad (20.10)$$

Since every operator $\widehat{p}_\mu$ contains the factor i , one has $\widehat{p}_\mu^* = -\widehat{p}_\mu$. In taking the complex conjugate of both sides of equations (20.5) one must also take into account that, by the Hermiticity of the matrices σ ($\sigma^* = \widetilde{\sigma}$),

$$(\sigma f)_\alpha^* = \sigma_{\alpha\beta}^* f_\beta^* = f_\beta^* \sigma_{\beta\alpha} = (f^* \sigma)_\alpha,$$

and one obtains equations of the form

$$\eta^* (\widehat{p}_0 + \widehat{\mathbf{p}} \boldsymbol{\sigma}) = -m \xi^*, \quad \xi^* (\widehat{p}_0 - \widehat{\mathbf{p}} \boldsymbol{\sigma}) = -m \eta^*. \quad (20.11)$$

In this way of writing it is tacitly assumed that the operator $\widehat{p}_\mu$ acts on the function placed to its left. The display of ξ^* and η^* as horizontal rows reflects the matrix multiplication convention adopted in these equations: the row f is contracted with the columns of the σ matrices:

$$(f^* \sigma)_\alpha = f_\beta^* \sigma_{\beta\alpha}. \quad (20.12)$$

The inversion transformation for ξ^* , η^* is defined as the complex conjugate of the transformation (20.4):

$$P : \quad \xi^{\alpha*} \rightarrow -i \eta_{\dot{\alpha}}^*, \quad \eta_{\dot{\alpha}}^* \rightarrow -i \xi^{\alpha*}. \quad (20.13)$$

§ 21. Symmetric form of the Dirac equation

Among all ways of writing the Dirac equation the spinor form stands out as the one that displays relativistic invariance most transparently. For practical calculations, though, it is often advantageous to pass to a different representation by selecting another quartet of independent components of the wave function.

Let ψ stand for the four-component wave function (its entries being ψ_i , $i = 1, 2, 3, 4$). Written in spinor language this object is a bispinor:

$$\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}. \quad (21.1)$$

With equal right, however, one may choose as the independent components of ψ any linear combinations of the components of the spinors ξ and η ⁸. We shall restrict the admissible linear transformations solely by the requirement of unitarity; such transformations leave invariant the bilinear forms composed of ψ and ψ^* (see § 28).

For a completely general selection of the components of ψ one can cast the Dirac equation as

$$\widehat{p}_\mu \gamma_{ik}^\mu \psi_k = m \psi_i,$$

where γ^μ ($\mu = 0, 1, 2, 3$) are certain 4×4 matrices (the *Dirac matrices*). It is customary to write this equation in symbolic form, suppressing the matrix indices:

$$(\gamma \widehat{p} - m) \psi = 0, \quad (21.2)$$

⁸For brevity we shall speak of the four-component quantity ψ as a bispinor also in its non-spinor representations.

where

$$\gamma \widehat{p} \equiv \gamma^\mu \widehat{p}_\mu = \widehat{p}_0 \gamma^0 - \widehat{\mathbf{p}} \boldsymbol{\gamma} = i\gamma^0 \frac{\partial}{\partial t} + i\boldsymbol{\gamma} \nabla, \quad \boldsymbol{\gamma} = (\gamma^1, \gamma^2, \gamma^3).$$

In the spinor form of the equation with the components ψ from (21.1) the corresponding matrices are⁹

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\gamma} = \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad (21.3)$$

as is readily seen by writing equations (20.5) in the form

$$\begin{pmatrix} 0 & \widehat{p}_0 + \widehat{\mathbf{p}} \boldsymbol{\sigma} \\ \widehat{p}_0 - \widehat{\mathbf{p}} \boldsymbol{\sigma} & 0 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = m \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$

and comparing with (21.2).

In the general case the matrices γ need only satisfy the conditions ensuring that $\widehat{p}^2 = m^2$. To determine these conditions, we multiply equation (21.2) on the left by $\gamma \widehat{p}$. We have

$$(\gamma^\mu \widehat{p}_\mu) (\gamma^\nu \widehat{p}_\nu) \psi = m (\widehat{p}_\mu \gamma^\mu) \psi = m^2 \psi.$$

Since $\widehat{p}_\mu \widehat{p}_\nu$ is a symmetric tensor (all operators $\widehat{p}_\mu$ commute), this equation can be rewritten as

$$\frac{1}{2} \widehat{p}_\mu \widehat{p}_\nu (\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) = m^2 \psi,$$

from which it is clear that one must have

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}. \quad (21.4)$$

Thus all pairs of distinct matrices γ^μ anticommute, and the square of each is

$$(\gamma^1)^2 = (\gamma^2)^2 = (\gamma^3)^2 = -1, \quad (\gamma^0)^2 = 1. \quad (21.5)$$

Under an arbitrary unitary transformation of the components of ψ ($\psi' = U\psi$, with U a unitary 4×4 matrix) the matrices γ transform according to

$$\gamma' = U\gamma U^{-1} = U\gamma U^+ \quad (21.6)$$

(so that the equation $(\gamma \widehat{p} - m)\psi = 0$ goes over into $(\gamma' \widehat{p} - m)\psi' = 0$). The commutation relations (21.4), of course, remain unchanged.

The matrix γ^0 of (21.3) is Hermitian, and the matrices $\boldsymbol{\gamma}$ are anti-Hermitian. These properties are preserved under every unitary transformation (21.6), so that we shall always have¹⁰:

$$\gamma^+ = -\boldsymbol{\gamma}, \quad \gamma^{0+} = \gamma^0. \quad (21.7)$$

⁹Here and below a compact notation is used for 4×4 matrices in terms of 2×2 blocks: each symbol in the expressions (21.3) represents a 2×2 matrix.

¹⁰These relations can be written together as

$$\gamma^{\lambda+} = \gamma^0 \gamma^\lambda \gamma^0. \quad (21.7 \text{ a})$$

We now derive the equation satisfied by ψ^* . Complex-conjugating both sides of (21.2) and invoking (21.7) gives

$$(\widehat{p}_0\gamma^0 - \widehat{\mathbf{p}}\widetilde{\boldsymbol{\gamma}} - m)\psi^* = 0.$$

We rearrange ψ^* by $\widetilde{\gamma}^\mu\psi^* = \psi^*\gamma^\mu$ and then multiply the equation on the right by γ^0 ; noting that $\boldsymbol{\gamma}\gamma^0 = -\gamma^0\boldsymbol{\gamma}$ and introducing the new bispinor

$$\bar{\psi} = \psi^*\gamma^0, \quad \psi^* = \bar{\psi}\gamma^0, \quad (21.8)$$

we obtain

$$\bar{\psi}(\gamma\widehat{p} + m) = 0. \quad (21.9)$$

Just as in (20.11), $\widehat{p}$ is understood to differentiate the function standing to its left. One calls $\bar{\psi}$ the *Dirac conjugate* (or *relativistic conjugate*) of ψ . The purpose of the γ^0 factor is to swap ξ^* and η^* (in the spinor representation), so that within $\bar{\psi} = (\eta^*, \xi^*)$ the undotted spinor occupies the first position—matching the ordering in ψ —while the dotted spinor occupies the second. This is why $\bar{\psi}$, rather than ψ^* , serves as the natural companion of ψ whenever the two enter bilinear combinations together (see § 28).

Under spatial inversion the wave function transforms according to

$$P: \quad \psi \rightarrow i\gamma^0\psi, \quad \bar{\psi} \rightarrow -i\bar{\psi}\gamma^0. \quad (21.10)$$

In the spinor representation of ψ the matrix γ^0 interchanges, as it should under inversion, the components ξ and η . The invariance of the Dirac equation under the transformation (21.10) is in the general case obvious and can be verified directly: replacing $\widehat{\mathbf{p}} \rightarrow -\widehat{\mathbf{p}}$ and simultaneously $\psi \rightarrow i\gamma^0\psi$ in equation (21.2), we get

$$(\widehat{p}_0\gamma^0 + \widehat{\mathbf{p}}\boldsymbol{\gamma} - m)\gamma^0\psi = 0.$$

Multiplying this equation on the left by γ^0 and using the anticommutativity of γ^0 and $\boldsymbol{\gamma}$, we recover the original equation.

Left-multiplying $(\gamma\widehat{p} - m)\psi = 0$ by $\bar{\psi}$, right-multiplying $\bar{\psi}(\gamma\widehat{p} + m) = 0$ by ψ , and summing the two expressions yields

$$\bar{\psi}\gamma^\mu(\widehat{p}_\mu\psi) + (\widehat{p}_\mu\bar{\psi})\gamma^\mu\psi = \widehat{p}_\mu(\bar{\psi}\gamma^\mu\psi) = 0,$$

where parentheses mark the function acted upon by $\widehat{p}$. Because this has the structure of a continuity equation $\partial_\mu j^\mu = 0$, the quantity

$$j^\mu = \bar{\psi}\gamma^\mu\psi = (\psi^*\psi, \psi^*\gamma^0\boldsymbol{\gamma}\psi) \quad (21.11)$$

is the particle current density 4-vector. We note that its time component $j^0 = \psi^*\psi$ is positive definite.

The Dirac equation can be written in a form explicitly solved for the time derivative:

$$i\frac{\partial\psi}{\partial t} = \widehat{H}\psi, \quad (21.12)$$

where $\widehat{H}$ is the particle Hamiltonian¹¹. To this end it suffices to multiply equation (21.2) on the left by γ^0 . For the Hamiltonian one obtains the expression

$$\widehat{H} = \alpha \widehat{\mathbf{p}} + \beta m, \quad (21.13)$$

where the standard notation has been introduced for the matrices appearing here:

$$\alpha = \gamma^0 \boldsymbol{\gamma}, \quad \beta = \gamma^0. \quad (21.14)$$

We note that

$$\alpha_i \alpha_k + \alpha_k \alpha_i = 2\delta_{ik}, \quad \beta \alpha + \alpha \beta = 0, \quad \beta^2 = 1, \quad (21.15)$$

i.e. all the matrices α, β anticommute with one another and their squares are unity; all of them are Hermitian. In the spinor representation

$$\alpha = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & -\boldsymbol{\sigma} \end{pmatrix}, \quad \beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (21.16)$$

When velocities are small the particle should reduce, as in non-relativistic quantum mechanics, to a single two-component spinor. And indeed, taking $\mathbf{p} \rightarrow 0$, $\varepsilon \rightarrow m$ in (20.5) yields $\xi = \eta$: the two spinors inside the bispinor coincide. A drawback of the spinor representation shows up at this point, however—all four entries of ψ stay non-vanishing despite only two being genuinely independent. One would prefer a representation in which two of the four components of ψ go to zero in the non-relativistic limit.

Accordingly we introduce, in place of ξ and η , their linear combinations φ and χ :

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}, \quad \varphi = \frac{1}{\sqrt{2}}(\xi + \eta), \quad \chi = \frac{1}{\sqrt{2}}(\xi - \eta). \quad (21.17)$$

Then for a particle at rest $\chi = 0$. This representation of ψ will be called the *standard* representation. Under inversion φ and χ transform into themselves:

$$P: \quad \varphi \rightarrow i\varphi, \quad \chi \rightarrow -i\chi. \quad (21.18)$$

The equations for φ and χ are obtained by adding and subtracting equations (20.5):

$$\widehat{p}_0 \varphi - \widehat{\mathbf{p}} \boldsymbol{\sigma} \chi = m\varphi, \quad -\widehat{p}_0 \chi + \widehat{\mathbf{p}} \boldsymbol{\sigma} \varphi = m\chi. \quad (21.19)$$

From these one sees that the standard representation corresponds to the matrices

$$\gamma^0 \equiv \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}, \quad \alpha = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}. \quad (21.20)$$

Since in (21.17) the first and second components of ξ and η are combined separately, in the standard representation—as in the spinor one—the components ψ_1 and ψ_3 correspond to

¹¹For a particle of spin 0 the wave equation could not be cast in this form: equation (10.5) for the scalar ψ is of second order in time, while the system (10.4) of first-order equations for the five-component quantity (ψ, ψ_k) contains time derivatives not of all of its components.

the eigenvalue $+1/2$ of the spin projection, while ψ_2 and ψ_4 correspond to the projection $-1/2$. In both representations, therefore, the matrix $\frac{1}{2}\Sigma$ with

$$\Sigma = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix}, \quad (21.21)$$

is the three-dimensional spin operator: the action of $\frac{1}{2}\Sigma_z$ on a bispinor containing only the components ψ_1, ψ_3 or ψ_2, ψ_4 multiplies the bispinor by $+1/2$ or $-1/2$, respectively. In an arbitrary representation this matrix can be written as

$$\Sigma = -\alpha\gamma^5 = -\frac{i}{2}[\mathbf{aa}] \quad (21.22)$$

(for the definition of γ^5 see below, (22.14)).

Problems

1. Find the transformation formulae for the wave function under an infinitesimal Lorentz transformation and an infinitesimal spatial rotation.

Solution. In the spinor representation of ψ , under an infinitesimal Lorentz transformation

$$\xi' = \left(1 - \frac{1}{2}\sigma\delta\mathbf{V}\right)\xi, \quad \eta' = \left(1 + \frac{1}{2}\sigma\delta\mathbf{V}\right)\eta$$

(see (18.8), (18.8 a), (18.12)). Both formulae can be written together as

$$\psi' = \left(1 - \frac{1}{2}\alpha\delta\mathbf{V}\right)\psi. \quad (1)$$

Similarly, the transformation law under an infinitesimal rotation is

$$\psi' = \left(1 + \frac{i}{2}\Sigma\delta\theta\right)\psi. \quad (2)$$

In this form the formulae are valid in any representation of ψ , provided one understands α and Σ as the matrices in the same representation.

It is easy to verify that the matrices α and Σ constitute the components of the antisymmetric “matrix 4-tensor”

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\alpha, i\Sigma)$$

(the listing of the components follows the rule (19.15)). Let us also introduce the infinitesimal antisymmetric tensor $\delta\varepsilon^{\mu\nu} = (\delta\mathbf{V}, \delta\theta)$. Then

$$\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu} = 2i\Sigma\delta\theta - 2\gamma\delta\mathbf{V}$$

and both formulae (1), (2) can be written in the unified form:

$$\psi' = \left(1 + \frac{1}{4}\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu}\right)\psi. \quad (3)$$

2. Write the Dirac equation in a representation that contains no imaginary coefficients (*E. Majorana, 1937*).

Solution. In the standard representation, in the equation

$$\left(\frac{\partial}{\partial t} + \alpha_x \frac{\partial}{\partial x} + \alpha_y \frac{\partial}{\partial y} + \alpha_z \frac{\partial}{\partial z} + im\beta \right) \psi = 0$$

the only imaginary matrices are α_y and $i\beta$. This imaginary character can be eliminated by performing a transformation $\psi' = U\psi$ that interchanges the imaginary matrix α_y with the real matrix β . To this end one sets

$$U = \frac{1}{\sqrt{2}}(\alpha_y + \beta) = U^{-1}.$$

Then

$$\alpha'_x = U\alpha_x U = -\alpha_x, \quad \alpha'_y = \beta, \quad \alpha'_z = -\alpha_z, \quad \beta' = \alpha_y,$$

and the Dirac equation takes the form

$$\left(\frac{\partial}{\partial t} - \alpha_x \frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y} - \alpha_z \frac{\partial}{\partial z} + im\alpha_y \right) \psi' = 0,$$

in which all the coefficients are real.

§ 22. Algebra of Dirac matrices

Calculations involving the Dirac equation make extensive use of the γ matrices without reference to their explicit form in any particular representation. All the algebraic rules governing these matrices follow entirely from the anticommutation relations

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}, \quad \mu, \nu = 0, 1, 2, 3, \quad (22.1)$$

which encode every general property that the γ matrices possess.

In this section we collect a number of formulae and algebraic identities for the γ matrices that prove useful in various computations.

The “scalar square” of γ with itself is $g_{\mu\nu} \gamma^\mu \gamma^\nu = 4$. For compact notation we introduce, in analogy with the covariant components of a 4-vector, the symbol $\gamma_\mu = g_{\mu\nu} \gamma^\nu$. Then

$$\gamma_\mu \gamma^\mu = 4. \quad (22.2)$$

When γ_μ and γ^μ are separated by one or more γ -factors, one can bring them next to each other by applying the commutation rule (22.1) one or more times, after which the contraction over μ is performed using (22.2). In this way one obtains the identities

$$\begin{aligned} \gamma_\mu \gamma^\nu \gamma^\mu &= -2\gamma^\nu, \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\mu &= 4g^{\lambda\nu}, \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\mu &= -2\gamma^\rho \gamma^\nu \gamma^\lambda, \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^\mu &= 2(\gamma^\sigma \gamma^\lambda \gamma^\nu \gamma^\rho + \gamma^\rho \gamma^\nu \gamma^\lambda \gamma^\sigma). \end{aligned} \quad (22.3)$$

The γ^μ factors usually appear contracted with various 4-vectors, forming “scalar products”¹²

$$\gamma a \equiv \gamma^\mu a_\mu. \quad (22.4)$$

For products of this kind the relations (22.1) become

$$(a\gamma)(b\gamma) + (b\gamma)(a\gamma) = 2(ab), \quad (a\gamma)(a\gamma) = a^2 \quad (22.5)$$

and the identities (22.3) take the form

$$\begin{aligned} \gamma_\mu(a\gamma)\gamma^\mu &= -2(a\gamma), \\ \gamma_\mu(a\gamma)(b\gamma)\gamma^\mu &= 4(ab), \\ \gamma_\mu(a\gamma)(b\gamma)(c\gamma)\gamma^\mu &= -2(c\gamma)(b\gamma)(a\gamma), \\ \gamma_\mu(a\gamma)(b\gamma)(c\gamma)(d\gamma)\gamma^\mu &= 2[(d\gamma)(a\gamma)(b\gamma)(c\gamma) + (c\gamma)(b\gamma)(a\gamma)(d\gamma)]. \end{aligned} \quad (22.6)$$

A widely used operation is the evaluation of the trace of a product of several γ matrices. Consider the quantities

$$T^{\mu_1\mu_2\ldots\mu_n} \equiv \frac{1}{4} \text{Sp}(\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_n}). \quad (22.7)$$

By the well-known cyclic property of the matrix trace, the tensor T is symmetric under cyclic permutations of the indices $\mu_1\mu_2\ldots\mu_n$.

Because the γ matrices take the same form in every reference frame, the quantities T are likewise frame-independent. They therefore constitute a tensor expressible solely through the metric tensor $g_{\mu\nu}$, which is the only tensor sharing this property.

From the rank-2 tensor $g_{\mu\nu}$ one can build only tensors of even rank. It follows at once that the trace of any product containing an odd number of γ factors vanishes. In particular, the trace of each individual γ is zero¹³:

$$\text{Sp} \gamma^\mu = 0. \quad (22.8)$$

The trace of the 4×4 unit matrix (which is implicitly present on the right-hand side of the anticommutation relation (22.1)) equals 4. Taking the trace of both sides of (22.1) therefore gives

$$T^{\mu\nu} = g^{\mu\nu}. \quad (22.9)$$

The trace of a product of four matrices is

$$T^{\lambda\mu\nu\rho} = g^{\lambda\mu}g^{\nu\rho} - g^{\lambda\nu}g^{\mu\rho} + g^{\lambda\rho}g^{\mu\nu}. \quad (22.10)$$

This formula can be derived, for instance, by “dragging” γ^λ to the right through $\text{Sp}(\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho)$ with the help of the anticommutation relation (22.1); each commutation produces one of the terms appearing in (22.10):

$$\begin{aligned} T^{\lambda\mu\nu\rho} &= 2g^{\lambda\mu}T^{\nu\rho} - T^{\mu\lambda\nu\rho} = 2g^{\lambda\mu}g^{\nu\rho} - T^{\mu\lambda\nu\rho}, \\ T^{\mu\lambda\nu\rho} &= 2g^{\lambda\nu}T^{\mu\rho} - T^{\mu\nu\lambda\rho} = 2g^{\lambda\nu}g^{\mu\rho} - T^{\mu\nu\lambda\rho}, \\ T^{\mu\nu\lambda\rho} &= 2g^{\lambda\rho}T^{\mu\nu} - T^{\mu\nu\rho\lambda} \end{aligned} \quad (22.11)$$

¹²In this edition we do not use any special notation for such a product. The literature frequently employs letters with a hat or with a slash (Feynman slash notation).

¹³The trace is invariant under similarity transformations $\gamma = U\gamma U^{-1}$. Hence (22.8) is also obvious from the explicit matrices (21.3).

and so on. After all permutations the right-hand side contains $-T^{\mu\rho\lambda} = -T^{\lambda\mu\nu\rho}$, which is brought to the left. By the same method the trace of a product of six γ 's reduces to traces of products of four, etc. Thus,

$$T^{\lambda\mu\nu\rho\sigma\tau} = g^{\lambda\mu}T^{\nu\rho\sigma\tau} - g^{\lambda\nu}T^{\mu\rho\sigma\tau} + g^{\lambda\rho}T^{\mu\nu\sigma\tau} - g^{\lambda\sigma}T^{\mu\nu\rho\tau} + g^{\lambda\tau}T^{\mu\nu\rho\sigma}. \quad (22.11)$$

We remark that every trace $T^{\lambda\mu\dots}$ is real, and that it is non-vanishing only when each of the matrices $\gamma^0, \gamma^1, \dots$

occurs an even number of times in the product; both statements are plain from the formulae just obtained. From this one easily deduces that the trace is unchanged when the order of all factors is reversed:

$$T^{\lambda\mu\dots\rho\sigma} = T^{\sigma\rho\dots\mu\lambda}. \quad (22.12)$$

As already noted, the γ factors usually appear as scalar products with various 4-vectors. In that case formulae (22.9) and (22.10), for example, read

$$\begin{aligned} \frac{1}{4} \text{Sp}(a\gamma)(b\gamma) &= ab, \\ \frac{1}{4} \text{Sp}(a\gamma)(b\gamma)(c\gamma)(d\gamma) &= (ab)(cd) - (ac)(bd) + (ad)(bc). \end{aligned} \quad (22.13)$$

A special role is played by the product $\gamma^0\gamma^1\gamma^2\gamma^3$, for which the standard notation is

$$\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (22.14)$$

One readily verifies that

$$\gamma^5\gamma^\mu + \gamma^\mu\gamma^5 = 0, \quad (\gamma^5)^2 = 1, \quad (22.15)$$

so that γ^5 anticommutes with every γ^μ . With respect to the matrices α and β the rules are

$$\alpha\gamma^5 - \gamma^5\alpha = 0, \quad \beta\gamma^5 + \gamma^5\beta = 0 \quad (22.16)$$

(the commutativity with α follows from the fact that $\alpha = \gamma^0\boldsymbol{\gamma}$ is a product of two γ^μ matrices).

The matrix γ^5 is Hermitian; indeed,

$$\gamma^{5+} = i\gamma^{3+}\gamma^{2+}\gamma^{1+}\gamma^{0+} = -i\gamma^3\gamma^2\gamma^1\gamma^0,$$

and because the sequence 3210 is obtained from 0123 by an even number of transpositions,

$$\gamma^{5+} = \gamma^5. \quad (22.17)$$

We also give the explicit form of this matrix in the two concrete representations:

$$\begin{aligned} \text{spinor} \quad \gamma^5 &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}; \\ \text{standard} \quad \gamma^5 &= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}. \end{aligned} \quad (22.18)$$

The trace of γ^5 vanishes:

$$\text{Sp } \gamma^5 = 0 \quad (22.19)$$

(as is also directly apparent from (22.18)). Likewise zero are the traces of $\gamma^5 \gamma^\mu \gamma^\nu$. For the product of γ^5 with four γ^μ factors, however, one has

$$\frac{1}{4} \text{Sp } \gamma^5 \gamma^\lambda \gamma^\mu \gamma^\nu \gamma^\rho = i e^{\lambda\mu\nu\rho}. \quad (22.20)$$

We note one further identity:

$$\gamma N = i \gamma^5 (\gamma a)(\gamma b)(\gamma c), \quad N^\lambda = e^{\lambda\mu\nu\rho} a_\mu b_\nu c_\rho, \quad (22.21)$$

valid for mutually orthogonal 4-vectors a, b, c : $ab = ac = bc = 0$.

There are situations (problems with non-relativistic particles) where one must compute traces of products containing γ^0 and the spatial “vector” $\boldsymbol{\gamma}$ as separate entities. A non-vanishing trace requires that γ^0 as well as $\boldsymbol{\gamma}$ each enter an even number of times. Every γ^0 factor then simplifies to the identity, and for products carrying two or four $\boldsymbol{\gamma}$ factors the traces read

$$\begin{aligned} \frac{1}{4} \text{Sp}(\mathbf{a}\boldsymbol{\gamma})(\mathbf{b}\boldsymbol{\gamma}) &= -\mathbf{a}\mathbf{b}, \\ \frac{1}{4} \text{Sp}(\mathbf{a}\boldsymbol{\gamma})(\mathbf{b}\boldsymbol{\gamma})(\mathbf{c}\boldsymbol{\gamma})(\mathbf{d}\boldsymbol{\gamma}) &= (\mathbf{a}\mathbf{b})(\mathbf{c}\mathbf{d}) - (\mathbf{a}\mathbf{c})(\mathbf{b}\mathbf{d}) + (\mathbf{a}\mathbf{d})(\mathbf{b}\mathbf{c}). \end{aligned} \quad (22.22)$$

§ 23. Plane waves

A free particle with definite momentum and energy is described by a plane wave, which we write as

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}} u_p e^{-ipx}. \quad (23.1)$$

The subscript p labels the 4-momentum; the wave amplitude u_p is a bispinor normalised in a specific way.

When carrying out the second quantisation later on, we shall need, alongside the wave functions (23.1), also the “negative-frequency” solutions that arise in relativistic theory, as explained in § 11, because of the two-valued square root $\pm\sqrt{\mathbf{p}^2 + m^2}$. Following the convention of § 11 we shall everywhere take ε to be the positive quantity $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$, so that the “negative frequency” is $-\varepsilon$; reversing also the sign of $\mathbf{p}$ we arrive at a function that is naturally denoted ψ_{-p} :

$$\psi_{-p} = \frac{1}{\sqrt{2\varepsilon}} u_{-p} e^{ipx}. \quad (23.2)$$

The physical meaning of these functions will become clear in § 26. Below we shall write formulae for ψ_p and ψ_{-p} side by side.

The components of the bispinor amplitudes u_p and u_{-p} satisfy the algebraic systems

$$(\gamma p - m)u_p = 0, \quad (\gamma p + m)u_{-p} = 0, \quad (23.3)$$

obtained by substituting (23.1), (23.2) into the Dirac equation

(which amounts to replacing the operator $\widehat{p}$ by $\pm p$)¹⁴. The relation $p^2 = m^2$ serves here as the consistency condition for each system. We shall always normalise the bispinor amplitudes by the invariant conditions

$$\bar{u}_p u_p = 2m, \quad \bar{u}_{-p} u_{-p} = -2m \quad (23.4)$$

(the bar denotes, as throughout, Dirac conjugation: $\bar{u} = u^* \gamma^0$). Left-multiplying equations (23.3) by $\bar{u}_{\pm p}$ gives $(\bar{u}_{\pm p} \gamma u_{\pm p})p = 2m^2 = 2p^2$, whence

$$\bar{u}_p \gamma u_p = \bar{u}_{-p} \gamma u_{-p} = 2p. \quad (23.5)$$

We note that the formulae for u_{-p} are obtained from those for u_p by changing the sign of m .

The current-density 4-vector is

$$j = \bar{\psi}_{\pm p} \gamma \psi_{\pm p} = \frac{1}{2\varepsilon} \bar{u}_{\pm p} \gamma u_{\pm p} = \frac{p}{\varepsilon}, \quad (23.6)$$

i.e. $j^\mu = (1, \mathbf{v})$, where $\mathbf{v} = \mathbf{p}/\varepsilon$ is the particle velocity. This shows that the functions ψ_p are normalised to “one particle per unit volume $V = 1$ ”.

Because of equations (23.3) the components of the wave amplitude are related to one another; the explicit form of these relations depends, of course, on the choice of representation for ψ . Let us find them for the standard representation.

From equations (21.19) we have, for a plane wave,

$$(\varepsilon - m)\varphi - \mathbf{p}\boldsymbol{\sigma}\chi = 0, \quad (\varepsilon + m)\chi - \mathbf{p}\boldsymbol{\sigma}\varphi = 0. \quad (23.7)$$

These relations yield the connection between φ and χ in two equivalent forms:

$$\varphi = \frac{\mathbf{p}\boldsymbol{\sigma}}{\varepsilon - m}\chi, \quad \chi = \frac{\mathbf{p}\boldsymbol{\sigma}}{\varepsilon + m}\varphi \quad (23.8)$$

(the equivalence is plain: multiplying the first relation on the left by $\mathbf{p}\boldsymbol{\sigma}/(\varepsilon + m)$ and using $(\mathbf{p}\boldsymbol{\sigma})^2 = \mathbf{p}^2$, $\varepsilon^2 - m^2 = \mathbf{p}^2$ reproduces the second). The overall factor in φ and χ is chosen so as to satisfy the normalisation condition (23.4). The result for u_p (and likewise for u_{-p}) is

$$u_p = \begin{pmatrix} \sqrt{\varepsilon + m} w \\ \frac{\mathbf{p}\boldsymbol{\sigma}}{\sqrt{\varepsilon + m}} w \end{pmatrix}, \quad u_{-p} = \begin{pmatrix} \frac{-\mathbf{p}\boldsymbol{\sigma}}{\sqrt{\varepsilon + m}} w' \\ \sqrt{\varepsilon + m} w' \end{pmatrix}, \quad (23.9)$$

(the second expression is obtained from the first by reversing the sign of m and relabelling $w \rightarrow (\mathbf{n}\boldsymbol{\sigma})w'$). Here $\mathbf{n}$ is the unit vector along $\mathbf{p}$, and w is an arbitrary two-component quantity subject only to the normalisation

$$w^* w = 1. \quad (23.10)$$

¹⁴We note also the analogous systems that follow from the Dirac equation (21.9) for the complex-conjugate function:

$$\bar{u}_p (\gamma p - m) = 0, \quad \bar{u}_{-p} (\gamma p + m) = 0. \quad (23.3 \text{ a})$$

For $\bar{u} = u^* \gamma^0$ (with γ^0 from (21.20)) we have

$$\begin{aligned}\bar{u}_p &= (\sqrt{\varepsilon + m} w^*, -\sqrt{\varepsilon - m} w^*(\mathbf{n}\sigma)), \\ \bar{u}_{-p} &= (\sqrt{\varepsilon - m} w'^*(\mathbf{n}\sigma), -\sqrt{\varepsilon + m} w'^*),\end{aligned}\quad (23.11)$$

and direct multiplication confirms that $\bar{u}_{\pm p} u_{\pm p} = \pm 2m$.

In the rest frame, i.e. when $\varepsilon = m$, we have

$$u_p = \sqrt{2m} \begin{pmatrix} w \\ 0 \end{pmatrix}, \quad u_{-p} = \sqrt{2m} \begin{pmatrix} 0 \\ w' \end{pmatrix}, \quad (23.12)$$

i.e. w is precisely the 3-spinor that each wave amplitude becomes in the slow-motion limit. Observe that for the bispinor u_{-p} the components that vanish at rest are the upper two, not the lower two. This behaviour of the negative-frequency Dirac solutions is easy to understand: putting $\mathbf{p} = 0$ and $\varepsilon \rightarrow -m$ in (23.7) yields $\varphi = 0$ ¹⁵.

The plane-wave amplitude contains one arbitrary two-component quantity. In other words, for a given momentum there exist two distinct independent states, corresponding to the two possible values of the spin projection. The projection of the spin on an arbitrary axis z cannot, however, possess a definite value. This is because the Hamiltonian of a particle with definite $\mathbf{p}$ (i.e. the matrix $H = \alpha \mathbf{p} + \beta m$) does not commute with $\Sigma_z = -i\alpha_x \alpha_y$. In agreement with the general statements of § 16, however, the helicity λ —the projection of the spin onto the direction of $\mathbf{p}$ —is conserved: the Hamiltonian commutes with $\mathbf{n}\Sigma$.

Helicity eigenstates correspond to plane waves whose 3-spinor $w = w^{(\lambda)}(\mathbf{n})$ is an eigenfunction of the operator $\mathbf{n}\sigma$:

$$\frac{1}{2}(\mathbf{n}\sigma)w^{(\lambda)} = \lambda w^{(\lambda)}. \quad (23.13)$$

The explicit form of these spinors is

$$w^{(\lambda=1/2)} = \begin{pmatrix} e^{-i\varphi/2} \cos \frac{\theta}{2} \\ e^{i\varphi/2} \sin \frac{\theta}{2} \end{pmatrix}, \quad w^{(\lambda=-1/2)} = \begin{pmatrix} -e^{-i\varphi/2} \sin \frac{\theta}{2} \\ e^{i\varphi/2} \cos \frac{\theta}{2} \end{pmatrix}, \quad (23.14)$$

where θ and φ are the polar angle and azimuth of the direction $\mathbf{n}$ with respect to the fixed axes xyz ¹⁶.

An alternative choice of two independent states of a free particle with given $\mathbf{p}$ (simpler, though less transparent physically) corresponds to the two values of the z -projection of the spin in the rest frame; we denote it by σ . The associated spinors are

$$w^{(\sigma=1/2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad w^{(\sigma=-1/2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (23.15)$$

As the two linearly independent negative-frequency solutions we choose plane waves with 3-spinors

$$w^{(\sigma)'} = -\sigma_y w^{(-\sigma)} = 2\sigma i w^{(\sigma)} \quad (23.16)$$

¹⁵In the spinor representation one finds $\xi = -\eta$ rather than the relation $\xi = \eta$ that holds at rest for positive-frequency solutions.

¹⁶The solution of equations (23.13) admits multiplication by an arbitrary phase factor, reflecting the freedom of an arbitrary rotation about $\mathbf{n}$.

(the rationale behind this choice will become clear in § 26).

One can construct a representation of the plane wave in which, in every frame (not only in the rest frame), it has just two non-vanishing components that correspond to definite values of the same physical observable—the spin projection in the rest frame (*L. Foldy, S. A. Wouthuysen, 1950*).

Taking u_p in the standard representation (23.9) as the starting point, we look for a unitary map into such a representation, writing

$$u'_p = U u_p, \quad U = e^{W\gamma\mathbf{n}},$$

where W is real; since $\gamma^+ = -\gamma$ the unitarity $U^+ = U^{-1}$ is automatic. Expanding in a series and noting that $(\gamma\mathbf{n})^2 = -1$, we can write U as

$$U = \cos W + \gamma\mathbf{n} \sin W$$

(cf. the passage from (18.13) to (18.14)). Requiring that the last two components of the transformed amplitude u'_p vanish, we find

$$\operatorname{tg} W = \frac{|\mathbf{p}|}{m + \varepsilon},$$

so that

$$U = \frac{m + \varepsilon + (\gamma\mathbf{n})|\mathbf{p}|}{\sqrt{2\varepsilon(\varepsilon + m)}}.$$

In the new representation

$$u'_p = \sqrt{2\varepsilon} \begin{pmatrix} w \\ 0 \end{pmatrix}. \quad (23.17)$$

The particle Hamiltonian in this representation becomes

$$\hat{H}' = U(\alpha\mathbf{p} + \beta m)U^{-1} = \beta\varepsilon \quad (23.18)$$

(here β, α, γ are the standard-representation matrices). This Hamiltonian commutes with

$$\Sigma = -\alpha\gamma^5 = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix},$$

which, in the Foldy–Wouthuysen representation, plays the role of the operator for the conserved rest-frame spin.

§ 24. Spherical waves

The wave functions of a free spin-1/2 particle with definite values of the total angular momentum j are spinor spherical waves. We shall determine their form, first recalling the analogous expressions from non-relativistic theory.

The non-relativistic wave function is a 3-spinor $\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \end{pmatrix}$. For a state with definite energy ε (and hence definite magnitude of the momentum p ¹⁷), orbital angular momentum l , total angular momentum j , and projection m , it takes the form

$$\psi = R_{pl}(r)\Omega_{jlm}(\theta, \varphi). \quad (24.1)$$

¹⁷Throughout this section p denotes $|\mathbf{p}|$.

The angular factor Ω_{jlm} is a 3-spinor whose components (for the two values $j = l \pm 1/2$ possible at a given l) are

$$\Omega_{l+1/2, l, m} = \begin{pmatrix} \sqrt{\frac{j+m}{2j}} Y_{l, m-1/2} \\ \sqrt{\frac{j-m}{2j}} Y_{l, m+1/2} \end{pmatrix}, \quad (24.2)$$

$$\Omega_{l-1/2, l, m} = \begin{pmatrix} -\sqrt{\frac{j-m+1}{2j+2}} Y_{l, m-1/2} \\ \sqrt{\frac{j+m+1}{2j+2}} Y_{l, m+1/2} \end{pmatrix}$$

(see III, § 106, problem). We shall call Ω_{jlm} the *spherical spinors*. Their normalisation is

$$\int \Omega_{jlm}^* \Omega_{j'l'm'} d\omega = \delta_{jj'} \delta_{ll'} \delta_{mm'}. \quad (24.3)$$

The radial functions R_{pl} are a common factor in both spinor components of ψ and are given by the formula (33.10) (see III):

$$R_{pl} = \sqrt{\frac{2\pi p}{r}} J_{l+1/2}(pr). \quad (24.4)$$

They are normalised according to

$$\int_0^\infty r^2 R_{p'l} R_{pl} dr = 2\pi \delta(p' - p). \quad (24.5)$$

Turning now to the relativistic case, we recall first of all that for a moving particle no separate conservation laws for spin and orbital angular momentum exist: the operators $\widehat{\mathbf{S}}$ and $\widehat{\mathbf{L}}$ individually fail to commute with the Hamiltonian. The parity of the state, however, remains conserved (for a free particle). Consequently the quantum number l ceases to specify a definite value of the orbital angular momentum, but it does determine (see below) the parity of the state.

We shall work with the desired wave function (a bispinor) in the standard representation: $\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}$. Under rotations φ and χ transform as 3-spinors, so their angular dependence is again carried by the spherical spinors Ω_{jlm} . Let $\varphi \propto \Omega_{jlm}$, where l is one (definite) of the two values $j + 1/2$ or $j - 1/2$. Under inversion $\varphi(\mathbf{r}) \rightarrow i\varphi(-\mathbf{r})$ (see (21.18)), and since $\Omega_{jlm}(-\mathbf{n}) = (-1)^l \Omega_{jlm}(\mathbf{n})$,

$$\varphi(\mathbf{r}) \rightarrow i(-1)^l \varphi(\mathbf{r}).$$

The lower component χ , on the other hand, transforms under inversion as $\chi(\mathbf{r}) \rightarrow -i\chi(-\mathbf{r})$. For the state to possess a definite parity (i.e. for inversion to multiply every component by the same phase), the angular dependence of χ must involve $\Omega_{j'l'm}$ with the other value of l : because the two allowed values differ by unity, $(-1)^{l'} = -(-1)^l$.

Furthermore, the radial dependence of φ and χ is governed by the same functions R_{pl} and $R_{p'l'}$ (with l and l' matching the order of the spherical harmonics entering Ω_{jlm}).

This follows from the fact that each component of ψ satisfies the second-order equation $(\hat{p}^2 - m^2)\psi = 0$, which for a given $|\mathbf{p}|$ reads

$$(\Delta + \mathbf{p}^2)\psi = 0,$$

coinciding in form with the free-particle Schrödinger equation of the non-relativistic theory.

Thus

$$\varphi = AR_{pl}\Omega_{jlm}, \quad \chi = BR_{pl'}\Omega_{jl'm}, \quad (24.6)$$

and it remains to fix the constant coefficients A and B . To this end we examine the far zone, where the spherical wave may be treated as a plane wave. By the asymptotic formula (33.12) (see III)

$$R_{pl} \approx \frac{1}{ir} \left\{ e^{i\left(pr - \frac{\pi l}{2}\right)} - e^{-i\left(pr - \frac{\pi l}{2}\right)} \right\}, \quad (24.7)$$

so that φ is a superposition of two plane waves propagating along $\pm \mathbf{n}$ ($\mathbf{n} = \mathbf{r}/r$). For each of these, by (23.8),

$$\chi = \frac{p}{\varepsilon + m}(\pm \mathbf{n}\boldsymbol{\sigma})\varphi.$$

From what has been said above (formulae (24.6)) it is clear that $(\mathbf{n}\boldsymbol{\sigma})\Omega_{jlm} = a\Omega_{jl'm}$ with a a constant. This constant is easily found by evaluating both sides at $m = 1/2$ with $\mathbf{n}$ along the z -axis. Using (7.2a) one obtains

$$(\mathbf{n}\boldsymbol{\sigma})\Omega_{jlm} = i^{l'-l}\Omega_{jl'm}. \quad (24.8)$$

Collecting the formulae and comparing with (24.6), we find

$$B = -\frac{p}{\varepsilon + m} A.$$

Finally, the coefficient A is fixed by the overall normalisation of ψ . Adopting the condition

$$\int \psi_{pjlm}^* \psi_{p'j'l'm'} d^3x = 2\pi \delta_{jj'} \delta_{ll'} \delta_{mm'} \delta(p - p'), \quad (24.9)$$

we arrive at the final result

$$\psi_{pjlm} = \frac{1}{\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon + m} R_{pl} \Omega_{jlm} \\ -\sqrt{\varepsilon - m} R_{pl'} \Omega_{jl'm} \end{pmatrix}, \quad l' = 2j - l. \quad (24.10)$$

At fixed j and m (and energy ε) there are therefore two states that differ in parity. Which parity a state carries is uniquely determined by l ($= j \pm 1/2$): inversion multiplies the bispinor (24.10) by $i(-1)^l$. The upper and lower components of this bispinor do, however, involve spherical harmonics of both orders l and l' —a direct manifestation of the fact that the orbital angular momentum is not separately well-defined.

As $r \rightarrow \infty$ one can, in any small region of space, regard the spherical waves (24.7) as plane waves with momentum $\mathbf{p} = \pm p\mathbf{n}$. It is therefore clear that the momentum-space wave functions differ from (24.10) mainly in the

absence of radial factors, with $\mathbf{n}$ now denoting the direction of the momentum.

To pass directly to the momentum representation one performs the Fourier transform:

$$\psi(\mathbf{p}') = \int \psi(\mathbf{r}) e^{-i\mathbf{p}'\mathbf{r}} d^3x, \quad (24.11)$$

The integral is evaluated with the aid of the plane-wave expansion in spherical harmonics (see III, (34.3)):

$$e^{i\mathbf{p}\mathbf{r}} = \frac{2\pi}{p} \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l R_{pl}(r) Y_{lm}^*\left(\frac{\mathbf{p}}{p}\right) Y_{lm}\left(\frac{\mathbf{r}}{r}\right). \quad (24.12)$$

Substituting $e^{-i\mathbf{p}'\mathbf{r}}$ into (24.11) via this expansion and using (24.5), the Fourier components of the function

$$\psi(\mathbf{r}) = R_{pl}(r) \Omega_{jlm}\left(\frac{\mathbf{r}}{r}\right)$$

are found to be

$$\psi(\mathbf{p}') = \frac{(2\pi)^2}{p} \delta(p' - p) i^{-l} Y_{lm'}\left(\frac{\mathbf{p}'}{p'}\right) \int \Omega_{jlm}\left(\frac{\mathbf{r}}{r}\right) Y_{lm'}^*\left(\frac{\mathbf{r}}{r}\right) d\Omega.$$

The integral here equals the coefficients of the spherical harmonics in the definition of the spherical spinors (24.2), and together with the factor $Y_{lm'}(\mathbf{p}'/p')$ it reconstructs the same spherical spinor, now depending on $\mathbf{p}'/p'$:

$$\psi(\mathbf{p}') = \frac{(2\pi)^2}{p} \delta(p' - p) i^{-l} \Omega_{jlm}\left(\frac{\mathbf{p}'}{p'}\right).$$

Applying this result to the bispinor wave function (24.10), we obtain its momentum-space form

$$\psi_{pjlm}(\mathbf{p}') = \delta(p' - p) \frac{(2\pi)^2}{p\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon + m} i^{-l} \Omega_{jlm}(\mathbf{p}'/p') \\ -\sqrt{\varepsilon - m} i^{-l'} \Omega_{jl'm}(\mathbf{p}'/p') \end{pmatrix}. \quad (24.13)$$

The states $|pjlm\rangle$ coincide with the states $|pjm|\lambda\rangle$ considered in § 16 (with $|\lambda| = 1/2$): both sets possess definite values of pjm and parity. The spherical spinors Ω_{jlm} are therefore expressible through the functions $D_m^{(j)}$ (both depending on the argument $\mathbf{p}/p$). In the limit $p \rightarrow 0$ the wave functions (24.13) reduce to 3-spinors Ω_{jlm} with parity $P = \eta(-1)^l$ (where $\eta = i$ is the “intrinsic parity” of the spinor). Comparison with the results of § 16 leads to the formula

$$\Omega_{jlm} = i^l \sqrt{\frac{2j+1}{8\pi}} \left(w^{(-1/2)} D_{-1/2,m}^{(j)} \pm w^{(1/2)} D_{1/2,m}^{(j)} \right) \quad (24.14)$$

(with $l = j \mp 1/2$, where $w^{(\lambda)}$ are the 3-spinors (23.14)).

§ 25. The connection between spin and statistics

Second quantisation of the field of spin-1/2 particles (the spinor field) proceeds in exactly the same way as was done in § 11 for the scalar field.

Without repeating the entire argument, we write down at once the expressions for the field operators, fully analogous to (11.2):

$$\begin{aligned}\hat{\psi} &= \sum_{\mathbf{p}\sigma} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}\sigma} u_{\mathbf{p}\sigma} e^{-i\mathbf{p}x} + \hat{b}_{\mathbf{p}\sigma}^+ u_{-\mathbf{p}-\sigma} e^{i\mathbf{p}x}), \\ \hat{\bar{\psi}} &\equiv \hat{\psi}^+ \gamma^0 = \sum_{\mathbf{p}\sigma} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}\sigma}^+ \bar{u}_{\mathbf{p}\sigma} e^{i\mathbf{p}x} + \hat{b}_{\mathbf{p}\sigma} \bar{u}_{-\mathbf{p}-\sigma} e^{-i\mathbf{p}x});\end{aligned}\quad (25.1)$$

the summation runs over all values of $\mathbf{p}$ and over $\sigma = \pm 1/2$. The antiparticle annihilation operators $\hat{b}_{\mathbf{p}\sigma}$ (like the particle annihilation operators $\hat{a}_{\mathbf{p}\sigma}$) stand as coefficients of functions whose coordinate dependence ($e^{i\mathbf{p}x}$) corresponds to a state with momentum $\mathbf{p}$ ¹⁸.

To construct the Hamiltonian of the spinor field there is no need to determine its energy–momentum tensor (as we did for the scalar field), because here the single-particle Hamiltonian exists and can be used to write down the wave equation (the Dirac equation) (21.12). The mean energy of a particle in a state with wave function ψ is the integral

$$\int \psi^* \hat{H} \psi d^3x = \int \psi^* \frac{\partial \psi}{\partial t} d^3x = i \int \bar{\psi} \gamma^0 \frac{\partial \psi}{\partial t} d^3x. \quad (25.2)$$

We draw attention to the fact that the “energy density” (the integrand) is not positive definite here.

Replacing ψ and $\bar{\psi}$ in (25.2) by ψ -operators, taking into account the mutual orthogonality of wave functions with different $\mathbf{p}$ or σ , and also the relation $\bar{u}_{\pm\mathbf{p}\sigma} \gamma^0 u_{\pm\mathbf{p}\sigma} = 2\varepsilon$ for the wave amplitudes, we obtain the field Hamiltonian

$$\hat{H} = \sum_{\mathbf{p}\sigma} \varepsilon (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} - \hat{b}_{\mathbf{p}\sigma} \hat{b}_{\mathbf{p}\sigma}^+). \quad (25.3)$$

It follows that quantisation must here be carried out according to Fermi statistics:

$$\{\hat{a}_{\mathbf{p}\sigma}, \hat{a}_{\mathbf{p}\sigma}^+\}_+ = 1, \quad \{\hat{b}_{\mathbf{p}\sigma}, \hat{b}_{\mathbf{p}\sigma}^+\}_+ = 1, \quad (25.4)$$

while all other pairs of operators $\hat{a}$, $\hat{a}^+$, $\hat{b}$, $\hat{b}^+$ anticommute (see III, § 65). Indeed, the Hamiltonian (25.3) then takes the form

$$\hat{H} = \sum_{\mathbf{p}\sigma} \varepsilon (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} + \hat{b}_{\mathbf{p}\sigma}^+ \hat{b}_{\mathbf{p}\sigma} - 1),$$

and the energy eigenvalues (after subtracting, as always, the infinite additive constant) are

$$E = \sum_{\mathbf{p}\sigma} \varepsilon (N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma}), \quad (25.5)$$

which are, as required, positive definite. Had we quantised according to Bose statistics, equation (25.3) would have given the meaningless eigenvalues

$$\sum_{\mathbf{p}\sigma} \varepsilon (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma}),$$

¹⁸Both types of function also correspond to the same value of σ , the rest-frame spin projection: for the functions $\bar{\psi}_{-\mathbf{p}, -\sigma}$ this will be demonstrated in § 26, see (26.10).

which are not positive definite.

The analogous expression

$$\mathbf{P} = \sum_{\mathbf{p}\sigma} \mathbf{p}(N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma}) \quad (25.6)$$

is obtained for the momentum of the system—the eigenvalues of the operator $\int \hat{\psi}^+ \hat{\mathbf{p}} \hat{\psi} d^3x$.

The 4-current operator is

$$\hat{j}^\mu = \hat{\psi} \gamma^\mu \hat{\psi}, \quad (25.7)$$

and for the “charge” operator of the field we get

$$\hat{Q} = \int \hat{\psi} \gamma^0 \hat{\psi} d^3x = \sum_{\mathbf{p}\sigma} (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} + \hat{b}_{\mathbf{p}\sigma} \hat{b}_{\mathbf{p}\sigma}^+) = \sum_{\mathbf{p}\sigma} (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} - \hat{b}_{\mathbf{p}\sigma}^+ \hat{b}_{\mathbf{p}\sigma} + 1), \quad (25.8)$$

with eigenvalues

$$Q = \sum_{\mathbf{p}\sigma} (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma}). \quad (25.9)$$

We thus arrive once more at the picture of particles and antiparticles; everything said about them in § 11 applies here as well.

Whereas spin-0 particles are bosons, however, spin-1/2 particles turn out to be fermions. Tracing this difference to its formal origin, one finds that it stems from a difference in the character of the “energy density” for the scalar and spinor fields. In the scalar case the energy density is positive definite, so that both terms in the Hamiltonian (11.3)

($\hat{a}^+ \hat{a}$ and $\hat{b} \hat{b}^+$) enter with a plus sign. To guarantee positive energy eigenvalues, the replacement of $\hat{b} \hat{b}^+$ by $\hat{b}^+ \hat{b}$ must then proceed without a change of sign, i.e. by the Bose commutation rule. For the spinor field, on the other hand, the energy density is not positive definite, so that the $\hat{b} \hat{b}^+$ term in the Hamiltonian (25.3) carries a minus sign; obtaining positive eigenvalues now requires that the rearrangement $\hat{b} \hat{b}^+ \rightarrow \hat{b}^+ \hat{b}$ be accompanied by a sign change, i.e. that it follow the Fermi commutation rule.

On the other hand, the form of the energy density is directly tied to the transformation properties of the wave function and to the demands of relativistic invariance. One can therefore say that the connection between a particle’s spin and the statistics it obeys is itself a direct consequence of these requirements.

From the fact that spin-1/2 particles are fermions one can also draw a general conclusion: every particle with half-integer spin is a fermion, and every particle with integer spin is a boson (the latter includes the spin-0 result proved in § 11)¹⁹.

This becomes obvious once one notes that a particle of spin s may be thought of as “assembled” from $2s$ particles of spin 1/2. When s is half-integer the number $2s$ is odd, while for integer s it is even. Now a composite object built from an even count

¹⁹The origin of the connection between the spin of a particle and the statistics it obeys was elucidated by Pauli (W. Pauli, 1940).

of fermions behaves as a boson, whereas one containing an odd count of fermions is a fermion²⁰.

When a system contains particles of several species, each species must be equipped with its own creation and annihilation operators. Operators belonging to different bosons, or to a boson and a fermion, commute with one another. As for operators belonging to different fermions, in non-relativistic theory they could be taken either commuting or anticommuting (III, § 65). In a relativistic theory, however, which allows mutual transformations of particles, the creation and annihilation operators of distinct fermion species must be regarded as anticommuting—just

like operators referring to different states of the same fermion.

Problem

Find the Lagrangian of the spinor field.

Solution. A real, Lorentz-scalar Lagrangian that yields the Dirac equation is

$$L = \frac{i}{2} (\bar{\psi} \gamma^\mu \partial_\mu \psi - \partial_\mu \bar{\psi} \cdot \gamma^\mu \psi) - m \bar{\psi} \psi. \quad (1)$$

Treating the components of ψ and $\bar{\psi}$ as “generalised coordinates” q , one easily verifies that the corresponding Euler–Lagrange equations (10.10) coincide with the Dirac equations for $\bar{\psi}$ and ψ . The overall sign of the Lagrangian (as well as the overall numerical coefficient) is a matter of convention here. Because L is linear in the derivatives of ψ and $\bar{\psi}$, the action $S = \int L d^4x$ can possess neither a minimum nor a maximum. The condition $\delta S = 0$ determines in this case only a stationary point, not an extremum, of the integral.

The field-theoretic Lagrangian of the spinor field is obtained by replacing ψ in (1) with the operator $\hat{\psi}$. Applying to this Lagrangian the formula (12.12), one recovers the current operator (25.7).

§ 26. Charge conjugation and time reversal of spinors

The factors $\psi_{p\sigma} = u_{p\sigma} e^{-ipx}$ that multiply the operators $\hat{a}_{p\sigma}$ in (25.1) are the wave functions of free particles (which we shall call “electrons”) carrying momentum $\mathbf{p}$ and polarisation σ :

$$\psi_{p\sigma}^{(e)} = \psi_{p\sigma}.$$

The factors $\bar{\psi}_{-p-\sigma}$ accompanying the operators $\hat{b}_{p\sigma}$, on the other hand, should be regarded as positron wave functions for the same $\mathbf{p}$ and σ . It turns out, however, that the electron and positron functions are expressed in different bispinor representations. This is apparent from the fact that ψ and $\bar{\psi}$ have distinct transformation properties, their components obeying different systems of equations. To eliminate this drawback one must carry out a specific unitary transformation of the components of $\bar{\psi}_{-p-\sigma}$ —one that

²⁰This reasoning tacitly assumes that all particles of the same spin obey the same statistics, regardless of how they are “assembled”. That this is indeed so can be seen from an analogous argument. If spin-0 fermions existed, then combining a spin-0 fermion with a spin-1/2 fermion would produce a spin-1/2 particle that is a boson—in contradiction with the general result established above for spin 1/2.

ensures the resulting four-component function satisfies the same equation as $\psi_{p\sigma}$ ²¹. It is precisely such a function that we shall call the positron wave function (with momentum $\mathbf{p}$ and polarisation σ). Denoting the matrix of the required unitary transformation by U_C , we write

$$\psi_{p\sigma}^{(p)} = U_C \bar{\psi}_{-p-\sigma}. \quad (26.1)$$

The operation C by which this function is constructed from $\psi_{-p-\sigma}$ is called the *charge conjugation* of the wave function (*H. A. Kramers*, 1937). This notion is by no means limited to plane waves. For any function ψ whatsoever there exists a “charge-conjugate” function

$$\widehat{C}\psi(t, \mathbf{r}) = U_C \bar{\psi}(t, \mathbf{r}), \quad (26.2)$$

which transforms in the same way as ψ and obeys the same equation.

The properties of U_C follow from this definition. If ψ solves the Dirac equation $(\gamma\widehat{p} - m)\psi = 0$, then $\bar{\psi}$ satisfies

$$\bar{\psi}(\gamma\widehat{p} + m) = 0, \quad \text{i.e.} \quad (\bar{\gamma}\widehat{p} + m)\bar{\psi} = 0.$$

Multiplying this equation on the left by U_C :

$$U_C \bar{\gamma}\widehat{p}\bar{\psi} + mU_C \bar{\psi} = 0,$$

and demanding that $U_C \bar{\psi}$ obey the same equation as ψ :

$$(\gamma\widehat{p} - m)U_C \bar{\psi} = 0,$$

we compare the two equations and obtain the following “commutation relation” between U_C and the matrices $\gamma^{\mu-1}$:

$$U_C \bar{\gamma}^{\mu} = -\gamma^{\mu} U_C. \quad (26.3)$$

We shall henceforth assume that the wave functions are given in the spinor or standard representation (the case of an arbitrary representation will be discussed only at the end of this section). In these representations

$$\begin{aligned} \gamma^{0,2} &= \bar{\gamma}^{0,2}, & \gamma^{1,3} &= -\bar{\gamma}^{1,3}, \\ (\gamma^{0,1,3})^* &= \gamma^{0,1,3}, & \gamma^{2*} &= -\gamma^2. \end{aligned} \quad (26.4)$$

The conditions (26.3) are then met by the matrix $U_C = \eta_C \gamma^2 \gamma^0$ with an arbitrary constant η_C . The requirement $\widehat{C}^2 = 1$ implies $|\eta_C|^2 = 1$, so U_C is fixed up to a phase factor. We shall set $\eta_C = 1$, giving

$$U_C = \gamma^2 \gamma^0 = -\alpha_y. \quad (26.5)$$

²¹For spin-0 particles this question did not arise, since the scalar functions ψ and ψ^* both satisfy the same equation, and ψ_{-p}^* simply coincides with ψ_p .

Noting further that $\bar{\psi} = \psi^* \gamma^0 = \bar{\gamma}^0 \psi^* = \gamma^0 \psi^*$, one can write the action of $\widehat{C}$ as

$$\widehat{C}\psi = \gamma^2 \gamma^0 \bar{\bar{\psi}} = \gamma^2 \psi^*. \quad (26.6)$$

Written out explicitly, the transformation (26.6) reads in the spinor representation

$$\begin{aligned} C : \quad \xi^\alpha &\rightarrow -i\eta^{\alpha*}, & \eta_\alpha &\rightarrow -i\xi_\alpha^*, \\ C : \quad \xi_\alpha &\rightarrow i\xi^{\alpha*}, & \eta^\alpha &\rightarrow -i\eta_\alpha^*. \end{aligned} \quad (26.7a, b)$$

The charge-conjugation transformation for the plane waves $\psi_{\pm p\sigma}$ is easily carried out using their explicit forms (23.9) together with the matrix U_C in the standard representation:

$$U_C = \begin{pmatrix} 0 & -\sigma_y \\ -\sigma_y & 0 \end{pmatrix}. \quad (26.8)$$

Noting that

$$\sigma_y \sigma^* = -\sigma \sigma_y,$$

and using the definition of $w^{(\sigma)'} from (23.16), we obtain$

$$U_C \bar{u}_{-p-\sigma} = u_{p\sigma}, \quad U_C u_{-p-\sigma} = \bar{u}_{p\sigma}. \quad (26.9)$$

Consequently,

$$\widehat{C}\psi_{-p-\sigma} = \psi_{p\sigma}, \quad (26.10)$$

confirming that the functions $\psi_{-p-\sigma}$ appearing in the ψ -operators (25.1) alongside $\widehat{b}_{p\sigma}$ do indeed describe states of a particle with momentum $\mathbf{p}$ and polarisation σ . We also see that the electron and positron states are described by the same functions:

$$\psi_{p\sigma}^{(e)} = \psi_{p\sigma}^{(p)} = \psi_{p\sigma}.$$

This is entirely natural, since the functions $\psi_{p\sigma}$ encode only the momentum and polarisation of the particle.

The operation of time reversal can be treated in an analogous manner. Reversing the sign of time must be accompanied by complex conjugation of the wave function. In order for the resulting “time-reversed” wave function ($\widehat{T}\psi$) to belong to the same representation as the original ψ , one must

perform a further unitary transformation on the components of ψ^* (or $\bar{\psi}$). Proceeding in analogy with (26.2), we represent the action of $\widehat{T}$ on ψ as

$$\widehat{T}\psi(t, \mathbf{r}) = U_T \bar{\psi}(-t, \mathbf{r}), \quad (26.11)$$

where U_T is a unitary matrix.

Writing once more the Dirac equation for ψ :

$$\left(i\gamma^0 \frac{\partial}{\partial t} + i\boldsymbol{\gamma} \nabla - m \right) \psi(t, \mathbf{r}) = 0,$$

and the corresponding equation for $\bar{\psi}$:

$$\left(i\bar{\gamma}^0 \frac{\partial}{\partial t} + i\bar{\boldsymbol{\gamma}} \nabla + m \right) \bar{\bar{\psi}}(t, \mathbf{r}) = 0.$$

Replacing $t \rightarrow -t$ in the latter and multiplying on the left by $-U_T$:

$$\left(iU_T\tilde{\gamma}^0\frac{\partial}{\partial t} - iU_T\tilde{\gamma}\nabla\right)\tilde{\psi}(-t, \mathbf{r}) - mU_T\tilde{\psi}(-t, \mathbf{r}) = 0.$$

We require $U_T\tilde{\psi}(-t, \mathbf{r})$ to satisfy the same equation as $\psi(t, \mathbf{r})$:

$$\left(i\gamma^0\frac{\partial}{\partial t} + i\gamma\nabla\right)U_T\tilde{\psi}(-t, \mathbf{r}) - mU_T\tilde{\psi}(-t, \mathbf{r}) = 0.$$

Comparison yields the conditions

$$U_T\tilde{\gamma}^0 = \gamma^0 U_T, \quad U_T\tilde{\gamma} = -\gamma U_T. \quad (26.12)$$

In the spinor and standard representations, these conditions are satisfied by the matrix²²

$$U_T = i\gamma^3\gamma^1\gamma^0. \quad (26.13)$$

The action of $\hat{T}$ is therefore given by

$$\hat{T}\psi(t, \mathbf{r}) = i\gamma^3\gamma^1\gamma^0\tilde{\psi}(-t, \mathbf{r}) = i\gamma^3\gamma^1\psi^*(-t, \mathbf{r}). \quad (26.14)$$

Explicitly, in the spinor representation this transformation reads

$$T : \quad \xi^\alpha \rightarrow -i\xi_\alpha^*, \quad \eta_{\dot{\alpha}} \rightarrow i\eta^{\dot{\alpha}*} \quad (26.15 \text{ a})$$

or

$$T : \quad \xi_\alpha \rightarrow i\xi^{\alpha*}, \quad \eta^\alpha \rightarrow -i\eta_{\dot{\alpha}}^*. \quad (26.15 \text{ b})$$

In the standard representation

$$U_T = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix}. \quad (26.16)$$

Let us now find the result of applying all three operations P , T , and C to ψ . Writing them out in succession:

$$\begin{aligned} \hat{T}\psi(t, \mathbf{r}) &= -i\gamma^1\gamma^3\psi^*(-t, \mathbf{r}), \\ \hat{P}\hat{T}\psi(t, \mathbf{r}) &= i\gamma^0(\hat{T}\psi) = \gamma^0\gamma^1\gamma^3\psi^*(-t, -\mathbf{r}), \\ \hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r}) &= \gamma^2(\gamma^0\gamma^1\gamma^3\psi^*)^* = \gamma^2\gamma^0\gamma^1\gamma^3\psi(-t, -\mathbf{r}), \end{aligned}$$

which gives

$$\hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r}) = i\gamma^5\psi(-t, -\mathbf{r}). \quad (26.17)$$

In the spinor representation

$$CPT : \quad \xi^\alpha \rightarrow -i\xi_\alpha^*, \quad \eta_{\dot{\alpha}} \rightarrow i\eta^{\dot{\alpha}*}, \quad (26.18)$$

²²The choice of phase factor in (26.13) is tied to the choice made in (26.5) by considerations discussed below in the footnote on p. 124.

as can also be verified directly from the transformation rules (20.4), (26.7), (26.15)²³.

The expressions for the matrices U_C and U_T written above assumed the spinor or standard representation of ψ . Let us finally clarify which properties of these expressions remain valid for an arbitrary representation.

If ψ undergoes a unitary transformation:

$$\psi' = U\psi, \quad \gamma' = U\gamma U^{-1}, \quad \bar{\psi}' = \psi'^* \gamma'^0 = \bar{\psi} U^+ = \bar{\psi} \tilde{U}^{-1}, \quad (26.19)$$

then in the new representation

$$(\hat{C}\psi)' = U(\hat{C}\psi) = UU_C\tilde{\psi} = UU_C(\widetilde{\psi'U}) = UU_C\tilde{U}\tilde{\psi}'.$$

Comparing with the definition of U'_C in the new representation ($(\hat{C}\psi)' = U'_C\tilde{\psi}'$), we find

$$U'_C = UU_C\tilde{U}. \quad (26.20)$$

The transformation (26.20) coincides with the transformation law of the γ matrices only when U is real. For this reason the expression (26.5) holds only in representations obtainable from the spinor or standard representation by a real transformation.

The matrix (26.5) is unitary, and transposition reverses its sign:

$$U_C U_C^+ = 1, \quad \tilde{U}_C = -U_C. \quad (26.21)$$

These properties are invariant under the transformation (26.20) and hence hold in every representation. The matrix (26.5) is also Hermitian ($U_C = U_C^+$), but this property is in general not preserved by (26.20).

Everything just said (including (26.21)) applies equally to the properties of the matrix U_T .

Within the second-quantisation formalism, the transformations C , P , T for ψ -operators must be formulated as transformation rules for the creation and annihilation operators of the particles. These rules can be established (similarly to how it was done in § 13 for spin-0 particles) by requiring that the transformed ψ -operators admit a representation of the form

$$\begin{aligned} \hat{\psi}^C(t, \mathbf{r}) &= U_C \tilde{\hat{\psi}}(t, \mathbf{r}), \\ \hat{\psi}^P(t, \mathbf{r}) &= i\gamma^0 \hat{\psi}(t, -\mathbf{r}), \\ \hat{\psi}^T(t, \mathbf{r}) &= U_T \tilde{\hat{\psi}}(-t, \mathbf{r}). \end{aligned} \quad (26.22)$$

Problem

Find the charge-conjugation operator in the Majorana representation (see Problem 2, § 21).

Solution. The matrix U'_C in the Majorana representation is obtained from $U_C = -\alpha_y$ in the standard representation via the transformation (26.20) with $U = (\alpha_y + \beta)/\sqrt{2}$,

²³The notation $\hat{C}\hat{P}\hat{T}$ implies that the operators act from right to left. The overall sign in (26.17), (26.18) depends on this ordering because $\hat{T}$ does not commute with $\hat{C}$ and $\hat{P}$ (in their action on a bispinor).

and equals $U'_C = \alpha_y$ (α_y and β denote the standard-representation matrices). Denoting quantities in the Majorana representation by a prime, we have $\widehat{C}\psi' = U'_C(\psi'^*\beta')$, and since $\beta' = \alpha_y$,

$$\widehat{C}\psi' = \alpha_y(\psi'^*\alpha_y) = \alpha_y\widetilde{\alpha}_y\psi'^* = \psi'^*,$$

i.e. charge conjugation reduces to complex conjugation.

§ 27. Internal symmetry of particles and antiparticles

In its rest frame the wave function of a spin-1/2 particle reduces to a single 3-spinor (which we denote Φ^α). The behaviour of this spinor under inversion is linked to the notion of the particle's intrinsic parity. However (as already noted in § 19), although the two possible transformation laws for 3-spinors ($\Phi^\alpha \rightarrow \pm i\Phi^\alpha$) are inequivalent, assigning a definite parity to a spinor has no absolute meaning. It is therefore meaningless to speak of the intrinsic parity of a spin-1/2 particle by itself. One can, however, speak of the *relative* intrinsic parity of two such particles.

From two (three-dimensional) spinors $\Phi^{(e)}$ and $\Phi^{(p)}$ one can form the scalar $\Phi^{(e)}_\alpha \Phi^{(p)\alpha}$. If this quantity is a true scalar, the particles described by these spinors are said to have the same parity; if it is a pseudoscalar, their intrinsic parities are said to be opposite.

We shall show that the intrinsic parities of a particle and its antiparticle (with spin 1/2) are opposite (V. B. Berestetskii, 1948).

To this end, note that if the operation C (26.7) is applied to both sides of the P -transformation (19.5) (in the spinor representation)

$$P : \quad \xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha, \quad (27.1)$$

one obtains

$$\eta^{c\alpha*} \rightarrow i\xi_{\dot{\alpha}}^*, \quad \xi_{\dot{\alpha}}^{c*} \rightarrow i\eta^{c\alpha*},$$

where the superscript c labels the components of the bispinor $\psi^C = \begin{pmatrix} \xi^c \\ \eta^c \end{pmatrix}$ that is charge-conjugate to $\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$. Taking the complex conjugate and rearranging the indices, we find

$$P : \quad \eta_{\dot{\alpha}}^c \rightarrow i\xi^{c\alpha}, \quad \xi^{c\alpha} \rightarrow i\eta_{\dot{\alpha}}^c. \quad (27.2)$$

We see that charge-conjugate bispinors transform under inversion according to the same law.

Let $\psi^{(e)}$ be the wave function of the particle (electron) and $\psi^{(p)}$ that of the antiparticle (positron). The latter is a bispinor obtained by charge-conjugating a certain “negative-frequency” solution of the Dirac equation. In the rest frame each reduces to a 3-spinor:

$$\xi^{(e)\alpha} = \eta_{\dot{\alpha}}^{(e)} = \Phi^{(e)\alpha}, \quad \xi^{(p)\alpha} = \eta_{\dot{\alpha}}^{(p)} = \Phi^{(p)\alpha}.$$

According to (27.1) and (27.2), these spinors transform under inversion as

$$\Phi^\alpha \rightarrow i\Phi^\alpha, \quad (27.3)$$

with the same rule for both $\Phi^{(e)}$ and $\Phi^{(p)}$. The product $\Phi^{(e)}\Phi^{(p)}$, however, changes sign—which proves the assertion.

A particle that coincides with its own antiparticle is called truly neutral (see § 12). The ψ -operator of the field of such particles satisfies the condition

$$\widehat{\psi}(t, \mathbf{r}) = \widehat{\psi}^C(t, \mathbf{r}).$$

For spin-1/2 particles this yields the conditions (in the spinor representation)²⁴

$$\xi^\alpha = -i\bar{\eta}^{\dot{\alpha}+}, \quad \widehat{\eta}_{\dot{\alpha}} = -i\widehat{\xi}_{\alpha}^+ \quad (27.4)$$

Like any relations expressing physical properties, these conditions are invariant under the *CPT* transformation²⁵. It is readily verified that they are in fact invariant not only under *CPT* as a whole, but also under each of the three transformations individually.

In § 19 we agreed to define spinor inversion as the transformation for which $\widehat{P}^2 = -1$, and we have followed this convention throughout. It is easy to see that the result obtained above concerning the relative parity of particle and antiparticle is independent, as it should be, of the particular convention adopted for the definition of inversion.

If inversion is defined by the condition $\widehat{P}^2 = 1$, then in place of (27.1) we have

$$P: \quad \xi^\alpha \rightarrow \eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow \xi^\alpha. \quad (27.5)$$

The charge-conjugate function then transforms according to the rule

$$\xi^{c\alpha} \rightarrow -\eta_{\dot{\alpha}}^c, \quad \eta_{\dot{\alpha}}^c \rightarrow -\xi^{c\alpha},$$

which differs from (27.5) by a sign. Accordingly, the three-dimensional spinors Φ transform as

$$\Phi^{(e)\alpha} \rightarrow \Phi^{(e)\alpha}, \quad \Phi^{(p)\alpha} \rightarrow -\Phi^{(p)\alpha},$$

so that the product $\Phi^{(e)}\Phi^{(p)}$ remains a pseudoscalar, as before.

The only possible physical distinction between the two conventions for inversion is that under the definition (27.5) the true-neutrality condition for the field would not be invariant under this transformation (or under *CP*): it would reverse the relative sign of the two sides of the equalities (27.4). In practice, truly neutral particles with spin 1/2 are unknown, and at present one cannot say whether the difference between the two definitions of inversion carries any real physical significance²⁶.

Problem

Determine the charge parity of positronium (a hydrogen-like system of an electron and a positron).

Solution. The wave function of two fermions must be antisymmetric under the simultaneous interchange of the coordinates, spins, and charge variables of the particles

²⁴In the Majorana representation, true neutrality simply means Hermiticity of the operator $\widehat{\psi}'$ (see the problem to § 26).

²⁵More precisely, the *CPT* transformation must be defined in this case so as to leave relations of the type (27.4) invariant. This is achieved by an appropriate choice of the phase factor in the definition of U_T (see the footnote on p. 121).

²⁶The incomplete equivalence of the two definitions of inversion was pointed out by *Racah* (G. Racah, 1937).

(cf. the problem to § 13). Interchanging the first multiplies the function by $(-1)^l$, the second by $(-1)^{l+S}$ (where $S = 0$ or 1 is the total spin of the system), and the third by the sought quantity C . From the condition $(-1)^l(-1)^{l+S}C = -1$ we find

$$C = (-1)^{l+S}.$$

Since the intrinsic parities of electron and positron are opposite, the spatial parity of the system is $P = (-1)^{l+1}$. The combined parity is $CP = (-1)^{S+1}$.

§28. Bilinear forms

Let us examine the transformation properties of the various bilinear forms that can be constructed from the components of ψ and ψ^* . Such forms are of great importance throughout quantum mechanics; among them is the 4-current density (21.11).

Since ψ and ψ^* each have four components, one can construct $4 \cdot 4 = 16$ independent bilinear combinations. Their classification by transformation properties follows immediately from the multiplication rules for two arbitrary bispinors listed in § 19 (which in the present case are ψ and ψ^*). Specifically, one can form a scalar (denoted S), a pseudoscalar (P), a mixed spinor of rank two equivalent to a true 4-vector V^μ (four independent quantities), a mixed rank-two spinor equivalent to a 4-pseudovector A^μ (four quantities), and a rank-two bispinor equivalent to an antisymmetric 4-tensor $T^{\mu\nu}$ (six quantities).

In a representation-independent notation these combinations are written as follows:

$$S = \bar{\psi}\psi, \quad P = i\bar{\psi}\gamma^5\psi, \quad (\text{III.2})$$

$$V^\mu = \bar{\psi}\gamma^\mu\psi, \quad A^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi, \quad T^{\mu\nu} = i\bar{\psi}\sigma^{\mu\nu}\psi, \quad (\text{28.1})$$

where

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma}) \quad (\text{28.2})$$

(the listing of components in (28.2) follows (19.15))²⁷. All the expressions written above are real.

The scalar and pseudoscalar character of S and P is evident from their spinor-representation forms: $S = \xi^*\eta + \eta^*\xi$, $P = i(\xi^*\eta - \eta^*\xi)$, which are precisely the expressions (19.7) and (19.8). That V^μ is a vector then follows from the Dirac equation: multiplying $\hat{p}_\mu\gamma^\mu\psi = m\psi$ on the left by $\bar{\psi}$ gives

$$(\bar{\psi}\hat{p}_\mu\gamma^\mu\psi) = m\bar{\psi}\psi;$$

the right-hand side is a scalar, so the left-hand side must be one as well.

The pattern governing the construction of the quantities (28.1) is straightforward: they are assembled as though the matrices γ^μ formed a 4-vector, γ^5 were a pseudoscalar, and the flanking factors $\bar{\psi}$ and ψ combined into a scalar²⁸.

²⁷Under a unitary change of representation $\psi \rightarrow U\psi$, $\gamma \rightarrow U\gamma U^{-1}$, $\bar{\psi} \rightarrow \bar{\psi}U^{-1}$, so the invariance of the bilinear forms is obvious.

²⁸The absence of bilinear forms having the character of a symmetric 4-tensor, obvious from the spinor representation, is clear from this rule as well: the symmetric combination $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$ would reduce the form to a scalar.

Second-quantised bilinear forms are obtained by replacing the ψ -functions in (28.1) with ψ -operators. For greater generality we shall assume that the two ψ -operators refer to the fields of different particles, distinguishing them by subscripts a and b . Let us determine how these operator forms transform under charge conjugation. Noting that ²⁹

$$\widehat{\psi}^C = U_C \widetilde{\widehat{\psi}}, \quad \widehat{\bar{\psi}}^C = U_C^\dagger \widehat{\bar{\psi}}, \quad (28.3)$$

and employing (26.3) and (26.21):

$$\begin{aligned} \widehat{\bar{\psi}}_a^C \widehat{\psi}_b^C &= \widehat{\bar{\psi}}_a U_C^\dagger U_C \widetilde{\widehat{\psi}}_b = -\widehat{\bar{\psi}}_a U_C^\dagger U_C \widehat{\bar{\psi}}_b = -\widehat{\bar{\psi}}_a \widehat{\bar{\psi}}_b, \\ \widehat{\bar{\psi}}_a^C \gamma^\mu \widehat{\psi}_b^C &= \widehat{\bar{\psi}}_a U_C^\dagger \gamma^\mu U_C \widetilde{\widehat{\psi}}_b = -\widehat{\bar{\psi}}_a U_C^\dagger \gamma^\mu U_C \widehat{\bar{\psi}}_b = \widehat{\bar{\psi}}_a \gamma^\mu \widehat{\bar{\psi}}_b. \end{aligned}$$

Restoring the operators to their original order ($\widehat{\bar{\psi}}$ to the left of $\widehat{\psi}$) reverses the sign of the product by the Fermi commutation rules (25.4) (and also generates state-independent terms, which we discard as in the analogous derivation of § 13). We thus obtain

$$\widehat{\bar{\psi}}_a^C \widehat{\psi}_b^C = \widehat{\bar{\psi}}_b \widehat{\psi}_a, \quad \widehat{\bar{\psi}}_a^C \gamma^\mu \widehat{\psi}_b^C = -\widehat{\bar{\psi}}_b \gamma^\mu \widehat{\psi}_a.$$

Treating the remaining forms in the same way, we find the following charge-conjugation rules ³⁰:

$$\begin{aligned} C : \quad \widehat{S}_{ab} &\rightarrow \widehat{S}_{ba}, \quad \widehat{P}_{ab} \rightarrow \widehat{P}_{ba}, \quad (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} \rightarrow (\widehat{V}^0, -\widehat{\mathbf{V}})_{ba}, \\ (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} &\rightarrow (\widehat{A}^0, -\widehat{\mathbf{A}})_{ba}, \quad \widehat{T}_{ab}^{\mu\nu} \rightarrow (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} \rightarrow (\widehat{\mathbf{p}}, -\widehat{\mathbf{a}})_{ba} \end{aligned} \quad (28.4)$$

($\widehat{\mathbf{p}}, \widehat{\mathbf{a}}$ are the three-dimensional vectors equivalent to the components of $\widehat{T}^{\mu\nu}$ according to (19.15)).

The behaviour of the same forms under time reversal is found analogously. One must bear in mind (see § 13) that this operation entails a reversal of the operator ordering, so that, for instance,

$$(\widehat{\bar{\psi}}_a \widehat{\psi}_b)^T = \widehat{\bar{\psi}}_b^T \widehat{\psi}_a^T.$$

Substituting

$$\widehat{\psi}^T = U_T \widetilde{\widehat{\psi}}, \quad \widehat{\bar{\psi}}^T = -U_T^\dagger \widehat{\bar{\psi}}, \quad (28.5)$$

we get

$$(\widehat{\bar{\psi}}_a \widehat{\psi}_b)^T = -\widehat{\bar{\psi}}_b U_T U_T^\dagger \widehat{\psi}_a = \widehat{\bar{\psi}}_b \widehat{\psi}_a.$$

Carrying out the same procedure for all the other forms yields

$$\begin{aligned} T : \quad \widehat{S}_{ab} &\rightarrow \widehat{S}_{ba}, \quad \widehat{P}_{ab} \rightarrow -\widehat{P}_{ba}, \quad (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} \rightarrow (\widehat{V}^0, -\widehat{\mathbf{V}})_{ba}, \\ (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} &\rightarrow (\widehat{A}^0, -\widehat{\mathbf{A}})_{ba}, \quad \widehat{T}_{ab}^{\mu\nu} \rightarrow (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} \rightarrow (\widehat{\mathbf{p}}, -\widehat{\mathbf{a}})_{ba} \end{aligned} \quad (28.6)$$

²⁹To derive the second relation from the first, write $\bar{\psi}^C = [U_C (\widehat{\bar{\psi}}^* \gamma^{0*})] \gamma^0 = -\bar{\gamma}^0 U_C^\dagger \gamma^0 \widehat{\bar{\psi}} = \bar{\gamma}^0 \gamma^{0*} U_C^\dagger \widehat{\bar{\psi}} = U_C^\dagger \widehat{\bar{\psi}}$ (using (26.3), (26.21) and the Hermiticity of γ^0).

³⁰Note that for bilinear forms built from ψ -functions (rather than ψ -operators), the transformations (28.4) would carry the opposite sign, since restoring the original ordering of the factors $\bar{\psi}$ and ψ would not entail a sign change.

($\hat{\mathbf{p}}, \hat{\mathbf{a}}$ are the three-dimensional vectors equivalent to the components of $\hat{T}^{\mu\nu}$ according to (19.15)).

Under spatial inversion, on the other hand, in accordance with the tensor character of the quantities³¹,

$$\begin{aligned} P: \quad \hat{S}_{ab} &\rightarrow \hat{S}_{ab}, \quad \hat{P}_{ab} \rightarrow -\hat{P}_{ab}, \quad (\hat{V}^0, \hat{\mathbf{V}})_{ab} \rightarrow (\hat{V}^0, -\hat{\mathbf{V}})_{ab}, \\ (\hat{A}^0, \hat{\mathbf{A}})_{ab} &\rightarrow (-\hat{A}^0, \hat{\mathbf{A}})_{ab}, \quad \hat{T}_{ab}^{\mu\nu} \rightarrow (\hat{\mathbf{p}}, \hat{\mathbf{a}})_{ab} \rightarrow (-\hat{\mathbf{p}}, \hat{\mathbf{a}})_{ab}. \end{aligned} \quad (28.7)$$

Finally, the joint application of all three operations leaves $\hat{S}_{ab}, \hat{P}_{ab}, \hat{T}_{ab}^{\mu\nu}$ unchanged and reverses the sign of every $\hat{V}_{ab}^\mu, \hat{A}_{ab}^\mu$. This is exactly what one expects from the interpretation of this combined transformation as a 4-inversion: since a 4-inversion is equivalent to a rotation of the four-dimensional coordinate system, it draws no distinction between true tensors and pseudotensors of any rank.

Consider now pairwise products of bilinear forms constructed from four distinct functions $\psi^a, \psi^b, \psi^c, \psi^d$. Different results are obtained depending on which pairs of functions are multiplied together. It turns out, however, that any such product can be reduced to products of bilinear forms with fixed pairings of the factors (*W. Pauli, M. Fierz, 1936*). We shall derive the identity underlying this reduction.

Consider the set of 4×4 matrices

$$1, \quad \gamma^5, \quad \gamma^\mu, \quad i\gamma^\mu\gamma^5, \quad i\sigma^{\mu\nu} \quad (28.8)$$

(1 being the unit matrix). Numbering these 16 ($= 1 + 1 + 4 + 4 + 6$) matrices in some definite order, we denote them by γ^A ($A = 1, \dots, 16$), and the same matrices with the 4-tensor indices (μ, ν) lowered by γ_A . They possess the following properties:

$$\text{Sp } \gamma^A = 0 \quad (\gamma^A \neq 1), \quad \gamma^A \gamma_A = 1, \quad \frac{1}{4} \text{Sp } \gamma^A \gamma_B = \delta_B^A. \quad (28.9)$$

By virtue of the last of these properties, the matrices γ^A are linearly independent. Since their number equals the number of elements (4×4) of a 4×4 matrix, the γ^A form a complete set in terms of which any 4×4 matrix Γ can be expanded:

$$\Gamma = \sum_A c_A \gamma^A, \quad c_A = \frac{1}{4} \text{Sp } \gamma_A \Gamma, \quad (28.10)$$

or, written out explicitly with matrix indices ($i, k = 1, 2, 3, 4$):

$$\Gamma_{ik} = \frac{1}{4} \sum_A \Gamma_{lm} \gamma_{ml}^A \gamma_{Ai k}.$$

Supposing, in particular, that Γ has only one non-vanishing element (Γ_{lm}), we obtain the desired identity (the “completeness relation”)

$$\delta_{il} \delta_{km} = \frac{1}{4} \sum_A \gamma_{Aik} \gamma_{ml}^A. \quad (28.11)$$

³¹To avoid misunderstanding, recall that the T and P transformations also require a change in the arguments of the functions on the right-hand sides (the transformed forms in (28.6), (28.7)) to $x^T = (-t, \mathbf{r})$ and $x^P = (t, -\mathbf{r})$, respectively, if the left-hand sides are functions of $x = (t, \mathbf{r})$.

Multiplying both sides by $\bar{\psi}_i^a \psi_k^b \bar{\psi}_m^c \psi_l^d$ gives

$$(\bar{\psi}^a \psi^b)(\bar{\psi}^c \psi^d) = \frac{1}{4} \sum_A (\bar{\psi}^a \gamma_A \psi^b)(\bar{\psi}^c \gamma^A \psi^d). \quad (28.12)$$

This is one identity of the kind described above: it expresses the product of two scalar bilinear forms in terms of products of forms built from different pairings of the factors³².

Further identities of this type are obtained from (28.12) by substituting

$$\psi^d \rightarrow \gamma^B \psi^d, \quad \psi^b \rightarrow \gamma^C \psi^b$$

and using the expansion (see the problem)

$$\gamma^A \gamma^B = \sum_R c_R \gamma^R, \quad c_R = \frac{1}{4} \text{Sp } \gamma^A \gamma^B \gamma_R.$$

We also record here, for later reference, the analogue of (28.11) for 2×2 matrices. A complete set of linearly independent 2×2 matrices σ^A ($A = 1, \dots, 4$) is formed by

$$1, \quad \sigma_x, \quad \sigma_y, \quad \sigma_z. \quad (28.13)$$

For these

$$\text{Sp } \sigma^A = 0 \quad (\sigma^A \neq 1), \quad \frac{1}{2} \text{Sp } \sigma^A \sigma^B = \delta_{AB}. \quad (28.14)$$

The completeness relation reads:

$$\delta_{\alpha\gamma} \delta_{\beta\delta} = (1/2) \sum_A \sigma_{\alpha\beta}^A \sigma_{\delta\gamma}^A = (1/2) \delta_{\alpha\beta} \delta_{\delta\gamma} + (1/2) \delta_{\alpha\delta} \delta_{\beta\gamma} \quad (28.15)$$

($\alpha, \beta, \dots = 1, 2$), or equivalently:

$$\sigma_{\alpha\beta} \sigma_{\delta\gamma} = -(1/2) \sigma_{\alpha\gamma} \sigma_{\delta\beta} + (3/2) \delta_{\alpha\gamma} \delta_{\delta\beta}. \quad (28.16)$$

Problem

Derive formulae analogous to (28.12) for the scalar products of two bilinear forms of types P, V, A, T .

Solution. Denote:

$$\begin{aligned} J_S &= (\bar{\psi}^a \psi^b)(\bar{\psi}^c \psi^d), & J_P &= (\bar{\psi}^a \gamma^5 \psi^b)(\bar{\psi}^c \gamma^5 \psi^d), \\ J_V &= (\bar{\psi}^a \gamma^\mu \psi^b)(\bar{\psi}^c \gamma_\mu \psi^d), & J_A &= (\bar{\psi}^a i \gamma^\mu \gamma^5 \psi^b)(\bar{\psi}^c i \gamma_\mu \gamma^5 \psi^d), \\ J_T &= (\bar{\psi}^a i \sigma^{\mu\nu} \psi^b)(\bar{\psi}^c i \sigma_{\mu\nu} \psi^d), \end{aligned}$$

³²We recall, to avoid confusion, that the forms here are built from ψ -functions. For forms built from anticommuting ψ -operators the sign of the rearrangement would be reversed.

and let primed letters denote the same products with ψ^b and ψ^d interchanged. Applying the procedure described in the text yields

$$\begin{aligned} 4J'_S &= J_S + J_V + J_T + J_A + J_P, \\ 4J'_V &= 4J_S - 2J_V + 2J_A - 4J_P, \\ 4J'_T &= 6J_S - 2J_T + 6J_P, \\ 4J'_A &= 4J_S + 2J_V - 2J_A - 4J_P, \\ 4J'_P &= J_S - J_V - J_T - J_A + J_P \end{aligned}$$

(the first row is formula (28.12)).

§ 29. Polarisation density matrix

The coordinate dependence of the wave function ψ describing free motion with momentum $\mathbf{p}$ (a plane wave) reduces to the common factor $e^{i\mathbf{p}\mathbf{r}}$, while the amplitude u_p plays the role of a spin wave function. In such a (pure) state the particle is fully polarised (see III, § 59). In the non-relativistic theory

this means that the particle's spin has a definite spatial direction (more precisely, there exists a direction along which the spin projection takes the definite value $+1/2$). In a relativistic theory such a characterisation of the state is impossible in an arbitrary reference frame, owing to the non-conservation of the spin vector (already noted in § 23). Purity of the state signifies only that the spin has a definite direction in the particle's rest frame.

In a state of partial polarisation no definite amplitude exists; there is only a *polarisation density matrix* ρ_{ik} ($i, k = 1, 2, 3, 4$ being bispinor indices). We define this matrix so that in a pure state it reduces to the product

$$\rho_{ik} = u_{pi} \bar{u}_{pk}. \quad (29.1)$$

The corresponding normalisation condition for ρ is

$$\text{Sp } \rho = 2m \quad (29.2)$$

(cf. (23.4)).

In a pure state the mean value of the spin is given by

$$\mathbf{s} = \frac{1}{2} \int \psi^* \boldsymbol{\Sigma} \psi d^3x = \frac{1}{4\varepsilon} u_p^+ \boldsymbol{\Sigma} u_p = \frac{1}{4\varepsilon} \bar{u}_p \gamma^0 \boldsymbol{\Sigma} u_p. \quad (29.3)$$

The corresponding expression for a partially polarised state is

$$\mathbf{s} = \frac{1}{4\varepsilon} \text{Sp}(\rho \gamma^0 \boldsymbol{\Sigma}) = \frac{1}{4\varepsilon} \text{Sp}(\rho \gamma^5 \boldsymbol{\gamma}). \quad (29.4)$$

The amplitudes u_p and $\bar{u}_p$ satisfy the algebraic equations

$$(\gamma p - m)u_p = 0, \quad \bar{u}_p(\gamma p - m) = 0.$$

The matrix (29.1) therefore obeys

$$(\gamma p - m)\rho = 0, \quad \rho(\gamma p - m) = 0. \quad (29.5)$$

The density matrix must satisfy the same linear equations in the general case of a mixed (in spin) state as well (cf. the analogous argument in III, § 14).

When a free particle is considered in its rest frame, the non-relativistic theory applies. In that theory the state of partial polarisation is completely specified by three parameters—the components of the mean spin vector $\mathbf{s}$ (see III, § 59). It is therefore clear that the same parameters will determine the polarisation state after any Lorentz boost, i.e. for a moving particle.

Denote the doubled mean spin vector in the rest frame by ζ (in a pure state $|\zeta| = 1$; in a mixed state $|\zeta| < 1$). For a four-dimensional description of the polarisation state it is convenient to introduce a 4-vector a^μ that coincides in the rest frame with the three-vector ζ ; since ζ is an axial vector, a^μ is a 4-pseudovector. This 4-vector is orthogonal to the 4-momentum in the rest frame (where $a^\mu = (0, \zeta)$ and $p^\mu = (m, 0)$), and hence in every frame:

$$a_\mu p^\mu = 0. \quad (29.6)$$

In an arbitrary frame one also has

$$a_\mu a^\mu = -\zeta^2. \quad (29.7)$$

The components of a^μ in a frame where the particle moves with velocity $\mathbf{v} = \mathbf{p}/\varepsilon$ are obtained by Lorentz-boosting from the rest frame:

$$a^0 = \frac{|\mathbf{p}|}{m} \zeta_{\parallel}, \quad \mathbf{a}_{\perp} = \zeta_{\perp}, \quad a_{\parallel} = \frac{\varepsilon}{m} \zeta_{\parallel}, \quad (29.8)$$

where the subscripts $\parallel$ and $\perp$ denote the components of ζ and $\mathbf{a}$ parallel and perpendicular to $\mathbf{p}$ ³³. These formulae can be written in vector form:

$$\mathbf{a} = \zeta + \frac{\mathbf{p}(\zeta \cdot \mathbf{p})}{m(\varepsilon + m)}, \quad a^0 = \frac{\mathbf{a} \cdot \mathbf{p}}{\varepsilon} = \frac{\mathbf{p} \cdot \zeta}{m}, \quad \mathbf{a}^2 = \zeta^2 + \frac{(\mathbf{p} \cdot \zeta)^2}{m^2}. \quad (29.9)$$

Consider first the unpolarised state ($\zeta = 0$). The density matrix can then depend only on the 4-momentum p . The unique matrix of this kind satisfying the equations (29.5) is

$$\rho = \frac{1}{2}(\gamma p + m) \quad (29.10)$$

(I. E. Tamm, 1930; H. B. G. Casimir, 1933). The numerical coefficient is fixed by the normalisation condition (29.2).

³³By their transformation properties, the components of the mean spin vector $\mathbf{s}$ (like those of any angular momentum) are, in relativistic mechanics, the spatial components of an antisymmetric tensor $S^{\lambda\mu}$. The 4-vector a^λ is related to this tensor by

$$S^{\lambda\mu} = \frac{1}{2m} \epsilon^{\lambda\mu\nu\rho} a_\nu p_\rho, \quad a^\lambda = -\frac{2}{m} \epsilon^{\lambda\mu\nu\rho} S_{\mu\nu} p_\rho.$$

Note that in an arbitrary frame the spatial part of a^λ does not coincide with $2\mathbf{s}$. One easily finds $2\mathbf{s}_{\parallel} = \frac{1}{m}(a_{\parallel}\varepsilon - a^0|\mathbf{p}|) = \zeta_{\parallel}$, $2\mathbf{s}_{\perp} = \frac{\varepsilon}{m}\mathbf{a}_{\perp} = \frac{\varepsilon}{m}\zeta_{\perp}$.

In the general case of partial polarisation ($\zeta \neq 0$) we seek the density matrix in the form

$$\rho = \frac{1}{4m}(\gamma p + m)\rho'(\gamma p + m), \quad (29.11)$$

which automatically satisfies (29.5). When $\zeta \neq 0$ the auxiliary matrix ρ' must reduce to the identity; since

$$(\gamma p + m)^2 = 2m(\gamma p + m),$$

equation (29.11) then coincides with (29.10). Furthermore, ρ' must contain the 4-vector a linearly as a parameter, i.e. it must have the form

$$\rho' = 1 - A\gamma^5(\gamma a); \quad (29.12)$$

the second term involves the scalar product of the pseudovector a with the “matrix 4-pseudovector” $\gamma^5\gamma$. To fix the coefficient A , write the density matrix in the rest frame:

$$\rho = \frac{m}{4}(1 + \gamma^0)(1 + A\gamma^5\gamma \cdot \zeta)(1 + \gamma^0) = \frac{m}{2}(1 + \gamma^0)(1 + A\gamma^5\gamma \cdot \zeta),$$

and compute the mean spin via (29.4). Using the trace rules listed in § 22, one readily finds that the only non-vanishing contribution to the required trace is

$$2s = \frac{1}{2m} \text{Sp}(\rho\gamma^5\gamma) = -\frac{A}{4} \text{Sp}((\gamma \cdot \zeta)\gamma) = A\zeta.$$

Setting this equal to ζ gives $A = 1$. To obtain the final expression for ρ , substitute (29.12) into (29.11) and commute ρ' past $(\gamma p + m)$; because a and p are orthogonal, γp anticommutes with γa :

$$(\gamma a)(\gamma p) = 2a \cdot p - (\gamma p)(\gamma a) = -(\gamma p)(\gamma a),$$

and therefore commutes with $\gamma^5(\gamma a)$.

The density matrix of a partially polarised electron is thus

$$\rho = \frac{1}{2}(\gamma p + m)[1 - \gamma^5(\gamma a)] \quad (29.13)$$

(*L. Michel, A. S. Wightman, 1955*). If ρ is known, the 4-vector a characterising the state (and with it the vector ζ) can be recovered from

$$a^\mu = \frac{1}{2m} \text{Sp}(\rho\gamma^5\gamma^\mu). \quad (29.14)$$

The formulae for the positron density matrix are analogous to those for the electron. Had we described a positron (with

4-momentum p) by a positron amplitude $u_p^{(\text{pos})}$ and a correspondingly defined density matrix $\rho^{(\text{pos})}$, there would be no difference from the electron case and $\rho^{(\text{pos})}$ would be given by the same formula (29.13). In actual calculations of scattering cross sections involving positrons, however, one deals (as we shall see later) not with $u_p^{(\text{pos})}$ but with the “negative-frequency” amplitudes u_{-p} . Accordingly, the polarisation density matrix (which we denote $\rho^{(-)}$) should be defined so that for a pure state it reduces to $u_{-p}i\bar{u}_{-p}k$.

By (26.1) the positron amplitude $u_p^{(\text{pos})} = U_C \bar{u}_{-p}$. Inversely:

$$u_{-p} = U_C u_p^{(\text{pos})}, \quad \bar{u}_{-p} = U_C^+ u_p^{(\text{pos})} = u_p^{(\text{pos})} U_C^*$$

(cf. (28.3)). If

$$\rho_{ik}^{(-)} = u_{-p i} \bar{u}_{-p k}, \quad \rho_{ik}^{(\text{pos})} = u_{pi}^{(\text{pos})} \bar{u}_{pk}^{(\text{pos})},$$

then these relations give

$$\rho^{(-)} = U_C \tilde{\rho}^{(\text{pos})} U_C^*. \quad (29.15)$$

Substituting (29.13) for $\rho^{(\text{pos})}$ and carrying out straightforward algebra (using (26.3) and (26.21)), we obtain

$$\rho^{(-)} = \frac{1}{2}(\gamma p - m)[1 - \gamma^5(\gamma a)]. \quad (29.16)$$

In particular, for an unpolarised state

$$\rho^{(-)} = \frac{1}{2}(\gamma p - m). \quad (29.17)$$

Henceforth, when speaking of the positron density matrix we shall mean $\rho^{(-)}$ and drop the superscript $(-)$ (the matrix $\rho^{(\text{pos})}$ is in practice never needed).

In various calculations one frequently has to average over spin states expressions of the form $\bar{u} F u (\equiv \bar{u}_i F_{ik} u_k)$, where F is some (4×4) matrix and u is the bispinor amplitude for a state with definite 4-momentum p . Such averaging is equivalent to replacing the products $u_k \bar{u}_i$ by the density matrix ρ_{ik} of a partially polarised state.

In particular, a complete average over the two independent spin states amounts to passing to the unpolarised state; by (29.10) we then have

$$\frac{1}{2} \sum_{\text{pol}} \bar{u}_p F u_p = \frac{1}{2} \text{Sp}(\gamma p + m) F. \quad (29.18)$$

Similarly, for the negative-frequency wave functions

$$\frac{1}{2} \sum_{\text{pol}} \bar{u}_{-p} F u_{-p} = \frac{1}{2} \text{Sp}(\gamma p - m) F. \quad (29.19)$$

If one is summing over spin states rather than averaging, the result is twice as large.

Let us trace how the density matrix (29.13) passes to its non-relativistic limit. To this end we go to the electron rest frame. In the standard representation the amplitudes u_p become two-component in this frame, and the density matrix must correspondingly become a 2×2 matrix. Indeed, in the rest frame

$$\rho = \frac{m}{2}(\gamma^0 + 1)(1 + \gamma^5 \boldsymbol{\gamma} \cdot \boldsymbol{\zeta}),$$

and using the explicit forms of the γ matrices (21.20) and (22.18) we find

$$\rho = \begin{pmatrix} \rho_{\text{nr}} & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho_{\text{nr}} = m(1 + \boldsymbol{\sigma} \cdot \boldsymbol{\zeta}) \quad (29.20)$$

(the zeros denote 2×2 null matrices). Adopting the normalisation $\text{Sp } \rho_{\text{nr}} = 1$ conventional in the non-relativistic theory (instead of $2m$), one must divide this expression by $2m$, yielding

$$\frac{1}{2}(1 + \boldsymbol{\sigma} \cdot \boldsymbol{\zeta}),$$

in agreement with formula (59.6) (see III).

The non-relativistic limit of the positron density matrix is obtained in a similar way:

$$\rho = \begin{pmatrix} 0 & 0 \\ 0 & \rho_{\text{nr}} \end{pmatrix}, \quad \rho_{\text{nr}} = -m(1 + \boldsymbol{\sigma} \cdot \boldsymbol{\zeta}).$$

Finally, let us write down a simplified expression for the density matrix in the ultrarelativistic case. Setting $|\mathbf{p}| \approx \varepsilon$ in (29.8) (thereby neglecting quantities of relative order $(m/\varepsilon)^2$), substituting into (29.13) or (29.16), and choosing the direction of $\mathbf{p}$ as the x -axis, we have

$$\rho = \frac{1}{2}[\varepsilon(\gamma^0 - \gamma^1) \pm m] \left[1 - \gamma^5 \left(\frac{\varepsilon}{m}(\gamma^0 - \gamma^1)\zeta_{\parallel} - \zeta_{\perp}\gamma_{\perp} \right) \right],$$

where the upper sign refers to the electron and the lower to the positron. On expanding the product the leading terms cancel, and the next-order terms give

$$\rho = \frac{1}{2}\varepsilon(\gamma^0 - \gamma^1)[1 + \gamma^5(\pm\zeta_{\parallel} + \zeta_{\perp}\gamma_{\perp})]$$

or, writing $\varepsilon(\gamma^0 - \gamma^1)$ as γp :

$$\rho = \frac{1}{2}(\gamma p)[1 + \gamma^5(\pm\zeta_{\parallel} + \boldsymbol{\zeta}_{\perp} \cdot \boldsymbol{\gamma}_{\perp})]. \quad (29.21)$$

This is the desired ultrarelativistic density matrix. Note that all components of the polarisation vector $\boldsymbol{\zeta}$ enter on an equal footing, as terms of the same order. Recall that $\zeta_{\parallel}$ is the component of this vector parallel ($\zeta_{\parallel} > 0$) or antiparallel ($\zeta_{\parallel} < 0$) to the particle's momentum. In particular, for a helicity eigenstate $\zeta_{\parallel} = 2\lambda = \pm 1$; the density matrix then takes the especially simple form

$$\rho = \frac{1}{2}(\gamma p)(1 \pm 2\lambda\gamma^5), \quad (29.22)$$

which coincides, as it should, with the density matrix of a neutrino or antineutrino—a particle with zero mass and definite helicity (see § 30).

§ 30. Two-component fermions

We saw in § 20 that the need to describe a spin-1/2 particle by two spinors (ξ and η) is tied to the particle having a mass. This requirement disappears when the mass vanishes. The wave equation for such a particle can be formulated in terms of a single spinor—say the dotted spinor η :

$$\widehat{p}^{\alpha\beta}\eta_{\beta} = 0, \quad (30.1)$$

or equivalently,

$$(\widehat{p}_0 + \widehat{\mathbf{p}} \cdot \boldsymbol{\sigma})\eta = 0. \quad (30.2)$$

In § 20 it was also noted that the wave equation containing the mass m is automatically symmetric under inversion (the transformation (20.4)). When the particle is described by a single spinor this symmetry is lost. There is, however, no necessity for it, since symmetry under inversion is not a universal law of nature.

The energy and momentum of a particle with $m = 0$ are related by $\varepsilon = |\mathbf{p}|$. For a plane wave ($\eta_p \propto e^{-ipx}$) the equation (30.2) therefore gives

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\eta_p = -\eta_p, \quad (30.3)$$

where $\mathbf{n}$ is the unit vector along $\mathbf{p}$. The same equation

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\eta_{-p} = -\eta_{-p} \quad (30.4)$$

holds also for the “negative-frequency” waves ($\eta_{-p} \propto e^{ipx}$). The second-quantised ψ -operator is

$$\widehat{\eta} = \sum_{\mathbf{p}} (\eta_p \widehat{a}_{\mathbf{p}} + \eta_{-p} \widehat{b}_{\mathbf{p}}^+), \quad \widehat{\eta}^+ = \sum_{\mathbf{p}} (\eta_p^* \widehat{a}_{\mathbf{p}}^+ + \eta_{-p}^* \widehat{b}_{\mathbf{p}}). \quad (30.5)$$

From this it follows, as usual, that η_{-p}^* are the antiparticle wave functions.

From the definition of the operators $\widehat{p}^{\alpha\beta}$ (20.1) it is clear that $\widehat{p}^{\alpha\beta*} = -\widehat{p}^{\dot{\alpha}\dot{\beta}}$. Hence the complex-conjugate spinor η^* satisfies $\widehat{p}^{\dot{\alpha}\dot{\beta}}\eta_{\dot{\beta}}^* = 0$, i.e.

$$\widehat{p}_{\dot{\alpha}\dot{\beta}}\eta^{\dot{\beta}*} = 0.$$

Denoting $\eta^{\dot{\beta}*} = \xi^{\dot{\beta}}$ (thus expressing the fact that complex conjugation converts a dotted spinor into an undotted one), we see that the antiparticle wave functions satisfy

$$\widehat{p}_{\dot{\alpha}\dot{\beta}}\xi^{\dot{\beta}} = 0, \quad (30.6)$$

or

$$(\widehat{p}_0 - \widehat{\mathbf{p}} \cdot \boldsymbol{\sigma})\xi = 0. \quad (30.7)$$

For a plane wave this gives

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\xi_p = \xi_p. \quad (30.8)$$

Now $\frac{1}{2}(\mathbf{n} \cdot \boldsymbol{\sigma})$ is the operator of the spin projection on the direction of motion. Equations (30.3) and (30.8) therefore mean that states with a definite momentum are automatically helicity eigenstates—the spin projection along the direction of motion takes a definite value. If the particle’s spin is antiparallel to its momentum (helicity $-1/2$), then the antiparticle’s spin is aligned with its momentum (helicity $+1/2$). Particles with these properties are, perhaps, realised in nature as **neutrinos**. The particle with helicity $-1/2$ is conventionally called the neutrino, while the one with helicity $+1/2$ is called the antineutrino³⁴.

³⁴The existence of the neutrino was predicted theoretically by *Pauli* to explain β -decay properties (1931). Equation (30.1) was first considered by *Weyl* (*H. Weyl*, 1929). A neutrino theory based on these equations was formulated by *L. D. Landau*; *Lee, Yang and Salam* (*T. D. Lee, C. N. Yang, A. Salam*) in 1957.

The experimental question of whether the neutrino mass is exactly zero has not been settled definitively up to the present time. We shall henceforth use the term “neutrino” conventionally to designate a particle described by equation (30.3).

In connection with the non-degeneracy of the neutrino states with respect to the spin direction, recall the remark made in § 8 that a massless particle (with spin 1/2) possesses only an axial symmetry about the direction of its momentum. For a truly neutral particle—the photon—this symmetry includes both rotations about the axis and reflections in planes passing through it. For the neutrino, however, there is no symmetry under reflections, and one is left only with the group of rotations about the axis, which preserves the momentum projection of the spin but does not reverse its sign. Symmetry under reflections exists only on the condition that particle and antiparticle are simultaneously interchanged.

It should also be noted that the obligatory longitudinal polarisation means that, for the neutrino, spin is altogether inseparable from orbital angular momentum (just as, for the photon, it is inseparable from the obligatory transversality of the field; see § 6).

From a single spinor η (or ξ) one can form only four bilinear combinations, which together make up a 4-vector

$$j^\mu = (\eta^* \eta, \eta^* \boldsymbol{\sigma} \eta). \quad (30.9)$$

It is easily verified that, by virtue of the equations

$$(\widehat{p}_0 + \widehat{\mathbf{p}} \cdot \boldsymbol{\sigma}) \eta = 0, \quad \eta^* (\widehat{p}_0 - \widehat{\mathbf{p}} \cdot \boldsymbol{\sigma}) = 0,$$

the continuity equation $\partial_\mu j^\mu = 0$ is satisfied, i.e. j^μ serves as the particle current-density 4-vector.

The neutrino plane waves are conveniently normalised in the same fashion as was done in § 23 for massive particles:

$$\eta_p = \frac{1}{\sqrt{2\varepsilon}} u_p e^{-ipx}, \quad \eta_{-p} = \frac{1}{\sqrt{2\varepsilon}} u_{-p} e^{ipx}, \quad (30.10)$$

with the spinor amplitudes satisfying the invariant normalisation condition

$$u_{\pm p}^* (1, \boldsymbol{\sigma}) u_{\pm p} = 2(\varepsilon, \mathbf{p}). \quad (30.11)$$

The current density is then $j^0 = 1$, $\mathbf{j} = \mathbf{p}/\varepsilon = \mathbf{n}$.

Since a free neutrino with a given momentum is always fully polarised, the concept of a mixed spin state does not arise. It may nonetheless be useful to introduce a 2×2 polarisation “density matrix” defined simply as a rank-two spinor:

$$\rho_{\dot{\alpha}\dot{\beta}} = u_{\dot{\alpha}} u_{\dot{\beta}}^* \quad (30.12)$$

(with $\text{Sp } \rho = 2\varepsilon$). The expression for this matrix is obtained by noting that it must satisfy

$$(\varepsilon + \mathbf{p} \cdot \boldsymbol{\sigma}) \rho = \rho (\varepsilon + \mathbf{p} \cdot \boldsymbol{\sigma}) = 0.$$

It follows that

$$\rho = \varepsilon - \mathbf{p} \cdot \boldsymbol{\sigma}. \quad (30.13)$$

When considering various interaction processes, the neutrino may appear alongside other particles (with spin 1/2) that have a mass and are therefore described by four-component wave functions. In such cases it is convenient to maintain a uniform notation

by formally introducing a “bispinor” wave function for the neutrino as well, $\psi = \begin{pmatrix} 0 \\ \eta \end{pmatrix}$, two of whose components are, however, identically zero. But such a form of ψ would, generally speaking, be spoiled by a change to a different (non-spinor) representation. This difficulty is circumvented by observing that in the spinor representation one has the identity

$$\frac{1 + \gamma^5}{2} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} 0 \\ \eta \end{pmatrix}, \quad (\eta^* \eta^*) \frac{1 - \gamma^5}{2} = (\eta^* 0),$$

where ξ is an arbitrary “ballast” spinor that drops out thanks to the matrix γ^5 from (22.18). The condition of true “two-componentness” of the neutrino is therefore maintained when describing it by a four-component ψ in any representation, provided one means by ψ a solution of the Dirac equation with $m = 0$:

$$(\gamma p)\psi = 0, \quad (30.14)$$

subject to the additional constraint $\frac{1}{2}(1 + \gamma^5)\psi = \psi$, i.e.

$$\gamma^5\psi = \psi. \quad (30.15)$$

This condition can be taken into account, once and for all, in every formula involving ψ and $\bar{\psi}$ by making the substitution

$$\psi \rightarrow \frac{1 + \gamma^5}{2}\psi, \quad \bar{\psi} \rightarrow \bar{\psi} \frac{1 - \gamma^5}{2}. \quad (30.16)$$

The current-density 4-vector then takes the form (cf. the replacement (30.16) in the expression $\bar{\psi}\gamma^\mu\psi$)

$$j^\mu = \frac{1}{4}\bar{\psi}(1 - \gamma^5)\gamma^\mu(1 + \gamma^5)\psi = \frac{1}{2}\bar{\psi}\gamma^\mu(1 + \gamma^5)\psi. \quad (30.17)$$

In accordance with the same rule, the 4×4 density matrix of the neutrino should be written as

$$\rho = \frac{1}{4}(1 + \gamma^5)(\gamma p)(1 - \gamma^5) = \frac{1}{2}(1 + \gamma^5)(\gamma p). \quad (30.18)$$

In the spinor representation it reduces, as it should, to the 2×2 matrix (30.13):

$$\rho = \begin{pmatrix} 0 & 0 \\ \varepsilon - \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix}.$$

The corresponding formulae for the antineutrino differ from those written above by a change of sign in front of γ^5 .

The neutrino is an electrically neutral particle. A neutrino with the properties described above is not, however, a truly neutral particle. We note in this connection that the “neutrino field” described by a two-component spinor is, by the number of particle states it permits (but not, of course, by any other physical property), equivalent to a truly neutral field described by a four-component bispinor. In place of particles and antiparticles with definite helicities, there would be the same number of states of a

single particle with two possible helicity values, and symmetry under inversion would be automatically preserved.

Note, however, that the vanishing of the “four-component” neutrino mass would be, so to speak, “accidental”—it would not be related to any symmetry property of the wave equation (which permits a non-zero mass as well). For this reason the account of various interactions of such a particle would automatically generate a mass that, while small, would not be strictly zero.

§ 31. Wave equation for a spin-3/2 particle

A spin-3/2 particle is described in its rest frame by a symmetric 3-spinor of rank three (with $2s + 1 = 4$ independent components). Accordingly, in an arbitrary frame the description may involve the 4-spinors $\xi^{\alpha\beta\dot{\gamma}}$, $\eta_{\dot{\alpha}\beta\gamma}$, $\zeta^{\alpha\beta\gamma}$, and $\chi_{\dot{\alpha}\beta\dot{\gamma}}$, each symmetric under interchange of all identical (dotted or undotted) indices; under inversion the spinors in the first pair, and likewise those in the second pair, transform into each other.

For the rest-frame 4-spinors $\xi^{\alpha\beta\dot{\gamma}}$ and $\eta_{\dot{\alpha}\beta\gamma}$, symmetric over all three indices, to reduce to 3-spinors they must satisfy the conditions

$$\widehat{p}^{\dot{\alpha}\beta} \eta_{\dot{\alpha}\beta\gamma} = 0, \quad \widehat{p}_{\alpha\dot{\beta}} \xi^{\alpha\beta\dot{\gamma}} = 0. \quad (31.1)$$

Indeed, in the rest frame

$$\widehat{p}^{\dot{\alpha}\beta} \rightarrow \widehat{p}_0 \delta_{\alpha}^{\beta} = m \delta_{\alpha}^{\beta}$$

(as is clear from (20.1)). The conditions (31.1) therefore lead to the equalities

$$\delta_{\alpha}^{\beta} \eta'^{\alpha}{}_{\beta\gamma} = 0, \quad \delta_{\alpha}^{\beta} \xi'^{\alpha}{}_{\beta\gamma} = 0,$$

where the primed letters denote the corresponding 3-spinors; in other words, these spinors vanish upon contraction over the in-

indices $\alpha\beta$, which means that they are symmetric in those indices, and hence in all three.

The differential relation between the spinors ξ and η is established by the equations

$$\widehat{p}^{\dot{\beta}\gamma} \eta_{\dot{\alpha}\beta}^{\beta} = m \xi^{\beta\gamma}{}_{\dot{\alpha}}, \quad \widehat{p}_{\dot{\beta}\gamma} \xi^{\beta\gamma}{}_{\dot{\alpha}} = m \eta_{\dot{\alpha}\beta}^{\beta}. \quad (31.2)$$

The symmetry of the left-hand sides of these equations (in the indices β , $\dot{\gamma}$ or α , $\dot{\beta}$) is ensured by the conditions (31.1), by virtue of which they vanish upon contraction over all indices. In the rest frame the 3-spinors ξ' and η' coincide because of equations (31.2). Eliminating η (or ξ) from (31.2), one finds that each component of the spinors ξ and η satisfies the second-order equation

$$(\widehat{p}^2 - m^2) \xi^{\alpha\beta\dot{\gamma}} = 0. \quad (31.3)$$

The system of equations (31.1), (31.2) constitutes the complete set of wave equations for a spin-3/2 particle³⁵. Adding the spinors ζ , χ would contribute nothing new. They are constructed according to

$$m \zeta^{\alpha\beta\gamma} = \widehat{p}^{\alpha\dot{\delta}} \eta_{\dot{\delta}}^{\beta\gamma}, \quad m \chi_{\dot{\alpha}\beta\dot{\gamma}} = \widehat{p}_{\dot{\alpha}\delta} \xi^{\delta}{}_{\beta\dot{\gamma}}.$$

³⁵For the Lagrangian formulation of these equations see the article by *Fierz* and *Pauli* cited on p. 76.

The equations for spin-3/2 particles can also be cast in a different form that exploits the vector properties of the spinors (W. Rarita, J. Schwinger, 1941; A. S. Davydov, I. E. Tamm, 1942). A pair of spinor indices $\alpha\dot{\beta}$ is associated with a single four-dimensional vector index μ . The components of a rank-three spinor $\xi^{\alpha\beta\dot{\gamma}}$ can therefore be brought into correspondence with the components of “mixed” quantities ψ^μ_l carrying one vector and one spinor index. Likewise, the spinor $\eta^{\dot{\beta}\alpha\gamma}$ corresponds to quantities ψ^μ_l , and the totality of both spinors to a “vector” bispinor ψ_μ (with the bispinor index suppressed). The wave equation is then written as a “Dirac equation” for each of the vector components ψ_μ :

$$(\gamma\hat{p} - m)\psi_\mu = 0 \quad (31.4)$$

with the supplementary condition

$$\gamma^\mu\psi_\mu = 0. \quad (31.5)$$

Using the expressions for the matrices γ^μ in the spinor representation (18.6), (18.7), it is straightforward to verify that the equations (31.2) are contained in (31.4), and the condition (31.5) is equivalent to the symmetry requirement on the spinors $\xi^{\alpha\beta\dot{\gamma}}$ and $\eta^{\dot{\alpha}\beta\gamma}$ in the indices $\beta\dot{\gamma}$ or $\beta\gamma$. Multiplying equation (31.4) on γ^μ and using (31.5):

$$\gamma^\mu\gamma^\nu\hat{p}_\nu\psi_\mu = 0$$

or, applying the commutation rules for the γ matrices,

$$2g^{\mu\nu}\hat{p}_\nu\psi_\mu - \gamma^\nu\hat{p}_\nu\gamma^\mu\psi_\mu = 0. \quad (31.6)$$

The second term vanishes by (31.5), and the first gives

$$\hat{p}^\mu\psi_\mu = 0. \quad (31.7)$$

It is easy to see that this condition, which follows automatically from (31.4) and (31.5), is equivalent to (31.1).

Finally, there is yet another way to formulate the wave equation. It consists in introducing quantities ψ_{ikl} ($i, k, l = 1, 2, 3, 4$) with three bispinor indices, symmetric under permutation (V. Bargmann, E. P. Wigner, 1948). The totality of these quantities is equivalent to the set of all four spinors ξ, η, ζ, χ . The wave equation is written as a system of “Dirac equations”

$$\hat{p}_\mu\gamma_{im}^\mu\psi_{mkl} = m\psi_{ikl}. \quad (31.8)$$

It is readily seen that these equations already yield the correct number of independent components (four), and no additional constraints are needed. Indeed, in the rest frame (31.8) reduces to

$$\gamma_{im}^0\psi_{mkl} = \psi_{ikl},$$

by virtue of which all components with $i, k, l = 3, 4$ (in the standard representation) vanish, i.e. ψ_{ikl} reduces to the components of a rank-three 3-spinor.

The results presented here generalise in an obvious way to particles of any half-integer spin s . In the description via equations of the type (31.4), (31.5) the wave function is a symmetric 4-tensor of rank $(2s - 1)/2$ carrying one bispinor index. In the description via equations of the type (31.8) the wave function has $2s$ bispinor indices over which it is symmetric.

A Particle in an External Field

§ 32. The Dirac equation for an electron in an external field

The wave equations for free particles express, in essence, only those properties that follow from the general requirements of space–time symmetry. The physical processes that occur with particles depend on the nature of their interactions.

A description of the electromagnetic interactions of particles in a relativistic quantum theory turns out to be possible by generalising the method employed for this purpose in both classical and non-relativistic quantum theories.

This method is, however, applicable only to the electromagnetic interactions of particles that are incapable of strong interactions. Electrons (and positrons) belong to this class, and thus the entire vast domain of quantum electrodynamics of electrons becomes accessible to the present theory. Muons are likewise incapable of strong interactions; they too are described by quantum electrodynamics in the regime of phenomena whose time scales are short compared with the muon lifetime (which is governed by weak interactions).

In this chapter we consider the class of problems in quantum electrodynamics that fall within the framework of the single-particle theory, where the number of particles is fixed and the interaction can be described in terms of an external electromagnetic field. Besides the conditions that permit the external field to be treated as given, the range of applicability of such a theory is further restricted by considerations connected with the so-called radiative corrections.

The wave equation of an electron in a given external field is obtained in the same way as in the non-relativistic theory (see III, § 111). Let $A^\mu = (\Phi, \mathbf{A})$ be the 4-potential of the external electromagnetic field ($\mathbf{A}$ is the vector potential, Φ the scalar potential). The desired equation is obtained by replacing in the Dirac equation the 4-momentum operator $\hat{p}$ with the difference $\hat{p} - eA$, where e is

the charge of the particle¹:

$$[\gamma(\hat{p} - eA) - m]\psi = 0. \quad (32.1)$$

The Hamiltonian corresponding to this equation is obtained from (21.13) by the same substitution:

$$\hat{H} = \alpha(\hat{\mathbf{p}} - e\mathbf{A}) + \beta m + e\Phi. \quad (32.2)$$

The invariance of the Dirac equation under a gauge transformation of the electromagnetic potentials is expressed by the fact that its form remains unchanged when the transformation $A \rightarrow A + i\hat{p}\chi$ (where χ is an arbitrary function) is accompanied by the

¹The charge is understood together with its sign, so that for the electron $e = -|e|$.

transformation of the wave function²

$$\psi \rightarrow \psi e^{ie\chi} \quad (32.3)$$

(cf. the analogous transformation for the Schrödinger equation in III, § 111).

The current density, expressed in terms of the wave function, is given by the same formula (21.11) $j = \bar{\psi} \gamma \psi$ as in the absence of an external field. It is easily verified that, on substituting (32.1) (together with equation (32.4) below) into the same calculation that was used to derive (21.11), the external field drops out and the continuity equation remains valid in its previous form.

Let us apply to equation (32.1) the operation of charge conjugation. To this end, write the equation as

$$\bar{\psi}[\gamma(\hat{p} + eA) + m] = 0, \quad (32.4)$$

which is obtained by complex conjugation from (32.1) in the same way as was done previously for equation (19.9) (bearing in mind that the 4-vector A is real). Rewriting this equation as

$$[\bar{\gamma}(\hat{p} + eA) + m]\bar{\psi} = 0,$$

multiplying on the left by the matrix U_C and using the relations (26.3), we find

$$[\gamma(\hat{p} + eA) - m](\hat{C}\psi) = 0. \quad (32.5)$$

Thus the charge-conjugate wave function satisfies an equation differing from the original one only by a

reversal of the sign of the charge. On the other hand, the operation of charge conjugation signifies a transition from particles to antiparticles. We see, therefore, that if the particles carry an electric charge, the charges of electron and positron are automatically opposite.

The first-order equation (32.1) can be converted into a second-order equation by applying the operator $\gamma(\hat{p} - eA) + m$:

$$[\gamma^\mu \gamma^\nu (\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) - m^2]\psi = 0.$$

Replacing $\gamma^\mu \gamma^\nu$ by

$$\gamma^\mu \gamma^\nu = \frac{1}{2}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) + \frac{1}{2}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) = g^{\mu\nu} + \sigma^{\mu\nu},$$

where $\sigma^{\mu\nu}$ is the antisymmetric “matrix 4-tensor” (28.2). On multiplying by $\sigma^{\mu\nu}$ one may antisymmetrise, i.e.

$$\begin{aligned} (\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) &\rightarrow \frac{1}{2}\{(\hat{p}_\mu - eA_\mu), (\hat{p}_\nu - eA_\nu)\}_- = \\ &= \frac{1}{2}e(-A_\mu \hat{p}_\nu + \hat{p}_\nu A_\mu - \hat{p}_\mu A_\nu + A_\nu \hat{p}_\mu) = \\ &= \frac{1}{2}ie(\partial_\nu A_\mu - \partial_\mu A_\nu) = -\frac{ie}{2}F_{\mu\nu} \end{aligned}$$

²The transformation (32.3) with a function $\chi(t, \mathbf{r})$ is sometimes called a “local gauge transformation”, as distinct from the “global gauge transformation” (12.10) with a constant phase α .

($F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ being the electromagnetic field tensor). The resulting second-order equation is

$$\left[(\hat{p} - eA)^2 - m^2 - \frac{i}{2} e F_{\mu\nu} \sigma^{\mu\nu} \right] \psi = 0. \quad (32.6)$$

The product $F_{\mu\nu} \sigma^{\mu\nu}$ can be written in three-dimensional form by expressing it in terms of its components:

$$\sigma^{\mu\nu} = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma}), \quad F^{\mu\nu} = (-\mathbf{E}, \mathbf{H}).$$

We then obtain the second-order equation as

$$[(\hat{p} - eA)^2 - m^2 + e\boldsymbol{\Sigma} \cdot \mathbf{H} - ie\boldsymbol{\alpha} \cdot \mathbf{E}] \psi = 0, \quad (32.7)$$

or, in ordinary units,

$$\left[\left(\frac{i\hbar}{c} \frac{\partial}{\partial t} - \frac{e}{c} \Phi \right)^2 - \left(i\hbar \nabla + \frac{e}{c} \mathbf{A} \right)^2 - m^2 c^2 + \frac{e\hbar}{c} \boldsymbol{\Sigma} \cdot \mathbf{H} - i \frac{e\hbar}{c} \boldsymbol{\alpha} \cdot \mathbf{E} \right] \psi = 0. \quad (32.7a)$$

The appearance in these equations of the terms containing $\mathbf{E}$ and $\mathbf{H}$ is connected with the spin of the particle; we shall return to their discussion in the next section.

Among the solutions of the second-order equation there are, of course, “spurious” ones—solutions that do not satisfy the original first-order equation (32.1) (they represent solutions

of equation (32.1) with the opposite sign in front of m). Selecting the correct solutions in specific cases is usually straightforward. A systematic selection procedure is the following: if φ is an arbitrary solution of the second-order equation, the correct solution of the first-order equation is

$$\psi = [\gamma(\hat{p} - eA) + m] \varphi. \quad (32.8)$$

Indeed, multiplying this relation by $\gamma(\hat{p} - eA) - m$, one sees that the right-hand side vanishes if φ satisfies equation (32.6).

It should be emphasised that the method of introducing an external field into the relativistic wave equation—by replacing $\hat{p}$ with $\hat{p} - eA$ —is not entirely self-evident. In carrying it out we essentially relied on an additional principle: the indicated replacement should be performed in the first-order equation. It is precisely for this reason that in equation (32.6) there appeared extra terms that would not have arisen had the substitution been made directly in the second-order equation.

Among the stationary solutions of the Dirac equation in an external field there may exist states belonging to both the continuous and the discrete spectrum. As in the non-relativistic theory, the continuous spectrum corresponds to infinite motion, in which the particle can be found at arbitrarily large distances where it may be regarded as free. Since the eigenvalues of the free-particle Hamiltonian are $\pm\sqrt{\mathbf{p}^2 + m^2}$, it is clear that the continuous spectrum of eigenvalues lies at $\varepsilon > m$ and at $\varepsilon \leq -m$. For $-m < \varepsilon < m$ the particle cannot escape to infinity, so that the motion is finite and the state belongs to the discrete spectrum.

As for free particles, so also for particles in an external field the wave functions with “positive frequency” ($\varepsilon > 0$) and “negative frequency” ($\varepsilon < 0$) enter the scheme of second quantisation in a definite way. For a particle in an external field this scheme is naturally generalised by replacing the plane waves in the formulae (25.1) with the correspondingly normalised eigenfunctions of the Dirac equation $\psi_n^{(+)}$ and $\psi_n^{(-)}$, belonging to the positive ($\varepsilon_n^{(+)}$) and negative ($-\varepsilon_n^{(-)}$) frequencies:

$$\begin{aligned}\widehat{\psi} &= \sum_n \{ \widehat{a}_n \psi_n^{(+)} \exp(-i\varepsilon_n^{(+)} t) + \widehat{b}_n^+ \psi_n^{(-)} \exp(i\varepsilon_n^{(-)} t) \}, \\ \widehat{\bar{\psi}} &= \sum_n \{ \widehat{a}_n^+ \bar{\psi}_n^{(+)} \exp(i\varepsilon_n^{(+)} t) + \widehat{b}_n \bar{\psi}_n^{(-)} \exp(-i\varepsilon_n^{(-)} t) \}.\end{aligned}\quad (32.9)$$

One must bear in mind here that, as the potential well deepens, the energy levels may cross the boundary $\varepsilon = 0$, i.e. the energies that were

positive may become negative (or, for a potential of the opposite sign, initially negative energies may become positive). Nonetheless, on grounds of continuity one must continue to regard these levels as electronic (rather than positronic). In other words, an electron includes all states which, in the limit of an infinitely slow removal of the field, tend to the positive boundary of the continuous spectrum ($\varepsilon = m$).

Although the Dirac equation for an electron in an external field does, as stated above, permit the solution of a wide class of quantum-electrodynamic problems, one must at the same time stress that the applicability of the external-field concept in relativistic theory is itself limited. This limitation is connected with the spontaneous creation of electron–positron pairs that arises in sufficiently strong fields (see below, §§ 35, 36).

We shall not consider in this book the introduction of an external field into the wave equations of particles with spin other than $1/2$, since this has no direct physical relevance—the real particles with such spins are hadrons, and their electromagnetic interactions cannot be described by wave equations. In this connection it should be noted that for spin-0 particles the wave equation has complex eigenvalues (with imaginary parts of both signs) in a sufficiently deep potential well. The wave equation for a spin-3/2 particle leads to a violation of causality, manifested in the appearance of solutions that propagate faster than light.

Problem

Determine the energy levels of an electron in a constant magnetic field.

Solution. Vector potential: $A_x = A_z = 0$, $A_y = Hx$ (the field H directed along the z -axis). The quantities that are conserved (along with the energy) are the components p_y and p_z of the generalised momentum.

We use the second-order equation for the auxiliary function φ (see (32.8)) and assume that φ is an eigenfunction of the operator Σ_z (with eigenvalue $\sigma = \pm 1$), as well as of the operators $\widehat{p}_y$ and $\widehat{p}_z$. The equation for φ then takes the form

$$\left\{ -\frac{d^2}{dx^2} + (eHx - p_y)^2 - eH\sigma \right\} \varphi = (\varepsilon^2 - m^2 - p_z^2) \varphi.$$

This equation is identical in form to the Schrödinger equation for a linear harmonic oscillator. Its eigenvalues are given by the formula

$$\varepsilon^2 - m^2 - p_z^2 = |e|H(2n+1) - eH\sigma, \quad n = 0, 1, 2, \dots$$

(cf. III, § 112). Note that the wave function ψ , which should be determined from φ by formula (32.8), is not an eigenfunction of the operator Σ_z , in accordance with the fact that for a moving particle the spin is not a conserved quantity.

§ 33. Expansion in powers of $1/c$

We saw (see § 21) that in the non-relativistic limit ($v \rightarrow 0$) the two components (χ) of the bispinor $\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}$ vanish. At low electron velocities, therefore, $\chi \ll \varphi$, and it becomes possible to derive an approximate equation involving only the two-component quantity φ , by formally expanding the wave function in powers of $1/c$ ³.

Starting from the Dirac equation for an electron in an external field written in the form

$$i\hbar \frac{\partial \psi}{\partial t} = \left\{ c\alpha \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 + e\Phi \right\} \psi. \quad (33.1)$$

The relativistic energy of the particle includes the rest energy mc^2 . To pass to the non-relativistic approximation this must be removed, which we do by introducing in place of ψ a function ψ' defined by

$$\psi = \psi' e^{-imc^2 t / \hbar}.$$

Then

$$\left(i\hbar \frac{\partial}{\partial t} + mc^2 \right) \psi' = \left\{ c\alpha \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 + e\Phi \right\} \psi'.$$

Writing $\psi' = \begin{pmatrix} \varphi' \\ \chi' \end{pmatrix}$ yields the system of equations:

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \varphi' = c\sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \chi', \quad (33.2)$$

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi + 2mc^2 \right) \chi' = c\sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \varphi' \quad (33.3)$$

(hereafter we drop the primes on φ and χ , since this will cause no confusion given that in this section we work exclusively with the transformed function ψ').

In the leading approximation we retain only the term $2mc^2\chi$ on the left of equation (33.3) and obtain

$$\chi = \frac{1}{2mc} \sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \varphi \quad (33.4)$$

(note that $\chi \sim \varphi/c$). Substituting this expression into (33.2) gives

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \varphi = \frac{1}{2m} \left(\sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \right)^2 \varphi.$$

³In this section we use ordinary units.

For the Pauli matrices one has the identity

$$(\sigma \mathbf{a})(\sigma \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\sigma \cdot [\mathbf{a} \times \mathbf{b}], \quad (33.5)$$

where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors (see (20.9)). Here $\mathbf{a} = \mathbf{b} = \hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}$, but the vector product $[\mathbf{a} \times \mathbf{b}]$ does not vanish because $\hat{\mathbf{p}}$ and $\mathbf{A}$ do not commute:

$$\left[\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A} \right) \times \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A} \right) \right] \varphi = i \frac{e\hbar}{c} \{ [\mathbf{A} \cdot \nabla] + [\nabla \cdot \mathbf{A}] \} \varphi = i \frac{e\hbar}{c} \text{rot } \mathbf{A} \cdot \varphi.$$

Thus

$$\left(\sigma \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A} \right) \right)^2 = \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A} \right)^2 - \frac{e\hbar}{c} \sigma \cdot \mathbf{H} \quad (33.6)$$

(where $\mathbf{H} = \text{rot } \mathbf{A}$ is the magnetic field), and for φ we obtain the equation

$$i\hbar \frac{\partial \varphi}{\partial t} = \hat{H}\varphi = \left[\frac{1}{2m} \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A} \right)^2 + e\Phi - \frac{e\hbar}{2mc} \sigma \cdot \mathbf{H} \right] \varphi. \quad (33.7)$$

This is the so-called *Pauli equation*. It differs from the non-relativistic Schrödinger equation by the presence in the Hamiltonian of the last term, which represents the potential energy of a magnetic dipole in an external field (cf. III, § 111). In the leading ($1/c$) approximation, therefore, the electron behaves as a particle that possesses, in addition to its charge, a magnetic moment⁴:

$$\boldsymbol{\mu} = \frac{e}{mc} \hbar \mathbf{s}. \quad (33.8)$$

The gyromagnetic ratio (e/mc) is twice as large as it would be for a magnetic moment associated with orbital motion.

The density is $\rho = \psi^* \psi = \varphi^* \varphi + \chi^* \chi$. In the leading approximation the second term is to be dropped, so that $\rho = |\varphi|^2$, as it should be for the Schrödinger equation.

The current density is:

$$\mathbf{j} = c\psi^* \boldsymbol{\alpha} \psi = c(\varphi^* \boldsymbol{\sigma} \chi + \chi^* \boldsymbol{\sigma} \varphi).$$

Substituting (33.4):

$$\chi = \frac{1}{2mc} \boldsymbol{\sigma} \left(-i\hbar \nabla - \frac{e}{c}\mathbf{A} \right) \varphi, \quad \chi^* = \frac{1}{2mc} \left(i\hbar \nabla - \frac{e}{c}\mathbf{A} \right) \varphi^* \boldsymbol{\sigma},$$

and using the products involving two $\boldsymbol{\sigma}$ factors, which are rearranged with the help of (33.5) in the form

$$(\sigma \mathbf{a}) \boldsymbol{\sigma} = \mathbf{a} + i[\boldsymbol{\sigma} \times \mathbf{a}], \quad \boldsymbol{\sigma} (\sigma \mathbf{a}) = \mathbf{a} + i[\mathbf{a} \times \boldsymbol{\sigma}]. \quad (33.9)$$

The result is

$$\mathbf{j} = \frac{i\hbar}{2m} (\varphi \nabla \varphi^* - \varphi^* \nabla \varphi) - \frac{e}{mc} \mathbf{A} \varphi^* \varphi + \frac{\hbar}{2m} \text{rot}(\varphi^* \boldsymbol{\sigma} \varphi), \quad (33.10)$$

⁴This remarkable result was obtained by Dirac in 1928. The two-component wave function satisfying equation (33.7) had been introduced by Pauli (1927) even before Dirac discovered his equation.

in agreement with expression (115.4) (see III) from the non-relativistic theory.

Let us now find the second approximation, continuing the expansion to terms $\sim 1/c^2$ ⁵. We shall assume that only an electric field is present ($\mathbf{A} = 0$).

First of all, we note that with the inclusion of terms $\sim 1/c^2$ the density

$$\rho = |\varphi|^2 + |\chi|^2 = |\varphi|^2 + \frac{\hbar^2}{4m^2c^2} |\boldsymbol{\sigma} \cdot \nabla \varphi|^2.$$

This expression differs from the Schrödinger form. To find (in the second approximation) a wave equation analogous to the Schrödinger equation, we must introduce in place of φ a different (two-component) function φ_{Sch} , for which the time-independent integral $\int |\varphi_{\text{Sch}}|^2 d^3x$ has the same form as in the Schrödinger equation.

To determine the required transformation we write the condition

$$\int \varphi_{\text{Sch}}^* \varphi_{\text{Sch}} d^3x = \int \left\{ \varphi^* \varphi + \frac{\hbar^2}{4m^2c^2} (\nabla \varphi^* \cdot \boldsymbol{\sigma})(\boldsymbol{\sigma} \cdot \nabla \varphi) \right\} d^3x$$

and integrate by parts:

$$\int (\nabla \varphi^* \cdot \boldsymbol{\sigma})(\boldsymbol{\sigma} \cdot \nabla) \varphi d^3x = - \int \varphi^* (\boldsymbol{\sigma} \cdot \nabla)(\boldsymbol{\sigma} \cdot \nabla) \varphi d^3x = - \int \varphi^* \Delta \varphi d^3x$$

(or the same with φ and φ^* interchanged). Thus

$$\int \varphi_{\text{Sch}}^* \varphi_{\text{Sch}} d^3x = \int \left\{ \varphi^* \varphi - \frac{\hbar^2}{8m^2c^2} (\varphi^* \Delta \varphi + \varphi \Delta \varphi^*) \right\} d^3x,$$

from which one sees that

$$\varphi_{\text{Sch}} = \left(1 + \frac{\hat{\mathbf{p}}^2}{8m^2c^2} \right) \varphi, \quad \varphi = \left(1 - \frac{\hat{\mathbf{p}}^2}{8m^2c^2} \right) \varphi_{\text{Sch}}. \quad (33.11)$$

For simplicity let us assume a stationary state, i.e. replace the operator $i\hbar\partial/\partial t$ by the energy ε (with the rest energy subtracted). In the next (after (33.4)) approximation we have from (33.3):

$$\chi = \frac{1}{2mc} \left(1 - \frac{\varepsilon - e\Phi}{2mc^2} \right) (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}) \varphi.$$

This expression is to be substituted into (33.2), after which we replace φ by φ_{Sch} according to (33.11), dropping throughout all terms of order higher than $1/c^2$. A straightforward computation yields the

equation $\varepsilon \varphi_{\text{Sch}} = \hat{H} \varphi_{\text{Sch}}$, with the Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + e\Phi - \frac{\hat{\mathbf{p}}^4}{8m^3c^2} + \frac{e}{4m^2c^2} (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}) \Phi (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}) - \frac{1}{2} (\hat{\mathbf{p}}^2 \Phi + \Phi \hat{\mathbf{p}}^2). \quad (33.12)$$

The expression in braces is rearranged using the formulae

$$(\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}) \Phi (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}) + (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \Phi) (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}) = \Phi \hat{\mathbf{p}}^2 + i\hbar (\boldsymbol{\sigma} \cdot \mathbf{E}) (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}),$$

⁵Here we follow the method of V. B. Berestetskii and L. D. Landau (1949).

$$\hat{\mathbf{p}}^2\Phi - \Phi\hat{\mathbf{p}}^2 = -\hbar^2\Delta\Phi + 2i\hbar\mathbf{E} \cdot \hat{\mathbf{p}},$$

where $\mathbf{E} = -\nabla\Phi$ is the electric field. The final expression for the Hamiltonian is:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + e\Phi - \frac{\hat{\mathbf{p}}^4}{8m^3c^2} - \frac{e\hbar}{4m^2c^2}\boldsymbol{\sigma} \cdot [\mathbf{E} \times \hat{\mathbf{p}}] - \frac{e\hbar^2}{8m^2c^2}\text{div } \mathbf{E}. \quad (33.12)$$

The last three terms are the sought-for corrections of order $1/c^2$. The first of them is a consequence of the relativistic dependence of kinetic energy on momentum (the expansion of $\sqrt{\hat{\mathbf{p}}^2 + m^2c^2} - mc^2$). The second, which may be called the *spin-orbit interaction* energy, represents the interaction of the magnetic moment of a moving charge with the electric field⁶. The last term differs from zero only at those points where the field-producing charges are located; for the Coulomb field of a point charge Ze : $\Delta\Phi = -4\pi Ze\delta(\mathbf{r})$ (C. G. Darwin, 1928).

If the electric field is centrally symmetric,

$$\mathbf{E} = -\frac{\mathbf{r}}{r} \frac{d\Phi}{dr}$$

and the spin-orbit interaction operator can be written as

$$\frac{e\hbar}{4m^2c^2r}\boldsymbol{\sigma} \cdot [\mathbf{r} \times \hat{\mathbf{p}}] \frac{d\Phi}{dr} = \frac{\hbar^2}{2m^2c^2r} \frac{dU}{dr} \hat{\mathbf{I}} \cdot \hat{\mathbf{s}}. \quad (33.13)$$

Here $\hat{\mathbf{I}}$ is the orbital angular momentum operator, $\hat{\mathbf{s}} = \frac{1}{2}\boldsymbol{\sigma}$ is the electron spin operator, and $U = e\Phi$ is the potential energy of the electron in the field.

§ 34. Fine structure of hydrogen atom levels

Let us determine the relativistic corrections to the energy levels of the hydrogen atom—an electron in the Coulomb field of a stationary nucleus⁷. The velocity of the electron in the hydrogen atom $v/c \sim \alpha \ll 1$. Therefore the desired corrections can be calculated by perturbation theory—as an average over the unperturbed state (i.e. the non-relativistic wave function) of the relativistic terms in the approximate Hamiltonian (33.12). For somewhat greater generality we set the nuclear charge equal to Ze , assuming, however, that $Z\alpha \ll 1$.

The field of the nucleus $\mathbf{E} = Ze\mathbf{r}/r^3$, and its potential satisfies the equation $\Delta\Phi = -4\pi Ze\delta(\mathbf{r})$. Substituting this into (33.12) (the last three terms), and taking into account the negativity of the electron's charge, we obtain the perturbation operator:

$$\hat{V} = -\frac{\hat{\mathbf{p}}^4}{8m^3} + \frac{Z\alpha}{2r^3m^2}\hat{\mathbf{I}} \cdot \hat{\mathbf{s}} + \frac{Z\alpha\pi}{2m^2}\delta(\mathbf{r}). \quad (34.1)$$

⁶Introducing the magnetic moment (33.8) and the velocity $\mathbf{v} = \mathbf{p}/m$, one can write this energy as $-\frac{1}{2}\boldsymbol{\mu} \cdot [\mathbf{E} \times \mathbf{v}]$. At first sight this result may seem surprising, since on passing to the rest frame of the particle the magnetic field $\mathbf{H} = \frac{1}{c}[\mathbf{E} \times \mathbf{v}]$ arises, relative to which the magnetic moment should carry energy $-\boldsymbol{\mu} \cdot \mathbf{H}$. In reality the factor $1/2$ ("Thomas precession", L. Thomas, 1926) is connected with the general requirements of relativistic invariance in conjunction with the specific properties of the electron as a "spinor" particle with its associated value of the gyromagnetic ratio (see § 41).

⁷The effect of nuclear motion on the values of these corrections is of higher-order smallness, which does not interest us here.

Since according to the non-relativistic Schrödinger equation

$$\hat{\mathbf{p}}^2 \psi = 2m \left(\varepsilon_0 + \frac{Z\alpha}{r} \right) \psi$$

($\varepsilon_0 = -mZ^2\alpha^2/2n^2$ is the unperturbed level, n the principal quantum number), the mean value

$$\overline{\mathbf{p}^4} = 4m^2 \overline{\left(\varepsilon_0 + \frac{Z\alpha}{r} \right)^2}.$$

This quantity, as well as the mean value of the second term in (34.1), is computed using the formulae (see III, § 36)

$$\overline{r^{-1}} = \frac{m\alpha Z}{n^2}, \quad \overline{r^{-2}} = \frac{(m\alpha Z)^2}{n^3(l + \frac{1}{2})}, \quad \overline{r^{-3}} = \frac{(m\alpha Z)^3}{n^3 l(l + \frac{1}{2})(l + 1)} \quad (34.2)$$

(the last formula applies to $l \neq 0$); the eigenvalue

$$\mathbf{l} \cdot \mathbf{s} = \begin{cases} \frac{1}{2}[j(j+1) - l(l+1) - \frac{3}{4}], & l \neq 0, \\ 0, & l = 0. \end{cases}$$

Finally, the averaging of the third term is performed using the formula

$$|\psi(0)|^2 = \begin{cases} \frac{1}{\pi} \left(\frac{Z\alpha m}{n} \right)^3, & l = 0, \\ 0, & l \neq 0. \end{cases} \quad (34.3)$$

The result of a simple calculation using the formulae written above can be presented in all cases (for all j and l) in the form

$$\Delta\varepsilon = -\frac{m(Z\alpha)^4}{2n^3} \left(\frac{1}{j + \frac{1}{2}} - \frac{3}{4n} \right). \quad (34.4)$$

Formula (34.4) gives the desired relativistic correction to the energy of the hydrogen levels—the fine-structure energy⁸. Recall that in the non-relativistic theory degeneracy occurs both with respect to the direction of spin and the Coulomb degeneracy with respect to l . The fine structure (spin–orbit interaction) removes this degeneracy, but not completely: there remain doubly degenerate levels with the same n, j , but different $l = j \pm \frac{1}{2}$ (non-degenerate turn out to be only the levels with the largest possible $j = j_{\max} = l_{\max} + \frac{1}{2} = n - \frac{1}{2}$ for a given n). Thus the sequence of hydrogen levels with fine structure taken into account is as follows:

$$\begin{aligned} &1s_{1/2}; \\ &2s_{1/2}, \quad 2p_{1/2}, \quad 2p_{3/2}; \\ &3s_{1/2}, \quad 3p_{1/2}, \quad 3p_{3/2}, \quad 3d_{3/2}, \quad 3d_{5/2}. \end{aligned}$$

⁸This formula (and the more exact formula (36.10)) was obtained by A. Sommerfeld from the old Bohr theory even before the creation of quantum mechanics.

A level with principal quantum number n splits into n components of fine structure.

Recall that in non-relativistic mechanics the “accidental” degeneracy of energy levels in the Coulomb field is connected with the existence of a conservation law specific to this field: the quantity $\mathbf{A}$ is conserved, whose operator is

$$\hat{\mathbf{A}} = \frac{\mathbf{r}}{r} + \frac{1}{2m\alpha} \{[\hat{\mathbf{l}}\hat{\mathbf{p}}] - [\hat{\mathbf{p}}\hat{\mathbf{l}}]\}$$

(see III, (36.30)). A specific conservation law is connected with, and persists in, the relativistic case as a twofold degeneracy: the Hamiltonian of the Dirac equation $\hat{H} = \alpha\hat{\mathbf{p}} + \beta m - e^2/r$ commutes with the operator

$$\hat{\Gamma} = \frac{\mathbf{r}}{r} \cdot \boldsymbol{\Sigma} + \frac{i}{m\alpha} \beta (\boldsymbol{\Sigma} \cdot \hat{\mathbf{l}} + 1) \gamma^5 (\hat{H} - m\beta)$$

(*M. H. Johnson, B. A. Lippmann, 1950*). In the non-relativistic limit this operator $\hat{\Gamma} \rightarrow \boldsymbol{\Sigma}\hat{\mathbf{A}}$.

We shall see later (§ 123) that this remaining degeneracy is lifted by so-called radiative corrections (the *Lamb shift*), not accounted for by the Dirac equation for the single-electron problem.

Looking ahead, let us note here that the order of magnitude of these corrections is $\sim mZ^4\alpha^5 \ln(1/\alpha)$. The second-order correction due to the spin-orbit interaction was $\sim m(Z\alpha)^6$, so that its ratio to the radiative corrections is $\sim Z^2\alpha/\ln(1/\alpha)$. For hydrogen ($Z = 1$) this ratio is quite small, and therefore the problem of an exact solution of the Dirac equation in this case is of no practical interest. This problem may, however, have meaning for the energy levels of an electron in the field of a nucleus with large Z (see § 36).

§ 35. Motion in a centrally symmetric field

Consider the motion of an electron in a centrally symmetric electric field. Since angular momentum and parity are conserved in motion in a central field (relative to the centre of the field, chosen as the coordinate origin), everything said in § 24 about the angular dependence of wave functions of spherical waves of free particles applies to this motion as well. Only the radial functions change. Accordingly we seek the wave function of stationary states (in the standard representation) in the form

$$\varphi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \begin{pmatrix} f(r) \Omega_{jlm} \\ (-1)^{(1+l-l')/2} g(r) \Omega_{jl'm} \end{pmatrix}, \quad (35.1)$$

where $l = j \pm \frac{1}{2}$, $l' = 2j - l$, and the factor -1 is introduced to simplify the subsequent formulae.

The Dirac equation in the standard representation gives the following system of equations for φ and χ :

$$(\varepsilon - m - U)\varphi = \sigma\hat{\mathbf{p}}\chi, \quad (\varepsilon + m - U)\chi = \sigma\hat{\mathbf{p}}\varphi, \quad (35.2)$$

where $U(r) = e\Phi(r)$ is the potential energy of the electron in the field. The calculation of the result of substituting the expressions (35.1) into these equations reduces to computing the right-hand sides.

Expressing the spherical spinor $\Omega_{jl'm}$ through Ω_{jlm} according to

$$\Omega_{jl'm} = i^{l-l'} \left(\frac{\boldsymbol{\sigma} \cdot \mathbf{r}}{r} \right) \Omega_{jlm}$$

(see (24.8)), we write:

$$(\boldsymbol{\sigma} \hat{\mathbf{p}}) \chi = -i(\boldsymbol{\sigma} \nabla)(\boldsymbol{\sigma} \mathbf{r}) \frac{g}{r} \Omega_{jlm}.$$

Transforming now the product $(\boldsymbol{\sigma} \hat{\mathbf{p}})(\boldsymbol{\sigma} \mathbf{r})$ with the aid of formula (33.5), after expanding the vector operations we find

$$\begin{aligned} (\boldsymbol{\sigma} \hat{\mathbf{p}}) \chi &= -i\{\hat{p}_r + i\boldsymbol{\sigma}[\hat{\mathbf{p}} \hat{\mathbf{r}}]\} \frac{g}{r} \Omega_{jlm} = \\ &= \{-\operatorname{div} \mathbf{r} - (\mathbf{r} \nabla) - \boldsymbol{\sigma}[\mathbf{r} \hat{\mathbf{r}}]\} \frac{g}{r} \Omega_{jlm} = -\left\{g' + \frac{2}{r}g + \frac{g}{r} \boldsymbol{\sigma} \cdot \hat{\mathbf{I}}\right\} \Omega_{jlm}, \end{aligned}$$

where $\hat{\mathbf{I}} = [\mathbf{r} \hat{\mathbf{p}}]$ is the orbital angular momentum operator; the prime denotes differentiation with respect to r . The eigenvalues of the product $\boldsymbol{\sigma} \cdot \hat{\mathbf{I}} = 2\hat{\mathbf{l}} \cdot \hat{\mathbf{s}}$ are

$$2\hat{\mathbf{l}} \cdot \hat{\mathbf{s}} = \mathbf{j}^2 - \mathbf{l}^2 - \mathbf{s}^2 = j(j+1) - l(l+1) - \frac{3}{4} = \begin{cases} j - \frac{1}{2}, & l = j - \frac{1}{2}, \\ -j - \frac{3}{2}, & l = j + \frac{1}{2}. \end{cases}$$

For uniformity of notation in both cases ($l = j \pm \frac{1}{2}$) it is convenient to introduce the notation

$$\kappa = \begin{cases} -(j + \frac{1}{2}) = -(l + 1), & j = l + \frac{1}{2}, \\ +(j + \frac{1}{2}) = l, & j = l - \frac{1}{2}. \end{cases} \quad (35.3)$$

The number κ takes all integer values excluding zero (positive numbers correspond to $j = l - \frac{1}{2}$, and negative ones to $j = l + \frac{1}{2}$). Then $\hat{\mathbf{l}} \cdot \boldsymbol{\sigma} = -(1 + \kappa)$, so that

$$(\boldsymbol{\sigma} \hat{\mathbf{p}}) \chi = -\left(g' + \frac{1 - \kappa}{r}g\right) \Omega_{jlm}.$$

On substituting this expression into the first of equations (35.2), the spherical spinor Ω_{jlm} cancels on both sides of the equation. Proceeding analogously with the second equation, we obtain the following system for the radial functions:

$$\begin{aligned} f' + \frac{1 + \kappa}{r}f - (\varepsilon + m - U)g &= 0, \\ g' + \frac{1 - \kappa}{r}g + (\varepsilon - m - U)f &= 0, \end{aligned} \quad (35.4)$$

or

$$\begin{aligned} (fr)' + \frac{\kappa}{r}(fr) - (\varepsilon + m - U)gr &= 0, \\ (gr)' - \frac{\kappa}{r}(gr) + (\varepsilon - m - U)fr &= 0. \end{aligned} \quad (35.5)$$

Let us investigate the behaviour of f and g at small distances, assuming that the field $U(r)$ grows as $r \rightarrow 0$ faster than $1/r$. Then in the region of small r equations (35.4) take the form

$$f' + Ug = 0, \quad g' - Uf = 0.$$

They have real solutions of the form

$$f = \text{const} \cdot \sin\left(\int U dr + \delta\right), \quad g = \text{const} \cdot \cos\left(\int U dr + \delta\right), \quad (35.6)$$

where δ is an arbitrary constant. These functions oscillate as $r \rightarrow 0$, without tending to any limit. It is easy to see that this situation corresponds in the non-relativistic theory to “falling” of the particle to the centre.

First of all we note that the region of small distances imposes in this case no restrictions on the choice of solution: the condition at $r = 0$ for the oscillating function is absent, and the choice of the constant δ remains arbitrary (the correct behaviour of the wave function at large r can be achieved for any ε by an appropriate choice of δ). This indeterminacy can be removed by considering the singular (at $r = 0$) potential as the limit $r_0 \rightarrow 0$ of the potential “cut off” at some small r_0 (i.e. equal to $U(r)$ for $r > r_0$ and $U(r_0)$ for $r < r_0$). With a finite r_0 one obtains, of course, a definite system of energy levels. However, the energy of the ground state tends to $-\infty$ as $r_0 \rightarrow 0$.

In the non-relativistic theory this is precisely the “falling” to the centre, since a particle at a deep level is localised in a small region around $r = 0$. In the relativistic theory, however, such a situation is altogether inadmissible, since it signifies the instability of the system with respect to the spontaneous creation of electron–positron pairs. Indeed, if in vacuum the energy needed to create such a pair exceeds $2m$, in the field a smaller energy already suffices. If there exists a bound state of the electron with energy $\varepsilon < m$, pair creation is possible at the cost of only the energy $\varepsilon + m < 2m$, with a free positron created and the electron in the bound state. If the energy of the bound state $\varepsilon < -m$, such a field can create positrons (with energy $-\varepsilon > m$) spontaneously, without any input of energy from an external source. In the field being considered, as $r_0 \rightarrow 0$ there is an infinite number of such “anomalous” levels with $\varepsilon < -m$. Therefore a potential $\Phi(r)$ growing as $r \rightarrow 0$ faster than $1/r$ cannot at all be considered in the Dirac theory. Let us stress that this applies to potentials of either sign. The “falling” occurs, of course, only in the case of attraction, since the sign of $U = e\Phi$ depends also on the sign of the charge; in one case the electronic, and in the other the positronic levels behave anomalously; in the second case the field creates free electrons.

Let us consider further the behaviour of wave functions at large distances. If the field $U(r)$ falls off sufficiently rapidly at $r \rightarrow \infty$, then in determining the asymptotic form of the wave functions at large distances the field can be completely neglected in the equations. For $\varepsilon > m$, i.e. in the continuous-spectrum region, we return then to the free-motion equation, so that the asymptotic form of the wave functions (spherical waves) differs from that for a free particle only by the appearance of additional “phase shifts”, whose values are determined by the form of the field at close distances⁹. These shifts depend on the values of j and l , or, equivalently, on the number κ (and also, of course, on the energy ε). Denoting them by δ_κ and using the expression for the free spherical

⁹Cf. III, § 33. As in the non-relativistic theory, $U(r)$ must fall off faster than $1/r$. The case $U \sim 1/r$ will be considered separately in § 36.

wave (24.7), we can immediately write the desired asymptotic formula

$$\psi \approx \frac{2}{r} \frac{1}{\sqrt{2\varepsilon}} \left(\begin{array}{l} \sqrt{\varepsilon + m} \Omega_{jlm} \sin\left(pr - \frac{\pi l}{2} + \delta_\kappa\right) \\ -\sqrt{\varepsilon - m} \Omega_{jl'm} \sin\left(pr - \frac{\pi l'}{2} + \delta_\kappa\right) \end{array} \right), \quad (35.7)$$

or, taking into account the definition (35.1):

$$\left. \begin{array}{l} f \\ g \end{array} \right\} = \frac{\sqrt{2}}{r} \sqrt{\frac{\varepsilon \pm m}{\varepsilon}} \sin\left(pr - \frac{\pi l}{2} + \delta_\kappa\right), \quad (35.8)$$

where $p = \sqrt{\varepsilon^2 - m^2}$. The overall coefficient here corresponds to normalisation of the radial functions according to (24.5).

The wave functions of the discrete spectrum ($\varepsilon < m$) decay exponentially at $r \rightarrow \infty$ according to the law

$$f = -\sqrt{\frac{m + \varepsilon}{m - \varepsilon}} g = \frac{A_0}{r} \exp\left(-r\sqrt{m^2 - \varepsilon^2}\right), \quad (35.9)$$

where A_0 is a constant.

As in the non-relativistic theory, the phase shifts δ_κ (more precisely, the quantities $e^{2i\delta_\kappa} - 1$) determine the scattering amplitude in the given field (this will be discussed in more detail in § 37). We shall not investigate here the analytic properties of these quantities (cf. III, § 128). We note only that $e^{2i\delta_\kappa}$ as a function of the energy still has poles at the points corresponding to the bound-state energy levels of the particle. The residue of the function $e^{2i\delta_\kappa}$ at such a pole is related in a definite way to the coefficient in the asymptotic expression of the corresponding wave function of the discrete spectrum. Let us find this relation, which generalises the non-relativistic formula (128.17) (see III). The necessary calculations are entirely analogous to those performed in III, § 128.

Differentiate equations (35.5) with respect to the energy:

$$\begin{aligned} \left(\frac{\partial r f}{\partial \varepsilon}\right)' + \frac{\kappa}{r} \frac{\partial r f}{\partial \varepsilon} - (\varepsilon + m - U) \frac{\partial r g}{\partial \varepsilon} &= r g, \\ \left(\frac{\partial r g}{\partial \varepsilon}\right)' - \frac{\kappa}{r} \frac{\partial r g}{\partial \varepsilon} + (\varepsilon - m - U) \frac{\partial r f}{\partial \varepsilon} &= -r f. \end{aligned}$$

Multiply these two equations respectively by rg and $-rf$, and the two equations (35.5) respectively by $-rg$ and rf ; then add all four equations term by term. After all cancellations we get

$$\frac{\partial}{\partial r} \left[r^2 \left(g \frac{\partial f}{\partial \varepsilon} - f \frac{\partial g}{\partial \varepsilon} \right) \right] = r^2 (f^2 + g^2).$$

Integrate this relation over r :

$$r^2 \left(g \frac{\partial f}{\partial \varepsilon} - f \frac{\partial g}{\partial \varepsilon} \right) = \int_0^r (f^2 + g^2) r^2 dr,$$

and then pass to the limit $r \rightarrow \infty$. By the normalisation condition the integral on the right-hand side becomes unity. On the left-hand side we take into account that in the asymptotic region the functions f and g are related by

$$rg = \frac{(rf)'}{\varepsilon + m},$$

obtained from (35.5) by neglecting the terms with U and $1/r$. As a result we get

$$\frac{1}{\varepsilon + m} \left[(rf)' \frac{\partial rf}{\partial \varepsilon} - rf \left(\frac{\partial rf}{\partial \varepsilon} \right)' \right] = 1. \quad (35.10)$$

This formula differs from the analogous non-relativistic formula (for the function χ) only by the coefficient ($\varepsilon + m$ instead of $2m$). There is therefore no need to repeat all the subsequent calculations, and we immediately give the final formula, valid near the point $\varepsilon = \varepsilon_0$ (ε_0 being the energy level):

$$e^{2i\delta_\kappa} = (-1)^l \frac{2A_0^2}{\varepsilon - \varepsilon_0} \sqrt{\frac{m - \varepsilon_0}{m + \varepsilon_0}}, \quad (35.11)$$

where A_0 is the coefficient in the asymptotic expression (35.9).

Problem

Find the limiting form of the wave function at small r in the field $U \sim r^{-s}$, $s < 1$.

Solution. For a free particle we have at small r : $f \sim r^l$, $g \sim r^{l'}$, so that for $l < l'$: $f \gg g$, and for $l > l'$: $f \ll g$. We make the assumption (justified by the result) that such a relation is preserved also in the field being considered. For $l < l'$ (i.e. $l = j - \frac{1}{2}$, $\kappa = -l - 1$) the g term in the first of equations (35.4) can be dropped, so that $f \sim r^l$ as before. From the second equation we then have $g \sim rfU$, i.e. $g \sim r^{l+1-s}$. The case $l > l'$ is treated analogously. As a result we find:

$$\begin{aligned} \text{for } l < l': \quad f &\sim r^l, & g &\sim r^{l'-s}; \\ \text{for } l > l': \quad f &\sim r^{l-s}, & g &\sim r^{l'}. \end{aligned}$$

§ 36. Motion in a Coulomb field

The study of the properties of motion in the most important case of the Coulomb field begins with an investigation of the behaviour of wave functions at small distances. Let us speak for definiteness of the field of attraction: $U = -Z\alpha/r$ ¹⁰. For small r in equations (35.5) one may drop the terms $\varepsilon \pm m$; then

$$(fr)' + \frac{\kappa}{r} fr - \frac{Z\alpha}{r} gr = 0,$$

$$(gr)' - \frac{\kappa}{r} gr + \frac{Z\alpha}{r} fr = 0.$$

¹⁰In ordinary units $U = -Ze^2/r$. In the transition to relativistic units e^2 is replaced by the dimensionless α .

The functions fr and gr enter into each of these equations on an equal footing. We therefore seek both in the form of equal powers of r : $fr = ar^\gamma$, $gr = br^\gamma$. Substitution into the equations gives

$$a(\gamma + \kappa) - bZ\alpha = 0, \quad aZ\alpha + b(\gamma - \kappa) = 0,$$

whence

$$\gamma^2 = \kappa^2 - (Z\alpha)^2. \quad (36.1)$$

Suppose $(Z\alpha)^2 < \kappa^2$. Then γ is real, and of the two values the positive one must be chosen: the corresponding solution either does not diverge at $r = 0$, or diverges less rapidly. Such a choice can be justified by considering the potential, cut off at some small r_0 , with a subsequent passage to the limit $r_0 \rightarrow 0$ (cf. analogous arguments in III, § 35). Thus,

$$f = \frac{Z\alpha}{\gamma + \kappa} g = \text{const} \cdot r^{-1+\gamma}. \quad (36.2)$$

Although the wave function may diverge at $r = 0$ to infinity (if $\gamma < 1$), the integral of $|\psi|^2$ remains, of course, convergent. If $(Z\alpha)^2 > \kappa^2$, both values of γ from (36.1) are imaginary. The corresponding solutions oscillate as $r \rightarrow 0$ (like $r^{-1} \cos(|\gamma| \ln r)$), which, as was already explained above, is an inadmissible situation in the relativistic theory of “falling” to the centre. Since $\kappa^2 \geq 1$, this means that the pure Coulomb field can be considered in the Dirac theory only when $Z\alpha < 1$, i.e. $Z < 137$.

Let us dwell on a qualitative description of the situation that arises for $Z > 137$. Again, to avoid the indeterminacy in the boundary condition at $r = 0$, one should consider the potential cut off at some small r_0 (*I. Ya. Pomeranchuk, Ya. A. Smorodinsky, 1945*). This has not only a formal, but also a direct physical meaning. A charge $Z > 137$ can in fact be concentrated only in a “superheavy” nucleus of some finite radius. Let us therefore consider how the disposition of levels changes with increasing Z for a given r_0 .

In the “uncut” Coulomb field the energy ε_1 of the lowest level is unity at $Z\alpha = 1$ and the curve of the dependence $\varepsilon_1(Z)$ terminates at a minimum (see (36.10)). In the “cut” field, at a given $r_0 \neq 0$, the level ε_1 passes through zero only at some value $Z\alpha > 1$. But the value $\varepsilon_1 = 0$ is in no way physically distinguished, and it is not formally distinguished either—the curve of the dependence $\varepsilon_1(Z)$ does not break off here. With further increase of Z the levels continue to descend, and at some “critical” value $Z = Z_c(r_0)$ the energy ε_1 reaches the boundary $(-m)$ of the lower continuum of levels. As was explained in the preceding section, this means that the energy required for the creation of a free positron vanishes. Therefore the critical value Z_c is the maximum charge that a “bare” nucleus can possess at the given r_0 .

For $Z > Z_c$ the level $\varepsilon_1 < -m$ and it becomes energetically favourable to create two electron–positron pairs. The positrons escape to infinity, carrying away the kinetic energy $2(|\varepsilon_1| - m)$, and two electrons fill the level ε_1 . The result is an “ion” with a filled K -shell and charge $Z_{\text{eff}} = Z - 2$ (*S. S. Gershtein, Ya. B. Zel'dovich, 1969*). This system is stable for $Z > Z_c$, up to values of Z at which the boundary $-m$ is reached by the next level¹¹.

¹¹Thus, if the nuclear charge is distributed uniformly in a sphere of radius $r_0 = 1.2 \cdot 10^{-13}$ cm, the critical

Let us note finally that even in the case of a point charge the behaviour of the potential at small distances is distorted by radiative corrections. Their account leads, however, only to corrections $\sim \alpha$ to the value $Z_c \alpha$.

Let us now turn to the exact solution of the wave equation (C. G. Darwin, 1928; W. Gordon, 1928).

Discrete spectrum ($\varepsilon < m$). We seek the functions f and g in the form

$$\begin{aligned} f &= \sqrt{m + \varepsilon} e^{-\rho/2} \rho^{\gamma-1} (Q_1 + Q_2), \\ g &= -\sqrt{m - \varepsilon} e^{-\rho/2} \rho^{\gamma-1} (Q_1 - Q_2), \end{aligned} \quad (36.3)$$

where we have introduced the notation

$$\rho = 2\lambda r, \quad \lambda = \sqrt{m^2 - \varepsilon^2}, \quad \gamma = \sqrt{\kappa^2 - Z^2 \alpha^2}. \quad (36.4)$$

This form is of course natural in view of the already known behaviour of the functions at $\rho \rightarrow 0$ (36.2) and their exponential decay ($\sim e^{-\rho/2}$) at $\rho \rightarrow \infty$. Since at $\rho \rightarrow \infty$ the first relation (35.9) should hold and in the case of the Coulomb field one should expect $Q_1 \gg Q_2$.

Substituting (36.3) into (35.4), we obtain the equations

$$\rho(Q_1 + Q_2)' + (\gamma + \kappa)(Q_1 + Q_2) - \rho Q_2 + Z\alpha \sqrt{\frac{m - \varepsilon}{m + \varepsilon}} (Q_1 - Q_2) = 0,$$

$$\rho(Q_1 - Q_2)' + (\gamma - \kappa)(Q_1 - Q_2) + \rho Q_2 - Z\alpha \sqrt{\frac{m + \varepsilon}{m - \varepsilon}} (Q_1 + Q_2) = 0$$

(the prime denotes differentiation with respect to ρ). Their sum and difference give

$$\rho Q_1' + \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda} \right) Q_1 + \left(\kappa - \frac{Z\alpha m}{\lambda} \right) Q_2 = 0, \quad (36.5)$$

$$\rho Q_2' + \left(\gamma + \frac{Z\alpha\varepsilon}{\lambda} - \rho \right) Q_2 + \left(\kappa + \frac{Z\alpha m}{\lambda} \right) Q_1 = 0,$$

or, after eliminating Q_1 or Q_2 ,

$$\rho Q_1'' + (2\gamma + 1 - \rho) Q_1' - \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda} \right) Q_1 = 0, \quad (36.6)$$

$$\rho Q_2'' + (2\gamma + 1 - \rho) Q_2' - \left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda} \right) Q_2 = 0$$

(one should take into account that $\gamma^2 - (Z\alpha\varepsilon/\lambda)^2 = \kappa^2 - (Z\alpha m/\lambda)^2$). The solutions of these equations, finite at $\rho = 0$:

$$\begin{aligned} Q_1 &= A F\left(\gamma - \frac{Z\alpha\varepsilon}{\lambda}, 2\gamma + 1, \rho\right), \\ Q_2 &= B F\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}, 2\gamma + 1, \rho\right), \end{aligned} \quad (36.6)$$

value $Z_c = 170$, and the next level reaches the boundary $-m$ at $Z = 185$ (V. S. Popov, 1970). For a more detailed exposition of the quantitative theory see the review article by Ya. B. Zel'dovich and V. S. Popov (UFN, 1971, Vol. 105, p. 403).

where $F(\alpha, \beta, z)$ is the confluent hypergeometric function. Setting $\rho = 0$ in either of equations (36.5), we find the relation between A and B :

$$B = -\frac{\gamma - Z\alpha\varepsilon/\lambda}{\kappa - Z\alpha m/\lambda} A. \quad (36.7)$$

Both hypergeometric functions in (36.6) must reduce to polynomials (otherwise they would grow as e^ρ at $\rho \rightarrow \infty$, and with them the whole wave function would grow as $e^{\rho/2}$). A function $F(\alpha, \beta, z)$ reduces to a polynomial if its first parameter equals a non-positive integer. Denoting

$$\gamma - Z\alpha\varepsilon/\lambda = -n_r. \quad (36.8)$$

If $n_r = 1, 2, \dots$, both hypergeometric functions reduce to polynomials. If, however, $n_r = 0$, only one of them reduces to a polynomial. But then the equality $n_r = 0$ means that $\gamma = Z\alpha\varepsilon/\lambda$, and then, as is easily verified, $Z\alpha m/\lambda = |\kappa|$. If $\kappa < 0$, the coefficient B (36.7) vanishes, so that $Q_2 = 0$, and the required condition is not violated. If, however, $\kappa > 0$, then $B = -A$, and Q_2 remains a divergent function. Thus the admissible values of the quantum number n_r are:

$$n_r = \begin{cases} 0, 1, 2, \dots & \text{for } \kappa < 0; \\ 1, 2, 3, \dots & \text{for } \kappa > 0. \end{cases} \quad (36.9)$$

From the definition (36.8) we now find the following expression for the discrete energy levels:

$$\frac{\varepsilon}{m} = \left[1 + \frac{(Z\alpha)^2}{(\sqrt{\kappa^2 - (Z\alpha)^2} + n_r)^2} \right]^{-1/2}. \quad (36.10)$$

In particular, the energy of the ground level $1s_{1/2}$ ($|\kappa| = 1$, $n_r = 0$):

$$\varepsilon_1 = m\sqrt{1 - (Z\alpha)^2}.$$

For $Z\alpha \ll 1$ the first terms of the expansion of formula (36.10) give

$$\frac{\varepsilon}{m} - 1 = -\frac{(Z\alpha)^2}{2(|\kappa| + n_r)^2} - \frac{(Z\alpha)^2}{(|\kappa| + n_r)} \left[\frac{1}{|\kappa|} - \frac{3}{4(|\kappa| + n_r)} \right] \Bigg\}.$$

Denoting $n_r + |\kappa| = n$ ($n = 1, 2, \dots$) and noting that $|\kappa| = j + \frac{1}{2}$, we recover formula (34.4), obtained earlier by perturbation theory. As was already noted at the end of § 34, the further terms of this expansion are meaningless, since they are inevitably exceeded by the radiative corrections. Formula (36.10), however, makes sense in its exact form when $Z\alpha \sim 1$.

Note that the twofold degeneracy discovered from the approximate formula (34.4) is preserved in the exact formula as well: since only $|\kappa|$ enters into it, levels with different l at the same n and j still coincide.

It remains to determine the overall normalisation coefficient A in the wave function. As always, the wave function of the discrete spectrum must be normalised by the condition $\int |\psi|^2 d^3x = 1$; for the functions f and g this means the condition

$$\int_0^\infty (f^2 + g^2) r^2 dr = 1.$$

The coefficient A is most easily found from the asymptotic behaviour of the functions at $r \rightarrow \infty$. Using the asymptotic formula

$$F(-n_r, 2\gamma + 1, \rho) \approx \frac{\Gamma(2\gamma + 1)}{\Gamma(n_r + 2\gamma + 1)} (-\rho)^{n_r}$$

(see III, (d.14)) we find

$$f \approx (-1)^{n_r} A \sqrt{m + \varepsilon} \frac{\Gamma(2\gamma + 1)}{\Gamma(n_r + 2\gamma + 1)} e^{-\lambda r} (2\lambda r)^{\gamma + n_r - 1}.$$

Comparing this formula with the expression (36.22), which will be found below, we determine A . Collecting then the formulae obtained, we write the final expressions for the normalised wave functions:

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = \frac{\pm (2\lambda)^{\gamma+1/2}}{\Gamma(2\gamma + 1)} \left[\frac{(m \pm \varepsilon) \Gamma(2\gamma + n_r + 1)}{4m (Z\alpha m/\lambda) (Z\alpha m/\lambda - \kappa) n_r!} \right]^{1/2} (2\lambda r)^{\gamma-1} e^{-\lambda r} \times \quad (36.11)$$

$$\times \left\{ \left(\frac{Z\alpha m}{\lambda} - \kappa \right) F(-n_r, 2\gamma + 1, 2\lambda r) \mp n_r F(1 - n_r, 2\gamma + 1, 2\lambda r) \right\}$$

(the upper signs refer to f , the lower to g).

Continuous spectrum ($\varepsilon > m$). There is no need to solve the wave equation anew for the continuous-spectrum states. The wave functions in this case are obtained from those of the discrete spectrum by the substitution¹²

$$\sqrt{m - \varepsilon} \rightarrow -i\sqrt{\varepsilon - m}, \quad \lambda \rightarrow -ip, \quad -n_r \rightarrow \gamma - i \frac{Z\alpha\varepsilon}{p} \quad (36.12)$$

(regarding the choice of sign in the analytic continuation of the root $\sqrt{m - \varepsilon}$, see III, § 128).

Carrying out the indicated substitution in (36.11), we express the functions f and g in the form

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = \frac{\sqrt{\varepsilon + m}}{i\sqrt{\varepsilon - m}} \cdot A' e^{ipr} (2pr)^{\gamma-1} \times$$

$$\times [e^{i\xi} F(\gamma - i\nu, 2\gamma + 1, -2ipr) \mp e^{-i\xi} F(\gamma + 1 - i\nu, 2\gamma + 1, -2ipr)],$$

where A' is a new normalisation constant and we have introduced the notation

$$\nu = \frac{Z\alpha\varepsilon}{p}, \quad e^{-2i\xi} = \frac{\gamma - i\nu}{\kappa - i\nu m/\varepsilon} \quad (36.13)$$

¹²Hereafter in this section p denotes $|\mathbf{p}| = \sqrt{\varepsilon^2 - m^2}$, and in the indices of the amplitude we write ε and $\mathbf{p}$ separately.

(the quantity ξ is real, since $\gamma^2 + (Z\alpha\varepsilon/p)^2 = \kappa^2 + (Z\alpha m/p)^2$).

According to the well-known formula

$$F(\alpha, \beta, z) = e^z F(\beta - \alpha, \beta, -z)$$

(see III, (d.10)) we have

$$\begin{aligned} F(\gamma + 1 - i\nu, 2\gamma + 1, -2ipr) &= e^{-2ipr} F(\gamma + i\nu, 2\gamma + 1, 2ipr) = \\ &= e^{-2ipr} F^*(\gamma - i\nu, 2\gamma + 1, -2ipr), \end{aligned}$$

therefore

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = 2iA' \sqrt{\varepsilon \pm m} (2pr)^{\gamma-1} \frac{\text{Im}}{\text{Re}} \left\{ e^{i(pr+\xi)} F(\gamma - i\nu, 2\gamma + 1, -2ipr) \right\}. \quad (36.14)$$

The normalisation coefficient A' is determined by comparing the asymptotic expression for this function with the general formula (35.7) for a normalised spherical wave. We write at once the resulting expression for the wave functions of the continuous spectrum (and then verify it)¹³:

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = 2^{3/2} \sqrt{\frac{m \pm \varepsilon}{\varepsilon}} \frac{e^{\pi\nu/2} |\Gamma(\gamma + 1 + i\nu)|}{\Gamma(2\gamma + 1)} \frac{(2pr)^\gamma}{r} \times \frac{\text{Im}}{\text{Re}} \left\{ e^{i(pr+\xi)} F(\gamma - i\nu, 2\gamma + 1, -2ipr) \right\}. \quad (36.15)$$

The asymptotic expression for this function is found using the formula III, (d.14), in which in the present case only the first term is essential (the second falls off with a higher power of $1/r$):

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = \frac{\sqrt{2}}{r} \sqrt{\frac{\varepsilon \pm m}{\varepsilon}} \sin \left(pr - \frac{\pi l}{2} + \delta_\kappa + \nu \ln 2pr - \frac{\pi l}{2} \right), \quad (36.16)$$

where

$$\delta_\kappa = \xi - \arg \Gamma(\gamma + 1 + i\nu) - \frac{\pi\gamma}{2} + \frac{\pi l}{2}, \quad (36.17)$$

or

$$e^{2i\delta_\kappa} = \frac{\kappa - i\nu m/\varepsilon}{\gamma - i\nu} \frac{\Gamma(\gamma + 1 - i\nu)}{\gamma + 1 + i\nu} e^{i\pi(l-\gamma)}. \quad (36.18)$$

Let us note for future reference the expression for the phases in the ultra-relativistic case ($\varepsilon \gg m, \nu \rightarrow Z\alpha$):

$$e^{2i\delta_\kappa} = \frac{\kappa}{\gamma - iZ\alpha} \frac{\Gamma(\gamma + 1 - iZ\alpha)}{\Gamma(\gamma + 1 + iZ\alpha)} e^{i\pi(l-\gamma)}. \quad (36.19)$$

Expression (36.16) differs from (35.8) only by a logarithmic term in the argument of the trigonometric function. As in the case of the Schrödinger equation, the slow fall-off

¹³The wave functions in a repulsive field are obtained from these by changing the sign before $Z\alpha$, i.e. by changing the sign of ν .

of the Coulomb potential leads to a distortion of the phase of the wave, which becomes a slowly varying function of r .

On analytic continuation into the region $\varepsilon < m$ expression (36.18) takes the form

$$e^{2i\delta_\kappa} = \frac{\kappa - Z\alpha m/\lambda}{\gamma - Z\alpha\varepsilon/\lambda} \frac{\Gamma(\gamma + 1 - Z\alpha\varepsilon/\lambda)}{\Gamma(\gamma + 1 + Z\alpha\varepsilon/\lambda)} e^{i\pi(l-\gamma)}. \quad (36.20)$$

It has poles at the points where $\gamma + 1 - Z\alpha\varepsilon/\lambda = 1 - n_r$, $n_r = 1, 2, \dots$ (poles of the Γ -function in the numerator), and also at the point $\gamma - Z\alpha\varepsilon/\lambda = -n_r = 0$ (if $\kappa < 0$); as was to be expected, these points coincide with the discrete energy levels.

Near one of the poles with $n_r \neq 0$ we have

$$e^{2i\delta_\kappa} \approx \frac{\left(\frac{Z\alpha m}{\lambda} - \kappa\right) e^{i\pi(l-\gamma)}}{n_r \Gamma(2\gamma + 1 + n_r)} \Gamma\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right).$$

The form of the Γ -function near its pole is found using the well-known formula $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$:

$$\begin{aligned} \Gamma\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right) &\approx \frac{\pi}{\Gamma(n_r) \sin \pi(\gamma + 1 - Z\alpha\varepsilon/\lambda)}, \\ \sin \pi\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right) &\approx \pi \cos \pi n_r \frac{d}{d\varepsilon} \left(\frac{Z\alpha\varepsilon}{\lambda}\right) (\varepsilon - \varepsilon_0) = \\ &= (-1)^{n_r} \frac{\pi Z\alpha m^2}{\lambda^3} (\varepsilon - \varepsilon_0) \end{aligned}$$

(ε_0 being the energy level). Thus¹⁴,

$$e^{2i\delta_\kappa} \approx (-1)^{l+n_r} \frac{e^{-i\pi\left(\frac{Z\alpha m}{\lambda} - \kappa\right)}}{n_r! \Gamma(2\gamma + 1 + n_r)} \frac{\lambda^3}{Z\alpha m^2} \frac{1}{\varepsilon - \varepsilon_0}. \quad (36.21)$$

At the end of the preceding section formula (35.11) was obtained, relating the residue of $e^{2i\delta_\kappa}$ at its pole to the coefficient in the asymptotic expression of the wave function of the corresponding bound state. In the case of the Coulomb field this formula must be somewhat modified in connection with the fact that instead of the constant phase shift δ_κ (as in (35.7) and (36.16)) one has the sum $\delta_\kappa + \nu \ln(2pr)$. On the left-hand side of (35.11) one should therefore write not $e^{2i\delta_\kappa}$, but

$$\exp[2i\delta_\kappa + 2i\nu \ln(2pr)] \rightarrow e^{2i\delta_\kappa} (2i\lambda r)^{2(n_r+\gamma)}.$$

Using (36.21) and determining from (35.11) the coefficient A_0 (which is now a power function of r), we find the asymptotic normalised wave function of the discrete spectrum:

$$f = \left[\frac{(Z\alpha m/\lambda - \kappa)(m + \varepsilon) \lambda^2}{2n_r! Z\alpha m^2 \Gamma(2\gamma + 1 + n_r)} \right]^{1/2} \frac{e^{-\lambda r}}{r} (2\lambda r)^{n_r+\gamma}. \quad (36.22)$$

This formula has already been used to determine the coefficient A in (36.11).

¹⁴It is easy to verify that this formula remains valid for $n_r = 0$ as well.

§ 37. Scattering in a centrally symmetric field

Let us write the asymptotic expression for the wave function of a particle scattered in the field of a stationary force centre, in the form¹⁵

$$\psi = u_{\varepsilon \mathbf{p}} e^{i p z} + u'_{\varepsilon \mathbf{p}'} \frac{e^{i p r}}{r}. \quad (37.1)$$

Here $u_{\varepsilon \mathbf{p}}$ is the bispinor amplitude of the incident plane wave. The bispinor $u'_{\varepsilon \mathbf{p}'}$ is a function of the scattering direction $\mathbf{n}'$, and for each given value of $\mathbf{n}'$ coincides in form (but, of course, not in normalisation!) with the bispinor amplitude of a plane wave propagating in the direction $\mathbf{n}'$.

We saw in § 24 that the bispinor amplitude of a plane wave is completely determined by specifying a two-component quantity—a 3-spinor w , representing the non-relativistic wave function in the rest frame of the particle. Through this spinor is expressed the flux density: it is proportional to $w^* w$ (with a coefficient of proportionality depending only on the energy ε) and, consequently, equal for the incident and scattered

particles. The scattering cross section is therefore $d\sigma = (w'^* w' / w^* w) d\omega$, or, if (as in § 24) we normalise the incident wave by the condition $w^\dagger w = 1$,

$$d\sigma = w'^* w' d\omega.$$

We introduce the scattering operator $\hat{f}$ by the definition

$$w' = \hat{f} w. \quad (37.2)$$

By virtue of the two-component character of w , w' , the operator defined in this way is analogous to the operator scattering amplitude that appears in the non-relativistic theory of scattering with spin taken into account (see III, § 140). Therefore one can transfer directly the formulae obtained there, expressing the operator through the phase shifts of the wave functions in the scattering field. One need only re-express the notation of these phases, expressing the shifts δ_l^+ and δ_l^- introduced in III, § 140 through the shift δ_κ that appears in the relativistic formula (35.7). Recall that the phases δ_l^+ and δ_l^- referred to the states with orbital angular momentum l and total angular momentum $j = l + \frac{1}{2}$ and $j = l - \frac{1}{2}$. According to the definition (35.3), $\kappa = -l - 1$ for $j = l + \frac{1}{2}$ and $\kappa = l$ for $j = l - \frac{1}{2}$. We must therefore re-label

$$\delta_l^+ \rightarrow \delta_{-(l+1)}, \quad \delta_l^- \rightarrow \delta_l$$

(and bear in mind that the index on δ now refers to the value of κ !). Thus we obtain the following formulae:

$$\hat{f} = A + B \mathbf{v} \cdot \boldsymbol{\sigma}, \quad (37.3)$$

$$A = \frac{1}{2ip} \sum_{l=0}^{\infty} [(l+1)(e^{2i\delta_{-l-1}} - 1) + l(e^{2i\delta_l} - 1)] P_l(\cos \theta), \quad (37.4)$$

$$B = \frac{1}{2p} \sum_{l=1}^{\infty} (e^{2i\delta_{-l-1}} - e^{2i\delta_l}) P_l^1(\cos \theta), \quad (37.5)$$

¹⁵In §§ 37, 38 p denotes $|\mathbf{p}|$, and in the indices of the amplitude we write ε and $\mathbf{p}$ separately.

where $\mathbf{v}$ is the unit vector in the direction $[\mathbf{n} \mathbf{n}']$.

Since w is the spinor wave function in the rest frame, the polarisation properties of the scattering are described with the aid of $\hat{f}$ by the same formulae as in III, § 140.

In the case of the Coulomb field it turns out to be possible to express both functions $A(\theta)$ and $B(\theta)$ through a single one. Let us indicate briefly the course of the corresponding calculations¹⁶.

For the Coulomb field the phases δ_κ are given by formula (36.18), which we present in the form

$$e^{2i\delta_\kappa} = - \left(\kappa - i \frac{Ze^2m}{p} \right) \frac{\kappa}{|\kappa|} C_\kappa, \quad (37.6)$$

$$C_\kappa = - \frac{\Gamma(\gamma - i\nu)}{\Gamma(\gamma + 1 + i\nu)} e^{i\pi(|\kappa| - \gamma)}$$

(noting that $e^{i\pi l} = e^{i\pi\kappa}$ for $\kappa > 0$ and $e^{i\pi l} = -e^{i\pi\kappa}$ for $\kappa < 0$). With the help of the quantities C_κ introduced in this way the series (37.4), (37.5) can be presented in the form

$$A(\theta) = \frac{1}{p} G(\theta) - i \frac{Ze^2m}{p^2} F(\theta), \quad (37.7)$$

$$B(\theta) = -\frac{i}{p} \tan \frac{\theta}{2} G(\theta) + \frac{Ze^2m}{p^2} \cot \frac{\theta}{2} F(\theta),$$

where

$$G(\theta) = \frac{i}{2} \sum_{l=1}^{\infty} l^2 C_l (P_l + P_{l-1}), \quad F(\theta) = \frac{i}{2} \sum_{l=1}^{\infty} l C_l (P_l - P_{l-1}). \quad (37.8)$$

In transforming the series for $B(\theta)$ the following recurrence relations among the Legendre polynomials have been used:

$$P_l^1 + P_{l-1}^1 = \cot \frac{\theta}{2} \cdot l (P_l - P_{l-1}), \quad (37.9)$$

$$P_l^1 - P_{l-1}^1 = \tan \frac{\theta}{2} \cdot l (P_l + P_{l-1}). \quad (37.10)$$

On the other hand, by virtue of the identity

$$(1 + \cos \theta) \frac{d}{d \cos \theta} [P_l(\cos \theta) - P_{l-1}(\cos \theta)] = l [P_l(\cos \theta) + P_{l-1}(\cos \theta)] \quad (37.11)$$

the functions $F(\theta)$ and $G(\theta)$ are related to each other by

$$G = (1 - \cos \theta) \frac{dF}{d \cos \theta} = -\cot \frac{\theta}{2} \frac{dF}{2 d\theta}. \quad (37.12)$$

Both $A(\theta)$ and $B(\theta)$ are thereby expressed through a single function $F(\theta)$ ¹⁷.

¹⁶R. L. Gluckstern, S. R. Lin // J. Math. Phys. 1964, Vol. 5, p. 1594.

¹⁷The function $F(\theta)$ cannot be expressed in closed form through elementary functions. It can, however, be written in the form of a certain double integral—see the article cited above.

§ 38. Scattering in the ultra-relativistic case

Let us consider in more detail the scattering in the ultra-relativistic case ($\varepsilon \gg m$). In the first approximation we completely neglect in the wave equation the mass m . It is then convenient to use the spinor representation for $\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$, since the equations for ξ and η at $m = 0$ separate:

$$-i\sigma\nabla\xi = (\varepsilon - U)\xi, \quad -i\sigma\nabla\eta = -(\varepsilon - U)\eta \quad (38.1)$$

(acquiring a “neutrino-like” form, see § 30).

A helical state of the electron, polarised in the direction $\mathbf{p}$, is described by the wave function $\psi = \begin{pmatrix} \xi \\ 0 \end{pmatrix}$, and one polarised against $\mathbf{p}$: $\psi = \begin{pmatrix} 0 \\ \eta \end{pmatrix}$. By virtue of the independence of the equations for ξ and η it is clear that this property is preserved under scattering. In other words, in ultra-relativistic scattering of electrons helicity is conserved. From considerations of symmetry (longitudinal polarisation) it is obvious that in the scattering of helical particles there is no azimuthal asymmetry. One can also assert that the scattering cross section for helical electrons does not depend on the sign of the helicity; this follows from the fact that the central field is invariant under inversion, and the sign of helicity reverses under inversion.

In the ultra-relativistic case the formulae (37.3)–(37.5) can be significantly simplified (*D. R. Yennie, D. G. Ravenhall, R. N. Wilson, 1954*).

Let the incident electron be polarised, say, along the direction of motion $\mathbf{n}$. For a plane wave with the helicity value $\mathbf{n}\sigma$ the spinor $\zeta = (\varphi + \chi)/\sqrt{2}$ is proportional to the same 3-spinor w that appeared in the standard representation of the wave. Therefore the relation between the spinor amplitudes of the incident and scattered waves is given by the same operator $\hat{f}$.

As a result of scattering the polarisation vector rotates together with the momentum, acquiring the direction $\mathbf{n}'$. The action of the operator $\hat{f}$ on the spinor wave function of the electron therefore reduces to a rotation of the spin through the angle θ (the angle between $\mathbf{n}$ and $\mathbf{n}'$) about the axis $\mathbf{v}$. A rotation by $-\theta$ is in turn equivalent to a rotation of the coordinate system about the same axis in the opposite direction, i.e. to the replacement $\theta \rightarrow -\theta$. It follows that the operator $\hat{f}$ must coincide (up to a coefficient) with the operator effecting the transformation of the wave function under the indicated change of coordinates, i.e. with the operator (18.17) with the substitution $\theta \rightarrow -\theta$.

Comparing (37.3) with (18.17), we find that it should be

$$\frac{B}{A} = -i \tan \frac{\theta}{2}. \quad (38.2)$$

Thus, in the ultra-relativistic limit,

$$\hat{f} = A(\theta) \left[1 - i \tan \frac{\theta}{2} \mathbf{v} \cdot \boldsymbol{\sigma} \right]. \quad (38.3)$$

The expression for $A(\theta)$ (37.4) can also be simplified, exploiting the relation between the phases δ_κ and $\delta_{-\kappa}$ that arises in the same limit. For its derivation we note that the equations (35.4) for the functions f and g , after striking out the terms containing m , become invariant with respect to the substitution

$$\kappa \rightarrow -\kappa, \quad f \rightarrow g, \quad g \rightarrow -f,$$

not affecting the parameters of the particle or of the field. It should therefore be that $f_{\kappa}/g_{\kappa} = -g_{-\kappa}/f_{-\kappa}$, and after substitution of the asymptotic expressions we find

$$\tan\left(pr - \frac{l\pi}{2} + \delta_{\kappa}\right) = -\cot\left(pr - \frac{l'\pi}{2} + \delta_{-\kappa}\right),$$

$$\delta_{\kappa} = \delta_{-\kappa} - (l' - l) \frac{\pi}{2} + \left(n + \frac{1}{2}\right) \pi,$$

whence

$$e^{2i\delta_l} = e^{2i\delta_{-l}}. \quad (38.4)$$

Using this relation (and replacing in the first term of the sum in (37.4) the summation index l by $l - 1$), we obtain

$$A(\theta) = \frac{1}{2ip} \sum_{l=1}^{\infty} l (e^{2i\delta_l} - 1) [P_l(\cos \theta) + P_{l-1}(\cos \theta)]. \quad (38.5)$$

From (38.2) it follows that $\text{Re}(AB^*) = 0$. This means that in the approximation being considered the cross section does not depend on the initial polarisation of the particle, and a non-polarised beam remains non-polarised after scattering (see the formulae III, (140.8)–(140.10)). We note also that as $\theta \rightarrow \pi$ the expression $A(\theta)$ (38.5) tends to zero as $(\pi - \theta)^2$ (recall that $P_l(-1) = (-1)^l$). Together with it, B also tends to zero, and so does the cross section

$$\frac{d\sigma}{d\theta} = |A|^2 + |B|^2 = \frac{|A(\theta)|^2}{\cos^2(\theta/2)}. \quad (38.6)$$

The listed properties disappear, of course, in the subsequent approximations in the small quantity m/ε . In particular, the analysis shows that as $\theta \rightarrow \pi$ the cross section tends to a limit proportional to $(m/\varepsilon)^2$.

For the Coulomb field in the ultra-relativistic case the phases δ_{κ} do not depend on the energy, as is evident from (36.19). Therefore in a pure Coulomb field the scattering cross section for $\varepsilon \gg m$ has the form

$$d\sigma = \frac{\tau(\theta)}{\varepsilon^2} d\theta, \quad (38.7)$$

where τ is a function of the angle only.

§39. System of wave functions of the continuous spectrum for scattering in a Coulomb field

Later on (see §§ 95, 96) various inelastic processes will be considered that occur in the scattering of ultra-relativistic electrons in the field of a heavy ($Z\alpha \sim 1$) nucleus. For the computation of the corresponding matrix elements we shall need the wave functions, whose asymptotic form (as $r \rightarrow \infty$) consists of a plane wave and a spherical wave.

We saw that in the ultra-relativistic case (energy of the electron $\varepsilon \gg m$) the main role in the scattering is played by momentum transfers (from the electron to the nucleus)

$q = |\mathbf{p}' - \mathbf{p}| \sim m$. These values of q correspond to “impact parameters” $\rho \sim 1/q \sim 1/m$, and the electron is deflected through angles ^{18, 19}

$$\theta \sim \frac{q}{p} \sim \frac{m}{\varepsilon}. \quad (39.1)$$

In terms of the coordinates r (distance from the centre) and $z = r \cos \theta$ this means the region

$$\rho \equiv r \sin \theta \sim 1/m, \quad p(r - z) = pr(1 - \cos \theta) \sim 1. \quad (39.2)$$

Since $r \sim \varepsilon/m^2$, i.e. we are dealing with the region of large distances.

Let us write the Dirac equation in the form

$$(\varepsilon - U - m\beta + i\alpha\nabla)\psi = 0, \quad U = -Z\alpha/r. \quad (39.3)$$

We convert it into a second-order equation by applying to (39.3) the operator $(\varepsilon - U + m\beta - i\alpha\nabla)$:

$$(\Delta + p^2 - 2\varepsilon U)\psi = (-i\alpha\nabla U - U^2)\psi. \quad (39.4)$$

Since in the region under consideration $r \gg Z\alpha/\varepsilon$, we have $U \ll \varepsilon$. In the first approximation one may neglect the right-hand side of (39.4). The remaining equation

$$\left(\Delta + p^2 + \frac{2\varepsilon Z\alpha}{r}\right)\psi = 0 \quad (39.5)$$

coincides in form with the non-relativistic Schrödinger equation in the Coulomb field

$$\left(\frac{1}{2m}\Delta + \frac{p^2}{2m} + \frac{Z\alpha}{r}\right)\psi = 0, \quad (39.5a)$$

differing from it only by an obvious change in the notation of the parameters (the “potential energy” has a superfluous factor ε/m). We can therefore immediately write its solution, which has the required asymptotic form (see III, § 136).

Thus, the wave function containing asymptotically a plane wave ($\propto e^{i\mathbf{p}\mathbf{r}}$) and a diverging spherical wave has the form

$$\psi_{\varepsilon\mathbf{p}}^{(+)} = C \frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} F\left(\frac{iZ\alpha\varepsilon}{p}, 1, i(pr - \mathbf{p}\mathbf{r})\right), \quad (39.6)$$

$$C = e^{\pi Z\alpha\varepsilon/(2p)} \Gamma\left(1 - i\frac{Z\alpha\varepsilon}{p}\right),$$

where F is the confluent hypergeometric function, and $u_{\varepsilon\mathbf{p}}$ is a constant bispinor amplitude of the plane wave, normalised by the condition we have adopted (23.4)

$$\bar{u}_{\varepsilon\mathbf{p}} u_{\varepsilon\mathbf{p}} = 2m. \quad (39.7)$$

¹⁸This is clear directly from equations (38.1), since for the Coulomb field the energy ε can be altogether eliminated from the equations by the substitution $\mathbf{r} \rightarrow \mathbf{r}'/\varepsilon$.

¹⁹In this section p denotes $|\mathbf{p}|$.

The wave function (39.6) is normalised in such a way that the plane wave in its asymptotic expression has the usual form

$$\frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}},$$

corresponding to one particle in a unit volume. In the ultra-relativistic case $p \approx \varepsilon$, and in (39.6) one can set $Z\alpha\varepsilon/p \approx Z\alpha$:

$$\begin{aligned} \psi_{\varepsilon\mathbf{p}}^{(+)} &= C \frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} F(iZ\alpha, 1, i(pr - \mathbf{p}\mathbf{r})), \\ C &= e^{\pi Z\alpha/2} \Gamma(1 - iZ\alpha). \end{aligned} \quad (39.8)$$

Let us draw attention to the fact that although the distances we consider are so large that $pr \gg 1$, one cannot replace the hypergeometric function in (39.8) by its asymptotic expression: the argument of the function F is not pr , but $pr(1 - \cos \theta)$, a quantity that is not assumed to be large²⁰.

In the applications it turns out to be necessary also to include the next approximation in ψ , which has a spinor structure different from (39.8) (reducing to the factor $u_{\varepsilon\mathbf{p}}$). For this computation we write ψ in the form

$$\psi = \frac{C}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} (u_{\varepsilon\mathbf{p}} F + \varphi).$$

On the right-hand side of equation (39.4) we keep the term linear in U , and for the function φ we obtain the equation

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)\varphi = -iu_{\varepsilon\mathbf{p}}(\boldsymbol{\alpha}\nabla U)F. \quad (39.9)$$

Its solution can be found by noting that the function F satisfies the equation

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)F = 0$$

(as can be verified by substituting (39.6) into (39.5)). Applying to this equation the operation ∇ , we get

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)\nabla F = 2\varepsilon F \nabla U.$$

Comparing with equation (39.9), we find

$$\varphi = -\frac{i}{2\varepsilon} (\boldsymbol{\alpha}\nabla) u_{\varepsilon\mathbf{p}} F.$$

Let us write the final expression for $\psi^{(+)}$ and for a similar function $\psi^{(-)}$, which contains in its asymptotic expression a converging spherical wave:

$$\begin{aligned} \psi_{\varepsilon\mathbf{p}}^{(+)} &= \frac{C}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} \left(1 - \frac{i\boldsymbol{\alpha}}{2\varepsilon} \nabla \right) F(iZ\alpha, 1, i(pr - \mathbf{p}\mathbf{r})) u_{\varepsilon\mathbf{p}}, \\ \psi_{\varepsilon\mathbf{p}}^{(-)} &= \frac{C^*}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} \left(1 - \frac{i\boldsymbol{\alpha}}{2\varepsilon} \nabla \right) F(-iZ\alpha, 1, -i(pr - \mathbf{p}\mathbf{r})) u_{\varepsilon\mathbf{p}}, \end{aligned} \quad (39.10)$$

²⁰In III, § 135 we were interested in arbitrarily large r , and therefore such a replacement was possible for all angles θ .

$$C = e^{\pi Z\alpha/2} \Gamma(1 - iZ\alpha)$$

(*W. H. Furry*, 1934). Let us write also analogous functions $\psi_{-\varepsilon, -\mathbf{p}}$ with “negative frequency”, which will be needed in considering processes involving positrons. They can be obtained from the functions $\psi_{\varepsilon\mathbf{p}}$ by the substitution $\mathbf{p} \rightarrow -\mathbf{p}$, $\varepsilon \rightarrow -\varepsilon$, where $p = |\mathbf{p}|$ does not change (and by virtue of the last circumstance the parameter $iZ\alpha$ of the hypergeometric function changes sign, as is clear from the original expression (39.6), in which this parameter appears in the form $iZ\alpha\varepsilon/p$). Thus,

$$\begin{aligned} \psi_{-\varepsilon, -\mathbf{p}}^{(+)} &= \frac{C}{\sqrt{2\varepsilon}} e^{-i\mathbf{p}\mathbf{r}} \left(1 + \frac{i\alpha}{2\varepsilon} \nabla \right) F(-iZ\alpha, 1, i(pr + \mathbf{p}\mathbf{r})) u_{-\varepsilon, -\mathbf{p}}, \\ \psi_{-\varepsilon, -\mathbf{p}}^{(-)} &= \frac{C^*}{\sqrt{2\varepsilon}} e^{-i\mathbf{p}\mathbf{r}} \left(1 + \frac{i\alpha}{2\varepsilon} \nabla \right) F(iZ\alpha, 1, -i(pr + \mathbf{p}\mathbf{r})) u_{-\varepsilon, -\mathbf{p}}, \\ C &= e^{-\pi Z\alpha/2} \Gamma(1 + iZ\alpha). \end{aligned} \quad (39.11)$$

Regarding the calculations performed, one further remark is in order. The asymptotic condition that we imposed is by itself certainly not sufficient to determine the solution of the wave equation uniquely (this is clear already from the fact that one can always add to ψ , without violating this condition, any Coulomb diverging spherical wave). In writing the solution of equation (39.5) in the form (39.6), we thereby tacitly assumed the choice of a solution finite at $r = 0$. Such a requirement was necessary in III, §§ 135, 136, where the exact Schrödinger equation was considered, valid everywhere in space²¹. In the present case, however, equation (39.5) applies only to large distances, and therefore the selection of solutions performed requires additional justification.

This is provided by the fact that for large “impact parameters” $\rho = r \sin \theta$ large orbital angular momenta l and small scattering angles θ are relevant: at $\rho \sim 1/m$ we have

$$l \sim \rho p \sim \rho \varepsilon \sim \varepsilon/m \gg 1,$$

and the angle θ can be estimated quasiclassically:

$$\theta \sim \frac{1}{p} \int \frac{dU}{dr} dt \sim \frac{U'(\rho)\rho}{p} \sim \frac{m}{\varepsilon} \ll 1.$$

This means that in the decomposition of ψ into spherical waves the dominant waves (in the region of r and θ being considered) are precisely those with the indicated large values of l . But a spherical wave with large l is necessarily small in approaching the origin of coordinates at the “classically inaccessible” (thanks to the centrifugal barrier) distances $r \ll l/\varepsilon$. Therefore, in “sewing together” the solution of equation (39.5) with the solution of the exact equation (39.4) at small distances at $r \sim r_1$, where $l/\varepsilon \gg r_1 \gg Z\alpha/\varepsilon$, the boundary condition for the solution of equation (39.5) will consist in requiring its smallness, and the selection made is justified.

Problem

²¹As explained in III, § 135, this condition was ensured by the choice of a partial integral of the type (135.1), instead of the sum of integrals over the general contour with different values of β_1, β_2 .

For the Coulomb field of attraction with $Z\alpha \ll 1$ find the correction (of relative order $Z\alpha$) to the non-relativistic wave function of the discrete spectrum.

Solution. The velocity of the electron in a bound state $v \sim Z\alpha$, so that for $Z\alpha \ll 1$ the wave function is non-relativistic in the zeroth approximation, i.e.

$$\psi = u\psi_{\text{nr}},$$

where ψ_{nr} is a function satisfying the non-relativistic Schrödinger equation, and u is a bispinor of the form $u = \begin{pmatrix} w \\ 0 \end{pmatrix}$, where w is a spinor describing the polarisation state of the electron. In the next approximation we write: $\psi = u\psi_{\text{nr}} + \psi^{(1)}$ and, substituting into (39.4), find for $\psi^{(1)}$ the equation

$$\left(\frac{1}{2m} \Delta + |\varepsilon_n| + \frac{Z\alpha}{r} \right) \psi^{(1)} = i \frac{Z\alpha}{2m} \left(\nabla \frac{1}{r} \right) (\alpha u) \psi_{\text{nr}},$$

where ε_n is the non-relativistic discrete energy level. Here the terms of relative order $(Z\alpha)^2$ have been dropped (one should bear in mind that in the non-relativistic case the essential distances are of the order of the Bohr radius: $r \sim 1/mZ\alpha$). The solution of this equation: $\psi^{(1)} = -\frac{i}{2m} \alpha u \nabla \psi_{\text{nr}}$, so that

$$\psi = \left(1 - \frac{i}{2m} \alpha \nabla \right) u \psi_{\text{nr}}.$$

§ 40. Electron in the field of a plane electromagnetic wave

The Dirac equation can be solved exactly for an electron moving in the field of a plane electromagnetic wave (*D. M. Volkov*, 1937).

The field of a plane wave with wave 4-vector k ($k^2 = 0$) depends on the 4-coordinates only through the combination $\varphi = kx$, so that the 4-potential

$$A^\mu = A^\mu(\varphi), \quad (40.1)$$

which satisfies the Lorentz gauge condition

$$\partial_\mu A^\mu = k_\mu A'^\mu = 0 \quad (40.2)$$

(the prime denotes differentiation with respect to φ). Since the constant term in A is inessential, in this gauge condition one may drop it and write it in the form

$$kA = 0.$$

We start from the second-order equation (32.6), in which the field tensor

$$F_{\mu\nu} = k_\mu A'_\nu - k_\nu A'_\mu. \quad (40.3)$$

When expanding the square $(i\partial - eA)^2$ one should take into account that by virtue of (40.2) $\partial_\mu (A^\mu \psi) = A^\mu \partial_\mu \psi$. As a result we obtain the equation

$$[-\partial^2 - 2ie(A\partial) + e^2 A^2 - m^2 - ie(\gamma k)(\gamma A')] \psi = 0 \quad (40.4)$$

$$(\partial^2 = \partial_\mu \partial^\mu).$$

We seek the solution of this equation in the form

$$\psi = e^{-ipx} F(\varphi), \quad (40.5)$$

where p is a constant 4-vector. Adding to p any vector of the form $\text{const} \cdot k$ does not change the form of the function ψ (only a corresponding redefinition of the function $F(\varphi)$ is needed). One may therefore impose on p one additional condition without loss of generality. Let

$$p^2 = m^2. \quad (40.6)$$

Then when the field is switched off the quantum numbers p^μ become the components of the 4-momentum of a free particle. The meaning of the components of the 4-vector p in the presence of the field is more transparent in a special reference frame, chosen so that $A_0 = 0$. Let in this frame the vector $\mathbf{A}$ be directed along the x^1 axis, and $\mathbf{k}$ along the x^3 axis (i.e. the electric field of the wave is polarised along x^1 , the magnetic along x^2 , and the wave propagates along x^3). Then (40.5) takes the form of eigenfunctions of the operators

$$\widehat{p}_1 = i \frac{\partial}{\partial x^1}, \quad \widehat{p}_2 = i \frac{\partial}{\partial x^2}, \quad \widehat{p}_0 - \widehat{p}_3 = i \left(\frac{\partial}{\partial x_0} - \frac{\partial}{\partial x^3} \right)$$

with eigenvalues $p_1, p_2, p_0 - p_3$ (the same operators, as is easily verified, are commutative with the Hamiltonian of the Dirac equation). Thus, in this reference frame, p^1, p^2 are the components of the generalised momentum along the axes x^1, x^2 , and $p^0 - p^3$ is the difference between the total energy and the component of the generalised momentum along the axis x^3 .

On substituting (40.5) into (40.4) we note that

$$\partial^\mu F = k^\mu F', \quad \partial_\mu \partial^\mu F = k^2 F'' = 0,$$

and find for $F(\varphi)$ the equation

$$2i(kp)F' + [-2e(pA) + e^2 A^2 - ie(\gamma k)(\gamma A)']F = 0.$$

The formal solution of this equation is

$$F = \exp \left\{ -i \int_0^{kx} \left[\frac{e}{(kp)} (pA) - \frac{e^2}{2(kp)} A^2 \right] d\varphi + \frac{e(\gamma k)(\gamma A)}{2(kp)} \right\} \frac{u}{\sqrt{2p_0}},$$

where $u/\sqrt{2p_0}$ is an arbitrary constant bispinor (its form will be discussed below).

All powers of $(\gamma k)(\gamma A)$ above the first vanish, since

$$\begin{aligned} & (\gamma k)(\gamma A)(\gamma k)(\gamma A) = \\ & = -(\gamma k)(\gamma k)(\gamma A)(\gamma A) + 2(kA)(\gamma k)(\gamma A) = -k^2 A^2 = 0. \end{aligned}$$

One can therefore replace

$$\exp \frac{e(\gamma k)(\gamma A)}{2(kp)} = 1 + \frac{e}{2(kp)} (\gamma k)(\gamma A),$$

so that ψ takes the form

$$\psi_p = \left[1 + \frac{e}{2(kp)} (\gamma k)(\gamma A) \right] \frac{u}{\sqrt{2p_0}} e^{iS}, \quad (40.7)$$

where²²

$$S = -px - \int_0^{kx} \left[\frac{e}{(kp)} (pA) - \frac{e^2}{2} A^2 \right] d\varphi. \quad (40.8)$$

To establish the conditions imposed on the constant bispinor u , one should consider the wave as infinitely slowly “switched on”, starting from $t = -\infty$. Then $A \rightarrow 0$ at $kx \rightarrow -\infty$ and ψ should go over into the solution of the free Dirac equation. For this $u = u(p)$ must satisfy the equation

$$(\gamma p - m)u = 0. \quad (40.9)$$

This condition discards the “spurious” solutions of the second-order equation. Since it does not depend on time, this condition remains in force at finite kx as well. Thus, $u(p)$ coincides with the bispinor amplitude of a free plane wave; we shall take it normalised by the same condition (23.4): $\bar{u}u = 2m$.

The arguments set forth also immediately allow one to find the normalisation of the wave functions (40.7). An infinitely slowly switched on field does not change the normalisation integral. It follows that the functions (40.7) satisfy the same normalisation condition

$$\int \psi_{p'}^* \psi_p d^3x = \int \bar{\psi}_{p'} \gamma^0 \psi_p d^3x = (2\pi)^3 \delta(\mathbf{p}' - \mathbf{p}), \quad (40.10)$$

as free plane waves.

Let us find the current density corresponding to the functions (40.7). Noting that

$$\bar{\psi}_p = \frac{\bar{u}}{\sqrt{2p_0}} \left[1 + \frac{e}{2(kp)} (\gamma A)(\gamma k) \right] e^{-iS},$$

by direct multiplication we get

$$j^\mu = \bar{\psi}_p \gamma^\mu \psi_p = \frac{1}{p_0} \left\{ p^\mu - eA^\mu + k^\mu \left(\frac{e(pA)}{(kp)} - \frac{e^2 A^2}{2(kp)} \right) \right\}. \quad (40.11)$$

If $A^\mu(\varphi)$ are periodic functions and their average (over time) vanishes, the mean value of the current density is

$$\bar{j}^\mu = \frac{1}{p_0} \left(p^\mu - \frac{e^2 \overline{A^2}}{2(kp)} k^\mu \right). \quad (40.12)$$

Let us find also the mean density of kinetic momentum in the state ψ_p . The operator of kinetic momentum is the difference $\hat{p} - eA = i\partial - eA$. By direct computation we find

$$\psi_p^* (\hat{p}^\mu - eA^\mu) \psi_p = \bar{\psi}_p \gamma^0 (\hat{p}^\mu - eA^\mu) \psi_p =$$

²²The expression for S coincides with the classical action for a particle moving in the field of a wave (cf. II, § 47, problem 2).

$$= p^\mu - eA^\mu + k^\mu \left(\frac{e(pA)}{(kp)} - \frac{e^2 A^2}{2(kp)} \right) + k^\mu \frac{ie}{8(kp)p_0} F_{\mu\nu} (u^* \sigma^{\lambda\nu} u). \quad (40.13)$$

The time-averaged value of this 4-vector, which we denote by q^μ , is

$$q^\mu = p^\mu - \frac{e^2 \overline{A^2}}{2(kp)} k^\mu. \quad (40.14)$$

Its square:

$$q^2 = m_*^2, \quad m_* = m \sqrt{1 - \frac{e^2}{m^2} \overline{A^2}}; \quad (40.15)$$

m_* plays the role of “effective mass” of the electron in the field. Comparing (40.14) and (40.12), we see that

$$\bar{j}^\mu = q^\mu / p_0. \quad (40.16)$$

Let us note also that the normalisation condition (40.10), expressed with the aid of the vector $\mathbf{q}$, has the form

$$\int \psi_p^* \psi_p d^3x = (2\pi)^3 \frac{q_0}{p_0} \delta(\mathbf{q}' - \mathbf{q}) \quad (40.17)$$

(the transition from (40.10) to (40.17) is most easily carried out in the special reference frame indicated above).

§41. Spin motion in an external field

The passage to the quasiclassical approximation in the Dirac equation is carried out in the same way as in the non-relativistic theory. In the equation of second order (32.7a) we substitute²³ ψ in the form

$$\psi = u e^{iS/\hbar},$$

where S is a scalar, u a slowly varying bispinor. It is assumed that the usual quasiclassical condition is satisfied: the momentum of the particle should change little over distances of the order of the wavelength $\hbar/|\mathbf{p}|$.

In the zeroth approximation in $\hbar$ one obtains the usual classical relativistic Hamilton–Jacobi equation for the action S . All terms containing spin (and proportional to $\hbar$) drop out of the equations of motion. The spin would appear only in the next approximation in $\hbar$. In other words,

the effect of the magnetic moment of the electron on its motion is always of the same order of magnitude as the quantum corrections. This is quite natural in view of the purely quantum nature of the spin angular momentum, which is proportional to $\hbar$.

In this connection the problem acquires physical meaning of the behaviour of the spin of an electron performing a given quasiclassical motion in an external field. The solution of this problem is contained in the next approximation in $\hbar$ of the Dirac equation. We shall, however, apply another, more intuitive method that is not directly connected with the Dirac equation. It has the advantage that it permits one to consider the motion of any

²³Here we again use ordinary units.

particle, including those possessing an “anomalous” gyromagnetic ratio, not described by the Dirac equation.

Our goal is to establish “equations of motion” for the spin in an arbitrary (given) motion of the particle. Let us begin with the non-relativistic case.

The non-relativistic Hamiltonian of a particle in an external field is

$$\hat{H} = \hat{H}' - \mu \boldsymbol{\sigma} \cdot \mathbf{H}, \quad (41.1)$$

where in $\hat{H}'$ all terms not containing the spin are included (see III, § 111); μ is the magnetic moment of the particle. This form of the Hamiltonian is not tied to a specific sort of particle. For electrons $\mu = e\hbar/2mc$ (the charge of the electron $e = -|e|$), and for nucleons μ contains also an “anomalous part”²⁴

$$\mu' = \mu - \frac{e\hbar}{2mc}. \quad (41.2)$$

According to the general rules of quantum mechanics, the operator equation of motion for the spin is obtained from the formula

$$\dot{\hat{\mathbf{s}}} = \frac{i}{\hbar} (\hat{H}\hat{\mathbf{s}} - \hat{\mathbf{s}}\hat{H}) = \frac{i}{2\hbar} (\hat{H}\boldsymbol{\sigma} - \boldsymbol{\sigma}\hat{H}). \quad (41.3)$$

Substituting here (41.1), we find

$$\dot{\hat{s}}_i = -\frac{i\mu}{2\hbar} H_k (\sigma_k \sigma_i - \sigma_i \sigma_k) = -\frac{\mu}{\hbar} e_{ikl} H_k \sigma_l,$$

or

$$\dot{\hat{\mathbf{s}}} = \frac{2\mu}{\hbar} [\hat{\mathbf{s}}\mathbf{H}]. \quad (41.4)$$

We now average this operator equation over the state of a quasiclassical wave packet moving along a given trajectory. This operation reduces to replacing the operator of spin by its

average value $\bar{\mathbf{s}}$, and the vector $\mathbf{H}$ by the function $\mathbf{H}(t)$ representing the value of the magnetic field at the point of the particle's (wave packet's) location at its given motion along the trajectory. In the non-relativistic approximation, within the framework of the Pauli equation, $\hat{\mathbf{s}} = \boldsymbol{\sigma}/2$ is the spin operator of the particle in its rest frame, and the average value we denoted in § 29 as $\boldsymbol{\zeta}/2$. Thus, we arrive at the equation²⁵

$$\frac{d\boldsymbol{\zeta}}{dt} = \frac{2\mu}{\hbar} [\boldsymbol{\zeta}\mathbf{H}(t)]. \quad (41.5)$$

In this form the equation has an essentially classical character. It says that the vector of magnetic moment precesses about the direction of the field $\mathbf{H}$ with the angular velocity $-2\mu\mathbf{H}/\hbar$. In the non-relativistic case, if $\mu' = 0$, then $\mu = e\hbar/2mc$, and this angular

²⁴Including radiative corrections, a very small “anomalous part” is present also in the magnetic moment of the electron.

²⁵Classically equation (41.5) is obtained directly from the equality $d\mathbf{M}/dt = [\boldsymbol{\mu}\mathbf{H}]$, where $\mathbf{M}$ is the angular momentum of the system, $\boldsymbol{\mu}$ its magnetic moment; $[\boldsymbol{\mu}\mathbf{H}]$ is the torque on the system. Setting $\mathbf{M} = \frac{1}{2}\hbar\boldsymbol{\zeta}$, $\boldsymbol{\mu} = \frac{\mu}{\hbar}\boldsymbol{\zeta} = \mu\boldsymbol{\zeta}$, we obtain (41.5).

velocity coincides with the velocity $-2\mu\mathbf{H}/\hbar$ of rotation of the vector $\boldsymbol{\zeta}$; in other words, the polarisation vector maintains a constant angle with the direction of motion (we shall see below that this result remains valid in the relativistic case as well).

In the same non-relativistic case the velocity $\mathbf{v}$ of the particle changes according to the equation

$$\frac{d\mathbf{v}}{dt} = \frac{e}{mc} [\mathbf{v} \mathbf{H}],$$

i.e. the vector $\mathbf{v}$ rotates about the direction of $\mathbf{H}$ with the angular velocity $-e\mathbf{H}/mc$. If $\mu' = 0$, this angular velocity coincides with the velocity $-2\mu\mathbf{H}/\hbar$ of rotation of the vector $\boldsymbol{\zeta}$; in other words, the polarisation vector maintains a constant angle with the direction of motion (we shall see below that this result remains valid in the relativistic case as well).

Let us now carry out the relativistic generalisation of equation (41.5). For the covariant description of the polarisation one should use the 4-vector a introduced in § 29, and the equation of spin motion should define the derivative $da^\mu/d\tau$ with respect to the proper time τ ²⁶.

The form of this equation can be established already from considerations of relativistic invariance, if we take into account that its right-hand side must be linear and homogeneous in the tensor of the electromagnetic field $F^{\mu\nu}$ and, besides it, can contain only the 4-velocity $u^\mu = p^\mu/m$. This condition is satisfied only by an equation of the form

$$\frac{da^\mu}{d\tau} = \alpha F^{\mu\nu} a_\nu + \beta u^\mu F^{\nu\lambda} u_\nu a_\lambda, \quad (41.6)$$

where α, β are constants. It is easy to see that by virtue of the condition $a_\mu u^\mu = 0$ and the antisymmetry of the tensor $F^{\mu\nu}$ (so that $F^{\mu\nu} u_\mu u_\nu = 0$) no other expressions of the required form can be constructed.

For $v \rightarrow 0$ this equation should coincide with (41.5). Setting $a^\mu = (0, \boldsymbol{\zeta})$, $u^\mu = (1, 0)$, $\tau = t$, we get

$$\frac{d\boldsymbol{\zeta}}{dt} = \alpha [\boldsymbol{\zeta} \mathbf{H}].$$

Comparing with (41.5), we find: $\alpha = 2\mu$.

To determine β we note that $a^\mu u_\mu = 0$. Differentiating this relation with respect to τ and using the classical equation of motion of a charge in a field

$$m \frac{du^\mu}{d\tau} = e F^{\mu\nu} u_\nu$$

(see II, § 23), we get

$$u_\mu \frac{da^\mu}{d\tau} = -a_\mu \frac{du^\mu}{d\tau} = -a_\mu \frac{e}{m} F^{\mu\nu} u_\nu = \frac{e}{m} F^{\mu\nu} u_\mu a_\nu.$$

On the other hand, multiplying equation (41.6) on both sides by u_μ , taking into account that $u_\mu u^\mu = 1$ and cancelling the common factor $F^{\mu\nu} u_\mu a_\nu$, we get

$$\beta = -2\left(\mu - \frac{e}{2m}\right) = -2\mu'.$$

²⁶Hereafter we again set $c = 1$, $\hbar = 1$.

Thus, we find the final relativistic equation of spin motion

$$\frac{da^\mu}{d\tau} = 2\mu F^{\mu\nu} a_\nu - 2\mu' u^\mu F^{\nu\lambda} u_\nu a_\lambda \quad (41.7)$$

(V. Bargmann, L. Michel, V. Telegdi, 1959)²⁷.

Let us go over from the 4-vector a to the quantity ζ , which directly characterises the polarisation of the particle in its “instantaneous” rest frame; the connection between a and ζ is given by formulae (29.7)–(29.9). We note immediately that from (41.7) it automatically follows that $a_\mu da^\mu/d\tau = 0$, i.e. $a_\mu a^\mu = \text{const}$. Since $a_\mu a^\mu = -\zeta^2$, this is a natural result: in the motion of the particle its polarisation remains unchanged in magnitude.

The equation determining the change in the direction of the polarisation is obtained by going over in (41.7) to three-dimensional notation. Expanding the spatial components of this equation, we find

$$\begin{aligned} \frac{d\mathbf{a}}{dt} = & \frac{2\mu m}{\varepsilon} [\mathbf{a} \mathbf{H}] + \frac{2\mu m}{\varepsilon} (\mathbf{a} \mathbf{v}) \mathbf{E} - \frac{2\mu' \varepsilon}{m} \mathbf{v} (\mathbf{a} \mathbf{E}) + \\ & + \frac{2\mu' \varepsilon}{m} \mathbf{v} (\mathbf{v} [\mathbf{a} \mathbf{H}]) + \frac{2\mu' \varepsilon}{m} \mathbf{v} (\mathbf{a} \mathbf{v}) (\mathbf{v} \mathbf{E}). \end{aligned}$$

Substituting here (29.9), taking into account on differentiating the equalities $\mathbf{p} = \varepsilon \mathbf{v}$, $\varepsilon^2 = \mathbf{p}^2 + m^2$ and the equations of motion

$$\frac{d\mathbf{p}}{dt} = e\mathbf{E} + e[\mathbf{v} \mathbf{H}], \quad \frac{d\varepsilon}{dt} = e(\mathbf{v} \mathbf{E}). \quad (41.8)$$

An elementary, although rather lengthy, computation leads to the following equation²⁸:

$$\frac{d\zeta}{dt} = \frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon} [\zeta \mathbf{H}] + \frac{2\mu' \varepsilon}{\varepsilon + m} (\mathbf{v} \mathbf{H}) [\mathbf{v} \zeta] + \frac{2\mu m + 2\mu' \varepsilon}{\varepsilon + m} [\zeta [\mathbf{E} \mathbf{v}]]. \quad (41.9)$$

Of particular interest is not so much the change in the absolute direction of the polarisation in space, as the change of its direction relative to the direction of motion. Representing ζ in the form

$$\zeta = \mathbf{n} \zeta_{\parallel} + \zeta_{\perp} \quad (41.10)$$

(where $\mathbf{n} = \mathbf{v}/v$) and writing an equation for the projection $\zeta_{\parallel}$ of the polarisation on the direction of motion. The computation using (41.8), (41.9) leads to the following result²⁹:

$$\frac{d\zeta_{\parallel}}{dt} = 2\mu'(\zeta_{\perp} [\mathbf{H} \mathbf{n}]) + \frac{2}{v} \left(\frac{\mu m^2}{\varepsilon^2} - \mu' \right) (\zeta_{\perp} \mathbf{E}). \quad (41.11)$$

A number of examples illustrating the application of the obtained equations are considered in the problems to this section. Here we merely note that in the motion in a pure magnetic field the polarisation of a particle without an anomalous magnetic moment

²⁷In another form a similar equation was first found by Ya. I. Frenkel' (1926).

²⁸This equation can be obtained somewhat more briefly by expanding the time component of equation (41.7).

²⁹This equation can also be obtained somewhat more briefly by expanding the time component of equation (41.7).

maintains a constant angle with the direction of motion (i.e. $\dot{\zeta}_{\parallel} = \text{const}$). Thus, this result, indicated above for the non-relativistic case, is in fact of a general character.

Let us refine the conditions of applicability of the obtained equations. The condition mentioned at the outset that the momentum of the particle change slowly amounts to a definite requirement of smallness of the fields $\mathbf{E}$ and $\mathbf{H}$; in particular, the Larmor radius in a magnetic field ($\sim p/eH$) should be large compared with the wavelength of the particle. Besides this, however, a condition must also hold that the fields do not change too rapidly in space, strictly speaking: the field should change little over the dimensions of the quasiclassical wave packet. Thus, the field should change little over distances of the order of the wavelength of the particle ($1/p$), and also of the Compton wavelength, $1/m$ ³⁰.

However, in practical problems of motion in macroscopic fields the condition of slow change of the fields is automatically fulfilled, so that only a sufficient smallness of them is required.

In § 33 the first relativistic corrections to the Hamiltonian of an electron moving in an external field were found. For an electron in an electric field the approximate Hamiltonian has the form (see (33.12))

$$\hat{H} = \hat{H}' - \frac{e}{4m} \left(\boldsymbol{\sigma} \left[\mathbf{E} \frac{\hat{\mathbf{p}}}{m} \right] \right), \quad \hat{\mathbf{p}} = -i\nabla, \quad (41.12)$$

where $\hat{H}'$ contains terms not involving the spin. In our case, because the field changes slowly, in $\hat{H}'$ one should neglect the term with derivatives of $\mathbf{E}$ (i.e. $\text{div } \mathbf{E}$); one may also drop the term with $\hat{\mathbf{p}}^4$, which has no relation to the effects of the field that interest us here, so that $\hat{H}'$ (in the absence of a magnetic field) reduces to the non-relativistic Hamiltonian $\hat{H}' = \hat{\mathbf{p}}^2/(2m) + e\Phi$.

Formula (41.12) can also be obtained directly from equation (41.9), without invoking the Dirac equation. The generalisation achieved in this way (in the quasiclassical case) applies to particles with an anomalous magnetic moment.

To within terms of first order in the velocity v , the equation for the spin motion in an electric field is obtained from

(41.9) in the form

$$\frac{d\zeta}{dt} = (\mu + \mu') [\zeta [\mathbf{E} \mathbf{v}]] = \left(\frac{e}{2m} + 2\mu' \right) [\zeta [\mathbf{E} \mathbf{v}]].$$

If we demand that this equation be obtained quantum-mechanically by commuting the spin operator with the Hamiltonian (in accordance with (41.3)), one must set

$$\hat{H} = \hat{H}' = \left(\mu' + \frac{e}{4m} \right) \left(\boldsymbol{\sigma} \left[\mathbf{E} \frac{\hat{\mathbf{p}}}{m} \right] \right). \quad (41.13)$$

This is the desired expression. For $\mu' = 0$ we return to (41.12). Note that the “normal” magnetic moment $e/2m$ enters with an extra factor $\frac{1}{2}$ as compared with the anomalous moment μ' ³¹.

³⁰The last requirement arises from the condition that the spread of velocities in the wave packet be small compared with c ; in the contrary case it would be impossible to use the non-relativistic formulae.

³¹This is the “Thomas half”, mentioned in the footnote on p. 151. The derivation presented here clearly demonstrates its origin.

Problems

1. Determine the change in the direction of polarisation of a particle in its motion in a plane perpendicular to a uniform magnetic field ($\mathbf{v} \perp \mathbf{H}$).

Solution. On the right-hand side of equation (41.9) only the first term remains, i.e. the vector $\boldsymbol{\zeta}$ precesses about the direction of $\mathbf{H}$ (the z -axis) with the angular velocity

$$-\frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon} \mathbf{H} = -\left(\frac{e}{\varepsilon} + 2\mu'\right) \mathbf{H}.$$

The projection of $\boldsymbol{\zeta}$ onto this plane we denote by ζ_1 . The vector $\mathbf{v}$ rotates in the same plane with the angular velocity $-e\mathbf{H}/\varepsilon$ (as follows from the equations of motion: $\dot{\mathbf{p}} = \varepsilon \dot{\mathbf{v}} = e[\mathbf{v} \mathbf{H}]$). It is clear that ζ_1 rotates relative to the direction of $\mathbf{v}$ with the angular velocity $-2\mu'\mathbf{H}$.

2. The same for motion along the direction of the magnetic field.

Solution. For coinciding directions of $\mathbf{v}$ and $\mathbf{H}$ equation (41.9) reduces to the form

$$\frac{d\boldsymbol{\zeta}}{dt} = \frac{2\mu m}{\varepsilon} [\boldsymbol{\zeta} \mathbf{H}].$$

i.e. $\boldsymbol{\zeta}$ precesses about the common direction of $\mathbf{v}$ and $\mathbf{H}$ with the angular velocity $-2\mu m \mathbf{H}/\varepsilon$.

3. The same for motion in a uniform electric field.

Solution. Let the field $\mathbf{E}$ be directed along the x -axis, and the motion takes place in the xy -plane (with $p_y = \text{const}$). From (41.9) we find that the vector $\boldsymbol{\zeta}$ precesses about the z -axis with the instantaneous angular velocity

$$-\left(\frac{e}{\varepsilon + m} + 2\mu'\right) E \frac{p_y}{\varepsilon}.$$

From (41.11) we find that ζ_1 rotates relative to $\mathbf{v}$ with the instantaneous angular velocity

$$\frac{2v_y}{v^2} \left(\frac{\mu m^2}{\varepsilon^2} - \mu' \right) E = \frac{p_y}{\varepsilon} \left(\frac{em}{p^2} - 2\mu' \right) E.$$

§ 42. Scattering of neutrons in an electric field

In collisions of neutrons with nuclei, scattering at large angles is determined by the principal interaction—nuclear forces. At scattering to small angles, however, the electromagnetic interaction of the magnetic moment of the neutron with the electric field of the nucleus becomes significant (*J. Schwinger*, 1948).

We shall assume the neutron non-relativistic, so that the interaction to be considered is described by the approximate Hamiltonian (41.13). The entire magnetic moment of an electrically neutral particle is “anomalous”, and the operator $\hat{H}'$ reduces in this case to the operator of kinetic energy³²:

$$\hat{H} = -\frac{\hbar^2}{2m} \Delta + i \frac{\mu \hbar}{mc} \boldsymbol{\sigma} [\mathbf{E} \nabla]. \quad (42.1)$$

³²In this section we use ordinary units, and the letter m denotes the neutron mass.

In view of the smallness of the electromagnetic interaction of the neutron, the scattering amplitude f_{em} caused by it can be computed in the Born approximation:

$$f_{\text{em}} = -\frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{p}' \cdot \mathbf{r}/\hbar} \left(i \frac{\mu\hbar}{mc} \boldsymbol{\sigma} [\mathbf{E}\nabla] \right) e^{i\mathbf{p} \cdot \mathbf{r}/\hbar} d^3x \quad (42.2)$$

(see III, § 126), or

$$f_{\text{em}} = \frac{\mu}{qc\hbar^3} \boldsymbol{\sigma} [\mathbf{E}_{\mathbf{q}} \mathbf{p}], \quad \mathbf{E}_{\mathbf{q}} = \int \mathbf{E}(\mathbf{r}) e^{-i\mathbf{q} \cdot \mathbf{r}} d^3x$$

($\mathbf{p}, \mathbf{p}'$ are the momenta of the neutron before and after the scattering, $\hbar\mathbf{q} = \mathbf{p}' - \mathbf{p}$). In the notation written, the scattering amplitude f_{em} is an operator with respect to the spin variable.

Before proceeding with the further calculation, let us make the following remark. Formula (42.1) was derived in the preceding section for slowly varying fields (which in fact meant the neglect of derivatives of the field in the Hamiltonian, terms containing them). In applications to the Coulomb field of a nucleus this means that the wavelength $\hbar/p$ must be small compared with the distances essential in the integral $\mathbf{E}_{\mathbf{q}}$, i.e. with the

distances $r \sim 1/q$. Hence $\hbar q \ll p$, so that the scattering angle $\theta \sim \hbar q/p \ll 1$. Thus, the required condition is indeed satisfied precisely for scattering at small angles.

For the Coulomb field with potential $\Phi = Ze/r$ the Fourier component of the field strength

$$\mathbf{E}_{\mathbf{q}} = -i\mathbf{q}\Phi_q = -i\mathbf{q} \cdot 4\pi Ze/q^2$$

(see II, (51.5)). Substitution into (42.2) gives

$$f_{\text{em}} = i \frac{2Ze\mu}{q^2 c \hbar^3} (\boldsymbol{\sigma} [\mathbf{p} \mathbf{p}']).$$

At small scattering angles $\hbar q \approx p\theta$, $[\mathbf{p} \mathbf{p}'] \approx p^2\theta\mathbf{v}$, where $\mathbf{v}$ is the unit vector in the direction $[\mathbf{p} \mathbf{p}']$. Thus,

$$f_{\text{em}} = i \frac{2Ze\mu}{\theta \hbar c} \boldsymbol{\sigma} \mathbf{v}.$$

To this expression one should add the amplitude of nuclear scattering. Owing to the rapid fall-off of the nuclear forces with distance, this amplitude tends at small angles to a finite (depending on the energy) complex value, which we denote by a . Therefore the total scattering amplitude

$$f = a + i \frac{b}{\theta} \boldsymbol{\sigma} \mathbf{v}, \quad b = \frac{2Ze\mu}{\hbar c} = 2Z\alpha \frac{\mu}{e}. \quad (42.3)$$

We see that the electromagnetic scattering indeed becomes dominant at sufficiently small angles. The form of expression (42.3) coincides with that considered in III, § 140. We can therefore directly use the formulae derived there. The cross section, summed over all possible final polarisation states:

$$\frac{d\sigma}{d\Omega} = |a|^2 + \frac{b^2}{\theta^2} + 2b \operatorname{Im} a \cdot \mathbf{v} \boldsymbol{\zeta}, \quad (42.4)$$

where ζ is the initial polarisation of the neutron beam (P see III, § 140). If the initial state is non-polarised ($\zeta = 0$), the polarisation after scattering is

$$\zeta' = \frac{2b \operatorname{Im} a \cdot \theta}{|a|^2 \theta^2 + b^2} \nu. \quad (42.5)$$

This polarisation is maximal at $\theta = b/|a|$, with $\zeta'_{\max} = \operatorname{Im} a/|a|$.

Radiation

§ 43. The operator of electromagnetic interaction

The interaction of electrons with the electromagnetic field can, as a rule, be treated by perturbation theory. This circumstance is connected with the comparative weakness of the electromagnetic interaction, expressed in the smallness of the corresponding dimensionless “coupling constant”—the fine-structure constant $\alpha = e^2/\hbar c = 1/137$. This smallness plays a fundamental role in quantum electrodynamics.

In classical electrodynamics (see II, §§ 16, 28) the electromagnetic interaction is described by the term

$$-e j^\mu A_\mu \quad (43.1)$$

in the Lagrangian density of the system “field + charges” (A is the 4-potential of the field, j the 4-vector of current density of the particles). The current density satisfies the continuity equation

$$\partial_\mu j^\mu = 0, \quad (43.2)$$

expressing the law of conservation of charge. Recall (see II, § 29) that the gauge invariance of the theory is closely connected with this law. Indeed, under the replacement $A_\mu \rightarrow A_\mu + \partial_\mu \chi$ (4.1) the quantity $-e j^\mu \partial_\mu \chi$ is added to the Lagrangian density (43.1), which by virtue of (43.2) can be written in the form of a 4-divergence

$$-e \partial_\mu (\chi j^\mu)$$

and therefore drops out upon integration over d^4x in the action $S = \int L d^4x$.

In quantum electrodynamics the 4-vectors j and A are replaced by the corresponding secondarily quantised operators. The operator of the current is expressed through the ψ -operators according to $\hat{j} = \hat{\bar{\psi}} \gamma \hat{\psi}$. The role of the generalised “coordinates” q in the Lagrangian

$$\int \hat{L}_{\text{int}} d^3x = -e \int (\hat{j} \hat{A}) d^3x$$

is played by the values of $\hat{\psi}, \hat{\bar{\psi}}, \hat{A}$ at each point of space. Since the Lagrangian density turns out to depend only on the “coordinates” q themselves (and not on their derivatives with respect to x), the transition from the Lagrangian density to the Hamiltonian density according to formula (10.11) reduces merely to a change of sign of the Lagrangian density¹. Thus, the operator of electromagnetic interaction (the integral over space of the Hamiltonian density of the interaction) has the form

$$\hat{V} = e \int (\hat{j} \hat{A}) d^3x. \quad (43.3)$$

¹Independently of these arguments we note that if one is speaking only of a first-order correction, then any small perturbation enters the Hamiltonian with the same sign as the Lagrangian (see I, § 40).

The operator of the free electromagnetic field is a sum

$$\hat{A} = \sum_n [\hat{c}_n A_n(x) + \hat{c}_n^\dagger A_n^*(x)], \quad (43.4)$$

containing the creation and annihilation operators of photons in various states (labelled by the index n). Each of them has matrix elements only for increasing or decreasing the corresponding occupation number N_n by 1 (with all remaining occupation numbers unchanged). Therefore the operator $\hat{A}$ also has matrix elements only for transitions with a change in the number of photons by 1. In other words, in the first approximation of perturbation theory only processes of single emission or absorption of a photon arise.

According to (2.15) the matrix elements

$$\langle N_n - 1 | \hat{c}_n | N_n \rangle = \langle N_n | \hat{c}_n^\dagger | N_n - 1 \rangle = \sqrt{N_n}. \quad (43.5)$$

If in the initial state of the field photons (of sort n) are absent, then $\langle 1 | \hat{c}_n^\dagger | 0 \rangle = 1$. The matrix element of the operator (43.3) for the emission of a photon is

$$V_{fi}(t) = e \int (j_{fi} A_n^*) d^3x, \quad (43.6)$$

where $A_n(x)$ is the wave function of the emitted photon, and j_{fi} is the matrix element of the operator $\hat{j}$ for the transition of the emitter from the initial state i to the final state f ². The 4-vector $j_{fi}^\mu = (\rho_{fi}, \mathbf{j}_{fi})$ is called the *transition current*.

Analogously, the matrix element for the absorption of a photon is

$$V_{fi}(t) = e \int (j_{fi} A_n) d^3x. \quad (43.7)$$

It differs from (43.6) only in that $A_n^*(x)$ is replaced by $A_n(x)$.

By indicating the argument t in V_{fi} we emphasise that the time-dependent matrix element is meant. Separating in the wave functions the time factors, one can pass in the usual way to the time-independent matrix elements:

$$V_{fi}(t) = V_{fi} e^{-i(E_i - E_f \mp \omega)t}, \quad (43.8)$$

(E_i, E_f are the initial and final energies of the emitting system; the signs $\mp$ correspond to emission and absorption of the photon ω).

The wave function of a photon with a definite momentum $\mathbf{k}$ and definite polarisation is

$$A^\mu = \sqrt{4\pi} \sqrt{\frac{1}{2\omega}} e^\mu e^{i\mathbf{k}\mathbf{r}} \quad (43.9)$$

(see (4.3); the time factor is omitted). Substituting in (43.6), we find the matrix element for the emission of such a photon in the form

$$V_{fi} = e \sqrt{4\pi} \sqrt{\frac{1}{2\omega}} e_\mu^* j_{fi}^\mu(\mathbf{k}), \quad (43.10)$$

²The notation in (43.6) contains a certain inconsistency: the indices of V_{fi} refer to the states of the entire system "emitter + field", while those of j_{fi} refer to the states of the emitter alone.

where $j_{fi}(\mathbf{k})$ is the transition current in the momentum representation, i.e. the Fourier components

$$j_{fi}(\mathbf{k}) = \int j_{fi}(\mathbf{r}) e^{-i\mathbf{k}\mathbf{r}} d^3x. \quad (43.11)$$

The analogous formula for the absorption of a photon:

$$V_{fi} = e\sqrt{4\pi} \sqrt{\frac{1}{2\omega}} e_\mu j_{fi}^\mu(-\mathbf{k}). \quad (43.12)$$

The law of conservation of current in the momentum representation is written as the condition of 4-transversality of the transition currents:

$$k_\mu j_{fi}^\mu(\mathbf{k}) = \omega \rho_{fi}(\mathbf{k}) - \mathbf{k} \cdot \mathbf{j}_{fi}(\mathbf{k}) = 0. \quad (43.13)$$

The formulae written in this section, in which the form of the current operator is not specified, have a general character and are valid for electromagnetic processes involving any charged particles. The existing theory provides the means of establishing the form of the current operator (and thereby of computing in principle its matrix elements) only for electrons. In application to strongly interacting particle systems (including nuclei) we shall confine ourselves to a phenomenological theory, in which the transition currents appear as quantities taken from experiment, satisfying only the general requirements of space–time symmetry and the continuity equation.

§ 44. Emission and absorption

The probability of a transition under the influence of a perturbation $\hat{V}$ in the first approximation is given by known formulae of perturbation theory (III, § 42). Let the initial and final states of the emitting system belong to the discrete spectrum³. Then the probability (per unit time) of the transition $i \rightarrow f$ with emission of a photon is

$$dw = 2\pi |V_{fi}|^2 \delta(E_i - E_f - \omega) d\nu, \quad (44.1)$$

where ν conventionally denotes the set of quantities characterising the state of the photon and spanning a continuous range of values (the wave function of the photon being normalised to a δ -function “on the ν scale”). If a photon with a definite value of the angular momentum is emitted, the only continuously varying quantity is the frequency ω . Integration of formula (44.1) over $d\nu \equiv d\omega$ eliminates the δ -function (replacing ω by $E_i - E_f$), and the transition probability becomes

$$w = 2\pi |V_{fi}|^2. \quad (44.2)$$

If instead the emission of a photon with a given momentum $\mathbf{k}$ is considered, then $d\nu = d^3k/(2\pi)^3 = \omega^2 d\omega d\Omega/(2\pi)^3$. It is assumed that the wave function of the photon (a plane wave) is normalised to one photon in a volume $V = 1$, as is always adopted

³By this we mean, in any event, neglect of the recoil: the emitter as a whole remains stationary.

in this book; $d\nu$ is the number of states in the phase-space volume $V d^3k$. Thus, the probability of emission of a photon with a given momentum is

$$dw = 2\pi |V_{fi}|^2 \delta(E_i - E_f - \omega) \frac{d^3k}{(2\pi)^3}, \quad (44.3)$$

or after integrating over $d\omega$:

$$dw = \frac{1}{4\pi^2} |V_{fi}|^2 \omega^2 d\omega. \quad (44.4)$$

Here the matrix element V_{fi} from (43.10) should be substituted.

In the following sections we shall use these formulae for computing the probability of radiation in various concrete cases. Here we shall consider some general relations among different kinds of radiative processes.

If in the initial state of the field there already existed a non-zero number N_n of the given photons, the matrix element of the transition is multiplied further by

$$\langle N_n + 1 | \hat{c}_n^+ | N_n \rangle = \sqrt{N_n + 1}, \quad (44.5)$$

i.e. the probability of the transition increases by a factor of $N_n + 1$. The unity in this factor corresponds to *spontaneous* emission, occurring also when $N_n = 0$. The term N_n gives rise to *stimulated* (or *induced*) emission: we see that the presence of photons in the initial state of the field stimulates additional emission of photons of the same kind.

The matrix element V_{if} for the reverse transition of the system ($f \rightarrow i$) differs from the element V_{fi} by the replacement of (43.5) by

$$\langle N_n - 1 | \hat{c}_n | N_n \rangle = \sqrt{N_n}$$

(and the replacement of all remaining quantities by their complex conjugates). This reverse transition represents the absorption of a photon by the system, going from level E_f to level E_i . Between the probabilities of emission and absorption of a photon (for a given pair of states i, f) the following important relation therefore holds⁴:

$$\frac{w^{(\text{emis})}}{w^{(\text{abs})}} = \frac{N_n + 1}{N_n} \quad (44.6)$$

(first pointed out by *A. Einstein* in 1916).

Let us relate the number of photons to the intensity of the radiation incident on the system from outside. Let

$$I_{\mathbf{ke}} d\omega d\Omega \quad (44.7)$$

be the energy of radiation falling per unit area per unit time, with polarisation $\mathbf{e}$, in the frequency interval $d\omega$ and in the direction of the wave vector within the solid angle $d\Omega$.

Let us note the analogous probabilities for induced emission and absorption. According to (44.6) and (44.8) these probabilities are related to each other by the following equations:

$$dw_{\mathbf{ke}}^{(\text{abs})} = dw_{\mathbf{ke}}^{(\text{ind})} = dw_{\mathbf{ke}}^{(\text{sp})} \frac{8\pi^2 c^2}{\hbar \omega^3} I_{\mathbf{ke}}. \quad (44.9)$$

⁴By this we mean, in any event, neglect of the recoil: the emitter as a whole remains stationary.

If the incident radiation is isotropic and unpolarised ($I_{\mathbf{k}\mathbf{e}}$ does not depend on the directions of $\mathbf{k}$ and $\mathbf{e}$), integrating (44.9) over $d\omega$ and summing over $\mathbf{e}$ yield analogous relations between total probabilities of radiative transitions (between given states i and f of the system):

$$w^{(\text{abs})} = w^{(\text{ind})} = w^{(\text{sp})} \frac{\pi^2 c^2}{\hbar \omega^3} I, \quad (44.10)$$

where $I = 2 \cdot 4\pi I_{\mathbf{k}\mathbf{e}}$ is the total spectral intensity of the incident radiation.

If the states i and f of the emitting (or absorbing) system are degenerate, the total probability of emission (or absorption) of given photons is obtained by summation over all mutually degenerate final states and averaging over all possible initial states. Denoting the multiplicities of degeneracy (statistical weights) of the states i and f by g_i and g_f respectively, for the processes of spontaneous and induced emission the initial states are the states i , and for absorption the states f . Assuming in each case that all g_i or g_f initial states are equally probable, we obtain, obviously, instead of (44.10), the following relations:

$$g_f w^{(\text{abs})} = g_i w^{(\text{ind})} = g_i w^{(\text{sp})} \frac{\pi^2 c^2}{\hbar \omega^3} I. \quad (44.11)$$

In the literature the so-called *Einstein coefficients* are frequently used, defined as

$$A_{if} = w^{(\text{sp})}, \quad B_{if} = w^{(\text{ind})} \frac{c}{I}, \quad B_{fi} = w^{(\text{abs})} \frac{c}{I} \quad (44.12)$$

(the quantity I/c is the spatial spectral density of the energy of the radiation). They are related to each other by the equations

$$g_f B_{fi} = g_i B_{if} = g_i A_{if} \frac{\pi^2 c^3}{\hbar \omega^3}. \quad (44.13)$$

§ 45. Dipole radiation

Let us apply the formulae obtained to the emission of a photon by a relativistic electron in a given external field. The transition current in this case is the matrix element of the operator

$$\hat{j} = \hat{\bar{\psi}} \gamma \hat{\psi},$$

in which the ψ -operators are assumed to be expanded over the system of wave functions of stationary states of the electron in the given field (see § 32). The transition of the electron from state i to state f is described by the matrix element $\langle 0_i 1_f | j | 1_i 0_f \rangle$. The change in the occupation numbers is effected by the operator $\hat{a}_f^+ \hat{a}_i$, and for the transition current we obtain

$$j_{fi}^\mu = \bar{\psi}_f \gamma^\mu \psi_i = (\psi_f^* \alpha \psi_i), \quad (45.1)$$

where ψ_i and ψ_f are the wave functions of the initial and final states of the electron.

We choose the wave function of the photon in the three-dimensionally transverse gauge (4-vector of polarisation $e = (0, \mathbf{e})$). Then from (43.10) the product $j_{fi} e^* = -\mathbf{j}_{fi} \cdot \mathbf{e}^*$.

Substituting V_{fi} from (44.4), we obtain the following formula for the probability of radiation (per unit time) into the element of solid angle do of a photon with polarisation $\mathbf{e}$:

$$dw_{\mathbf{en}} = e^2 \frac{\omega}{2\pi} |\mathbf{e}^* \cdot \mathbf{j}_{fi}(\mathbf{k})|^2 do, \quad (45.2)$$

where

$$\mathbf{j}_{fi}(\mathbf{k}) = \int \psi_f^* \boldsymbol{\alpha} \psi_i e^{-i\mathbf{k}\mathbf{r}} d^3x. \quad (45.3)$$

Summation over the polarisations of the photon is carried out by averaging over the directions of $\mathbf{e}$ (in the plane perpendicular to the given direction $\mathbf{n} = \mathbf{k}/\omega$), after which the result is multiplied by 2 to account for the two independent possibilities for transverse polarisation of the photon⁵. Thus we obtain the formula

$$dw_{\mathbf{n}} = e^2 \frac{\omega}{2\pi} |[\mathbf{n} \mathbf{j}_{fi}(\mathbf{k})]|^2 do. \quad (45.4)$$

A very important case is when the wavelength of the photon λ is large compared with the dimensions of the radiating system a . This situation is usually connected with the smallness of the velocities of particles compared with the speed of light. In the first approximation in a/λ (corresponding to *dipole radiation*—cf. II, § 67) in the transition current (45.3) one can replace the slowly varying factor $e^{-i\mathbf{k}\mathbf{r}}$ by unity in the region where ψ_i or ψ_f are appreciably different from zero. This replacement means, in other words, the neglect of the momentum of the photon compared with the momenta of the particles in the system. In the same approximation the integral $\mathbf{j}_{fi}(0)$ may be replaced by its non-relativistic expression, i.e. simply by the matrix element $\mathbf{v}_{fi}$ of the velocity of the electron with respect to the eigenfunction wave functions. In turn this element $\mathbf{v}_{fi} = -i\omega \mathbf{r}_{fi} = \mathbf{d}_{fi}$, where $\mathbf{d}$ is the dipole moment of the electron (in its orbital motion). Thus, we find the following formula for the probability of dipole radiation:

$$dw_{\mathbf{en}} = \frac{\omega^3}{2\pi} |\mathbf{e}^* \cdot \mathbf{d}_{fi}|^2 do \quad (45.5)$$

(the direction $\mathbf{n}$ appears here in an implicit form: the vector $\mathbf{e}$ must be perpendicular to $\mathbf{n}$). Summing over polarisations, we get

$$dw_{\mathbf{n}} = \frac{\omega^3}{2\pi} |[\mathbf{n} \mathbf{d}_{fi}]|^2 do. \quad (45.6)$$

In view of the non-relativistic (with respect to the electron) character of these formulae, their generalisation to arbitrary electronic systems is obvious: under $\mathbf{d}_{fi}$ one should understand the matrix element of the total dipole moment of the system.

⁵For the averaging one uses the formula

$$\overline{e_i e_k^*} = \frac{1}{2} (\delta_{ik} - n_i n_k) \quad (45.4a)$$

or

$$(\mathbf{ae})(\mathbf{be}^*) = \frac{1}{2} \{ \mathbf{ab} - (\mathbf{an})(\mathbf{bn}) \} = \frac{1}{2} [\mathbf{an}][\mathbf{bn}], \quad (45.4b)$$

where $\mathbf{a}$, $\mathbf{b}$ are constant vectors.

Integrating formula (45.6) over all directions, we find the total probability of radiation:

$$w = \frac{4\omega^3}{3} |\mathbf{d}_{fi}|^2, \quad (45.7)$$

or in ordinary units:

$$w = \frac{4\omega^3}{3\hbar c^3} |\mathbf{d}_{fi}|^2. \quad (45.7a)$$

The intensity I of the radiation is obtained by multiplying the probability by $\hbar\omega$:

$$I = \frac{4\omega^4}{3c^3} |\mathbf{d}_{fi}|^2. \quad (45.8)$$

This formula reveals a direct analogy with the classical formula (see II, (67.11)) for the intensity of dipole radiation by a system of periodically moving particles: the radiation has the frequency $\omega_s = s\omega$ (where ω is the frequency of the motion of the particles, s an integer), and the intensity is

$$I_s = \frac{4\omega_s^4}{3c^3} |\mathbf{d}_s|^2, \quad (45.9)$$

where $\mathbf{d}_s$ are the Fourier components of the dipole moment, i.e. the coefficients of the expansion

$$\mathbf{d}(t) = \sum_{s=-\infty}^{\infty} \mathbf{d}_s e^{-is\omega t}. \quad (45.10)$$

The quantum formula (45.8) is obtained from (45.9) by replacing these Fourier components by the matrix elements of the corresponding transitions. This rule (expressing the *correspondence principle* of Bohr) is a special case of the general correspondence between Fourier components of classical quantities and quantum matrix elements of the corresponding transitions (see III, § 48). The radiation is quasiclassical for transitions between states with large quantum numbers; the frequency $\hbar\omega = E_i - E_f$ is then small compared with the energies E_i and E_f . This circumstance, however, does not lead to any changes in the form of formula (45.8), valid for any transitions. By this is explained the (in a certain sense accidental) fact that the correspondence principle for the intensity of radiation turns out to be valid not only in the quasiclassical, but also in the general quantum case.

§ 46. Electric multipole radiation

Instead of considering the emission of a photon in a given direction (i.e. with a given momentum), let us now consider the emission of a photon with definite values of the angular momentum j and its projection m onto some chosen direction z . We saw in § 6 that such photons can be of two types—electric and magnetic; we begin with the emission of photons of the electric type.

It is convenient here to carry out the calculations using the wave functions of the photon in the momentum representation, i.e. representing the 4-vector $A^\mu(\mathbf{r})$ in the form

of a Fourier integral. Then the matrix element

$$V_{fi} = e \int d^3x \cdot j_{fi}^\mu(\mathbf{r}) \int \frac{d^3k}{(2\pi)^3} A^*(\mathbf{k}) e^{-i\mathbf{k}\mathbf{r}}. \quad (46.1)$$

(to simplify the notation we omit the indices ωjm of the wave functions of the photon).

For an Ej -photon we take the wave function from (7.10), choosing the arbitrary constant C to be

$$C = -\sqrt{\frac{j+1}{j}}.$$

The purpose of this choice is so that in the spatial components of the wave function (**A**) the terms containing spherical functions of order $j-1$ cancel out (as can be seen from formulae (7.16)). The vector **A** will then contain only spherical functions of order $j+1$, as a result of which the corresponding contribution to V_{fi} turns out (as will be clear from the subsequent calculation) to be of higher order of smallness (in a/λ) than the contribution from the component $A^0 \equiv \Phi$, containing spherical functions of lower order j .

Thus, we put

$$A^\mu = (\Phi, 0), \quad \Phi = -\sqrt{\frac{j+1}{j}} \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) Y_{jm}(\mathbf{n})$$

($\mathbf{n} = \mathbf{k}/\omega$). Substituting this into (46.1) and integrating over $|\mathbf{k}|$, we get

$$V_{fi} = -e \sqrt{\frac{j+1}{j}} \frac{\sqrt{\omega}}{2\pi} \int d^3x \cdot \rho_{fi}(\mathbf{r}) \int d\mathbf{o}_n e^{-i\mathbf{k}\mathbf{r}} Y_{jm}^*(\mathbf{n}). \quad (46.2)$$

For computing the inner integral we use the expansion (24.12), writing it in the form

$$e^{i\mathbf{p}\mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l g_l(kr) Y_{lm}^*\left(\frac{\mathbf{k}}{k}\right) Y_{lm}\left(\frac{\mathbf{r}}{r}\right), \quad (46.3)$$

where

$$g_l(kr) = \sqrt{\pi/(2kr)} J_{l+1/2}(kr) \quad (46.4)$$

(see III, (33.3))⁶. Substituting this expansion into (46.2), we get

$$\int e^{-i\mathbf{k}\mathbf{r}} Y_{jm}^*(\mathbf{n}) d\mathbf{o}_n = 4\pi i^{-j} g_j(kr) Y_{jm}^*\left(\frac{\mathbf{r}}{r}\right),$$

(the remaining terms vanish by the orthogonality of the spherical functions). In view of the condition $a/\lambda \ll 1$ the integral in (46.2) is restricted to distances r for which $kr \ll 1$.

⁶The normalisation of the functions g_l is such that their asymptotic form at $kr \rightarrow \infty$ is:

$$g_l(kr) \approx \frac{\sin(kr - \pi l/2)}{kr}. \quad (46.4a)$$

We can therefore replace the functions $g_j(kr)$ by the first terms of their expansions in powers of kr ⁷:

$$g_j(kr) \approx \frac{(kr)^j}{(2j+1)!!}. \quad (46.5)$$

As a result we obtain

$$V_{fi} = (-1)^{m+1} i^j \sqrt{\frac{(2j+1)(j+1)}{\pi j}} \frac{\omega^{j+1/2}}{(2j+1)!!} e(Q_{j,-m}^{(\mathcal{E})})_{fi}, \quad (46.6)$$

where we have introduced the quantities

$$(Q_{jm}^{(\mathcal{E})})_{fi} = \sqrt{\frac{4\pi}{2j+1}} \int \rho_{fi}(\mathbf{r}) r^j Y_{jm}\left(\frac{\mathbf{r}}{r}\right) d^3x \quad (46.7)$$

(recall that $Y_{j,-m} = (-1)^{j-m} Y_{jm}^*$). The quantities (46.7) are called the 2^j -pole electric moments of the transition of the system, by analogy with the corresponding classical quantities (II, § 41)⁸.

For an electron in an external field $\rho_{fi} = \psi_f^* \psi_i$, and the quantities (46.7) are then computed as matrix elements of the classical quantity

$$Q_{jm}^{(\mathcal{E})} = \sqrt{\frac{4\pi}{2j+1}} r^j Y_{jm}.$$

In the non-relativistic (in terms of particle velocity) case, the transition moment can in principle be computed analogously for any system of N interacting particles. The transition charge density is then expressed through the wave functions of the system in the form

$$\rho_{fi}(\mathbf{r}) = \int \psi_f^*(\mathbf{r}_1, \dots, \mathbf{r}_N) \psi_i(\mathbf{r}_1, \dots, \mathbf{r}_N) \sum_{n=1}^N \delta(\mathbf{r} - \mathbf{r}_n) \cdot d^3x_1 \dots d^3x_N, \quad (46.8)$$

where the integral is taken over the entire configuration space⁹.

The wave function of the photon we have used corresponds (in the coordinate representation) to normalisation on a δ -function on the ω -scale, as assumed in formula (44.2). Substituting into it (46.6), we obtain the probability of Ej -radiation¹⁰

$$w_{jm}^{(\mathcal{E})} = \frac{2(2j+1)(j+1)}{j[(2j+1)!!]^2} \omega^{2j+1} e^2 |(Q_{j,-m}^{(\mathcal{E})})_{fi}|^2. \quad (46.9)$$

⁷The power of kr coincides with the order of the spherical function Y_{jm} in the product with which g_j appears, which justifies the neglect of the terms in $\mathbf{A}$ containing spherical functions of higher order.

⁸We define the multipole moments without the factor e , in correspondence with the fact that in this book the current is also defined without the charge factor.

⁹A situation is possible when the transition probability given by approximate selection rules vanishes; in such a case, to obtain a result different from zero one must use wave functions with a relativistic correction that takes into account the spin-orbit interaction.

¹⁰At first glance it might appear that by virtue of the spatial isotropy the total probability of emission of a photon should not depend on the value of m . This is not so, as is easy to see from the fact that for emission of photons with different values of m the final states of the system must be different (for a given initial state); cf. selection rule (46.16) below.

In particular, for $j = 1$ we have

$$w_{1m}^{(\mathcal{E})} = \frac{4\omega^3}{3} e^2 |(Q_{1,-m}^{(\mathcal{E})})_{fi}|^2. \quad (46.10)$$

The quantities $Q_{1m}^{(\mathcal{E})}$ are related to the components of the vector of the electric dipole moment by the formulae

$$eQ_{10}^{(\mathcal{E})} = id_z, \quad eQ_{1,\pm 1}^{(\mathcal{E})} = \mp \frac{i}{\sqrt{2}} (d_x \pm id_y). \quad (46.11)$$

Summing (46.10) over the values of m , we return, as expected, to the already known formula (45.7) for the total probability of dipole radiation.

The angular distribution of multipole radiation is determined by formula (7.11). Normalising it to the total probability of emission w_{jm} , we have

$$dw_{jm} = |\mathbf{Y}_{jm}^{(\mathcal{E})}(\mathbf{n})|^2 w_{jm} d\Omega = \frac{w_{jm}}{j(j+1)} |\nabla_{\mathbf{n}} Y_{jm}|^2 d\Omega. \quad (46.12)$$

In particular, for $j = 1$

$$Y_{10} = i\sqrt{\frac{3}{4\pi}} \cos \theta, \quad Y_{1,\pm 1} = \mp i\sqrt{\frac{3}{8\pi}} \sin \theta \cdot e^{\pm i\varphi},$$

where θ, φ are the polar angle and the azimuth of the direction $\mathbf{n}$ relative to the z -axis. Computing the gradient, we find that the angular distribution of dipole radiation with definite values of m is given by the expressions

$$dw_{10} = w_{10} \frac{3}{8\pi} \sin^2 \theta d\theta, \quad dw_{1,\pm 1} = w_{1,\pm 1} \frac{3}{8\pi} \frac{1 + \cos^2 \theta}{2} d\Omega. \quad (46.13)$$

These could, of course, also be obtained directly from formula (45.6), setting in it once (for $m = 0$): $d_x = d_y = 0$, $d_z = d$, and another time ($m = \pm 1$): $d_y = \mp id_x = d/\sqrt{2}$, $d_z = 0$.

If the order of magnitude of the dimensions of the system (atom or nucleus) is a , then the order of magnitude of the electric multipole moments is, generally speaking, $Q_{jm}^{(\mathcal{E})} \sim a^j$. The probability of multipole radiation is therefore

$$w_{jm}^2 \sim \alpha k (ka)^{2j}. \quad (46.14)$$

Increasing the multipole order by 1 diminishes the probability of radiation by a factor $\sim (ka)^2$.

The laws of conservation of angular momentum and parity lead to definite *selection rules*, restricting the possible changes in the state of the radiating system. If the initial angular momentum of the system equals J_i , then after emission of a photon with angular momentum j the angular momentum

of the system can take only the values J_f determined by the rule of addition of angular momenta ($\mathbf{J}_i - \mathbf{J}_f = \mathbf{j}$):

$$|J_i - J_f| \leq j \leq J_i + J_f. \quad (46.15)$$

For given values of J_i and J_f the same rule (46.15) determines the possible values of the photon angular momentum j . But since the probability of radiation rapidly diminishes with increasing j , the radiation occurs predominantly with the lowest possible multipole order.

The projections M_i and M_f of the angular momenta $\mathbf{J}_i$ and $\mathbf{J}_f$ together with the projection m of the photon angular momentum satisfy the obvious (from the same law of addition of angular momenta) rule

$$M_i - M_f = m. \quad (46.16)$$

The parities P_i and P_f of the initial and final states of the radiating system must satisfy the condition $P_f P_\Phi = P_i$, where P_Φ is the parity of the emitted photon; since the parities can only take the values ± 1 , this condition can also be written in the form

$$P_i P_f = P_\Phi. \quad (46.17)$$

For a photon of the electric type $P_\Phi = (-1)^j$, so that the parity selection rule for electric multipole radiation is:

$$P_i P_f = (-1)^j. \quad (46.18)$$

The selection rules for angular momentum and parity are completely rigorous and must be obeyed in radiation by any systems. Alongside these rules there may exist other, more restrictive ones, connected with specific structural features of particular radiating systems. Such rules inevitably have a more or less approximate character; we shall consider them in subsequent sections of this chapter.

The dependence of the probability of emission on the quantum numbers m , M_i , M_f is determined entirely by the tensor character of the multipole moments. The quantities Q_{jm} with a given j constitute a spherical tensor of rank j . The dependence of their matrix elements on the indicated quantum numbers is given by the formula

$$|\langle n_f J_f M_f | Q_{j,-m} | n_i J_i M_i \rangle|^2 = \begin{pmatrix} J_f & j & J_i \\ M_f & m & -M_i \end{pmatrix}^2 |\langle n_f J_f || Q_j || n_i J_i \rangle|^2 \quad (46.19)$$

(see III, (107.6)), where the letter n conventionally denotes the set of remaining quantum numbers, besides J and M , of the state of the system. The reduced matrix elements standing on the right-hand side of (46.19) do not depend on the numbers m , M_i , M_f . Substituting into (46.9) this formula, we determine the desired dependence, which turns out to be proportional to

$$\begin{pmatrix} J_f & j & J_i \\ M_f & m & -M_i \end{pmatrix}^2$$

(it is assumed, of course, that the emitter is not located in an external field; then the transition frequency ω does not depend on the numbers M_i and M_f).

Summing the probability over all values M_f (for given M_i), we obtain the total probability of emission of a photon of given frequency from the initial level $n_i J_i$. By

virtue of the spatial isotropy it is obvious that this quantity will not depend also on the initial value of M_i . The summation is carried out using the formula

$$\sum_{M_f} |\langle n_f J_f M_f | Q_{j,-m} | n_i J_i M_i \rangle|^2 = \frac{1}{2J_i + 1} |\langle n_f J_f || Q_j || n_i J_i \rangle|^2 \quad (46.20)$$

(see III, (107.11)).

§ 47. Magnetic multipole radiation

The wave function of a photon of the magnetic type $A^\mu = (0, \mathbf{A})$, where $\mathbf{A}$ is given by formula (7.6). Substituting it into (46.1), we obtain for the matrix element of the transition

$$V_{fi} = -e \frac{\sqrt{\omega}}{2\pi} \int d^3x \cdot \mathbf{j}_{fi}(\mathbf{r}) \int d\mathbf{o}_n \cdot e^{-i\mathbf{k}\mathbf{r}} \mathbf{Y}_{jm}^{(M)*}(\mathbf{n}). \quad (47.1)$$

The components of the vector $\mathbf{Y}_{jm}^{(M)}$ are expressed according to (7.16) through spherical functions of order j . Substituting the expansion (46.3) into (46.2) and integrating over $d\mathbf{o}_n$, we get

$$\int e^{-i\mathbf{k}\mathbf{r}} \mathbf{Y}_{jm}^{(M)*}(\mathbf{n}) d\mathbf{o}_n = 4\pi i^{-j} g_j(kr) \mathbf{Y}_{jm}^{(M)*}\left(\frac{\mathbf{r}}{r}\right),$$

and after substituting g_j from (46.5) ¹¹

$$V_{fi} = -ei^{-j} \frac{2\omega^{j+1/2}}{(2j+1)!!} \int \mathbf{j}_{fi}(\mathbf{r}) r^j \mathbf{Y}_{jm}^{(M)*}\left(\frac{\mathbf{r}}{r}\right) d^3x. \quad (V.1)$$

We transform under the integral sign

$$r^j \mathbf{j}_{fi} \cdot [\mathbf{r} \nabla Y_{jm}^*] = -[\mathbf{r} \mathbf{j}_{fi}] \nabla (r^j Y_{jm}^*)$$

and obtain

$$V_{fi} = (-1)^m i^j \sqrt{\frac{(2j+1)(j+1)}{\pi j}} \frac{\omega^{j+1/2}}{(2j+1)!!} e(Q_{j,-m}^{(M)})_{fi}, \quad (47.2)$$

where we have introduced the quantities

$$(Q_{jm}^{(M)})_{fi} = \frac{1}{j+1} \sqrt{\frac{4\pi}{2j+1}} \int [\mathbf{r} \mathbf{j}_{fi}] \nabla (r^j Y_{jm}) d^3x. \quad (47.3)$$

They are called the 2^j -pole magnetic moments of the transition. In view of the analogy between expressions (47.2) and (46.6), the formula for the probability of emission is obtained from (46.9) merely by replacing the electric moments by magnetic ones. The angular distribution formula (46.12) also remains in force (as was already noted in connection with (7.11)).

¹¹Do not confuse the current $\mathbf{j}$ with the angular momentum j !

Let us examine the structure of expression (47.3) for $j = 1$. In this case the functions

$$\sqrt{\frac{4\pi}{3}} rY_{10} = iz, \quad \sqrt{\frac{4\pi}{3}} rY_{1,\pm 1} = \mp \frac{i}{\sqrt{2}} (x \pm iy),$$

and their gradients are simply the circular unit vectors $\mathbf{e}^{(0)}, \mathbf{e}^{(\pm 1)}$ (7.14). The quantities $e(Q_{1m}^{(M)})_{fi}$ therefore represent the spherical components of the vector

$$\boldsymbol{\mu}_{fi} = \frac{e}{2} \int [\mathbf{r} \mathbf{j}_{fi}] d^3x, \quad (47.4)$$

which by its structure is analogous to the classical magnetic moment (see II, § 44). The total probability of $M1$ -radiation (in ordinary units) is

$$w = \frac{4\omega^3}{3\hbar c^3} |\boldsymbol{\mu}_{fi}|^2. \quad (47.5)$$

Let us show how formula (47.4) is connected with the usual non-relativistic expression for the operator of magnetic moment.

The expression for the transition current (see III, § 115):

$$\mathbf{j}_{fi} = -\frac{i}{2m} (\psi_f^* \nabla \psi_i - \psi_i \nabla \psi_f^*) + \frac{\mu}{e s} \text{rot}(\psi_f^* \widehat{\mathbf{s}} \psi_i), \quad (47.6)$$

where μ is the magnetic moment of the particle, s its spin. Therefore

$$\boldsymbol{\mu}_{fi} = -\frac{ie}{4m} \int \psi_f^* [\mathbf{r} \nabla] \psi_i d^3x + \frac{ie}{4m} \int \psi_i [\mathbf{r} \nabla] \psi_f^* d^3x + \frac{\mu}{2s} \int [\mathbf{r} \text{rot}(\psi_f^* \widehat{\mathbf{s}} \psi_i)] d^3x. \quad (47.7)$$

In the second term we write

$$\int \psi_i [\mathbf{r} \nabla] \psi_f^* d^3x = - \int \psi_f^* [\mathbf{r} \nabla] \psi_i d^3x + \int \text{rot}(\mathbf{r} \psi_f^* \psi_i) d^3x.$$

The last integral transforms into an integral over an infinitely remote surface and vanishes. Thus, the first two terms in (47.7) are the same. In the third term we transform the integral as follows (temporarily denoting $\mathbf{F} = \psi_f^* \widehat{\mathbf{s}} \psi_i$):

$$\int [\mathbf{r} [\nabla \mathbf{F}]] d^3x = \oint [\mathbf{r} [d\mathbf{f} \cdot \mathbf{F}]] - \int [[\mathbf{F} \nabla] \mathbf{r}] d^3x.$$

The integral over the surface vanishes, and in the last integral we have: $[[\mathbf{F} \nabla] \mathbf{r}] = -\mathbf{F} \text{div} \mathbf{r} + \mathbf{F} = -2\mathbf{F}$. Thus,

$$\int [\mathbf{r} \text{rot} \mathbf{F}] d^3x = 2 \int \mathbf{F} d^3x.$$

As a result the expression for $\boldsymbol{\mu}_{fi}$ takes the form

$$\boldsymbol{\mu}_{fi} = \int \psi_f^* \left(\frac{e}{2m} \widehat{\mathbf{L}} + \frac{\mu}{s} \widehat{\mathbf{s}} \right) \psi_i d^3x, \quad (47.8)$$

where $\widehat{\mathbf{L}} = -i[\mathbf{r}\nabla]$ is the orbital angular momentum operator of the particle. As it should be, μ_{fi} turns out to be the matrix element of the operator

$$\widehat{\boldsymbol{\mu}} = \frac{e}{2m} \widehat{\mathbf{L}} + \frac{\mu}{s} \widehat{\mathbf{s}}, \quad (47.9)$$

composed of the operators of the orbital and intrinsic magnetic moments of the particle.

The selection rules for magnetic multipole radiation are analogous to those for the electric case: for the total angular momentum the same rules (46.15), (46.16) hold, and for parity the rule is

$$P_i P_f = (-1)^{j+1}, \quad (47.10)$$

obtained by substituting into (46.17) the parity of the Mj -photon: $P_\Phi = (-1)^{j+1}$.

§48. Angular distribution and polarisation of radiation

The formulae derived in §§ 46 and 47 referred to the emission of a photon with definite values of the angular momentum j and its projection m . Correspondingly, it was assumed that the radiating system (say, a nucleus) both before and after emission possesses not only definite values of the angular momentum J , but also definite polarisations, i.e. values of M .

Let us now consider the more general case of radiation by a partially polarised nucleus (whose dimensions, as before, are assumed small compared with the wavelength). The emitted photon still possesses a definite angular momentum j , but may be partially polarised. We shall find the probability of emission as a function of the direction $\mathbf{n}$ of the photon. It must be expressed through the density matrices describing the polarisation states of the nucleus and the photon.

To this end let us first write the emission probability as a function of the direction $\mathbf{n}$ and the helicity λ of the photon ($\lambda = \pm 1$) for the case when both the initial and final nuclei have definite values: $J_i M_i$ and $J_f M_f$.

The matrix element of a photon with definite jm is proportional to the matrix element of the 2^j -pole (electric or magnetic) moment of the nucleus:

$$\langle J_f M_f; jm | V | J_i M_i \rangle \propto (-1)^m \langle J_f M_f | Q_{j,-m} | J_i M_i \rangle. \quad (48.1)$$

The wave function of the emitted photon (in the momentum representation) is proportional to $\mathbf{Y}_{jm}^{(\mathcal{E})}(\mathbf{n})$ or $\mathbf{Y}_{jm}^{(M)}(\mathbf{n})$. The wave function of a photon with momentum in the direction $\mathbf{n}$ and helicity λ is proportional to the polarisation vector $\mathbf{e}^{(\lambda)}$. The matrix element of emission of a photon $\mathbf{n}\lambda$ is obtained by multiplying (48.1) with the projection of the wave function of the state $|jm\rangle$ onto the wave function of the state $|\mathbf{n}\lambda\rangle$:

$$\langle J_f M_f; \mathbf{n}\lambda | V | J_i M_i \rangle \propto (-1)^m \langle J_f M_f | Q_{j,-m} | J_i M_i \rangle (\mathbf{e}^{(\lambda)*} \mathbf{Y}_{jm}).$$

According to (16.23) for photons of both types

$$\mathbf{e}^{(\lambda)*} \mathbf{Y}_{jm}(\mathbf{n}) \propto D_{\lambda m}^{(j)}(\mathbf{n}). \quad (48.2)$$

The matrix element of the multipole moment is expressed in the usual way through the reduced element. As a result we obtain for the transition amplitude the form

$$\langle J_f M_f; \mathbf{n} \lambda | V | J_i M_i \rangle \propto (-1)^{J_f - M_f + m} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} Q D_{\lambda m}^{(j)}(\mathbf{n}), \quad (48.3)$$

where Q denotes $\langle J_f || Q || J_i \rangle$.

Now we can proceed to the general case of mixed polarisation states. According to the general rules of quantum mechanics the transition probability will be proportional to the expression¹²

$$\begin{aligned} & \sum_{(m)} (-1)^{2J_i - M_i - M'_i} D_{\lambda m}^{(j)} D_{\lambda' m'}^{(j)*} \times \\ & \times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle \times \\ & \times \langle M'_f | \rho^{(f)} | M_f \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle, \end{aligned} \quad (48.4)$$

where $\rho^{(i)}$, $\rho^{(f)}$, $\rho^{(\gamma)}$ are the density matrices of the initial nucleus, the final nucleus and the emitted photon; the symbol (m) under the summation sign means that the summation is taken over all doubly repeated m -indices $(M_i M'_i M_f M'_f \lambda \lambda')$. In (48.4) one must substitute (48.3)¹³.

Let us denote the probability of emission of a photon into the solid angle do by $w(\mathbf{n})do$. The total probability of emission over all directions and over all polarisations of the photon and the recoil nucleus does not depend, obviously, on the initial polarisation state of the nucleus. It is given by formulae already known to us and does not interest us

¹²If the initial and final states of the system are described by superpositions

$$\psi^{(i)} = \sum_n a_n \psi_n^{(i)}, \quad \psi^{(f)} = \sum_m b_m \psi_m^{(f)},$$

then the matrix element

$$\langle f | V | i \rangle = \sum_{mn} b_m^* a_n V_{mn},$$

and its square

$$|\langle f | V | i \rangle|^2 = \sum_{nn' mm'} V_{mn} V_{m'n'}^* a_n a_{n'}^* b_m^* b_{m'}.$$

The transition to mixed states is effected by the replacements

$$a_n a_{n'}^* \rightarrow \rho_{nn'}^{(i)}, \quad b_m b_m^* \rightarrow \rho_{m'm}^{(f)},$$

so that

$$|\langle f | V | i \rangle|^2 \rightarrow \sum_{nn' mm'} V_{mn} V_{m'n'}^* \rho_{nn'}^{(i)} \rho_{m'm}^{(f)}.$$

¹³In the transformations of sign factors one may use the fact that the numbers $2J_i$, $2J_f$, $2M_i$, $2M_f$ are of the same parity. Recall also that the numbers j , m are integers, while $\lambda = \pm 1$.

here. We shall therefore normalise the probability $w(\mathbf{n})$ to 1. For this we obtain²

$$\begin{aligned}
 w(\mathbf{n}) = & \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{2J_i-M_i-M'_i} D_{\lambda m}^{(j)} D_{\lambda' m'}^{(j)*} \times \\
 & \times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle \times \\
 & \times \langle M'_f | \rho^{(f)} | M_f \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle
 \end{aligned} \quad (48.5)$$

(we shall verify the correctness of the normalisation below). We transform this formula by expanding the product of two D -functions into a series (110.2) (see III):

$$\begin{aligned}
 D_{\lambda m}^{(j)} D_{\lambda' m'}^{(j)*} &= (-1)^{\lambda'+m'} D_{\lambda m}^{(j)} D_{-\lambda', -m'}^{(j)} = \\
 &= (-1)^{\lambda+m} \sum_L (2L+1) \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} D_{\Lambda \mu}^{(L)}
 \end{aligned} \quad (V.2)$$

(indices $\Lambda = \lambda - \lambda'$, $\mu = m - m'$; L are integers, $L \geq 2j$). Thus, we obtain finally

$$\begin{aligned}
 w(\mathbf{n}) = & \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{2J_i-M_i-M'_i+m+1} (2L+1) \times \\
 & \times \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times \\
 & \times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} D_{\Lambda \mu}^{(L)}(\mathbf{n}) \langle M_i | \rho^{(i)} | M'_i \rangle \langle M'_f | \rho^{(f)} | M_f \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle.
 \end{aligned} \quad (48.5)$$

As before, $\sum_{(m)}$ denotes summation over all (doubly repeated) m -indices. In this connection it should be recalled that, unlike the indices λ , λ' , the remaining m -indices are summed not over all $2j+1$ possible (for a given j -index) values, but only over two values: λ , $\lambda' = \pm 1$, corresponding to the two polarisation states of the photon.

Formula (48.5) contains all the necessary information about the angular distribution of the emitted photons and their polarisation, as well as about the polarisation of the secondary (i.e. the ones that have emitted photons) nuclei. It is assumed that the initial density matrix is given.

Angular distribution. The angular distribution of photons is obtained by summation over all polarisations of the photon and the recoil nucleus. The averaging over polarisations is effected by substituting the density matrices of unpolarised states:

$$\langle \lambda | \rho^{(\gamma)} | \lambda' \rangle = \frac{1}{2} \delta_{\lambda \lambda'}, \quad \langle M_f | \rho^{(f)} | M'_f \rangle = \frac{1}{2J_f+1} \delta_{M_f M'_f}, \quad (48.6)$$

after which the summation reduces to multiplication by 2 (for the photon) or by $2J_f+1$ (for the nucleus). In other words, the summation is effected simply by the replacement

$$\langle \lambda | \rho^{(\gamma)} | \lambda' \rangle \rightarrow \delta_{\lambda \lambda'}, \quad \langle M_f | \rho^{(f)} | M'_f \rangle \rightarrow \delta_{M_f M'_f}. \quad (48.7)$$

²In the transformations of sign factors one may use the fact that the numbers $2J_i$, $2J_f$, $2M_i$, $2M_f$ are of the same parity. Recall also that the numbers j , m are integers, while $\lambda = \pm 1$.

Thus, the angular distribution

$$\begin{aligned} \bar{w}(\mathbf{n}) &= \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{m'+1} (2L+1) D_{0\mu}^{(L)}(\mathbf{n}) \times \\ &\times \begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times \\ &\times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle. \end{aligned} \quad (\text{V.3})$$

This formula can be substantially simplified by carrying out the summation over the m -indices.

First of all we note that

$$\begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} = (-1)^L \begin{pmatrix} j & j & L \\ -\lambda & \lambda & 0 \end{pmatrix}, \quad (48.8)$$

and therefore

$$\sum_{\lambda=\pm 1} \begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} = \begin{cases} 2 \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} & \text{for even } L, \\ 0 & \text{for odd } L. \end{cases}$$

Thus, in the sum over L only terms with even L remain, i.e. it contains spherical functions ($D_{0\mu}^{(L)}$) of only even orders. This result could have been anticipated: by virtue of parity conservation the probability distribution must be invariant under inversion, i.e. under the replacement $\mathbf{n} \rightarrow -\mathbf{n}$.

Thus,

$$\begin{aligned} \bar{w}(\mathbf{n}) &= \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L (2L+1) \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} D_{0\mu}^{(L)}(\mathbf{n}) \times \\ &\times \sum_{(m)} (-1)^{m'+1} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times \\ &\times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle. \end{aligned} \quad (\text{V.4})$$

Let us note that it is easy here to verify the normalisation: by virtue of the formula

$$\int D_{0\mu}^{(L)}(\mathbf{n}) \frac{d\omega}{4\pi} = \delta_{L0} \delta_{\mu 0}$$

after integration over directions only the term with $L = \mu = 0$ remains; and with the help of the formulae

$$\begin{pmatrix} j & j & 0 \\ m & -m & 0 \end{pmatrix} = (-1)^{j-m} \frac{1}{\sqrt{2j+1}},$$

$$\sum_{M_f m} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix}^2 = \frac{1}{2J_i + 1}, \quad \text{Sp } \rho^{(i)} = 1$$

we verify that the integral equals 1.

The further summation over $mm' M_f$ in the inner sum in $\bar{w}(\mathbf{n})$ is carried out with the help of formula (108.4) (see III). As a result we obtain for the angular distribution of photons the following final formula:

$$\begin{aligned} \bar{w}(\mathbf{n}) = & (-1)^{1+J_i+J_f} \frac{(2j+1)\sqrt{2J_i+1}}{4\pi} \sum_{\text{even } L} (-i)^L \sqrt{2L+1} \times \\ & \times \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \begin{Bmatrix} J_i & J_i & L \\ j & j & J_f \end{Bmatrix} \sum_{\mu} \mathcal{P}_{L\mu}^{(i)} D_{0\mu}^{(L)}(\mathbf{n}), \end{aligned} \quad (48.9)$$

where we have introduced the notation

$$\begin{aligned} \mathcal{P}_{L\mu}^{(i)} = & i^L \sqrt{(2L+1)(2J_i+1)} \sum_{M_i M'_i} (-1)^{J_i-M'_i} \begin{pmatrix} J_i & L & J_i \\ -M'_i & \mu & M_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle, \\ & \mathcal{P}_{L\mu}^{(i)*} = (-1)^{L-\mu} \mathcal{P}_{L,-\mu}^{(i)}. \end{aligned} \quad (48.10)$$

The inner sum in (48.9) is taken over all $|\mu| \leq L$, and the outer sum over all even values of L satisfying the conditions

$$L \leq 2j, \quad L \leq 2J_i \quad (48.11)$$

(these conditions are the triangle rules which the j -indices in the $3j$ -symbols appearing in (48.9), (48.10) must satisfy). By virtue of these conditions the number of terms in the sum is usually not large. Thus, for $J_i = 0$ or $\frac{1}{2}$ only the term with $L = 0$ remains, i.e. the radiation is isotropic (it is easy to verify that the term with $L = 0$ equals $1/4$, as it should be by the normalisation condition). For $J_i = 1, \frac{3}{2}$ or for $j = 1$ two terms remain in the sum over L : $L = 0, 2$. Let us also note that if the density matrix $\rho^{(i)}$ is diagonal ($M_i = M'_i$), then $\mu = 0$, and the distribution function (48.9) takes the form of an expansion in Legendre polynomials (according to (16.5) and (58.23) (see III) the functions $D_{00}^{(L)}$ reduce to the polynomials $P_L(\cos \theta)$). Finally, if

$$\langle M_i | \rho^{(i)} | M'_i \rangle = 1/(2J_i + 1) \delta_{M_i M'_i},$$

i.e. the initial nucleus is unpolarised, then all the $\mathcal{P}_{L\mu}^{(i)} = 0$ except $\mathcal{P}_{00}^{(i)} = 1$ ¹⁴.

¹⁴Indeed, noting that

$$\begin{pmatrix} J & & J \\ -M' & 0 & M \end{pmatrix} = (-1)^{J-M} \frac{1}{\sqrt{2J+1}} \delta_{MM'},$$

we have

$$\begin{aligned} & \sum_{MM'} (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \delta_{MM'} = \\ & = \sqrt{2J+1} \sum_{MM'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \begin{pmatrix} J & 0 & J \\ -M' & 0 & M \end{pmatrix} = \sqrt{2J+1} \delta_{L0} \delta_{\mu 0}, \end{aligned}$$

from which using the definition (48.10) we find the stated result.

The quantities $\mathcal{P}_{L\mu}$ are convenient characteristics of the polarisation state of the nucleus; we shall call them *polarisation moments*. Formula (48.10) determines these quantities through the density matrix elements $\rho_{MM'}$. By direct verification one can easily establish the validity of the inverse formula, expressing the density matrix through the polarisation moments:

$$\rho_{MM'} = \sum_{L\mu} \sqrt{\frac{2L+1}{2J+1}} i^{-L} (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \mathcal{P}_{L\mu}. \quad (48.12)$$

Let $f_{L\mu}$ be some spherical tensor of rank L depending on the polarisation state of the nucleus. According to the general rules (see III, (14.8)) its mean value in a state with density matrix $\rho_{MM'}$ equals

$$\bar{f}_{L\mu} = \sum_{MM'} \rho_{MM'} \langle JM' | f_{L\mu} | JM \rangle. \quad (48.13)$$

Expressing the matrix elements of the quantity $f_{L\mu}$ through the reduced element $\langle J || f_L || J \rangle$ according to

$$\langle JM' | f_{L\mu} | JM \rangle = i^L (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \langle J || f_L || J \rangle$$

and introducing the polarisation moments according to the definition (48.10), we obtain

$$\bar{f}_{L\mu} = \frac{\langle J || f_L || J \rangle}{\sqrt{(2L+1)(2J+1)}} \mathcal{P}_{L\mu}. \quad (48.14)$$

Polarisation of the photon. With given (along with $\rho^{(i)}$) matrices $\rho^{(\gamma)}$ and $\rho^{(f)}$ formula (48.5) determines the probability of a transition in which the emitted photon and the nucleus are found in definite polarisation states. These states are essentially not a characteristic of the emission process as such, but rather of those detectors which register the photon and the nucleus, selecting from the emitted radiation definite polarisations. A more natural formulation of the problem is the other one, in which the final state of the system “nucleus + photon” is not fixed in advance, and it is required to determine the polarisation density matrix of this state for a given direction of emission of the photon.

The answer to this question is given by the same formula (48.5). If we represent it in the form

$$w = \bar{w}(\mathbf{n}) \sum_{(m)} \langle M_f; \mathbf{n}\lambda | \rho | M'_f; \mathbf{n}\lambda' \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle \langle M'_f | \rho^{(f)} | M_f \rangle, \quad (48.15)$$

then the expression $\langle M_f; \mathbf{n}\lambda | \rho | M'_f; \mathbf{n}\lambda' \rangle$ will be the required density matrix, being the “projection” onto the given $\rho^{(\gamma)} \rho^{(f)}$. The factor $\bar{w}(\mathbf{n})$ is separated out in (48.15) in order that this matrix should be normalised by the usual condition

$$\sum_{\lambda M_f} \langle M_f; \mathbf{n}\lambda | \rho | M_f; \mathbf{n}\lambda \rangle = 1.$$

If we are interested in the polarisation of only the photon, we must sum over $M_f = M'_f$:

$$\langle \mathbf{n}\lambda | \rho | \mathbf{n}\lambda' \rangle = \sum_{M_f} \langle M_f; \mathbf{n}\lambda | \rho | M_f; \mathbf{n}\lambda' \rangle.$$

Quite analogously to the derivation of formula (48.9) we obtain

$$\begin{aligned} \langle \mathbf{n}\lambda | \rho | \mathbf{n}\lambda' \rangle &= (-1)^{1+J_i+J_f} \frac{(2j+1)\sqrt{2J_i+1}}{8\pi\bar{w}(\mathbf{n})} \times \\ &\times \sum_L (-i)^L \sqrt{2L+1} \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{Bmatrix} J_i & J_i & L \\ j & j & J_f \end{Bmatrix} \sum_{\mu} \mathcal{P}_{L\mu}^{(i)*} D_{\Lambda\mu}^{(L)}(\mathbf{n}) \end{aligned} \quad (48.16)$$

($\Lambda = \lambda - \lambda'$), where the summation is taken over all integer values of L satisfying conditions (48.11).

In particular, the circular polarisation is determined by the Stokes parameter

$$\xi_2 = \langle \mathbf{n}1 | \rho | \mathbf{n}1 \rangle - \langle \mathbf{n}, -1 | \rho | \mathbf{n}, -1 \rangle$$

(see the problem to § 8). By virtue of the relations (48.8) in this difference all terms with even L drop out, and for ξ_2 one obtains a formula differing from the expression (48.9) only in that the summation is carried out over odd (instead of even) values of L .

Polarisation of recoil nuclei. Finally, if we are interested only in the final polarisation of the nuclei, we must put $\rho^{(\gamma)} \rightarrow \delta$. If we also integrate over the directions of the photon, then the density matrix of the recoil nucleus will be

$$\begin{aligned} \langle M_f | \rho | M'_f; \mathbf{n} \rangle &= \int \bar{w}(\mathbf{n}) \langle M_f; \mathbf{n} | \rho | M'_f; \mathbf{n} \rangle d\mathbf{o} = \\ &= (2J_i+1) \sum_{mM_iM'_i} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times \\ &\times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle. \end{aligned} \quad (\text{V.5})$$

The polarisation moments computed from this density matrix are

$$\mathcal{P}_{L\mu}^{(f)} = (-1)^{J_i+J_f+L+j} \sqrt{(2J_i+1)(2J_f+1)} \begin{Bmatrix} J_i & J_i & L \\ J_f & J_f & j \end{Bmatrix} \mathcal{P}_{L\mu}^{(i)}. \quad (48.17)$$

If the initial nucleus is unpolarised, then the final nucleus will also be unpolarised. However, even in this case there will exist a correlation polarisation, i.e. the polarisation of the nucleus after emission of radiation in a given direction. Setting $\rho^{(i)} \rightarrow \delta/(2J_i+1)$ (and correspondingly $\bar{w}(\mathbf{n}) = 1/(4\pi)$) and carrying out a calculation analogous to the derivation of (48.9), we obtain for the density matrix describing this polarisation

$$\begin{aligned} \langle M_f; \mathbf{n} | \rho | M'_f; \mathbf{n} \rangle &= (2j+1)(-1)^{J_i+M'_f+1} \sum_{\text{even } L} (2L+1) \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} J_f & L & J_f \\ -M'_f & \mu & M_f \end{pmatrix} \times \\ &\times \begin{Bmatrix} J_f & J_f & L \\ j & j & J_i \end{Bmatrix} D_{0\mu}^{(L)}(\mathbf{n}). \end{aligned} \quad (48.18)$$

The polarisation moments corresponding to this density matrix are

$$\mathcal{P}_{L\mu}^{(f)} = i^L (-1)^{1+J_i+J_f} (2j+1) \sqrt{(2L+1)(2J_f+1)} \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \begin{Bmatrix} J_f & J_f & L \\ j & j & J_i \end{Bmatrix} D_{0\mu}^{(L)}(\mathbf{n}). \quad (48.19)$$

Only moments of even order arise (which is again a consequence of parity conservation).

If the secondary nucleus in turn radiates, then, being polarised, it gives a non-isotropic distribution of photons. Since the polarisation moments (48.19) depend on the direction $\mathbf{n}$ of the photon emitted in the first decay, there arises a definite angular correlation between the directions of successively emitted photons (with an unpolarised primary nucleus). In an analogous way one can consider other correlation phenomena in cascade emissions (correlation of polarisations, etc.)¹⁵.

Problem

Express the polarisation moments $\mathcal{P}_{1\mu}$ and $\mathcal{P}_{2\mu}$ through the mean values of the angular momentum vector $\mathbf{J}$ and the quadrupole moment tensor Q_{ik} .

Solution. The reduced elements of the vector $\mathbf{J}$ and the tensor Q_{ik} are determined from the equalities (cf. III, (107.10), (107.11)). The operator $\hat{Q}_{ik}$ is expressed through the angular momentum operators by formula (75.2) (see III):

$$\hat{Q}_{ik} = \frac{3Q}{2J(2J-1)} \left(\hat{J}_i \hat{J}_k + \hat{J}_k \hat{J}_i - \frac{2}{3} J^2 \delta_{ik} \right).$$

From this we find the mean value

$$\overline{Q_{ik}^2} = \frac{3Q^2}{2J^2(2J-1)^2} J^2(4J^2-3) = Q^2 \frac{3(J+1)(2J+3)}{2J(2J-1)}.$$

The reduced matrix elements:

$$\begin{aligned} \langle J \| J \| J \rangle &= \sqrt{J(J+1)(2J+1)}, \\ \langle J \| Q \| J \rangle &= Q \sqrt{\frac{3(2J+1)(J+1)(2J+3)}{2J(2J-1)}}. \end{aligned}$$

From (48.14) we now see that the polarisation moments $\mathcal{P}_{1\mu}$ coincide with the spherical components of the vector

$$\sqrt{\frac{3}{J(J+1)}} \bar{\mathbf{J}},$$

and the moments $\mathcal{P}_{2\mu}$ with the spherical components of the tensor

$$\sqrt{\frac{10J(2J-1)}{3(J+1)(2J+3)}} \bar{Q}_{ik}.$$

¹⁵A detailed exposition of these questions can be found in a paper by A. Z. Dolginov in the book "Gamma Rays" (Izd. AN SSSR, 1961).

§ 49. Radiation of atoms. Electric type¹⁶

The energies of the outer electrons of an atom (participating in optical radiative transitions) are in rough estimate of the order of $E \sim me^4/\hbar^2$, so that the wavelengths radiated are $\lambda \sim \hbar c/E \sim \hbar^2/(\alpha me^2)$. The dimensions of the atom are $a \sim \hbar^2/me^2$. Therefore in optical spectra of atoms, as a rule, the inequality $a/\lambda \sim \alpha \ll 1$ is satisfied. The same order of magnitude is the ratio $v/c \sim \alpha$, where v is the speed of the optical electrons.

Thus, in optical spectra of atoms the condition is fulfilled by virtue of which the probability of electric dipole radiation (if it is allowed by selection rules) substantially exceeds the probability of multipole transitions¹⁷. In connection with this, in the spectroscopy of atoms the most important role is played by precisely the electric dipole transitions.

As has already been indicated, such transitions are subject to strict selection rules for the total angular momentum J of the atom and for parity P ¹⁸:

$$|J' - J| \leq 1 \leq J + J', \quad (49.1)$$

$$PP' = -1. \quad (49.2)$$

The inequality $|J' - J| \leq 1$ means that the angular momentum J can change only by 0, ± 1 ; by virtue of the inequality $J + J' \geq 1$ the transition $0 \rightarrow 0$ is additionally forbidden. The parities of the initial and final states must be opposite.

The probability of radiation with the transition $nJM \rightarrow n'J'M'$ is determined by the corresponding matrix element of the dipole moment of the atom according to

$$w(nJM \rightarrow n'J'M') = \frac{4\omega^3}{3\hbar c^3} |\langle n'J'M' | \mathbf{d}_{-m} | nJM \rangle|^2, \quad (49.3)$$

$$\omega = \omega(nJ \rightarrow n'J').$$

Summing (49.3) over all values $M' = M - m$ (for a given M), we obtain the total probability of emission of radiation of a given frequency from the atomic level nJ . The summation is carried out with the help of (46.20) and gives

$$w(nJ \rightarrow n'J') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2J+1} |\langle n'J' || d || nJ \rangle|^2. \quad (49.4)$$

The square of the modulus of the reduced matrix element standing here is sometimes called the *line strength of the transition*; this quantity is symmetric with respect to the initial and final states.

The observed intensity of emission is obtained by multiplying w by $\hbar\omega$ and by the number of atoms in the source found on the given excited level ($N_{n,J}$). Thus, in a gas with temperature T the number $N_{n,J} \propto (2J+1) \exp(-E_{n,J}/T)$; the factor $(2J+1)$ is the statistical weight of the level with angular momentum J .

¹⁶In §§ 49, 51, 53–55 we use ordinary units.

¹⁷Typical values of the probability of dipole transitions in the optical region of atomic spectra are of the order 10^8 s^{-1} .

¹⁸The selection rule for parity was established by *Laporte* (O. Laporte, 1924).

Further conclusions about the probabilities of transitions in atomic spectra can be made only on specifying the character of the states of the atom. We shall not dwell here on the methods of computing matrix elements, the degree of approximation of which does not have a sharp theoretical character. We shall derive only certain relations for a fairly broad (and especially in light

atoms) category of states, constructed according to the LS -coupling scheme (see III, § 72). Such states are characterised, besides the total angular momentum, also by definite conserved values of the orbital angular momentum L and the spin S in this case.

Since the dipole moment is a purely orbital quantity, its operator commutes with the spin operator, i.e. its matrix is diagonal in the number S . By the number L for the dipole moment the same selection rules apply as for any orbital vector (see III, § 29). Thus, transitions between states constructed according to the LS -scheme are subject to additional (besides (49.1), (49.2)) selection rules:

$$S' - S = 0, \quad (49.5)$$

$$|L' - L| \leq 1 \leq L + L'. \quad (49.6)$$

Let us emphasise once more that these rules are approximate and break down when the spin-orbit interaction is taken into account. Let us note that the rule (49.5) (the forbiddenness of transitions between terms of different multiplicity) holds not only for dipole but for all transitions of the electric type: the electric multipole moments of all orders are orbital tensors, so that their matrices are diagonal in spin. Thus, for electric quadrupole transitions, besides the general rules

$$|J' - J| \leq 2 \leq J + J', \quad PP' = 1, \quad (49.7)$$

in the case of LS -coupling the additional selection rules hold:

$$S' - S = 0, \quad |L' - L| \leq 2 \leq L + L'. \quad (49.8)$$

The dependence of the probability of radiation on the numbers S, J, J' can be determined in explicit form. This question is solved directly with the help of general formulae for matrix elements of spherical tensors in the addition of angular momenta. According to formula (109.3) (see III) we have ¹⁹:

$$|\langle n' L' S J' \| d \| n L S J \rangle|^2 = (2J + 1)(2J' + 1) \begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}^2 |\langle n' L' \| d \| n L \rangle|^2. \quad (49.9)$$

Substituting this into (49.4), we get

$$w(n L S J \rightarrow n' L' S J') = \frac{4\omega^3}{3\hbar c^3} (2J' + 1) \begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}^2 |\langle n' L' \| d \| n L \rangle|^2, \quad (49.10)$$

¹⁹In the formulae of vol. III, § 109 under “angular momenta of subsystems 1 and 2” one should now understand the orbital angular momentum and the spin of the atom, the interaction between which we neglect. The role of the quantity $f_q^{(1)}$ is played by the orbital vector d_q .

where $\omega = \omega(nLS \rightarrow n'L'S)$ ²⁰.

For these probabilities one can obtain a definite sum rule. For the squares of the $6j$ -symbols the summation formula holds (see III, (108.7)):

$$\sum_{J'} (2J' + 1) \left\{ \begin{matrix} L' & J' & S \\ J & L & 1 \end{matrix} \right\}^2 = \frac{1}{2L + 1}. \quad (49.11)$$

With its help we find from (49.10)

$$\sum_{J'} w(nLSJ \rightarrow n'L'SJ') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2L + 1} |\langle n'L' || d || nL \rangle|^2. \quad (49.12)$$

Let us note that this quantity turns out to be independent of the initial value of J .

If we are dealing with the radiation of a gas with temperature T , much larger than the intervals of the fine structure of the atomic term nSL , then the states with different J are populated uniformly, i.e. all values of J are equally probable. The probability that the atom is on a level with a certain definite value of J in such a case is

$$\frac{2J + 1}{(2L + 1)(2S + 1)}, \quad (49.13)$$

i.e. the ratio of the statistical weight of this level to the full statistical weight of the term nSL . The averaging of the expressions (49.10) or their sums (49.12) over these probabilities reduces to multiplication by the factor (49.13); we shall denote this averaging by a bar over the letter. The total probability of radiation from all lines of the spectral multiplet (formed by all possible transitions between the fine-structure components of the two terms nSL and $n'SL'$) is the sum:

$$\bar{w}(nLS \rightarrow n'L'S) = \sum_J \sum_{J'} \bar{w}(nLSJ \rightarrow n'L'SJ'). \quad (49.14)$$

Since, of course, $\sum_J (2J + 1) = (2S + 1)(2L + 1)$, for the full probability we obtain an expression coinciding with (49.12).

Therefore for the relative probability (or, which is the same, the relative intensity) of an individual line we obtain

$$\frac{\bar{w}(nLSJ \rightarrow n'L'SJ')}{\bar{w}(nLS \rightarrow n'L'S)} = \frac{(2J + 1)(2J' + 1)}{(2S + 1)} \left\{ \begin{matrix} L' & J' & S \\ J & L & 1 \end{matrix} \right\}^2. \quad (49.15)$$

Analysis of the numerical values given by this formula reveals that among the lines of the multiplet the most intense are those for which $\Delta J = \Delta L$ (they are called *principal lines*, as distinct from the remaining components of the multiplet, called *satellites*). The intensity of the principal lines is the greater the larger the initial value of J .

Summing (49.15) over J or over J' gives

$$\frac{\sum_{J'} \bar{w}(nLSJ \rightarrow n'L'SJ')}{\sum_J \bar{w}(nLSJ \rightarrow n'L'SJ')} = \frac{2J + 1}{(2L + 1)(2S + 1)}, \quad (49.16)$$

²⁰Neglecting the spin-orbit interaction in the computation of matrix elements, we neglect also the dependence of frequencies on J and J' , i.e. the fine structure of the initial and final levels of the atom.

$$\frac{\sum_J \bar{w}(nLSJ \rightarrow n'L'SJ')}{\sum_{J'} \bar{w}(nLSJ \rightarrow n'L'SJ')} = \frac{2J' + 1}{(2L + 1)(2S + 1)}.$$

Thus, the sum of intensities of all lines of a spectral multiplet having one and the same initial (or final) level is proportional to the statistical weight of the initial (or final) level.

Let us dwell also on the hyperfine structure of spectral lines of the atom. Recall that the hyperfine splitting of atomic levels arises as a result of the interaction of the electrons with the spin of the nucleus, if the latter is different from zero (see III, § 122). The total angular momentum of the atom (including the nucleus) $\mathbf{F}$ is composed of the total angular momentum of the electrons $\mathbf{J}$ and the moment of the nucleus $\mathbf{I}$. Each component of the hyperfine structure of the level nJ is characterised by its value of the quantum number F .

The strict law of conservation of angular momentum leads now to a strict selection rule for the total angular momentum F ; for electric dipole radiation

$$|F' - F| \leq 1 \leq F + F'. \quad (49.17)$$

But by virtue of the extreme weakness of the interaction of the electrons with the spin of the nucleus it can be entirely neglected in the computation of the matrix elements of the electric (and magnetic) moments of the electron shell of the atom. Therefore the previous selection rules for the electronic angular momentum J and for the electronic parity also remain valid. In particular, by virtue of the latter, electric dipole transitions between components of the hyperfine structure of one and the same term are impossible: all these levels have the same parity, whereas the indicated transitions are possible only between states of different parity.

Since the operator of the dipole moment commutes with the spin of the nucleus, the dependence of the matrix elements on the numbers I and F

can be found in explicit form; the calculations differ from those given above only by an obvious change of notation. The probability of radiation, summed over the final values of the projection of the total angular momentum F :

$$w(nJIF \rightarrow n'J'IF') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2F + 1} |\langle n'J'IF' || d || nJIF \rangle|^2, \quad (49.18)$$

$$\omega = \omega(nJ \rightarrow n'J'),$$

where the square of the reduced matrix element

$$|\langle n'J'IF' || d || nJIF \rangle|^2 = (2F + 1)(2F' + 1) \begin{Bmatrix} J' & F' & I \\ F & J & 1 \end{Bmatrix}^2 |\langle n'J' || d || nJ \rangle|^2. \quad (49.19)$$

Problem

The majority of lines in the spectra of alkali metals can be described as a result of transitions of one outer (optical) electron in the self-consistent field of the atomic core forming a closed configuration; the state of the atom is constructed according to the

LS-coupling scheme. Under these assumptions determine the relative intensities of the fine-structure components of spectral lines.

Solution. The total angular momenta L and $S = \frac{1}{2}$ of the atom coincide with the orbital angular momentum and spin of the optical electron. The parity of the state is $(-1)^l$ (the parity of the closed configuration of the atomic core is positive). The selection rules for parity therefore forbid a dipole transition with $L' = L$, so that only transitions with $L' - L = \pm 1$ are possible. The transitions between components of the doublet levels n, L and $n', L - 1$ give in all (by virtue of the selection rules for J) just three lines (Fig. 1).

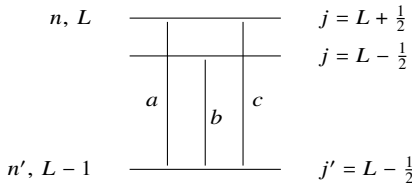


Figure 1

Their relative intensities (denoting them a, b, c) can be determined, without resorting to formula (49.15), from the rule (49.16). Composing the ratios of the summed intensities of lines with one and the same initial (or final) level, we obtain two equalities:

$$\frac{b+c}{a} = \frac{2L}{2L+2}, \quad \frac{a+b}{c} = \frac{2L}{2L-2},$$

from which

$$a : b : c = [(L+1)(2L-1)] : 1 : [(L-1)(2L+1)].$$

If $L = 1$, the lower level is not split, line c is absent, and $a/b = 2$.

§ 50. Radiation of atoms. Magnetic type

The magnetic moment of an atom is in order of magnitude given by the Bohr magneton: $\mu \sim e\hbar/mc$. This estimate differs by a factor α from the order of magnitude of the electric dipole moment of the atom: $d \sim ea \sim \hbar^2/(me)$ (since $v/c \sim \alpha$, we have $\mu \sim d\omega/c$, as one would expect). It follows therefore that the probability of magnetic dipole ($M1$) radiation by an atom is approximately α^2 times smaller than the probability of electric dipole radiation (of the same frequency). Magnetic radiation therefore plays a role in practice only for transitions forbidden by the selection rules of the electric type.

As regards the electric quadrupole ($E2$) radiation, the ratio of its probability to the probability of $M1$ -radiation is in order of magnitude

$$\frac{E2}{M1} \sim \frac{(ea^2)^2\omega^2/c^2}{\mu^2} \sim \frac{a^4m^2\omega^2}{\hbar^2} \sim \left(\frac{\Delta E}{E}\right)^2 \quad (50.1)$$

(quadrupole moment $\sim ea^2$, $E \sim \hbar^2/ma^2$ is the energy of the atom, ΔE is the energy change in the transition). We see that for medium atomic frequencies (i.e. for $\Delta E \sim E$)

the probabilities of $E2$ - and $M1$ -radiation are of the same order (provided, of course, that both are allowed by the selection rules). If $\Delta E \ll E$ (for example, for transitions between fine-structure components of one and the same term), then $M1$ -radiation is more probable than $E2$ -radiation.

Magnetic dipole transitions are subject to the strict selection rules

$$|J' - J| \leq 1 \leq J + J', \quad (50.2)$$

$$PP' = 1. \quad (50.3)$$

In the case of LS -coupling additional selection rules arise, even more restrictive than in the electric case. The latter circumstance is connected with the specific property of the magnetic moment of the atom, arising as a result of the equality of all particles in the system (electrons). Namely, the operator of the magnetic moment of the atom is expressed through the operators of its orbital and spin angular momenta:

$$\hat{\mu} = -\mu_0(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) = -\mu_0(\hat{\mathbf{J}} + \hat{\mathbf{S}}), \quad (50.4)$$

where $\mu_0 = |e|\hbar/(2mc)$ is the Bohr magneton (see III, § 113). By virtue of the conservation of the total angular momentum the operator $\hat{\mathbf{J}}$ does not at all have off-diagonal (in energy) matrix elements; so that for the theory of radiation it suffices to write $\hat{\mu} = -\mu_0\hat{\mathbf{S}}^{23}$.

When the spin-orbit interaction is neglected, the spin operator is diagonal in all the quantum numbers nSL characterising the unsplit term. For any transition at all to take place, the number J must necessarily change. Thus, we have the selection rules:

$$n' = n, \quad S' = S, \quad L' = L, \quad J' - J = \pm 1, \quad (50.5)$$

i.e. transitions are possible only between fine-structure components of one and the same term.

The computation of the probability of radiation in this case can be carried through to the end. Changing the designations correspondingly in formula (49.10), we find

$$w(nLSJ \rightarrow nLSJ') = \frac{4\omega^3\mu_0^2}{3\hbar c^3} (2J' + 1) \begin{Bmatrix} S & J' & L \\ J & S & 1 \end{Bmatrix}^2 |\langle S||S||S \rangle|^2.$$

The reduced matrix element of the spin with respect to the eigenfunction of the spin itself is given by the formula

$$\langle S||S||S \rangle = \sqrt{S(S+1)(2S+1)} \quad (50.6)$$

(see III, (29.13)). The $6j$ -symbol we need equals

$$\begin{Bmatrix} S & J-1 & L \\ J & S & 1 \end{Bmatrix}^2 = \frac{(L+S+J+1)(L+S-J+1)(L-S+J)(S-L+J)}{S(2S+1)(2S+2)(2J-1)2J(2J+1)} \quad (50.7)$$

²³An exception is the case when the electronic angular momentum $\mathbf{J}$ of the atom is not conserved: when the hyperfine structure is taken into account, in the presence of an external field, etc. (see the Problems).

(see the table in III, § 108). As a result we obtain

$$\begin{aligned}
 w(nLSJ \rightarrow nLS, J-1) &= \frac{2J+1}{2J-1} w(nLS, J-1 \rightarrow nLSJ) = \\
 &= \frac{\omega^3 \mu_0^2}{3\hbar c^3 (2J+1)J} (L+S+J+1)(L+S-J+1)(L-S+J) \times \\
 &\quad \times (J+L-S).
 \end{aligned} \tag{50.8}$$

Transitions between components of the hyperfine structure of one and the same level (whose frequencies lie in the radio-wave region) cannot occur as electric-dipole transitions, since all these components possess the same parity. Without change of parity the transitions can occur as $E2$ and $M1$. But in view of the very small magnitude of the intervals of the hyperfine structure the $E2$ emission is highly improbable in comparison with $M1$ (cf. (50.1)), so that the indicated transitions are realised as magnetic-dipole transitions.

Problems

1. Find the probability of the $M1$ -transition between components of the hyperfine structure of one and the same level.

Solution. The probability of the transition is given by formulae (49.18), (49.19), in which there will now figure the diagonal reduced matrix element of the magnetic moment: $\langle nJ || \mu || nJ \rangle$. Its value can

be written at once if we notice that the full (unreduced) matrix element $\langle nJM | \mu_\lambda | nJM \rangle$ is precisely the splitting of the given level in the Zeeman effect (see III, § 113) and equals $-\mu_0 g M$, where g is the Landé g -factor. The reduced matrix element (see III, (29.7))

$$\langle nJ || \mu || nJ \rangle = \frac{1}{M} \sqrt{J(J+1)(2J+1)} \langle nJM | \mu_\lambda | nJM \rangle = -\mu_0 g \sqrt{J(J+1)(2J+1)}.$$

As a result we find for the desired probability²⁴

$$\begin{aligned}
 w(nJIF \rightarrow nJI, F-1) &= \frac{2F+1}{2F-1} w(nJI, F-1 \rightarrow nJIF) = \\
 &= \frac{\omega^3 \mu_0^2 g^2}{3\hbar c^3 (2F+1)F} (J+I+F+1)(J+I-F+1)(F+J-I)(F-J+I).
 \end{aligned} \tag{V.6}$$

²⁴An interesting example is the transition between the components of the hyperfine structure of the ground level of the hydrogen atom ($1s_{1/2}$), strictly forbidden not only as $E1$ but also as $E2$ (the latter by the rule forbidding the quadrupole transition with $J+J'=1$). This transition has the frequency $\omega = 2\pi \cdot 1.42 \cdot 10^9 \text{ s}^{-1}$ (wavelength $\lambda = 21 \text{ cm}$). Setting $g = 2$, $I = \frac{1}{2}$, $J = \frac{1}{2}$, $F = 1$, $F' = 0$, we obtain

$$w = \frac{4\omega^3 \mu_0^2}{3\hbar c^3} = 2.85 \cdot 10^{-15} \text{ s}^{-1}.$$

This expression differs from (50.8) only by an obvious change of notation and an additional factor g^2 .

2. Find the probability of the $M1$ -transition between Zeeman components of one and the same atomic level.

Solution. We are considering the transition $M \rightarrow M - 1$ at unchanged nJ ; the frequency of the transition (see below, (51.3)): $\hbar\omega = \mu_0 g H$ (g is the Landé factor). The matrix element of the spherical component μ_{-1} of the vector μ :

$$\begin{aligned} |\langle nJ, M-1 | \mu_{-1} | nJM \rangle| &= \sqrt{\frac{(J-M+1)(J+M)}{2J(J+1)(2J+1)}} |\langle nJ || \mu || nJ \rangle| = \\ &= -\mu_0 g \sqrt{\frac{1}{2}(J-M+1)(J+M)} \end{aligned}$$

(see III, (27.12) and the preceding Problem). The transition probability

$$w = \frac{4\omega^3}{3\hbar c^3} |\langle nJ, M-1 | \mu_{-1} | nJM \rangle|^2 = \frac{2\mu_0^2 H^3}{3\hbar^4 c^3} (J-M+1)(J+M).$$

§ 51. Radiation of atoms. Zeeman and Stark effects

In an external magnetic field H (which we assume weak) each atomic level with total angular momentum J splits into $2J + 1$ levels

$$E_M = E^{(0)} + \mu_0 g M H, \quad (51.1)$$

where $E^{(0)}$ is the unperturbed level, μ_0 is the Bohr magneton, g is the Landé factor, M is the projection of the angular momentum J onto the direction of the field (see III, § 113). The degeneracy with respect to the directions of the angular momentum is thus completely removed.

Correspondingly, the spectral lines, arising from transitions between two split levels, also split. The number of components of the line is determined by the selection rule for the number M , according to which for dipole radiation it must be

$$m = M - M' = 0, \pm 1. \quad (51.2)$$

Additionally the transitions with $M = M' = 0$ are forbidden if at the same time $J' = J$. This is seen directly from the general expressions (29.7) (see III) for the matrix elements of an arbitrary vector.

The components arising from transitions with $m = 0, \pm 1$ are called respectively π - and σ -components. Their frequencies:

$$\hbar\omega_\pi = \hbar\omega^{(0)} + \mu_0 H(g - g')M, \quad (51.3)$$

$$\hbar\omega_\sigma = \hbar\omega^{(0)} + \mu_0 H[gM - g'(M \pm 1)].$$

In the particular case, when $g = g'$, we have

$$\hbar\omega_\pi = \hbar\omega^{(0)}, \quad \hbar\omega_\sigma = \hbar\omega^{(0)} \mp \mu_0 g H, \quad (51.4)$$

independently of the value of M ; in other words, in this case the line splits into a triplet with an undisplaced π -component and two σ -components symmetrically situated on either side of it (the so-called *normal Zeeman effect*).

The total (over all directions) probability of radiation is proportional to the square of the modulus $|\langle n' J' M' | \mathbf{d}_{-m} | n J M \rangle|^2$. Therefore, by virtue of formula (46.19) with $j = 1$, the relative probability of emission of each of the Zeeman components of the spectral line is

$$\begin{pmatrix} J' & 1 & J \\ M' & m & -M \end{pmatrix}^2. \quad (51.5)$$

In the particular case of the normal Zeeman effect there are in all just three components, each of which arises from transitions with all initial M for a given m . Since

$$\sum_{MM'} \begin{pmatrix} J' & 1 & J \\ M' & m & -M \end{pmatrix}^2 = \frac{1}{3} \quad (51.6)$$

(see III, (106.12)), in this case the radiation of all three components is equiprobable.

Of much greater interest, however, are the relative intensities of the Zeeman components when observed in a definite direction (relative to the direction of the applied magnetic field). According to (45.5) the probability of

radiation (and with it the intensity of the line) in a given direction $\mathbf{n}$ is proportional to $\sum |\mathbf{e}^* \mathbf{d}_{fi}|^2$, where the summation is carried out over the two independent polarisations $\mathbf{e}$ possible for the given $\mathbf{n}$.

In observation along the field (axis z) the sum is

$$|(d_x)_{fi}|^2 + |(d_y)_{fi}|^2.$$

Passing to the spherical components, we obtain

$$|(d_1)_{fi}|^2 + |(d_{-1})_{fi}|^2.$$

This means that in the longitudinal (along the field) direction only the two σ -components ($m = \pm 1$) are observed. Their intensities are proportional to

$$\begin{pmatrix} J' & 1 & J \\ M \mp 1 & \pm 1 & -M \end{pmatrix}^2. \quad (51.7)$$

Possessing definite values of the angular momentum projection m along the direction of propagation, these lines have right ($m = 1$) and left ($m = -1$) circular polarisations (see § 8).

In observation in the perpendicular-to-field direction (let this be the axis x) the intensity is proportional to the sum

$$|(d_z)_{fi}|^2 + |(d_y)_{fi}|^2 = |(d_0)_{fi}|^2 + \frac{1}{2} \{ |(d_1)_{fi}|^2 + |(d_{-1})_{fi}|^2 \}.$$

Thus, in the transverse direction two σ -components and a π -component are observed, with intensities proportional respectively to

$$\frac{1}{2} \begin{pmatrix} J' & 1 & J \\ M \mp 1 & \pm 1 & -M \end{pmatrix}^2 \quad \text{and} \quad \begin{pmatrix} J' & 1 & J \\ M & 0 & -M \end{pmatrix}^2. \quad (51.8)$$

(the intensities of the σ -components are half those in longitudinal observation). The π -component is polarised linearly along the axis z , and the σ -components are observed in this direction polarised linearly along the axis y .

Let us note that the relative intensities of the Zeeman components are determined entirely by the initial and final values of J and M without any dependence on other characteristics of the levels.

The selection rules forbid electric-dipole transitions between Zeeman components of one and the same level, since all of them have the same parity. For the same reason that was indicated at the end of the preceding paragraph for transitions between components of the hyperfine structure of a level, the indicated transitions are realised as magnetic-dipole ones. By virtue of the selection rule for the number M the transitions occur only between adjacent components ($M' - M = \pm 1$)²⁵.

The splitting of levels of an atom in a weak electric field (*Stark effect*), unlike the splitting in a magnetic field, does not lead to complete removal of the degeneracy with respect to the directions of the angular momentum. All levels, with the exception of the level $M = 0$, remain doubly degenerate: to each belong two states with projections of the angular momentum M and $-M$.

The computation of the relative intensities of the Stark components of a spectral line is analogous to that presented above for the Zeeman effect. In this connection one must bear in mind that the π -component receives contributions (for $M \neq 0$) from the transitions $M \rightarrow M$ and $-M \rightarrow -M$, and the σ -component receives contributions from the transitions $M \rightarrow M \pm 1$ and $-M \rightarrow -(M \pm 1)$. Therefore, for example, upon transverse observation the intensities of the π -components are proportional to

$$2 \begin{pmatrix} J' & 1 & J \\ M & 0 & -M \end{pmatrix}^2,$$

and the intensities of the σ -components are proportional to the sums

$$\frac{1}{2} \begin{pmatrix} J' & 1 & J \\ M \pm 1 & \mp 1 & -M \end{pmatrix}^2 + \frac{1}{2} \begin{pmatrix} J' & 1 & J \\ -M \mp 1 & \pm 1 & M \end{pmatrix}^2 = \begin{pmatrix} J' & 1 & J \\ M \pm 1 & \mp 1 & -M \end{pmatrix}^2$$

(recall that on changing the sign of all numbers in the second row the $3j$ -symbols can only change sign, while their squares do not change).

In an external field, even a weak one, strictly speaking the total angular momentum $\mathbf{J}$ ceases to be conserved; in a uniform field only the conservation of the projection of the angular momentum M is exactly preserved. Therefore upon radiative transitions in a weak field the conservation of angular momentum is no longer strictly obligatory, and in the spectrum of the atom lines may appear that are forbidden by the usual selection rules.

The computation of the intensities of these lines reduces to the computation of corrections to the matrix of the dipole moment, which in turn requires the determination of corrections to the wave functions of stationary states. In the first approximation of perturbation theory (in the weak external field) the wave function acquires “ad-

²⁵These transitions usually have frequencies in the centimetre range and are observed in absorption and stimulated emission (electronic paramagnetic resonance): the absorbing atoms are placed in a strong static magnetic field (producing Zeeman splitting) and a weak radio-frequency field of the resonance frequency.

mixtures" of states connected with the original by non-zero matrix elements of perturbation ($-\mathbf{E}\mathbf{d}$ in an electric field): the admixture of a certain state ψ_2 to the state ψ_1 is

$$\frac{-\mathbf{E}\mathbf{d}_{21}}{E_1 - E_2} \psi_2.$$

As a result, in the matrix element of the "forbidden" transition there will appear a term

$$\frac{-(\mathbf{E}\mathbf{d}_{21})\mathbf{d}_{32}}{E_1 - E_2},$$

different from zero if the transitions from the "intermediate" state 2 into both the initial and final states 1 and 3 are allowed.

§ 52. Radiation of atoms. Hydrogen atom

The hydrogen atom represents a unique case in which the computation of the matrix elements of the transition can be carried through to the end in analytical form (*W. Gordon, 1929*).

The parity of the state of the hydrogen atom is $(-1)^l$, i.e. it is unambiguously determined by the orbital angular momentum of the electron (recall that the number l as the quantity determining the parity of the state retains its meaning also for the exact relativistic wave functions, i.e. when the spin-orbit interaction is taken into account). Therefore the selection rule for parity strictly forbids electric-dipole transitions without change of l ; only transitions with $l \rightarrow l \pm 1$ are possible. Changes of the principal quantum number n , however, are not restricted.

The dipole moment of the hydrogen atom reduces to the radius vector of the electron: $\mathbf{d} = e\mathbf{r}$. Since the wave function of the electron in the hydrogen atom is a product of the angular part and the radial function R_{nl} , the reduced matrix elements of the radius vector are also represented as a product

$$\langle n', l-1 || r || nl \rangle = \langle l-1 || \mathbf{v} || l \rangle \int_0^\infty R_{n', l-1} r R_{nl} r^2 dr, \quad (52.1)$$

where $\langle l-1 || \mathbf{v} || l \rangle$ are the reduced matrix elements of the unit vector $\mathbf{v}$ in the direction of $\mathbf{r}$. The latter are

$$\langle l-1 || \mathbf{v} || l \rangle = \langle l || \mathbf{v} || l-1 \rangle^* = i\sqrt{l}$$

(see III, (29.14)). Thus,

$$\langle n', l-1 || r || nl \rangle = -\langle nl || r || n', l-1 \rangle = i\sqrt{l} \int_0^\infty R_{n', l-1} R_{nl} r^3 dr. \quad (52.1)$$

The non-relativistic radial functions of the discrete spectrum of the hydrogen atom are given by formula (36.13) (see III)²⁶:

$$R_{nl} = \frac{2}{n^{l+2}(2l+1)!} \sqrt{\frac{(n+l)!}{(n-l-1)!}} (2r)^l e^{-r/n} \times F(-n+l+1, 2l+2, 2r/n). \quad (52.2)$$

²⁶In this paragraph we use atomic units. In ordinary units the expressions given below for matrix elements of coordinates should be multiplied by $\hbar^2/(me^2)$ (if a hydrogen-like ion with atomic number Z is under consideration, then by $\hbar^2/(mZe^2)$).

The integral (52.1) with the product of two hypergeometric functions is evaluated with the help of formulae in III, § f²⁷. The computation leads to the result

$$\begin{aligned}
 \langle n', l-1 \| r \| nl \rangle = & \\
 = i\sqrt{l} \frac{(-1)^{n'-l}}{4(2l-1)!} \sqrt{\frac{(n+l)!(n'+l-1)!}{(n-l-1)!(n'-l)!}} \frac{(4nn')^{l+1}(n-n')^{n+n'-2l-2}}{(n+n')^{n+n'}} \times \\
 & \times \left\{ F(-n+l+1, -n'+l, 2l, -\frac{4nn'}{(n-n')^2}) - \right. \\
 & \left. - \left(\frac{n-n'}{n+n'}\right)^2 F(-n+l-1, -n'+l, 2l, -\frac{4nn'}{(n-n')^2}) \right\}, \quad (52.3)
 \end{aligned}$$

where $F(\alpha, \beta, \gamma, z)$ is the hypergeometric function. Since the parameters α, β in this case are negative integers (or zero), the functions reduce to polynomials²⁸.

For reference we give the expressions obtained from (52.3) in some particular cases (the value of l is indicated by the spectroscopic symbol $s, p, d, \dots$):

$$\begin{aligned}
 |\langle 1s \| r \| np \rangle|^2 &= \frac{2^8 n^7 (n-1)^{2n-5}}{(n+1)^{2n+5}}, \\
 |\langle 2s \| r \| np \rangle|^2 &= \frac{2^{17} n^7 (n^2-1)(n-2)^{2n-6}}{(n+2)^{2n+6}}, \\
 |\langle 2p \| r \| nd \rangle|^2 &= \frac{2^{19} n^9 (n^2-1)(n-2)^{2n-7}}{3(n+2)^{2n+7}}, \\
 |\langle 2p \| r \| ns \rangle|^2 &= \frac{2^{15} n^5 (n-2)^{2n-6}}{3(n+2)^{2n+6}}. \quad (52.4)
 \end{aligned}$$

Formula (52.3) is inapplicable for transitions without change of the principal quantum number n (transitions between fine-structure components of a level). In this case ($n = n'$) for carrying out the integration we start from the representation of the radial functions through the generalised Laguerre polynomials:

$$R_{nl} = -\frac{2}{n^2} \sqrt{\frac{(n-l-1)!}{[(n+l)!]^3}} e^{-r/n} \left(\frac{2r}{n}\right)^l L_{n+l}^{2l+1}\left(\frac{2r}{n}\right). \quad (52.5)$$

In the integral

$$\int_0^\infty R_{n,l-1} R_{nl} r^3 dr \propto \int_0^\infty e^{-\rho} \rho^{2l+2} L_{n+l}^{2l+1}(\rho) L_{n+l-1}^{2l+1}(\rho) d\rho$$

²⁷The computation involves evaluating the integral $J_{l_2+2}^{l_2}(-n+l+1, -n'+l)$. It is carried out with the help of the formulae (f. 12) (f. 16).

²⁸Numerical tables of matrix elements and transition probabilities for hydrogen can be found in the book: *H. Bethe, E. Salpeter, Quantum Mechanics of One- and Two-Electron Atoms*. M.: IL, 1960.

we replace one of the polynomials by its expression through the generating function (see III, § d):

$$L_{n+l}^{2l+1}(\rho) = -\frac{(n+l)!}{(n-l-1)!} e^{\rho} \rho^{-2l-1} \left(\frac{d}{d\rho}\right)^{n-l-1} e^{-\rho} \rho^{n+l}.$$

After $(n-l-1)$ -fold integration by parts we obtain an integral of the form

$$\int_0^{\infty} e^{-\rho} \rho^{n+l} \left(\frac{d}{d\rho}\right)^{n-l-1} \rho L_{n+l-1}^{2l+1}(\rho) d\rho,$$

in which we replace the Laguerre polynomial by its explicit expression according to the formula

$$L_n^m(\rho) = (-1)^m n! \sum_{k=0}^{n-m} \binom{n}{m+k} \frac{(-\rho)^k}{k!}.$$

After carrying out the differentiation only three terms remain in the sum, after which the integration is elementary. The computation leads to the simple result:

$$\langle n, l-1 || r || nl \rangle = i\sqrt{l} \cdot \frac{3}{2} n \sqrt{n^2 - l^2}. \quad (52.6)$$

The integral

$$\int_0^{\infty} R_{n', l-1} R_{nl} r^3 dr = \int_0^{\infty} \chi_{n', l-1}(r) \chi_{nl}(r) dr$$

(where $\chi_{nl} = r R_{nl}$) is the coefficient of the expansion of the function $r \chi_{nl}$ in the system of orthogonal functions $\chi_{n', l-1}$ ($n' = 1,$

$2, \dots$). The sum of squares of the moduli of these coefficients equals the integral of the square of the function being expanded²⁹. Therefore

$$\sum_{n'} |\langle n', l-1 || r || nl \rangle|^2 = l \int_0^{\infty} r^2 \chi_{nl}^2 dr. \quad (52.7)$$

Making use of the known expression for the mean value of r^2 in the state nl (see III, (36.16)), we find the following sum rule:

$$\sum_{n'} |\langle n', l-1 || r || nl \rangle|^2 = l \frac{n^2}{2} [5n^2 + 1 - 3l(l+1)]. \quad (52.8)$$

For given values of n, l and large values of n' the matrix element of the transition $nl \rightarrow n'l'$ decreases according to the law

$$|\langle n'l' || r || nl \rangle|^2 \propto \frac{3}{n'^3}, \quad (52.9)$$

as one can verify both from the particular expressions (52.4) and from the general formula (52.3). This result is entirely natural: the Coulomb levels $E' = -1/2n'^2$ for

²⁹The summation is over both discrete and continuous spectrum states.

large n' are spaced quasi-continuously, and the probability of a transition to any level in the interval dE' is proportional to the density of levels, which itself is $\propto n'^{-3}$.

The Stark effect in hydrogen has, as is well known, a specific character (see III, § 77)—the splitting is proportional to the first power of the electric field. At the same time it is assumed that the field is not strong (condition of applicability of perturbation theory), but at the same time such that the splitting of levels is large compared with the fine structure (i.e. large compared to the Lamb shift). Under these conditions the angular momentum is not conserved at all and the levels must be classified by the parabolic quantum numbers n_1, n_2, m . The last of these—the magnetic quantum number m —as before determines the projection of the orbital angular momentum onto the axis z (direction of the field), which under the given conditions (neglect of the spin-orbit interaction) is conserved. Therefore the usual selection rule applies:

$$m' - m = 0, \pm 1. \quad (52.10)$$

For the change of the numbers n_1, n_2 there are no restrictions.

The matrix elements of the dipole moment in parabolic coordinates can also be computed analytically. The

resulting formulae are, however, very cumbersome, and we shall not present them here³⁰.

Problems

1. Find the Stark splitting of the levels of hydrogen in the case when the splitting is small compared with the fine-structure intervals (but large compared with the Lamb shift).

Solution. Under the stated conditions the doubly degenerate levels with $l = j \pm \frac{1}{2}$ remain unperturbed, and the Stark splitting Δ is determined from the secular equation

$$\begin{vmatrix} -\Delta & -E(d_z)_{12} \\ -E(d_z)_{21} & -\Delta \end{vmatrix} = 0, \quad \Delta = \pm E|(d_z)_{12}|$$

(indices 1, 2 refer to the states with $l = j \pm \frac{1}{2}$ and a given magnetic quantum number m ; the perturbation $V = -Ed_z$ is diagonal in m and does not have diagonal (in l) matrix elements). The matrix element of the orbital quantity d_z is computed with the help of formulae (29.7) and (109.3) (see III), according to which

$$\langle j, l-1, m | d_z | jlm \rangle = \frac{m}{\sqrt{j(j+1)(2j+1)}} \langle j, l-1 || d || j, l \rangle,$$

$$\langle j, l-1 || d || j, l \rangle = -(2j+1) \begin{Bmatrix} l-1 & j & \frac{1}{2} \\ j & l & 1 \end{Bmatrix} \langle l-1 || d || l \rangle,$$

where one should put $l = j + \frac{1}{2}$; the quantity $\langle l-1 || d || l \rangle$ is taken from (52.6). As a result we obtain $\Delta = \pm \frac{3}{2} \sqrt{n^2 - (j + \frac{1}{2})^2} \frac{m}{j(j+1)} E$.

³⁰These formulae and corresponding numerical tables can be found in the book cited above by *H. Bethe* and *E. Salpeter*.

2. Determine the probability of emission of a photon in the transition between the states $2s_{1/2} \rightarrow 1s_{1/2}$ of the hydrogen atom (*G. Breit, E. Teller, 1940*).

Solution. The process under consideration is strictly forbidden for $E1$ -radiation by parity, and for $E2$ -radiation by the rule (46.15). We should therefore compute the probability of $M1$ -radiation, given by formula (47.5). In the given case ($l = 0$), however, the magnetic moment is a purely spin quantity, and its matrix element in the neglect of the spin-orbit interaction vanishes by virtue of the mutual orthogonality of the orbital wave functions with different principal quantum numbers, so that one must start from the full Dirac equation.

In the standard representation of the wave functions the transition current (see § 35)

$$j_{fi} = \psi_f^\dagger \boldsymbol{\alpha} \psi_i = \varphi_f^\dagger \boldsymbol{\sigma} \chi_i + \chi_f^\dagger \boldsymbol{\sigma} \varphi_i.$$

According to (35.1), (24.2), (24.8), for states with $l = 0$, $j = \frac{1}{2}$, the wave functions have the form

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \frac{1}{4\pi} \begin{pmatrix} f(r) w(m) \\ -ig(r)(\boldsymbol{\sigma}\mathbf{n}) w(m) \end{pmatrix},$$

where $\mathbf{n} = \mathbf{r}/r$, and $w(m)$ is a real unit 3-spinor corresponding to the spin projection m . Thus,

$$j_{fi} = \frac{1}{4\pi} \{ f_f g_i w_f \boldsymbol{\sigma}(\boldsymbol{\sigma}\mathbf{n}) w_i - g_f f_i w_f(\boldsymbol{\sigma}\mathbf{n}) \boldsymbol{\sigma} w_i \}.$$

Substituting this expression into (47.4) and integrating over directions $\mathbf{n}$, we get

$$\mu_{fi} = -\frac{e}{6i} w_f [\boldsymbol{\sigma}\boldsymbol{\sigma}] w_i I = -\frac{e}{3} w_f \boldsymbol{\sigma} w_i I$$

(by virtue of the commutation relations of the Pauli matrices $[\boldsymbol{\sigma}\boldsymbol{\sigma}] = 2i\boldsymbol{\sigma}$); here

$$I = \int_0^\infty (f_f g_i + f_i g_f) r^3 dr. \quad (1)$$

The probability of emission of a photon (47.5), summed over the values of m_f , is

$$w = \frac{4e^2\omega^3}{27} w_i \boldsymbol{\sigma}^2 w_i I^2 = \frac{4e^2\omega^3}{9} I^2. \quad (2)$$

From (35.4) we have (taking $\kappa = -1$)

$$g = \frac{f'}{\varepsilon + m + \alpha/r} \approx \frac{f'}{2m} - \left(\varepsilon - m + \frac{\alpha}{r} - \frac{R'}{4m^2} \right),$$

where in the second term the exact function f is replaced by the non-relativistic radial function R . If we restrict ourselves to the approximation $g = R'/2m$, the integral

$$I = \frac{1}{2m} \int_0^\infty (R_f R'_i + R'_f R_i) r^3 dr = -\frac{3}{2m} \int_0^\infty R_f R_i r^2 dr = 0 \quad (3)$$

by the orthogonality of the functions R_f and R_i . In the next approximation, taking into account (3),

$$I = \frac{1}{2m} \int_0^\infty (f_f f'_i)' r^3 dr + \frac{1}{4m^2} \int_0^\infty \left\{ R'_f R_i (\varepsilon_i - \varepsilon_f) - \frac{\alpha}{r} (R_f R_i)' \right\} r^3 dr. \quad (4)$$

Noting that by the orthogonality of the exact functions ψ_i and ψ_f we have (for $\kappa_i = \kappa_f$)

$$\int_0^\infty (f_i f_f + g_i g_f) r^2 dr = 0,$$

the first term in (4) can be rewritten after integration by parts as

$$-\frac{3}{2m} \int_0^\infty f_f f_i r^2 dr \approx \frac{3}{8m^3} \int_0^\infty R'_f R'_i r^2 dr,$$

with account of (3).

The computation of the integral with the functions

$$R_f = 2(m\alpha)^3 e^{-mar}, \quad R_i = \frac{(m\alpha)^3}{\sqrt{2}} \left(1 - \frac{mar}{2}\right) e^{-mar/2}$$

(see III, § 36) and the difference of energies

$$\omega = \varepsilon_i - \varepsilon_f = \frac{m\alpha^2}{2} \left(1 - \frac{1}{2^2}\right) = \frac{3}{8} m\alpha^2$$

gives $I = 2^{3/2} \alpha^2 / (9m)$. Whence the probability of the transition (ordinary units)

$$w = \frac{2^5 \alpha^7 \hbar^2 \omega^3}{3^5 m^2 c^4} = \frac{mc^2 \alpha^{13}}{2^4 \cdot 3^5 \hbar} = 5.6 \cdot 10^{-6} \text{ s}^{-1}.$$

The corresponding lifetime of the $2s_{1/2}$ state is very large, and in practice the decay occurs much more probably through the simultaneous emission of two photons (see the footnote on p. 263).

§ 53. Radiation of diatomic molecules. Electronic spectra

The specificity of molecular spectra is connected first of all with the separation of the energy of the molecule into electronic, vibrational and rotational parts, of which each succeeding one is small compared with the preceding. The structure of levels of diatomic molecules was studied in detail in vol. III, Ch. XI. Here we shall concern ourselves with clarifying the picture of the spectrum that arises and with computing the intensities of lines in it³¹.

Let us start with the general case, when in a transition the electronic state of the molecule changes (and along with it, generally speaking, the vibrational and rotational states as well). The frequencies of these transitions lie in the visible and ultraviolet regions of the spectrum. The totality of them is spoken of as the *electronic spectrum* of the molecule. We shall always have in mind electric-dipole transitions; transitions of other types are generally of little significance in molecular spectroscopy.

As for dipole transitions in any system, the selection rule for the total angular momentum J of the molecule holds:

$$|J' - J| \leq 1 \leq J + J'. \quad (53.1)$$

³¹The subsequent exposition is based on the material in vol. III, §§ 78, 82–88. To avoid cluttering the text, we shall refrain from constant references to these paragraphs.

The strict selection rule for the parity of the system corresponds in this case to a selection rule for the *sign* of the level; recall that in the molecular spectroscopy terminology adopted, states with wave functions not changing or changing sign on inversion (change of sign of the coordinates of electrons and nuclei) are called *positive* or *negative*. Thus, we have the strict rule:

$$+ \rightarrow -, \quad - \rightarrow +. \quad (53.2)$$

If the molecule consists of identical atoms (with nuclei of one and the same isotope), there arises the classification of levels according to the behaviour of the wave functions under permutation of the nuclear coordinates: *symmetric* (*s*) levels with wave functions not changing sign under this transformation, and *antisymmetric* (*a*) levels with functions changing sign. Since the operator of the electronic dipole moment is not affected by this transformation at all, its matrix elements are different from zero only for transitions without change of this symmetry³²:

$$s \rightarrow s, \quad a \rightarrow a. \quad (53.3)$$

This rule, however, is not absolutely strict. The point is that the existence in levels of a given symmetry property is connected with the existence in the molecule of one or another definite value of the total nuclear spin *I*. By virtue of the extreme weakness of the interaction of the nuclear spins with the electrons, *I* is conserved with great accuracy, but still not strictly. When this interaction is taken into account, *I* will not have a definite value, the symmetry property (*s* or *a*) is not conserved, and the selection rule (53.3) is lost.

The electronic terms of molecules of identical atoms are characterised also by their *parity* (*g* or *u*)—the behaviour of the wave functions (as measured from the centre of the molecule) under change of sign of the electron coordinates at unchanged nuclear coordinates. There exists a close connection between this property of the electronic term, on the one hand, and the nuclear symmetry and the sign of the rotational levels belonging to this term, on the other. The levels belonging to an even (*g*) electronic term may have characteristics *s* + or *a* −, and those to an odd one *s* − or *a* +. From the rules (53.2) and (53.3) there follows therefore also the rule

$$g \rightarrow u, \quad u \rightarrow g. \quad (53.4)$$

As an approximate rule (53.4) remains valid also for molecules from different isotopes of one and the same element. Since the nuclear charges are identical, on considering the electronic term with the nuclei at rest we are dealing in such a case with a system of electrons in an electric field possessing a centre of symmetry (at the point dividing in half the distance between the nuclei). The symmetry of the electronic wave function with respect to inversion at this point determines the parity of the term, and since the vector of the electric dipole moment changes sign under this transformation, the parity must change in a transition; hence we arrive at the rule (53.4). The approximateness of the rule, based on this derivation alone, is connected with the necessity of treating the nuclei

³²This rule applies, obviously, also to transitions of any multipolarity.

as stationary. It therefore breaks down when the interaction between the electronic state and the rotation of the molecule is taken into account.

Further selection rules are connected with one or another concrete assumptions about the comparative magnitudes of the various interactions in the molecule (i.e. about its coupling type). These rules can therefore be only approximate.

The majority of electronic terms of diatomic molecules belong to coupling types *a* or *b*. Both these types are characterised by the fact that the coupling of the orbital angular momentum with the axis of the molecule (the electric interaction of both atoms in the molecule) is large compared with all other interactions. In this connection there exist

the quantum numbers Λ and S (the projection of the electronic orbital angular momentum onto the axis of the molecule and the total spin of the electrons). The operator of the orbital quantity—the electronic orbital angular momentum—commutes with the spin operator, so that

$$S' - S = 0 \quad (\text{case } a, b). \quad (53.5)$$

The number Λ is subject to the selection rule

$$\Lambda' - \Lambda = 0, \pm 1 \quad (\text{case } a, b), \quad (53.6)$$

and for transitions between states with $\Lambda = 0$ (Σ -terms) the additional rule applies

$$\Sigma^+ \rightarrow \Sigma^+, \quad \Sigma^- \rightarrow \Sigma^- \quad (\text{case } a, b) \quad (53.7)$$

(recall that the states Σ^+ and Σ^- are distinguished by their behaviour under reflection in a plane passing through the axis of the molecule). The rules (53.6), (53.7) are obtained in the system of coordinates rigidly connected with the nuclei (see III, § 87); the rule (53.6) is analogous to the selection rule for the magnetic quantum number in the atomic case.

The coupling types *a* and *b* differ from each other by the ratio between the energy of the “spin–axis” interaction and the energy of rotation (the spacings of the rotational levels). In case *a* the first of these is larger, while in case *b* the other way round.

Case *a*. In this case there exists the quantum number Σ —the projection of the total spin onto the axis of the molecule (and with it the number $\Omega = \Sigma + \Lambda$ —the projection of the total angular momentum). If both (initial and final) states belong to case *a*, the rule

$$\Sigma' - \Sigma = 0 \quad (\text{case } a) \quad (53.8)$$

is valid (following from the already mentioned commutativity of the dipole moment operator with the spin). From (53.6) and (53.8) follows

$$\Omega' - \Omega = 0, \pm 1. \quad (53.9)$$

If $\Omega = \Omega' = 0$, then additionally to the general rule (53.1) transitions with $J' = J$ are forbidden³³:

$$J' - J = \pm 1 \quad \text{at} \quad \Omega = \Omega' = 0 \quad (\text{case } a). \quad (53.10)$$

³³This rule is analogous to the prohibition of transitions with $J = J'$ at $M = M' = 0$ for atoms (see § 5), where, however, it could be of interest only in the presence of an external field. In the present case the rule follows directly from the formula (53.12) given below; the $3j$ -symbol $\begin{pmatrix} J' & 1 & J \\ 0 & 0 & 0 \end{pmatrix}$ vanishes at $J' = J$, when the sum $J' + J + 1$ is odd.

Let us consider transitions between the rotational components of any two definite vibrational levels belonging to two different electronic terms (of type *a*). Taking into account the fine structure of the electronic term, each of these levels splits into several components, the number of which $(2S + 1)$ must by virtue of rule (53.5) be the same in both. According to rule (53.8) each component of one level combines only with a single component of the other with the same value of Σ .

Let us take further a pair of levels with identical Σ ; their values of Ω and Ω' can differ (together with Λ and Λ') by 0 or ± 1 . When the rotation is taken into account each of them splits into a sequence of levels, distinguished by consecutive values of J and J' running from values $J \geq |\Omega|$, $J' \geq |\Omega'|$. The dependence of the transition probabilities on these numbers can be established in the general form (H. Hönl, *F. London*, 1925).

The matrix element of the transition $n\Lambda\Omega JM_J \rightarrow n'\Lambda'\Omega'J'M'_J$ (n denoting the remaining quantum numbers besides Ω and Λ , characteristics of the electronic term) equals

$$\begin{aligned} |\langle n'\Lambda'\Omega'J'M'_J | d_q | n\Lambda\Omega JM_J \rangle| = \\ = \sqrt{(2J+1)(2J'+1)} \begin{pmatrix} J' & 1 & J \\ -\Omega' & q' & \Omega \end{pmatrix} \begin{pmatrix} J' & 1 & J \\ -M'_J & q & M_J \end{pmatrix} \times \\ \times |\langle n'\Lambda' | \bar{d}_{q'} | n\Lambda \rangle|, \end{aligned} \quad (53.11)$$

where d_q and $\bar{d}_{q'}$ are the spherical components of the vector of the dipole moment respectively in the fixed system of coordinates xyz and in the “moving” system $\xi\eta\zeta$ with the axis ζ along the axis of the molecule (this formula is obtained with the help of formula (110.6) (see III)). The matrix elements $\langle n'\Lambda' | \bar{d}_{q'} | n\Lambda \rangle$ do not depend on the rotational quantum numbers J, J' , but depend only on the characteristics of the electronic terms³⁴; therefore in the designation of the matrix element the indices Ω and Ω' are omitted.

The probability of the transition $n\Lambda\Omega J \rightarrow n'\Lambda'\Omega'J'$ is proportional to the square of the matrix element (53.11), summed over M'_J . By virtue of formula (106.12) (see III) we have

$$\sum_{M'_J} \begin{pmatrix} J' & 1 & J \\ -M'_J & q & M_J \end{pmatrix}^2 = \frac{1}{2J+1},$$

so that

$$w(n\Lambda\Omega J \rightarrow n'\Lambda'\Omega'J') = (2J'+1) \begin{pmatrix} J' & 1 & J \\ -\Omega' & \Omega' - \Omega & \Omega \end{pmatrix}^2 B(n', n; \Lambda', \Lambda), \quad (53.12)$$

where the coefficients B do not depend on J, J' (we neglect, of course, the small difference in frequencies of transitions between different J, J')³⁵.

³⁴One can verify this in the same way as was done at the beginning of III, § 29 for a scalar quantity f . In the present case the operator of the vectorial quantity $\mathbf{d}$ commutes with the operator of the conserved (in the zeroth approximation) vector $\mathbf{S}$ on the axis ζ , and Σ is the projection of $\mathbf{S}$ onto the axis of the rotating system of coordinates, in which one has to consider the condition of commutativity of $\mathbf{d}$ and $\mathbf{S}$.

³⁵If we sum (53.12) over J' , then (by the orthogonality property of the $3j$ -symbols (see III, (106.13))) we get simply $B(n', n; \Lambda', \Lambda)$. In other words, the total probability of the transition from the rotational level J of the state Ω to all levels J' of the state Ω' does not depend on J .

Case *b*. In this case there exists, alongside the total angular momentum J , also the quantum number K —the angular momentum of the molecule without the spin. The selection rules for this number coincide with the general selection rules for any orbital vectorial quantity (which the electric dipole moment is):

$$|K' - K| \leq 1 \leq K + K' \quad (\text{case } b) \quad (53.13)$$

with the additional prohibition of the transition with $K = K'$ at $\Lambda = \Lambda' = 0$ (analogously to (53.10)):

$$K' - K = \pm 1 \quad \text{at} \quad \Lambda = \Lambda' = 0. \quad (53.14)$$

Let us consider the transitions between the rotational components of definite vibrational levels of two electronic states belonging to type *b*. The probabilities of transitions between them are determined by the same formulae (53.12), in which one has to write K , Λ instead of J , Ω . When the fine structure is taken into account (for $S \neq 0$) each rotational level K splits into $2S + 1$ components with $J = |K - S|, \dots, K + S$, as a result of which instead of a single line $J \rightarrow J'$ a multiplet arises. Since in the present case we are dealing with the composition of the free (not connected with the molecular axis) angular momenta $\mathbf{K}$ and $\mathbf{S}$, the formulae for the relative probabilities of the various lines of the multiplet coincide with the analogous formulae (49.15) for the fine-structure components of atomic spectral lines, where the same role is played by the angular momenta $\mathbf{L}$ and $\mathbf{S}$ in the case of LS -coupling³⁶.

Thus, we have examined the selection rules determining the possible lines of the spectrum in all the main cases that can arise in diatomic molecules.

The set of lines from transitions between rotational components of two given electronic-vibrational levels constitutes, as one says in spectroscopy, a *band*; by virtue of the smallness of the rotational intervals the lines in the band are very closely spaced. The frequencies of these lines are given by the expressions

$$\hbar\omega_{J,J'} = \text{const} + BJ(J + 1) - B'J'(J' + 1), \quad (53.15)$$

where B , B' are the rotational constants in both electronic states (to avoid superfluous complications we assume the electronic terms to be singlet). For $J' = J$, $J \pm 1$ formula (53.15) is represented graphically (Fig. 2) by three branches (parabolas), whose points for integer values of J give the frequencies. (The arrangement of branches in Fig. 2 corresponds to the case $B' < B$; for $B' > B$ they are opened in the direction of small ω , and the upper branch is the curve for $J' = J - 1$ ³⁷.) The presence of a turning branch leads, as is clear from the figure, to a bunching of lines towards a definite limiting position (*band edges*).

Speaking of intensities of lines, one should also mention a peculiar phenomenon of *intensity alternation* in certain bands of the electronic spectrum of molecules of atoms of

³⁶Each of the considered rotational levels J when the Λ -doubling is taken into account splits further into two levels, of which one is positive and the other negative. Therefore instead of a single transition $J \rightarrow J'$ one must take into account, taking into account the selection rule (53.2), two transitions: from the positive (negative) component of level J to the negative (positive) component of level J' . The probabilities of these transitions are equal.

³⁷Series of lines corresponding to transitions with $J' = J + 1$, J , $J - 1$ are called respectively *P*-, *Q*- and *R*-branches.

one and the same isotope (*W. Heisenberg, F. Hund, 1927*). The symmetry requirements connected with the nuclear spins lead to the fact that in the electronic Σ -terms the rotational components with even and odd values of K possess opposite symmetry with respect to the nuclei and correspondingly different nuclear statistical weights. According to rule (53.14), when the transitions between two different Σ -terms are considered, by virtue of rule (53.4) one of the Σ -terms must be even and the other odd. As a result, for a given value of $J' - J$ the transitions alternate between pairs of symmetric and pairs of antisymmetric levels (as is illustrated by the diagram in Fig. 3 for the case of the states Σ_u^+ and Σ_g^+). On the other hand, the observed intensity of a line is proportional to the number of molecules found in a given initial state, and thereby to its statistical weight. Therefore the intensity of consecutive lines ($J = 0, 1, 2, \dots$) will alternate between larger and smaller, proportional alternately to g_a and

g_s (besides the monotonic variation predictable from formulae (53.12))³⁸.

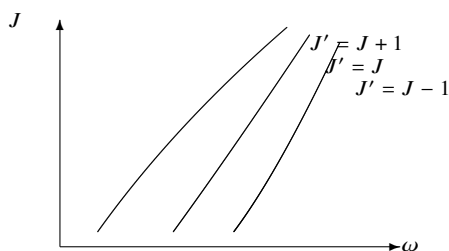


Fig. 2

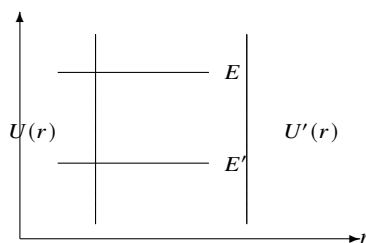


Fig. 4

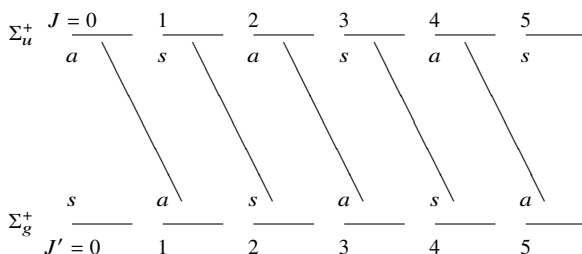


Fig. 3

For changes in the vibrational quantum number in transitions between two different electronic terms there are no strict selection rules. There exists, however, a rule (the *Franck-Condon principle*), allowing one to predict the most probable change of the vibrational state. It is based on the quasiclassical character of the motion of the nuclei, connected with their large mass (cf. what was said about predissociation in III, § 90)³⁹.

In the integral determining the matrix element of the transition between vibrational states with energies E and E' of electronic terms $U(r)$ and $U'(r)$, the main role is played

³⁸It is assumed that all states with different values of the total nuclear spin are populated uniformly.

³⁹Strictly speaking, it is also necessary that the vibrational quantum number be sufficiently large.

by the neighbourhood of the point $r = r_0$, at which

$$U(r_0) - U'(r_0) = E - E' \quad (53.16)$$

(i.e. the momenta of the relative motion of the nuclei in both states are the same: $p = p'$). For a given value of E the probability of the transition (as a function of the final energy E') is the greater the smaller each of the differences $E - U$ and $E' - U'$. It is maximal when

$$E - U(r_0) = E' - U'(r_0) = 0, \quad (53.17)$$

i.e. when the “transition point” r_0 (the root of equation (53.16)) coincides with the classical turning point of the nuclei (Fig. 4 illustrates this connection between E and the most probable E' graphically). For clarity one can say that the most probable transition occurs near the point at which the nuclei come to rest and in the neighbourhood of which they spend, consequently, the most time.

§ 54. Radiation of diatomic molecules. Vibrational and rotational spectra

The selection rules and formulae for the transition probabilities listed in the preceding paragraph retain their validity also for transitions in which the electronic state of the molecule does not change⁴⁰. We shall dwell here only on some specific features of these transitions.

First of all, by the rule (53.4) (dipole) transitions without change of the electronic state are altogether forbidden in molecules of identical atoms, since in such a transition the parity of the electronic term would remain unchanged. As follows from what was said in § 53, this prohibition would be violated only when the interaction of nuclear spins with electrons is taken into account, while for molecules from different isotopes of one and the same element it is already violated by the influence of the rotation on the electronic state.

The computation of matrix elements of the dipole moment reduces (according to formulae of § 87 (see III)) to their computation in the system of coordinates rotating with the molecule. The wave function of the molecule in this system is a product of the electronic wave function at a given distance r between the nuclei and the wave function of the vibrational motion of the nuclei in the effective field $U(r)$ of the electrons and the nuclei. With complete neglect of the influence of the motion of the nuclei on the electronic state, the initial and final electronic wave functions in the transitions under consideration are the same. Integration over the electronic coordinates gives therefore in the matrix element simply the average dipole moment of the molecule $\bar{d}$ (directed, obviously, along the axis of the molecule)⁴¹. By virtue of the smallness of the vibrations the function $\bar{d}(r)$ can be expanded in powers of the vibrational coordinate $q = r - r_0$.

⁴⁰Transitions with change of vibrational (and with it rotational) state form, as one says, the *vibrational spectrum* of the molecule; it lies in the near infrared region (wavelengths less than $20 \mu\text{m}$). Transitions with change of the rotational state only form the *rotational spectrum*, lying in the far infrared region (wavelengths greater than $20 \mu\text{m}$).

⁴¹In a molecule of identical atoms $\bar{d} = 0$, which is obvious from symmetry considerations.

For transitions involving a change of the vibrational state, the zeroth-order term of the expansion drops out of the matrix element by virtue of the orthogonality of the wave functions of the vibrational motion in one and the same field $U(q)$, so that there remains the term proportional to q . If the vibrations are treated as harmonic, then according to the well-known properties of the linear oscillator (see III, § 23) the matrix elements are non-zero only for transitions between adjacent vibrational states; in other words, for the vibrational quantum number ν the selection rule

$$\nu' - \nu = \pm 1. \quad (54.1)$$

is valid. This rule is violated, however, when the anharmonicity of the vibrations is taken into account, as well as by the following terms of the expansion of the function $\bar{d}(q)$.

For a purely rotational transition (without change of the vibrational and electronic states) the matrix element of the projection of the dipole moment onto the moving axis ζ can be set simply equal to the average dipole moment of the molecule $\bar{d} = \bar{d}(0)$ ⁴². For the probability of the transition $J \rightarrow J - 1$ we obtain as a result the formula

$$w(nJ \rightarrow n, J - 1) = \frac{4\omega^3}{3\hbar c^3} \bar{d}^2 \frac{J^2 - \Omega^2}{J(2J + 1)}, \quad (54.2)$$

which makes it possible to compute not only the relative (as in (53.12)) but also the absolute values of the probabilities. (Formula (54.2) is written for case a ; in case b one has to write K, Λ instead of J, Ω .)

The frequencies of purely rotational transitions are given by the differences of the rotational energies $BJ(J + 1)$ and equal

$$\hbar\omega_{J,J-1} = 2BJ. \quad (54.3)$$

The successive lines are equally spaced (at intervals $2B$) from each other.

§ 55. Radiation of nuclei

For γ -radiation of nuclei, as a rule, the condition of smallness of the dimensions of the system (the nuclear radius R) compared with the wavelength of the photon is satisfied. However, the spacings between nuclear levels (and thereby the energy of the γ -quantum) are usually small compared with the energy per nucleon in the nucleus. Therefore the quantity R/λ is not connected directly with the speed v/c of nucleons in the nucleus and, generally speaking, is significantly smaller. Correspondingly

the probability of $M1$ -radiation is, as a rule, greater than that of $E, l + 1$ radiation (cf. the beginning of § 50).

The general selection rules for the total angular momentum ("spin") of the nucleus and for parity are the same as for radiation by any system. A characteristic feature of nuclear radiation is the prevalence of transitions of higher multipolarity. In contrast to atoms, whose radiation is usually electric-dipole, in nuclei at small energies such transitions are comparatively rare, being forbidden by the selection rules.

⁴²In a molecule of identical atoms $\bar{d} = 0$, which is obvious from symmetry considerations.

If a radiative transition of a nucleus can be regarded as a single-particle process—a change of state of one nucleon with the state of the nuclear “core” remaining unchanged—then the selection rules for the angular momentum of this nucleon are added. The accuracy of observance of such “single-particle” selection rules turns out, however, to be very low.

Specific for nuclei are the selection rules for the isotopic spin. Recall that the projection T_3 of the isotopic spin is determined already by the mass and number of the nucleus:

$$T_3 = \frac{1}{2}(Z - N) = Z - \frac{A}{2}.$$

For a given value of T_3 the absolute magnitude of the isotopic spin can have any values $T \geq |T_3|$. The selection rule for the number T for radiative transitions arises in connection with the fact that the operators of the electric and magnetic moments of the nucleus, expressed through the operators of isotopic spins of the nucleons, represent sums of a scalar and the x_3 -component of a vector in isotopic space (see III, § 116). Therefore their matrix elements are non-zero only under the condition

$$T' - T = 0, \pm 1. \quad (55.1)$$

In itself this rule, however, does not impose any particular restrictions on transitions in light nuclei (for which only one can speak with sufficient accuracy about the conservation of isotopic spin); the fact is that among the lowest levels of these nuclei there are practically no levels with $T > 1$.

But for $E1$ -transitions there is an additional rule in connection with the fact that for the electric dipole moment the isotopically scalar part drops out, and its operator reduces to the x_3 -component of the isotopic vector (see III, § 116). Therefore if $T_3 = 0$, the transitions with $\Delta T = 0$ are additionally forbidden. In other words, in nuclei with equal numbers of neutrons and protons ($N = Z$, $A = 2Z$) $E1$ -transitions are possible only when

$$T' - T = \pm 1 \quad (T_3 = 0). \quad (55.2)$$

Of course, the accuracy of observance of this rule depends on the accuracy with which the isotopic spin of the nucleus is conserved.

The probability of $E1$ -transitions in the nucleus is also influenced by the effect of recoil of the nuclear core in the motion of individual nucleons. This effect leads to the fact that in forming the dipole moment the protons participate with an effective charge $e(1 - Z/A)$ instead of e , and the neutrons with a charge $-eZ/A$ instead of 0 (see

III, § 118). The reduction of the effective charge of the proton leads to a certain suppression of the probability of $E1$ -transitions.

The energy levels of non-spherical nuclei possess rotational structure. In connection with this there appears a specific rotational structure of the spectrum of γ -radiation for such nuclei. The symmetry of the field in which the nucleons move in a “stationary” non-spherical (axially symmetric) nucleus coincides with the symmetry of the field in which the electrons move in a “stationary” diatomic molecule of identical atoms (point group $C_{\infty h}$). Therefore the symmetry properties of levels of the non-spherical nucleus (and with them the selection rules for matrix elements) are analogous to the symmetry

properties of levels of a diatomic molecule (see III, § 119). In particular, as for a diatomic molecule of identical atoms, electric-dipole transitions are forbidden within one and the same rotational band (i.e. without change of the internal state of the nucleus)—cf. § 54. These transitions are therefore realised as $E2$ - or $M1$ -transitions. In the first case the total angular momentum J of the nucleus can change by 2 or 1, and in the second by 1.

According to (46.9) the probability of the quadrupole transition, summed over the final values of the projection M' of the total angular momentum of the nucleus, is

$$w_{E2} = \frac{\omega^5}{15\hbar c^5} \sum_{M'} |\langle J' \Omega M' | Q_{2,-m}^{(\mathcal{E})} | J \Omega M \rangle|^2$$

(J is the total angular momentum of the nucleus; Ω is its projection onto the axis of the nucleus; $m = M - M'$). With the help of formula (110.8) (see III) this sum can be expressed through the squares of the given quantities—the diagonal (with respect to the internal state of the nucleus) quadrupole moments of the transition $\bar{Q}_{2\lambda}$, defined in the coordinate system $\xi\eta\zeta$ associated with the nucleus. Here $\lambda = \Omega - \Omega'$, so that in the given case ($\Omega' = \Omega$) only the component $\bar{Q}_{20}$ appears. By definition the quantity $e\bar{Q}_0$ is simply called the quadrupole moment of the nucleus:

$$eQ_0 = e \int \rho_{ii} (2\zeta^2 - \xi^2 - \eta^2) d\xi d\eta d\zeta = -2e(\bar{Q}_{20})_{ii}.$$

We therefore obtain

$$w_{E2}(\Omega J \rightarrow \Omega J') = \frac{\omega^5}{60\hbar c^5} Q_0^2 (2J' + 1) \begin{pmatrix} J' & 2 & J \\ -\Omega & 0 & \Omega \end{pmatrix}^2. \quad (55.3)$$

Concerning these formulae, however, the following remark should be made. In them the matrix elements computed with wave functions of the form

$$\psi_{J\Omega M} = \text{const} \cdot \chi_{\Omega} D_{\Omega M}^{(J)}(\mathbf{n})$$

(χ is the wave function of the internal state of the nucleus) have been used. These functions correspond to definite (in magnitude and sign) values of the projection of the angular momentum onto the axis ζ . In nuclei, however, we are dealing with states possessing only definite parity and magnitude of the projection of the angular momentum (the latter quantity is usually and will be understood under Ω). Therefore as initial and final wave functions one should take the combinations

$$\frac{1}{\sqrt{2}} (\psi_{J\Omega M} \pm \psi_{J,-\Omega,M}), \quad \frac{1}{\sqrt{2}} (\psi_{J'\Omega M'} \pm \psi_{J',-\Omega,M'}).$$

The products of first and second terms will give the same value of the matrix element of the quadrupole transition. The “cross” terms, however, will lead to non-vanishing integrals if $2\Omega \leq 2^{43}$. Therefore formula (55.3), strictly speaking, is inapplicable for

⁴³For the matrix elements of the l -pole transition the sub-integral expressions will contain products of the form

$$D_{-J,\Omega M'}^{(J')} D_{q'q}^{(l)} D_{\Omega M}^{(J)}.$$

The integral over angles will be non-zero only when $q' = -2\Omega$, while q' runs only from $-l$ to l ; therefore it is necessary that $2\Omega \leq l$.

$\Omega = \frac{1}{2}, 1$; in these cases an additional term appears in the probability of the transition, not expressible through the mean value of the quadrupole moment⁴⁴.

Analogously to the derivation of formula (55.3) for the probability of the $M1$ -transition we obtain the formula

$$w_{M1}(\Omega J \rightarrow \Omega, J-1) = \frac{4\omega^3}{3\hbar c^3} \mu^2 (2J-1) \begin{pmatrix} J-1 & 1 & J \\ -\Omega & 0 & \Omega \end{pmatrix}^2 = \frac{4\omega^3}{3\hbar c^3} \mu^2 \frac{J^2 - \Omega^2}{J(2J+1)}, \quad (55.4)$$

where μ is the magnetic moment of the nucleus (this formula is inapplicable for $\Omega = \frac{1}{2}$).

§ 56. Photoeffect. Non-relativistic case

In §§ 49–52 we considered radiative transitions (emission or absorption of a photon) between levels of the discrete spectrum. The photoeffect differs from such a process only in that the final state belongs to the continuous spectrum.

The cross section of the photoeffect can be computed to the end in analytical form for the hydrogen atom or for a hydrogen-like ion (with nuclear charge $Z \ll 137$).

In the initial state we have an electron on a discrete level $\varepsilon_i \equiv -I$ (I is the ionisation potential of the atom) and a photon with a definite momentum $\mathbf{k}$. In the final state the electron has momentum $\mathbf{p}$ (and energy $\varepsilon_f \equiv \varepsilon$). Since $\mathbf{p}$ runs through a continuous range of values, the cross section of the photoeffect is given by the formula

$$d\sigma = 2\pi |V_{fi}|^2 \delta(-I + \omega - \varepsilon) \frac{d^3p}{(2\pi)^3}, \quad (56.1)$$

(cf. (44.3)), where the wave function of the final state of the electron is assumed normalised to one particle in a volume $V = 1$. Thus the photon wave function is also normalised to one photon in a volume $V = 1$; for the cross section $d\sigma$ the probability $d\sigma$ should be divided by the flux density of photons (equal to $c/V = c$), but in relativistic units this does not change the form of formula (56.1).

Let us choose, as in (45.2), the three-dimensionally transverse gauge of the wave function of the photon. Then

$$V_{fi} = -e\mathbf{A}\mathbf{j}_{fi} = -e\sqrt{4\pi} \frac{1}{\sqrt{2\omega}} M_{fi},$$

where we have denoted

$$M_{fi} = \int \psi'^* (\alpha\mathbf{e}) e^{i\mathbf{k}\mathbf{r}} \psi d^3x \quad (56.2)$$

($\psi \equiv \psi_i$ and $\psi' \equiv \psi_f$ are the initial and final wave functions of the electron). Replacing in (56.1) $d^3p \rightarrow \mathbf{p}^2 d|\mathbf{p}| \varepsilon d\varepsilon d\omega$ and integrating the δ -function over $d\varepsilon$, we rewrite this formula in the form

$$d\sigma = e^2 \frac{|\mathbf{p}|}{2\pi\omega} |M_{fi}|^2 d\omega. \quad (56.3)$$

We shall carry out the computation in two cases, distinguished by the value of the photon energy: for $\omega \gg I$ and for $\omega \ll m$. Since $I \sim me^4 Z^2 \ll m$, these two regions

⁴⁴In practice this correction gives a substantial correction only for $\Omega = \frac{1}{2}$, when the connection between the rotation and the internal state of the nucleus is especially strong (see III, § 119).

partially overlap (for $I \ll \omega \ll m$), so that an investigation of these cases gives essentially a complete description of the photoeffect.

Let us begin with the case

$$\omega \ll m. \quad (56.4)$$

In this case the speed of the electron is small both in the initial and the final state, so that as regards the electron the problem is entirely non-relativistic. Correspondingly, we replace in

(56.2) the non-relativistic operator of velocity $\hat{\mathbf{v}} = -i\nabla/m$ (cf. § 45). Moreover, we can go over to the dipole approximation—set $e^{i\mathbf{kr}} \approx 1$, i.e. neglect the photon momentum compared with the electron momentum. Then

$$d\sigma = e^2 \frac{m|\mathbf{p}|}{2\pi\omega} |\mathbf{e}\mathbf{v}_{fi}|^2 d\omega, \quad \mathbf{v}_{fi} = -\frac{i}{m} \int \psi'^* \nabla \psi \cdot d^3x. \quad (56.5)$$

We shall consider the photoeffect from the ground level of the hydrogen atom (or hydrogen-like ion). Then

$$\psi = \frac{(Ze^2m)^{3/2}}{\sqrt{\pi}} e^{-Ze^2mr}, \quad (56.6)$$

(in ordinary units $me^2 \rightarrow 1/a_0$, where $a_0 = \hbar^2/me^2$ is the Bohr radius).

As the ψ' we must take the wave function, asymptotically containing a plane wave ($e^{i\mathbf{pr}}$) and along with it a converging spherical wave (see III, § 136, where such a function was denoted $\psi_{\mathbf{p}}^{(-)}$). By the selection rule for l the transition from an s -state is possible only to a p -state (dipole case). Therefore in the expansion⁴⁵

$$\psi_{\mathbf{p}}^{(-)} = \frac{1}{2p} \sum_{l=0}^{\infty} i^l (2l+1) e^{-i\delta_l} R_{pl}(r) P_l(\mathbf{nn}_1) \quad (56.7)$$

($\mathbf{n} = \mathbf{p}/p$, $\mathbf{n}_1 = \mathbf{r}/r$) it suffices to retain only the term with $l = 1$. Omitting inessential phase factors, we get

$$\psi' = \frac{3}{2p} (\mathbf{nn}_1) R_{p1}(r). \quad (56.8)$$

With the functions ψ and ψ' from (56.6), (56.8) we have

$$\begin{aligned} \mathbf{e}\mathbf{v}_{fi} &= \frac{3(Ze^2m)^{5/2}}{\sqrt{\pi}m} \iint (\mathbf{nn}_1)(\mathbf{n}_1\mathbf{e}) e^{-Ze^2mr} R_{p1}(r) d\omega_1 \cdot r^2 dr = \\ &= \frac{\sqrt{2\pi} (Ze^2m)^{5/2}}{pm} \int r^2 e^{-Ze^2mr} R_{p1}(r) r^2 dr. \end{aligned}$$

According to formulae (36.18) and (36.24) (see III) the radial function (in the units adopted here⁴⁶)

$$R_{pl} = \frac{\sqrt{8\pi} Ze^2m}{3} \sqrt{\frac{1+\nu^2}{\nu(1-e^{-2\pi\nu})}} pr e^{-ipr} F(2+i\nu, 4, 2ipr),$$

⁴⁵Below in this paragraph p denotes $|\mathbf{p}|$.

⁴⁶In this paragraph we use atomic units.

where we have denoted:

$$\nu = \frac{Ze^2 m}{p} \quad \left(= \frac{Ze^2}{\hbar v} \right). \quad (56.9)$$

The integral we need is evaluated with the help of the formula

$$\int_0^\infty e^{-\lambda z} z^{\gamma-1} F(\alpha, \gamma, kz) dz = \Gamma(\gamma) \lambda^{\alpha-\gamma} (\lambda - k)^{-\alpha}$$

(see III, (f.3)). Noting also that

$$\left(\frac{\nu + i}{\nu - i} \right)^{i\nu} = e^{-2\nu \operatorname{arctg} \nu},$$

we obtain

$$\mathbf{e} \nu_{fi} = \frac{2^{7/2} \pi \nu^3 (\mathbf{ne})}{\sqrt{p m} (1 + \nu^2)^{3/2}} \frac{e^{-2\nu \operatorname{arctg} \nu}}{\sqrt{1 - e^{-2\pi\nu}}}.$$

The ionisation energy from the ground level of the hydrogen atom (or hydrogen-like ion) $I = Z^2 e^4 m / 2$. Therefore

$$\omega = \frac{p^2}{2m} + I = \frac{p^2}{2m} (1 + \nu^2). \quad (56.10)$$

Taking this relation into account, we write the final expression for the cross section of the photoeffect with ejection of the electron into the solid angle do :

$$d\sigma = 2^7 \pi \alpha a^2 \left(\frac{I}{\hbar \omega} \right)^4 \frac{e^{-4\nu \operatorname{arctg} \nu}}{1 - e^{-2\pi\nu}} (\mathbf{ne})^2 do, \quad (56.11)$$

where $a = \hbar^2 / (mZe^2) = a_0 / Z$ (here and below, ordinary units). Let us note that the angular distribution of photoelectrons is determined by the factor $(\mathbf{ne})^2$. It is maximal in the directions parallel to the polarisation vector $\mathbf{e}$ of the incident photons, and vanishes in the directions perpendicular to $\mathbf{e}$, including the direction of incidence. For unpolarised photons formula (56.11) should be averaged over directions of $\mathbf{e}$, which amounts to the replacement

$$(\mathbf{ne})^2 \rightarrow \frac{1}{2} [\mathbf{n}_0 \mathbf{n}]^2,$$

where $\mathbf{n}_0 = \mathbf{k} / k$ (see (45.4b)).

Integration of formula (56.11) over angles gives the total cross section of the photoeffect:

$$\sigma = \frac{2^9 \pi^2}{3} \alpha a^2 \left(\frac{I}{\hbar \omega} \right)^4 \frac{e^{-4\nu \operatorname{arctg} \nu}}{1 - e^{-2\pi\nu}} \quad (56.12)$$

(*M. Stobbe*, 1930).

The limiting value of σ when $\hbar \omega \rightarrow I$ (i.e. $\nu \rightarrow \infty$):

$$\sigma = \frac{2^9 \pi^2}{3e^4} \alpha a^2 = \frac{2^9 \pi^2}{3e^4} \alpha a_0^2 = 0.23 \frac{a_0^2}{Z^2} \quad (56.13)$$

(in the denominator $e = 2.71 \dots$!). As should be the case for reactions with formation of charged particles (see III, § 147), the cross section of the photoeffect near its threshold tends to a constant limit.

The case $\hbar\omega \gg I$ (while still $\hbar\omega \ll mc^2$) corresponds to the Born approximation ($\nu = Ze^2/(\hbar\nu) \ll 1$). Formula (56.12) takes the form

$$\sigma = \frac{2^8 \pi}{3} \alpha a_0^2 Z^5 \left(\frac{I_0}{\hbar\omega} \right)^{7/2} \quad (56.14)$$

($I_0 = e^4 m / (2\hbar^2)$ is the ionisation energy of the hydrogen atom).

The process inverse to the photoeffect is radiational recombination of an electron with a stationary ion. The cross section of this process (σ_{rec}) can be found from the cross section of the photoeffect (σ_{Φ}) with the help of the principle of detailed balance (III, § 144). According to this principle the cross sections of the processes $i \rightarrow f$ and $f \rightarrow i$ (with two particles in each of the states i and f) are related by the equation

$$g_i p_i^2 \sigma_{i \rightarrow f} = g_f p_f^2 \sigma_{f \rightarrow i},$$

where p_i, p_f are the momenta of relative motion of the particles, and g_i, g_f are the spin statistical weights of states i and f . Taking into account also that for the photon $g = 2$ (two directions of polarisation), and the statistical weight of the free electron and ion is the same as the statistical weight of the ground state of the hydrogen atom, we get

$$\sigma_{\text{rec}} = \sigma_{\Phi} \frac{2\mathbf{k}^2}{\mathbf{p}^2} \quad (56.15)$$

($\mathbf{p} = m\mathbf{v}$ is the momentum of the incident electron, $\mathbf{k}$ is the momentum of the emitted photon).

Problems

1. Derive formula (56.14) by direct use of the Born approximation in the non-relativistic case.

Solution. In the Born approximation as the ψ' in formula (56.5) we must write simply a plane wave $\psi' = e^{i\mathbf{p}\mathbf{r}}$, and ψ is as before the function (56.6). Then

$$\mathbf{v}_{fi} = \frac{1}{m} \int \psi'^* \hat{\mathbf{p}} \psi d^3x = \frac{\mathbf{p}}{m} \frac{(Ze^2 m)^{3/2}}{\sqrt{\pi}} (e^{-Ze^2 m r})_{\mathbf{p}}.$$

The Fourier component is given by formula (57.6b), so that

$$\mathbf{v}_{fi} \approx 8\sqrt{\pi} p^{-3} m^{3/2} (Ze^2)^{5/2} \mathbf{n}.$$

Substituting in (56.5) and integrating over $d\omega$, we get (56.14) (with, to sufficient accuracy, $p^2/(2m) \approx \omega$).

2. Determine the total cross section of radiational recombination of a fast non-relativistic electron ($I \ll mv^2 \ll mc^2$) with a nucleus (charge $Z \ll 137$).

Solution. The cross section of capture onto the K -shell (principal quantum number $n = 1$) is obtained by substituting (56.14) into (56.15):

$$\sigma_1^{\text{rec}} = \frac{2^7 \pi}{3} Z^5 \alpha^3 a_0^2 \left(\frac{I_0}{\varepsilon} \right)^{5/2}$$

($\varepsilon = mv^2/2$ is the energy of the incident electron; $\hbar\omega \approx \varepsilon$). Of the other states of the atom being formed only the s -states are essential: in computing the matrix element in the Born approximation the important region is the values of the wave function of the bound state at small r (as will be seen from what follows in § 57), and for $l > 0$ these values are small by comparison with the values of functions with $l = 0$; it suffices to take into account only the first four terms of the expansion of ψ by powers of r . For states with $l = 0$ and arbitrary n the expansion

$$\psi = \frac{1}{\sqrt{\pi} a^{3/2} n^{3/2}} \left(1 - \frac{r}{a} \right)$$

i.e. contains only the common factor $n^{-3/2}$ (the above expression is obtained using the expansion (36.13) (see III)). Therefore the total recombination cross section

$$\sigma^{\text{rec}} = \sum_{n=1}^{\infty} \sigma_n^{\text{rec}} = \sigma_1^{\text{rec}} \sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3) \sigma_1^{\text{rec}}$$

(the value of the ζ -function: $\zeta(3) = 1.202$).

§ 57. Photoeffect. Relativistic case

Let us now turn to the case

$$\omega \gg I. \quad (57.1)$$

In this case also $\varepsilon = \omega - I \gg I$, and therefore the influence of the Coulomb field of the nucleus on the wave function of the photoelectron (ψ') can be taken into account with the help of perturbation theory. We write ψ' in the form

$$\psi' = \frac{1}{\sqrt{2\varepsilon}} (u' e^{i\mathbf{p}\mathbf{r}} + \psi^{(1)}). \quad (57.2)$$

The photoelectron may be relativistic; therefore the unperturbed function in (57.2) is written in the form of a relativistic plane wave (23.1).

Although in the initial state the electron is non-relativistic, its wave function ψ must none the less include (for reasons to be clarified below) a relativistic correction ($\sim Ze^2$). Such a function is given by the formula (see the problem to § 39)

$$\psi = \left(1 - \frac{i}{2m} \gamma^0 \gamma \nabla \right) \frac{u}{\sqrt{2m}} \psi_{\text{nr}}, \quad (57.3)$$

where ψ_{nr} is the non-relativistic function of the bound state (56.6), and u is a bispinor amplitude of the electron at rest, normalised by the condition we have adopted $\bar{u}u = 2m$.

Substituting the functions (57.2), (57.3) into the matrix element (56.2)⁴⁷:

$$M_{fi} = \frac{1}{\sqrt{2m\varepsilon}} \int \left\{ \bar{u}'(\boldsymbol{\gamma}\mathbf{e}) \left[\left(1 - \frac{i}{2m} \boldsymbol{\gamma}^0 \boldsymbol{\gamma} \nabla \right) u \psi_{\text{nr}} \right] e^{-i(\mathbf{p}-\mathbf{k})\mathbf{r}} + \bar{\psi}_{\mathbf{k}}^{(1)}(\boldsymbol{\gamma}\mathbf{e}) e^{i\mathbf{k}\mathbf{r}} u \psi_{\text{nr}} \right\} d^3x. \quad (57.4)$$

Having in mind extracting the first term of the expansion of this quantity in powers of Ze^2 , we can in the second term in the curly brackets replace ψ_{nr} simply by the constant $(Ze^2m)^{3/2}/\sqrt{\pi}$. The first term of the result of such a replacement would vanish (for $\mathbf{p} - \mathbf{k} \neq 0$), and it is precisely for this reason that in ψ one must also include the first relativistic correction, proportional to Ze^2 ; for $v \sim 1$ this correction contributes to the cross section terms of the same order as the next term of the expansion of ψ_{nr} in powers of Ze^2 .

In the first term of (57.4) we carry out the integration by parts, transferring the action of the operator ∇ from ψ_{nr} onto the exponential factor. As a result we obtain

$$M_{fi} = \frac{(Ze^2m)^{3/2}}{2(\pi m\varepsilon)^{1/2}} \left\{ \bar{u}'(\boldsymbol{\gamma}\mathbf{e}) \left[1 + \frac{1}{2m} \boldsymbol{\gamma}^0 \boldsymbol{\gamma}(\mathbf{p} - \mathbf{k}) \right] u(e^{-Ze^2mr})_{\mathbf{p}-\mathbf{k}} + \bar{\psi}_{-\mathbf{k}}^{(1)}(\boldsymbol{\gamma}\mathbf{e}) u \right\}, \quad (57.5)$$

where the vector index denotes the spatial Fourier component. To the accuracy of terms $\sim Ze^2$ ⁴⁸

$$(e^{-Ze^2mr})_{\mathbf{p}-\mathbf{k}} = \frac{8\pi Ze^2m}{(\mathbf{p} - \mathbf{k})^4}. \quad (57.6)$$

To compute the Fourier component $\bar{\psi}_{\mathbf{k}}^{(1)}$ we write the equation satisfied by the function $\psi^{(1)}$:

$$(\boldsymbol{\gamma}^0 \varepsilon + i\boldsymbol{\gamma} \nabla - m)\psi^{(1)} = e(\boldsymbol{\gamma}^\mu A_\mu) u' e^{i\mathbf{p}\mathbf{r}} = -\frac{Ze^2}{r} \boldsymbol{\gamma}^0 u' e^{i\mathbf{p}\mathbf{r}}$$

(this is obtained by substituting (57.2) into (32.1)). Applying to both sides of this equation the operator $(\boldsymbol{\gamma}^0 \varepsilon + i\boldsymbol{\gamma} \nabla + m)$, we get

$$(\Delta + \mathbf{p}^2)\psi^{(1)} = -Ze^2(\boldsymbol{\gamma}^0 \varepsilon - \boldsymbol{\gamma}\mathbf{k} + m)(\boldsymbol{\gamma}^0 u') \frac{1}{r} e^{i\mathbf{p}\mathbf{r}}.$$

⁴⁷Formula (57.3) was derived for distances $r \sim 1/(mZe^2)$, at which the correction term in it and the relativistic correction to the purely exponential function (56.6) are always proportional to Ze^2 . But for the ground state (and indeed for all s -states) formula (57.3) is suitable for any r , since the derivative of the purely exponential function (56.6) (and the correction term in (57.3)) is always proportional to Ze^2 .

⁴⁸Taking the Fourier component from both sides of the equality

$$(\Delta - \lambda^2) \frac{e^{-\lambda r}}{r} = -4\pi \delta(\mathbf{r}),$$

we obtain

$$\left(\frac{e^{-\lambda r}}{r} \right)_{\mathbf{q}} = \frac{4\pi}{\mathbf{q}^2 + \lambda^2}. \quad (57.6a)$$

Differentiating this expression with respect to the parameter λ gives

$$(e^{-\lambda r})_{\mathbf{q}} = \frac{8\pi\lambda}{(\mathbf{q}^2 + \lambda^2)^2}. \quad (57.6b)$$

Multiplying this equation by $e^{-i\mathbf{k}\mathbf{r}}$ and integrating over d^3x , with the terms containing Δ and ∇ being integrated by parts in the usual way:

$$\begin{aligned} (\mathbf{p}^2 - \mathbf{k}^2)\bar{\psi}_{\mathbf{k}}^{(1)} &= -Ze^2(\gamma^0\varepsilon - \boldsymbol{\gamma}\mathbf{k} + m)(\gamma^0 u') \left(\frac{1}{r} \right)_{\mathbf{k}-\mathbf{p}} = \\ &= -Ze^2(2\varepsilon\gamma^0 - \boldsymbol{\gamma}(\mathbf{k} - \mathbf{p}))(\gamma^0 u') \frac{4\pi}{(\mathbf{k} - \mathbf{p})^2}. \end{aligned}$$

In the last line we have taken into account that the amplitude u' satisfies the equation

$$(\varepsilon\gamma^0 - \boldsymbol{\gamma}\mathbf{p} - m)u' = 0, \quad \text{or} \quad (\varepsilon\gamma^0 + \boldsymbol{\gamma}\mathbf{p} - m)\gamma^0 u' = 0.$$

From this we find

$$\bar{\psi}_{-\mathbf{k}}^{(1)}\gamma^0 = 4\pi Ze^2 \bar{u}' \frac{2\varepsilon\gamma^0 + \boldsymbol{\gamma}(\mathbf{k} - \mathbf{p})}{(\mathbf{k}^2 - \mathbf{p}^2)(\mathbf{k} - \mathbf{p})^2} \gamma^0. \quad (57.7)$$

Substituting (57.6), (57.7) into the matrix element (57.5), we represent it in the form

$$M_{fi} = \frac{4\pi^{1/2}(Ze^2m)^{5/2}}{(\varepsilon m)^{1/2}(\mathbf{k} - \mathbf{p})^2} \bar{u}' A u,$$

where

$$\begin{aligned} A &= a(\boldsymbol{\gamma}\mathbf{e}) + (\boldsymbol{\gamma}\mathbf{e})\gamma^0(\boldsymbol{\gamma}\mathbf{b}) + (\boldsymbol{\gamma}\mathbf{c})\gamma^0(\boldsymbol{\gamma}\mathbf{e}), \\ a &= \frac{1}{(\mathbf{k} - \mathbf{p})^2} + \frac{\varepsilon}{m} \frac{1}{\mathbf{k}^2 - \mathbf{p}^2}, \quad \mathbf{b} = -\frac{\mathbf{k} - \mathbf{p}}{2m(\mathbf{k} - \mathbf{p})^2}, \quad \mathbf{c} = \frac{\mathbf{k} - \mathbf{p}}{2m(\mathbf{k}^2 - \mathbf{p})^2}. \end{aligned}$$

The cross section

$$d\sigma = \frac{8e^2(Ze^2m)^5|\mathbf{p}|}{m\omega(\mathbf{k} - \mathbf{p})^4} (\bar{u}' A u)(\bar{u} \bar{A} u') d\Omega,$$

where $\bar{A} = \gamma^0 A^+ \gamma^0$ (see § 65). This expression must still be summed over the final and averaged over the initial directions of the spin of the electron. These operations are carried out according to the rules described below in § 65, with the help of the polarisation density matrices of the initial and final states:

$$\rho = \frac{m}{2}(\gamma^0 + 1), \quad \rho' = \frac{1}{2}(\gamma^0\varepsilon - \boldsymbol{\gamma}\mathbf{p} + m)$$

(in the initial state $\mathbf{p} = 0$, $\varepsilon = m$). They lead to the expression

$$d\sigma = \frac{16e^2(Ze^2m)^5|\mathbf{p}|}{m\omega(\mathbf{k} - \mathbf{p})^4} \text{Sp}(\rho' A \rho \bar{A}) d\Omega.$$

The computation of the trace (using formulae (22.22)) is a purely algebraic operation and leads to the following result:

$$\text{Sp}(\rho' A \rho \bar{A}) = \frac{m}{\varepsilon + m} [a\mathbf{p} - (\mathbf{b} - \mathbf{c})(\varepsilon + m)]^2 + 4m(\mathbf{b}\mathbf{e})[(\varepsilon + m)(\mathbf{c}\mathbf{e}) + a(\mathbf{p}\mathbf{e})]$$

(vector $\mathbf{e}$ is assumed to be real linear polarisation of the photon).

Let us bring the formula for the cross section of the photoeffect to its final form, introducing the polar angle θ and the azimuth φ of the direction $\mathbf{p}$ relative to the direction $\mathbf{k}$, and $\mathbf{e}$ in the plane xz (so that $\mathbf{pe} = |\mathbf{p}| \cos \varphi \sin \theta$). For $\omega \gg I$ the conservation of energy can be written as $\varepsilon - m = \omega$ (instead of $\varepsilon - m = \omega - I$). It is easy to verify that

$$\mathbf{k}^2 - \mathbf{p}^2 = -2m(\varepsilon - m), \quad (\mathbf{k} - \mathbf{p})^2 = 2\varepsilon(\varepsilon - m)(1 - v \cos \theta),$$

where $\mathbf{v} = \mathbf{p}/\varepsilon$ is the velocity of the photoelectron. After straightforward transformations we finally obtain

$$d\sigma = Z^5 \alpha^4 r_e^2 \frac{v^3 (1 - v^2)^{3/2} \sin^2 \theta}{(1 - \sqrt{1 - v^2})^5 (1 - v \cos \theta)^4} \times \\ \times \left\{ \frac{(1 - \sqrt{1 - v^2})^2}{2(1 - v^2)^{3/2}} (1 - v \cos \theta) + \left[2 - \frac{(1 - \sqrt{1 - v^2})(1 - v \cos \theta)}{1 - v^2} \right] \cos^2 \varphi \right\} d\varphi, \quad (57.8)$$

where $r_e = e^2/m$.

In the ultra-relativistic case ($\varepsilon \gg m$) the cross section of the photoeffect has a sharp maximum at small angles ($\theta \sim \sqrt{1 - v^2}$), i.e. the electrons are emitted predominantly in the direction of incidence of the photon. Near the maximum we write

$$1 - v \cos \theta \approx \frac{1}{2}[(1 - v^2) + \theta^2],$$

and the leading terms in (57.8) give

$$d\sigma \approx 4Z^5 \alpha^4 r_e^2 \frac{(1 - v^2)^{3/2}}{(1 - v^2 + \theta^2)^3} \theta d\theta d\varphi. \quad (57.9)$$

The elementary, though rather lengthy, integration of expression (57.8) over angles leads to the following formula for the total cross section (*F. Sauter*, 1931):

$$\sigma = 2\pi Z^5 \alpha^4 r_e^2 \frac{(\gamma^2 - 1)^{3/2}}{(\gamma - 1)^5} \times \\ \times \left\{ \frac{3}{4} + \frac{\gamma(\gamma - 2)}{\gamma + 1} \left(1 - \frac{1}{2\gamma\sqrt{\gamma^2 - 1}} \ln \frac{\gamma + \sqrt{\gamma^2 - 1}}{\gamma - \sqrt{\gamma^2 - 1}} \right) \right\}, \quad (57.10)$$

where for brevity we have introduced the "Lorentz factor"

$$\gamma = \frac{1}{\sqrt{1 - v^2}} = \frac{\varepsilon}{m} \approx \frac{m + \omega}{m}. \quad (57.11)$$

In the ultra-relativistic case this formula reduces to the simple expression

$$\sigma = 2\pi Z^5 \alpha^4 r_e^2 / \gamma. \quad (57.12)$$

In the case $I \ll \omega \ll m$ formula (57.10) goes to the limit of small $\gamma - 1$ and leads to the result (56.14), already known to us.

§ 58. Photodisintegration of the deuteron

A characteristic feature of the deuteron is the smallness of its binding energy (compared with the depth of the potential well). This circumstance allows one to describe the origin of the deuteron reaction without detailed knowledge of the course of the nuclear forces, using only the binding energy (see III, § 133). It is assumed at the same time that the wavelengths of the colliding particles are large compared with the radius of action of the nuclear forces a .

This also applies to the disintegration of the deuteron by γ -quanta, for which $ka \ll 1$. It is assumed also that $pa \ll 1$, where $\mathbf{p}$ is the momentum of relative motion of the freed neutron and proton (this condition is even stronger than the preceding⁴⁹).

Starting from the non-relativistic formula for the cross section of the photoeffect (56.5), integrating it over the directions of the photon:

$$\sigma = \frac{e^2 p}{3M\omega} \frac{M}{2} \frac{4\pi}{3} |\mathbf{v}_p)_{fi}|^2. \quad (58.1)$$

Here $\mathbf{p}$ is the momentum of relative motion of the proton and neutron⁵⁰, and m in (56.5) is replaced by their reduced mass $M/2$ (where M is the nucleon mass). The matrix element is taken from the velocity of the proton $\mathbf{v}_p$, since only the proton interacts with the photon. Expressing $\mathbf{v}_p$ through the momentum $\mathbf{p}$ ($\mathbf{v}_p = \mathbf{v}/2 = \mathbf{p}/M$), we have $e\mathbf{p}/M = e\mathbf{v}_p = \dot{\mathbf{d}}$, so that $e\mathbf{p}_{fi}/M = i\omega\mathbf{d}_{fi}$.

The index ($\mathcal{E}$) indicates that this formula corresponds to electric-dipole transitions: $e\mathbf{p}/M = e\mathbf{v}_p = \dot{\mathbf{d}}$, so that $e\mathbf{p}_{fi}/M = i\omega\mathbf{d}_{fi}$.

The normalised wave function of the initial (ground) state of the deuteron:

$$\psi = \sqrt{\frac{\kappa}{2\pi}} \frac{e^{-\kappa r}}{r}, \quad \kappa = \sqrt{MI}, \quad (58.2)$$

where $I = 2.23$ MeV is the binding energy (see III, § 133)⁵¹. As the wave function of the final state we can take the function of free motion, i.e. a plane wave

$$\psi' = e^{i\mathbf{p}\mathbf{r}}. \quad (58.3)$$

The reason lies in the fact that in the theory under consideration the “size of the deuteron” $1/\kappa$ is considered large compared with the effective radius of interaction a . Therefore the interaction between the proton and the neutron should be taken into account

⁴⁹The photon energy for which $pa \approx 1$ ($a = 1.5 \cdot 10^{-13}$ cm) is about 15 MeV.

⁵⁰In this paragraph p denotes $|\mathbf{p}|$.

⁵¹This function can be refined by introducing the correction connected with the finiteness of a . This is achieved by replacing the normalisation coefficient in (58.2) by the coefficient

$$\sqrt{\frac{\kappa}{2\pi(1 - a\kappa)}}$$

(see III, (133.13)). Correspondingly there will appear a factor $1/(1 - a\kappa)$ in the formulae for cross sections. In practice this correction is not so small: for the ground state of the deuteron $a\kappa \approx 0.4$.

The ground state of the deuteron is the state 3S_1 with a small “admixture” of the state 3D_1 , associated with the action of tensor nuclear forces (see III, § 117). This admixture, and thereby the tensor forces, we shall neglect.

only in S -states, neglecting the wave functions in states with $l \neq 0$ at small distances. Meanwhile, according to the selection rules, the electric-dipole transitions between two S -states (the ground state and an S -state of the continuous spectrum) are forbidden. This gives us the possibility of neglecting the interaction of the nucleons in the final state as well.

By integration by parts we find for the matrix element

$$\mathbf{p}_{fi} = -i\sqrt{\frac{\kappa}{2\pi}} \int e^{-i\mathbf{p}\mathbf{r}} \nabla \frac{e^{-\kappa r}}{r} d^3x = \sqrt{\frac{\kappa}{2\pi}} \mathbf{p} \left(\frac{e^{-\kappa r}}{r} \right)_{\mathbf{p}} = \sqrt{\frac{\kappa}{2\pi}} \frac{4\pi\mathbf{p}}{p^2 + \kappa^2}$$

(see the footnote on p. 246).

Noting also the relation

$$\frac{1}{M} (\kappa^2 + \mathbf{p}^2) = I + \frac{\mathbf{p}^2}{M} = \omega,$$

expressing conservation of energy, we finally obtain the cross section of photodisintegration (in ordinary units)

$$\sigma^{(\varepsilon)} = \frac{8\pi}{3} \alpha \frac{\hbar^2}{M} \frac{\sqrt{I(\hbar\omega - I)}(\sqrt{I})^2}{(\hbar\omega)^3} \quad (58.4)$$

(*H. A. Bethe, R. Peierls, 1935*). It has a maximum at $\hbar\omega = 2I$ and vanishes at $\hbar\omega \rightarrow I$ and $\hbar\omega \rightarrow \infty$.

The electric-dipole absorption described by formula (58.4) does not, however, give the main contribution to the cross section near the threshold of the photoeffect ($\hbar\omega$ close to I). The point is that in this region the main effect must arise from transitions into an S -state, which in electric-dipole absorption are forbidden. There is no such restriction in electric-quadrupole absorption either. But in view of the very small magnitude of the intervals of the hyperfine structure the $E2$ emission is highly improbable in comparison with $M1$ (cf. (50.1)), so that near the threshold of photodisintegration it is necessary to consider the magnetic-dipole absorption, whose selection rules permit transitions between S -states (*E. Fermi, 1935*).

Replacing in formula (58.1) the electric moment by the magnetic one, we have

$$\sigma^{(\mathcal{M})} = \frac{1}{3} \omega M p |\boldsymbol{\mu}_{fi}|^2. \quad (58.5)$$

The magnetic moment of the orbital motion does not contribute to $\boldsymbol{\mu}_{fi}$, since the orbital angular momentum $\mathbf{L}$ has no matrix elements for transitions between S -states. The spin magnetic moment

$$\boldsymbol{\mu} = 2\mu_p \mathbf{s}_p + 2\mu_n \mathbf{s}_n = 2(\mu_p - \mu_n) \mathbf{s}_p + 2\mu_n \mathbf{S},$$

where $\mathbf{S} = \mathbf{s}_p + \mathbf{s}_n$, and μ_p, μ_n are the magnetic moments of the proton and neutron. In the neglect of tensor nuclear forces the total spin is conserved, so that its operator does not give rise to transitions. Therefore

$$\boldsymbol{\mu}_{fi} = 2(\mathbf{s}_p)_{fi}(\mu_p - \mu_n).$$

In the same approximation (without tensor forces) the spin and coordinate variables separate. Together with the wave functions representing a product of the spin and coordinate parts, the matrix element also

$$\mu_{fi} = 2(\mu_p - \mu_n) \langle s_p S' M' | s_p | s_p S M \rangle \int \psi'^* \psi d^3x.$$

But the presence of spin-spin nuclear forces leads to the fact that the wave equation for the coordinate functions $\psi(r)$ contains

as a parameter the value of the spin S . If $S' = S$, then $\psi'(r)$ and $\psi(r)$ are eigenfunctions of one and the same operator and are therefore orthogonal. Thus, from the initial state 3S the photodisintegration will occur only to a state of the continuous spectrum 1S .

The square $|\mu_{fi}|^2$ in (58.5) must, of course, be averaged over the projections M of the spin S in the initial state. Thus, the problem reduces to the computation of the quantity

$$\frac{1}{2S+1} \sum_M |\langle s_p S' M' | s_p | s_p S M \rangle|^2,$$

where $s_p = s_n = \frac{1}{2}$, $S = 1$, $S' = 0$. By the general formulae for matrix elements in the addition of angular momenta this quantity is

$$\frac{1}{(2S+1)(2S'+1)} \begin{Bmatrix} s_p & S' & s_n \\ s_p & S & 1 \end{Bmatrix}^2 |\langle s_p || s_p || s_p \rangle|^2 = \frac{1}{6} |\langle s_p || s_p || s_p \rangle|^2$$

(formulae III, (107.11), (109.3) have been used). The reduced matrix element

$$\langle s_p || s_p || s_p \rangle = \sqrt{s_p(s_p+1)(2s_p+1)} = \sqrt{3/2}.$$

Formula (58.5) takes as a result the form

$$\sigma^{(\mathcal{M})} = \frac{1}{3} \omega M p (\mu_p - \mu_n)^2 \left| \int \psi'^* \psi d^3x \right|^2. \quad (58.6)$$

The initial function ψ is given by (58.2). The final function is

$$\psi' = \frac{1}{2p} R_{p0}(r).$$

This is the first ($l = 0$) term of the expansion (56.7) of the function containing asymptotically a plane wave and a converging spherical wave; an inessential phase factor is omitted. Since the integration is carried out over the region outside the radius of action of the nuclear forces, the radial function

$$R_{p0}(r) = 2 \frac{\sin(pr + \delta)}{r}.$$

The phase δ is related to the energy of the virtual level ($I_1 = 0.067$ MeV) of the system "proton + neutron" at $S = 0$:

$$\operatorname{ctg} \delta = \frac{\kappa_1}{p}, \quad \kappa_1 = \sqrt{M I_1}$$

(see III, § 133). Now

$$\int \psi'^* \psi d^3x = (2\pi)^{3/2} \frac{\sqrt{\kappa}}{p\pi} \operatorname{Im} \int e^{-\kappa r + i p r} e^{i\delta} dr = (2\pi)^{3/2} \frac{\sqrt{\kappa}}{p\pi} \operatorname{Im} \frac{e^{i\delta}}{\kappa - ip}.$$

After simple algebraic transformations we obtain the following expression for the cross section of photodisintegration (in ordinary units):

$$\sigma^{(\mathcal{M})} = \frac{8\pi}{3\hbar c} (\mu_p - \mu_n)^2 \frac{\sqrt{I(\hbar\omega - I)}(\sqrt{I} + \sqrt{I_1})^2}{\hbar\omega(\hbar\omega - I + I_1)}. \quad (58.7)$$

At $\hbar\omega \rightarrow I$ this cross section vanishes as $\sqrt{\hbar\omega - I}$ —in accordance with the general behaviour of cross sections near the threshold of a reaction (see III, § 147).

The process inverse to photodisintegration is the radiational capture of a proton by a neutron. The cross section of capture (σ_3) is found from the cross section of the photoeffect (σ_Φ) with the help of the principle of detailed balance (cf. the derivation of (56.15)). The spin statistical weight of the neutron and proton is $2 \cdot 2 = 4$. The statistical weight of the deuteron (in the state with $S = 1$) and the photon is $3 \cdot 2 = 6$. Therefore

$$\sigma_3 = \frac{3}{2} \frac{(\hbar\omega)^2}{c^2 p^2} \sigma_\Phi = \frac{3(\hbar\omega)^2}{2Mc^2(\hbar\omega - I)} \sigma_\Phi. \quad (58.8)$$

Scattering of Light

§ 59. Scattering tensor

Scattering of a photon by an electronic system (for definiteness we shall speak of an atom) consists of the absorption of the initial photon $\mathbf{k}$ with simultaneous emission of another photon $\mathbf{k}'$. In this process the atom may remain either in the initial state, or in some other discrete energy level. In the first case the frequency of the photon does not change (*Rayleigh*, or *unshifted scattering*), and in the second it changes by the amount

$$\omega' - \omega = E_1 - E_2, \quad (59.1)$$

where E_1, E_2 are the initial and final energies of the atom (*combination*, or *Raman scattering*)¹. If the initial state of the atom is the ground state, then for combination scattering $E_2 > E_1$, so that $\omega' < \omega$ —the scattering occurs with a decrease in frequency (the so-called *Stokes* case). For scattering on an excited atom both Stokes and *anti-Stokes* ($\omega' > \omega$) cases are possible.

Since the operator of the electromagnetic perturbation has no matrix elements for transitions with a simultaneous change of two photon occupation numbers, the scattering effect appears only in the second approximation of perturbation theory. It should be regarded as occurring via definite intermediate states, which may be of two types:

I. Photon $\mathbf{k}$ is absorbed, the atom goes into one of its possible states E_n ; in the subsequent transition to the final state photon $\mathbf{k}'$ is emitted.

II. Photon $\mathbf{k}'$ is emitted, the atom goes into a state E_n ; in the transition to the final state photon $\mathbf{k}$ is absorbed.

The role of the matrix element for the process under consideration is played by a sum (see III, (43.7))

$$V_{21} = \sum_n' \left(\frac{V'_{2n} V_{n1}}{\mathcal{E}_1 - \mathcal{E}_n^I} + \frac{V_{2n} V'_{n1}}{\mathcal{E}_1 - \mathcal{E}_n^{II}} \right), \quad (59.2)$$

where the energy of the system “atom + photons” is $\mathcal{E}_1 = E_1 + \omega$, and the energies of the intermediate states

$$\mathcal{E}_n^I = E_n, \quad \mathcal{E}_n^{II} = E_n + \omega + \omega',$$

V are the matrix elements of absorption of photon $\mathbf{k}$, V' those of emission of photon $\mathbf{k}'$; the initial state is excluded from the summation over n (which is indicated by the prime on the summation sign). The cross section of scattering

$$d\sigma = 2\pi |V_{21}|^2 \frac{\omega'^2 d\omega'}{(2\pi)^3}, \quad (59.3)$$

¹In this chapter quantities pertaining to the initial and final states of the scattering system are labelled with the indices 1 and 2.

where do' is the element of solid angle for the directions $\mathbf{k}'$. The energy of light dI' , scattered (in 1 s) into the solid angle do' , is expressed through the intensity I (flux density of energy) of the incident light by the formula

$$dI' = I \frac{\omega'}{\omega} d\sigma.$$

We shall assume that the wavelengths of the initial and the final photon are large compared with the dimensions a of the scattering system. Correspondingly we consider all transitions in the dipole approximation. If the states of the photons are described by plane waves, then this approximation corresponds to the replacement of the factor $e^{i\mathbf{k}\mathbf{r}}$ by unity. Then the wave functions of the photons (in the three-dimensionally transverse gauge)

$$\mathbf{A}_{\mathbf{e}\omega} = \sqrt{4\pi} \frac{\mathbf{e}}{\sqrt{2\omega}} e^{-i\omega t}, \quad \mathbf{A}_{\mathbf{e}'\omega'} = \sqrt{4\pi} \frac{\mathbf{e}'}{\sqrt{2\omega'}} e^{-i\omega' t}.$$

In the conditions under consideration the operator of electromagnetic interaction can be written in the form

$$\hat{V} = -\hat{\mathbf{d}}\hat{\mathbf{E}}, \quad (59.4)$$

where $\hat{\mathbf{E}} = -\hat{\mathbf{A}}$ is the operator of the field strength, $\hat{\mathbf{d}}$ is the operator of the dipole moment of the atom (analogous to the classical expression for the energy of a system of small dimensions in an electric field—see II, § 42). Its matrix elements:

$$V_{n1} = -i\sqrt{2\pi\omega} (\mathbf{e}\mathbf{d}_{n1}), \quad V'_{2n} = i\sqrt{2\pi\omega'} (\mathbf{e}'^*\mathbf{d}_{2n}).$$

Substituting these expressions into (59.2), (59.3), we obtain the cross section of scattering (writing it in ordinary units)²:

$$d\sigma = \left| \sum_n \left\{ \frac{(\mathbf{d}_{2n}\mathbf{e}'^*)(\mathbf{d}_{n1}\mathbf{e})}{\omega_{n1} - \omega - i0} + \frac{(\mathbf{d}_{2n}\mathbf{e})(\mathbf{d}_{n1}\mathbf{e}'^*)}{\omega_{n1} + \omega' - i0} \right\} \right|^2 \frac{\omega\omega'^3}{\hbar^2 c^4} do', \quad (59.5)$$

$$\hbar\omega_{n1} = E_n - E_1, \quad \omega' - \omega = \omega_{12}.$$

The summation is carried out over all possible states of the atom, including states of the continuous spectrum (states 1 and 2 drop out of the summation automatically, since the diagonal matrix elements $\mathbf{d}_{11} = \mathbf{d}_{22} = 0$). The infinitely small imaginary additions in the denominators correspond to the usual rule of going around the poles in the perturbation theory (see III, § 43): to the energies of the intermediate states E_n over which the summation takes place a negative infinitely small imaginary part is added. The rule of going around is essential when the poles of the expression (59.5) in the variable E_n fall in the region of the continuous spectrum (i.e. if state 1 is the ground state, then for this $\hbar\omega$ must exceed the ionisation threshold of the atom)^{3 4}.

²This formula was first obtained by *Kramers and Heisenberg* (*H. A. Kramers, W. Heisenberg*, 1925) even before the creation of quantum mechanics.

³For molecules the role of the ionisation threshold in this context is played by the threshold of dissociation into atoms.

⁴The majority of the results presented in §§ 59–61 are due to *Placzek* (*G. Placzek*, 1931, 1933).

Let us introduce a notation (ordinary units)⁴:

$$(c_{ik})_{21} = \frac{1}{\hbar} \sum_n \left[\frac{(d_i)_{2n}(d_k)_{n1}}{\omega_{n1} - \omega - i0} + \frac{(d_k)_{2n}(d_i)_{n1}}{\omega_{n1} + \omega' - i0} \right] \quad (59.6)$$

($i, k = x, y, z$ are three-dimensional vector indices). With its help formula (59.5) can be rewritten in the form

$$d\sigma = \frac{\omega(\omega + \omega_{12})^3}{c^4} |(c_{ik})_{21} e_i'^* e_k|^2 d\omega'. \quad (59.7)$$

The notation (59.6) is justified by the fact that this sum can actually be represented as the matrix element of a certain tensor. This is simplest to verify by introducing a vector quantity $\mathbf{b}$, the operator of which satisfies the equation

$$\dot{\mathbf{b}} + \omega \hat{\mathbf{b}} = \hat{\mathbf{d}}.$$

Its matrix elements

$$\mathbf{b}_{n1} = \frac{\mathbf{d}_{n1}}{\omega - \omega_{n1}}, \quad \mathbf{b}_{2n} = \frac{\mathbf{d}_{2n}}{\omega + \omega_{n2}}$$

so that

$$(c_{ik})_{21} = (b_k d_i - d_i b_k)_{21}. \quad (59.8)$$

The matrix elements $(c_{ik})_{21}$ we shall call the *scattering tensor*.

From the above it follows that the selection rules for scattering coincide with the selection rules for the matrix elements of an arbitrary second-rank tensor. Let us note at once that if the system has a centre of symmetry (so that its states can

be classified by parity), then transitions are possible only between states of the same parity (including without change of state). This rule is opposite to the parity selection rule in emission (electric-dipole), so that there is an alternative prohibition: transitions allowed in emission are forbidden in scattering, and those allowed in scattering are forbidden in emission.

Let us decompose the tensor c_{ik} into irreducible parts:

$$c_{ik} = c^0 \delta_{ik} + c_{ik}^s + c_{ik}^a, \quad (59.9)$$

where

$$c^0 = \frac{1}{3} c_{ii}, \quad c_{ik}^s = \frac{1}{2} (c_{ik} + c_{ki}) - c^0 \delta_{ik}, \quad c_{ik}^a = \frac{1}{2} (c_{ik} - c_{ki}) \quad (59.10)$$

are respectively a scalar, a symmetric tensor (with zero trace), and an antisymmetric tensor. Their matrix elements:

$$(c^0)_{21} = \frac{1}{3} \sum_n \frac{\omega_{n1} + \omega_{n2}}{(\omega_{n1} - \omega)(\omega_{n2} + \omega)} (d_i)_{2n} (d_i)_{n1}, \quad (59.11)$$

$$(c_{ik}^s)_{21} = \frac{1}{2} \sum_n \frac{\omega_{n1} + \omega_{n2}}{(\omega_{n1} - \omega)(\omega_{n2} + \omega)} [(d_i)_{2n} (d_k)_{n1} + (d_k)_{2n} (d_i)_{n1}] - (c^0)_{21} \delta_{ik}, \quad (59.12)$$

$$(c_{ik}^a)_{21} = \frac{2\omega + \omega_{12}}{2} \sum_n \frac{(d_i)_{2n}(d_k)_{n1} - (d_k)_{2n}(d_i)_{n1}}{(\omega_{n1} - \omega)(\omega_{n2} + \omega)} \quad (59.13)$$

(the signs of going around the poles are omitted for brevity).

Let us consider some properties of the scattering tensor in the limiting cases of small and large photon frequencies⁵.

For unshifted scattering ($\omega_{12} = 0$) the antisymmetric part of the tensor vanishes at $\omega \rightarrow 0$ (because of the factor ω in front of the sum in (59.13)). The scalar and symmetric parts of the scattering tensor tend at $\omega \rightarrow 0$ to finite limits. Correspondingly the cross section at small ω is proportional to ω^4 .

In the opposite case, when the frequency ω is large compared with all the frequencies ω_{n1}, ω_{n2} appearing in (59.6) (but of course the wavelength is still $\gg a$), we should return to the formulae of the classical theory. The first term of the expansion of the scattering tensor in powers of $1/\omega$ is

$$\frac{1}{\omega} \sum_n [(d_k)_{2n}(d_i)_{n1} - (d_i)_{2n}(d_k)_{n1}] = \frac{1}{\omega} (\hat{d}_k \hat{d}_i - \hat{d}_i \hat{d}_k)_{21}$$

and vanishes by virtue of the commutativity of $\hat{d}_i, \hat{d}_k$. The next term of the expansion

$$\begin{aligned} (c_{ik})_{21} &= \frac{1}{\omega^2} \sum_n [\omega_{2n}(d_k)_{2n}(d_i)_{n1} - (d_i)_{2n}\omega_{n1}(d_k)_{n1}] = \\ &= \frac{1}{i\omega^2} (\dot{\hat{d}}_k \hat{d}_i - \hat{d}_i \dot{\hat{d}}_k)_{21}. \end{aligned}$$

Using the definition $\mathbf{d} = \sum \mathbf{er}$ (the sum is over all electrons in the atom) and the commutation rules between momenta and coordinates, we obtain

$$(c_{ik})_{11} = -\frac{Ze^2}{m\omega^2} \delta_{ik}, \quad (c_{ik})_{21} = 0, \quad (59.14)$$

where Z is the total number of electrons in the system, m is the electron mass. Thus, in the limit of large frequencies in the scattering tensor only the scalar part remains, and the scattering occurs without change of the state of the system (i.e. the scattering is entirely coherent—see below). The cross section of scattering in this case

$$d\sigma = r_e^2 Z^2 |\mathbf{e}'^* \mathbf{e}|^2 d\omega', \quad (59.15)$$

where $r_e = e^2/m$. After summing over polarisations of the final photon we obtain the formula

$$d\sigma = r_e^2 Z^2 \{1 - (\mathbf{en}')^2\} d\omega' = r_e^2 Z^2 \sin^2 \theta \cdot d\omega', \quad (59.16)$$

which indeed coincides with the classical Thomson formula (see II, (80.7)); θ is the angle between the direction of scattering and the polarisation vector of the incident photon).

Let us consider the scattering of light by a collection of N identical atoms, located in a volume whose dimensions are small compared with the wavelength. The scattering tensor of such a collection will be equal to the sum of the scattering tensors of each

⁵The case of resonance (when ω is close to one of the frequencies ω_{n1} or ω_{n2}) will be considered in § 63.

of the atoms. At the same time, however, one must note that the wave functions (with the help of which the matrix elements of the dipole moment are computed) for several identical atoms considered simultaneously cannot simply be taken as identical. The wave functions are by their very nature determined only to within an arbitrary phase factor, and these factors are individual for each atom. The cross section of scattering must be averaged over the phase factor of each atom independently.

The scattering tensor $(c_{ik})_{21}$ of each atom contains the factor $e^{i(\varphi_1 - \varphi_2)}$, where φ_1, φ_2 are the phases of the wave functions of the initial and final states. For shifted scattering states 1 and 2 are different, and this factor differs from unity. In the square of the modulus

$$|e_i'^* e_k \sum (c_{ik})_{21}|^2$$

(the sum is over all N atoms) the products of terms of the sums, pertaining to different atoms, will contain phase factors which go to zero upon independent averaging over the phases of the atoms; only the squares of the moduli of each of the terms will remain. This means that the total cross section of scattering of N atoms is obtained by multiplying by N the cross section of scattering on one atom (incoherent scattering).

If the initial and final states of the atom coincide, then the factors $e^{i(\varphi_1 - \varphi_2)} = 1$. The multiplier N can differ from unity: in this case the amplitude of scattering of the collection of atoms is proportional to N , and the cross section of scattering is proportional to N^2 (coherent scattering)⁶. If the energy level of the atom is non-degenerate, then the unshifted scattering will be entirely coherent. If the energy level is degenerate, then alongside coherent unshifted scattering there will also be incoherent unshifted scattering arising from transitions between different mutually degenerate states. Let us note that the latter represents a purely quantum effect: in the classical theory scattering without change of frequency is always coherent.

The tensor of coherent scattering is given by the diagonal matrix element $(c_{ik})_{11}$; let us denote it through α_{ik} (dropping for simplification of notation the index that should indicate the state of the atom). According to (59.6)

$$\alpha_{ik}(\omega) \equiv (c_{ik})_{11} = \sum_n \left[\frac{(d_i)_{1n}(d_k)_{n1}}{\omega_{n1} - \omega - i0} + \frac{(d_k)_{1n}(d_i)_{n1}}{\omega_{n1} + \omega + i0} \right]. \quad (59.17)$$

This expression can also be represented in the form

$$\alpha_{ik}(\omega) = \frac{e^2}{m\omega^2} \left\{ -Z\delta_{ik} + \frac{1}{m} \sum_n \left[\frac{(p_i)_{1n}(p_k)_{n1}}{\omega_{n1} - \omega - i0} + \frac{(p_k)_{1n}(p_i)_{n1}}{\omega_{n1} + \omega + i0} \right] \right\}, \quad (59.18)$$

where the highlighted limiting expression (59.14) is exhibited. Here $\mathbf{p}$ is the total momentum of the electrons in the atom; the equivalence of these formulae is easy to verify, noting that the matrix elements of the momentum and the dipole moment are related to each other by the relation

$$e\mathbf{p}_{1n}/m = i\omega_{1n}\mathbf{d}_{1n},$$

⁶Let us note that the factor Z^2 in formulae (59.15), (59.16) has the same nature: the cross section of coherent scattering on Z electrons of one atom is Z^2 times greater than the cross section of scattering on one electron.

and taking account of the relation used in the derivation of (59.14).

If neither the sum nor the difference $E_1 \pm \omega$ coincides with one of the energy levels E_n of the atom (including the continuous

spectrum), then one can omit the terms $i0$ in the denominators. Noting that $p_{1n}^* = p_{n1}$, we find then that the tensor α_{ik} is Hermitian ⁷:

$$\alpha_{ik} = \alpha_{ki}^*. \quad (59.19)$$

This means that its scalar and symmetric parts are real, and the antisymmetric part is imaginary. Let us note that the antisymmetric part vanishes automatically if the atom is in a non-degenerate state; the wave function of such a state is real, and so the diagonal matrix elements are also real.

The tensor α_{ik} is related to the polarisability of the atom in an external electric field. To establish this connection, we compute the correction to the mean value of the dipole moment of the system, if the system is placed in an external electric field

$$\frac{1}{2}(\mathbf{E}e^{-i\omega t} + \mathbf{E}^*e^{i\omega t}). \quad (59.20)$$

This can be done using the familiar formulae of the perturbation theory (see III, § 40): if the system is acted upon by the perturbation

$$\hat{V} = \hat{F}e^{-i\omega t} + \hat{F}^+e^{i\omega t},$$

then the correction of first order to the diagonal matrix elements of a certain quantity f is

$$\begin{aligned} f_{11}^{(1)}(t) = & - \sum_n \left\{ \left[\frac{f_{1n}^{(0)} F_{n1}}{\omega_{n1} - \omega - i0} + \frac{f_n^{(0)} F_{1n}^*}{\omega_{n1} + \omega + i0} \right] e^{-i\omega t} + \right. \\ & \left. + \left[\frac{f_{1n}^{(0)*} F_{n1}^*}{\omega_{n1} + \omega - i0} + \frac{f_n^{(0)} F_{1n}}{\omega_{n1} - \omega + i0} \right] e^{i\omega t} \right\} \end{aligned}$$

(the perturbation $\hat{V}$ should be considered as one slowly switched on from $t = -\infty$, so that in the first term ω should be understood as $\omega + i0$, and in the second as $\omega - i0$; in accordance with this the imaginary additions in the denominators are also written).

In the case considered $\hat{F} = -\mathbf{d}\mathbf{E}/2$, and the correction to the diagonal matrix element of the dipole moment turns out to be

$$\mathbf{d}_{11}^{(1)} = \frac{1}{2}(\bar{\mathbf{d}}e^{-i\omega t} + \bar{\mathbf{d}}^*e^{i\omega t}), \quad (59.21)$$

where $\bar{\mathbf{d}}$ is a vector with components

$$\bar{d}_i = \alpha_{ik}^{(n)} E_k, \quad (59.22)$$

and the expression for the tensor $\alpha_{ik}^{(n)}(\omega)$ differs from the expression (59.17) for α_{ik} by the opposite sign of the imaginary addition in the denominator of the second

⁷This result is associated with the neglect of the natural linewidth, and thereby with the possibility of absorption of the incident light (see § 62).

term. By definition, $\alpha_{ik}^{(n)}(\omega)$ is the tensor of polarisability of the atom in a field with frequency ω . For frequencies at which the imaginary additions in the denominators may be dropped and the tensor α_{ik} is Hermitian, the tensors α_{ik} and $\alpha_{ik}^{(n)}$ simply coincide with each other. In particular, at $\omega = 0$ formula (59.22) goes over to the formula for the tensor of static polarisability (76.4) (see III), and the expression for the tensor of static polarisability (76.5) (see III) coincides with $\alpha_{ik}(0)$ from (59.17). Let us note also that if state 1 is the ground state⁸, then all $\omega_{n1} > 0$, and the rule of going around the pole in the first term of (59.17) is essential only at $\omega > 0$, and in the second at $\omega < 0$. In this case

$$\alpha_{ik}(\omega) = \alpha_{ik}^{(n)}(|\omega|). \quad (59.23)$$

In the sense of the theory of scattering the formulae with the scattering tensor imply that $\omega > 0$; then the tensor α_{ik} coincides with the tensor of polarisability.

In what follows we shall need, alongside the cross section, also the amplitude of scattering of the photon f . As usual in the perturbation theory, it coincides, to within a normalisation factor, with the matrix element (59.2) taken with the opposite sign. Having chosen this factor so as to be able to represent the cross section (59.7) in the form $d\sigma = |f|^2 d\omega'$, we find for the amplitude of elastic scattering

$$f = \omega^2 \alpha_{ik} e_i'^* e_k. \quad (59.24)$$

According to the optical theorem (see below formula (71.10)) the imaginary part of the amplitude of forward scattering (i.e. without change of momentum and polarisation of the photon) determines the total cross section σ_t of all possible elastic and inelastic processes for a given initial state of the photon:

$$\sigma_t = \frac{4\pi}{\omega} \text{Im}(\omega^2 \alpha_{ik} e_i^* e_k) = 4\pi\omega \frac{\alpha_{ik} - \alpha_{ki}^*}{2i} e_i^* e_k. \quad (59.25)$$

Thus, the total cross section is determined by the anti-Hermitian part of the scattering tensor.

Formula (59.25) has a simple classical meaning. The electric field $\mathbf{E}$ does work per unit time on the system of charges, equal to $\sum e \mathbf{v} \mathbf{E} = \dot{\mathbf{E}} \mathbf{d}$. Representing the field in the form (59.20), and the dipole moment in the form (59.21), (59.22) and averaging this work over time, we obtain

$$\frac{1}{2} \omega |E|^2 e_i^* e_k \frac{\alpha_{ik} - \alpha_{ki}^*}{2i}.$$

($\mathbf{E} = eE$). On the other hand, if $\mathbf{E}$ is the field of the incident light, then the mean flux density of energy in it is $|E|^2/(8\pi)$, and the energy absorbed by the atom is

$$\frac{1}{8\pi} |E|^2 \sigma_t.$$

Equating the two expressions, we obtain formula (59.25).

⁸Only this case (which we shall henceforth have in mind in the subsequent arguments) permits an entirely rigorous examination on account of the finite lifetime of excited states (see § 62).

If the angular momentum J of the ground state of the atom equals zero, then by virtue of spherical symmetry $\alpha_{ik} = \alpha \delta_{ik}$. Then

$$\sigma_t = 4\pi\omega \operatorname{Im} \alpha. \quad (59.26)$$

For systems with angular momentum such a relation is also valid for the quantities averaged over directions in space (see § 60). For photon energies above the ionisation threshold of the atom the main contribution to the total cross section σ_t is made by the process of ionisation—absorption at the photoeffect. The cross section of scattering is a quantity of higher order in e^2 (compare, for example, (56.13) with (59.16)).

If the photon energy lies below the ionisation threshold (i.e. is not close to any of the discrete excitation frequencies of the atom), then the cross section reduces to the cross section of scattering, and along with it also the imaginary part of the amplitude turns out to be of higher order of smallness than its real part. Neglecting the imaginary part, we recover (59.19).

Alongside scattering, the two-photon processes appearing in the second order of perturbation theory include also *double emission*—simultaneous emission by the atom of two quanta. The expression for the probability of this process differs from formula (59.5) only by the replacements $\omega \rightarrow -\omega$, $\mathbf{e} \rightarrow \mathbf{e}^*$ (emission of photon ω instead of absorption) and an additional factor

$$\frac{d^3 k}{(2\pi)^3} = \frac{\omega^2 d\omega d\Omega}{(2\pi)^3}$$

—the number of quantum states of the emitted photon in given intervals of frequency ω and directions $\mathbf{k}$; the frequency of the second photon is determined by ω from the equality $\omega + \omega' = \omega_{12}$. Thus, the probability of emission (per unit time)⁹

$$dw = |(b_{ik})_{21} e_i'^* e_k^*|^2 \frac{\omega^3 \omega'^3}{(2\pi)^3 c^6 \hbar^2} d\Omega d\Omega' d\omega, \quad (59.27)$$

where

$$(b_{ik})_{21} = \sum_n \left[\frac{(d_i)_{2n}(d_k)_{n1}}{\omega_{n1} + \omega - i0} + \frac{(d_k)_{2n}(d_i)_{n1}}{\omega_{n1} + \omega' - i0} \right]$$

differs from $(c_{ik})_{21}$ (59.6) only by the sign in front of ω . Having summed this expression over polarisations of the photons and integrated over the directions of their emission¹⁰, we obtain

$$dw = \frac{8\omega^3 \omega'^3}{9\pi \hbar^2 c^6} |(b_{ik})_{21}|^2 d\omega. \quad (59.28)$$

The probability of emission of two photons ω and ω' is usually very small compared with the probability of emission of one photon with frequency $\omega + \omega'$. An exception are cases when selection rules, forbidding the second process, allow the first. Such are, for example, transitions between two states with $J = 0$, for which all processes of one-photon emission are forbidden strictly. Another example is the transition from the first excited state of the hydrogen atom ($2s_{1/2}$) to the ground state ($1s_{1/2}$), forbidden both for $E1$ and for $M1$ emission (see § 2, Problem 52)¹¹.

⁹Here and below in this paragraph ordinary units.

¹⁰This operation reduces to the complete averaging over directions $\mathbf{e}$ and consists, in accordance with $e_i^* e_k^* = \delta_{ik}/3$, of subsequent multiplication by $2 \cdot 2 \cdot 4\pi \cdot 4\pi$.

¹¹The lifetime of the $2s_{1/2}$ level, due to spontaneous double emission, is 0.15 s.

If an atom is in the field of a flux of photons ω , $\mathbf{k}$, then alongside spontaneous double emission, the probability of which is (59.27), there also exists stimulated double emission: under the influence of the field one more photon is emitted, identical to a photon of the field, together with a photon ω' , $\mathbf{k}'$. The probability of this process differs from the probability of spontaneous double emission by the factor $N_{\mathbf{k}\mathbf{e}}$ —the density of photons of the incident light with given $\mathbf{k}$, $\mathbf{e}$. The density of incident photon flux is

$$dI = cN_{\mathbf{k}\mathbf{e}} \frac{d^3k}{(2\pi)^3} = N_{\mathbf{k}\mathbf{e}} \frac{\omega^2}{(2\pi)^3} d\omega d\mathbf{o}.$$

Having expressed from here $N_{\mathbf{k}\mathbf{e}}$ through dI and dividing by dI the probability of the process, we obtain its cross section

$$d\sigma = \frac{\omega\omega'^3}{\hbar^2 c^4} |(b_{ik})_{12} e_i'^* e_k^*|^2 d\mathbf{o}'. \quad (59.29)$$

Analogously, if the atom is in a field of photons ω' , $\mathbf{k}'$, then when a photon ω , $\mathbf{k}$ falls on it *stimulated Raman scattering* takes place, the cross section of which is proportional to the number density of photons ω' , $\mathbf{k}'$.

The computation of tensors $(c_{ik})_{12}$ or $(b_{ik})_{12}$ for specific atoms requires computing sums of the form

$$(M_{ik}^{(2)})_{21} = \sum_n \frac{(d_i)_{2n} (d_k)_{n1}}{E_n - E - i0}, \quad (59.30)$$

where E takes the values $E_1 \pm \hbar\omega$ or $E_1 \pm \hbar\omega'$. Let us, to simplify the notation, speak of the hydrogen atom. Let us write the sum (59.30) in the form of an integral

$$(M_{ik}^{(2)})_{21} = \int \psi_2^*(\mathbf{r}) \hat{d}_i G(\mathbf{r}, \mathbf{r}'; E) \hat{d}_k \psi_1(\mathbf{r}') d^3x d^3x', \quad (59.31)$$

where

$$G(\mathbf{r}, \mathbf{r}'; E) = \sum_n \frac{\psi_n(\mathbf{r}) \psi_n^*(\mathbf{r}')}{E_n - E - i0}. \quad (59.32)$$

Acting on the function G with the operator $\hat{H} - E$, where $\hat{H}$ is the Hamiltonian of the atom. Since $\hat{H}\psi_n = E_n\psi_n$, we obtain

$$(\hat{H} - E)G = \sum_n \psi_n(\mathbf{r}) \psi_n^*(\mathbf{r}').$$

But the sum standing here is, by virtue of the completeness of the system of functions ψ_n , the δ -function $\delta(\mathbf{r} - \mathbf{r}')$. Thus the function G satisfies the equation

$$(\hat{H} - E)G(\mathbf{r}, \mathbf{r}'; E) = \delta(\mathbf{r} - \mathbf{r}'), \quad (59.33)$$

i.e. is the Green's function of the Schrödinger equation (the rule of going around in (59.32) determines which of the solutions of this equation should be chosen). Thereby the problem of computing the sum (59.30) reduces to finding the Green's function of the atom. The exact solution of equation (59.33) is possible, however, only if the exact

solutions of the homogeneous Schrödinger equation are known, i.e. practically only for the hydrogen atom¹².

Problem

Compute the probability of elastic scattering of an electron (non-relativistic) on a nearly monochromatic standing light wave (*P. L. Kapitza, P. A. M. Dirac, 1933*).

Solution. A standing wave can be regarded as a collection of photons with momenta $\mathbf{k}$ and $-\mathbf{k}$ (and identical polarisations). Scattering of the electron can be treated as absorption of a photon with momentum $\mathbf{k}$ and stimulated emission of a photon with momentum $-\mathbf{k}$, as a result of which the momentum $\mathbf{p}$ of the electron receives an increment $2\hbar\mathbf{k}$, rotating (without change of magnitude) through the

angle ϑ : $|\mathbf{p}| \sin(\vartheta/2) = \hbar\omega/c$. The probability of this process can be obtained from the Thomson scattering cross section (59.15):

$$d\sigma = r_e^2 |\mathbf{e}'^* \mathbf{e}|^2 d\omega' = r_e^2 d\omega'$$

by multiplying by the photon flux density and the number of photons with momentum $-\mathbf{k}$.

The photon flux density with frequencies in the spectral interval $d\omega$ is

$$\frac{cU_\omega d\omega}{2\hbar\omega},$$

where U_ω is the spectral energy density in the standing wave in the spectral interval $d\omega$ (the factor $1/2$ takes into account that the energy of the wave is divided equally between photons of opposite directions). The momenta $\mathbf{k}$ of all photons forming a standing wave are parallel to a definite direction $\mathbf{n}$ (the “direction” of the standing wave). In other words, the energy density as a function of photon frequency and direction $\mathbf{n}'$: $U_{\omega\mathbf{n}'} = U_\omega \delta^{(2)}(\hat{\mathbf{n}}' - \mathbf{n})$. Correspondingly the number of photons with momentum $-\mathbf{k}$ per unit volume is (cf. (44.8)):

$$\int N_{-\mathbf{k}} d\omega' = \frac{8\pi^3 c^3}{\hbar\omega^3} \frac{U_\omega}{2}.$$

As a result we obtain the probability of scattering of the electron (in 1 s)

$$w = \frac{2\pi^3 e^4}{\hbar^2 m^2 \omega^4} \int U_\omega^2 d\omega.$$

The factor ω^{-4} is placed outside the integral sign, since the degree of monochromaticity $\Delta\omega$ is assumed small. The value of the integral is inversely proportional to $\Delta\omega$ (for a given total intensity).

§ 60. Scattering by freely orientable systems

If the energy level of the atom is not degenerate, then the polarisability and the intensity of coherent scattering are determined by one and the same tensor $\alpha_{ik} \equiv (c_{ik})_{11}$. If the

¹²See *Hostler L.* // *J. Math. Phys.* 1964. Vol. 5. P. 591. The application of this Green's function to the computation of the amplitude of scattering on the hydrogen atom: see *Granovskii Ya. I.* // *ZhETF*. 1969. Vol. 56. P. 605.

level is degenerate, then the observed values of the indicated quantities are obtained by averaging over all states belonging to a given level. The polarisability must be defined as the mean value

$$\alpha_{ik} = \overline{(c_{ik})_{11}},$$

and the observed intensity of scattering is determined by mean values of products

$$\overline{(c_{ik})_{11}(c_{lm})_{11}^*}.$$

Therefore the connection between polarisability and scattering becomes less direct.

Let us note that although each of the quantities $(c_{ik})_{11}$ may be complex, their mean values are real (it is assumed that there is no absorption, so that the tensor α_{ik} is Hermitian). Indeed, in the averaging one can arbitrarily choose a set of independent wave functions (corresponding to the given degenerate level), and one can always ensure that all functions are real.

For free (not in an external field) atoms or molecules, the degeneracy of levels is usually associated with free orientability in space with respect to the angular momentum. Let the initial state in the scattering have angular momentum J_1 , and the final J_2 . As usual, the cross section of scattering must be averaged over all values of the projection M_1 and summed over the values of M_2 . After the first averaging the cross section ceases to depend on M_2 , so that the further summation reduces to multiplication by $(2J_2 + 1)$. Thus, the averaged cross section of scattering

$$d\bar{\sigma} = \omega\omega'^3 c_{iklm}^{(21)} e_i'^* e_k e_l' e_m^* do', \quad (60.1)$$

where

$$c_{iklm}^{(21)} = \frac{1}{2J_1 + 1} \sum_{M_1 M_2} (c_{ik})_{21} (c_{lm})_{21}^* = (2J_2 + 1) \overline{(c_{ik})_{21} (c_{lm})_{21}^*}, \quad (60.2)$$

and the bar with index 1 denotes averaging over M_1 .

For unshifted scattering states 1 and 2 belong to one and the same level of energy ($\omega_{12} = 0$). If one speaks only of coherent scattering, then the states 1 and 2 must coincide completely, i.e. $M_1 = M_2$. Summation over M_2 and the multiplier $2J_2 + 1$ in (60.2) then drop out:

$$c_{iklm}^{(\text{coh})} = \overline{(c_{ik})_{11} (c_{lm})_{11}^*}. \quad (60.3)$$

The result of averaging can be written down without elaborate computation, if one takes into account that the averaging over M_1 is equivalent to averaging over all orientations of the system, after which the mean value can be expressed only through the unit tensor δ_{ik} . At the same time only the mean values of products of the scalar, symmetric, and antisymmetric parts of the scattering tensor can turn out to be different from zero: with the help of the unit tensor it is not possible to construct expressions which by their symmetry properties could correspond to cross products. Thus,

$$c_{iklm}^{(21)} = G_{21}^0 \delta_{ik} \delta_{lm} + c_{iklm}^{(21)s} + c_{iklm}^{(21)a}, \quad (60.4)$$

where

$$\begin{aligned} G_{21}^0 &= (2J_2 + 1) \overline{(c^0)_{21} (c^0)_{21}^*}, \\ c_{iklm}^{(21)s} &= (2J_2 + 1) \overline{(c_{ik}^s)_{21} (c_{lm}^s)_{21}^*}, \\ c_{iklm}^{(21)a} &= (2J_2 + 1) \overline{(c_{ik}^a)_{21} (c_{lm}^a)_{21}^*}. \end{aligned} \quad (60.5)$$

In other words, the cross section (and with it the intensity) of scattering of a freely orientable system decomposes into a sum of three

independent parts, which we shall call *scalar*, *symmetric*, and *antisymmetric scattering*.

Each of the three terms in (60.4) is expressed through just one independent quantity. Scalar scattering is expressed through G_{21}^0 , and for symmetric and antisymmetric scattering we have

$$\begin{aligned} c_{iklm}^{(21)s} &= \frac{1}{10} G_{21}^s (\delta_{il} \delta_{km} + \delta_{im} \delta_{kl} - \frac{2}{3} \delta_{ik} \delta_{lm}), \\ G_{21}^s &= (2J_2 + 1) \overline{(c_{ik}^s)_{21} (c_{ik}^s)_{21}^*}; \\ c_{iklm}^{(21)a} &= \frac{1}{6} G_{21}^a (\delta_{il} \delta_{km} - \delta_{im} \delta_{kl}), \\ G_{21}^a &= (2J_2 + 1) \overline{(c_{ik}^a)_{21} (c_{ik}^a)_{21}^*} \end{aligned} \quad (60.6)$$

(the combination of unit tensors is composed according to the properties of symmetry, after which the overall coefficient is found by contracting over pairs of indices il and km).

Substitution of formulae (60.4)–(60.6) into (60.1) leads to the following expression for the cross section of scattering:

$$\begin{aligned} d\bar{\sigma} = \omega\omega'^3 \left\{ G_{21}^0 |\mathbf{e}'^* \mathbf{e}|^2 + \frac{1}{10} G_{21}^s \left(1 + |\mathbf{e}'|^2 - \frac{2}{3} |\mathbf{e}'^* \mathbf{e}|^2 \right) + \right. \\ \left. + \frac{1}{6} G_{21}^a (1 - |\mathbf{e}' \mathbf{e}|^2) \right\} d\omega'. \end{aligned} \quad (60.7)$$

This formula determines in explicit form the angular dependence and the polarisation properties of scattering.

The total cross section of scattering over all directions, summed over polarisations of the final photon and averaged over polarisations and directions of incidence of the initial photon, can be obtained directly from (60.1). For this we note that

$$e_i^* e_k = \delta_{ik}/3,$$

if the averaging is carried out both over polarisations and over directions of propagation of the photon (the summation over the latter gives a result $2 \cdot 4\pi$ times larger). As a result we obtain

$$\bar{\sigma} = \frac{8\pi}{9} \omega\omega'^3 c_{ikik}^{(21)} = \frac{8\pi}{9} \omega\omega'^3 (3G_{21}^0 + G_{21}^s + G_{21}^a). \quad (60.8)$$

It was already indicated above that the selection rules for scattering coincide with the selection rules for matrix elements of an arbitrary second-rank tensor. In connection

with the decomposition of the intensity of scattering into three independent parts, it is useful to formulate these rules for each of the parts separately.

The selection rules for symmetric scattering coincide with the selection rules for electric-quadrupole emission, since the latter is also determined by an irreducible symmetric tensor (the tensor of quadrupole moments). For antisymmetric scattering the selection rules coincide with those for magnetic-dipole emission, since both are determined by an axial vector (recall that an antisymmetric tensor is equivalent (dual) to an axial vector)¹³. At the same time, however, there is a difference in that the diagonal matrix elements which in the case of emission give the mean values of electric or magnetic moments (and do not correspond to radiative transitions), in the case of scattering are significant—they refer to coherent scattering.

For scalar scattering the selection rules coincide with those for matrix elements of a scalar quantity. This means that transitions are possible only between states of identical angular momentum. In particular, the values of the total angular momentum J and its projection M must be identical (and the diagonal matrix elements of M do not depend on M —see III, (29.3)). For unshifted scattering, states 1 and 2 must thus coincide completely (not only by energy but also by M), so that unshifted scalar scattering is entirely coherent. Conversely, since in scalar scattering all states in any case combine with themselves, the scalar part is always present also in the coherent scattering.

Analogously to the averaging carried out above, the cross section of scattering of a freely orientable system must also be averaged over the direction of the angular momentum J_1 , and the tensor of polarisability. The averaging is quite simple: obviously,

$$\alpha_{ik} \equiv \overline{(c_{ik})_{11}} = \overline{(c^0)_{11}} \delta_{ik}.$$

The symmetric and antisymmetric parts of the scattering tensor drop out upon averaging: δ_{ik} is the only isotropic tensor of second rank.

It was already noted above that the diagonal matrix elements of the scalar do not depend on M_1 . Therefore the sign of averaging over $(c^0)_{11}$ can be omitted entirely (and $(c^0)_{11}$ computed at any value of M_1), so that the polarisability

$$\alpha_{ik} \equiv (c^0)_{11} \delta_{ik}. \quad (60.9)$$

For the same reason the sign of averaging can be omitted also from the quantity G_{11}^0 , which determines the scalar part of coherent scattering:

$$G_{11}^0 = |(c^0)_{11}|^2 = (c^0)_{11}^2. \quad (60.10)$$

(the multiplier $2J_2 + 1$ is omitted in correspondence with (60.3)). Thus there is a simple connection between the mean polarisability and the scalar part of coherent scattering. Both are determined by the quantity

$$(c^0)_{11} = \frac{2}{3} \sum_n \frac{\omega_{n1}}{\omega_{n1}^2 - \omega^2} |\mathbf{d}_{n1}|^2. \quad (60.11)$$

¹³The selection rules in question are, of course, those associated with symmetry, and not with a concrete type of axial vector: in the case of emission the vector of the magnetic moment contains the spin part, whereas in scattering the matrix elements considered are from quantities of orbital (coordinate) nature.

Problems

1. Find the angular distribution and the degree of depolarisation in scattering of linearly polarised light.

Solution. Let ϑ be the angle between the direction of scattering $\mathbf{n}'$ and the direction of polarisation of the incident light $\mathbf{e}$. The scattered light contains two independent components, polarised in the plane $\mathbf{n}'\mathbf{e}$ (intensity I_1) and perpendicular to it (intensity I_2); the degree of depolarisation is given by the ratio I_2/I_1 . The intensities I_1 and I_2 are determined by formula (60.7) with the corresponding directions of $\mathbf{e}'$.

For scalar scattering the light turns out to be completely polarised in the same plane ($I_2 = 0$), and the angular distribution of intensities

$$I = \frac{3}{2} \sin^2 \vartheta.$$

(Here and below the expressions for $I = I_1 + I_2$ are normalised so as to give 1 when averaged over directions.) For symmetric scattering

$$I = \frac{3}{20}(6 + \sin^2 \vartheta), \quad \frac{I_2}{I_1} = \frac{3}{3 + \sin^2 \vartheta}.$$

For antisymmetric scattering

$$I = \frac{3}{8}(2 + \sin^2 \vartheta), \quad \frac{I_2}{I_1} = 1 + \sin^2 \vartheta.$$

2. The same for scattering of natural light.

Solution. The transition in formula (60.7) to natural (unpolarised) incident light is effected by the replacement

$$e_i e_k^* \rightarrow \frac{1}{2}(\delta_{ik} - n_i n_k),$$

corresponding to averaging over directions of polarisation $\mathbf{e}$ at a given direction of incidence $\mathbf{n}$. The scattered light will be partially polarised, and from considerations of symmetry it is clear that its two independent components will be linearly polarised in the plane of scattering $\mathbf{nn}'$ (intensity $I_{\parallel}$) and perpendicular to it (intensity $I_{\perp}$). The angle of scattering (angle between $\mathbf{n}$ and $\mathbf{n}'$) is denoted by θ .

For scalar scattering

$$I = I_{\perp} + I_{\parallel} = \frac{3}{4}(1 + \cos^2 \theta), \quad \frac{I_{\parallel}}{I_{\perp}} = \cos^2 \theta;$$

for symmetric scattering

$$I = \frac{3}{40}(13 + \cos^2 \theta), \quad \frac{I_{\parallel}}{I_{\perp}} = \frac{6 + \cos^2 \theta}{7};$$

for antisymmetric scattering

$$I = \frac{3}{8}(2 + \sin^2 \theta), \quad \frac{I_{\parallel}}{I_{\perp}} = 1 + \sin^2 \theta.$$

3. For scattering of circularly polarised light, determine the reversal coefficient (ratio of the intensity of the component polarised in a circle in the “reverse” direction to the intensity of the component polarised in the “proper” direction).

Solution. With circularly polarised incident light the angular distribution and degree of depolarisation (the ratio $I_{\parallel}/I_{\perp}$) are the same as for scattering of natural light.

Let the vector $\mathbf{e}$ of the incident light have components $\mathbf{e} = (1, i, 0)/\sqrt{2}$ (in a coordinate system with the xz -plane coinciding with the plane of scattering, and the z -axis along the direction $\mathbf{n}$). Then for the “reverse” and “proper” circularly polarised components of the scattered light the polarisation vectors are

$$\mathbf{e}' = \frac{1}{\sqrt{2}}(\cos \theta, -i, -\sin \theta) \quad \text{and} \quad \mathbf{e}' = \frac{1}{\sqrt{2}}(\cos \theta, i, -\sin \theta).$$

Computing the intensity with the help of (60.7), we find the reversal coefficients P for all three types of scattering:

$$P^0 = \operatorname{tg}^4 \frac{\theta}{2}, \quad P^s = \frac{13 + \cos^2 \theta + 10 \cos \theta}{13 + \cos^2 \theta - 10 \cos \theta}, \quad P^a = \frac{1 - \cos^4(\theta/2)}{1 - \sin^4(\theta/2)}$$

(θ is the angle of scattering).

4. Compute the cross section of scattering of a photon of small frequency on the hydrogen atom in the ground state.

Solution. A photon of small frequency can be scattered only elastically. Since in the ground state the orbital angular momentum $L = 0$, the selection rules in the neglect of spin-orbit coupling allow only scalar scattering. The static polarisability of the atom (in ordinary units)

$$\alpha = \frac{9}{2} \left(\frac{\hbar^2}{me^2} \right)^3$$

(see III, § 76, Problem 4). Substituting in (60.8), we obtain the desired cross section:

$$\sigma_t = 54\pi \left(\frac{\omega}{e} \right)^4 \left(\frac{\hbar^2}{me^2} \right)^6.$$

5. Compute the cross section of elastic scattering of γ -radiation by the deuteron (H. A. Bethe, R. Peierls, 1935).

Solution. The wave functions of the ground state of the deuteron and of its continuous spectrum states (dissociated deuteron) are

$$\psi_0 = \sqrt{\frac{\kappa}{2\pi}} \frac{e^{-\kappa r}}{r}, \quad \psi_{\mathbf{p}} = e^{i\mathbf{p}\mathbf{r}}, \quad \kappa = \sqrt{M I}$$

(see (58.2), (58.3)). The matrix element of the dipole moment $\mathbf{d}_{\mathbf{p}0} = -ie\mathbf{p}_{1n}/(M\omega_{\mathbf{p}0})$ was computed in § 58:

$$\mathbf{d}_{\mathbf{p}0} = -\frac{4\pi ie}{M\omega_{\mathbf{p}0}} \sqrt{\frac{\kappa}{2\pi}} \frac{\mathbf{p}}{\kappa^2 + \mathbf{p}^2},$$

where the frequency $\omega_{\mathbf{p}0} = (\mathbf{p}^2 + \kappa^2)/M$. The scattering tensor of polarisability

$$\alpha_{ik} = \left\{ \frac{2}{3} \int \frac{\omega_{\mathbf{p}0}}{\omega_{\mathbf{p}0}^2 - \omega^2} |\mathbf{d}_{\mathbf{p}0}|^2 \frac{d^2 p}{(2\pi)^3} - \frac{e^2}{2M\omega^2} \right\} \delta_{ik}.$$

The first term is related to the virtual excitation of the internal degrees of freedom of the deuteron; it is written in the form (60.11). The second term is related to the action of the light wave on the translational motion of the deuteron as a whole. Since this is quasiclassical scattering, the corresponding part of the scattering tensor is given by formula (59.14) (with the mass of the deuteron $2M$ in place of m).

Computing α_{ik} reduces to taking the integral

$$J = \frac{1}{2} \int_{-\infty}^{\infty} \frac{z^4 dz}{(z^2 + 1)^3 [(z^2 + 1)^2 - \gamma^2]}, \quad z = \frac{p}{\kappa}, \quad \gamma = \frac{M\omega}{\kappa^2} = \frac{\hbar\omega}{I}.$$

We have

$$J = \frac{1}{8} \frac{d}{d\lambda} \left(\frac{1}{\lambda} \frac{dJ_0}{d\lambda} \right)_{\lambda=1}, \quad J_0 = \frac{1}{2} \int_{-\infty}^{\infty} \frac{z^4 dz}{(z^2 + \lambda^2) [(z^2 + 1)^2 - \gamma^2]}.$$

For $\gamma < 1$ the integrand has in the upper half-plane of the complex variable z poles at the points $i\lambda, i\sqrt{1+\gamma}, i\sqrt{1-\gamma}$; the integral J_0 is computed by the residues at these poles. As a result we obtain

$$J = \frac{\pi}{2} \left\{ \frac{(1+\gamma)^{3/2}}{2\gamma^4} + \frac{(1-\gamma)^{3/2}}{2\gamma^4} - \frac{3}{8\gamma^2} + \frac{1}{\gamma^4} \right\}.$$

The total cross section of scattering is expressed through α_{ik} from (60.8), (60.10):

$$\sigma = \frac{8\pi}{3} \left(\frac{e^2}{Mc^2} \right)^2 \left| -1 - \frac{4}{3\gamma^2} + \frac{2}{3\gamma^2} (1+\gamma)^{3/2} + (1-\gamma)^{3/2} \right| \quad \text{for } \gamma = \frac{\hbar\omega}{I} < 1.$$

The amplitude of scattering at $\gamma > 1$ (above the threshold of dissociation of the deuteron) is obtained from the amplitude at $\gamma < 1$ by analytic continuation, with the imaginary part appearing positive (in correspondence with the rule of going around in (59.17)):

$$\sigma = \frac{8\pi}{3} \left(\frac{e^2}{Mc^2} \right)^2 \left| -1 - \frac{4}{3\gamma^2} + \frac{2}{3\gamma^2} (\gamma+1)^{3/2} + i \frac{2}{3\gamma^2} (\gamma-1)^{3/2} \right| \quad \text{for } \gamma > 1.$$

For $\gamma \gg 1$ we get $\sigma = (8\pi/3)(e^2/Mc^2)^2$, which corresponds, as expected, to relativistic scattering on a free proton.

The angular distribution of the radiation

$$d\sigma = \sigma \frac{3}{4} (1 + \cos^2 \theta) \frac{d\theta}{4\pi},$$

where θ is the angle of scattering. Having determined the amplitude of scattering from (59.24), we should have

$$\text{Im } f(0) = \frac{2e^2}{3Mc^2} \frac{(\gamma-1)^{3/2}}{\gamma^2}, \quad \gamma > 1.$$

According to the optical theorem (59.26) this quantity coincides with $\omega\sigma_t/(4\pi)$, where σ_t is the total cross section of photodisintegration (58.4). Let us recall that the cross section of elastic scattering is of higher order ($\sim e^4$) than the cross section of dissociation ($\sim e^2$, see (58.4)), so that σ_t coincides with the cross section of dissociation. For the same reason, in the approximation considered, the amplitude of scattering at $\gamma < 1$ (below the threshold of dissociation) turned out to be real.

§61. Scattering on molecules

The specifics of molecular scattering are connected with the same properties of molecules that underlie their spectral theory—with the possibility of separate consideration of the electronic state at fixed nuclei and the motion of nuclei in a given effective field of electrons.

Let the frequency of the incident light ω be less than the energy ω_e of the first electronic excitation. Then upon scattering the electronic terms cannot be excited. The scattering will be either unshifted, or shifted on account of excitation of rotational or vibrational levels.

We shall further assume that the ground electronic term of the molecule is non-degenerate (and has no fine structure). In other words, it is assumed that the total spin of the electrons and the projection of their total orbital angular momentum onto the axis of the molecule are equal to zero (for a molecule of the symmetric top type). As is known, these conditions are satisfied for the ground states of the majority of molecules¹⁴.

Finally, we shall assume the frequency ω to be large compared with the intervals of the nuclear (rotational and vibrational) structure of the ground term, and the difference $\omega_e - \omega$ to be in the same relation to the nuclear structure of the excited electronic term. In other words, the frequency of the incident light must be sufficiently far from resonances. It is precisely these conditions that allow one to separate, in computing the scattering tensor, the electronic part from the motion of nuclei, treating the problem at a given nuclear configuration.

In this formulation the scattering tensor coincides with the tensor of polarisability $\alpha_{ik} \equiv (c_{ik})_{11}$ and is computed in principle by the general formula (59.17), in which the summation is carried out over all excited electronic terms. The quantities α_{ik} obtained in this way will be functions of the nuclear coordinates q (on which, as on parameters, the energies and wave functions of the electronic terms depend). By virtue of the non-degeneracy of the state, the tensor $\alpha_{ik}(q)$ will be real, and therefore symmetric.

The tensor $\alpha_{ik}(q)$ represents the electronic polarisability of a given nuclear configuration of the molecule. To solve the actual problem of scattering one must also take into account the motion of nuclei in the initial and final states. Let $\psi_{s_1}(q)$ and $\psi_{s_2}(q)$ be the nuclear wave functions of these states (so that s_1, s_2 are sets of vibrational and rotational quantum numbers). The desired matrix element of the scattering tensor is the matrix element of the tensor $\alpha_{ik}(q)$, computed with respect to these functions:

$$\langle s_2 | \alpha_{ik} | s_1 \rangle = \int \psi_{s_2}^*(q) \alpha_{ik} \psi_{s_1} dq. \quad (61.1)$$

By virtue of the symmetry of the tensor $\alpha_{ik}(q)$, the tensor (61.1) will also be symmetric (both for coinciding and for different s_1, s_2). Thus we arrive at the conclusion that in the conditions under consideration the antisymmetric part is absent both in shifted

¹⁴The results presented below may, however, also be valid (with a certain accuracy) in cases when the degeneracy of the ground electronic term is associated with non-zero spin, and the spin-orbit interaction is small (so that the fine structure it causes can be neglected). In this approximation the states with different spin directions do not combine and in this sense behave as if they were non-degenerate. Such is, for example, the case of the molecule O_2 with the ground term $^3\Sigma$.

and in unshifted scattering. The scattering will contain only the scalar and the symmetric parts.

The scalar part of the polarisability $\alpha^0(q)$ does not depend on the orientation of the molecule, but depends only on the internal arrangement of atoms in it. Let us denote via v the set of vibrational quantum numbers of the molecule, and via r the set of rotational quantum numbers, excluding the magnetic number m . Then the matrix elements

$$\langle v_2 r_2 m_2 | \alpha^0 | v_1 r_1 m_1 \rangle = \langle v_2 | \alpha^0 | v_1 \rangle \delta_{r_1 r_2} \delta_{m_1 m_2}. \quad (61.2)$$

The diagonality with respect to the numbers r, m is a general property of any scalar. The specific feature of (61.2) is that the matrix elements in this case do not depend on these numbers at all. Thus, scalar scattering occurs only for a number of vibrational transitions and does not depend on the rotational state.

The symmetric scattering is determined by the matrix elements of the tensor α_{ik}^s . Its components relative to a non-moving system of coordinates xyz are expressed through the components $\bar{\alpha}_{i'k'}^s$ in a system $\xi\eta\zeta$ attached to the molecule, according to

$$\alpha_{ik}^s = \sum_{i'k'} \bar{\alpha}_{i'k'}^s D_{i'i} D_{k'k}, \quad (61.3)$$

where $D_{i'i}$ are the direction cosines of the new axes relative to the old. The quantities $\bar{\alpha}_{i'k'}^s$ do not depend on the orientation of the molecule, and $D_{i'i}$ do not depend on its internal coordinates. Therefore

$$\langle v_2 r_2 m_2 | \alpha_{ik}^s | v_1 r_1 m_1 \rangle = \sum_{i'k'} \langle v_2 | \bar{\alpha}_{i'k'}^s | v_1 \rangle \langle r_2 m_2 | D_{i'i} D_{k'k} | r_1 m_1 \rangle.$$

The sum over $r_2 m_2, ik$ of the squared moduli of these quantities is, as is easy to verify¹⁵,

$$\sum_{r_2 m_2} \sum_{ik} |\langle v_2 r_2 m_2 | \alpha_{ik}^s | v_1 r_1 m_1 \rangle|^2 = \sum_{i'k'} |\langle v_2 | \bar{\alpha}_{i'k'}^s | v_1 \rangle|^2. \quad (61.4)$$

This means that the total intensity of scattering with transitions from a given vibrational-rotational level $v_1 r_1$ to all rotational levels of the vibrational state v_2 does not depend on r_1 .

For a molecule of the symmetric-top type one can go further and establish the dependence of the intensity of scattering on the rotational quantum numbers for each transition $v_1 r_1 \rightarrow v_2 r_2$. The numbers r in this case are the angular momentum J and its projection k onto the axis of the molecule. Let us introduce, in place of the Cartesian components of α_{ik}^s , the corresponding spherical tensor of second rank, whose

¹⁵In the transformation of the sums the unitarity of the matrix D_{ik} , which expresses the relation:

$$\begin{aligned} & \sum_{ik} \sum_{r_2 m_2} \langle r_1 m_1 | D_{il} D_{kk'} | r_2 m_2 \rangle \langle r_2 m_2 | D_{il'} D_{kk'} | r_1 m_1 \rangle = \\ & = \langle r_1 m_1 | \sum_{ik} D_{il} D_{kk'} D_{il'} D_{kk'} | r_1 m_1 \rangle = \langle r_1 m_1 | \delta_{ll'} \delta_{kk'} | r_1 m_1 \rangle = \delta_{ll'} \delta_{kk'}, \end{aligned}$$

is used.

components we denote by α_λ ($\lambda = 0, \pm 1, \pm 2$). According to formula (110.7) (see III) the squared moduli of its matrix elements

$$\begin{aligned} & |\langle v_2 J_2 k_2 m_2 | \alpha_\lambda | v_1 J_1 k_1 m_1 \rangle|^2 = \\ & = (2J_1 + 1)(2J_2 + 1) \begin{pmatrix} J_2 & 2 & J_1 \\ -k_2 & \lambda' & k_1 \end{pmatrix}^2 \begin{pmatrix} J_2 & 2 & J_1 \\ \lambda & m_1 & \end{pmatrix}^2 |\langle v_2 | \bar{\alpha}_\lambda | v_1 \rangle|^2, \end{aligned}$$

where $\bar{\alpha}_\lambda(q)$ is the spherical tensor of polarisation referred to the molecular axis, $\lambda' = k_2 - k_1$. Having summed over m_2 and λ (at given m_1), we obtain (cf. III, (110.8))

$$\sum_{m_2, \lambda} |\langle v_2 J_2 k_2 m_2 | \alpha_\lambda | v_1 J_1 k_1 m_1 \rangle|^2 = (2J_2 + 1) \begin{pmatrix} J_2 & 2 & J_1 \\ -k_2 & \lambda' & k_1 \end{pmatrix}^2 |\langle v_2 | \bar{\alpha}_{\lambda'} | v_1 \rangle|^2. \quad (61.5)$$

This quantity determines the intensity of scattering with the vibrational-rotational transition $v_1 J_1 k_1 \rightarrow v_2 J_2 k_2$. Since the rotational matrix elements of the molecule do not enter at all, the dependence of the intensity on the numbers J_1, J_2, k_1, k_2 is thereby determined. Let us note that only one spherical component of the tensor of polarisability enters the right-hand side of (61.5).

If one sums the equality (61.5) over J_2 and k_2 , then we obtain ¹⁶

$$\sum_{\lambda} \sum_{J_2 k_2 m_2} |\langle v_2 J_2 k_2 m_2 | \alpha_\lambda | v_1 J_1 k_1 m_1 \rangle|^2 = \sum_{\lambda'} |\langle v_2 | \bar{\alpha}_{\lambda'} | v_1 \rangle|^2,$$

i.e. we return to the sum rule (61.4).

A special case of the symmetric top is the rotator—a linear molecule (in particular, a diatomic one). The projection of the angular momentum onto the axis of such a molecule is zero (in the non-degenerate electronic state with zero electronic orbital angular momentum) ¹⁷. Therefore in (61.5) one must in this case put $k_1 = k_2 = 0$.

Finally, let us consider the question of selection rules in vibrational combination spectra of emission (or absorption) of molecules, together with the analogous question for the vibrational spectra of Raman scattering ¹⁸.

For scattering the question reduces to finding the conditions under which the matrix elements of the tensor $\alpha_{ik}(q)$, computed with respect to the vibrational wave functions $\psi_v(q)$, are different from zero; at the same time one should consider separately the scalar α^0 (for scalar scattering) and the irreducible symmetric tensor α_{ik}^s (for symmetric scattering). The analogous role in emission (or absorption) is played by the matrix elements of the vector $\mathbf{d}(q)$ —the dipole moment of the molecule, averaged over the

¹⁶In the summation over J_2 at given k_1 and λ' (and thereby $k_2 = k_1 + \lambda'$) we have

$$\sum_{J_2} (2J_2 + 1) \begin{pmatrix} J_2 & 2 & J_1 \\ -k_2 & \lambda' & k_1 \end{pmatrix}^2 = 1$$

by virtue of (106.13) (see III). After this the summation over k_2 (or, what is the same, over λ') at given k_1 is produced.

¹⁷We do not consider here effects connected with the interaction of vibrations and rotations of the molecule (see III, § 104).

¹⁸These spectra belong to the infra-red region and are observed usually in absorption.

electronic state at a given position of the nuclei (for diatomic molecules this was already indicated in § 54).

The vibrations of a polyatomic molecule are classified by types of symmetry—irreducible representations of the corresponding point group: D_a , a is the label of the representation (see III, § 100). By these representations the symmetry of the vibrational wave functions of the vibrational states of the molecule is also determined (see III, § 101). The symmetry of the wave functions of the first vibrational state (vibrational quantum number $v_a = 1$) coincides with the symmetry D_a

of the vibration. The symmetry of higher states ($v_a > 1$) is given by the representation $[D_a^{v_a}]$ —the symmetrical product of the representation D_a by itself v_a times. Finally, the symmetry of a state with simultaneously excited different vibrations a and b is the direct product $[D_a^{v_a}] \times [D_b^{v_b}]$ ¹⁹. The method of finding selection rules of various quantities (scalar, vector, tensor) by types of symmetry is presented in vol. III, § 97.

The selection rules, based on the symmetry properties of the molecule, are strict. Alongside them there also exist approximate rules, associated with the assumption of harmonicity of vibrations and with the expansion of the functions $\alpha_{ik}(q)$ or $\mathbf{d}(q)$ in powers of vibrational coordinates q . They arise as a consequence of the known selection rule for the harmonic oscillator, according to which the matrix elements of its coordinate q are different from zero only for transitions with a change of the vibrational quantum number $\Delta v = \pm 1$.

§ 62. Natural width of spectral lines

Up to now in the study of emission and scattering of light we have considered all energy levels of the system (of the atom, say) as strictly discrete. Among the excited levels, however, having a finite lifetime, there is a probability of de-excitation. In accordance with the general principles of quantum mechanics this leads to the fact that the levels become quasidiscrete, acquiring a finite (small) width (see III, § 134); they are written in the form $E - i\Gamma/2$, where $\Gamma (= \Gamma/\hbar)$ is the total probability (in 1 s) of all possible “decay” processes of the given state.

Let us consider the question of how this circumstance affects the process of emission (V. Weisskopf, E. Wigner, 1930). It is clear in advance that because of the finite width of the emitting level the emitted light will not be strictly monochromatic: the frequencies will be scattered in an interval $\Delta\omega \sim \Gamma (= \Gamma/\hbar)$. At the same time, on account of the finiteness of the lifetime of the initial state of the emitting system, the more natural formulation is that of finding the total probability of emission of a photon of given frequency, rather than the probability of emission per unit time. We compute this probability first of all for the case of the transition of the atom from some excited level

$$E_1 - \frac{i}{2} \Gamma_1$$

to the ground level E_2 , which has an infinite lifetime and is therefore strictly discrete.

¹⁹The symmetry properties of vibrational wave functions, of course, do not depend on the specific form of the vibrational potential energy; they follow, in particular, from the assumption made in vol. III, § 101 about the harmonicity of vibrations.

Let Ψ be the wave function of the atom and the photon field, $\hat{H} = \hat{H}^{(0)} + \hat{V}$ the Hamiltonian of this system, where $\hat{V}$ is the operator of the interaction of the atom with the photon field. We seek the solution of the Schrödinger equation

$$i \frac{\partial \Psi}{\partial t} = (\hat{H}^{(0)} + \hat{V})\Psi \quad (62.1)$$

in the form of an expansion over wave functions of the unperturbed states of the system

$$\Psi = \sum_{\nu} a_{\nu}(t) \Psi_{\nu}^{(0)} = \sum_{\nu} a_{\nu}(t) e^{-i\mathcal{E}_{\nu}t} \psi_{\nu}^{(0)}. \quad (62.2)$$

For the coefficients $a_{\nu}(t)$ we obtain the system of equations

$$i \frac{\partial a_{\nu}}{\partial t} = \sum_{\nu'} \langle \nu | V | \nu' \rangle a_{\nu'} \exp\{i(\mathcal{E}_{\nu} - \mathcal{E}_{\nu'})t\}. \quad (62.3)$$

Let $|\nu\rangle$ denote the state with energy $\mathcal{E}_{\nu} = E_2 + \omega$, in which the atom is in the ground level E_2 and there is one quantum with a definite frequency ω ; let us denote this state by the symbol $|\omega 2\rangle$. At the initial moment of time the system is in the state $|1\rangle$, in which the atom is excited on the level E_1 and there are no photons. In other words, at $t = 0$ it must be

$$a_1 = 1, \quad a_{\nu} = 0 \quad \text{for } |\nu\rangle \neq |1\rangle. \quad (62.4)$$

The solution found with this initial condition of the equation (62.3) will give (with the proper normalisation of the wave functions) the probability at the moment t of the transition of the atom $1 \rightarrow 2$ with emission of a photon in the frequency interval $d\omega$:

$$|a_{\omega 2}(t)|^2 d\omega. \quad (62.5)$$

We shall be interested in the probability at $t \rightarrow \infty$:

$$dw = |a_{\omega 2}(\infty)|^2 d\omega. \quad (62.5)$$

For a better understanding of the formulation of the question let us recall that when finding the usual probability of emission (in 1 s) with transition $1 \rightarrow 2$ (without taking into account the level width), one must solve equation (62.3) replacing in the right-hand side in the first approximation all $a_{\nu}(t)$ by the values (62.4). The solution obtained is then considered at large t (cf. III, § 42). We can now clarify the meaning of this procedure: it refers to times that are small compared with $1/\Gamma_1$, but still

large compared with the period $1/(E_1 - E_2)$ of oscillation of the excited level.

In our case, however, times comparable with $1/\Gamma_1$ are being considered, and the function $a_1(t)$ decreases with time according to the law

$$a_1(t) = e^{-\Gamma_1 t/2}. \quad (62.6)$$

The functions $a_{\nu}(t)$ for the states $|\nu'\rangle$, which can arise during de-excitation of the atom, increase with time. If the de-excitation from the level E_1 is possible to various (in addition to E_2) levels of the atom, then there will be many growing functions $a_{\nu}(t)$;

each of them corresponds to a state in which the atom is on one of its levels and there is one photon of the corresponding energy. But the right-hand side of equation (62.3) will nonetheless contain only one term—that for $|\nu'\rangle = |1\rangle$. Indeed, since the matrix elements of the perturbation may differ from zero only for transitions with a change of 1 in the number of photons of some given energy, they are zero for transitions between states each containing one photon of different energies.

Thus, for $a_{\omega 2}(t)$ we have the equation

$$i \frac{da_{\omega 2}}{dt} = \langle \omega 2 | V | 1 \rangle e^{i(E_2 + \omega - E_1)t} a_1 = \langle \omega 2 | V | 1 \rangle \exp \left\{ i(\omega - \omega_{12})t - \frac{\Gamma_1}{2} t \right\}, \quad (62.7)$$

where $\omega_{12} = E_1 - E_2$. Integrating with the condition $a_{\omega 2}(0) = 0$, we find

$$a_{\omega 2} = \langle \omega 2 | V | 1 \rangle \frac{1 - \exp\{i(\omega - \omega_{12})t - \Gamma_1 t/2\}}{\omega - \omega_{12} + i\Gamma_1/2}. \quad (62.8)$$

The sum

$$\Gamma_{1 \rightarrow 2} = 2\pi \sum |\langle \omega 2 | V | 1 \rangle|^2, \quad (62.9)$$

taken over the polarisations and directions of motion of the photon, is the total usual probability of emission. This is at the same time a part of the width of the level E_1 (the partial width), associated with the transition $1 \rightarrow 2$, in contrast to the total width Γ_1 , constituted of contributions from all possible decay modes of the given quasistationary state²⁰.

Having carried out also the summation of the probability dw , we obtain the following definitive formula for the frequency distribution of the emitted light:

$$dw = w_t \frac{\Gamma_1}{2\pi} \frac{d\omega}{(\omega - \omega_{12})^2 + \Gamma_1^2/4}, \quad (62.10)$$

where $w_t = \Gamma_{1 \rightarrow 2}/\Gamma_1$ is the total relative probability of the transition $1 \rightarrow 2$. This is a distribution of dispersion type. The shape of the spectral line (62.10) is characteristic of an isolated stationary atom; it is called *natural*²¹.

Let now the E_2 level of the atom be also excited, with finite width Γ_2 . For simplicity we shall assume that this width is associated with the transition of the atom to the ground state E_0 accompanied by emission of one photon (the final answer—formula (62.12) below—does not depend on this assumption). Then the process of the decay of state 1 can be treated as the process of emission of two photons, studied in § 59. The matrix element of this process—so long as the finiteness of the lifetime of state 2 is not taken into account—is given by formula (59.2)

$$\langle \omega \omega' 0 | V^{(2)} | 1 \rangle = \frac{\langle \omega' 0 | V | \omega 2 \rangle \langle \omega 2 | V | 1 \rangle}{E_0 - E_2 + \omega' + i0}, \quad (62.11)$$

(in formula (59.2) the labelling of states $2 \rightarrow 0$ has been changed, and in the sum over n only the one term, corresponding to finding the atom in state 2, that is resonantly large at

²⁰Formulae (62.6), (62.9) can be obtained, of course, by solving equation (62.7) also analytically for $a_1(t)$.

²¹In contrast to the broadening connected with the interaction of an atom with other atoms (broadening by collisions) or with the presence in the source of atoms moving with different velocities (Doppler broadening).

values of ω' close to $E_2 - E_0$, has been retained). Now taking into account the finite lifetime of

state 2, this will lead to (62.11) only the replacement $E_2 \rightarrow E_2 - i\Gamma_2/2$, so that

$$\langle \omega \omega' 0 | V^{(2)} | 1 \rangle = \frac{\langle \omega' 0 | V | \omega 2 \rangle \langle \omega 2 | V | 1 \rangle}{E_0 - E_2 + \omega' + i\Gamma_2/2}.$$

Substituting this value of the matrix element into the equation for $a_{\omega\omega'2}(t)$ (differing from (62.7) only in notation) and proceeding after the derivation quite analogously to (62.8):

$$a_{\omega\omega'0}(\infty) = \frac{\langle \omega \omega' 0 | V | \omega 2 \rangle \langle \omega 2 | V | 1 \rangle}{(\omega' - \omega_{20} + i\Gamma_2/2)(\omega + \omega' - \omega_{10} + i\Gamma_1/2)}.$$

The probability of emission of photons ω and ω' is

$$\begin{aligned} dw &= |a_{\omega\omega'0}(\infty)|^2 d\omega d\omega' = \\ &= \frac{\Gamma_{1 \rightarrow 2}}{2\pi} \frac{\Gamma_{2 \rightarrow 0}}{2\pi} \frac{d\omega d\omega'}{[(\omega' - \omega_{20})^2 + \Gamma_2^2/4][(\omega + \omega' - \omega_{10})^2 + \Gamma_1^2/4]}. \end{aligned} \quad (62.12)$$

As it should be, this expression has sharp maxima at $\omega' \approx \omega_{20}$ and $\omega \approx \omega_{12}$.

The desired shape of the spectral line, corresponding to the transition $1 \rightarrow 2$, is obtained by integrating (62.12) over $d\omega'$ (which can range over the whole interval from $-\infty$ to $+\infty$). The integral is most simply computed by closing the path of integration with a remote semicircle in the upper half-plane of the complex variable ω' , and is given by the sum of residues at the poles

$$\omega' = \omega_{20} + i\Gamma_2/2 \quad \text{and} \quad \omega' = \omega_{10} - \omega + i\Gamma_1/2.$$

As a result we obtain

$$dw = w_t \frac{\Gamma_1 + \Gamma_2}{2\pi} \frac{d\omega}{(\omega - \omega_{12})^2 + (\Gamma_1 + \Gamma_2)^2/4}, \quad (62.13)$$

where $w_t = \Gamma_{1 \rightarrow 2}\Gamma_{2 \rightarrow 0}/(\Gamma_1\Gamma_2)$ is the total probability of the double transition $1 \rightarrow 2 \rightarrow 0$ ²².

The shape of the line (62.13) differs from (62.10) only by the replacement of Γ_1 by $\Gamma_1 + \Gamma_2$ —the width of the line equals the sum of the widths of the initial and final states.

Let us note that the line width turns out to be, generally speaking, not proportional to the probability $\Gamma_{1 \rightarrow 2}$ of the transition $1 \rightarrow 2$ itself, i.e. not proportional to the intensity of the line (as it would be in classical theory). Since $\Gamma_1 + \Gamma_2 > \Gamma_{1 \rightarrow 2}$, the line can have a large width at comparatively small intensity.

§ 63. Resonance fluorescence

Taking into account the finite widths of levels in the problem of scattering of light is essential in cases when the frequency ω of the incident light is close to one of

²²In more complex cases w_t is the total probability of all cascades, beginning with the transition $1 \rightarrow 2$ and ending at level 0.

the “intermediate” frequencies ω_{n1} or ω_{n2} —the so-called *resonance fluorescence* (V. Weisskopf, 1931).

Let us consider unshifted scattering by a system (say, an atom) in the ground state, so that the initial and final levels coincide and are strictly discrete. Let the frequency of the light be close to a certain frequency ω_{n1} , where the level n is excited and therefore quasidiscrete.

This question could be solved by the method presented in the preceding paragraph. In this, however, there is no necessity, since the problem is completely analogous to that of the non-relativistic resonance scattering considered in vol. III, § 134. According to the results obtained there the resonance amplitude of scattering must contain the pole factor

$$\left[\omega - \left(E_n - i \frac{\Gamma_n}{2} \right) - E_1 \right]^{-1}.$$

On the other hand, for $|\omega - \omega_{n1}| \gg \Gamma_n$ the formula must go over to the non-resonance formula (59.5). From this it is clear that the desired cross section of scattering is obtained simply by the replacement of E_n by $E_n - i\Gamma_n/2$ in formula (59.5), and in the sum over n one may limit oneself to the resonance terms only:

$$d\sigma = \frac{|\sum M_n (\mathbf{d}_{2n} \mathbf{e}'^*) (\mathbf{d}_{n1} \mathbf{e})|^2}{(\omega_{n1} - \omega)^2 + \Gamma_n^2/4} \omega^4 d\omega'. \quad (63.1)$$

The summation is carried out over all states (with different projections M_n), corresponding to the resonance level E_n ; in the sum over n in formula (59.5) the states 1 and 2 may differ by different values of M_1 and M_2 .

The cross section (63.1) is maximal at $\omega = \omega_{n1}$. By order of magnitude the value at the maximum is $\sigma_{\max} \sim \omega^4 d^4 / \Gamma_n^2$. Since the probability of spontaneous transition $n \rightarrow 1$, and with it also the width Γ_n , are of order $\omega^3 d^2$, this value

$$\sigma_{\max} \sim \omega^{-2} \sim \lambda^2, \quad (63.2)$$

i.e. of order of the square of the wavelength of the light and does not depend on the constant of the fine structure—in place of the typical values $\sim r_e^2$ outside the region of resonance.

Let us emphasise that since the atom both before and after scattering is on a strictly discrete (ground) level, the frequencies of the primary and secondary photons strictly coincide. Therefore in

irradiation with monochromatic light the scattered light will also be monochromatic. If, however, the incident light has a spectral distribution of intensities $I(\omega)$, and the function $I(\omega)$ changes little over the width Γ_n , then the intensity of the scattered light will be proportional to

$$\frac{I(\omega_{n1}) d\omega}{(\omega - \omega_{n1})^2 + \Gamma_n^2/4}. \quad (63.3)$$

In other words, the line shape of the scattered light will coincide with the natural line shape in spontaneous emission from the level E_n .

The cross section (63.1) corresponds to the scattering tensor

$$(c_{ik})_{21} = \frac{\sum_{M_n} (d_i)_{2n} (d_k)_{n1}}{\omega_{n1} - \omega - i\Gamma_n/2}. \quad (63.4)$$

In particular, the tensor of polarisability

$$\frac{\sum_{M_n} (d_i)_{1n} (d_k)_{n1}}{\omega_{n1} - \omega - i\Gamma_n/2}. \quad (63.5)$$

Let us at once note that the addition of the imaginary part to the energies of intermediate excited states violates the Hermiticity of the tensor of polarisability and also at frequencies below the ionisation threshold. Its imaginary part appears, directly associated with the absorption of light. Having absorbed a quantum, the atom will sooner or later return to the ground state with emission of one or several photons. Therefore from the point of view of the absorption cross section the absorption cross section is simply the total cross section σ_t of all possible scattering processes²³. On the other hand, according to formula (59.25), expressing the optical theorem, this cross section is determined by the anti-Hermitian part of the tensor of polarisability.

Substituting in (59.25) the tensor α_{ik} from (63.5), we find the following formula for the absorption cross section of a photon of frequency ω close to ω_{n1} :

$$\sigma^{(\text{abs})} = 4\pi^2 \sum_{M_n} |\mathbf{d}_{n1}\mathbf{e}|^2 \omega \frac{\Gamma_n/2}{\pi[(\omega - \omega_{n1})^2 + \Gamma_n^2/4]}. \quad (63.6)$$

In the limit $\Gamma_n \rightarrow 0$ the last factor in this formula tends to the δ -function $\delta(\omega - \omega_{n1})$, in accordance with the fact that in this case only a photon of a strictly defined frequency can be absorbed. Let light fall on the atom with spectral and angular distribution given by the flux density of energy $I_{\mathbf{ke}}$ (cf. (44.7)). Then the photon number flux density is $I_{\mathbf{ke}}/\omega$, and the absorption probability

$$dW^{(\text{abs})} = \sigma^{(\text{abs})} \frac{I_{\mathbf{ke}}}{\omega} d\omega d\Omega. \quad (63.7)$$

If the function $I_{\mathbf{ke}}(\omega)$ changes little over the width Γ_n , then after integrating over frequencies we obtain

$$dW^{(\text{abs})} = 4\pi^2 \sum_{M_n} |\mathbf{d}_{n1}\mathbf{e}|^2 I_{\mathbf{ke}}(\omega_{n1}) d\omega d\Omega.$$

Noting, on the other hand, that according to (45.5),

$$dW^{(\text{sp})} = \frac{\omega^3}{2\pi} \sum_{M_n} |\mathbf{d}_{1n}\mathbf{e}^*|^2 d\Omega = \frac{\omega^3}{2\pi} \sum_{M_n} |\mathbf{d}_{n1}\mathbf{e}|^2 d\Omega$$

is the probability of spontaneous emission of a photon of frequency ω_{n1} , we return to formula (44.9).

²³Let us emphasise that we are speaking of the absorption by a system in a stable, ground state. On account of the finiteness of the lifetime of the excited state the formulation of the question would be different.

The Scattering Matrix

§ 64. Scattering amplitude

The general formulation of the problem of collisions consists in finding, for a given initial state of the system (a certain collection of free particles), the probabilities of various possible final states (other collections of free particles). If the symbol $|i\rangle$ denotes the initial state, then the result of the collision can be represented as the superposition

$$\sum_f |f\rangle \langle f|S|i\rangle, \quad (64.1)$$

where the summation is carried out over all possible final states $|f\rangle$. The coefficients of this expansion (or in abbreviated notation S_{fi}) constitute the *scattering matrix*, or *S-matrix*¹. The squares $|S_{fi}|^2$ give the probabilities of transitions to definite states $|f\rangle$.

In the absence of interaction between the particles the state of the system would not change, which would correspond to the *S*-matrix being the unit matrix (absence of scattering). It is convenient to always separate out this unit matrix, representing the scattering matrix in the form

$$S_{fi} = \delta_{fi} + i(2\pi)^4 \delta^{(4)}(P_f - P_i) T_{fi}, \quad (64.2)$$

where T_{fi} is a new matrix. In the second term a four-dimensional δ -function is separated out, expressing the law of conservation of 4-momentum (P_i and P_f are the sums of the 4-momenta of all particles in the initial and final states); the remaining factors are introduced for convenience in what follows. In non-diagonal matrix elements the first term in (64.2) drops out, so that for the transition $i \rightarrow f$ the elements of the matrices *S* and *T* are related to each other by the simple relation

$$S_{fi} = i(2\pi)^4 \delta^{(4)}(P_f - P_i) T_{fi}. \quad (64.3)$$

The matrix elements T_{fi} , remaining after the separation of the δ -function, we shall call the *scattering amplitudes*.

When squaring the modulus $|S_{fi}|$ the square of the δ -function will appear. It should be understood as follows. The δ -function

$$\delta^{(4)}(P_f - P_i) = \frac{1}{(2\pi)^4} \int e^{i(P_f - P_i)x} d^4x. \quad (64.4)$$

If we compute another such integral at $P_f = P_i$ (by virtue of the already present δ -function), extending the integration over a large volume *V* and time interval *t*, then we

¹From the English word "scattering" or the German "Streuung".

obtain $Vt/(2\pi)^4$ ². Therefore one may write

$$|S_{fi}|^2 = (2\pi)^4 \delta^{(4)}(P_f - P_i) |T_{fi}|^2 Vt.$$

Dividing by t , we obtain the probability of transition per unit time

$$w_{i \rightarrow f} = (2\pi)^4 \delta^{(4)}(P_f - P_i) |T_{fi}|^2 V. \quad (64.5)$$

Each of the free particles (both initial and final) is described by its own wave function—a plane wave with some amplitude u (for an electron this is a bispinor, for a photon a 4-vector, and so on). The amplitude of scattering T_{fi} has the structure of a product

$$T_{fi} = u_1'^* u_2'^* \dots Q u_1 u_2 \dots, \quad (64.6)$$

where on the left stand the amplitudes of the wave functions of the final particles, on the right those of the initial particles; Q is a certain matrix (with respect to the indices of the components of the amplitudes of all particles).

Most important are the cases when in the initial state there are only one or two particles. In the first case it is a question of decay, in the second of the collision of two particles.

Let us consider first the decay into an arbitrary number of other particles with momenta $\mathbf{p}'_a$ in the element of momentum space $\prod d^3 p'_a$ (the index a labels the particles in the final state). The number of states, falling within this element (on the normalisation volume V ³), is

$$\prod_a \frac{V d^3 p'_a}{(2\pi)^3}.$$

To this quantity we should multiply expression (64.5):

$$dw = (2\pi)^4 \delta^{(4)}(P_f - P_i) |M_{fi}|^2 \frac{1}{2\varepsilon} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.11)$$

At the same time the wave functions of all particles, used in the computation of the matrix element, should be normalised to one particle in the volume V . Thus, for the electron the plane wave will be

$$\psi = u e^{-ipx}, \quad \bar{u}u = 2m, \quad (64.8)$$

and the photon wave

$$A = \sqrt{4\pi} e e^{-ikx}, \quad e e^* = -1, \quad ek = 0. \quad (64.9)$$

All these functions contain the factor $1/\sqrt{2\varepsilon V}$, where ε is the energy of the particle. However in the sequel it will be convenient to agree to write in all computations the wave

²This can also be shown otherwise: compute the integral at $P_f = P_i$, extending the integration in (64.4) to each of the coordinates over a large volume V and time interval t , then $Vt/(2\pi)^4$.

³For greater clarity of the computations in this paragraph we shall not set the normalisation volume equal to unity.

functions of the particles without these factors (which we include into the expression for the probability). Thus, the electron plane wave will be

$$\psi = ue^{-ipx}, \quad \bar{u}u = 2m,$$

and the scattering amplitude calculated with such wave functions we denote (in contrast to T_{fi}) by M_{fi} . Obviously,

$$T_{fi} = \frac{M_{fi}}{(2\varepsilon_1 V \cdot 2\varepsilon_2 V \dots)^{1/2}}; \quad (64.10)$$

in the denominator stands one multiplier $\sqrt{2\varepsilon V}$ for each initial or final particle.

In particular, for the probability of decay (64.11) we get a more definitive form by eliminating the δ -function for the case when the decay is into two particles (with momenta $\mathbf{p}'_1, \mathbf{p}'_2$ and energies $\varepsilon'_1, \varepsilon'_2$). In the rest system of the decaying particle $\mathbf{p}'_1 = -\mathbf{p}'_2 \equiv \mathbf{p}'$, so that we have

$$dw = \frac{1}{(2\pi)^2} |M_{fi}|^2 \frac{1}{2m} \frac{1}{4\varepsilon'_1 \varepsilon'_2} \delta(\mathbf{p}'_1 + \mathbf{p}'_2) \delta(\varepsilon'_1 + \varepsilon'_2 - m) d^3 p'_1 d^3 p'_2.$$

The first δ -function is eliminated by integration over $d^3 p'_2$; writing $d^3 p' \rightarrow \mathbf{p}'^2 d|\mathbf{p}'| d\phi$

$$d^3 p' = |\mathbf{p}'|^2 d|\mathbf{p}'| d\phi = |\mathbf{p}'| d\phi \frac{\varepsilon'_1 \varepsilon'_2 d(\varepsilon'_1 + \varepsilon'_2)}{\varepsilon'_1 + \varepsilon'_2}, \quad (64.12)$$

and eliminating the second δ -function, we get

$$dw = \frac{1}{32\pi^2 m^2} |M_{fi}|^2 |\mathbf{p}'| d\phi'. \quad (64.13)$$

Let us consider now the collision of two particles (with momenta $\mathbf{p}_1$ and $\mathbf{p}_2$ and energies ε_1 and ε_2) with transformation into an arbitrary collection of particles with momenta $\mathbf{p}'_a$. Instead of (64.11) we now get

$$dw = (2\pi)^4 \delta^{(4)}(P_f - P_i) |M_{fi}|^2 \frac{1}{4\varepsilon_1 \varepsilon_2 V} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.14)$$

(it is easy to verify, noting that $\varepsilon_1'^2 - m_1'^2 = \varepsilon_2'^2 - m_2'^2 = \mathbf{p}'^2$). Integration over $d(\varepsilon'_1 + \varepsilon'_2)$ eliminates the second δ -function, and we get

$$dw = \frac{1}{32\pi^2 m^2} |M_{fi}|^2 |\mathbf{p}'| d\phi'.$$

The interesting quantity in this case is, however, not the probability, but the cross section $d\sigma$. The invariant (with respect to Lorentz transformations) cross section is obtained from dw by dividing by the quantity

$$j = \frac{I}{V\varepsilon_1 \varepsilon_2}, \quad (64.14)$$

where I denotes the 4-scalar

$$I = \sqrt{(p_1 p_2)^2 - m_1^2 m_2^2} \quad (64.15)$$

(see II, § 12)⁴. In the centre of inertia system ($\mathbf{p}_1 = -\mathbf{p}_2 \equiv \mathbf{p}$)

$$I = |\mathbf{p}|(\varepsilon_1 + \varepsilon_2), \quad (64.16)$$

so that

$$j = \frac{|\mathbf{p}|}{V} \left(\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} \right) = \frac{v_1 + v_2}{V}, \quad (64.17)$$

which coincides with the usual definition of the flux density of the colliding particles (v_1, v_2 are their velocities)⁵. Thus, we find for the cross section the formula

$$d\sigma = (2\pi)^4 \delta^{(4)}(P_f - P_i) |M_{fi}|^2 \frac{1}{4I} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.18)$$

Let us bring this formula to its definitive form, eliminating the δ -function for the case when in the final state there are also just two particles. We shall consider the process in the centre of inertia system. Let $\varepsilon = \varepsilon_1 + \varepsilon_2 = \varepsilon'_1 + \varepsilon'_2$ be the total energy; $\mathbf{p}_1 = -\mathbf{p}_2 \equiv \mathbf{p}$ and $\mathbf{p}'_1 = -\mathbf{p}'_2 \equiv \mathbf{p}'$ the initial and the final momenta. The elimination of the δ -functions is carried out in the same way as for the decay (64.13), and we get

$$d\sigma = \frac{1}{64\pi^2} |M_{fi}|^2 \frac{|\mathbf{p}'|}{|\mathbf{p}| \varepsilon^2} d\sigma' \quad (64.19)$$

(in the particular case of elastic scattering, when the type of particles is unchanged in the collision, $|\mathbf{p}'| = |\mathbf{p}|$).

Let us rewrite this formula also in another form, introducing the invariant quantity

$$t \equiv (p_1 - p'_1)^2 = m_1^2 + m_1'^2 - 2(p_1 p'_1) = m_1^2 + m_1'^2 - 2\varepsilon_1 \varepsilon'_1 + 2|\mathbf{p}| |\mathbf{p}'| \cos \theta, \quad (64.20)$$

where θ is the angle between $\mathbf{p}_1$ and $\mathbf{p}'_1$. In the centre of inertia system the momenta $|\mathbf{p}_1| \equiv |\mathbf{p}|$ and $|\mathbf{p}'_1| \equiv |\mathbf{p}'|$ are determined by the total energy ε alone, and for given ε

$$dt = 2|\mathbf{p}| |\mathbf{p}'| d \cos \theta. \quad (64.21)$$

Therefore in (64.19) one can replace

$$d\sigma' = -d\varphi d \cos \theta = \frac{d\varphi d(-t)}{2|\mathbf{p}| |\mathbf{p}'|},$$

⁴For future reference we also write another expression for I in the form

$$I^2 = \frac{1}{4} [s - (m_1 + m_2)^2] [s - (m_1 - m_2)^2], \quad (64.15a)$$

where $s = (p_1 + p_2)^2$.

⁵In an arbitrary system of reference

$$j = (1/V) \sqrt{(\mathbf{v}_1 - \mathbf{v}_2)^2 - [\mathbf{v}_1 \mathbf{v}_2]^2}.$$

This expression reduces to the usual flux density in all cases when $\mathbf{v}_1 \parallel \mathbf{v}_2$: $j = |\mathbf{v}_1 - \mathbf{v}_2|/V$.

where (φ is the azimuth of $\mathbf{p}'_1$ relative to $\mathbf{p}_1$)⁶. Thus,

$$d\sigma = \frac{1}{64\pi} |M_{fi}|^2 \frac{dt d\varphi}{I^2 2\pi}, \quad (64.22)$$

(we have again introduced the invariant I from (64.16)). The azimuth φ , and with it the cross section in the form (64.22), are invariant with respect to Lorentz transformations not changing the direction of relative motion of the particles. If the cross section does not depend on the azimuth, formula (64.22) takes the especially simple form

$$d\sigma = \frac{1}{64\pi} |M_{fi}|^2 \frac{dt}{I^2}. \quad (64.23)$$

If one of the colliding particles is sufficiently heavy (and its state is unchanged in the collision), then its role

in the process reduces to that of a stationary source of a constant field, in which the other particle is scattered. In accordance with the fact that in a constant field energy (but not momentum!) of the system is conserved, in this treatment the elements of the S -matrix should be represented in the form

$$S_{fi} = i \cdot 2\pi\delta(E_f - E_i) T_{fi}. \quad (64.24)$$

In the expression for $|S_{fi}|^2$ the square of the one-dimensional δ -function should be understood as

$$[\delta(E_f - E_i)]^2 \rightarrow \frac{1}{2\pi} \delta(E_f - E_i) t.$$

Proceeding then (as in the derivation of (64.11)) to the amplitudes M_{fi} instead of T_{fi} , we obtain the following expression for the probability of the process in which one particle, scattered in a constant field, creates in the final state some number of other particles:

$$dw = 2\pi\delta(E_f - \varepsilon) |M_{fi}|^2 \frac{1}{2\varepsilon V} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.25)$$

Here again $\varepsilon (= E_i)$ is the energy of the initial particle, $\mathbf{p}'_a$ and ε'_a are the momenta and energies of the final particles. The cross section of scattering is obtained by dividing dw by the flux density $j = v/V$, where $v = |\mathbf{p}|/\varepsilon$ is the velocity of the scattered particle. As a result the normalisation volume drops out of the answer and we get

$$d\sigma = 2\pi\delta(E_f - \varepsilon) |M_{fi}|^2 \frac{1}{2|\mathbf{p}|} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.25)$$

In the particular case of elastic scattering there is also just one particle in the final state with the same (in magnitude) momentum and energy. Replacing $d^3 p' \rightarrow \mathbf{p}'^2 d|\mathbf{p}'| d\theta' = |\mathbf{p}'| \varepsilon' d\varepsilon' d\theta'$ and eliminating $\delta(\varepsilon' - \varepsilon)$ by integration over $d\varepsilon'$, we obtain the cross section in the form

$$d\sigma = \frac{1}{16\pi^2} |M_{fi}|^2 d\theta'. \quad (64.26)$$

⁶Since the correct sign of the differential is obvious in such cases, we shall below for simplicity write dt in place of $d(-t)$, etc.

Finally, if the external field depends on time (say, the field of a system of particles performing a given motion), then in the S -matrix $S_{fi} = iT_{fi}$ and, after going from T_{fi} to M_{fi} according to (64.10), the probability of the process in which the field creates a definite collection of particles will be given by the formula

$$dw = |M_{fi}|^2 \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.27)$$

§ 65. Reactions with polarised particles

In this paragraph we shall show on simple examples how the polarisation state of the particles participating in the reaction is taken into account in the computation of the cross section of scattering.

Let there be in the initial and in the final state one electron each. Then the amplitude of scattering has the form

$$M_{fi} = \bar{u}' Au (\equiv \bar{u}'_i A_{ik} u_k), \quad (65.1)$$

where u and u' are the bispinor amplitudes of the initial and the final electrons, A is a certain matrix (depending on the momenta and polarisations of the remaining particles participating in the reaction, if there are any).

The cross section is proportional to $|M_{fi}|^2$. We have

$$(\bar{u} Au)^* = u'^+ \gamma^{0*} A^* u^* = u^* A^+ \gamma^{0+} u',$$

or

$$(\bar{u}' Au)^* = \bar{u} \bar{A} u', \quad (65.2)$$

where ⁷

$$\bar{A} = \gamma^0 A^+ \gamma^0.$$

Thus,

$$|M_{fi}|^2 = (\bar{u}' Au)(\bar{u} \bar{A} u') \equiv \bar{u}'_i \bar{u}_k A_{kl} \bar{A}_{mi} u_l \bar{u}_m. \quad (65.3)$$

If the initial electron was in a mixed (partially polarised) state with the density matrix ρ and if we are interested in the cross section of the process with the formation of the final electron in a definite predetermined polarisation state ρ' , then we should replace the products of the bispinor components of the amplitudes

$$u'_i \bar{u}'_k \rightarrow \rho'_{ik}, \quad u_l \bar{u}_m \rightarrow \rho_{lm}.$$

Then

$$|M_{fi}|^2 = \text{Sp}(\rho' A \rho \bar{A}). \quad (65.4)$$

⁷In connection with the necessity to form the matrix $\bar{A}$ let us note for future reference the following easily verified equalities:

$$\bar{\gamma}^\mu = \gamma^\mu, \quad \overline{\gamma^\mu \gamma^\nu \dots \gamma^\rho} = \gamma^\rho \dots \gamma^\nu \gamma^\mu, \quad \bar{\gamma}^5 = -\gamma^5, \quad \overline{\gamma^5 \gamma^\mu} = \gamma^5 \gamma^\mu. \quad (65.2a)$$

The density matrices ρ and ρ' are given by formula (29.13)

$$\rho = \frac{1}{2}(\gamma p + m)[1 - \gamma^5(\gamma a)] \quad (65.5)$$

(and analogously for ρ').

If the initial electron is not polarised, then

$$\rho = \frac{1}{2}(\gamma p + m). \quad (65.6)$$

The substitution of this expression is equivalent to averaging over polarisations of the electron. If it is required to determine the cross section of scattering with arbitrary polarisation of the final electron, then one should also put $\rho' = (\gamma p' + m)/2$ and double the result; this operation is equivalent to summation over polarisations of the electron. Thus, we obtain

$$\frac{1}{2} \sum_{\text{polar}} |M_{fi}|^2 = \frac{1}{2} \text{Sp} \{(\gamma p' + m)A(\gamma p + m)\bar{A}\}, \quad (65.7)$$

where $\sum_{\text{polar}}$ denotes summation over the initial and final polarisations, and the factor $1/2$ converts one of the summations into an averaging.

The density matrix ρ' in (65.4) is an auxiliary concept, characterising, in essence, the properties of the detector (selecting this or that polarisation of the final electron), and not the process of scattering as such. There arises the question of the polarisation state of the electron that is produced by the process of scattering itself. If $\rho^{(f)}$ is the density matrix of this state, then the probability of detecting the electron in state ρ' is obtained by projecting $\rho^{(f)}$ onto ρ' , i.e. by forming the trace $\text{Sp}(\rho^{(f)}\rho')$. This quantity will also be proportional to the corresponding cross section, i.e. to $|M_{fi}|^2$. Comparing with (65.4), we conclude that

$$\rho^{(f)} \sim A\rho\bar{A}. \quad (65.8)$$

Since it is known in advance that $\rho^{(f)}$ must have the form (65.5) with some 4-vector $a^{(f)}$, the matter reduces to the determination of the latter. This could be done from formula (29.14), but it is even simpler to proceed as will be indicated below.

We saw in § 29 that the components of the 4-vector a are expressed through the components of the 3-vector ζ —the mean (doubled) value of the spin of the electron in its rest frame. The polarisation states of electrons are completely determined by these vectors, and it is natural to express through them also the cross section.

Obviously, the square $|M_{fi}|^2$ will be linear in each of the vectors ζ and ζ' , pertaining to the initial and final electrons. As a function of ζ' it will have the form

$$|M_{fi}|^2 = \alpha + \beta\zeta', \quad (65.9)$$

where α and β are themselves linear functions of ζ . The vector ζ' in (65.9) is the given polarisation of the final electron, selected by the detector. The vector $\zeta^{(f)}$, corresponding to the density matrix $\rho^{(f)}$, is easy to find as follows. According to what was said above

$$|M_{fi}|^2 \sim \text{Sp}(\rho'\rho^{(f)}).$$

By virtue of the relativistic invariance of this quantity it can be computed in any system of reference. In the rest frame of the final electron, according to (29.20)

$$\rho' \rho^{(f)} \sim (1 + \sigma \zeta')(1 + \sigma \zeta^{(f)}).$$

Therefore

$$|M_{fi}|^2 \sim 1 + \zeta' \zeta^{(f)}$$

and, comparing with (65.9), we find that

$$\zeta^{(f)} = \beta/\alpha. \quad (65.10)$$

Thus, having computed the cross section as a function of the parameter ζ' , we thereby also determine the polarisation $\zeta^{(f)}$.

In more complex cases (more than one electron both in the initial and final state) the computations are carried out according to the scheme described analogously.

Thus, if in the beginning and end there are two electrons each, the amplitude of scattering acquires the form

$$M_{fi} = (\bar{u}'_1 A u_1)(\bar{u}'_2 B u_2) + (\bar{u}'_2 C u_1)(\bar{u}'_1 D u_2),$$

where u_1, u_2 are the bispinor amplitudes of the initial, and u'_1, u'_2 those of the final electrons. In forming the square $|M_{fi}|^2$ there will appear terms of the type

$$|\bar{u}'_1 A u_1|^2 |\bar{u}'_2 B u_2|^2$$

and of the type

$$(\bar{u}'_1 A u_1)(\bar{u}'_2 B u_2)(\bar{u}'_2 C u_1)^*(\bar{u}'_1 D u_2)^*.$$

The first ones reduce to products of two traces of the type (65.4), and the second to traces of the type

$$\text{Sp}(\rho'_1 A \rho_1 C \rho'_2 B \rho_2 D).$$

Positrons are described by amplitudes of “negative frequency” $u(-p)$. For reactions with the participation of positrons the difference from what was said above reduces to the fact that as density matrices one should use expressions differing from (65.5), (65.6) only by a change of sign in front of m (cf. (29.16), (29.17)).

Let us now turn to the polarisation states of photons participating in the reaction.

The polarisation of each initial photon enters the amplitude of scattering linearly in the form of the 4-vector e , and of each final photon in the form e^* . In both cases in the cross section (i.e. in the square $|M_{fi}|^2$) the 4-tensor $e_\mu e_\nu^*$ enters. For the transition to the case of an arbitrary partially polarised state this tensor should

be replaced by the four-dimensional density matrix—a 4-tensor $\rho_{\mu\nu}$:

$$e_\mu e_\nu^* \rightarrow \rho_{\mu\nu}. \quad (65.11)$$

In particular, for an unpolarised photon, according to (8.15),

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \quad (65.12)$$

Thus, averaging over polarisations of a photon reduces to the tensor contraction over the corresponding two tensor indices $\mu\nu$ in $|M_{fi}^2|$ ⁸.

If it is required to perform not averaging, but a summation over polarisations of the photon, one should replace $e_\mu e_\nu^*$ by a quantity twice as large:

$$e_\mu e_\nu^* \rightarrow -g_{\mu\nu}. \quad (65.13)$$

The density matrix of a polarised photon is given by formula (8.17). The choice of the 4-vectors $e^{(1)}$, $e^{(2)}$ appearing in this expression is dictated usually by the specific conditions of the problem. In some cases these vectors may be connected with definite spatial directions in a given system of reference. In others it is more convenient to connect them with the characteristic 4-vectors of the problem—the 4-momenta of the particles.

In (8.17) the polarisation of the photon is described by the Stokes parameters, constituting the “vector” $\xi = (\xi_1, \xi_2, \xi_3)$. As for the electron, it is necessary to distinguish the polarisation $\xi^{(f)}$ of the final photon as such from the polarisation ξ' of it selected by the detector. If the square of the amplitude of scattering is known as a function of the parameter ξ' :

$$|M_{fi}|^2 = \alpha + \beta\xi', \quad (\text{VII.1})$$

then the polarisation $\xi^{(f)} = \beta/\alpha$, analogously to formula (65.10).

§ 66. Kinematic invariants

Let us consider some kinematic relations for scattering processes, in which both in the initial and in the final states there are just two particles each. We have in mind relations which are consequences of the general conservation laws alone and are therefore valid irrespective of the nature of the particles and of the laws of their interaction.

Let us write the law of conservation of 4-momentum in a general form, not specifying which of the momenta refer to the initial particles and which to the final ones:

$$q_1 + q_2 + q_3 + q_4 = 0. \quad (66.1)$$

Here $\pm q_a$ are the 4-vectors of momenta, two of them corresponding to the incoming particles, and two to the scattered ones; for the latter the momenta are $-q_a$. In other words, two of the q_a have temporal component $q_a^0 > 0$, and two have $q_a^0 < 0$.

Alongside the law of conservation of 4-momentum, the law of conservation of charge should also be obeyed. By charge one should understand here not only the electric charge, but also other conserved quantities, having different sign for particles and antiparticles.

From the given species of particles participating in the process, the squares of the 4-vectors q_a are the given squared masses of the particles ($q_a^2 = m_a^2$). Depending on the values of the time components q_a^0 run through, and on the values of the charges, we

⁸Expression (65.12) as it were reduces averaging over two really possible polarisations of the photon to averaging over four independent directions of the 4-vector e .

obtain three different reactions. Let us write these three processes as:

$$\begin{aligned} \text{I.} \quad & 1 + 2 \rightarrow 3 + 4, \\ \text{II.} \quad & 1 + \bar{3} \rightarrow \bar{2} + 4, \\ \text{III.} \quad & 1 + \bar{4} \rightarrow \bar{2} + 3. \end{aligned} \tag{66.2}$$

Here the digit indicates the number of the particle, and a bar over it indicates the antiparticle. The transition from one of the reactions to another, i.e. the transfer of a particle from one side of the formula to the other, corresponds to a change of sign of the temporal component q_a^0 , and also of the sign of the charge, i.e. the replacement of the particle by the antiparticle. (Alongside the processes (66.2), the inverse reactions are possible, of course.)

The three processes (66.2) are referred to as three *crossed* (or *cross-channels*) of one (generalised) reaction.

Let us give several examples. If particles 1 and 3 are electrons, and 2 and 4 are photons, then channel I represents scattering of a photon by an electron; by virtue of the true neutrality of the photon channel III represents the same as I. Channel II is the conversion of an electron-positron pair into two photons. If all four particles are electrons, then channel I is electron-electron scattering, and channels II and III are electron-positron scattering. If particles 1 and 3 are electrons, and 2 and 4 are muons, then channel I is $e\mu$ scattering, channel III is $e\bar{\mu}$ scattering, and channel II is the conversion of the pair $e\bar{e}$ into a pair $\mu\bar{\mu}$.

In the study of scattering processes an especially important role is played by invariant quantities, which can be composed from the 4-momenta. They are functions of the invariant amplitudes of scattering (see § 70).

From the four 4-momenta one can compose two independent invariant quantities. Indeed, by virtue of (66.1) only three of the four 4-vectors q_a are independent; let these be q_1, q_2, q_3 . From them one can compose six invariants: three squares q_1^2, q_2^2, q_3^2 and three products q_1q_2, q_1q_3, q_2q_3 . But the first three are the given squared masses, and the latter three are connected by one relation, following from the equality

$$(q_1 + q_2 + q_3)^2 = q_4^2 = m_4^2.$$

For the sake of greater symmetry it is convenient, however, to consider not two but three invariants, which we choose as follows⁹:

$$\begin{aligned} s &= (q_1 + q_2)^2 = (q_3 + q_4)^2, \\ t &= (q_1 + q_3)^2 = (q_2 + q_4)^2, \\ u &= (q_1 + q_4)^2 = (q_2 + q_3)^2. \end{aligned} \tag{66.3}$$

⁹In the general case, when the reaction involves $n \geq 4$ particles, the number of functionally independent invariant variables is $3n - 10$. Indeed, there are in all $4n$ variables—components of n 4-momenta q_a . Among them there are n functional relations $q_a^2 = m_a^2$, and four more given by the law of conservation $\sum q_a = 0$. Arbitrary values can also be given to six parameters, determining the general Lorentz transformation (a general four-dimensional rotation). Therefore the number of independent invariant variables is: $4n - n - 4 - 6 = 3n - 10$.

They are connected, as is easy to see, by the relation

$$s + t + u = h, \quad (66.4)$$

where

$$h = m_1^2 + m_2^2 + m_3^2 + m_4^2. \quad (66.5)$$

In channel I the invariant s has a simple physical meaning. It is the square of the total energy of the colliding particles (1 and 2) in the system of their centre of inertia (with $\mathbf{p}_1 + \mathbf{p}_2 = 0$: $s = (\varepsilon_1 + \varepsilon_2)^2$). In channel II an analogous role is played by the invariant t , and in channel III the invariant u . In connection with this, channels I, II, III are often called the s -, t -, and u -channels.

It is not difficult to express the invariants s, t, u through the energies and momenta of the colliding particles in each of the channels. Let us consider the s -channel. In the centre of inertia system of particles 1 and 2 the temporal and spatial components of the 4-vectors q_a are given as follows:

$$\begin{aligned} q_1 &= p_1 = (\varepsilon_1, \mathbf{p}_s), & q_2 &= p_2 = (\varepsilon_2, -\mathbf{p}_s), \\ q_3 &= -p_3 = (-\varepsilon_3, -\mathbf{p}'_s) & q_4 &= -p_4 = (-\varepsilon_4, \mathbf{p}'_s) \end{aligned} \quad (66.6)$$

(the index s on $\mathbf{p}_s, \mathbf{p}'_s$ indicates that these momenta refer to the reaction in the s -channel). Then

$$s = \varepsilon_s^2, \quad \varepsilon_s = \varepsilon_1 + \varepsilon_2 = \varepsilon_3 + \varepsilon_4; \quad (66.7)$$

$$4\mathbf{p}_s^2 = [s - (m_1 + m_2)^2][s - (m_1 - m_2)^2], \quad (66.8)$$

$$4\mathbf{p}_s'^2 = [s - (m_3 + m_4)^2][s - (m_3 - m_4)^2]; \quad (\text{VII.2})$$

$$2t = h - s + 4\mathbf{p}_s \mathbf{p}'_s - \frac{1}{s}(m_1^2 - m_2^2)(m_3^2 - m_4^2), \quad (66.9)$$

$$2u = h - s - 4\mathbf{p}_s \mathbf{p}'_s + \frac{1}{s}(m_1^2 - m_2^2)(m_3^2 - m_4^2).$$

In the case of elastic scattering ($m_1 = m_3, m_2 = m_4$) we have $|\mathbf{p}_s| = |\mathbf{p}'_s|$, so that $\varepsilon_1 = \varepsilon_3, \varepsilon_2 = \varepsilon_4$. Instead of (66.9) the simpler formulae then hold:

$$\begin{aligned} t &= -(\mathbf{p}_s - \mathbf{p}'_s)^2 = -2\mathbf{p}_s^2(1 - \cos \theta_s), \\ u &= -2\mathbf{p}_s^2(1 + \cos \theta_s) + (\varepsilon_1 - \varepsilon_2)^2, \end{aligned} \quad (66.10)$$

where θ_s is the angle between $\mathbf{p}_s$ and $\mathbf{p}'_s$. Let us note that the invariant $-t$ represents the square of the (three-dimensional) momentum transferred in the collision.

Analogous formulae for other channels are obtained by a simple change of notation. For the transition to the t -channel one should make the substitution $s \leftrightarrow t, 2 \leftrightarrow 3$; for the transition to the u -channel the substitution $s \leftrightarrow u, 2 \leftrightarrow 4$.

§ 67. Physical regions

Considering the amplitudes of scattering as functions of the independent variables s, t, u (connected by the relation $s + t + u = h$), we encounter the necessity of distinguishing the physically admissible and inadmissible regions of their values. Values which can correspond to a physical scattering process must satisfy certain conditions which follow from the law of conservation of 4-momentum and from the fact that the square of each of the 4-vectors q_a is a given quantity $q_a^2 = m_a^2$.

The product of two 4-momenta

$$p_a p_b \geq m_a m_b. \quad (67.1)$$

Therefore $(q_a + q_b)^2 = (p_a + p_b)^2 \geq (m_a + m_b)^2$,

if $q_a = p_a, q_b = p_b$ (or $q_a = -p_a, q_b = -p_b$),

or $(q_a + q_b)^2 = (p_a - p_b)^2 \leq (m_a - m_b)^2$,

if $q_a = p_a, q_b = -p_b$.

From here it follows that for the reaction in the s -channel:

$$\begin{aligned} (m_1 + m_2)^2 &\leq s \leq (m_3 + m_4)^2, \\ (m_1 - m_3)^2 &\geq t \leq (m_2 - m_4)^2, \\ (m_1 - m_4)^2 &\geq u \leq (m_2 - m_3)^2 \end{aligned} \quad (67.2)$$

(analogous inequalities hold in the t - and u -channels).

To find the remaining conditions, let us form the 4-vector L , dual to the product of any three of the 4-vectors q_a , say

$$L_\lambda = e_{\lambda\mu\nu\rho} q_1^\mu q_2^\nu q_3^\rho. \quad (67.3)$$

In the rest frame of one of the particles (for example, particle 1) $q_1 = (q_1^0, 0)$. Here L has only spatial components: $L_i = e_{i0kl} q_1^k q_2^l q_3^0$. In other words, L is a spacelike vector, and in every system of reference $L^2 \leq 0$. Expanding the square L^2 , we obtain the condition

$$\begin{vmatrix} q_1^2 & q_1 q_2 & q_1 q_3 \\ q_2 q_1 & q_2^2 & q_2 q_3 \\ q_3 q_1 & q_3 q_2 & q_3^2 \end{vmatrix} \geq 0. \quad (67.4)$$

It can be expressed through the invariants s, t, u and in a unified form for all channels in the form

$$stu \geq as + bt + cu, \quad (67.5)$$

where

$$\begin{aligned} ah &= (m_1^2 m_2^2 - m_3^2 m_4^2)(m_1^2 + m_2^2 - m_3^2 - m_4^2), \\ bh &= (m_1^2 m_3^2 - m_2^2 m_4^2)(m_1^2 + m_3^2 - m_2^2 - m_4^2), \\ ch &= (m_1^2 m_4^2 - m_2^2 m_3^2)(m_1^2 + m_4^2 - m_2^2 - m_3^2) \end{aligned} \quad (67.6)$$

(T. W. B. Kibble, 1960).

For a graphical representation of the regions of variation of the variables s, t, u it is convenient to use the so-called triangular coordinates in the plane (the Mandelstam plane; *S. Mandelstam*, 1958). The coordinate axes in it are three straight lines, forming at their intersection an equilateral triangle. The coordinates s, t, u are measured along directions perpendicular to these three lines. (We take the positive directions to point inside the triangle, as indicated in Fig. 5 by arrows.) In other words, to every point of the plane there correspond values of s, t, u , represented (with appropriate signs) by the lengths of the perpendiculars dropped onto the three axes. The condition $s + t + u = h$ is ensured by a well-known geometrical theorem (if the height of the equilateral triangle is h)¹⁰.

Let us consider an important case when the basic (s) channel corresponds to elastic scattering; the masses of the particles are pairwise identical:

$$m_1 = m_3 \equiv m, \quad m_2 = m_4 \equiv \mu. \quad (67.7)$$

Let $m > \mu$. From the condition (67.5) we have

$$h = 2(m^2 + \mu^2), \quad a = c = 0, \quad b = (m^2 - \mu^2)^2,$$

so that

$$sut \geq (m^2 - \mu^2)^2 t. \quad (67.8)$$

The boundary of the region, determined by this inequality, consists of the line $t = 0$ and of the hyperbola

$$su = (m^2 - \mu^2)^2, \quad (67.9)$$

two branches of which lie in the sectors $u < 0, s < 0$ and $s > 0, u > 0$; the axes $s = 0$ and $u = 0$ are asymptotes of the hyperbola. Instead of (67.8) one can write

$$t > 0, \quad su > (m^2 - \mu^2)^2$$

or

$$t < 0, \quad su < (m^2 - \mu^2)^2.$$

Besides the conditions (67.2) one must additionally take into account the inequality $s > (m + \mu)^2$ in the s -channel and $u > (m + \mu)^2$ in the u -channel; the remaining inequalities are satisfied after this automatically. As a result we find that channels I, II, III (s, t, u) correspond to the *physical regions*, indicated by hatching in Fig. 6.

If $m = \mu$, then the boundaries of the region (67.8) degenerate into the coordinate axes and the physical regions are the three sectors shown in Fig. 8.

In the general case of four different masses the equation

$$stu = as + bt + cu \quad (67.10)$$

determines a curve of third order, whose branches delimit the physical regions of three channels, as shown in Fig. 9.

Let

$$m_1 \geq m_2 \geq m_3 \geq m_4.$$

¹⁰Connecting, for example, the point P (Fig. 5) with the three vertices of the triangle ABC , we decompose it into three triangles with heights s, t, u ; equating the sum of the areas of these triangles to the area of the triangle ABC , we find the required equality. Analogously it is proved also in the case when the point P lies outside the triangle ABC .

Then

$$a \geq b \geq c, \quad a > 0, \quad b > 0.$$

The curve (67.10) intersects the coordinate axes in points lying on the line

$$as + bt + cu = 0$$

(see the dashed lines in Fig. 9). Depending on the sign of c it passes as shown in Fig. 9. For $c < 0$ the physical region of the

u -channel covers part of the area of the coordinate triangle; in other words, in this case the quantities s, t, u can be simultaneously positive. All three branches of the boundary curve have as asymptotes the corresponding coordinate axes (in this it is easy to verify by eliminating from the equation (67.10) one of the variables with the help of the relation $s + t + u = h$ and sending one of the remaining variables to infinity). The conditions (67.2) introduce nothing essentially new compared with the boundaries established by the equation (67.10). The straight lines, corresponding to the signs of equality in (67.2), intersect the hatched physical regions in Fig. 9; some of them are tangent to the boundaries of these regions, corresponding to the extremal values of the variables s, t , or u in the corresponding channel.

In the case when the mass of one of the particles is larger than the sum of the three remaining masses ($m_1 > m_2 + m_3 + m_4$), alongside the channels I, II, III there is possible also a fourth channel of the reaction, corresponding to the decay:

$$\text{IV. } 1 \rightarrow \bar{2} + 3 + 4, \quad (67.11)$$

For this channel in the rest system of the decaying particle

$$q_1 = (m_1, 0), \quad q_2 = (-\varepsilon_2, -\mathbf{p}_2), \quad q_3 = (-\varepsilon_3, -\mathbf{p}_3), \quad q_4 = (-\varepsilon_4, -\mathbf{p}_4),$$

$$\varepsilon_2 + \varepsilon_3 + \varepsilon_4 = m_1, \quad \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 = 0.$$

The invariants:

$$\begin{aligned} s &= m_1^2 + m_2^2 - 2m_1\varepsilon_2, \\ t &= m_1^2 + m_3^2 - 2m_1\varepsilon_3, \\ u &= m_1^2 + m_4^2 - 2m_1\varepsilon_4. \end{aligned} \quad (67.12)$$

From (67.1) we now get:

$$\begin{aligned} (m_3 + m_4)^2 &\leq s \leq (m_1 - m_2)^2, \\ (m_2 + m_4)^2 &\leq t \leq (m_1 - m_3)^2, \\ (m_2 + m_3)^2 &\leq u \leq (m_1 - m_4)^2. \end{aligned} \quad (67.13)$$

Thus all three invariants are positive, i.e. the physical region of the decay channel is located inside the coordinate triangle.

Problems

1. Find the physical regions in the case of three identical masses: $m_1 \equiv m, m_2 = m_3 = m_4 \equiv \mu$ (for example, reaction $K + \pi \rightarrow \pi + \pi$).

Solution. The equation (67.10) takes the form

$$stu = \mu^2(m^2 - \mu^2)^2, \quad (1)$$

with $s + t + u = 3\mu^2 + m^2$.

The physical regions are bounded by identical in form curves (for I: $s > 0, t < 0, u < 0$, and analogously for II and III). If $m > 3\mu$, then (1) also has a branch (closed curve) with $s > 0, t > 0, u > 0$, bounding the region of channel IV (Fig. 10).

2. The same in the case $m_1 \equiv m, m_2 = \mu, m_3 = m_4 = 0, m > \mu$ (for example, reaction $\mu + \nu \rightarrow e + \nu$).

Solution. The condition (67.5) takes the form $stu \geq m^2\mu^2s$, with $s + t + u = m^2 + \mu^2$. The physical regions are bounded by the axes $s = 0$ and by two branches of the hyperbola $tu = m^2\mu^2$ (Fig. 11).

3. The same in the case $m_1 = m_3 \equiv m, m_2 = 0, m_4 \equiv \mu$, with $m > 2\mu$ (for example, reaction $p + \gamma \rightarrow p + \pi^0$).

Solution. The boundary equation (67.10) takes the form

$$\begin{aligned} stu &= a(s + u) + bt, & ah &= m^2\mu^4, \\ bh &= m^4(2m^2 - \mu^2), & h &= 2m^2 + \mu^2. \end{aligned}$$

Eliminating u , we obtain

$$t^2 + \left(\frac{b-a}{s} + s - h \right) t + \frac{ah}{s} = 0.$$

For a given s this is a quadratic equation for t . For $s > (m + \mu)^2$ (region of the s -channel) to each value of s there correspond two negative values of t . At $s = (m + \mu)^2$ the two roots of the quadratic equation merge into one: $t = -m\mu^2/(m + \mu)$. The boundary of the region of the s -channel has the form shown in Fig. 12. The lower branch of the boundary curve asymptotically approaches the axis $u = 0$, and the upper branch intersects this axis at the point $t = \mu^4/(\mu^2 - m^2)$.

The region of the u -channel is symmetric to the region of the s -channel, and the region of the t -channel is located as shown in the figure.

§ 68. Partial-wave expansion

An essential step in the analysis of a reaction of the type

$$a + b \rightarrow c + d \quad (68.1)$$

is the expansion of the amplitude of scattering into partial amplitudes, each of which corresponds (at a given total energy ε) to a definite value of the total angular momentum J of the particles in the system of their centre of inertia¹¹.

¹¹The greater part of the results presented in §§ 68, 69 is due to Jacob and Wick (M. Jacob, G. C. Wick, 1959).

These partial amplitudes are, in other words, elements of the S -matrix in the angular-momentum representation:

$$\langle \varepsilon J' M' | S | \varepsilon J M \rangle. \quad (\text{VII.3})$$

Since the total angular momentum J and its projection M on a given axis z are conserved, the S -matrix is diagonal in these numbers (as in the energy ε). By virtue of the isotropy of space the diagonal elements do not depend on the value of M . For given J, M, ε the scattering matrix remains a matrix with respect to the spin quantum numbers; we shall write the elements of this matrix in a more abbreviated form as

$$\langle \varepsilon J M \lambda' | S | \varepsilon J M \lambda \rangle \equiv \langle \lambda' | S^J(\varepsilon) | \lambda \rangle, \quad (68.2)$$

where λ and λ' are the sets of spin quantum numbers. As the latter the most natural choice here is the helicities of the particles. Recall that helicity (in contrast to the projection of spin onto an arbitrary axis in space) is conserved for a free particle, and also that it commutes both with the momentum and with the angular momentum of the particle (see § 16). Therefore helicities can be used both in the momentum and in the angular-momentum representations.

The elements of the S -matrix with respect to the helicity indices we shall call the *helicity amplitudes* of scattering; thus, by λ and λ' we shall mean the sets of helicities of the initial and final particles: $\lambda = (\lambda_a, \lambda_b)$, $\lambda' = (\lambda_c, \lambda_d)$.

In the momentum representation the elements of the scattering matrix are defined with respect to the states $|\varepsilon \mathbf{n} \lambda\rangle$ ($\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ is the direction of the momentum of relative motion in the system of the centre of inertia), and in the angular-momentum representation with respect to the states $|\varepsilon J M \lambda\rangle$. They are expressed through one another in the form of expansions

$$|J M \lambda\rangle = \int |\mathbf{n} \lambda\rangle \langle \mathbf{n} | J M \lambda \rangle d\mathbf{o}_{\mathbf{n}}, \quad (68.3)$$

where the integration is over the directions $\mathbf{n}$ (the energy ε in the symbols of states we shall for brevity omit). By virtue of the unitarity of this transformation (see III, § 12) the coefficients of the inverse transformation

$$\langle J M \lambda | \mathbf{n} \lambda \rangle = \langle \mathbf{n} \lambda | J M \lambda \rangle^*. \quad (68.4)$$

By the general rule of transformation of matrices these coefficients also determine the connection between elements of the S -matrix in both representations:

$$\langle \mathbf{n}' \lambda' | S | \mathbf{n} \lambda \rangle = \sum_{J M} \langle \mathbf{n}' \lambda' | J M \lambda' \rangle \langle J M \lambda' | S | J M \lambda \rangle \langle J M \lambda | \mathbf{n} \lambda \rangle. \quad (68.5)$$

The coefficients of the expansion (68.3) are easily found with the help of the results of § 16. Let the wave functions of all states be expressed in the momentum representation, i.e. as functions of the direction of the momentum (at given energy); this direction as an independent variable we denote by $\boldsymbol{\nu}$, in contrast to the direction $\mathbf{n}$ as a quantum number of the state. In this representation the wave function has the form (16.2)

$$\psi_{\mathbf{n} \lambda}(\boldsymbol{\nu}) = u^{(\lambda)} \delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}). \quad (68.6)$$

On substituting (68.6) into the expansion (68.3) the latter reduces to a single term:

$$\psi_{JM\lambda} = \langle \mathbf{v}\lambda | JM\lambda \rangle u^{(\lambda)}. \quad (68.7)$$

The helicity λ_a of each of the two particles is defined as the projection of its spin onto the direction of *its* momentum. If the momenta of the particles are $\mathbf{p}_a \equiv \mathbf{p}$, $\mathbf{p}_b = -\mathbf{p}$, then for the first particle this is the direction $\mathbf{n}$, and for the second the direction $-\mathbf{n}$. If we now regard the system as one particle with helicity Λ in the direction $\mathbf{n}$, then $\Lambda = \lambda_a - \lambda_b$. In this representation the wave function (in the momentum representation) can be represented according to (16.4) in the form

$$\psi_{JM\lambda} = u^{(\lambda)} D_{\Lambda M}^{(J)}(\mathbf{v}) \sqrt{\frac{2J+1}{4\pi}}. \quad (68.8)$$

Comparing the expressions (68.7), (68.8) (and changing the variable designation $\mathbf{v}$ to $\mathbf{n}$), we obtain for the desired coefficients

$$\langle \mathbf{n}\lambda | JM\lambda \rangle = \sqrt{\frac{2J+1}{4\pi}} D_{\Lambda M}^{(J)}(\mathbf{n}). \quad (68.9)$$

Substitution of these coefficients into (68.5) gives

$$\langle \mathbf{n}'\lambda' | S | \mathbf{n}\lambda \rangle = \sum_{JM} \frac{2J+1}{4\pi} D_{\Lambda' M}^{(J)*}(\mathbf{n}') D_{\Lambda M}^{(J)}(\mathbf{n}) \langle \lambda' | S^J | \lambda \rangle, \quad (68.10)$$

where the abbreviated notation (68.2) has been used. Let us choose the direction $\mathbf{n}$ as the z -axis; then

$$D_{\Lambda M}^{(J)}(\mathbf{n}) = \delta_{\Lambda M}$$

and (68.10) takes the form

$$\langle \mathbf{n}'\lambda' | S | \mathbf{n}\lambda \rangle = \sum_J \frac{2J+1}{4\pi} D_{\Lambda'\Lambda}^{(J)*}(\mathbf{n}') \langle \lambda' | S^J | \lambda \rangle. \quad (68.11)$$

We see that the expansion in partial amplitudes is effected with the functions $D_{\Lambda'\Lambda}^{(J)}$ as coefficients. For the reaction of the type (68.1) it is convenient to define the amplitude of scattering f in such a way that the cross section (in the centre of inertia system) would be

$$d\sigma = |\langle \mathbf{n}'\lambda' | f | \mathbf{n}\lambda \rangle|^2 d\Omega' \quad (68.12)$$

(comparing with (64.19) one can relate this amplitude to the matrix element M_{fi}). Its expansion in partial amplitudes we write in the form

$$\langle \mathbf{n}'\lambda' | f | \mathbf{n}\lambda \rangle = \sum_{JM} (2J+1) D_{\Lambda'\Lambda}^{(J)*}(\mathbf{n}') D_{\Lambda M}^{(J)}(\mathbf{n}) \langle \lambda' | f^J | \lambda \rangle, \quad (68.13)$$

or, choosing the z -axis along the direction $\mathbf{n}$:

$$\langle \mathbf{n}'\lambda' | f | \mathbf{n}\lambda \rangle = \sum_J (2J+1) D_{\Lambda'\Lambda}^{(J)*}(\mathbf{n}') \langle \lambda' | f^J | \lambda \rangle. \quad (68.14)$$

This formula represents a generalisation of the ordinary expansion in partial amplitudes for scattering of spinless particles (see III, (123.14)). Since $D_{00}^{(L)} = P_L(\cos \theta)$, for zero spins (68.14) reduces to the expansion in Legendre polynomials

$$f(\theta) = \sum_L (2L+1) f_L P_L(\cos \theta).$$

The cross section (68.12) refers to the case when all particles have definite helicities. If the particles are in mixed polarisation states, then the cross section is obtained by averaging the products

$$\langle \lambda_c \lambda_d | f | \lambda_a \lambda_b \rangle \langle \lambda'_c \lambda'_d | f | \lambda'_a \lambda'_b \rangle^*$$

over the polarisation density matrices of the particles

(see the footnote on p. 204). Thus, for the reaction between unpolarised particles a , b with formation of unpolarised particles c , d we obtain

$$d\sigma = \frac{do}{(2s_a+1)(2s_b+1)} \sum_{\{\lambda\}} \sum_{JJ'} (2J)(2J'+1) \langle \lambda_c \lambda_d | f^J | \lambda_a \lambda_b \rangle \times \quad (68.15)$$

$$\times \langle \lambda_c \lambda_d | f^{J'} | \lambda_a \lambda_b \rangle^* D_{\Lambda' \Lambda}^{(J)}(\mathbf{n}') D_{\Lambda' \Lambda}^{(J')*}(\mathbf{n}')$$

(the axis z is directed along $\mathbf{n}$, the sign $\sum_{\{\lambda\}}$ denotes summation over $\lambda_a \lambda_b \lambda_c \lambda_d$).

Replacing the function $D_{\Lambda' \Lambda}^{(J')*}$ according to formula (58.19) (see III) and then using the expansion (110.2) (see III), we obtain finally

$$d\sigma = \frac{do}{(2s_a+1)(2s_b+1)} \sum_{\{\lambda\}JJ'} (-1)^{\Lambda-\Lambda'} (2J+1)(2J'+1) \times \quad (68.16)$$

$$\times \langle \lambda_c \lambda_d | f^J | \lambda_a \lambda_b \rangle \langle \lambda_c \lambda_d | f^{J'} | \lambda_a \lambda_b \rangle^* \sum_L (2L+1) \times$$

$$\times \begin{pmatrix} J & J' & L \\ \Lambda & -\Lambda & 0 \end{pmatrix} \begin{pmatrix} J & J' & L \\ \Lambda' & -\Lambda' & 0 \end{pmatrix} P_L(\cos \theta)$$

(θ is the angle between $\mathbf{n}'$ and the axis z); summation over L is over all integer values arising in the vector addition of $\mathbf{J}$ and $\mathbf{J}'$.

The expansion of the scattering amplitude in partial amplitudes fully takes into account all the properties of the angular distribution of scattering, associated with the symmetry with respect to spatial rotations. It does not, however, explicitly take into account the properties associated with the symmetry with respect to spatial inversion. P -invariance (if the interaction possesses this symmetry) leads to the appearance of definite relations between different helicity amplitudes (see below, § 69).

§ 69. Symmetry of helicity amplitudes

The requirements imposed by symmetry with respect to the transformations P , C , T (if, of course, the given process of particle interaction possesses this symmetry), lead to the appearance of definite relations between different helicity amplitudes and thereby reduce the number of independent amplitudes¹².

¹²The number of independent amplitudes itself does not depend, of course, on the concrete form of the representation of the matrix S^J and remains the same for any choice of spin variables.

To establish these relations we shall first clarify the symmetry properties of the helicity states of two particles. Let us consider the particles in the system of their centre of inertia. One has momentum $\mathbf{p}_1 \equiv \mathbf{p}$ and helicity λ_1 relative to the direction $\mathbf{p}$, and the other has momentum $\mathbf{p}_2 = -\mathbf{p}$ and helicity λ_2 relative to the direction $-\mathbf{p}$. If we define the helicities of both particles relative to one and the same direction $\mathbf{p}$, they will be λ_1 and $-\lambda_2$. Correspondingly they will be described by plane waves with amplitudes $u_{\mathbf{p}}^{(\lambda_1)}$ and $u_{\mathbf{p}}^{(-\lambda_2)}$. The system of both particles is described by a function (multicomponent) $u_{\mathbf{p}}^{(\lambda_1, \lambda_2)}$, composed of products of $u_{\mathbf{p}}^{(\lambda_1)}$ and $u_{\mathbf{p}}^{(-\lambda_2)}$.

Regarding now the system as one particle with helicity Λ in the direction $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$, we may write the wave function (in the momentum representation), i.e. as a function of the direction $\mathbf{n}$ for states with definite values of $J, M, \lambda_1, \lambda_2$ (and also the total energy ε):

$$\psi_{JM\lambda_1\lambda_2} = u_{\mathbf{p}}^{(\lambda_1, \lambda_2)} D_{\Lambda M}^{(J)}(\mathbf{n}) \sqrt{\frac{2J+1}{4\pi}}, \quad \Lambda = \lambda_1 - \lambda_2 \quad (69.1)$$

(cf. (68.8)). Since Λ is the projection of the total angular momentum on $\mathbf{p}$, it must be

$$|\Lambda| \leq J. \quad (69.2)$$

According to (16.14) under inversion

$$\hat{P} u^{(\lambda_1, \lambda_2)}(\mathbf{n}) = \eta_1 \eta_2 (-1)^{s_1 + s_2 - \lambda_1 + \lambda_2} u^{(-\lambda_1, -\lambda_2)}(\mathbf{n}), \quad (69.3)$$

where η_1, η_2 are the intrinsic parities of the particles. Using also (16.10), we find the transformation law of the functions (69.1):

$$\hat{P} \psi_{JM\lambda_1\lambda_2} = \eta_1 \eta_2 (-1)^{s_1 + s_2 - J} \psi_{JM\lambda_2\lambda_1}, \quad (69.4)$$

where $\Lambda = \lambda_1 - \lambda_2$.

For identical particles only symmetric (for bosons) or only antisymmetric (for fermions) states with respect to permutation are admissible. Since the first case occurs for integer and the second for half-integer spin of the particle s , in both cases the helicity states of two identical particles can be written in the form of linear combinations

$$[1 + (-1)^{2s} \hat{P}_{12}] \psi_{JM\lambda_1\lambda_2},$$

or, according to (69.5)

$$\psi_{JM\lambda_1\lambda_2} + (-1)^J \psi_{JM\lambda_2\lambda_1}. \quad (69.6)$$

It is noteworthy that this combination has a single form for both bosons and fermions.

For a system of particle and antiparticle the result of the permutation is expressed by the same formula (69.5). However, unlike the case of identical particles, here both permutational symmetries are admissible, i.e. both combinations

$$\psi^{\pm} = \psi_{JM\lambda_1\lambda_2} \pm (-1)^J \psi_{JM\lambda_2\lambda_1}. \quad (69.7)$$

These states possess definite charge parities. The charge conjugation operation C can be represented as the result of a full permutation of all variables of the two particles

(spinor and charge), with a subsequent inverse permutation of spin variables (helicities). The result of the first operation must coincide with the result of the permutation in the system of two identical particles. From this it is clear that with the upper sign in (69.7) (coinciding with the permutational sign) the system will be charge-even, and with the lower charge-odd:

$$\hat{C}\psi^\pm = \pm\psi^\pm.$$

Finally, let us consider the time-reversal operation. The wave function of a particle at rest with spin s and its projection σ is transformed according to

$$\hat{T}\psi_{s\sigma} = (-1)^{s-\sigma} \psi_{s,-\sigma}$$

(see III, (60.2)). The wave function of two particles in the centre of inertia system can also be treated (with respect to its transformation properties) as the wave function of a “particle at rest” with angular momentum J and its projection M . As for the helicities λ_1 , λ_2 , they do not change: time reversal changes the signs of the vectors of momentum and angular momentum, and the product **jp** therefore does not change. Thus,

$$\hat{T}\psi_{JM\lambda_1\lambda_2} = (-1)^{J-M} \psi_{JM\lambda_1\lambda_2}. \quad (69.8)$$

Now we can directly write down the symmetry relations for the helicity amplitudes. If the interaction is P -invariant, then for the reaction

$$a + b \rightarrow c + d$$

the amplitudes of the transitions

$$|\lambda_a\lambda_b\rangle \rightarrow |\lambda_c\lambda_d\rangle \quad \text{and} \quad \hat{P}|\lambda_a\lambda_b\rangle \rightarrow \hat{P}|\lambda_c\lambda_d\rangle$$

should coincide (at given J and ε).

Using (69.4), we find therefore

$$\langle \lambda_c\lambda_d | S^J | \lambda_a\lambda_b \rangle = \frac{\eta_1\eta_c}{\eta_a\eta_b} (-1)^{s_a-s_b-s_c-s_d} \langle -\lambda_c, -\lambda_d | S^J | -\lambda_a, -\lambda_b \rangle. \quad (69.9)$$

If, instead of states with definite helicities, we choose states with definite parities, i.e. combinations

$$\frac{1}{\sqrt{2}} (\psi_{JM\lambda_1\lambda_2} \pm \hat{P}\psi_{JM\lambda_1\lambda_2})$$

(where $\lambda_1\lambda_2 = \lambda_a\lambda_b$ or $\lambda_c\lambda_d$), then the amplitudes of transitions not conserving parity vanish.

Time reversal interchanges initial and final states. Therefore T -invariance leads to the relations

$$\langle \lambda_c\lambda_d | S^J(\varepsilon) | \lambda_a\lambda_b \rangle = \langle \lambda_a\lambda_b | S^J(\varepsilon) | \lambda_c\lambda_d \rangle. \quad (69.10)$$

These two amplitudes, however, refer to different processes (direct and inverse reactions). Only in the case of elastic scattering both processes of scattering coincide, and (69.10) represents a definite connection between the helicity amplitudes of one and the same reaction.

At elastic scattering of two identical particles the number of distinct amplitudes is further reduced also by virtue of the permutation symmetry. We saw that for given J only either symmetric or only antisymmetric states with respect to λ_1, λ_2 are realised. Thereby conservation of angular momentum also automatically implies conservation of symmetry with respect to permutation of helicities.

An analogous situation occurs also at elastic scattering of a particle on an antiparticle (or upon conversion of such a pair into another pair, i.e. at a reaction of the type $a + \bar{a} \rightarrow b + \bar{b}$). For given J there exist both symmetric and antisymmetric states with respect to λ_1, λ_2 , but these states correspond to different charge parities of the system.

P -invariance upon decay is expressed by the relation

$$\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle = \frac{\eta_b \eta_c}{\eta_a} (-1)^{s_a - s_b - s_c} \langle -\lambda_b, -\lambda_c | S^J | -\lambda_a \rangle \quad (69.13)$$

(here (69.4) has been used alongside (69.3), and also the law of transformation of the wave function of one particle (16.16)).

When the primary particle is truly neutral, further restrictions arise if C -parity is conserved. Here one must distinguish three cases. If the products of the decay are also truly neutral, then $C_a = C_b C_c$; this condition either forbids the decay entirely, or is satisfied, leading to no new restrictions. If particles b and c are different, then C -invariance establishes a relation between the amplitudes of different processes: $a \rightarrow b + c$ and $a \rightarrow \bar{b} + \bar{c}$. Finally, for the decay $a \rightarrow b + \bar{b}$ there arises a restriction connected with the fact that at given charge parity C and given total angular momentum $J = s_a$ the system may be only in states either symmetric or antisymmetric with respect to helicity—in dependence on the parity of the number J and the sign of C .

By virtue of the universal CPT -invariance the existence of T -invariance implies also CP -invariance. The latter leads to the equality of the amplitudes of two reactions, of which one is obtained from the other by replacing all particles by antiparticles (and changing the sign of the helicities), with $\lambda_{\bar{\pi}} = -\lambda_a, \dots$ ¹³:

$$\langle \lambda_c \lambda_d | S^J | \lambda_a \lambda_b \rangle = \langle \bar{\lambda}_c \bar{\lambda}_d | S^J | \bar{\lambda}_a \bar{\lambda}_b \rangle. \quad (69.11)$$

The number of independent amplitudes is the same for all cross-channels of one and the same generalised reaction; therefore for the determination of this number one can consider any one of the channels. Thus, the same number of independent amplitudes describes elastic scattering $a + b \rightarrow a + b$ and annihilation $a + \bar{a} \rightarrow b + \bar{b}$. The restrictions imposed in the first case by T -invariance are equivalent to the restrictions imposed in the second case by C -invariance.

Let us dwell also on the reaction of decay of one particle into two: $a \rightarrow b + c$. In the centre of inertia system (the rest system of particle a) we have $\mathbf{p}_b = -\mathbf{p}_c$. Multiplying $\mathbf{p}_b$ by the equality $\mathbf{j}_a = \mathbf{j}_b + \mathbf{j}_c$, we obtain

$$\lambda_a = \lambda_b - \lambda_c \quad (69.12)$$

¹³ Since these two amplitudes refer to different reactions, the interference between which is impossible, the phase factor in (69.11) is meaningless and one may put it equal to 1. The real content is only the following from (69.11) equality of cross sections.

(the helicity λ_a of the primary particle is defined as the projection of its spin onto the direction of the momentum of one of the secondary particles). This relation is, one could say, a consequence of the additional symmetry that the given process possesses: axial symmetry around $\mathbf{p}_b$ and $\mathbf{p}_c$. If the spin of the primary particle $s_a < s_b + s_c$, then the relation (69.12) restricts the admissible sets of values $\lambda_a, \lambda_b, \lambda_c$ and thereby the number of independent helicity amplitudes of the decay. The total angular momentum J in this case coincides with the spin of the primary particle s_a , and so is a fixed quantity.

CP -invariance leads to the equality of the amplitudes of decays $a \rightarrow b + c$ and $\bar{a} \rightarrow \bar{b} + \bar{c}$:

$$\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle = \langle \bar{\lambda}_b \bar{\lambda}_c | S^J | \bar{\lambda}_a \rangle \quad (69.14)$$

where $\lambda_{\bar{a}} = -\lambda_a, \dots$, i.e. to the equality of probabilities of the decay of the particle and antiparticle. However, this result presupposes the fulfilment of CP -invariance, which is not a universal property of nature. The universal character is borne by CPT -invariance; this requirement alone would lead to only the equality

$$\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle = \langle \bar{\lambda}_b \bar{\lambda}_c | S^J | \bar{\lambda}_a \bar{\lambda}_b \rangle,$$

the right-hand side of which refers to the process inverse to the decay. We shall see below (see § 71) that the condition of CPT -invariance together with the requirements of unitarity does nevertheless lead to a certain, though more restricted, relation between the probabilities of decay of a particle and antiparticle.

Problems

1. With the help of (69.6) obtain the classification of possible states of a system of two photons.

Solution. In this case $\lambda_1, \lambda_2 = \pm 1$. For even J ($J > 0$), according to (69.6), the three symmetric in $\lambda_1 \lambda_2$ states are admissible:

$$\text{a) } \psi_{JM11} \quad \text{b) } \psi_{JM-1-1} \quad \text{c) } \psi_{JM1-1} + \psi_{JM-11}.$$

For odd J ($J > 1$) one antisymmetric state is admissible:

$$\text{d) } \psi_{JM1-1} - \psi_{JM-11}.$$

States c) and d) possess in the same time definite (+1) parity: according to (69.4)

$$\hat{P}(\psi_{JM1-1} \pm \psi_{JM-11}) = \pm(-1)^J (\psi_{JM1-1} \pm \psi_{JM-11});$$

the multiplier $\pm(-1)^J = 1$, since the upper sign corresponds to even and the lower to odd values of J . States a) and b) themselves do not possess definite parity, but composed of them combinations

$$\text{a') } \psi_{JM11} + \psi_{JM-1-1}, \quad \text{b') } \psi_{JM11} - \psi_{JM-1-1}$$

we get even and odd states. For $J = 0$ there is only the condition $|\lambda_1 - \lambda_2| \leq J$, so that only state c) survives, and there remains only one even and one odd state from

combinations a') and b'). Finally, for $J = 1$ the only admissible state at odd J is d), which is forbidden, since for it $\lambda = 2 > J$. Thus we arrive at the table of admissible states (9.5).

2. In the non-relativistic approximation the total angular momentum J of the system is the resultant of the spin S and the orbital angular momentum L . For the system of two particles find the connection between the states $|JLSM\rangle$ and $|JM\lambda_1\lambda_2\rangle$.

Solution. According to the rule of composition of wave functions at the addition of angular momenta we have

$$\psi_{JLSM} = \sum \{\psi_{s_1\sigma_1}\psi_{s_2\sigma_2}\langle\sigma_1\sigma_2|SMS_j\rangle\}\psi_{LM_L}\langle M_L M_S|JM\rangle. \quad (1)$$

Here $\psi_{s\sigma}$ are the eigenfunctions of spin s with projection σ (onto a fixed axis z), ψ_{LM_L} are the same for the orbital angular momentum L with projection M_L ; the expressions in brackets correspond to the addition of s_1 and s_2 into S , after which S is added with L into J ; the summation is over all m -indices. Let us express all functions in the momentum representation (the momentum $\mathbf{p} \equiv \mathbf{p}_1$), and the spin functions $\psi_{s\sigma}$ through the functions of helicity states $\psi_{s,\lambda}$, using formula (58.7) (see III) for the D -functions; then we get the coefficients

$$\langle JM\lambda_1\lambda_2|JLSM\rangle, \quad (3)$$

where

$$\psi_{JM\lambda_1\lambda_2} = \psi_{s_1\lambda_1}\psi_{s_2,-\lambda_2} D_{\Lambda M}^{(J)}(\mathbf{n}) \sqrt{\frac{2J+1}{4\pi}}, \quad \Lambda = \lambda_1 - \lambda_2,$$

and the coefficients

$$\langle JM\lambda_1\lambda_2|JLSM\rangle = (-i)^L (-1)^{s_1-s_2+S} \sqrt{(2L+1)(2S+1)} \begin{pmatrix} s_1 & s_2 & S \\ \lambda_1 & -\lambda_2 & -\Lambda \end{pmatrix} \begin{pmatrix} L & S & J \\ 0 & \Lambda & -\Lambda \end{pmatrix}. \quad (3)$$

By virtue of the unitarity of the transformation (2)

$$\langle JLSM|JM\lambda_1\lambda_2\rangle = \langle JM\lambda_1\lambda_2|JLSM\rangle^*.$$

§ 70. Invariant amplitudes

In the helicity amplitudes a definite reference frame is used—the system of the centre of inertia. Meanwhile, in computing scattering amplitudes with the aid of the invariant perturbation theory (and also for the study of their general analytic properties) it is convenient to write the amplitudes in an explicitly invariant form.

If the particles participating in the reaction have no spin, then the amplitude of scattering depends only on invariant products of the 4-momenta of the particles. For a reaction of the type

$$a + b \rightarrow c + d \quad (70.1)$$

as these invariants one can choose any two of the three quantities s, t, u defined in § 66. Then the amplitude of scattering reduces to a single function $M_{fi} = f(s, t)$.

If the particles possess spins, then besides the kinematic invariants s, t, u there also exist invariants which can be composed of the wave-function amplitudes of the particles (bispinors, 4-tensors, and so on). The amplitudes of scattering must then have the form

$$M_{fi} = \sum_n f_n(s, t) F_n, \quad (70.2)$$

where F_n are invariants, linearly dependent on the wave-function amplitudes of all participating particles (and also on their 4-momenta). The coefficients $f_n(s, t)$ are called *invariant amplitudes*.

Having chosen the wave amplitudes so that they correspond to particles with definite helicities, we obtain definite values of the invariants F_n . Then the helicity amplitudes of scattering are represented in the form of linear homogeneous combinations of invariant amplitudes f_n . From this it is clear that the number of independent functions f_n coincides with the number of independent helicity amplitudes. Since the number of the latter is easily determined (as was explained in § 69), thereby the task of constructing the invariants F_n is also facilitated—we know in advance how many of them there should be.

Let us consider several examples. In all examples we shall assume that the interaction is T - and P -invariant; the latter property implies that the invariants F_n must be true (and not pseudo) scalars.

Scattering of a particle with spin 0 on a particle with spin 1/2. For counting the number of invariants—or, what is the same, the number of elements of the matrix S^J (i.e. the number of different sets of numbers $\lambda_1, \lambda_2, \lambda'_1, \lambda'_2$)—in this case is equal to 4 ($\lambda_1 = \lambda'_1 = 0, \lambda_2, \lambda'_2 = \pm 1/2$). Taking into account P -invariance the number of independent elements reduces to two, after which the account of T -invariance no longer changes this number.

As the two independent invariants one can choose

$$F_1 = \bar{u}'u, \quad F_2 = \bar{u}'(\gamma K)u. \quad (70.3)$$

Here $u = u(p), u' = u(p')$ are the bispinor amplitudes of the initial and final fermions; $K = k + k'$, where k and k' are the 4-momenta of the initial and final bosons¹⁴.

T -invariance is obvious, noting that the products $\bar{u}'u$ and $\bar{u}'\gamma^\mu u$ are transformed under time reversal by the same law (28.6), as the operators $\hat{\psi}\hat{\psi}$ and $\hat{\psi}\gamma^\mu\hat{\psi}$, the matrix elements of which they are: the product $\bar{u}'u$ is invariant by itself, and the 4-vector $\bar{u}'\gamma^\mu u$ is transformed by the law

$$\bar{u}'\gamma^0 u \rightarrow \bar{u}'\gamma^0 u, \quad \bar{u}'\gamma u \rightarrow -\bar{u}'\gamma u.$$

In the same manner the 4-momenta are transformed: $(K^0, \mathbf{K}) \rightarrow (K^0, -\mathbf{K})$, and the scalar product $F_2 = K_\mu(\bar{u}'\gamma^\mu u)$ is, consequently, invariant.

¹⁴At first glance one could also construct an invariant of the form $\bar{u} \sigma_{\mu\nu} k^\mu k'^\nu u$ (the matrices $\sigma_{\mu\nu}$ are defined in (28.2)). It is easy, however, to verify its reducibility to the invariants (70.3), if one takes into account the law of conservation $k' = p + k - p'$ and the equations

$$(\gamma p)u = mu, \quad \bar{u}'(\gamma p') = m\bar{u}',$$

which the bispinor amplitudes satisfy.

Elastic scattering of two identical particles with spin 1/2. For the counting of the number of independent helicity amplitudes it is convenient to start from the linear combinations of helicity states:

$$\psi_{1g} = \psi_{++} + \psi_{--}, \quad \psi_{2g} = \psi_{+-} - \psi_{-+},$$

$$\psi_{3g} = \psi_{+-} + \psi_{-+}, \quad \psi_u = \psi_{+-} - \psi_{-+},$$

where the indices “+”, “−” indicate the values of the helicities ($\pm 1/2$) of the two particles. The states $1g$, $2g$, $3g$ are even, and the state u is odd with respect to permutation of the particles. Therefore transitions $g \leftrightarrow u$ are forbidden, so that out of the original $16 - 6 = 10$ matrix elements there remain

P the functions ψ_{1g} , ψ_{3g} and ψ_{2g} have opposite parities; the prohibition of transitions between them reduces the number of independent amplitudes to six. Finally, T -invariance leads to the coincidence of amplitudes of transitions $1g \rightarrow 3g$ and $3g \rightarrow 1g$, so that there remain altogether five independent amplitudes. As the five independent invariants one can choose

$$\begin{aligned} F_1 &= (\bar{u}'_1 u_1)(\bar{u}'_2 u_2), & F_2 &= (\bar{u}'_1 \gamma^5 u_1)(\bar{u}'_2 \gamma^5 u_2), \\ F_3 &= (\bar{u}'_1 \gamma^\mu u_1)(\bar{u}'_2 \gamma_\mu u_2), & F_4 &= (\bar{u}'_1 \gamma^\mu \gamma^5 u_1)(\bar{u}'_2 \gamma_\mu \gamma^5 u_2) \\ F_5 &= (\bar{u}'_1 \sigma^{\mu\nu} u_1)(\bar{u}'_2 \sigma_{\mu\nu} u_2), \end{aligned} \quad (70.4)$$

where u_1 , u_2 are the bispinor amplitudes of the initial, and u'_1 , u'_2 those of the final particles. Permutation of the initial (or final) particles does not lead to new invariants: the new invariants are expressed through the old ones (see § 28, Problem). But the expression (70.2) with F_n from (70.4) does not explicitly take into account the requirement, according to which the permutation of two identical fermions must change the sign of the amplitude of scattering. An expression satisfying this requirement can be written in the form

$$M_{fi} = [(\bar{u}'_1 u_1)(\bar{u}'_2 u_2) f_1(t, u) - (\bar{u}'_2 u_1)(\bar{u}'_1 u_2) f_1(u, t)] + \dots \quad (70.5)$$

Under permutation p'_1 and p'_2 (or p_1 and p_2) the kinematic invariants $s \rightarrow s$, $u \rightarrow u$, $u \rightarrow t$, so that the indicated requirement is satisfied automatically.

Elastic scattering of a photon on particles with spin 0 and 1/2. The amplitude of these processes should conveniently be expressed with the help of unit spacelike 4-vectors $e^{(1)}$, $e^{(2)}$, satisfying the conditions

$$\begin{aligned} e^{(1)2} &= e^{(2)2} = -1, & e^{(1)} e^{(2)} &= 0, \\ e^{(1)} k &= e^{(2)} k = 0, & e^{(1)} k' &= e^{(2)} k' = 0 \end{aligned} \quad (70.6)$$

(for each of the two photons these 4-vectors can serve as the 4-orthonormals, with the help of which the invariant description of its polarisation properties is effected—see § 8).

Let k and k' be the initial and the final 4-momenta of the photon, and p and p' the same for the scattering particle. Let us consider the 4-vectors

$$P^\lambda = p^\lambda + p'^\lambda - K^\lambda \frac{pK}{K^2}, \quad N^\lambda = e^{\lambda\mu\nu\rho} P_\mu q_\nu K_\rho, \quad (70.7)$$

where

$$K = k + k', \quad q = p - p' = k' - k.$$

They are obviously mutually orthogonal. They are also orthogonal to the 4-vectors K , q , and hence to k , k' . Being orthogonal to the timelike 4-vector K ($K^2 = 2kk' > 0$), they are spacelike (indeed, in the system of reference in which $\mathbf{K} = 0$, from $KP = 0$ it follows that $P_0 = 0$, and hence $P^2 < 0$). Having normalised P and N , i.e. formed

$$e^{(1)\lambda} = \frac{N^\lambda}{\sqrt{-N^2}}, \quad e^{(2)\lambda} = \frac{P^\lambda}{\sqrt{-P^2}}, \quad (70.8)$$

we obtain a pair of 4-vectors, possessing all the required properties. Let us note that $e^{(2)}$ is a true vector, and $e^{(1)}$ a pseudovector.

Let us represent the amplitude of scattering of the photon in the form

$$M_{fi} = F^{\lambda\mu} e'_\lambda{}^* e_\mu, \quad (70.9)$$

having separated out the 4-vectors e and e' of the initial and the final photons.

The helicity of the photon takes only two values (± 1). Therefore for scattering of the photon on a particle with spin 0 the number of independent helicity amplitudes is the same as for the interaction of the particle with spin 0 and with spin 1/2, i.e. is equal to two. The tensor $F^{\lambda\mu}$ in (70.9) must be constructed only from the 4-momenta of the particles. It can be represented in the form

$$F^{\lambda\mu} = f_1 e^{(1)\lambda} e^{(1)\mu} + f_2 e^{(2)\lambda} e^{(2)\mu}, \quad (70.10)$$

where f_1, f_2 are the invariant amplitudes. Let us call attention to the fact that in $F^{\lambda\mu}$ there cannot be a term containing the product $e^{(1)\lambda} e^{(2)\mu}$, since this product is a pseudotensor and upon substitution into (70.9) would give a pseudoscalar.

Finally, let us consider scattering of a photon on a particle with spin 1/2. For the counting of the number of independent helicity amplitudes we note that the total number of elements of the matrix S^J in this case is 16 (the helicity of each of the two initial and two final particles takes two values). The requirement of P -invariance reduces this number to 8, after which the account of T -invariance brings it down to 6.

Let us represent the tensor $F_{\lambda\mu}$ in this case in the form

$$\begin{aligned} F_{\lambda\mu} = & G_0(e_\lambda^{(1)} e_\mu^{(1)} + e_\lambda^{(2)} e_\mu^{(2)}) + G_1(e_\lambda^{(1)} e_\mu^{(1)} - e_\lambda^{(2)} e_\mu^{(2)}) + \\ & + G_2(e_\lambda^{(1)} e_\mu^{(2)} + e_\lambda^{(2)} e_\mu^{(1)}) + G_3(e_\lambda^{(1)} e_\mu^{(2)} - e_\lambda^{(2)} e_\mu^{(1)}), \end{aligned} \quad (70.11)$$

where G_0, G_3 are true, and G_1, G_2 are pseudo-scalars. These and other bilinear quantities relative to the bispinor amplitudes of the fermions $\bar{u}(p')$ and $u(p)$, i.e. they have the form

$$G_n = \bar{u}(p') Q_n u(p). \quad (70.12)$$

The general form of the matrices (with respect to the bispinor indices) Q_n :

$$\begin{aligned} Q_0 &= f_1 + f_2(\gamma K), & Q_1 &= \gamma^5 [f_3 + f_4(\gamma K)], \\ Q_2 &= \gamma^5 [f_5 + f_6(\gamma K)], & Q_3 &= f_7 + f_8(\gamma K), \end{aligned} \quad (70.13)$$

where $K = k + k'$. The coefficients $f_1, \dots, f_8$ are invariant amplitudes, whose number has turned out to be 8 (instead of the required 6) because T -invariance has not yet been taken into account.

Time reversal interchanges the initial and the final 4-momenta of the particles, also changing the signs of their spatial components:

$$(k_0, \mathbf{k}) \leftrightarrow (k'_0, -\mathbf{k}'), \quad (p_0, \mathbf{p}) \leftrightarrow (p'_0, -\mathbf{p}'); \quad (70.14)$$

the 4-vectors of polarisation of the photons are transformed according to

$$(e_0, \mathbf{e}) \leftrightarrow (e_0'^*, -\mathbf{e}'^*) \quad (70.15)$$

(cf. (8.11a)), so that

$$(e_0'^* e_0, e_i'^* e_0, e_0'^* e_k, e_i'^* e_k) \rightarrow (e_0'^* e_0, -e_0'^* e_i, e_k'^* e_i).$$

By virtue of the last transformation the condition of invariance of the amplitude of scattering (70.9) is equivalent to the requirement

$$(F_{00}, F_{i0}, F_{ik}) \rightarrow (F_{00}, -F_{0i}, F_{ik}).$$

On the other hand, as a consequence of the replacements (70.14), we have

$$(K_0, \mathbf{K}) \rightarrow (K_0, -\mathbf{K}), \quad (q_0, \mathbf{q}) \rightarrow (-q_0, \mathbf{q}), \quad (N_0, \mathbf{N}) \rightarrow (N_0, -\mathbf{N}),$$

so that

$$(e_0^{(1,2)}, \mathbf{e}^{(1,2)}) \rightarrow (e_0^{(1,2)}, -\mathbf{e}^{(1,2)}). \quad (70.16)$$

From the expression (70.11) it follows therefore that one should have

$$G_{0,1,3} \rightarrow G_{0,1,3}, \quad G_2 \rightarrow -G_2.$$

But under time reversal

$$\bar{u}' \gamma^5 u \rightarrow -\bar{u}' \gamma^5 u, \quad \bar{u}' \gamma^5 (\gamma K) u \rightarrow \bar{u}' \gamma^5 (\gamma K) u,$$

as is clear from the laws of transformation of the pseudoscalar and pseudovector bilinear forms in (28.6). Therefore from the expressions (70.12), (70.13) it is clear that by virtue of T -invariance of the amplitude of scattering it must be

$$f_3 = f_6 = 0. \quad (70.17)$$

§ 71. Unitarity condition

The scattering matrix must be unitary: $\hat{S}\hat{S}^+ = 1$, or in matrix elements:

$$(SS^+)_{fi} = \sum_n S_{fn} S_{in}^* = \delta_{fi}, \quad (71.1)$$

where the index n enumerates all possible intermediate states¹⁵. This is the most general property of the S -matrix, which

ensures the conservation of normalisation and orthogonality of the states in the reaction (cf. III, §§ 125, 144). In particular, the diagonal elements of the equality (71.1) express simply the fact that the sum of probabilities of transition from a given initial state to any final state equals unity:

$$\sum_n |S_{ni}|^2 = 1.$$

Substituting in (71.1) the matrix elements in the form (64.2), we get

$$\begin{aligned} T_{fi} - T_{if}^* &= i(2\pi)^4 \sum_n \delta^{(4)}(P_f - P_n) T_{fn} T_{in}^* = \\ &= i(2\pi)^4 \sum_n \delta^{(4)}(P_f - P_n) T_{nf}^* T_{ni}. \end{aligned} \quad (71.2)$$

The two equivalent forms of the right-hand side of the equality are obtained when writing the condition of unitarity correspondingly in the form $\hat{S}\hat{S}^+ = 1$ or $\hat{S}^+\hat{S} = 1$, with different orderings of the factors $\hat{S}$ and $\hat{S}^+$.

Let us note that the left-hand side of this equality is linear, while the right-hand side is quadratic in the matrix elements T . Therefore if the interaction (like, for example, the electromagnetic) contains a small parameter, then the left-hand side will be of the first order, and the right-hand side of the second order of smallness. In the first approximation the latter can therefore be neglected, and then

$$T_{fi} = T_{if}^*, \quad (71.3)$$

i.e. the matrix T is Hermitian.

To lend the condition of unitarity (71.2) a more concrete form it is necessary to specify what precisely is meant by the summation over n . Let us do this for the collision of two particles, assuming that the laws of conservation permit only elastic scattering; then we will also assume that in (71.2) the intermediate states are also two-particle (the so-called “two-particle” condition of unitarity). Making it for the collision of two particles, we find after summing over n :

$$\sum_n \rightarrow \int \frac{V^2 d^3 p_1'' d^3 p_2''}{(2\pi)^6} \sum_{\lambda''}.$$

Eliminating the δ -functions in the same way as was done in § 64, we obtain the “two-particle” condition of unitarity in the form

$$M_{fi} - M_{if}^* = \frac{i}{(4\pi)^2} \sum_{\lambda''} \frac{|\mathbf{p}|}{\varepsilon} \int M_{fn} M_{in}^* d\omega'', \quad (71.4)$$

where $\mathbf{p}$ is the momentum, ε is the total energy in the system of the centre of inertia. The normalisation volume disappears from this relation after the transition from the

¹⁵The meaning of the symbol δ_{fi} in (71.1) depends, of course, on the concrete choice of quantum numbers and on the normalisation of wave functions of the system. It must be defined so that $\sum_f \delta_{if} = 1$.

amplitudes T_{fi} to the amplitudes M_{fi} , according to (64.10):

$$M_{fi} - M_{if}^* = \frac{i}{(4\pi)^2} \sum_{\lambda''} \frac{|\mathbf{p}|}{\varepsilon} \int M_{fn} M_{in}^* d\omega''.$$

Let us define the amplitude of elastic scattering so that it would be

$$d\sigma = |\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle|^2 d\omega' \quad (71.5)$$

($\mathbf{n}$, $\mathbf{n}'$ are the directions of the initial and final momenta; λ , λ' are the initial and final spin quantum numbers). Comparison with (64.19) shows that

$$\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle = \frac{1}{8\pi\varepsilon} M_{fi}, \quad (71.6)$$

and the condition of unitarity (71.4) takes the form

$$\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle - \langle \mathbf{n} \lambda | f | \mathbf{n}' \lambda' \rangle^* = \frac{i|\mathbf{p}|}{2\pi} \sum_{\lambda''} \int \langle \mathbf{n}' \lambda' | f | \mathbf{n}'' \lambda'' \rangle \langle \mathbf{n} \lambda | f | \mathbf{n}'' \lambda'' \rangle^* d\omega'', \quad (71.7)$$

generalising the familiar formula of non-relativistic theory (see III, (125.8)).

The amplitude of elastic scattering at zero angle is called the diagonal matrix element T_{ii} , in which the final state coincides with the initial ¹⁶. For this amplitude the condition of unitarity (71.2) takes the form

$$2 \operatorname{Im} T_{ii} = (2\pi)^4 \sum_n |T_{in}|^2 \delta^{(4)}(P_i - P_n). \quad (71.8)$$

The right-hand side of this equality differs only by the multiplier from the total cross section of all possible scattering processes from the given initial state i ; let us denote this cross section through σ_t . Indeed, summing the probability (64.5) over the states f and dividing by the flux density j , we find

$$\sigma_t = \frac{(2\pi)^4 V}{j} \sum_n |T_{in}|^2 \delta^{(4)}(P_i - P_n),$$

so that

$$\frac{2V}{j} \operatorname{Im} T_{ii} = \sigma_t.$$

The normalisation volume disappears from here after the replacement $T_{ii} = M_{ii}/(2\varepsilon_1 V \cdot 2\varepsilon_2 V)$ (ε_1 , ε_2 are the energies of the particles in the system of the centre of inertia) and the substitution of j from (64.17):

$$\operatorname{Im} M_{ii} = 2|\mathbf{p}|\varepsilon\sigma_t. \quad (71.9)$$

This formula constitutes the content of the so-called *optical theorem*. If we introduce the amplitude of elastic scattering, it assumes its usual form

$$\operatorname{Im} \langle \mathbf{n} \lambda | f | \mathbf{n} \lambda \rangle = \frac{|\mathbf{p}|}{4\pi} \sigma_t \quad (71.10)$$

¹⁶Let us emphasise that this refers to elements of the matrix T , not S , i.e. the diagonal element is taken after the exclusion from S of the unit matrix.

(cf. III, (142.10)).

If the S -matrix is given in the angular-momentum representation (partial amplitudes), then by virtue of its diagonality with respect to J the condition of unitarity is written for each value of J separately.

Thus, if only elastic scattering is possible, the condition of unitarity has the form

$$\sum_{\lambda''} \langle \lambda' | S^J | \lambda'' \rangle \langle \lambda | S^J | \lambda'' \rangle^* = \delta_{\lambda\lambda'}. \quad (71.11)$$

By virtue of the T -invariance the matrix of elastic scattering is symmetric (see (69.10)) and therefore can be reduced to diagonal form. After this the condition of unitarity simply requires the equality of the moduli of the diagonal elements to unity; it is customary in such a case to write

$$S_{Jn}^J = \exp(2i\delta_{Jn}), \quad (71.12)$$

where δ_{Jn} are real constants—the scattering phases (the index n labels the diagonal elements at given J). In the general case, when the number N of independent amplitudes exceeds the rank of the (square) matrix S^J , the coefficients of the transformation effecting the diagonalisation of S^J depend on J and E (in these coefficients, alongside the principal values of the matrix, are also contained N independent quantities, equivalent to the original N independent helicity amplitudes). But if N coincides with the rank of the matrix S^J (and thereby with the number of its principal values), then the coefficients of diagonalisation are universal. In this case the diagonalising states are states with definite parities (since, of course, the determination has no definite helicities).

Condition (71.11), expressed with the help of partial amplitudes $\langle \lambda' | f^J | \lambda \rangle$, has the form

$$\langle \lambda' | f^J | \lambda \rangle - \langle \lambda | f^J | \lambda' \rangle^* = 2i|\mathbf{p}| \sum_{\lambda''} \langle \lambda' | f^J | \lambda'' \rangle \langle \lambda | f^J | \lambda'' \rangle^*, \quad (71.13)$$

as is easy to verify by substituting in (71.7) the expansion (68.13) and using the orthonormality of the D -functions. For T -invariance the matrix $\langle \lambda' | f^J | \lambda \rangle$ is symmetric, and (71.13) takes the form

$$\text{Im} \langle \lambda' | f^J | \lambda \rangle = |\mathbf{p}| \langle \lambda' | f^J f^{J+} | \lambda \rangle. \quad (71.14)$$

If the matrix is diagonalised, then its diagonal elements

$$f_n^J = \frac{1}{2i|\mathbf{p}|} (\exp(2i\delta_{Jn}) - 1) = \frac{1}{|\mathbf{p}|} \exp(i\delta_{Jn}) \sin \delta_{Jn}. \quad (71.15)$$

Finally, let us indicate some consequences arising from the condition of unitarity together with the requirement of CPT -invariance. By virtue of the latter

$$T_{fi} = T_{\bar{i}\bar{f}}, \quad (71.16)$$

where $\bar{i}$ and $\bar{f}$ are the states differing from i and f by the replacement of all particles by antiparticles (and also by the change of sign of the momentum vectors of all particles at unchanged momenta).

In particular, for diagonal elements

$$T_{ii} = T_{\bar{i}\bar{i}}.$$

From (71.8) or (71.9) it follows therefore that the total cross section of all possible processes (with a given initial state) is the same for reactions between particles and antiparticles. In particular, the total probabilities of decay (i.e. the lifetimes) of a particle and antiparticle are equal. These results (alongside the equality of masses of particle and antiparticle—§ 11) are the most important consequences of *CPT*-invariance of the interactions. Let us recall that this also applies to every individual decay channel separately, and requires also the fulfilment of *CP*-invariance.

Problem

Starting from the condition of unitarity, find the connection between the phases of the partial amplitudes of photoproduction of pions on nucleons ($\gamma + N \rightarrow \pi + N$) and of elastic scattering of pions on nucleons; it is assumed in this that πN -scattering is connected with strong interactions, and photoproduction and γN -scattering with electromagnetic interaction.

Solution. Let us denote the partial amplitudes: $\langle \pi N | S | \gamma N \rangle = S_{\pi\gamma}$, $\langle \gamma N | S | \gamma N \rangle = S_{\gamma\gamma}$, $\langle \pi N | S | \pi N \rangle = S_{\pi\pi}$ (indices J and helicities are omitted). Photoproduction is a first-order process with respect to charge e , and $S_{\pi\gamma} \sim e$. Pion scattering on the nucleon $S_{\pi\pi}$ does not contain e . The amplitude $S_{\gamma\gamma}$ is of second order by charge. With an accuracy up to $\sim e$ from the condition (71.1) we get

$$S_{\pi\gamma} S_{\pi\gamma}^* + S_{\pi\pi} S_{\pi\gamma}^* \approx S_{\pi\gamma} + S_{\pi\pi} S_{\pi\gamma}^* = 0, \quad (1)$$

$$S_{\pi\gamma} S_{\gamma\gamma}^* + S_{\pi\pi} S_{\pi\gamma}^* \approx S_{\pi\pi} S_{\pi\gamma}^* = 1 \quad (2)$$

(in the right-hand side of equation (2) one should understand 1 as the unit matrix in the spin indices). By virtue of *T*-invariance the matrix $S_{\pi\pi}$ is symmetric, and $S_{\pi\gamma} = S_{\gamma\pi}$. Let us choose the matrix $S_{\pi\pi}$ in diagonal form, i.e. with respect to states of the pion with definite parities; then from (2) it follows that the diagonal elements have the form $e^{2i\delta_\pi}$ with different scattering constants δ_π . From (1) we then find for each of the elements of the matrix $S_{\pi\gamma}$:

$$S_{\pi\gamma} / S_{\pi\gamma}^* = -e^{2i\delta_\pi},$$

whence

$$S_{\pi\gamma} = \pm |S_{\pi\gamma}| i e^{i\delta_\pi}.$$

Thus, the phase of the partial amplitude of photoproduction (in a state with definite parity) is determined by the phase of elastic πN -scattering.

Invariant Perturbation Theory

§ 72. Chronological product

The probabilities of various processes in collisions of particles, the interaction between which can be considered small, are computed with the help of perturbation theory. In its usual (for non-relativistic quantum mechanics) form the apparatus of this theory possesses, however, the drawback that it does not make manifest the requirement of relativistic invariance. Although in applying such an apparatus to relativistic problems the final result will ultimately satisfy this requirement, the non-invariant form of the intermediate formulae substantially complicates the calculation. The present chapter is devoted to the development, free from this drawback, of the consistent relativistic perturbation theory; it was constructed by *Feynman* (*R. P. Feynman*, 1948–1949).

Having in mind the second-quantised description of the system, let us denote by Φ its wave function in the representation of occupation numbers of states of free particles. The Hamiltonian of the system $\hat{H} = \hat{H}_0 + \hat{V}$, where $\hat{V}$ is the interaction operator. Let Φ_n be the eigenfunctions of the unperturbed Hamiltonian; each of them corresponds to a certain definite set of occupation numbers. An arbitrary function Φ is represented in the form of the expansion $\Phi = \sum C_n \Phi_n$. Then the exact wave equation

$$i \frac{\partial \Phi}{\partial t} = (\hat{H}_0 + \hat{V})\Phi \quad (72.1)$$

is represented in the form of a system of equations for the coefficients C_n :

$$i \dot{C}_n = \sum_m V_{nm} \exp[i(E_n - E_m)t] C_m, \quad (72.2)$$

where V_{nm} are the time-independent matrix elements of the operator $\hat{V}$, and E_n are the energy levels of the unperturbed system (cf. III, § 40).

By definition the operator $\hat{V}$ does not depend explicitly on time. But the quantities

$$V_{nm}(t) = V_{nm} \exp[i(E_n - E_m)t] \quad (72.3)$$

can be regarded as matrix elements of a time-dependent operator

$$\hat{V}(t) = \exp(i\hat{H}_0 t) \hat{V} \exp(-i\hat{H}_0 t). \quad (72.4)$$

One speaks of it as of the operator in the *interaction representation* (in contrast to the original time-independent Schrödinger operator $\hat{V}$ ¹). Denoting now by the former letter

¹Let us emphasise that in the definition (72.4) there figures the unperturbed Hamiltonian $\hat{H}_0$. In this it differs from the *Heisenberg representation* of operators, in which

$$\hat{V}^H(t) = \exp(i\hat{H}t) \hat{V} \exp(-i\hat{H}t)$$

(see III, § 13 and below, § 102).

Φ the wave function in this new representation, we write equation (72.2) in the symbolic form

$$i\dot{\Phi} = \hat{V}(t)\Phi. \quad (72.5)$$

The change of the wave function in this representation is connected only with the action of the perturbation, i.e. it corresponds to the processes occurring as a result of the interaction of the particles.

If $\Phi(t)$ and $\Phi(t + \delta t)$ are the values of Φ at two infinitely close moments of time, then by virtue of (72.5) they are connected with each other by

$$\Phi(t + \delta t) = [1 - i\delta t \hat{V}(t)]\Phi(t) = \exp[-i\delta t \cdot \hat{V}(t)]\Phi(t).$$

Correspondingly, the value of Φ at an arbitrary moment t_f can be expressed through the value at some initial moment t_i ($t_f > t_i$) as

$$\Phi(t_f) = \left\{ \prod_i^f \exp[-i\delta t_\alpha \cdot \hat{V}(t_\alpha)] \right\} \Phi(t_i), \quad (72.6)$$

where the sign $\prod$ denotes the limit of the product over all infinitely small intervals δt_α between t_i and t_f . If $V(t)$ were an ordinary function, then this limit would reduce simply to

$$\exp \left\{ -i \int_{t_i}^{t_f} V(t) dt \right\}.$$

But such a reduction relies on the commutativity of the factors (taken at different moments of time) implied in passing from the product in (72.6) to summation in the exponent. For the operator $\hat{V}(t)$ no such commutativity exists, and the reduction to an ordinary integral is impossible.

Let us write (72.6) in the symbolic form

$$\Phi(t_f) = T \exp \left\{ -i \int_{t_i}^{t_f} \hat{V}(t) dt \right\} \Phi(t_i), \quad (72.7)$$

where T is the symbol of *chronologisation*, denoting a definite (“chronological”) ordering of moments of time in the successive factors of the product (72.6). In particular, setting $t_i \rightarrow -\infty$, $t_f \rightarrow +\infty$, we obtain

$$\Phi(+\infty) = \hat{S}\Phi(-\infty), \quad (72.8)$$

where

$$\hat{S} = T \exp \left\{ -i \int_{-\infty}^{\infty} \hat{V}(t) dt \right\}. \quad (72.9)$$

The meaning of the expressions (72.7)–(72.9) as the formally exact solution of the wave equation consists in that such a representation makes it easy to write a series representing the expansion in powers of the perturbation:

$$\hat{S} = \sum_{k=0}^{\infty} \frac{(-i)^k}{k!} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \dots \int_{-\infty}^{\infty} dt_k T\{\hat{V}(t_1)\hat{V}(t_2) \dots \hat{V}(t_k)\}. \quad (72.10)$$

Here in each term the k -th power of the integral is written in the form of a k -fold integral, and the symbol T means that in each region of values of the variables $t_1, t_2, \dots, t_k$ one must place the corresponding operators in chronological order from right to left in the order of increasing values of t ².

From definition (72.8) it is clear that if before the collision the system was in the state Φ_i (a certain set of free particles), then the amplitude of probability of its transition into the state Φ_f (a different set of free particles) is the matrix element S_{fi} . In other words, these elements are what constitute the S -matrix.

The operator of electromagnetic interaction was written already in § 43:

$$\hat{V} = e \int (\hat{j}\hat{A}) d^3x. \quad (72.11)$$

Substituting it into (72.9), we get

$$\hat{S} = T \exp \left\{ -ie \int (\hat{j}\hat{A}) d^4x \right\}. \quad (72.12)$$

It is essential that the operator (72.12) is relativistically invariant. This is evident from the scalar character of the sub-integral expression, the invariant nature of the integration over d^4x , and the invariant character of the operation of chronologisation. The last circumstance, however, requires clarification.

As is well known, the ordering of two moments of time t_1 and t_2 (the sign of the difference $t_2 - t_1$) does not depend on the choice of the system of reference if these moments pertain to world points x_1 and x_2 separated by a timelike interval: $(x_2 - x_1)^2 > 0$. In this case the invariance of the chronologisation is automatic. But if $(x_2 - x_1)^2 < 0$ (spacelike interval), then in different systems of reference one can have both $t_2 > t_1$ and $t_2 < t_1$ ³. But such two points correspond to events between which there can be no causal connection. It is clear therefore that there can be no non-commutative operators of two physical quantities relating to such points: non-commutativity of operators physically signifies the joint immeasurability of the given quantities, which presupposes the existence of a physical connection between both measurements. Consequently, the chronological product remains invariant also in this case: although the Lorentz transformation may violate the ordering of moments of time, by virtue of the commutativity of the factors they can be rearranged back into chronological order⁴.

²The derivation of the rules of relativistic perturbation theory with the help of the expansion (72.10) is due to Dyson (*F. Dyson*, 1949).

³Instead of timelike and spacelike intervals one speaks for brevity of regions: all points x , separated from the point x' by intervals with $(x - x')^2 > 0$, lie within the light cone; all points x' , separated from point x by intervals with $(x - x')^2 < 0$, lie outside the light cone.

⁴As applied to the product $\hat{V}(t_1)\hat{V}(t_2)\dots$ this statement needs to be made more precise. Since the operator $\hat{V}$ does not possess gauge invariance (it changes together with $\hat{A}$), the factors $\hat{V}(t_1), \hat{V}(t_2)\dots$, commutative for one gauge of the potentials, may turn out to be non-commutative for another gauge. The statement made above should therefore be formulated as the possibility of such a choice of gauge of the potential for which $\hat{V}(t_1)$ and $\hat{V}(t_2)$ outside the light cone will be commutative. This reservation, of course, in no way affects the invariance of the S -matrix: the amplitudes of scattering as real physical quantities cannot depend on the gauge of the potential (formally this independence follows from the gauge invariance of the action integral noted in § 43).

It is easy to see that the definition of the S -matrix given in this section automatically satisfies the condition of unitarity. Representing $\hat{S}$ in the form of a chronological product, figuring in (72.6), and noting that the Hermiticity of $\hat{V}$ implies that $\hat{S}^+$ is expressed

(with the reversed sign in the exponent) in reverse chronological order. Therefore when multiplying $\hat{S}$ and $\hat{S}^+$ all factors cancel pairwise.

Let us call attention to the fact that the unitarity of the operator $\hat{S}$ is ensured in this case by the Hermiticity of the Hamiltonian. But the requirement of unitarity has in fact a more general character than the premises lying at the basis of the theory presented. It should hold also in quantum-mechanical description not employing the concepts of the Hamiltonian and wave functions.

§ 73. Feynman diagrams for electron scattering

Let us show by concrete examples how the computation of elements of the scattering matrix is carried out. These examples will facilitate the further formulation of the general rules of invariant perturbation theory.

The current operator $\hat{j}$ contains the product of two electron ψ -operators. Therefore in the first order of perturbation theory only processes could arise in which altogether (in the initial and final states) three particles participate—two electrons (operator $\hat{j}$) and one photon (operator $\hat{A}$). It is easy, however, to see that such processes between free particles are impossible—they are forbidden by the laws of conservation of energy and momentum. If p_1 and p_2 are the 4-momenta of the electrons, and k is the 4-momentum of the photon, then the conservation of 4-momentum would be expressed by the equality $k = p_2 - p_1$ or $k = p_2 + p_1$. But such equalities are impossible, since for the photon $k^2 = 0$, while the square $(p_2 \pm p_1)^2$ is certainly different from zero. Indeed, computing the value of the invariant $(p_2 \pm p_1)^2$ in the rest frame of one of the electrons, we get

$$(p_2 \pm p_1)^2 = 2(m^2 \pm p_1 p_2) = 2(m^2 \pm \varepsilon_1 \varepsilon_2 \mp \mathbf{p}_1 \mathbf{p}_2) = 2m(m \pm \varepsilon_2). \quad (73.1)$$

Since $\varepsilon_2 > m$, we have

$$(p_2 + p_1)^2 > 0, \quad (p_2 - p_1)^2 < 0.$$

Thus, the first non-vanishing (off-diagonal) elements of the S -matrix can appear only in the second order of perturbation theory. All relevant processes are contained in the operator of second order, obtained from expanding expression (72.12):

$$\hat{S}^{(2)} = -\frac{e^2}{2!} \iint d^4x d^4x' T(\hat{j}^\mu(x) \hat{A}_\mu(x) \hat{j}^\nu(x') \hat{A}_\nu(x')).$$

Since the electronic and photonic operators are commutative with each other, the T -product here can be

split into two T -products:

$$\hat{S}^{(2)} = -\frac{e^2}{2!} \iint d^4x d^4x' T(\hat{j}^\mu(x) \hat{j}^\nu(x')) T(\hat{A}_\mu(x) \hat{A}_\nu(x')). \quad (73.2)$$

As a first example let us consider elastic scattering of two electrons: in the initial state we have two electrons with 4-momenta p_1 and p_2 , and in the final state two other

electrons with 4-momenta p_3 and p_4 . It is also assumed that all electrons are in definite spin states; spin indices are for brevity everywhere omitted.

Since in both states photons are completely absent, the matrix element we need of the T-product of the photonic operators is the diagonal element $\langle 0 | \dots | 0 \rangle$, where the symbol $|0\rangle$ denotes the photon-vacuum state. This vacuum average of the T-product represents a certain definite (for each pair of indices $\mu\nu$) function of the two points x and x' . By virtue of the homogeneity of 4-space the coordinates can enter only in the form of the difference $x - x'$. The tensor

$$D_{\mu\nu}(x - x') = i\langle 0 | T A_\mu(x) A_\nu(x') | 0 \rangle \quad (73.3)$$

is called the *photon propagator* (or *photon Green's function*). An explicit expression for it will be obtained below in § 76. For the T-product of the electronic operators we must compute the matrix element

$$\langle 34 | T j^\mu(x) j^\nu(x') | 12 \rangle, \quad (73.4)$$

where the symbols $|12\rangle$, $|34\rangle$ denote states with pairs of electrons with the corresponding momenta. This matrix element too can be represented in the form of a vacuum average with the help of the obvious equality

$$\langle 2 | F | 1 \rangle = \langle 0 | a_2 F a_1^\dagger | 0 \rangle,$$

where F is an arbitrary operator, and $\hat{a}_1^\dagger$ and $\hat{a}_2$ are the operators of creation and annihilation of the second electron respectively. Therefore instead of (73.4) one can compute the quantity

$$\langle 0 | a_3 a_4 T(j^\mu(x) j^\nu(x')) a_2^\dagger a_1^\dagger | 0 \rangle \quad (73.5)$$

(indices 1, 2, ... for brevity replace $\mathbf{p}_1, \mathbf{p}_2, \dots$).

Each of the two current operators is a product $\hat{j} = \hat{\bar{\psi}} \gamma \hat{\psi}$, and each of the ψ -operators is represented by the sum

$$\hat{\psi} = \sum_{\mathbf{p}} (\hat{a}_{\mathbf{p}} \psi_{\mathbf{p}} + \hat{b}_{\mathbf{p}}^\dagger \bar{\psi}_{-\mathbf{p}}), \quad \hat{\bar{\psi}} = \sum_{\mathbf{p}} (\hat{a}_{\mathbf{p}}^\dagger \bar{\psi}_{\mathbf{p}} + \hat{b}_{\mathbf{p}} \bar{\psi}_{-\mathbf{p}}) \quad (73.6)$$

(the second terms contain positronic operators which in the present case “do not work”). Therefore the product $\hat{j}^\mu(x) \hat{j}^\nu(x')$ is represented in the form of a sum of terms, each of which contains the product of two operators $\hat{a}_{\mathbf{p}}$ and two $\hat{a}_{\mathbf{p}}^\dagger$. These operators must ensure the annihilation of electrons 1, 2 and the creation of electrons 3, 4. In other words, these must be the operators $\hat{a}_1, \hat{a}_2, \hat{a}_3^\dagger, \hat{a}_4^\dagger$, which are, as one says, *contracted* with the “external” operators $\hat{a}_1^\dagger, \hat{a}_2^\dagger, \hat{a}_3, \hat{a}_4$ in (73.5) and cancel out according to the equalities

$$\langle 0 | a_{\mathbf{p}} a_{\mathbf{p}}^\dagger | 0 \rangle = 1. \quad (73.7)$$

Depending on which of the ψ -operators the operators $\hat{a}_1, \hat{a}_2, \hat{a}_3^\dagger, \hat{a}_4^\dagger$ are taken from, four terms arise in (73.5):

$$\begin{aligned} (73.5) = & \overbrace{a_3 a_4 (\bar{\psi}' \gamma^\mu \psi)} (\bar{\psi}' \gamma^\nu \psi') \overbrace{a_2^\dagger a_1^\dagger} + \overbrace{a_3 a_4 (\bar{\psi} \gamma^\mu \psi)} (\bar{\psi}' \gamma^\nu \psi') \overbrace{a_2^\dagger a_1^\dagger} + \\ & + \overbrace{a_3 a_4 (\bar{\psi}' \gamma^\mu \psi')} (\bar{\psi}' \gamma^\nu \psi') \overbrace{a_2^\dagger a_1^\dagger} + \overbrace{a_3 a_4 (\bar{\psi} \gamma^\mu \psi')} (\bar{\psi}' \gamma^\nu \psi') \overbrace{a_2^\dagger a_1^\dagger}, \end{aligned} \quad (73.8)$$

where $\psi = \psi(x)$, $\psi' = \psi(x')$, and the braces join the contracted operators, i.e. those from which the pair of operators $\hat{a}$, $\hat{a}^+$ is taken for contraction according to (73.7). In each of these terms, by successive transpositions, we bring the conjugate operators into pairwise proximity ($\hat{a}_1 \hat{a}_1^+$ and so on), after which the vacuum average reduces to the product of pairwise vacuum averages (73.7). Noting that all these operators are anticommutative (1, 2, 3, 4 are different states!)⁵, we find that the matrix element (73.4) equals

$$\langle 34 | T j^\mu(x) j^\nu(x') | 12 \rangle = (\bar{\psi}_4 \gamma^\mu \psi_2) (\bar{\psi}_3 \gamma^\nu \psi'_1) + (\bar{\psi}_3 \gamma^\mu \psi_1) (\bar{\psi}_4 \gamma^\nu \psi'_2) - (\bar{\psi}_3 \gamma^\mu \psi_2) (\bar{\psi}_4 \gamma^\nu \psi'_1) - (\bar{\psi}_4 \gamma^\mu \psi_1) (\bar{\psi}_3 \gamma^\nu \psi'_2). \quad (73.9)$$

Let us note that the overall sign of this sum is conventional and depends on the order in which we have placed the “external” electronic operators in (73.5). This circumstance corresponds to the fact that the overall sign of the matrix element for scattering of identical fermions is altogether arbitrary. The relative sign of the various terms in (73.9) does not, of course, depend on the adopted order of placement of the external operators.

The two terms in the first and second lines of (73.9) differ from each other merely by the simultaneous permutation of indices μ , ν and of the arguments x , x' . Such a permutation is immaterial, and the matrix

element (73.3) (in which the ordering of factors is in any case established by the symbol T). Therefore after multiplying (73.3) and (73.9) and integrating over $d^4x d^4x'$ the four terms in (73.9) give pairwise the same result, so that the matrix element

$$S_{fi} = ie^2 \iint d^4x d^4x' D_{\mu\nu}(x - x') \{ (\bar{\psi}_4 \gamma^\mu \psi_2) (\bar{\psi}_3 \gamma^\nu \psi'_1) - (\bar{\psi}_4 \gamma^\mu \psi_1) (\bar{\psi}_3 \gamma^\nu \psi'_2) \} \quad (73.10)$$

(let us call attention to the disappearance of the factor 1/2!).

The electronic wave functions are plane waves (64.8), and therefore the expression in braces

$$\begin{aligned} \{ \dots \} &= (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma^\nu u_1) e^{-i(p_2 - p_4)x} e^{-i(p_1 - p_3)x'} - \\ &\quad - (\bar{u}_4 \gamma^\mu u_1) (\bar{u}_3 \gamma^\nu u_2) e^{-i(p_1 - p_4)x} e^{-i(p_2 - p_3)x'} \\ &= \{ (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma^\nu u_1) e^{-i[(p_2 - p_4) + (p_3 - p_1)]\xi/2} - \\ &\quad - (\bar{u}_4 \gamma^\mu u_1) (\bar{u}_3 \gamma^\nu u_2) e^{-i[(p_1 - p_4) + (p_3 - p_2)]\xi/2} \} e^{-i(p_1 + p_2 - p_3 - p_4)X}, \end{aligned}$$

where $X = (x + x')/2$, $\xi = x - x'$. The integration over $d^4x d^4x'$ is replaced by integration over $d^4\xi d^4X$. The integral over d^4X gives a δ -function (by virtue of which $p_1 + p_2 = p_3 + p_4$). Passing then from the matrix S to the matrix M (see § 64), we obtain finally the amplitude of scattering

$$M_{fi} = e^2 \{ (\bar{u}_4 \gamma^\mu u_2) D_{\mu\nu}(p_4 - p_2) (\bar{u}_3 \gamma^\nu u_1) - (\bar{u}_4 \gamma^\mu u_1) D_{\mu\nu}(p_4 - p_1) (\bar{u}_3 \gamma^\nu u_2) \}. \quad (73.11)$$

⁵In view of this anticommutativity of the operators $\hat{j}(x)$ and $\hat{j}(x')$, in the present case they can be considered (when computing the matrix element) commutative and the sign T omitted.

Here the photon propagator in the momentum representation

$$D_{\mu\nu}(k) = \int D_{\mu\nu}(\xi) e^{ik\xi} d^4\xi \quad (73.12)$$

has been introduced. Each of the two terms of the amplitude (73.11) can be symbolically represented in the form of the so-called *Feynman diagrams*. The first term is represented by the diagram

$$e^2(\bar{u}_4\gamma^\mu u_2)D_{\mu\nu}(k)(\bar{u}_3\gamma^\nu u_1) = \quad (73.13)$$

To each of the intersection points of the lines (*vertices* of the diagram) there corresponds a factor γ . “Incoming” solid lines, directed towards the vertex, correspond to initial electrons; they are associated with factors u —bispinor amplitudes of the corresponding electronic states. “Outgoing” solid lines,

directed from the vertex, are final electrons; to these lines there correspond factors $\bar{u}$. When “reading” the diagram the indicated factors are written from left to right in the order corresponding to traversal along the solid lines against the direction of the arrows. Both vertices are joined by a dashed line corresponding to the *virtual* (intermediate) photon, “emitted” at one vertex and “absorbed” at another; to this line there corresponds the factor $-iD_{\mu\nu}(k)$. The 4-momentum of the virtual photon k is determined by “conservation of the 4-momentum in the vertex”: by the equality of the total momenta of the incoming and outgoing lines; in the present case $k = p_1 - p_3 = p_4 - p_2$. Besides the listed factors, to the diagram as a whole there is added yet an overall factor $(-ie)^2$ (the power of the exponent equals the number of vertices in the diagram), and in this form it enters as a term in iM_{fi} . Analogously the second term in (73.11) is represented by the diagram

$$e^2(\bar{u}_4\gamma^\mu u_1)D_{\mu\nu}(k')(\bar{u}_3\gamma^\nu u_2) = \quad (73.14)$$

(one must bear in mind that $k' = p_1 - p_4 = p_3 - p_2$). It is immaterial whether we start reading the diagram from the end of p_3 or p_4 ; the resulting expressions in this case coincide with each other by virtue of the symmetry of the tensor $D_{\mu\nu}$. Equally arbitrary is the choice of direction of the line of the virtual photon: the change of this direction will lead merely to the change of sign of k , which is immaterial by virtue of the evenness of the function $D_{\mu\nu}(k)$ (see § 76).

Lines corresponding to initial and final particles are called *external lines* or *free ends* of the diagram. Diagrams (73.13) and (73.14) differ from each other by the exchange of two free electronic ends (p_3 and p_4). Such a permutation of two fermions changes the sign of the diagram; this rule corresponds to the fact that in the amplitude (73.11) both terms enter with opposite signs.

From now on we shall always use Feynman diagrams in the described, momentum, representation. Let us note, however, that these diagrams can be brought into correspondence also with terms of the amplitude of scattering in their original—coordinate—representation (integrals (73.10)). The role of electronic amplitudes is played then by the corresponding coordinate wave functions, and as propagators in the coordinate representation are taken.

Let us consider now the mutual scattering of an electron and a positron; let us denote their initial momenta by p_- and p_+ respectively, and the final ones by p'_- and p'_+ .

The operators of creation and annihilation of positrons enter in the ψ -operators (73.6) together with the operators of annihilation and creation of electrons. At the same time, while in the previous case the annihilation of both initial particles was ensured by the operator $\hat{\psi}$, and the creation of both final particles by the operator $\hat{\bar{\psi}}$, here the role of these operators is reversed with respect to electrons and positrons. Therefore the conjugate function $\bar{\psi}(-p_+)$ will now describe the initial positron, and the final positron will be described by the function $\psi(-p'_+)$ (both with the reversed sign of the 4-momentum). Taking into account this difference we obtain as a result the amplitude of scattering⁶

$$M_{fi} = -e^2 (\bar{u}(p'_-)\gamma^\mu u(p_-)) D_{\mu\nu}(p_- - p'_-) (\bar{u}(-p_+)\gamma^\nu u(-p'_+)) + \\ + e^2 (\bar{u}(-p_+)\gamma^\mu u(p_-)) D_{\mu\nu}(p_- + p_+) (\bar{u}(p'_-)\gamma^\nu u(-p'_+)). \quad (73.15)$$

The first and second terms in this expression are represented by the following diagrams:

The rules for drawing diagrams change only in the part relating to positrons. As before, to incoming solid lines there corresponds the factor u , and to outgoing ones the factor $\bar{u}$. But now the incoming lines correspond to final, and the outgoing ones to initial positrons, and the momenta of all positrons are taken with the reversed sign.

Let us call attention to the different character of the two diagrams (73.16). In the first diagram (of the “scattering” type) at one of the vertices the lines of the initial and final electron intersect, and at the other vertex the same for the positron. In the second diagram (of the “annihilation” type) at each of the vertices the lines of electrons and positrons—initial and final—intersect; at the upper vertex there occurs as it were the annihilation of the pair with emission of a virtual photon, and at the lower one the creation of a pair from the photon.

This difference is also reflected in the properties of the virtual photons in both diagrams. In the first diagram the 4-momentum of the virtual photon equals the difference of the 4-momenta of two electrons (or positrons); therefore $k^2 < 0$ (cf. (73.1)). In the second diagram, however, $k' = p_- + p_+$, and therefore $k'^2 > 0$. Let us note in this connection that for a virtual photon $k^2 \neq 0$ always, in contrast to a real photon for which $k^2 = 0$.

If the colliding particles are not identical and are not a particle and antiparticle (say, an electron and a muon), then the amplitude of scattering is represented by just one diagram:

A diagram of the annihilation or exchange type is in this case impossible. We obtain this result also analytically, having written the current operator as the sum of the electronic and muonic currents

$$\hat{j} = \hat{j}^{(e)} + \hat{j}^{(\mu)} = (\hat{\bar{\psi}}^{(e)}\gamma\hat{\psi}^{(e)}) + (\hat{\bar{\psi}}^{(\mu)}\gamma\hat{\psi}^{(\mu)})$$

⁶For scattering of non-identical particles the overall sign of the amplitude is unambiguous. It is determined by the requirement that in (73.5) the “external” operators should be placed so that both electronic operator stood at the edges:

$$\langle 0|a'b' \dots b^+a^+|0\rangle$$

(or both in the middle); by this condition one ensures the “same sign” of the initial and final vacuum states. The overall sign of the amplitude can be verified also by means of the non-relativistic limit: we shall see below (see § 81) that in this limit the second term in (73.15) tends to zero, and the first term tends to the Born amplitude of Rutherford scattering.

and taken in the product $\hat{j}^\mu(x)\hat{j}^\nu(x')$ the matrix elements from the terms producing the required annihilation and creation of particles.

Let us return to the processes forbidden in the first order, as was indicated at the beginning of this section, by the law of conservation of 4-momentum. The matrix elements of the operator

$$\hat{S}^{(1)} = -ie \int \hat{j}(x) \hat{A}(x) d^4x \quad (73.18)$$

for such transitions correspond to creation or annihilation “in one and the same point x ” of three real particles: two electrons and one photon. They arise as a result of contracting the operators

$\hat{\psi}(x)$ and $\hat{\bar{\psi}}(x)$ in one point x and are determined (for example, for the emission of a photon) by integrals of the form

$$S_{fi} = -ie \int \bar{\psi}_2(x) \psi_1(x) (\gamma A^*(x)) d^4x,$$

vanishing due to the presence in the sub-integral expression of the factor $\exp[-i(p_1 - p_2 - k)x]$ with a non-zero exponent. In the language of diagrams this means that diagrams with three free ends vanish:

For the same reason there are also impossible processes of second order in which six particles (in the initial and final states) would participate. In the matrix element S_{fi} the corresponding transitions over $d^4x d^4x'$ would break up into the product of two integrals vanishing to zero, each of three wave functions taken in one and the same point x or x' . In other words, the corresponding diagrams would break up into two independent diagrams of the type (73.19).

§ 74. Feynman diagrams for photon scattering

Let us consider another second-order effect—the scattering of a photon on an electron (*Compton effect*). Let in the initial state the photon and electron have 4-momenta k_1 and p_1 , and in the final state k_2 and p_2 (as well as definite polarisations, which for brevity we do not indicate).

The photonic matrix element

$$\langle 2 | T A_\mu(x) A_\nu(x') | 1 \rangle, \quad (74.1)$$

where

$$\hat{A} = \sum_{\mathbf{k}} (\hat{c}_{\mathbf{k}} A_{\mathbf{k}} + \hat{c}_{\mathbf{k}}^+ A_{\mathbf{k}}^*).$$

Contracting the external and internal operators, we obtain

$$(74.1) = c_2 A_\mu A'_\nu c_1^+ + c_2 A'_\mu A'_\nu c_1^+ = A_{2\mu}^* A'_{1\nu} + A_{1\mu} A_{2\nu}^* \quad (74.2)$$

(in this the commutativity of the operators $\hat{c}_1, \hat{c}_2^+$ has been taken into account; for this same reason the sign T in this case can be omitted). The electronic matrix element

$$\langle 2 | T j^\mu(x) j^\nu(x') | 1 \rangle = \langle 0 | a_2 T(\bar{\psi} \gamma^\mu \psi) (\bar{\psi}' \gamma^\nu \psi') a_1^+ | 0 \rangle. \quad (74.3)$$

In it there figure four ψ -operators. Only two of them will be engaged in annihilating electron 1 and creating electron 2, i.e. will be contracted with the operators $\hat{a}_1^+$ and $\hat{a}_2$. These may be the operators $\bar{\psi}'$, $\hat{\psi}$ or $\bar{\psi}$, $\hat{\psi}'$ (but not $\hat{\psi}$, $\bar{\psi}$ or $\hat{\psi}'$, $\bar{\psi}'$; creation and annihilation in one and the same point x or x' of two real electrons together with one real photon leads to a vanishing expression). Performing the contraction in two ways, we obtain in the matrix element (74.3) two terms; writing them first for $t > t'$:

$$(74.3) = a_2(\bar{\psi}'\gamma^\mu\psi)(\bar{\psi}'\gamma^\nu\psi')a_1^+ + a_2(\bar{\psi}\gamma^\mu\psi)(\bar{\psi}'\gamma^\nu\psi')a_1^+. \quad (74.4)$$

In the first term the contracted operators are

$$\hat{a}_2\hat{\psi} \rightarrow \hat{a}_2\hat{a}_2^+\bar{\psi}_2, \quad \hat{\psi}'\hat{a}_1^+ \rightarrow \hat{a}_1\hat{a}_1^+\psi'_1.$$

Since the operators $\hat{a}_2\hat{a}_2^+$ are diagonal and stand at the edges of the product, they are replaced by their vacuum value, i.e. by unity. For the analogous transformation of the second term in (74.4) one must first “push” the operator $\hat{a}_2^+$ to the left, and $\hat{a}_1$ to the right. This is accomplished with the help of the commutation rules for the operators $\hat{a}_p$, $\hat{a}_p^+$, by virtue of which

$$\{\hat{a}_p, \hat{\psi}\}_+ = \{\hat{a}_p^+, \hat{\psi}\}_+ = 0, \quad (74.5)$$

$$\{\hat{a}_p, \hat{\psi}\}_+ = \bar{\psi}_p, \quad \{\hat{a}_p^+, \hat{\psi}\}_+ = \psi_p.$$

As a result, expression (74.4) transforms to the form

$$\langle 0 | (\bar{\psi}_2\gamma^\mu\hat{\psi})(\bar{\psi}'\gamma^\nu\psi'_1) - (\bar{\psi}\gamma^\mu\psi_1)(\bar{\psi}_2\gamma^\nu\hat{\psi}') | 0 \rangle, \quad t > t' \quad (74.6)$$

(it is understood that the averaging is over only the operator factors). Analogously for $t < t'$ we obtain the expression differing by the permutation of the prime and the indices μ, ν :

$$\langle 0 | -(\hat{\psi}'\gamma^\nu\psi'_1)(\bar{\psi}_2\gamma^\mu\hat{\psi}) + (\bar{\psi}_2'\gamma^\nu\hat{\psi}')(\hat{\psi}\gamma^\mu\psi_1) | 0 \rangle, \quad t < t'. \quad (74.7)$$

Both expressions can be written in a unified form by introducing the *chronological product of ψ -operators* according to the definition

$$T\hat{\psi}_i(x)\hat{\psi}_k(x') = \begin{cases} \hat{\psi}_i(x)\hat{\psi}_k(x'), & t' < t, \\ -\hat{\psi}_k(x')\hat{\psi}_i(x), & t' > t \end{cases} \quad (74.8)$$

(i, k are bispinor indices). Then the first and second terms in (74.6), (74.7) can be written in a unified form:

$$\bar{\psi}_2\gamma^\mu\langle 0 | T\psi \cdot \bar{\psi}' | 0 \rangle \gamma^\nu\psi'_1 + i\bar{\psi}_2'\gamma^\nu\langle 0 | T\psi' \cdot \bar{\psi} | 0 \rangle \gamma^\mu\psi_1 \quad (74.9)$$

($\psi \cdot \bar{\psi}$ denotes the matrix $\psi_i\bar{\psi}_k$).

Let us call attention to the fact that in the definition (74.8) the products of operators at $t < t'$ and $t > t'$ are taken with different signs. In this it differs from the definition of the T-product to which we resorted for the operators $\hat{A}$ and $\hat{j}$. The origin of this difference is connected with the fact that the fermionic operators $\hat{\psi}$, $\hat{\psi}'$ anticommute outside the light cone (in contrast to the commuting bosonic operators $\hat{A}$), and also

with the bilinearity of the operators $\hat{j} = \hat{\bar{\psi}}\gamma\hat{\psi}$ ⁷. Thereby the relativistic invariance of definition (74.8) is ensured (the formal proof of the commutation rules for ψ -operators will be given in § 75)⁸.

Let us introduce the *electron propagator* (or *electron Green's function*)—a bispinor of second rank $G_{ik}(x - x')$ —according to the definition

$$G_{ik}(x - x') = -i\langle 0 | T \psi_i(x) \bar{\psi}_k(x') | 0 \rangle. \quad (74.10)$$

Then the electronic matrix element is written in the form

$$\langle 2 | T j^\mu(x) j^\nu(x') | 1 \rangle = i\bar{\psi}_2 \gamma^\mu G \gamma^\nu \psi'_1 + i\bar{\psi}'_2 \gamma^\nu G \gamma^\mu \psi_1. \quad (74.11)$$

After multiplication by the photonic matrix element (74.1) and integration over $d^4x d^4x'$, both terms in (74.11) give the same result, so that there arises

$$S_{fi} = -ie^2 \iint d^4x d^4x' \bar{\psi}_2(x) \gamma^\mu G(x - x') \gamma^\nu \psi_1(x') \times \{A_{2\mu}^*(x) A_{1\nu}(x') + A_{2\nu}^*(x') A_{1\mu}(x)\}. \quad (74.12)$$

Substituting for the electronic and photonic wave functions plane waves (64.8), (64.9) and isolating the δ -function as was done for (73.10), we obtain finally the amplitude of scattering

$$M_{fi} = -4\pi e^2 \bar{u}_2 \{ (\gamma e_2^*) G(p_1 + k_1) (\gamma e_1) + (\gamma e_1) G(p_1 - k_2) (\gamma e_2^*) \} u_1, \quad (74.13)$$

where e_1, e_2 are the 4-vectors of polarisation of the photons, $G(p)$ is the electron propagator in the momentum representation.

The two terms in this expression are represented by the following Feynman diagrams:

$$\begin{aligned} 4\pi e^2 \bar{u}_2 (\gamma e_2^*) G(f) (\gamma e_1) u_1 &= \\ 4\pi e^2 \bar{u}_2 (\gamma e_1) G(f') (\gamma e_2^*) u_1 &= \end{aligned} \quad (74.14)$$

The dashed free ends correspond to real photons; to the incoming lines (initial photon) there corresponds the factor $\sqrt{4\pi} e_\mu$, and to the outgoing lines (final photon) the factor $\sqrt{4\pi} e^*$, where e is the 4-vector of polarisation. In the first diagram the initial photon is absorbed together with the initial electron, and the final photon is emitted together with the final electron. In the second diagram the emission of the final photon occurs together with the annihilation of the initial electron, and the absorption of the initial photon together with the creation of the final electron.

The internal solid line (joining both vertices) corresponds to a virtual electron, whose 4-momentum is determined by conservation of 4-momentum in the vertices. To this line there corresponds the factor $iG(f)$. In contrast to the 4-momentum of a real particle, the square of the 4-momentum of the virtual electron is not equal to m^2 . Considering the invariant f^2 , for example, in the rest frame of the electron, it is easy to find that

$$f^2 = (p_1 + k_1)^2 > m^2, \quad f'^2 = (p_1 - k_2)^2 < m^2. \quad (74.15)$$

⁷Let us recall that the ψ -operators by themselves do not correspond to any measurable physical quantities and are therefore not required to be commutative outside the light cone.

⁸Analogously one can define the T-product of any number of ψ -operators. It is equal to the product of all these operators, placed in a row in the order of decreasing times, the sign being determined by the parity of the permutation which it is necessary to perform in order to obtain this order from the order indicated under the sign T.

§ 75. Electron propagator

The concept of propagators (Green's functions) introduced in the preceding sections plays a fundamental role in the apparatus of quantum electrodynamics. The photon propagator $D_{\mu\nu}$, occupying in it a position as the basic quantity characterising the interaction of two electrons, is the same role that the electron propagator plays in the interaction of the electron and the photon.

Let us now turn to the actual computation of the propagators, starting with the electronic case. We act on the function

$$G_{ik}(x - x') = -i\langle 0 | T \psi_i(x) \bar{\psi}_k(x') | 0 \rangle \quad (75.1)$$

(i, k are bispinor indices, where $\hat{p}_\mu = i\partial_\mu$). Since the operator $\hat{\psi}(x)$ satisfies the Dirac equation $(\gamma\hat{p} - m)\hat{\psi}(x) = 0$, we obtain zero at all points x , with the exception only of those for which $t = t'$. The point is that $G(x - x')$ tends to different limits when $t \rightarrow t' + 0$ and $t \rightarrow t' - 0$: according to definition (74.8) these limits are respectively

$$-i\langle 0 | \psi_i(\mathbf{r}, t) \bar{\psi}_k(\mathbf{r}', t) | 0 \rangle \quad \text{and} \quad +i\langle 0 | \bar{\psi}_k(\mathbf{r}', t) \psi_i(\mathbf{r}, t) | 0 \rangle$$

and, as we shall see, on the light cone these do not coincide. This leads to the appearance in the derivative $\partial G / \partial t$ of an additional term with a δ -function:

$$\frac{\partial G}{\partial t} = -i\langle 0 | T \frac{\partial \psi_i(x)}{\partial t} \bar{\psi}_k(x') | 0 \rangle + \delta(t - t')(G|_{t \rightarrow t' + 0} - G|_{t \rightarrow t' - 0}). \quad (75.2)$$

Noting that the operator $\gamma\hat{p} - m$ produces the derivative with respect to t in the form $i\gamma^0 \partial / \partial t$, we have therefore

$$(\gamma\hat{p} - m)_{ik} G_{kl}(x - x') = \delta(t - t') \gamma_{ik}^0 \langle 0 | \{\psi_k(\mathbf{r}, t), \bar{\psi}_l(\mathbf{r}', t)\}_+ | 0 \rangle. \quad (75.3)$$

We compute the anticommutator standing here. Multiplying the operators $\hat{\psi}(\mathbf{r}, t)$ and $\hat{\psi}(\mathbf{r}', t)$ (see (73.6)) and using the commutation rules for fermionic operators $\hat{a}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}$, we find

$$\{\hat{\psi}_i(\mathbf{r}, t), \hat{\psi}_k(\mathbf{r}', t)\}_+ = \sum_{\mathbf{p}} [\psi_{pi}(\mathbf{r}) \bar{\psi}_{pk}(\mathbf{r}') + \psi_{-pi}(\mathbf{r}) \bar{\psi}_{-pk}(\mathbf{r}')], \quad (75.4)$$

where $\psi_{\pm\mathbf{p}}(\mathbf{r})$ are the wave functions without the time factor (as in §§ 73, 74, for brevity we do not write out the polarisation indices). The totality of all functions $\psi_{\pm\mathbf{p}}(\mathbf{r})$ —the eigenfunctions of the Hamiltonian of the electron—constitutes a complete system of normalised functions, and according to the general properties of such systems (cf. III, (5.12)):

$$\sum_{\mathbf{p}} [\psi_{pi}(\mathbf{r}) \psi_{pk}^*(\mathbf{r}') + \psi_{-pi}(\mathbf{r}) \psi_{-pk}^*(\mathbf{r}')] = \delta_{ik} \delta(\mathbf{r} - \mathbf{r}'). \quad (75.5)$$

The sum on the right-hand side of (75.4) differs from the sum written above by the replacement of ψ_k^* by $(\psi^* \gamma^0)_k$, and equals $\gamma_{kk'}^0 \delta(\mathbf{r} - \mathbf{r}')$. Thus,

$$\{\hat{\psi}_i(\mathbf{r}, t), \hat{\psi}_k(\mathbf{r}', t)\}_+ = \delta(\mathbf{r} - \mathbf{r}') \gamma_{ik}^0. \quad (75.6)$$

Let us note that from this formula it follows, in particular, the assertion mentioned in § 74, about the anticommutativity of the operators $\hat{\psi}$ and $\hat{\bar{\psi}}$ outside the light cone. When $(x - x')^2 < 0$ there always exists such a system of reference in which $t = t'$; while at the same time $\mathbf{r} \neq \mathbf{r}'$, so that the anticommutator (75.6) is effectively equal to zero.

Substituting (75.6) into (75.3) (and dropping the bispinor indices), we find finally⁹

$$(\gamma \hat{p} - m) G(x - x') = \delta^{(4)}(x - x'). \quad (75.7)$$

Thus, the electron propagator satisfies the Dirac equation with a δ -function on the right-hand side. In other words, it is the *Green's function* of the Dirac equation.

In the following we shall have to deal not with the function $G(\xi)$ itself ($\xi = x - x'$), but with its Fourier components

$$G(p) = \int G(\xi) e^{ip\xi} d^4\xi \quad (75.8)$$

(the propagator in the momentum representation). Taking the Fourier component of both sides of (75.7), we find that $G(p)$ satisfies the system of algebraic equations

$$(\gamma p - m) G(p) = 1. \quad (75.9)$$

The solution of this system:

$$G(p) = \frac{\gamma p + m}{p^2 - m^2}. \quad (75.10)$$

The four components of the 4-vector p in $G(p)$ are independent variables (not connected with the relativistic equation $p^2 \equiv p_0^2 - \mathbf{p}^2 = m^2$). Writing the denominator in (75.10) in the form $p_0^2 - (\mathbf{p}^2 + m^2)$, we see that $G(p)$ as a function of p_0 at a given $\mathbf{p}$ has two poles: at $p_0 = \pm \varepsilon$, where $\varepsilon = \sqrt{\mathbf{p}^2 + m^2}$. In the integration over dp_0 in the integral

$$G(\xi) = \frac{1}{(2\pi)^4} \int e^{-ip\xi} G(p) d^4p = \frac{1}{(2\pi)^4} \int d^3p \cdot e^{i\mathbf{p}\mathbf{r}} \int dp_0 \cdot e^{-ip_0\tau} G(p) \quad (75.11)$$

($\tau = t - t'$) the question therefore arises about the manner of going around the poles; without indicating this manner expression (75.10) is essentially still indeterminate.

To elucidate this question let us return to the original definition (75.1). Substituting into it the ψ -operators in the form of the sums (73.6), and noting that from the vacuum average of zero only the following products of creation and annihilation operators are different from zero:

$$\langle 0 | a_{\mathbf{p}} a_{\mathbf{p}}^+ | 0 \rangle = 1, \quad \langle 0 | b_{\mathbf{p}} b_{\mathbf{p}}^+ | 0 \rangle = 1.$$

⁹In the explicit notation with bispinor indices

$$(\gamma \hat{p} - m)_{ik} G_{kl}(x - x') = \delta^{(4)}(x - x') \delta_{il}. \quad (75.7a)$$

(Since in the vacuum state there are no particles, before “annihilating” a particle by the operator $\hat{a}_{\mathbf{p}}$ or $\hat{b}_{\mathbf{p}}$, one must first “create” it by its operator $\hat{a}_{\mathbf{p}}^+$ or $\hat{b}_{\mathbf{p}}^+$.) We obtain

$$\begin{aligned} G_{ik}(x - x') &= -i \sum_{\mathbf{p}} \psi_{pi}(\mathbf{r}, t) \bar{\psi}_{pk}(\mathbf{r}', t) = \\ &= -i \sum_{\mathbf{p}} e^{-i\varepsilon(t-t')} \psi_{pi}(\mathbf{r}) \bar{\psi}_{pk}(\mathbf{r}'), \quad t - t' > 0; \\ G_{ik}(x - x') &= i \sum_{\mathbf{p}} \bar{\psi}_{-pk}(\mathbf{r}', t') \psi_{-pi}(\mathbf{r}, t) = \\ &= i \sum_{\mathbf{p}} e^{i\varepsilon(t-t')} \psi_{-pi}(\mathbf{r}) \bar{\psi}_{-pk}(\mathbf{r}') \end{aligned} \quad (75.12)$$

(when $t - t' > 0$ only the electronic terms contribute to G , and when $t - t' < 0$ only the positronic ones).

Imagining the summation over $\mathbf{p}$ to be replaced by integration over d^3p , and comparing (75.12) with (75.11), we see that the integral

$$\int e^{-ip_0\tau} G(p) dp_0 \quad (75.13)$$

should have the phase factor $e^{-i\varepsilon\tau}$ when $\tau > 0$ and $e^{i\varepsilon\tau}$ when $\tau < 0$. We satisfy this condition if we agree to go around the poles $p_0 = \varepsilon$ and $p_0 = -\varepsilon$ respectively from above and from below (in the plane of the complex variable p_0):

Indeed, when $\tau > 0$ we close the integration contour by an infinitely distant semicircle in the lower half-plane, so that the value of the integral (75.13) will be given by the residue in the pole $p_0 = +\varepsilon$; when $\tau < 0$ we close the contour in the upper half-plane, and the integral is determined by the residue in the pole $p_0 = -\varepsilon$. In both cases the required result is obtained.

This rule of going around (the *Feynman rule*) can be formulated otherwise: integration is performed everywhere along the real axis itself, but to the mass of the particle there is ascribed an infinitely small negative imaginary part:

$$m \rightarrow m - i0. \quad (75.15)$$

Indeed, then

$$\varepsilon \rightarrow \sqrt{\mathbf{p}^2 + (m - i0)^2} = \sqrt{\mathbf{p}^2 + m^2 - i0} = \varepsilon - i0.$$

In other words, the poles $p_0 = \pm\varepsilon$ shift downward and upward from the real axis: so that integration along the real axis becomes equivalent to integration along the contour (75.14)¹⁰. Taking into account the rule (75.15), the propagator (75.10) can be written in the form

$$G(p) = \frac{\gamma p + m}{p^2 - m^2 + i0}. \quad (75.17)$$

¹⁰It is useful to note that the rule of pole displacement corresponds to the fact that $G(x - x')$ acquires an infinitely small damping at $|\tau|$, where $\tau = t - t'$. Indeed, if one writes the value of p_0 in the displaced poles as $-(\varepsilon - i\delta)$ and $(\varepsilon - i\delta)$ (where $\delta \rightarrow +0$), then the temporal factor in the integral (75.13) will equal $\exp(-i\varepsilon|\tau| - \delta|\tau|)$.

The rule of integration under displacement of the poles is demonstrated by the following relation:

$$\frac{1}{x+i0} = P\frac{1}{x} - i\pi\delta(x). \quad (75.18)$$

It must be understood in the sense that upon multiplication by any function $f(x)$ and integration we have

$$\int_{-\infty}^{\infty} \frac{f(x) dx}{x+i0} = \int_{-\infty}^{\infty} \frac{f(x) dx}{x} - i\pi f(0), \quad (75.19)$$

where the crossed-out integral sign, or the symbol P, denotes the principal value.

The Green's function (75.10) represents the product of the bispinor factor $\gamma p + m$ and a scalar:

$$G^{(0)}(p) = \frac{1}{p^2 - m^2}. \quad (75.20)$$

The corresponding coordinate function $G^{(0)}(\xi)$ is, evidently, a solution of the equation

$$(\hat{p}^2 - m^2) G^{(0)}(x - x') = \delta^{(4)}(x - x'), \quad (75.21)$$

i.e. the Green's function of the equation $(\hat{p}^2 - m^2)\psi = 0$. In this sense one can say that $G^{(0)}(x - x')$ is the propagator of scalar particles. It is easy to verify by computation (analogous to the one performed above) that the propagator of a scalar field is expressed through the ψ -operators (11.2) by the formula

$$G^{(0)}(x - x') = -i\langle 0 | T\psi(x)\psi^+(x') | 0 \rangle. \quad (75.22)$$

analogous to definition (75.1). In this the chronological product is defined (as for all bosonic operators) in the following manner:

$$T\hat{\psi}(x)\hat{\psi}^+(x') = \begin{cases} \hat{\psi}(x)\hat{\psi}^+(x'), & t > t', \\ \hat{\psi}^+(x')\hat{\psi}(x), & t < t' \end{cases} \quad (75.23)$$

(with the same signs when $t > t'$ and $t < t'$).

§ 76. Photon propagator

Until now it was necessary (in §§ 43, 74) to use the explicit form of the operators of the electromagnetic field $\hat{A}$ when finding matrix elements only for transitions involving a change of the number of real photons. For this purpose it was sufficient to write the potentials of the free field in the form of the expansion (§ 2) in transverse plane waves.

Such a representation, however, does not by itself provide a complete description of the arbitrary field. This is already clear from the fact that the diagrams of scattering (73.13), (73.14) should describe also the Coulomb interaction of the electrons. The latter is described by the scalar potential Φ , and cannot obviously be reduced to the exchange of only transverse virtual photons.

These difficulties, however, have only a formal, not physical, character, and they can be circumvented by using some general properties of the propagator, obvious from the requirements of relativistic and gauge invariance.

The most general form of a tensor of second rank, depending only on the 4-vector $\xi = x - x'$, is

$$D_{\mu\nu}(\xi) = g_{\mu\nu}D(\xi^2) - \partial_\mu\partial_\nu D^{(l)}(\xi^2), \quad (76.2)$$

where $D, D^{(l)}$ are scalar functions of the invariant ξ^2 ¹¹. Let us note that the tensor automatically turns out to be symmetric. Correspondingly in the momentum representation we shall have

$$D_{\mu\nu}(k) = D(k^2)g_{\mu\nu} + k_\mu k_\nu D^{(l)}(k^2), \quad (76.3)$$

where $D(k^2), D^{(l)}(k^2)$ are the Fourier components of $D(\xi^2), D^{(l)}(\xi^2)$.

In the physical quantities—the amplitudes of scattering—the photon propagator enters multiplied by the transition currents of two electrons, i.e. in combinations of the form $j_{21}^\mu D_{\mu\nu} j_{43}^\nu$ (see, for example, (73.13)). But by virtue of conservation of the current ($\partial_\mu j^\mu = 0$) its matrix elements $j_{21} = \bar{\psi}_2 \gamma \psi_1$ satisfy the condition of 4-transversality

$$k_\mu (j^\mu)_{21} = 0, \quad (76.4)$$

where $k = p_2 - p_1$ (cf. (43.13)). It is clear therefore that the physical results will not change under the replacement

$$D_{\mu\nu} \rightarrow D_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \quad (76.5)$$

where χ_μ are any functions of $\mathbf{k}$ and k_0 . This arbitrariness in the choice of $D_{\mu\nu}$ corresponds to the arbitrariness of gauge of the potentials of the field. An arbitrary gauge transformation (76.5) may violate the relativistically invariant form of $D_{\mu\nu}$ assumed in (76.3) (if the quantities χ_μ do not constitute a 4-vector). But also remaining within the framework of relativistically invariant forms of the propagator, we see that the choice of the function $D^{(l)}(k^2)$ in (76.3) is entirely arbitrary; it does not affect the physical results and can be specified from considerations of convenience (*L. D. Landau, A. A. Abrikosov, I. M. Khalatnikov*, 1954).

The determination of the propagator reduces, thus, to the determination of one gauge-invariant

function $D(k^2)$. If one considers a given value of k^2 and chooses the z -axis along the direction of $\mathbf{k}$, then the transformations (76.5) will not affect the components $D_{xx} = D_{yy} = -D(k^2)$. It is therefore sufficient to compute just one component D_{xx} .

Let us use the gauge in which $\text{div } \hat{\mathbf{A}} = 0$ and the operator $\hat{A}$ is given by the expansion (2.17), (2.18):

$$\hat{A} = \sum_{\mathbf{k}\alpha} \sqrt{\frac{2\pi}{\omega}} (\hat{c}_{\mathbf{k}\alpha} \mathbf{e}^{(\alpha)} e^{-ikx} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{e}^{(\alpha)*} e^{ikx}), \quad \omega = |\mathbf{k}| \quad (76.6)$$

(the index $\alpha = 1, 2$ labels the polarisations). From all vacuum averages of products of creation and annihilation operators only the following is different from zero:

$$\langle 0 | c_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha}^+ | 0 \rangle = 1.$$

¹¹These functions are different in three regions of values of the argument, not transforming into one another under Lorentz transformations: outside the light cone ($\xi^2 < 0$), in the upper ($\xi^2 > 0, \xi_0 > 0$) and in the lower ($\xi^2 > 0, \xi_0 < 0$) halves of the light cone.

From definition (76.1) we obtain therefore

$$D_{ik}(\xi) = \frac{1}{(2\pi)^3} \int \frac{2\pi i d^3 k}{\omega} \left(\sum_{\alpha} e_i^{(\alpha)} e_k^{(\alpha)*} \right) e^{-i\omega|\tau| + i\mathbf{k}\xi} \quad (76.7)$$

(i, k are three-dimensional vector indices; from summation over $\mathbf{k}$ we have passed to integration over $d^3 k / (2\pi)^3$). The fact that in the exponent stands the absolute value of the difference $\tau = t - t'$ is a consequence of the chronologisation of the product of operators in (76.1).

From (76.7) it is seen that the sub-integral expression without the factor $e^{i\mathbf{k}\xi}$ is the Fourier component of the three-dimensional function $D_{ik}(\mathbf{r}, t)$. For $D_{xx} = -D$ it equals

$$\frac{2\pi i}{\omega} e^{-i\omega|\tau|} \sum_{\alpha} |e_x^{(\alpha)}|^2 = \frac{2\pi i}{\omega} e^{-i\omega|\tau|}.$$

It remains to decompose $D_{xx}(k^2)$ in the Fourier transform with respect to time:

$$\frac{2\pi i}{\omega} e^{-i\omega|\tau|} = -\frac{1}{2\pi} \int \frac{4\pi}{k_0^2 - \mathbf{k}^2 + i0} e^{-ik_0\tau} dk_0.$$

As was explained in the preceding section, such an integration implies going around the poles $k_0 = \pm|\mathbf{k}| = \pm\omega$ respectively from below and above; when $\tau > 0$ the integral is determined by the residue in the pole $k_0 = +\omega$, and when $\tau < 0$ by the residue in the pole $k_0 = -\omega$. In both cases the required result is obtained.

Thus we find finally

$$D(k^2) = \frac{4\pi}{k^2 + i0}. \quad (76.8)$$

The appearance of $+i0$ in the denominator, to which in the above derivation we arrived automatically, coincides with the rule (75.15): from (the zero) mass of the photon $i0$ is subtracted. From (76.8) it is seen that the corresponding coordinate function $D(\xi^2)$ satisfies the equation

$$-\partial_i \partial^i D(x - x') = 4\pi \delta^{(4)}(x - x'), \quad (76.9)$$

i.e. it is the Green's function of the wave equation.

We shall ordinarily use the propagator in the form

$$D_{\mu\nu} = g_{\mu\nu} D(k^2) = \frac{4\pi}{k^2 + i0} g_{\mu\nu} \quad (76.10)$$

(*Feynman gauge*).

Let us indicate also other gauges, which can present certain advantages in some applications.

Setting $D^{(l)} = -D/k^2$, we obtain the propagator in the form

$$D_{\mu\nu} = \frac{4\pi}{k^2} \left(g_{\mu\nu} - \frac{k_{\mu} k_{\nu}}{k^2} \right) \quad (76.11)$$

(*Landau gauge*). In this case $D_{\mu\nu} k^{\nu} = 0$. Such a choice is analogous to the Lorentz gauge of the potentials ($A_{\mu} k^{\mu} = 0$).

To the gauge of the potentials by the three-dimensional condition $\text{div } \mathbf{A} = 0$ the analogous gauge of the propagator is defined by the conditions

$$D_{il}k^l = 0, \quad D_{0l}k^l = 0.$$

Together with the equality $D_{xx} = -D = -4\pi/k^2$ these conditions give

$$D_{il} = \frac{4\pi}{\omega^2 - \mathbf{k}^2} \left(\delta_{il} - \frac{k_i k_l}{\mathbf{k}^2} \right). \quad (76.12)$$

In order to obtain also D_{il} , one must perform the transformation (76.5), setting

$$\chi_0 = -\frac{4\pi\omega}{2(\omega^2 - \mathbf{k}^2)\mathbf{k}^2}, \quad \chi_i = \frac{4\pi k_i}{2(\omega^2 - \mathbf{k}^2)\mathbf{k}^2}.$$

In this case for the remaining components of $D_{\mu\nu}$ we obtain

$$D_{00} = -4\pi/\mathbf{k}^2, \quad D_{0i} = 0. \quad (76.13)$$

Such a gauge is called the *Coulomb gauge* (E. Salpeter, 1952); let us note that D_{00} here is the Fourier component of the Coulomb potential.

Finally, the gauge of the potentials by the condition $\Phi = 0$ is analogous to the gauge of the propagator in which

$$D_{il} = -\frac{4\pi}{\omega^2 - \mathbf{k}^2} \left(\delta_{il} - \frac{k_i k_l}{\omega^2} \right), \quad D_{0i} = D_{00} = 0. \quad (76.14)$$

This form is found convenient for applications in non-relativistic problems (I. E. Dzyaloshinskii, L. P. Pitaevskii, 1959).

All the expressions given pertain to the momentum representation of the propagator. In some problems it is convenient to use the mixed frequency-coordinate representation, i.e. the function

$$D_{\mu\nu}(\omega, \mathbf{r}) = \int D_{\mu\nu}(\omega, \mathbf{k}) e^{i\mathbf{k}\mathbf{r}} \frac{d^3 k}{(2\pi)^3}. \quad (76.15)$$

In the Feynman gauge (76.10)

$$D_{\mu\nu}(\omega, \mathbf{r}) = g_{\mu\nu} D(\omega, \mathbf{r}),$$

where

$$D(\omega, \mathbf{r}) = 4\pi \int \frac{e^{i\mathbf{k}\mathbf{r}}}{(2\pi)^3} \frac{d^3 \mathbf{k}}{\omega^2 - \mathbf{k}^2 + i0} = -\frac{i}{\pi r} \int_0^\infty \frac{e^{ikr} - e^{-ikr}}{\omega^2 - k^2 + i0} k dk,$$

or, after the replacement $k \rightarrow -k$ in the second term of the sub-integral expression:

$$D(\omega, \mathbf{r}) = -\frac{i}{\pi r} \int_{-\infty}^\infty \frac{e^{ikr} k dk}{\omega^2 - k^2 + i0}.$$

The last integration is performed by closing the contour of integration by an infinitely distant semicircle in the upper half-plane of the complex variable k and reducing to the residue at the pole $k = |\omega| + i0$. We finally obtain

$$D(\omega, \mathbf{r}) = -e^{i|\omega|r}/r. \quad (76.16)$$

In connection with this expression let us make the following remark. The process described by diagrams (73.13), (73.14) can be regarded as scattering of electron 2 in the field created by electron 1 (or vice versa). The function (76.16) corresponds to the ordinary “retarded” potential $\propto e^{i\omega r}$ (see II, (64.1), (64.2)) only for $\omega > 0$. The sign ω , however, depends on the conventional choice of direction of the arrow k on the diagram. The noted property of the function $D(\omega, \mathbf{r})$ means that in quantum electrodynamics the source of the field should be considered that particle which emits energy, i.e. emits a virtual photon.

In conclusion let us dwell on the question of the propagator of a particle with spin 1 and with a mass different from zero. In this case the gauge arbitrariness is absent and the choice of the propagator is unambiguous.

Substituting the ψ -operators (14.16) into the definition

$$G_{\mu\nu} = -i\langle 0 | T \psi_\mu(x) \psi_\nu^+(x') | 0 \rangle, \quad (76.17)$$

we obtain an expression differing from (76.7) only by the replacement of the sum over polarisations in the sub-integral expression by

$$\sum_\alpha u_\mu^{(\alpha)} u_\nu^{(\alpha)*}.$$

The summation over polarisations is equivalent, with a subsequent multiplication by 3, to averaging over the number of independent polarisations. The averaging gives the density matrix of unpolarised particles (14.15). Thus we find the following expression for the propagator of massive vector particles:

$$G_{\mu\nu}(p) = -\frac{1}{p^2 - m^2 + i0} \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right). \quad (76.18)$$

Let us call attention to the analogous structure of the propagators (75.17) and (76.18): in the denominator stands the difference $p^2 - m^2$, and the numerator is, to within a factor, the density matrix of unpolarised particles with a given spin.

§ 77. General rules of diagram technique

The computation performed in §§ 73, 74 for a few simple cases of elements of the scattering matrix contains in itself all the principal elements of the general method. It presents no special difficulty to establish the corresponding generalisations by appropriate generalisation of the rules of computing matrix elements in any order of perturbation theory.

As has already been indicated, the matrix element of the scattering operator $\hat{S}$ for the transition between any initial and final states coincides with the vacuum average of the operator obtained by multiplying $\hat{S}$ from the right by operators of creation of all initial particles, and from the left by operators of annihilation of all final particles. The computation of such averages is accomplished with the help of the following statements constituting the content of *Wick's theorem* (G. C. Wick, 1950).

1. The vacuum average of a product of any number of bosonic operators $\hat{c}^+$, $\hat{c}$ is equal to the sum of products of all possible pairwise averages (contractions) of these

operators. In this each pair the factors must stand in the same order as in the original product.

2. For fermionic operators $\hat{a}^+, \hat{a}, \hat{b}^+, \hat{b}$ (of the same or different particles) the rule changes only in that each term enters the sum with a sign plus or minus depending on the evenness or oddness of the number of permutations of fermionic operators needed to place side by side all the pairwise averaged operators.

It is clear that the average can be different from zero only if alongside each factor $\hat{a}, \hat{b}, \hat{c}$ in the product there is also a factor $\hat{a}^+, \hat{b}^+, \hat{c}^+$. In this contractions should include only pairs of operators $(\hat{a}, \hat{a}^+), \dots$, relating to the same states, and only those in which $\hat{a}^+, \dots$ stands to the right of $\hat{a}, \dots$: a particle is first created and then annihilated ($\langle 0|a^+a|0\rangle = 0, \dots$).

If each pair $(\hat{a}, \hat{a}^+), \dots$ enters the product only once, then Wick's theorem is obvious (the average reduces to a single product of pairwise averages). It is also obvious in the case when all operators of annihilation stand to the right of the operators of creation in the product (such a product is called *normal-ordered*); the average is in this case equal to zero. From here it is easy by complete induction to prove Wick's theorem for the general case, when one and the same pair of operators enters the product several (k) times.

Let us consider the vacuum average $\langle 0|\dots cc^+\dots|0\rangle$, in which a pair of bosonic operators $\hat{c}, \hat{c}^+$ enters k times (for fermionic operators the further arguments are completely analogous). Rearranging the factors $\hat{c}, \hat{c}^+$ in some pair, we obtain on the basis of the commutation rules

$$\langle 0|\dots cc^+\dots|0\rangle = \langle 0|\dots c^+c\dots|0\rangle + \langle 0|\dots 1\dots|0\rangle. \quad (77.2)$$

The average $\langle 0|\dots 1\dots|0\rangle$ contains $k-1$ pairs, and for it Wick's theorem is assumed to hold. On the other hand, if one expands the average $\langle 0|\dots c^+c\dots|0\rangle$ according to Wick's theorem, it will differ from the average $\langle 0|\dots c^+c\dots|0\rangle$ by precisely the term $\langle 0|\dots 1\dots|0\rangle$

$$\langle 0|\dots 1\dots|0\rangle \langle 0|cc^+|0\rangle = \langle 0|\dots 1\dots|0\rangle$$

(in expanding $\langle 0|\dots c^+c\dots|0\rangle$ the analogous term $\langle 0|\dots 1\dots|0\rangle \times \langle 0|c^+c|0\rangle = 0$). Therefore from (77.2) it follows that if Wick's theorem is valid for the matrix element $\langle 0|\dots c^+c\dots|0\rangle$, then it remains valid also after the permutation of $\hat{c}$ and $\hat{c}^+$. Since for one definite (normal) ordering of the factors Wick's theorem holds trivially, it is thereby true in any case.

Being valid for products of operators $\hat{a}, \hat{b}, \dots$, Wick's theorem is valid also for any products containing alongside the $\hat{a}, \hat{b}, \dots$ also their linear combinations $\hat{\psi}, \hat{\bar{\psi}}, \hat{A}$. Applying this theorem to the matrix element (77.1), we represent it in the form of a sum of terms, each of which will be a product of some pairwise averages. Among the latter there will be contractions of the operators $\hat{\psi}, \hat{\bar{\psi}}, \hat{A}$ with "external" operators—operators of creation of initial or annihilation of final particles. These contractions are expressed through the wave functions of the initial and final particles according to the formulae:

$$\begin{aligned} \langle 0|Ac_{\mathbf{p}}^+|0\rangle &= A_p, & \langle 0|c_{\mathbf{p}}A|0\rangle &= A_p^*, \\ \langle 0|\psi a_{\mathbf{p}}^+|0\rangle &= \psi_p, & & \\ \langle 0|a_{\mathbf{p}}\bar{\psi}|0\rangle &= \bar{\psi}_p^*, & \langle 0|b_{\mathbf{p}}\psi|0\rangle &= \psi_{-p}, & \langle 0|\bar{\psi}b_{\mathbf{p}}^+|0\rangle &= \bar{\psi}_{-p}, \end{aligned} \quad (77.3)$$

where A_p , ψ_p are the photonic and electronic wave functions with momenta $\mathbf{p}$ (as in §§ 73, 74, for brevity we do not write out the polarisation indices). There will also be contractions of the operators $\hat{\psi}$, $\hat{\bar{\psi}}$, $\hat{A}$ standing under the sign of T-products. Since in applying Wick's theorem the ordering of factors in each contracting pair is preserved, in these contractions the chronological ordering of the operators is also preserved, so that they are replaced by the corresponding propagators¹².

Each of the terms of the sum into which the matrix element is decomposed as a result of its expansion according to Wick's theorem is represented by a definite Feynman diagram. A diagram of the n -th approximation contains n vertices, to each of which there corresponds one of the variables of integration—one of the 4-vectors $x_1, x_2, \dots$. To each vertex there come three rays—two solid (electronic) and one dashed (photonic), corresponding respectively to the electronic ($\hat{\psi}$ and $\hat{\bar{\psi}}$) and photonic ($\hat{A}$) operators as functions of one and the same variable x . In this the operator $\hat{\psi}$ corresponds to a line coming into the vertex, and $\hat{\bar{\psi}}$ to a line going out from it.

For illustration let us present several examples of correspondence between members of the matrix element of the third approximation and diagrams. Dropping the sign of the integral, the sign of the operators, and the sign T, and also the factors $-ie\gamma$ and not writing out the arguments x of the operators, let us write these terms symbolically in the form

$$\begin{aligned}
 a & (\bar{\psi} A \psi)(\bar{\psi} A \psi)(\bar{\psi} A \psi) \\
 b & (\bar{\psi} A \psi)(\bar{\psi} A \psi)(\bar{\psi} A \psi) \\
 c & (\bar{\psi} A \psi)(\bar{\psi} A \psi)(\bar{\psi} A \psi) \\
 d & (\bar{\psi} A \psi)(\bar{\psi} A \psi)(\bar{\psi} A \psi)
 \end{aligned} \tag{77.4}$$

For clarity, electronic and photonic contractions are depicted on the diagram as solid and dashed arcs respectively. The direction of the arrows on the electronic contractions (from $\bar{\psi}$ to ψ) corresponds to the direction of the electron propagator; for the internal photonic contractions the direction is immaterial (which is also manifested in the evenness of the photon propagator as a function of $x - x'$).

Among the terms obtained in this way there are equivalent ones, differing only by the permutation of vertex numbers, i.e. simply by the designation of the variables of integration $x_1, x_2, \dots$. The number of such permutations is $n!$. It cancels the factor $1/n!$

in (77.1), after which it is no longer necessary to count diagrams with permutation of vertices. We have already encountered this circumstance in §§ 73, 74. Thus, two diagrams of the second approximation are equivalent:

$$(\bar{\psi} A \psi)(\bar{\psi} A \psi) = \tag{77.5}$$

¹²Concerning the last assertion the following remark is in order. In proving Wick's theorem we used the commutation rules for the operators $\hat{c}$, $\hat{c}^+$, which have meaning only for real ("transverse") photons. The "external" operators $\hat{c}_f^+$, $\hat{c}_f$ also correspond to such (initial and final) photons. But the operators $\hat{A}$ (entering under the sign of the T-product) describe, as was indicated in § 76, not only transverse

photons. The situation is the same here as when computing $D_{\mu\nu}$ in § 76. By virtue of relativistic and gauge invariance it is sufficient to prove the theorem for the transverse (i.e. the component of the tensor $\langle 0|T A_\mu A_\nu|0\rangle$) which are defined by transverse parts of the potentials. It is thereby proved also for arbitrary products.

In (77.4) and (77.5) only the internal lines of the diagrams are depicted, which correspond to internal contractions (virtual electrons and photons). The remaining free operators are contracted with those or other external operators—operators of creation of initial or annihilation of final particles, and as a result there is established the correspondence between the free ends of the diagrams and those or other initial and final particles.

In this $\hat{\psi}$ (contracting with the operators $\hat{a}_f$ or $\hat{b}_f^+$) gives a line of the final electron or initial positron, and $\hat{\bar{\psi}}$ (contracting with $\hat{a}_f^+$ or $\hat{b}_f$) gives a line of the initial electron or final positron. The free operator $\hat{A}$ (contracting with $\hat{c}_f^+$ or $\hat{c}_f$) can correspond to both the initial and the final photon. Thus, there are obtained several topologically identical (i.e. consisting of an identical number of identically arranged lines) diagrams, differing only by the permutations of initial and final particles among the incoming and outgoing free ends.

Each such permutation is equivalent, obviously, to a definite permutation of external operators $\hat{a}, \hat{b}, \dots$ in (77.1). It is clear therefore that if among the initial or among the final particles there are identical fermions, then the relative signs of diagrams differing by an odd number of permutations of free ends must be opposite.

The continuous succession of solid lines on the diagrams, along which the arrows preserve an unbroken direction, constitutes an *electron line*. Such a line can either have two free ends or form a closed loop. Thus, the diagram

$$(\bar{\psi} A \psi)(\bar{\psi} A \psi) =$$

has a loop with two vertices. The conservation of direction along the

electron line is a graphical expression of conservation of charge: the charge “entering” each vertex equals the charge “leaving” it.

The arrangement of bispinor indices along a continuous electron line corresponds to the convention of writing the matrices from left to right in the direction of traversal against the arrows. Along an open line the bispinor indices never get confused. Along a closed loop, however, the succession of indices terminates at the free ends of electronic (or positronic) wave functions; while along a closed loop the succession also closes, i.e. the loop corresponds to a trace of the product of matrices arranged along it. It is easy to see that this trace must be taken with a minus sign.

Indeed, a loop with k vertices corresponds to a set of k contractions

$$\overbrace{(\bar{\psi} A \psi)(\bar{\psi} A \psi) \dots (\bar{\psi} A \psi)}$$

(or another equivalent, differing by a permutation of vertices). In the $(k-1)$ -th contraction the operators $\hat{\psi}$ and $\hat{\bar{\psi}}$ already stand side by side (the $\hat{\bar{\psi}}$ to the right of $\hat{\psi}$), in which order they should stand in the electron propagator. But the operators at the edges, having been brought to proximity with an even number of permutations with other ψ -operators, turn out to be arranged in the order $\bar{\psi}\psi$. Since

$$\langle 0 | T \bar{\psi} \psi | 0 \rangle = - \langle 0 | T \psi \bar{\psi} | 0 \rangle$$

(cf. the remark on p. 334), the replacement of this contraction by the corresponding propagator is connected with the change of sign of the whole expression.

The transition to the momentum representation in the general case is carried out completely analogously to how it was done in §§ 73, 74. Alongside the general law of conservation of 4-momentum, there must also be satisfied “conservation laws” in each vertex. However, all these conservation laws may turn out to be insufficient for the unambiguous determination of the 4-momenta of all internal lines of the diagram. In such cases, for all remaining undetermined internal momenta there remain integrations (over $d^4 p / (2\pi)^4$), carried out over the whole of p -space (including also over p_0 from $-\infty$ to $+\infty$).

In the above arguments it was assumed that the perturbation acts as the interaction between the particles themselves, “actively” participating in the reaction (i.e. between particles whose state changes as a result of the process). Analogously one considers also the case when the problem involves an external electromagnetic field, i.e. a field created by “passive” particles whose state does not change in the given process.

Let $A^{(e)}(x)$ be the 4-potential of the external field. It enters the Lagrangian of interaction

in the form of the sum $\hat{A} + A^{(e)}$ (which is also multiplied by the current operator $\hat{j}$). Since $A^{(e)}$ does not contain any operators, it cannot form contractions with other operators. In other words, to the external field there correspond in the Feynman diagrams only external lines.

Let us represent $A^{(e)}$ in the form of the Fourier integral:

$$A^{(e)}(x) = \int A^{(e)}(q) e^{-iqx} \frac{d^4 q}{(2\pi)^4}, \quad (77.6)$$

$$A^{(e)}(q) = \int A^{(e)}(x) e^{iqx} d^4 x.$$

In the expressions for matrix elements in the momentum representation the 4-vector q will figure alongside the 4-momenta of the other external lines corresponding to real particles. To each such line of the external field there corresponds the factor $A^{(e)}(q)$, the line being to be considered “incoming”—in accordance with the sign of the exponent in the factor e^{-iqx} with which $A^{(e)}(q)$ enters the Fourier integral (77.6). If thereby the law of conservation of 4-momentum (given the 4-momenta of all real particles) does not fix uniquely all 4-momenta q of all external field lines, then for the remaining “free” ones integration over $d^4 q / (2\pi)^4$ is carried out, as for all other unfixed internal 4-momenta of lines of the diagram.

If the external field does not depend on time, then

$$A^{(e)}(q) = 2\pi\delta(q^0) A^{(e)}(\mathbf{q}), \quad (77.7)$$

where $A^{(e)}(\mathbf{q})$ is the three-dimensional Fourier component:

$$A^{(e)}(\mathbf{q}) = \int A^{(e)}(\mathbf{r}) e^{-i\mathbf{q}\mathbf{r}} d^3 x. \quad (77.8)$$

In this case the external line is associated with the factor $A^{(e)}(\mathbf{q})$ and to it there is ascribed the 4-momentum $q^\mu = (0, \mathbf{q})$; the energies of the electronic lines intersecting

(together with the external field line) in the vertex are equal by virtue of the law of conservation. For all remaining unfixed three-dimensional momenta $\mathbf{p}$ of internal lines integration over $d^3p/(2\pi)^3$ is carried out. The computed amplitude M_{fi} determines, for example, the cross section of scattering (64.25).

Let us give a final summary of the rules of *diagram technique*, according to which the expression for the amplitude of scattering (more precisely, the expression for iM_{fi}) is compiled in the momentum representation.

1. To the n -th order of perturbation theory there correspond diagrams with n vertices, to each of which there come one incoming and one outgoing electronic (solid) and one photonic (dashed) line. The amplitude of the process of scattering includes all diagrams having free ends (external lines) in the number equal to the number of initial and final particles.

2. To each external incoming solid line there corresponds the amplitude of the initial electron $u(p)$ or of the final positron $\bar{u}(-p)$ (p is the 4-momentum of the particle). To each outgoing solid line there corresponds the amplitude of the final electron $\bar{u}(p)$ or of the initial positron $u(-p)$.

3. To each vertex there corresponds the 4-vector $-ie\gamma^\mu$.

4. To each external incoming dashed line there corresponds the amplitude of the initial photon $\sqrt{4\pi}e_\mu$, and to the outgoing line the amplitude $\sqrt{4\pi}e_\mu^*$ of the final photon (e is the 4-vector of polarisation).

5. To each internal solid line there corresponds the factor $iG(p)$, and to each internal dashed line the factor $-iD_{\mu\nu}(p)$. The tensor indices $\mu\nu$ coincide with the indices of the matrices γ^μ, γ^ν in the vertices joined by the dashed line.

6. Along each continuous succession of solid lines the arrows have an unchanged direction, and the arrangement of bispinor indices along it corresponds to the convention of writing the matrices from left to right in the direction of traversal against the arrows. A closed electron loop corresponds to a trace of the product of matrices arranged along it.

7. In each vertex the 4-momenta of lines intersecting in it satisfy the law of conservation, i.e. the sum of momenta of incoming lines equals the sum of momenta of outgoing lines. The momenta of the free ends are given (with the overall law of conservation of 4-momentum ensured), and the positronic line is ascribed the momentum $-p$.

8. To the external free end corresponding to the external field there corresponds the factor $A^{(e)}(q)$; the 4-vector q is related to the 4-momenta of other lines by the law of conservation in the vertex.

9. An additional factor -1 is introduced into the expression for iM_{fi} for each closed electron loop in the diagram and for each pair of positronic external ends, if these ends are the beginning and end of one continuous succession of solid lines. If among the initial or among the final particles there are several electrons or positrons, then the relative sign of diagrams differing by an odd number of permutations of free ends (i.e. diagrams which would be identical after the removal of all photonic lines) must be opposite.

For the last rule to be more precise, let us add that diagrams with identical signs should in any case have diagrams whose solid lines are identical, i.e. diagrams which would be identical after the removal of all photonic lines. Let us recall also that in the

presence of identical fermions the overall sign of the amplitude is arbitrary.

§ 78. Crossing invariance

The representation of amplitudes of scattering M_{fi} by Feynman integrals reveals their remarkable symmetry, consisting in the following.

Any of the incoming external lines of a Feynman diagram can be regarded (without changing the direction of its arrow) as a particle in the initial or antiparticle in the final state, and each outgoing line as a final particle or initial antiparticle. Simultaneously with the transition from particle to antiparticle there also changes the meaning of the 4-momentum ascribed to the line: $p_i \rightarrow -p_f$, where p_i is the 4-momentum of the initial particle (in one channel), and p_f is the 4-momentum of the final (in another channel) particle. The polarisation also changes: the initial wave amplitude u is replaced for a particle (say, electron) by $u = u_s$, and the outgoing $\bar{u}$ by u_s^* ; for a positron $u = u_s^*$. But the transition from u to u^* and from $\bar{u}$ to $\bar{u}^*$ signifies a change of sign of the projection of spin (or of its helicity).

For a photon, as a truly neutral particle, the change of the meaning of an external line means simply the transition from emission of the photon to its absorption or vice versa: an external photonic line with 4-momentum k corresponds either to absorption of a photon with momentum $k_{\text{abs}} = k$, or to emission of a photon with momentum $k_{\text{em}} = -k$ and with opposite sign of the helicity.

Such a change of meaning of external lines is equivalent to the transition from one crossed channel of the reaction to another. Depending on the channel, only the meaning of the arguments of the function changes: when going from particle to antiparticle the replacement $p_i \rightarrow -p_f$ is made. About this property of the amplitude of scattering one speaks as about *crossing symmetry*, or *crossing invariance*.

In terms of the invariant amplitudes introduced in § 70 and of the kinematic invariants, one can say that these are one and the same functions for all channels, but for each channel their arguments run through values in their own physical region. In other words, the Feynman integrals define the invariant amplitudes as analytic functions; their values in the various physical regions are the analytic continuation of a function given in one of the regions. Since the Feynman integral expressions contain singularities, the invariant amplitudes also have singularities; their values in different physical regions are defined by the appropriate rules of going around these integrals (taking into account the rule of going around the poles—cf. § 75).

Let us stress that crossing invariance is nothing more than a property of the scattering matrix, following from the general requirements of space-time symmetry. Subsequently the equality of amplitudes of processes obtained from one another by the permutation of initial and final states with the replacement of all particles by antiparticles (at unchanged momenta $\mathbf{p}$ of all particles and changed in sign projections of all their moments) is required. This is the requirement of *CPT*-invariance^{13 14}. Crossing invariance, however,

¹³Let us note that the formal description of transition from one of the indicated reactions to another by means of changing the sign of all 4-momenta on Feynman diagrams corresponds to the meaning of the *CPT*-operation as 4-inversion.

¹⁴If this or that channel is forbidden by conservation of 4-momentum, then the probability of the transition

allows this transformation to be made not for all particles simultaneously but also for any individual particle separately.

§ 79. Virtual particles

The internal lines of Feynman diagrams play in the invariant perturbation theory a role analogous to the role of intermediate states in the “ordinary” theory. The character of these states, however, is different in both theories. In the ordinary theory intermediate states are conserved (three-dimensional) momentum, but the energy is not

conserved; in this sense one speaks of them as *virtual states*. In the invariant theory, however, momentum and energy enter on equal footing: in the intermediate states all four components of the 4-momentum are conserved (this is the result of the fact that in the elements of the S -matrix the integration is carried out over both the coordinates and the time, by which the invariance of the theory is achieved). In this, however, in the intermediate states there is violated the relation between energy and momentum characteristic of real particles (expressed by the equality $p^2 = m^2$). In this sense one speaks of intermediate *virtual particles*. The ratio of momentum to energy of a virtual particle is arbitrary—it is whatever the conservation of 4-momentum in the vertices requires.

Let us consider some diagram consisting of two parts (I and II) connected by a single line. Not concerning ourselves with the internal structure of these parts, let us represent the diagram schematically in the form (the depicted lines can be both solid and dashed). By virtue of the overall law of conservation of 4-momentum, the sum of 4-momenta of external lines of parts I and II are equal; and by virtue of conservation in each vertex this same quantity is the 4-momentum p of the internal line joining parts I and II . In other words, in the matrix element there is no integration over this momentum.

Depending on the channel of the reaction the square p^2 can be both positive and negative. There always exists such a channel, in which $p^2 > 0$ ¹⁵. Then the virtual particle, by its formal properties, becomes entirely analogous to a real particle with a mass $M = \sqrt{p^2}$. For it one can introduce the rest frame and determine its spin and so on. The photon propagator (76.11) by its tensor structure coincides with the density matrix of an unpolarised particle with spin 1 and mass different from zero:

$$\rho_{\mu\nu} = -\frac{1}{3} \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right)$$

(see (14.15)). With the electron propagator (75.10) the situation is analogous. The propagator

$$G \propto \gamma p + m.$$

Here m is the mass of the real electron, whereas the “mass” of the virtual particle $M = \sqrt{p^2}$. Writing

$$\gamma p + m = \frac{M + m}{2M} (\gamma p + M) + \frac{M - m}{2M} (\gamma p - M), \quad (79.2)$$

automatically vanishes by the δ -function figuring in (64.5) as the common factor.

¹⁵Such is, for example, the channel in which all free ends of part I correspond to the initial, and all of part II to the final particles. Then $p = P_i$ (the sum of 4-momenta of all initial particles), and in the system of the centre of inertia $p = (P_i^0, 0)$, so that $p^2 > 0$.

we see that the first term corresponds to the density matrix of a particle with mass M and spin $1/2$, and the second to the density matrix of such an “antiparticle” (cf. (29.10) and (29.17)). Recalling that the particle and antiparticle have different internal parities (see § 27), we arrive at the conclusion that the virtual electron must be ascribed the same spin $1/2$, but a definite parity cannot be ascribed to it.

The characteristic feature of diagram (79.1) is that it can be cut into two parts not connected to each other by cutting only one internal line¹⁶. This line corresponds to a *one-particle* intermediate state—a state of only one virtual particle. The amplitude of scattering corresponding to such a diagram contains the characteristic (not subject to integration!) factor

$$\frac{1}{p^2 - M^2 + i0}$$

arising from the internal line p (where m is the mass of the electron if the line is electronic, or $m = 0$ if the line is photonic). In other words, the amplitude of scattering has a pole at those values of p for which the virtual particle becomes physical ($p^2 = M^2$). This situation is analogous to the fact that in non-relativistic quantum mechanics the amplitude of scattering has poles at energies corresponding to bound states of the colliding system (see III, § 128).

Let us consider diagram (79.1) for that channel of the reaction in which all right free ends correspond to the initial, and all left ones to the final particles; in this case $p^2 > 0$. Then one can say that in the intermediate state the system of initial particles transforms into one virtual particle. This is possible only if such a transformation does not contradict the necessary conservation laws (without taking into account conservation of 4-momentum): conservation of charge, charge parity, and so on. In this is contained the necessary condition of appearance of *pole diagrams*.

For example, the conservation laws listed do not prevent the appearance of a virtual electron according to $e + \gamma \rightarrow e$. This possibility corresponds to the pole of the Compton-effect amplitude, and at the same time of another channel of this reaction—two-photon annihilation of the electron pair. The appearance of a virtual photon according to $e^- + e^+ \rightarrow \gamma$ is also not prevented by any conservation law, and corresponds to the pole of the scattering amplitude of the electron on the positron, and also of the electron on the electron. From two photons a virtual electron cannot be produced, and the virtual photon cannot appear; accordingly, the amplitude of scattering of a photon on a photon does not contain pole singularities.

The origin of the pole singularities of the amplitude of scattering, which we have traced here from the Feynman integrals, has in fact a more general character, not connected with perturbation theory. Let us show that these singularities already arise as a consequence of the unitarity condition (71.2).

Let us assume that among the intermediate states figuring in (71.2) n is a one-particle state. The contribution of this state:

$$(T_{fi} - T_{if}^*)^{(\text{one-part.})} = i(2\pi)^4 \sum_{\lambda} \int \delta^{(4)}(P_f - p) T_{fn} T_{in}^* \frac{V d^3 p}{(2\pi)^3},$$

¹⁶This property is possessed by the diagrams of almost all processes in the first non-vanishing approximation.

where p and λ are the 4-momentum and helicity of the intermediate particle. The integration over d^3p is replaced by integration over d^4p (where $p^0 \equiv \varepsilon > 0$) according to

$$d^3p \rightarrow 2\varepsilon\delta(p^2 - M^2) d^4p$$

(M is the mass of the intermediate particle). The integration removes the δ -function $\delta^{(4)}(P_f - p)$, and we find

$$(M_{fi} - M_{if}^*)^{(\text{one-part.})} = 2\pi i\delta(p^2 - M^2) \sum_{\lambda} M_{fn} M_{in}^*. \quad (79.3)$$

Assuming T - and P -invariance, we shall have (to within a phase factor) $M_{if} = M_{fi'}$, where the states i' , f' differ from i , f only by the signs of the helicities of the particles (with the same momenta). Taking the sum of equality (79.3) and also of the same for $M_{f'i'} M_{f'n}^*$, we get

$$\text{Im } \overline{M}_{fi}^{(\text{one-part.})} = -\pi\delta(p^2 - M^2) R, \quad (79.4)$$

where it is denoted

$$\overline{M}_{fi} = M_{fi} + M_{f'i'}, \quad R = - \sum_{\lambda} (M_{fn} M_{in}^* + M_{f'n} M_{i'n}^*).$$

From this it follows that $\overline{M}_{fi}$ as an analytic function of $p^2 = P_i^2 = P_f^2$ has a pole at $p^2 = M^2$. According to (75.18) we have for the pole part

$$\overline{M}_{fi}^{(\text{one-part.})} = \frac{R}{p^2 - M^2 + i0}. \quad (79.5)$$

Real transitions to the one-particle state are possible only when the value $P_i^2 = P_f^2$ equals M^2 . Thus we have indeed obtained the structure of the amplitude of scattering of the type of diagram (79.1).

Finally, let us dwell on an important property of diagrams containing closed loops. This property can easily be obtained by the method of application to a virtual photon of the concept of charge parity: to the virtual photon, like the real one, there should be ascribed a definite (negative) charge parity¹⁷.

If some diagram contains a closed loop (with the number of vertices $N > 2$), then alongside this diagram in the amplitude of the process being considered there must also figure another diagram, differing from the first only by the direction of traversal of the loop (for $N = 2$ the concept of direction of traversal has, obviously, no meaning). "Cutting out" these loops from their neighbouring dashed lines, we obtain two loops Π_I and Π_{II} : which can be regarded as diagrams defining the amplitude of the process of transformation of one set of photons (real or virtual) into another: the number N is the total number of initial and final photons. But the conservation of charge parity forbids the transformation of an even number of photons into an odd one. Therefore when N is odd the sum of expressions corresponding to the loops (79.6) must vanish. The contributions

¹⁷This follows from the same considerations about the action of the operator of electromagnetic interaction in each vertex, which were indicated in § 13 for the real photon.

of both loops also vanish in the amplitude of scattering of two diagrams containing these loops, i.e.

Let us trace in more detail the mechanism of the indicated mutual cancellation of diagrams. A closed electron loop corresponds to the expression (given the 4-momenta of the photonic lines $k_1, k_2, \dots, k_N$)

$$\int d^4p \operatorname{Sp}[(\gamma e_1)G(p)(\gamma e_2)G(p+k_1)\dots], \quad (79.7)$$

where $p, p+k_1, \dots$ are the 4-momenta of the electronic lines (not fully determined even after taking into account the conservation laws in the vertices). The charge conjugation operation is performed on all matrices γ^μ and G in the trace: they are replaced by $U_C^{-1}\gamma^\mu U_C$ and $U_C^{-1}GU_C$. The expression (79.7) is unchanged, since the trace of a matrix is invariant with respect to such a transformation. On the other hand, according to (26.3),

$$U_C^{-1}\gamma^\mu U_C = -\tilde{\gamma}^\mu, \quad (79.8)$$

and therefore

$$U_C^{-1}G(p)U_C = \frac{-\tilde{p} + m}{p^2 - m^2} = \tilde{G}(-p). \quad (79.9)$$

But the replacement of $G(p)$ by the transposed matrix with a changed sign of p signifies, obviously, the change of direction of traversal of the loop, in which the direction of all arrows is reversed. In other words, the transposition transforms one loop into the other, and there appears the factor $(-1)^N$ arising from the replacement (79.8) in each vertex. Thus,

$$\Pi_I = (-1)^N \Pi_{II}, \quad (79.10)$$

i.e. the contributions of both loops are equal for an even and opposite in sign for an odd number of vertices.

Interaction of Electrons

§ 80. Scattering of an electron in an external field

Elastic scattering of an electron in a constant external field represents the simplest process, existing already in the first approximation of perturbation theory (first Born approximation). To it there corresponds a diagram with one vertex

$$(80.1)$$

where p and p' are the initial and final 4-momenta of the electron, and $q = p' - p$. Since the energy of the electron is conserved in scattering in a constant field ($\varepsilon = \varepsilon'$), we have $q = (0, \mathbf{q})$ ¹.

The corresponding scattering amplitude

$$M_{fi} = -e\bar{u}(p')[\gamma A^{(e)}(\mathbf{q})]u(p), \quad (80.2)$$

where $A^{(e)}(\mathbf{q})$ is the component of the spatial Fourier expansion of the external field. The scattering cross section, according to (64.26),

$$d\sigma = \frac{1}{16\pi^2} |M_{fi}|^2 d\Omega'. \quad (80.3)$$

For an electrostatic field $A^{(e)} = (A_0^{(e)}, 0)$, so that

$$M_{fi} = -e\bar{u}(p')\gamma^0 u(p)A_0^{(e)}(\mathbf{q}) = -eu^*(p')u(p)A_0^{(e)}(\mathbf{q}). \quad (80.4)$$

In the non-relativistic case the bispinor amplitudes of the plane waves $u(p)$ reduce to the non-relativistic (two-component) amplitudes. For scattering without change of polarisation this is a quantity independent of p , and by virtue of the normalisation condition we have adopted $u^*u = 2m$. Taking this into account, we obtain

$$d\sigma = \left| -\frac{m}{2\pi} U(\mathbf{q}) \right|^2 d\Omega',$$

where $U(\mathbf{q}) = eA_0^{(e)}(\mathbf{q})$ is the component of the Fourier expansion of the potential energy of the electron in the field; this expression coincides with the well-known Born formula (see III, (126.7)).

¹ In the case of an external field such a diagram is of course not forbidden by the law of conservation of 4-momentum (as it was in diagram (73.19) with a real photon): the square q^2 , in contrast to the square of the 4-momentum of a real photon, need not be equal to zero; from the Fourier integral representing the external field, the component with the required q is automatically selected.

In the general relativistic case the scattering cross section of unpolarised electrons is obtained by averaging the square of the modulus $|M_{fi}|^2$ over the initial and summing over the final polarisations, i.e. by forming the quantity

$$\frac{1}{2} \sum_{\text{polar}} |M_{fi}|^2,$$

where the summation is carried out over the spin directions of the initial and final electrons; the factor $1/2$ converts one of these summations into an averaging. By the rules set out in § 65 we obtain

$$\begin{aligned} \frac{1}{2} \sum_{\text{polar}} |M_{fi}|^2 &= 2 \text{Sp} \rho' (\gamma A_0^{(e)}) \rho (\gamma A_0^{(e)*}) = \\ &= \frac{1}{2} |A_0^{(e)}(\mathbf{q})|^2 \text{Sp}(m + \gamma p') \gamma^0 (m + \gamma p) \gamma^0. \end{aligned}$$

For computing the trace we note that $\gamma^0 (\gamma p) \gamma^0 = \gamma \tilde{p}$, where $\tilde{p} = (\varepsilon, -\mathbf{p})$, and therefore

$$\begin{aligned} \frac{1}{4} \text{Sp}(m + \gamma p') \gamma^0 (m + \gamma p) \gamma^0 &= \frac{1}{4} \text{Sp}(m + \gamma p') (m + \gamma \tilde{p}) = \\ &= m^2 + p' \tilde{p} = \varepsilon^2 + m^2 + \mathbf{p} \mathbf{p}' = 2\varepsilon^2 - \mathbf{q}^2/2. \end{aligned}$$

Hence the cross section

$$d\sigma = \frac{e^2 |A_0^{(e)}(\mathbf{q})|^2}{4\pi^2} \varepsilon^2 \left(1 - \frac{\mathbf{q}^2}{4\varepsilon^2}\right) d\omega'. \quad (80.5)$$

For the field created by a static distribution of charges with density $\rho(\mathbf{r})$, we have

$$A_0^{(e)}(\mathbf{q}) = \frac{4\pi\rho(\mathbf{q})}{\mathbf{q}^2}, \quad (80.6)$$

where $\rho(\mathbf{q})$ is the Fourier transform of the distribution $\rho(\mathbf{r})$ (form factor). In particular, for the Coulomb field of a point charge Ze we have: $\rho(\mathbf{q}) = Ze$. Then the scattering cross section

$$d\sigma = d\omega' \frac{4(Ze^2)^2 \varepsilon^2}{\mathbf{q}^4} \left(1 - \frac{\mathbf{q}^2}{4\varepsilon^2}\right) \quad (80.7)$$

(*N. F. Mott*, 1929). The square

$$\mathbf{q}^2 = 4\mathbf{p}^2 \sin^2(\theta/2),$$

where θ is the scattering angle. Therefore the expression in front of the bracket by its angular dependence may be called the Rutherford cross section:

$$d\sigma_{\text{Ru}} = d\omega' \frac{4(Ze^2)^2 \varepsilon^2}{\mathbf{q}^4} = d\omega' \frac{4(Ze^2)^2 \varepsilon^2}{4\mathbf{p}^4} \sin^{-4} \frac{\theta}{2} \quad (80.8)$$

(in the non-relativistic limit the coefficient $\varepsilon^2/\mathbf{p}^4 \rightarrow 1/(m^2v^4)$). Thus²,

$$d\sigma = d\sigma_{\text{Ru}} \left(1 - v^2 \sin^2 \frac{\theta}{2} \right). \quad (80.9)$$

Let us note that in the ultrarelativistic case the angular distribution differs from the non-relativistic one by a strong suppression of backward scattering (when $\theta \rightarrow \pi$: $d\sigma/d\sigma_{\text{Ru}} \rightarrow m^2/\varepsilon^2$).

In the ultrarelativistic case for scattering at small angles (80.7) gives

$$d\sigma = \frac{4(Ze^2)^2}{\varepsilon^4 \theta^4} d\theta'. \quad (80.10)$$

Although we have obtained this formula in the Born approximation (i.e. assuming $Ze^2 \ll 1$), it nevertheless remains valid (for angles $\theta \lesssim m/\varepsilon$) also when $Ze^2 \sim 1$. One can verify this with the help of the ultrarelativistic exact (in Ze^2) wave function $\psi_{\mathbf{p}}^{(+)}$ (39.10). This solution, valid in the region (39.2), remains, of course, valid also in the asymptotic region for arbitrarily large r . Here

$$F \propto 1 + \text{const} \cdot e^{i(pr - \mathbf{p}\mathbf{r})}, \quad \frac{\alpha \nabla F}{\varepsilon} \sim 1 - \cos \theta \sim \theta^2 \ll 1,$$

so that the correction term remains, as one would expect, small. The wave function is of the type $e^{i\mathbf{p}\mathbf{r}} F$, coinciding in form with the non-relativistic function (with an obvious change of parameters), has the same asymptotic form, and therefore also for the cross section one obtains the Rutherford expression.

For computing the cross section of scattering of arbitrarily polarised electrons one could use the general rules of density matrices (see § 29.13). In the present case, however,

the result can be obtained in a less cumbersome way, by representing the bispinor amplitudes $u(p')$ and $u(p)$ in the form (23.9); multiplying them out, we obtain

$$u^*(p')u(p) = w'^* \{ \varepsilon + m + (\varepsilon - m)(\mathbf{n}'\boldsymbol{\sigma})(\mathbf{n}\boldsymbol{\sigma}) \} w,$$

or, using formula (33.5),

$$u^*(p')u(p) = w'^* \hat{f} w, \quad (80.11)$$

where³

$$\hat{f} = A + B\boldsymbol{\nu}\boldsymbol{\sigma}, \quad (80.12)$$

$$A = (\varepsilon + m) + (\varepsilon - m) \cos \theta, \quad B = -i(\varepsilon - m) \sin \theta, \quad \boldsymbol{\nu} = \frac{[\mathbf{nn}']}{\sin \theta}.$$

The two-component quantity (3-spinor) w represents the non-relativistic spinor wave function of the electron. The transition to partially polarised states is accomplished

²The fact that this formula expresses the difference of $d\sigma$ from $d\sigma_{\text{Ru}}$ is specific to particles with spin 1/2. For the scattering of particles with spin 0 (if their motion in the electromagnetic field were described by the wave equation) we would obtain $d\sigma = d\sigma_{\text{Ru}}$. At first glance it appears strange that this purely quantum effect does not contain $\hbar$. One must, however, keep in mind that the condition of applicability of the Born approximation ($e^2/(\hbar v) \ll 1$) is the opposite of the condition of quasiclassicality for the motion in a Coulomb field and therefore the passage to the classical case in formula (80.9) is impossible.

³The definitions of $\hat{f}$ here and in § 37, § 38 differ by a common factor.

by replacing products $w_\alpha w_\beta^*$ (α, β are spinor indices) by the non-relativistic two-row density matrix $\rho_{\alpha\beta}$. Thus, one must replace

$$|M_{fi}|^2 \rightarrow e^2 |A_0^{(e)}(\mathbf{q})|^2 \text{Sp } \rho' (A - B\boldsymbol{\nu}\boldsymbol{\sigma}) \rho' (A + B\boldsymbol{\nu}\boldsymbol{\sigma}),$$

where

$$\rho = \frac{1}{2}(1 + \boldsymbol{\sigma}\boldsymbol{\zeta}), \quad \rho' = \frac{1}{2}(1 + \boldsymbol{\sigma}\boldsymbol{\zeta}'),$$

and $\boldsymbol{\zeta}$ and $\boldsymbol{\zeta}'$ are the vectors of the initial and final polarisations, the latter selected by a detector. Computing the trace leads to the result

$$d\sigma = d\sigma_0 \left\{ 1 + \frac{(A^2 - |B|^2)\boldsymbol{\zeta}\boldsymbol{\zeta}' + 2|B|^2(\boldsymbol{\nu}\boldsymbol{\zeta})(\boldsymbol{\nu}\boldsymbol{\zeta}') + 2A|B|[\boldsymbol{\nu}\boldsymbol{\zeta}\boldsymbol{\zeta}']} {A^2 + |B|^2} \right\}, \quad (80.13)$$

where $d\sigma_0$ is the cross section of scattering of unpolarised electrons.

Representing the curly brackets in (80.13) in the form $\{1 + \boldsymbol{\zeta}^{(f)}\boldsymbol{\zeta}'\}$, we find the polarisation of the final electron as such (in contrast to the detectable final polarisation $\boldsymbol{\zeta}'$ —see § 65)⁴:

$$\boldsymbol{\zeta}^{(f)} = \frac{(A^2 - |B|^2)\boldsymbol{\zeta} + 2|B|^2(\boldsymbol{\nu}\boldsymbol{\zeta})\boldsymbol{\nu} + 2A|B|[\boldsymbol{\nu}\boldsymbol{\zeta}]} {A^2 + |B|^2}. \quad (80.14)$$

We see that the scattered electrons are polarised only if the incident electrons are polarised. This is a general property of the first Born approximation (cf. III, § 140).

In the non-relativistic case ($\varepsilon \rightarrow m$) from (80.14) we obtain $\boldsymbol{\zeta}^{(f)} = \boldsymbol{\zeta}$, i.e. the electron preserves its polarisation upon scattering (a natural consequence of neglecting spin-orbit interaction).

In the opposite, ultrarelativistic, case we have

$$A = \varepsilon(1 + \cos \theta), \quad B = -i\varepsilon \sin \theta$$

(in accordance with the general formula (38.2)).

If furthermore the incident electron has a definite helicity ($\boldsymbol{\zeta} = 2\lambda\mathbf{n}$, $\lambda = \pm 1/2$), then from (80.14) it follows after simple reduction that

$$\boldsymbol{\zeta}^{(f)} = 2\lambda\mathbf{n}'.$$

In other words, after scattering the electron remains helical, preserving the former value (λ) of the helicity.

This property, as was already noted in § 38, is connected with the fact that on neglecting mass the Dirac equation in the spinor representation separates into two independent equations for the functions ξ and η . This result has also a more general significance, since the current

$$\hat{j} = (\xi^*\xi + \eta^*\eta, \xi^*\boldsymbol{\sigma}\xi - \eta^*\boldsymbol{\sigma}\eta),$$

⁴Formula (80.14) corresponds to the formula found in Problem 1, (see III, § 140) and is obtained from it with real A and imaginary B .

and with it the operator of electromagnetic perturbation $\hat{V} = e\hat{j}\hat{A}$, does not contain mixed terms in ξ and η , and therefore has no matrix elements for transitions between ξ - and η -states. From this it follows that if an ultrarelativistic electron has a definite helicity (i.e. differs from zero either ξ or η), then in processes of interaction this helicity will be conserved in the approximation corresponding to the complete neglect of the electron mass.

§ 81. Scattering of electrons and positrons on an electron

Let us consider the scattering of an electron on an electron: two electrons with 4-momenta p_1, p_2 collide, acquiring 4-momenta p'_1, p'_2 . Conservation of 4-momentum is expressed by the equality

$$p_1 + p_2 = p'_1 + p'_2. \quad (81.1)$$

Below we shall use the kinematic invariants introduced in § 66, defined as

$$\begin{aligned} s &= (p_1 + p_2)^2 = 2(m^2 + p_1 p_2), \\ t &= (p_1 - p'_1)^2 = 2(m^2 - p_1 p'_1), \\ u &= (p_1 - p'_2)^2 = 2(m^2 - p_1 p'_2), \\ s + t + u &= 4m^2. \end{aligned} \quad (81.2)$$

The process under consideration is depicted by two Feynman diagrams (73.13), (73.14), and its amplitude equals⁵

$$M_{fi} = 4\pi e^2 \left\{ \frac{1}{t} (\bar{u}'_2 \gamma^\mu u_2) (\bar{u}'_1 \gamma_\mu u_1) - \frac{1}{u} (\bar{u}'_1 \gamma^\nu u_2) (\bar{u}'_2 \gamma_\nu u_1) \right\}. \quad (81.3)$$

According to the rules stated in § 65 for the states of initial and final particles, described by the polarisation density matrices $\rho_1, \rho'_1, \dots$, we replace

$$\begin{aligned} |M_{fi}|^2 \rightarrow 16\pi^2 \alpha^2 e^4 \left\{ \frac{1}{t^2} \text{Sp}(\rho'_2 \gamma^\mu \rho_2 \gamma^\nu) \text{Sp}(\rho'_1 \gamma_\mu \rho_1 \gamma_\nu) + \right. \\ \left. + \frac{1}{u^2} \text{Sp}(\rho'_1 \gamma^\mu \rho_2 \gamma^\nu) \text{Sp}(\rho'_2 \gamma_\mu \rho_1 \gamma_\nu) - \right. \\ \left. - \frac{1}{tu} \text{Sp}(\rho'_2 \gamma^\mu \rho_2 \gamma^\nu \rho'_1 \gamma_\mu \rho_1 \gamma_\nu) - \right. \\ \left. - \frac{1}{tu} \text{Sp}(\rho'_1 \gamma^\mu \rho_2 \gamma^\nu \rho'_2 \gamma_\mu \rho_1 \gamma_\nu) \right\}. \end{aligned} \quad (81.4)$$

For the scattering of unpolarised electrons (not concerning ourselves with their polarisation after scattering) we must set for all density matrices $\rho = \frac{1}{2}(\gamma p + m)$, multiplying the result by $2 \cdot 2 = 4$ (averaging over the polarisations of the two initial and summing over the polarisations of the two final electrons). The scattering cross

⁵This form of M_{fi} is in agreement with the general expression (70.5). In the first non-vanishing approximation of perturbation theory only one of the five invariant amplitudes differs from zero: $f_3(t, u) = 4\pi e^2/t$.

section is determined by formula (64.23), in which one must set, according to (64.15a), $I^2 = \frac{1}{4}s(s - 4m^2)$. Let us represent the cross section in the form

$$d\sigma = dt \frac{4\pi\alpha^2}{s(s - 4m^2)} \{f(t, u) + g(t, u) + f(u, t) + g(u, t)\}, \quad (81.5)$$

$$f(t, u) = \frac{1}{16t^2} \text{Sp}[(\gamma p'_2 + m)\gamma^\mu(\gamma p_2 + m)\gamma^\nu] \times \\ \times \text{Sp}[(\gamma p'_1 + m)\gamma_\mu(\gamma p_1 + m)\gamma_\nu], \quad (\text{IX.1})$$

$$g(t, u) = -\frac{1}{16tu} \text{Sp}[(\gamma p'_2 + m)\gamma^\mu(\gamma p_2 + m)\gamma^\nu(\gamma p'_1 + m)\gamma_\mu(\gamma p_1 + m)\gamma_\nu]. \quad (\text{IX.2})$$

In $f(t, u)$ the traces are first computed (using (22.9), (22.10)), and then the summation over μ and ν is carried out⁶; in $g(t, u)$ the summation over μ and ν is first carried out (using formula (22.6)).

As a result we obtain

$$f(t, u) = \frac{2}{t^2} [(p_1 p_2)^2 + (p_1 p'_2)^2 + 2m^2(m^2 - p_1 p'_1)],$$

$$g(t, u) = \frac{2}{tu} (p_1 p_2 - 2m^2)(p_1 p_2),$$

or, expressing the functions f and g through the invariants (81.2),

$$f(t, u) = \frac{1}{t^2} \left[\frac{s^2 + u^2}{2} + 4m^2(t - m^2) \right], \quad (81.6)$$

$$g(t, u) = g(u, t) = \frac{2}{tu} \left(\frac{s}{2} - m^2 \right) \left(\frac{s}{2} - 3m^2 \right).$$

Thus, the cross section

$$d\sigma = r_e^2 \frac{4\pi m^2 dt}{s(s - 4m^2)} \left\{ \frac{1}{t^2} \left[\frac{s^2 + u^2}{2} + 4m^2(t - m^2) \right] + \right. \\ \left. + \frac{1}{u^2} \left[\frac{s^2 + t^2}{2} + 4m^2(u - m^2) \right] + \frac{4}{tu} \left(\frac{s}{2} - m^2 \right) \left(\frac{s}{2} - 3m^2 \right) \right\}, \quad (81.7)$$

where $r_e = e^2/m$.

Let us apply this formula in the centre-of-inertia system. Here

$$s = 4\varepsilon^2, \quad t = -4\mathbf{p}^2 \sin^2 \frac{\theta}{2}, \quad u = -4\mathbf{p}^2 \cos^2 \frac{\theta}{2}, \quad (81.8)$$

$$-dt = -2\mathbf{p}^2 d \cos \theta = \frac{\mathbf{p}^2}{\pi} d\phi$$

⁶Let us note for future reference the formula

$$\frac{1}{4} \text{Sp}(\gamma p_1 + m)\gamma^\mu(\gamma p_2 + m)\gamma^\nu = g^{\mu\nu}(m^2 - p_1 p_2) + p_1^\mu p_2^\nu + p_1^\nu p_2^\mu.$$

($|\mathbf{p}|$, ε are the magnitude of the momentum and the energy of the electrons, which do not change upon scattering; θ is the scattering angle). In the non-relativistic case ($\varepsilon \approx m$)⁷ we obtain

$$\begin{aligned} d\sigma &= r_e^2 \frac{2\pi m^4 dt}{\mathbf{p}^2} \left(\frac{1}{t^2} + \frac{1}{u^2} - \frac{1}{tu} \right) = \\ &= \left(\frac{e^2}{mv^2} \right)^2 \left(\frac{1}{\sin^4 \frac{\theta}{2}} + \frac{1}{\cos^4 \frac{\theta}{2}} - \frac{1}{\sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2}} \right) d\theta = \\ &= \left(\frac{e^2}{mv^2} \right)^2 \frac{4(1 + 3 \cos^2 \theta)}{\sin^4 \theta} d\theta \quad (\text{n. r.}) \end{aligned} \quad (81.9)$$

(where $\mathbf{v} = 2\mathbf{p}/m$ is the relative velocity of the electrons) which agrees with non-relativistic theory (see III, § 137).

In the general case of arbitrary velocities formula (81.7) after substitution of (81.8) and simple transformations can be brought to the form

$$d\sigma = r_e^2 \frac{m^2(\varepsilon^2 + \mathbf{p}^2)^2}{4\mathbf{p}^4 \varepsilon^2} \left[\frac{4}{\sin^4 \theta} - \frac{3}{\sin^2 \theta} + \left(\frac{\mathbf{p}^2}{\varepsilon^2 + \mathbf{p}^2} \right)^2 \left(1 + \frac{4}{\sin^2 \theta} \right) \right] d\theta \quad (81.10)$$

(Ch. Möller, 1932). In the ultrarelativistic case ($\mathbf{p}^2 \approx \varepsilon^2$)

$$d\sigma = r_e^2 \frac{m^2}{\varepsilon^2} \frac{(3 + \cos^2 \theta)^2}{4 \sin^4 \theta} d\theta \quad (\text{u. r.}) \quad (81.11)$$

In the laboratory system of reference, in which one of the electrons (say, the second) was at rest before the collision, we express the cross section through the quantity

$$\Delta = \frac{\varepsilon_1 - \varepsilon'_1}{m} = \frac{\varepsilon'_2 - m}{m} \quad (81.12)$$

—the energy (in units of m) transferred by the first (incident) electron to the second⁸. The invariants

$$\begin{aligned} s &= 2m(m + \varepsilon_1), \quad t = -2m^2 \Delta, \\ u &= -2m(\varepsilon_1 - m - m\Delta). \end{aligned} \quad (81.13)$$

Substitution of these expressions in (81.7) leads to the following formula for the distribution over energies of the secondary electrons (or, as one says, δ -electrons), arising in the scattering of fast primary electrons:

$$d\sigma = 2\pi r_e^2 \frac{d\Delta}{\gamma^2 - 1} \left\{ \frac{(\gamma - 1)^2 \gamma^2}{\Delta^2 (\gamma - 1 - \Delta)^2} - \frac{2\gamma^2 + 2\gamma - 1}{\Delta(\gamma - 1 - \Delta)} + 1 \right\}, \quad (81.14)$$

where $\gamma = \varepsilon_1/m$; $m\Delta$ and $m(\gamma - 1 - \Delta)$ are the kinetic energies of the two electrons after the collision; the identity of both particles manifests itself here in the symmetry

⁷The velocity v is assumed to be small ($v \ll 1$), but such that the condition of applicability of perturbation theory is still satisfied: $e^2/v (= e^2/(\hbar v)) \ll 1$.

⁸The kinematic relations for elastic collisions in different systems of reference see in II, § 13.

of the formula with respect to these quantities. If one agrees to call the recoil electron the one which has the smaller energy, then Δ ranges from 0 to $(\gamma - 1)/2$. For small Δ formula (81.14) takes the form

$$d\sigma = 2\pi r_e^2 \frac{\gamma^2}{\gamma^2 - 1} \frac{d\Delta}{\Delta^2} = \frac{2\pi r_e^2}{\varepsilon_1^2} \frac{d\Delta}{\Delta^2}, \quad \Delta \ll \gamma - 1. \quad (81.15)$$

Let us note that this formula, expressed through the velocity of the incident electron ($v_1 = |\mathbf{p}_1|/\varepsilon_1$), preserves its form in the transition to the non-relativistic case. It is natural therefore that it agrees in form with the result of non-relativistic theory (cf. III, (148.17)).

Let us now consider the scattering of a positron on an electron (*H. Bhabha*, 1936). This is another cross channel of the same generalised reaction, to which the scattering of an electron on an electron pertains. If p_- , p_+ are the initial, and p'_- , p'_+ the final 4-momenta of the electron and positron, then the transition from one case to the other is accomplished by the substitution

$$p_1 \rightarrow -p'_+, \quad p_2 \rightarrow p_-, \quad p'_1 \rightarrow -p_+, \quad p'_2 \rightarrow p'_-.$$

The kinematic invariants (81.2) thereby acquire the following meaning:

$$s = (p_- - p'_+)^2, \quad t = (p_+ - p'_-)^2, \quad u = (p_- - p_+)^2. \quad (81.16)$$

If the ee -scattering was the s -channel, then the $\bar{e}e$ -scattering is the u -channel of the reaction. The square of the scattering amplitude, expressed through s , t , u , remains the same, while in the denominator of formula (81.5) one must replace $s \rightarrow u$. Thus, for the cross section of scattering of the positron on the electron we obtain instead of (81.7)

$$d\sigma = r_e^2 \frac{4\pi m^2 dt}{u(u - 4m^2)} \left\{ \frac{1}{t^2} \left[\frac{s^2 + u^2}{2} + 4m^2(t - m^2) \right] + \right. \\ \left. + \frac{1}{u^2} \left[\frac{s^2 + t^2}{2} + 4m^2(u - m^2) \right] + \frac{4}{tu} \left(\frac{s}{2} - m^2 \right) \left(\frac{s}{2} - 3m^2 \right) \right\}. \quad (81.17)$$

In the centre-of-inertia system the values of the invariants s , t , u differ from (81.8) by the permutation of s and u :

$$s = -4\mathbf{p}^2 \cos^2 \frac{\theta}{2}, \quad t = -4\mathbf{p}^2 \sin^2 \frac{\theta}{2}, \quad u = 4\varepsilon^2. \quad (81.18)$$

In the non-relativistic limit formula (81.17) reduces to the Rutherford formula

$$d\sigma = \left(\frac{e^2}{mv^2} \right)^2 \frac{d\omega}{\sin^4(\theta/2)} \quad (\text{n. r.}), \quad (81.19)$$

where $\mathbf{v} = 2\mathbf{p}/m$ is the relative velocity of the electron and positron. It is obtained from the first term in the curly brackets in (81.17), originating from the diagram of the “scattering” type (see § 73). The contributions of the “annihilation” diagram (the second term in (81.17)) and of its interference with the scattering diagram (the third term) vanish in the non-relativistic limit⁹.

⁹The passage to the non-relativistic limit in the scattering and annihilation parts of the scattering amplitude see below (83.4) and (83.20). The annihilation term (83.20) contains the factor $1/c^2$ and therefore vanishes in this limit.

In the general case of arbitrary velocities the contributions of all three terms in (81.17) are of the same order of magnitude (only in the region

of small angles the first term predominates thanks to the factor $t^{-2} \propto \sin^{-4}(\theta/2)$). After the reduction of similar terms one can represent the cross section of positron scattering on the electron (in the centre-of-inertia system) in the form

$$d\sigma = d\sigma \frac{r_e^2}{16} \frac{m^2}{\varepsilon^2} \left\{ \frac{(\varepsilon^2 + \mathbf{p}^2)^2}{\mathbf{p}^2} \frac{1}{\sin^4(\theta/2)} - \frac{8\varepsilon^4 - m^4}{\mathbf{p}^2 \varepsilon^2} \frac{1}{\sin^2(\theta/2)} + \right. \\ \left. + \frac{12\varepsilon^4 + m^4}{\varepsilon^4} - \frac{4\mathbf{p}^2(\varepsilon^2 + \mathbf{p}^2)}{\varepsilon^4} \sin^2 \frac{\theta}{2} + \frac{4\mathbf{p}^4}{\varepsilon^4} \sin^2 \frac{\theta}{2} \right\}. \quad (81.20)$$

The symmetry with respect to the substitution $\theta \rightarrow \pi - \theta$, characteristic of the scattering of identical particles, is, of course, absent in the scattering of the positron on the electron. In the ultrarelativistic limit expression (81.20) differs from the electron-electron scattering cross section by the factor $\cos^4(\theta/2)$ alone:

$$d\sigma_{\bar{e}e} = \cos^4 \frac{\theta}{2} d\sigma_{ee} \quad (\text{u. r.}). \quad (81.21)$$

In the laboratory system, in which one of the particles (say, the electron) was initially at rest, we again introduce the quantity

$$\Delta = \frac{\varepsilon_+ - \varepsilon'_+}{m} = \frac{\varepsilon'_- - m}{m}, \quad (81.22)$$

i.e. the energy transferred by the positron to the electron. Analogously to (81.13) we now have

$$s = -2m(\varepsilon_+ - m - m\Delta), \quad t = -2m^2\Delta, \quad u = 2m(m + \varepsilon_+).$$

Substituting these expressions in (81.17), after simple transformations we obtain the following formula for the distribution of secondary electrons by energy:

$$d\sigma = 2\pi r_e^2 \frac{d\Delta}{\gamma^2 - 1} \left\{ \frac{\gamma^2}{\Delta^2} - \frac{2\gamma^2 + 4\gamma + 1}{\gamma + 1} \frac{1}{\Delta} + \frac{3\gamma^2 + 6\gamma + 4}{(\gamma + 1)^2} - \frac{2\gamma}{(\gamma + 1)^2} \Delta + \frac{1}{(\gamma + 1)^2} \Delta^2 \right\}, \quad (81.23)$$

where $\gamma = \varepsilon_+/m$; Δ ranges from 0 to $\gamma - 1$. For $\Delta \ll \gamma - 1$ from (81.23) one obtains the same formula (81.15), as for the scattering of electrons as well.

Polarisation effects in the scattering of electrons or positrons are computed by the general rules set out in § 65. In any general case the computations lead to cumbersome formulas. Here we limit ourselves to only a few remarks¹⁰.

In the approximation being considered (first, non-vanishing approximation of perturbation theory) there are absent in the cross section terms linear in the polarisation vectors of the initial or final particles. As in the non-relativistic theory (see III, § 140), such terms are forbidden by the requirements stemming from the hermiticity of the scattering matrix. Therefore the scattering cross section does not change if only one of the colliding

¹⁰More detailed information on this topic can be found in the review article *McMaster W. H* Rev. Mod. Phys. 1961. V. 133. P. 8.

particles is polarised, and the scattering of unpolarised particles does not lead to their polarisation.

These same requirements also forbid the correlation terms in the cross section containing products of polarisations of three of the four participating (initial and final) particles in the process. The cross section contains, however, double and quadruple correlation terms. In the scattering of non-identical particles (electron and positron, electron and muon) in the non-relativistic limit these terms vanish, since the spin-orbit interaction is absent. In the collision of identical particles correlation terms arise already in the non-relativistic case thanks to the exchange effect.

Problems

1. Determine the cross section of scattering of polarised electrons in the non-relativistic case.

Solution. In the non-relativistic case the bispinor amplitudes in the standard representation become two-component, and the density matrices become two-row matrices (29.20). In the scattering amplitude (81.3) there remain only the terms with $\mu = \nu = 0$, containing the diagonal (in the standard representation) matrices γ^0 . Instead of (81.4) we shall have

$$\sum_{\text{polar}} |M_{fi}|^2 = 16\pi^2 \alpha^2 e^4 \cdot 4m^4 \left\{ \left(\frac{1}{t^2} + \frac{1}{u^2} \right) \text{Sp}(1 + \sigma \zeta_1) \text{Sp}(1 + \sigma \zeta_2) - \frac{2}{tu} \text{Sp}(1 + \sigma \zeta_1) \text{Sp}(1 + \sigma \zeta_2) \right\} = 16\pi^2 \alpha^2 e^4 \cdot 4m^4 \left[\frac{1}{t^2} + \frac{1}{u^2} - \frac{1}{tu} (1 + \zeta_1 \zeta_2) \right].$$

(summation over the polarisations of the final electrons only). Hence the cross section of scattering

$$d\sigma = d\sigma_0 \left[1 - \zeta'_1 \zeta_2 \frac{\sin^2 \theta}{1 + 3 \cos^2 \theta} \right].$$

Whence for the vector of polarisation of the scattered electron we have

$$\zeta_1^{(f)} = -\zeta_2 \frac{2 \cos \theta (1 - \cos \theta)}{1 + 3 \cos^2 \theta}.$$

From this, for the probability of reversal of the direction of the spin of a completely polarised electron on scattering by an unpolarised electron

$$\frac{d\sigma}{d\sigma_0} = \frac{(1 - \cos \theta)^2}{2(1 + 3 \cos^2 \theta)}.$$

2. In the non-relativistic case, determine the cross section for the given initial and final polarisations ζ_1 and ζ'_1 and also as in Problem 1 we obtain

$$d\sigma = \frac{1}{2} d\sigma_0 \left[1 + \zeta_1 \zeta'_1 \frac{2 \cos \theta (1 + \cos \theta)}{1 + 3 \cos^2 \theta} \right].$$

Setting $\zeta_1 \zeta'_1 = -1$, we find from this the probability of spin reversal:

$$\frac{d\sigma}{d\sigma_0} = \frac{(1 - \cos \theta)^2}{2(1 + 3 \cos^2 \theta)}.$$

3. In the non-relativistic case, determine the probability of reversal of the direction of the spin of a completely polarised electron on scattering by an unpolarised electron.

Solution. Analogously we find the cross section for the given initial polarisations ζ_1 and ζ'_1 :

$$d\sigma = \frac{1}{2}d\sigma_0 \left[1 + \zeta_1 \zeta'_1 \frac{2 \cos \theta (1 + \cos \theta)}{1 + 3 \cos^2 \theta} \right].$$

Setting $\zeta_1 \zeta'_1 = -1$, we find from this the probability of spin reversal:

$$\frac{d\sigma}{d\sigma_0} = \frac{(1 - \cos \theta)^2}{2(1 + 3 \cos^2 \theta)}.$$

4. Determine the ratio of scattering cross sections of helical electrons with parallel and antiparallel spins in the ultrarelativistic case.

Solution. In (81.4) one must set, according to (29.22),

$$\rho_1 = \frac{1}{2}(\gamma p_1)(1 - 2\lambda_1 \gamma^5), \quad \rho_2 = \frac{1}{2}(\gamma p_2)(1 - 2\lambda_2 \gamma^5);$$

$$\rho'_1 = \frac{1}{2}\gamma p'_1, \quad \rho'_2 = \frac{1}{2}\gamma p'_2,$$

where $\lambda_1, \lambda_2 = \pm 1/2$. The computation of traces is carried out by the formulas cited in § 22; in particular,

$$\begin{aligned} & \text{Sp}[\gamma^5(\gamma a)\gamma^\mu(\gamma b)\gamma^\nu] \text{Sp}[\gamma^5(\gamma c)\gamma_\mu(\gamma d)\gamma_\nu] = \\ & = i^2(e^{\mu\lambda\kappa\sigma}a_\kappa b_\lambda)(e_{\mu\nu\rho\tau}c^\rho d^\tau) = 2(\delta_\nu^\lambda \delta_\rho^\kappa \delta_\tau^\sigma - \delta_\rho^\lambda \delta_\tau^\kappa \delta_\nu^\sigma)a_\kappa b_\lambda c^\rho d^\tau = 2(ac)(bd) - 2(ad)(bc). \end{aligned}$$

From this, for the ratio of cross sections one obtains the result coinciding with the formula found in Problem 4 of III, § 137 (taking $\mathbf{p}^2 \approx \varepsilon^2$), we find the desired ratio of cross sections:

$$\frac{d\sigma_{\uparrow\uparrow}}{d\sigma_{\uparrow\downarrow}} = \frac{1}{8}(1 + 6 \cos^2 \theta + \cos^4 \theta). \quad (1)$$

This ratio is minimal (1/8) at $\theta = \pi/2$.

Since the momenta of the colliding electrons (in the centre-of-inertia system) are mutually opposite, equal helicities ($\lambda_1 = \lambda_2$) correspond to antiparallel spins, and different helicities ($\lambda_1 = -\lambda_2$) correspond to parallel spins. Substituting s, t, u from (81.8) (taking $\mathbf{p}^2 \approx \varepsilon^2$), we find for the desired ratio of cross sections

$$\frac{d\sigma_{\uparrow\uparrow}}{d\sigma_{\uparrow\downarrow}} = \frac{1}{8}(1 + 6 \cos^2 \theta + \cos^4 \theta).$$

This ratio is minimal (1/8) at $\theta = \pi/2$.

5. The same for the scattering of positrons on electrons.

Solution. Instead of (81.4) one must compute

$$\begin{aligned} |M_{fi}|^2 \rightarrow & 16\pi^2 e^4 \left\{ \frac{1}{t^2} \text{Sp}(\rho'_- \gamma^\mu \rho_- \gamma^\nu) \text{Sp}(\rho_+ \gamma_\mu \rho'_+ \gamma_\nu) - \right. \\ & \left. - \frac{1}{tu} \text{Sp}(\rho'_- \gamma^\mu \rho_- \gamma^\nu \rho_+ \gamma_\mu \rho'_+ \gamma_\nu) + \dots \right\} \end{aligned}$$

(the remaining terms are obtained from those written by the permutation ρ_+ and ρ'_-). The density matrices:

$$\rho_- = \frac{1}{2}(\gamma p_-)(1 - 2\lambda_- \gamma^5), \quad \rho_+ = \frac{1}{2}(\gamma p_+)(1 + 2\lambda_+ \gamma^5),$$

$$\rho'_- = \frac{1}{2}\gamma p'_-, \quad \rho'_+ = \frac{1}{2}\gamma p'_+,$$

where $\lambda_+, \lambda_- = \pm 1/2$ (note that for the positron, as for the electron, $\lambda = 1/2$ denotes a spin directed along the momentum). The computation gives

$$\frac{d\sigma}{dt} \propto \left(\frac{s^2 + u^2}{t^2} + \frac{s^2 + t^2}{u^2} + \frac{2s^2}{tu} \right) - 4\lambda_+ \lambda_- \left(\frac{s^2 - u^2}{t^2} + \frac{s^2 - t^2}{u^2} + \frac{2s^2}{tu} \right).$$

From this for the ratio of cross sections one obtains a result coinciding with the formula of Problem 4.

6. Determine the cross section of scattering of muons on electrons.

Solution. The process is described by only one diagram (73.17). Instead of (81.5) we have

$$d\sigma = \frac{\pi e^4 dt}{(p_e p_\mu)^2 - m^2 \mu^2} f(t, u), \quad (1)$$

$$f(t, u) = \frac{1}{16t^2} \text{Sp}[(\gamma p'_\mu + \mu)\gamma^\lambda(\gamma p_\mu + \mu)\gamma^\nu] \text{Sp}[(\gamma p'_e + m)\gamma_\lambda(\gamma p_e + m)\gamma_\nu]$$

(p_e, p_μ and p'_e, p'_μ are the initial and final 4-momenta of the electron and muon; m, μ are their masses). The invariants:

$$s = (p_e + p_\mu)^2 = m^2 + \mu^2 + 2p_e p_\mu,$$

$$t = (p_e - p'_e)^2 = 2(m^2 - p_e p'_e) = 2(\mu^2 - p_\mu p'_\mu),$$

$$u = (p_e - p'_\mu)^2 = m^2 + \mu^2 - 2p_e p'_\mu,$$

$$s + t + u = 2(m^2 + \mu^2).$$

The computation leads to the result

$$f = \frac{1}{t^2} \left\{ \frac{s^2 + u^2}{2} + \frac{1}{2}(m^2 + \mu^2)(2t - m^2 - \mu^2) \right\} = \quad (\text{IX.3})$$

$$= \frac{1}{t^2} \left\{ \frac{s^2 + u^2}{2} + (m^2 + \mu^2)(2t - m^2 - \mu^2) \right\}. \quad (2)$$

Formulas (1) and (2) solve the problem posed. In the centre-of-inertia system

$$d\sigma = \frac{e^4 do}{8(\varepsilon_e + \varepsilon_\mu)^2 \mathbf{p}^4 \sin^4(\theta/2)} \times \\ \times [(\varepsilon_e \varepsilon_\mu + \mathbf{p}^2)^2 + (\varepsilon_e \varepsilon_\mu + \mathbf{p}^2 \cos \theta)^2 - 2(m^2 + \mu^2) \mathbf{p}^2 \sin^2(\theta/2)],$$

where $do = 2\pi \sin \theta d\theta$; $\varepsilon_e, \varepsilon_\mu$ are the energies of the electron and muon; $\mathbf{p}^2 = \varepsilon_e^2 - m^2 = \varepsilon_\mu^2 - \mu^2$.

When $\mathbf{p}^2 \ll \mu^2$ we return to formula (80.9) for the scattering of an electron on a stationary Coulomb centre.

7. Determine the ratio of cross sections of mutual scattering of helical electrons and muons with parallel and antiparallel spins in the ultrarelativistic case ($\varepsilon_\mu \gg \mu$, $\varepsilon_e \gg m$).

Solution. Analogously to Problem 4 we find

$$\frac{d\sigma_{\uparrow\downarrow}}{d\sigma_{\uparrow\uparrow}} = \cos^4 \frac{\theta}{2}$$

(θ is the scattering angle in the centre-of-inertia system).

8. Determine the cross section of conversion of an electron pair into a muon pair (B. B. Berestetskii, I. Ya. Pomeranchuk, 1955).

Solution. This is another cross-channel of the reaction, which pertains to μe -scattering. In this channel:

$$s = (p_e - \bar{p}_\mu)^2, \quad t = (p_\mu - \bar{p}_\mu)^2, \quad u = (p_e - p_\mu)^2,$$

where $p_e, \bar{p}_e$ are the 4-momenta of the electron and positron, and $p_\mu, \bar{p}_\mu$ those of the muon and antimuon. The reaction threshold corresponds to the energy of the electron pair, equal (in the centre-of-inertia system) to 2μ , so that $t > 4\mu^2$ must hold. In the laboratory system, in which the electron before the collision rests, the positron has energy ε_+ , $t = 2m(\varepsilon_+ + m) \approx 2m\varepsilon_+$, so that $\varepsilon_+ > \varepsilon_{\text{th}}$, where the threshold energy $\varepsilon_{\text{th}} = 2\mu^2/m$. In the laboratory system in which the electron before the collision rests the positron has energy ε_+ ; $t = 2m(\varepsilon_+ + m)$.

The differential cross section (instead of (1), in Problem 6):

$$d\sigma = \frac{4\pi e^4 ds}{[s - (m + \mu)^2][s - (m - \mu)^2]} f(t, u), \quad (1)$$

$$f(t, u) = \frac{1}{16t^2} \text{Sp}[(\gamma p'_\mu + \mu)\gamma^\lambda(\gamma p_\mu + \mu)\gamma^\nu] \text{Sp}[(\gamma p'_e + m)\gamma_\lambda(\gamma p_e + m)\gamma_\nu].$$

At a given t , the quantity s ranges between limits determined by the equations $su \approx \mu^2$, $s + t + u \approx 2\mu^2$, i.e.

$$\mu^2 - \frac{t}{2} - \frac{1}{2}\sqrt{t(t - 4\mu^2)} \leq s \leq \mu^2 - \frac{t}{2} + \frac{1}{2}\sqrt{t(t - 4\mu^2)}.$$

Elementary integration yields the result:

$$\sigma = \frac{4\pi}{3} r_e^2 \frac{m^2}{t} \sqrt{1 - \frac{4\mu^2}{t}} \left(1 + \frac{2\mu^2}{t}\right), \quad r_e = \frac{e^2}{m}. \quad (1)$$

(in the laboratory system $t = 2m\varepsilon_+$). This formula is inapplicable in the immediate vicinity of the threshold: when $\varepsilon_+ - \varepsilon_{\text{th}} \sim \mu e^4$, the emerging muons cannot be considered free particles (taking into account the Coulomb interaction between them, the cross section will not tend to zero but to a constant; see III, § 147).

The cross section (1) is maximal at $\varepsilon_+ = 1.7\varepsilon_{\text{th}}$. Its maximum value is approximately 20 times smaller than the cross section of two-photon annihilation at the same energy.

§ 82. Ionization losses of fast particles

Let us consider the collisions of a fast relativistic particle with an atom, accompanied by excitation and ionisation of the latter. In the non-relativistic case such inelastic collisions were considered in III, §§ 144–150; here we give a relativistic generalisation of the results obtained there (*H. A. Bethe*, 1933).

The velocity of the incident particle is assumed to be large in comparison with the velocities of the atomic electrons (thereby $Z\alpha \ll 1$, i.e. the atomic number is not too large). This condition ensures the applicability of the Born approximation to the process being considered.

The solution of the problem is somewhat different depending on whether the fast particle is a light (electron, positron) or heavy (meson, proton, α -particle and so on) one. We shall consider here the latter case, which is simpler.

Let $p = (\varepsilon, \mathbf{p})$ and $p' = (\varepsilon', \mathbf{p}')$ be the initial and final 4-momenta of the fast particle in the laboratory system of reference, in which the atom before the collision was at rest; the difference $q = p' - p$ gives the energy and momentum transferred by the particle to the atom. Let us divide the whole range of possible momentum transfers to the atom into two regions:

$$\text{I) } \frac{\mathbf{q}^2}{m} \ll m, \quad \text{II) } \frac{\mathbf{q}^2}{m} \gg I, \quad (82.1)$$

where m is the mass of the electron, I is some average atomic energy (the ionisation potential of the atom). Both regions overlap each other when $I \ll \mathbf{q}^2/m \ll m$; this circumstance will permit making a precise matching of the results obtained for each of the regions. We shall speak of the values of $\mathbf{q}$ in the first and in the second regions respectively as small and large momentum transfers.

Small momentum transfers. In this region the atomic electrons can be considered non-relativistic both in the initial and in the final states of the atom. The amplitude of the process is given by the expression

$$M_{fi}^{(n)} = e^2 J_{n0}^\mu (-q) J_{p'p}^\nu (q) D_{\mu\nu}(q), \quad (82.2)$$

where J_{n0} is the 4-current of transition of the atom from the initial (0) to the final (n) state, $J_{p'p}$ is the 4-current of transition of the fast particle; these currents, which replace here the expressions $(\bar{u}'\gamma u)$ that would stand, for example, in the amplitude of scattering of two “elementary” particles—the electron and the muon (73.17) (cf. also (139.3)). The transition currents are taken in the momentum representation (see (43.11)). The cross section of the process in the laboratory system of reference:

$$d\sigma_n = 2\pi\delta(\varepsilon - \varepsilon' - \omega_{n0}) \left| M_{fi}^{(n)} \right|^2 \frac{d^3 p'}{2|\mathbf{p}|2\varepsilon'(2\pi)^3}, \quad (82.3)$$

where $\omega_{n0} = E_n - E_0$ is the frequency of transition between states of the atom. The final state may correspond both to a discrete and to a continuous spectrum; the first case corresponds to excitation, and the second to ionisation of the atom.

The photon propagator is conveniently chosen in the present case in the Coulomb gauge (76.14), in which only its spatial components are different from zero:

$$D_{ik}(q) = -\frac{4\pi}{\omega^2 - \mathbf{q}^2} \left(\delta_{ik} - \frac{q_i q_k}{\mathbf{q}^2} \right). \quad (82.4)$$

Then in (82.2) only the spatial components of the 4-currents of transition are needed.

The atomic transition current $\mathbf{J}_{n0}(\mathbf{q})$ in the present case is the Fourier component of the usual non-relativistic expression:

$$\mathbf{J}_{n0}(\mathbf{q}) = \frac{i}{2m} \int e^{-i\mathbf{q}\mathbf{r}} (\psi_0 \nabla \psi_n^* - \psi_n^* \nabla \psi_0) d^3x, \quad (82.5)$$

where ψ_0, ψ_n are the atomic wave functions (with the summation sign over the atomic electrons, which for brevity of notation we omit here and below, to be understood).

Prointegrating over the first member by parts, we may rewrite this expression in the form of the matrix element:

$$\mathbf{J}_{n0}(\mathbf{q}) = \frac{1}{2} (\mathbf{v} e^{-i\mathbf{q}\mathbf{r}} + e^{-i\mathbf{q}\mathbf{r}} \mathbf{v})_{n0}, \quad (82.6)$$

where $\hat{\mathbf{v}} = -\frac{i}{m} \nabla$ is the operator of the velocity of the electron.

As regards the current of transition of the scattered particle, in view of the relative smallness of its recoil ($|\mathbf{q}| \ll |\mathbf{p}|$) it can simply be replaced by the diagonal element

$$\mathbf{J}_{pp}(0) = 2\mathbf{p}z, \quad (82.7)$$

corresponding to a classical rectilinear motion (cf. below (99.5)); here is also introduced a factor z that takes into account the possible difference of the charge of the particle (ze) from the electron charge.

The smallness of $\mathbf{q}$ means also the smallness of the deflection angle of the particle ϑ . Furthermore, the longitudinal (with respect to $\mathbf{p}$) and transverse components of $\mathbf{q}$ are

$$-q_{\parallel} \approx \frac{dp}{d\varepsilon} \omega_{n0} = \frac{\omega_{n0}}{v}, \quad q_{\perp} \approx |\mathbf{p}| \vartheta, \quad (82.8)$$

so that $\mathbf{q}\mathbf{p} \approx -\varepsilon\omega_{n0}$.

Substitution of ((82.4)–(82.8)) in (82.2) gives, taking into account that

$$M_{fi}^{(n)} = -\frac{4\pi ze^2}{q^2} \left\langle n \left| \frac{\varepsilon}{\omega_{n0}} (\mathbf{q}\mathbf{v} e^{-i\mathbf{q}\mathbf{r}} + e^{-i\mathbf{q}\mathbf{r}} \mathbf{q}\mathbf{v}) + (\mathbf{p}\mathbf{v} e^{-i\mathbf{q}\mathbf{r}} + e^{-i\mathbf{q}\mathbf{r}} \mathbf{p}\mathbf{v}) \right| 0 \right\rangle.$$

In the first member we note that

$$\mathbf{q}\hat{\mathbf{v}}f + f\mathbf{q}\hat{\mathbf{v}} = 2i\hat{f},$$

where $f \equiv e^{-i\mathbf{q}\mathbf{r}}$ (see III, § 149); therefore the matrix element of this operator coincides with the matrix element $2i(\hat{f})_{n0} = 2\omega_{n0}f_{n0}$. In the second member it suffices to replace, in view of the smallness of $\mathbf{q}$, $e^{-i\mathbf{q}\mathbf{r}}$ by unity. Then

$$M_{fi}^{(n)} = -\frac{8\pi ze^2}{q^2} \{ \varepsilon (e^{-i\mathbf{q}\mathbf{r}})_{n0} - i\mathbf{p}\mathbf{r}_{n0}\omega_{n0} \}.$$

The square of the modulus of this expression:

$$\begin{aligned} |M_{fi}^{(n)}|^2 = \frac{64\pi^2 (ze)^2}{(q^2)^2} & \left\{ \varepsilon^2 |(e^{-i\mathbf{q}\mathbf{r}})_{n0}|^2 + 2(\mathbf{q}\mathbf{r}_{n0})(\mathbf{p}\mathbf{r}_{n0})\varepsilon\omega_{n0} + \right. \\ & \left. + (\mathbf{p}\mathbf{r}_{n0})^2 \omega_{n0}^2 \right\} \end{aligned} \quad (82.9)$$

(in the second member it is assumed that $e^{-i\mathbf{q}\mathbf{r}} \approx 1 - i\mathbf{q}\mathbf{r}$; in the first member this cannot be done for the reason which will be clarified below—see remark on p. 378).

The energy losses of a fast particle as a result of its inelastic collisions with atoms¹¹ are determined by the quantity

$$\kappa = \sum_n \int \omega_{n0} d\sigma_n = \frac{1}{16\pi^2} \sum_n \int \omega_{n0} \left| M_{fi}^{(n)} \right|^2 d\vartheta', \quad (82.10)$$

where the summation is carried out over all possible final states of the atom, and the integration over the directions of the scattered particle; we shall call this quantity the *effective retardation* (the ratio κ/ε is called the *cross section of energy losses*).

The integration in (82.10) can be carried out in two stages: as an averaging over the direction of $\mathbf{p}'$ relative to $\mathbf{p}$ and then integration over $d\vartheta' \approx 2\pi\vartheta d\vartheta$, where ϑ is the small angle of deflection. The first operation replaces $\mathbf{q}\mathbf{r}_{n0}$ by

$$\mathbf{q}\mathbf{r}_{n0} \rightarrow q_{\parallel} x_{n0} = -\frac{\omega_{n0}}{v} x_{n0},$$

where x_{n0} is the matrix element of one of the Cartesian coordinates of the atomic electrons¹². The integration over ϑ can be replaced by integration over q^2 , noting that

$$-q^2 = -\omega_{n0}^2 + \mathbf{q}^2 \approx -\omega_{n0}^2 + \frac{\omega_{n0}^2}{v^2} + \mathbf{p}^2 \vartheta^2 = \frac{\omega_{n0}^2 M^2}{\mathbf{p}^2} + \mathbf{p}^2 \vartheta^2 \quad (82.11)$$

and therefore $2\vartheta d\vartheta = d|q^2|/\mathbf{p}^2$ (M is the mass of the fast particle). As a result we obtain

$$\kappa = 4\pi(z e)^2 \sum_n \int \left\{ |(e^{-i\mathbf{q}\mathbf{r}})_{n0}|^2 \frac{\omega_{n0}}{v^2} - \omega_{n0}^3 |x_{n0}|^2 \left| \frac{M^2}{\mathbf{p}^2} + \frac{1}{v^2} \right| \right\} \frac{d|q^2|}{|q^2|^2}. \quad (82.12)$$

The lower limit of integration over q^2 :

$$|q^2|_{\min} = \frac{M^2}{\mathbf{p}^2} \omega_{n0}^2. \quad (82.13)$$

As the upper limit we shall choose a certain value $|q^2|_1$ such that

$$I \ll \frac{|q^2|_1}{m} \ll m, \quad (82.14)$$

i.e. lying in the region of overlap of regions I and II (see (82.1)).

The integration and summation in (82.12) is accomplished similarly to what was done in III, § 149 for the non-relativistic case. Let us divide the whole interval of integration into two parts still: a) from $|q^2|_{\min}$ to $|q^2|_0$ and b) from $|q^2|_0$ to $|q^2|_1$, where the value $|q^2|_0$ is such that

$$\frac{IM}{|\mathbf{p}|} \ll \sqrt{|q^2|_0} \ll m\alpha \quad (82.15)$$

¹¹These losses are frequently called ionisation losses, although they are connected not only with ionisation but also with excitation of atoms.

¹²It is immaterial which one: after the summation over the direction of the momentum, the matrix element x_{n0} arising below no longer depends on the direction of the axis x .

(the quantity $m\alpha$ is of the order of the magnitudes of the momenta of the atomic electrons). In region a) one can expand $e^{-i\mathbf{q}\mathbf{r}} \approx 1 - i\mathbf{q}\mathbf{r}$, and the contribution of this region to κ takes the form

$$4\pi(ze)^2 \sum_n \int_{|q^2|_{\min}}^{|q^2|_0} \left\{ \frac{1}{v^2} \omega_{n0} |x_{n0}|^2 \frac{1}{|q^2|} - \frac{M^2}{\mathbf{p}^2} \omega_{n0}^3 |x_{n0}|^2 \frac{1}{|q^2|^2} \right\} d|q^2| \approx$$

$$\approx \frac{4\pi(ze)^2}{v^2} \sum_n \omega_{n0} |x_{n0}|^2 \left[\ln \frac{|q^2|_0 \mathbf{p}^2}{M^2 \omega_{n0}^2} - v^2 \right].$$

(The integration in the second member can be extended to infinity.)

The summation is accomplished with the help of the formula

$$\sum_n \omega_{n0} |x_{n0}|^2 = \frac{Z}{2m}, \quad (82.16)$$

where Z is the number of electrons in the atom (see III, (149.10)). The result can be represented in the form

$$\frac{2\pi(ze)^2 Z}{mv^2} \left[\ln \frac{|q^2|_0 \mathbf{p}^2}{M^2 I^2} - v^2 \right], \quad (82.17)$$

where I is a certain average atomic energy, defined by the formula

$$\ln I = \frac{\sum_n \omega_{n0} |x_{n0}|^2 \ln \omega_{n0}}{\sum_n \omega_{n0} |x_{n0}|^2} = \frac{2m}{Z} \sum_n \omega_{n0} |x_{n0}|^2 \ln \omega_{n0}. \quad (82.18)$$

In region b) we have, according to (82.11), $|q^2| \approx \mathbf{p}^2 \vartheta^2$, i.e. $|q^2|$ does not depend on the number n of the final state of the atom; n also does not affect the limits of integration. Therefore the summation over n in (82.12) can be carried out under the integral sign. In the first member it is accomplished by the formula

$$\sum_n |(e^{-i\mathbf{q}\mathbf{r}})_{n0}|^2 \omega_{n0} = \frac{Z}{2m} |q^2| \quad (82.19)$$

(see III, (149.5)), and the integral from it is equal to ¹³

$$\frac{2\pi Z(ze)^2}{mv^2} \ln \frac{|q^2|_1}{|q^2|_0}.$$

The integral from the second member in (82.12) in this region gives a negligible contribution to κ .

Combining the last formula with (82.17), we find the contribution to κ from the whole region of small momentum transfers:

$$\frac{2\pi Z(ze)^2}{mv^2} \left[\ln \frac{|q^2|_1 \mathbf{p}^2}{M^2 I^2} - v^2 \right]. \quad (82.20)$$

¹³The logarithmic divergence of the integral at the upper limit is precisely the reason why in the first member in (82.12) it was not possible to expand $e^{-i\mathbf{q}\mathbf{r}}$ in powers of $\mathbf{q}$.

Large momentum transfers. Let us turn to collisions with momentum transfer large in comparison with the momentum of the atomic electrons ($\mathbf{q}^2 \gg mI$). In this region one can, obviously, neglect the binding of the electrons in the atom, i.e. consider their scattering free. Correspondingly, the collision of the fast particle with the atom will represent in this case the elastic scattering on each of the Z atomic electrons individually. Due to the large velocity of the fast particle the atomic electrons can be considered initially at rest.

Let us denote by $m\Delta$ the energy transferred by the fast particle to the atomic electron, and let $d\sigma_\Delta$ be the cross section of elastic scattering with such transfer. The differential effective retardation on each atom will then be

$$d\kappa = Zm\Delta d\sigma_\Delta. \quad (82.21)$$

The maximum energy that can be transferred to the resting electron by a collision with a particle of mass $M \gg m$ is

$$m\Delta_{\max} = \frac{2m\mathbf{p}^2}{m^2 + M^2 + 2m\varepsilon} \approx \frac{2m\mathbf{p}^2}{M^2 + 2m\varepsilon}, \quad (82.22)$$

where ε and $\mathbf{p}$ are the energy and momentum of the incident particle (see II, (13.13)). We shall assume further that the energy ε although it can be ultrarelativistic ($\varepsilon \gg M$), is at the same time

$$\varepsilon \ll \frac{M^2}{m}. \quad (82.23)$$

Then the maximal transferred energy

$$m\Delta_{\max} \approx \frac{2m\mathbf{p}^2}{M^2} = 2mv^2\gamma^2, \quad \gamma = \frac{1}{\sqrt{1-v^2}} \quad (82.24)$$

remains small in comparison with the original kinetic energy of the incident particle ($m\Delta_{\max} \ll \varepsilon - M$). Correspondingly the momentum transfer $\mathbf{q}$ also remains always small in comparison with the initial momentum of the particle $\mathbf{p}$. This circumstance permits one to consider the motion of the latter not changing during the collision, i.e. to consider the incident particle as infinitely heavy. Then the scattering cross section is obtained simply by transformation of the cross section of scattering of the electron on a stationary Coulomb centre (80.7) to the laboratory system of reference in which the electron

was initially at rest. This is easy to do, noting that in the indicated approximation

$$-q^2 \approx \mathbf{q}^2 = 4\mathbf{p}^2 \sin^2 \frac{\theta}{2}, \quad d\sigma' = \frac{\pi d|q^2|}{\mathbf{p}^2},$$

and the relative velocity v is the same in both systems. Formula (80.7) takes the form

$$d\sigma = \frac{4\pi(ze)^2}{v^2} \left(1 - \frac{|q^2|}{4m^2\gamma^2} \right) \frac{d|q^2|}{|q^2|^2}.$$

The energy transfer Δ is expressed through the same invariant q^2 : according

to $-q^2 = 2m^2\Delta$. Therefore we have¹⁴

$$d\sigma_\Delta = \frac{2\pi(ze)^2}{mv^2} \left(1 - v^2 \frac{\Delta}{\Delta_{\max}}\right) \frac{d\Delta}{\Delta^2}. \quad (82.25)$$

The contribution to the effective retardation from the region of momentum transfers under consideration is obtained by integrating (82.21) from the previously introduced boundary $|q^2|_1$ to $|q^2|_{\max} = 2m^2\Delta_{\max}$. It equals

$$\frac{2\pi(ze)^2 Z}{mv^2} \left(\ln \frac{2\Delta_{\max} m^2}{|q^2|_1} - v^2 \right). \quad (82.26)$$

Finally, adding the contributions (82.20) and (82.25), we obtain the following result for the total ionisation losses of a fast heavy particle:

$$\kappa = \frac{4\pi Z(ze)^2}{mv^2} \left(\ln \frac{2mv^2}{I(1 - v^2/c^2)} - \frac{v^2}{c^2} \right) \quad (82.27)$$

(in ordinary units). In the non-relativistic limit from this one recovers the previous formula (150.10) (see III):

$$\kappa = \frac{4\pi Z(ze)^2}{mv^2} \ln \frac{2mv}{I}, \quad (82.28)$$

and in the ultrarelativistic case

$$\kappa = \frac{4\pi Z(ze)^2}{mc^2} \left(\ln \frac{2mc^2}{I(1 - v^2/c^2)} - 1 \right). \quad (82.29)$$

The retardation depends only on the velocity (but not on the mass) of the fast particle. The decrease of the retardation with increasing velocity of the fast particle, according to (82.27), is replaced in the ultrarelativistic region by a slow (logarithmic) growth.

Problems

1. Determine the effective retardation of a relativistic electron.

Solution. For the region of small momentum transfers the result is the same as the expression (82.20). For the region of large transfers, instead of (82.24), one should use formula (81.14), taking into account the exchange effects. Integrating $\Delta d\sigma_\Delta$ from $d\Delta$ over $|q^2|_1/2m^2$ to $(\gamma - 1)/2$ and combining with (82.20), we find

$$\kappa = \frac{2\pi Ze^4}{mv^2} \left[\ln \frac{m^2(\gamma^2 - 1)(\gamma - 1)}{2I^2} - \left(\frac{2}{\gamma} - \frac{1}{\gamma^2} \right) \ln 2 + \frac{1}{\gamma^2} + \frac{(\gamma - 1)^2}{8\gamma^2} \right]. \quad (1)$$

In the non-relativistic case from this one recovers the formula from Problem to § 149 in III, and in the ultrarelativistic ($\gamma \gg 1$)

$$\kappa = \frac{2\pi Ze^4}{mc^2} \left(\ln \frac{m^2 c^4 \gamma^3}{2I^2} + \frac{1}{8} \right). \quad (2)$$

¹⁴In this formula, of course, specific effects of strong interactions are not taken into account, if the heavy particle is a hadron. These effects (the hadronic form factor), however, become important only for $|q^2| \propto 1/M^2$, and for the condition (82.22) such momentum transfers are excluded.

2. The same for the positron.

Solution. For $d\sigma_\Delta$ in the region of large transfers one should use (81.23), with the upper limit of integration over Δ being $\gamma - 1$. The answer in the ultrarelativistic case:

$$\kappa = \frac{2\pi Ze^4}{mc^2} \left(\ln \frac{2m^2 c^4 \gamma^3}{I^2} - \frac{23}{12} \right).$$

§ 83. Breit equation

As is well known, in classical electrodynamics a system of interacting particles can be described with the help of a Lagrangian function dependent only on the coordinates and velocities of the particles themselves and correct to terms $\sim 1/c^2$ (see II, § 65). This circumstance is connected with the fact that radiation appears only as an effect of order $1/c^2$.

In quantum theory this situation corresponds to the possibility of description of the system by a Schrödinger equation, with perturbation terms of second order. For an electron moving in an external electromagnetic field, such an equation was established in § 33. Now we turn to deriving an analogous equation describing a system of two interacting particles.

We shall proceed from the relativistic expression for the amplitude of scattering of two particles. In the non-relativistic approximation it goes over to the ordinary Born scattering amplitude, proportional to the Fourier component of the potential of electrostatic interaction of two charges. Computing the amplitude to accuracy of terms of second order in $1/c^2$, we shall be able to establish the form of the potential corresponding to it.

Let us suppose first that the two particles are different, with masses m_1 and m_2 (say, electron and muon). Then the scattering is depicted by a single diagram

To it there corresponds the amplitude

$$M_{fi} = e^2 (\bar{u}'_1 \gamma^\mu u_1) D_{\mu\nu}(q) (\bar{u}'_2 \gamma^\nu u_2), \quad q = p'_1 - p_1 = p_2 - p'_2 \quad (83.1)$$

(here it is assumed that the charges of the particles are of the same sign; in the opposite case e^2 is replaced by $-e^2$).

Further calculations are substantially simplified if we choose the photon propagator $D_{\mu\nu}$ not in the ordinary but in the Coulomb gauge (76.12), (76.13)¹⁵:

$$D_{00} = -\frac{4\pi}{\mathbf{q}^2}, \quad D_{0i} = 0, \quad D_{ik} = \frac{4\pi}{\mathbf{q}^2 - \omega^2/c^2} \left(\delta_{ik} - \frac{q_i q_k}{\mathbf{q}^2} \right). \quad (83.2)$$

Then the amplitude of scattering

$$M_{fi} = e^2 \{ (\bar{u}'_1 \gamma^0 u_1) (\bar{u}'_2 \gamma^0 u_2) D_{00} + (\bar{u}'_1 \gamma^k u_1) (\bar{u}'_2 \gamma^k u_2) D_{ik} \}. \quad (83.3)$$

¹⁵In this paragraph we write out the factors c in all intermediate formulas, and in the final formulas also $\hbar$.

To the neglect of all terms containing the factor $1/c$, the second member in the curly brackets drops out entirely, and the first gives

$$M_{fi} = -2m_1 \cdot 2m_2 \begin{pmatrix} w_1^{(0)'} \\ w_1^{(0)} \end{pmatrix} \begin{pmatrix} w_2^{(0)'} \\ w_2^{(0)} \end{pmatrix} U(\mathbf{q}), \quad (83.4)$$

where

$$U(\mathbf{q}) = \frac{4\pi e^2}{\mathbf{q}^2}, \quad (83.5)$$

and by $w_1^{(0)}$, $w_2^{(0)}$, ... are denoted the 3-spinors of the non-relativistic plane waves introduced in §23 (two-component amplitudes). The function $U(\mathbf{q})$ represents the Fourier component of the potential energy of the Coulomb interaction: $U(r) = e^2/r$.

In the next (by $1/c$) approximation the “Schrödinger” wave function of the free particle φ_{Schr} (normalised by the integral $\int |\varphi_{\text{Schr}}|^2 d^3x$) satisfies the equation

$$\hat{H}^{(0)} \varphi_{\text{Schr}} = (\varepsilon - mc^2) \varphi_{\text{Schr}}, \quad \hat{H}^{(0)} = \frac{\hat{\mathbf{p}}^2}{2m} - \frac{\hat{\mathbf{p}}^4}{8m^3 c^2}, \quad \hat{\mathbf{p}} = -i\hbar \nabla, \quad (83.6)$$

in which the next-order term of the relativistic expansion for the kinetic energy has been taken into account. The amplitude of the (spinor) part of such a plane wave we denote by w (when $1/c \rightarrow 0$ it goes over to $w^{(0)}$). It is precisely through these amplitudes that the desired scattering amplitude should be expressed so that from its form one could determine the “Schrödinger” interaction potential in the approximation being considered.

In accordance with formula (33.11) the bispinor amplitude of the free particle u is expressed through the “Schrödinger” amplitude w —with the accuracy required here—in the form

$$u = \sqrt{2m} \begin{pmatrix} \left(1 - \frac{\mathbf{p}^2}{8m^2 c^2}\right) w \\ \frac{\boldsymbol{\sigma} \mathbf{p}}{2mc} w \end{pmatrix}. \quad (83.7)$$

With the help of this formula we find

$$\begin{aligned} \bar{u}'_1 \gamma^0 u_1 &= u'^*_1 \alpha_1 u_1 = 2m_1 \left(1 - \frac{\mathbf{p}_1'^2 + \mathbf{p}_1^2}{8m_1^2 c^2}\right) w'^*_1 w_1 + \\ &+ \frac{1}{2m_1 c^2} w'^*_1 (\boldsymbol{\sigma} \mathbf{p}'_1) (\boldsymbol{\sigma} \mathbf{p}_1) w_1 = 2m_1 w'^*_1 \left\{1 - \frac{\mathbf{q}^2}{8m_1^2 c^2} + \frac{i\boldsymbol{\sigma} [\mathbf{q} \mathbf{p}_1]}{4m_1^2 c^2}\right\} w_1, \\ \bar{u}'_1 \gamma u_1 &= u'^*_1 \alpha_1 u_1 = \frac{1}{c} w'^*_1 \{\boldsymbol{\sigma} (\boldsymbol{\sigma} \mathbf{p}_1) + (\boldsymbol{\sigma} \mathbf{p}'_1) \boldsymbol{\sigma}\} w_1 = \\ &= \frac{1}{c} w'^*_1 \{i[\boldsymbol{\sigma} \mathbf{q}] + 2\mathbf{p}_1 + \mathbf{q}\} w_1 \end{aligned}$$

(where $\mathbf{q} = \mathbf{p}'_1 - \mathbf{p}_1 = \mathbf{p}_2 - \mathbf{p}'_2$). Analogous expressions for $(\bar{u}'_2 \gamma^0 u_2)$ and $(\bar{u}'_2 \gamma u_2)$ differ by the replacement of indices 1 by 2 and correspondingly the replacement of q by $-q$.

Substituting these expressions in (83.3). Since the product $(\bar{u}'_1 \gamma u_1)(\bar{u}'_2 \gamma u_2)$ already contains the factor $1/c^2$, in it one can neglect the term ω^2/c^2 in the denominator of D_{ik} . As a result we obtain the scattering amplitude in the form

$$M_{fi} = -2m_1 \cdot 2m_2 (w_1'^* w_2'^* U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}) w_1 w_2), \quad (83.8)$$

where

$$\begin{aligned} U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}) = 4\pi e^2 \left\{ \frac{1}{\mathbf{q}^2} - \frac{1}{8m_1^2 c^2} - \frac{1}{8m_2^2 c^2} + \frac{(\mathbf{q}\mathbf{p}_1)(\mathbf{q}\mathbf{p}_2)}{m_1 m_2 c^2 \mathbf{q}^4} - \frac{\mathbf{p}_1 \mathbf{p}_2}{m_1 m_2 c^2 \mathbf{q}^2} + \right. \\ \left. + \frac{i\sigma_1[\mathbf{q}\mathbf{p}_1]}{4m_1^2 c^2 \mathbf{q}^2} - \frac{i\sigma_1[\mathbf{q}\mathbf{p}_2]}{2m_1 m_2 c^2 \mathbf{q}^2} - \frac{i\sigma_2[\mathbf{q}\mathbf{p}_2]}{4m_2^2 c^2 \mathbf{q}^2} + \frac{i\sigma_2[\mathbf{q}\mathbf{p}_1]}{2m_1 m_2 c^2 \mathbf{q}^2} + \right. \\ \left. + \frac{(\sigma_1 \mathbf{q})(\sigma_2 \mathbf{q})}{4m_1 m_2 c^2 \mathbf{q}^2} - \frac{\sigma_1 \sigma_2}{4m_1 m_2 c^2} \right\} \end{aligned} \quad (83.9)$$

(indices 1, 2 of the Pauli matrices indicate on which spinor indices they act: σ_1 acts on w_1 , and σ_2 on w_2).

The function $U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q})$ is the operator of particle interaction in the momentum representation. It is related to the operator $\hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r})$ in the coordinate representation by the formula

$$\int e^{-i(\mathbf{p}'_1 \mathbf{r}_1 + \mathbf{p}'_2 \mathbf{r}_2)} \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}) e^{i(\mathbf{p}_1 \mathbf{r}_1 + \mathbf{p}_2 \mathbf{r}_2)} d^3 x_1 d^3 x_2 = \quad (IX.4)$$

$$= (2\pi)^3 \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}). \quad (83.10)$$

If the operator $\hat{U}$ represents simply a function $U(\mathbf{r})$ ($\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$) that does not depend on $\mathbf{p}_1$ and $\mathbf{p}_2$, then formula (83.10) reduces to the ordinary definition of the Fourier components:

$$\int e^{-i\mathbf{q}\mathbf{r}} U(\mathbf{r}) d^3 x = U(\mathbf{q}).$$

From this it is clear that to find $\hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r})$ one must compute the integral

$$\int e^{i\mathbf{q}\mathbf{r}} U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}) \frac{d^3 q}{(2\pi)^3}$$

and then replace $\mathbf{p}_1, \mathbf{p}_2$ by operators $\hat{\mathbf{p}}_1 = -i\nabla_1, \hat{\mathbf{p}}_2 = -i\nabla_2$, placing them to the right of all other factors.

The necessary integrals are computed by differentiating the formula

$$\int e^{i\mathbf{q}\mathbf{r}} \frac{4\pi}{\mathbf{q}^2} \frac{d^3 q}{(2\pi)^3} = \frac{1}{r}. \quad (83.11)$$

Thus, taking the gradient we find

$$\int e^{i\mathbf{q}\mathbf{r}} \cdot \frac{4\pi \mathbf{q}}{\mathbf{q}^2} \frac{d^3 q}{(2\pi)^3} = -i\nabla \frac{1}{r} = \frac{i\mathbf{r}}{r^3}. \quad (83.12)$$

Further ($\mathbf{a}$, $\mathbf{b}$ are constant vectors)

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^4} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = \frac{i}{2} \left(\mathbf{a} \frac{\partial}{\partial \mathbf{r}} \right) \int e^{i\mathbf{q}\mathbf{r}} \left(\mathbf{b} \frac{\partial}{\partial \mathbf{q}} \right) \frac{1}{q^2} \frac{d^3q}{(2\pi)^3}; \quad (83.13)$$

the resulting integral after integration by parts reduces to (83.12) and gives

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^4} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = \frac{1}{2} (\mathbf{a}\nabla) \frac{\mathbf{b}\mathbf{r}}{r} = \frac{1}{2r} \left[\mathbf{a}\mathbf{b} - \frac{(\mathbf{a}\mathbf{r})(\mathbf{b}\mathbf{r})}{r^2} \right].$$

Finally,

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^2} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = -(\mathbf{a}\nabla)(\mathbf{b}\nabla) \frac{1}{r}.$$

In expanding the derivatives one must bear in mind that this expression contains the δ -function $\delta(\mathbf{r})$. To separate it we note that after averaging over the directions of $\mathbf{r}$:

$$-(\mathbf{a}\nabla)(\mathbf{b}\nabla) \frac{1}{r} = -\frac{1}{3} (\mathbf{a}\mathbf{b}) \Delta \frac{1}{r} = \frac{4\pi}{3} (\mathbf{a}\mathbf{b}) \delta(\mathbf{r}).$$

Expanding now the derivatives in the ordinary way, we find

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^2} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = \frac{1}{r^3} \left\{ \mathbf{a}\mathbf{b} - 3 \frac{(\mathbf{a}\mathbf{r})(\mathbf{b}\mathbf{r})}{r^2} \right\} + \frac{4\pi}{3} (\mathbf{a}\mathbf{b}) \delta(\mathbf{r}) \quad (83.14)$$

(on averaging over the directions of $\mathbf{r}$ the first term on the right vanishes and there remains only the term with the δ -function).

With the help of these formulas we obtain the following final expression for the operator of particle interaction:

$$\begin{aligned} \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}) &= \frac{e^2}{r} - \frac{\pi e^2 \hbar^2}{2c^2} \left(\frac{1}{m_1^2} + \frac{1}{m_2^2} \right) \delta(\mathbf{r}) - \\ &- \frac{e^2}{2m_1 m_2 c^2 r} \left[\hat{\mathbf{p}}_1 \hat{\mathbf{p}}_2 + \frac{\mathbf{r}(\mathbf{r} \hat{\mathbf{p}}_1) \hat{\mathbf{p}}_2}{r^2} \right] - \frac{e^2 \hbar}{4m_1^2 c^2 r^3} [\mathbf{r} \hat{\mathbf{p}}_1] \sigma_1 + \frac{e^2 \hbar}{4m_2^2 c^2 r^3} [\mathbf{r} \hat{\mathbf{p}}_2] \sigma_2 - \\ &- \frac{e^2}{2m_1 m_2 c^2 r} \{ -(\sigma_1 + 2\sigma_2) [\mathbf{r} \hat{\mathbf{p}}_1] + (\sigma_2 + 2\sigma_1) [\mathbf{r} \hat{\mathbf{p}}_2] \} + \\ &+ \frac{e^2 \hbar^2}{4m_1 m_2 c^2} \left\{ \frac{\sigma_1 \sigma_2}{r^3} - 3 \frac{(\sigma_1 \mathbf{r})(\sigma_2 \mathbf{r})}{r^5} - \frac{8\pi}{3} \sigma_1 \sigma_2 \delta(\mathbf{r}) \right\}. \end{aligned} \quad (83.15)$$

The full Hamiltonian of the system of two particles in this approximation

$$\hat{H} = \hat{H}_1^{(0)} + \hat{H}_2^{(0)} + \hat{U}, \quad (83.16)$$

where $\hat{H}^{(0)}$ are the Hamiltonians of free particles (83.6).

Two electrons. If both particles are identical (two electrons), then in the scattering amplitude there appears a second term, depicted by the “exchange” diagram

Computing its contribution to the interaction operator, however, is not necessary. The point is that the description of a system of identical particles by the Schrödinger equation

can be accomplished with the help of the same interaction operator as for non-identical particles, if one takes care of the proper symmetrisation of the

wave functions. In particular, in the consideration of the scattering of identical particles such symmetrisation automatically takes into account the contributions to the amplitude corresponding to both Feynman diagrams.

Thus, the Hamiltonian of a system of two electrons is obtained from formulas (83.15), (83.16), if one simply puts $m_1 = m_2$ ¹⁶:

$$\hat{H} = \frac{1}{2m}(\hat{\mathbf{p}}_1^2 + \hat{\mathbf{p}}_2^2) - \frac{1}{8m^3c^2}(\hat{\mathbf{p}}_1^4 + \hat{\mathbf{p}}_2^4) + \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}), \quad (\text{IX.5})$$

$$\begin{aligned} \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}) = & \frac{e^2}{r} - \pi \left(\frac{e\hbar}{mc} \right)^2 \delta(\mathbf{r}) - \frac{e^2}{2m^2c^2r} \left(\hat{\mathbf{p}}_1 \hat{\mathbf{p}}_2 + \frac{\mathbf{r}(\mathbf{r}\hat{\mathbf{p}}_1)\hat{\mathbf{p}}_2}{r^2} \right) + \\ & + \frac{e^2\hbar}{4m^2c^2r^3} \{ -(\boldsymbol{\sigma}_1 + 2\boldsymbol{\sigma}_2)[\mathbf{r}\hat{\mathbf{p}}_1] + (\boldsymbol{\sigma}_2 + 2\boldsymbol{\sigma}_1)[\mathbf{r}\hat{\mathbf{p}}_2] \} + \\ & + \frac{1}{4} \left(\frac{e\hbar}{mc} \right)^2 \frac{1}{r^3} \left\{ \frac{\boldsymbol{\sigma}_1 \boldsymbol{\sigma}_2}{r^3} - 3 \frac{(\boldsymbol{\sigma}_1 \mathbf{r})(\boldsymbol{\sigma}_2 \mathbf{r})}{r^5} - \frac{8\pi}{3} \boldsymbol{\sigma}_1 \boldsymbol{\sigma}_2 \delta(\mathbf{r}) \right\}. \end{aligned} \quad (83.17)$$

Let us note that the presence of terms with $\delta(\mathbf{r})$ does not, of course, signify the existence of a special strong interaction. The integral magnitude of all correction terms is the same, and in the spirit of the expansion performed they must all be considered small in comparison with the first term—the Coulomb interaction.

The various groups of terms in the interaction operator (83.17) have different character. The terms of the first row in $\hat{U}$ have a purely orbital origin. In the second row stand the terms, linear in the spin operators of the particles; they correspond to the *spin-orbit interaction*. Finally, the terms of the third row, quadratic in the spin operators, describe the *spin-spin interaction*¹⁷.

Electron and positron. The system of electron and positron requires special consideration. The scattering amplitude in this case is composed of two terms:

$$\begin{aligned} M_{fi} = & -e^2 [\bar{u}(p'_-)\gamma^\mu u(p_-)] D_{\mu\nu}(p_- - p'_-) [\bar{u}(-p_+)\gamma^\nu u(-p'_+)] + \\ & + e^2 [\bar{u}(-p_+)\gamma^\mu u(p_-)] D_{\mu\nu}(p_- + p_+) [\bar{u}(p'_-)\gamma^\nu u(-p'_+)] \end{aligned} \quad (83.18)$$

(the first corresponds to the scattering diagram, and the second to the annihilation diagram). Since the wave function of the “electron +

positron” system should not be antisymmetric, both terms give independent contributions to the operator of interaction.

The first term (whose structure coincides with the structure of the scattering amplitude (83.1)) leads, naturally, to an operator differing from (83.17) only by the overall sign.

¹⁶The wave equation with Hamiltonian (83.17) was first established by *Breit* (*G. Breit*, 1929), and its consistent quantum-mechanical derivation was given by *L. D. Landau* (1932).

¹⁷This interaction was mentioned in III, § 72 in connection with the fine structure of atomic levels, and the spin-spin interaction of electrons with the nucleus was treated in III, § 121, in connection with the superhyperfine structure of levels. In particular, formula (121.9) (see III) corresponds to the δ -function term in the spin-spin interaction operator.

Let us turn to the transformation of the second term.

We use here the photon propagator in the ordinary gauge:

$$D_{\mu\nu} = \frac{4\pi}{k^2} g_{\mu\nu} = \frac{4\pi}{\omega^2/c^2 - \mathbf{k}^2} g_{\mu\nu}. \quad (83.19)$$

In the present case $k = p_+ + p_-$, and since the particles are “almost non-relativistic”, we have

$$\frac{\omega^2}{c^2} \equiv \frac{(\varepsilon_+ + \varepsilon_-)^2}{c^2} \approx 4m^2 c^2 \gg (\mathbf{p}_+ + \mathbf{p}_-)^2 \equiv \mathbf{k}^2.$$

Therefore for the photon propagator it suffices to write

$$D_{\mu\nu} \approx \frac{\pi}{m^2 c^2} g_{\mu\nu}.$$

This already contains the factor $1/c^2$. Therefore the amplitudes $u(p)$ can be taken in the zeroth approximation:

$$u(p_-) = \sqrt{2m} \begin{pmatrix} w_-^{(0)} \\ 0 \end{pmatrix}, \quad u(-p_+) = \sqrt{2m} \begin{pmatrix} 0 \\ w_+^{(0)} \end{pmatrix},$$

where $w_-^{(0)}$, $w_+^{(0)}$ are the 3-spinors figuring in (23.12) (below we omit the indices (0) on them). With these amplitudes

$$\bar{u}(-p_+) \gamma^0 u(p_-) = u^*(-p_+) u(p_-) = 0,$$

$$\bar{u}(-p_+) \gamma u(p_-) = u^*(-p_+) \alpha u(p_-) = 2m(w_-^* \sigma w_-),$$

where $w_-^{(0)}$, $w_+^{(0)}$ are the 3-spinors figuring in (23.12). With these amplitudes

$$M_{fi}^{(\text{ann})} = -e^2 \frac{\pi}{m^2 c^2} (2m)^2 (w_-^* \sigma w_-) (w_+^{*\prime} \sigma w_+'). \quad (83.20)$$

From this, however, one cannot yet draw direct conclusions about the form of the operator of interaction. First, the scattering amplitude should be brought to the form in which the electronic (w_- and w_-') and positronic (w_+ and w_+') spinors are grouped together. This is achieved with the help of the formula

$$(w_-^* \sigma w_-) (w_+^{*\prime} \sigma w_+') = \frac{3}{2} (w_-^{*\prime} w_-) (w_+^* w_+') - \frac{1}{2} (w_-^{*\prime} \sigma w_-) (w_+^* \sigma w_+'), \quad (83.22)$$

which itself follows from (28.16).

Finally, expressing w and w' through w_+ and w_+' , according to (83.21), we find, as is easy to verify,

$$(w_-^* w_+') = (w_+^{*\prime} w_+), \quad (w_-^* \sigma w_+') = -(w_+^{*\prime} \sigma w_+). \quad (83.23)$$

Substituting (83.23) in (83.22) and then in (83.20), we obtain the final expression for the annihilation part of the scattering amplitude

$$M_{fi}^{(\text{ann})} = -4m^2 \left\{ w_-^{*\prime} w_+^{*\prime} \left[\frac{\pi e^2}{2m^2 c^2} (3 + \sigma_+ \sigma_-) \right] w_- w_+ \right\} \quad (\text{IX.6})$$

(the matrices σ_- and σ_+ act respectively on w_- and w_+). The expression in the square brackets represents the interaction operator in the momentum representation. The corresponding coordinate operator

$$\hat{U}^{(\text{ann})}(\mathbf{r}) = \frac{\pi\hbar^2 e^2}{2m^2 c^2} (3 + \sigma_+ \sigma_-) \delta(\mathbf{r}), \quad \mathbf{r} = \mathbf{r}_- - \mathbf{r}_+ \quad (83.24)$$

(Pirenne, 1947; B. B. Berestetskii and L. D. Landau, 1949). The full interaction operator of the electron and positron is

$$-\hat{U} + \hat{U}^{(\text{ann})}$$

with $\hat{U}$ from (83.17).

§ 84. Positronium

The formulas obtained in the preceding paragraph can be applied to positronium—a hydrogen-like system of electron and positron.

In the centre-of-inertia system the operators of momenta of the electron and positron: $\hat{\mathbf{p}}_- = -\hat{\mathbf{p}}_+ \equiv \hat{\mathbf{p}}$, where $\hat{\mathbf{p}} = -i\hbar\nabla$ is the operator of the momentum of relative motion corresponding to the relative radius vector $\mathbf{r} = \mathbf{r}_- - \mathbf{r}_+$. The full Hamiltonian of positronium¹⁸

$$\begin{aligned} \hat{H} &= \frac{\hat{\mathbf{p}}^2}{m} - \frac{e^2}{r} + \hat{V}_1 + \hat{V}_2 + \hat{V}_3, \\ \hat{V}_1 &= \frac{\hat{\mathbf{p}}^4}{4m^3 c^2} + 4\pi\mu_0^2 \delta(\mathbf{r}) - \frac{e^2}{2m^2 c^2 r} \left\{ \hat{\mathbf{p}}^2 + \frac{\mathbf{r}(\mathbf{r}\hat{\mathbf{p}})\hat{\mathbf{p}}}{r^2} \right\}, \\ \hat{V}_2 &= 6\mu_0^2 \frac{1}{r^3} \hat{\mathbf{I}}\hat{\mathbf{S}}, \\ \hat{V}_3 &= 6\mu_0^2 \frac{1}{r^3} \left\{ \frac{(\hat{\mathbf{S}}\mathbf{r})(\hat{\mathbf{S}}\mathbf{r})}{r^2} - \frac{1}{3} \hat{\mathbf{S}}^2 \right\} + 4\pi\mu_0^2 \left(\frac{7}{3} \hat{\mathbf{S}}^2 - 2 \right) \delta(\mathbf{r}). \end{aligned} \quad (84.1)$$

Here $\mu_0 = e\hbar/(2mc)$ is the Bohr magneton, $\hat{\mathbf{I}} = [\mathbf{r}\hat{\mathbf{p}}]$ is the operator of orbital angular momentum, $\hat{\mathbf{S}} = (\sigma_+ + \sigma_-)/2$ is the operator of total spin of the system (its square $\hat{\mathbf{S}}^2 = \frac{1}{2}(3 + \sigma_+ \sigma_-)$). In $\hat{V}_1$ are included all correction terms of purely orbital character; $\hat{V}_2$ is the spin-orbit interaction; $\hat{V}_3$ includes the spin-spin and “annihilation” interactions.

The “unperturbed” Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{m} - \frac{e^2}{r}$$

differs, naturally, from the Hamiltonian of the hydrogen atom only by the replacement of the electron mass by the reduced mass $m/2$. The energy levels of positronium are therefore half (in absolute value) of the levels of the hydrogen atom:

$$E = -\frac{me^4}{4\hbar^2 n^2} \quad (84.2)$$

¹⁸In ordinary units.

(n is the principal quantum number).

The remaining terms in (84.1) lead to a splitting of the levels (84.2)—the appearance of fine structure. The arising levels are classified first of all by the values of the total angular momentum j . We also see that the spin operators of the particles enter in the Hamiltonian (84.1) only in the form of the sum $\hat{\mathbf{S}}$. This means that the Hamiltonian commutes with the operator of the square of the total spin $\hat{\mathbf{S}}^2$, i.e. the total spin continues to be conserved also in the approximation under consideration (second in $1/c$). Therefore the energy levels of positronium can also be classified by the total spin, taking the values $S = 0$ and $S = 1$. The levels with spin 0 are called levels of *parapositronium*, and the levels with spin 1—levels of *orthopositronium*.

Let us emphasise that the conservation of total spin in positronium is in fact an exact law, not connected with this or that approximation by $1/c$; it follows from the

CP-invariance of electromagnetic interactions. Positronium represents a truly neutral system, and therefore its states are characterised by definite charge and combined parities. The latter is equal to $(-1)^{S+1}$ (see Problem to § 27); since S can take only the two values, 0 and 1, the conservation of combined parity is equivalent to conservation of total spin.

For $S = 0$ the total moment j coincides with the orbital. For spin $S = 1$ and given j the number l takes the values $j, j \pm 1$, correspondingly each level (n, j) of orthopositronium splits, generally speaking, into three levels. Since the two values $l = j$ and $l = j \pm 1$ correspond to different parities, the Hamiltonian has no matrix elements connecting these states. However the operator of perturbation (the first member in $\hat{V}_3$), generally speaking, has non-diagonal matrix elements connecting states with $l = j + 1$ and $l = j - 1$, so that when this number l loses, of course, its strict meaning of the orbital angular momentum.

Specific properties are possessed by the Zeeman effect in positronium (*B. B. Berestetskii, I. Ya. Pomeranchuk, 1949*). The orbital magnetic moment of positronium is always zero: since in positronium $[\mathbf{r}_+ \hat{\mathbf{p}}_+] = [\mathbf{r}_- \hat{\mathbf{p}}_-]$, the operator

$$\hat{\mu}_l = \mu_0([\mathbf{r}_+ \hat{\mathbf{p}}_+] - [\mathbf{r}_- \hat{\mathbf{p}}_-]) = 0.$$

The operator of the spin magnetic moment

$$\hat{\mu}_s = \mu_0(\boldsymbol{\sigma}_+ - \boldsymbol{\sigma}_-) \quad (84.3)$$

is not proportional to the operator of total spin $\hat{\mathbf{S}} = \frac{1}{2}(\boldsymbol{\sigma}_+ + \boldsymbol{\sigma}_-)$, and the operators $\hat{\mathbf{S}}^2$ and $\hat{\boldsymbol{\mu}}^2$ are not commutative. Therefore the states with definite values of the total spin S and its projection S_z are not eigenstates of the magnetic moment.

The states with given S and S_z are described by the spin functions χ_{SS_z} of the form

$$\begin{aligned} \chi_{11} &= \alpha_+ \alpha_-, & \chi_{1,-1} &= \beta_+ \beta_-, \\ \chi_{10} &= \frac{1}{\sqrt{2}}(\alpha_+ \beta_- + \alpha_- \beta_+), \\ \chi_{00} &= \frac{1}{\sqrt{2}}(\alpha_+ \beta_- - \alpha_- \beta_+), \end{aligned} \quad (84.4)$$

where α and β are the spin functions of one particle, corresponding to the spin projections $+1/2$ and $-1/2$ (the indices “+” or “-” indicate whether the function refers to the positron

or the electron). Of these the first two (χ_{11} and $\chi_{1,-1}$) are also eigenfunctions of the operator μ_z , with eigenvalue $\mu_z = 0$. The func-

tions χ_{10} and χ_{00} , however, are not eigenfunctions of μ_z ; such are the combinations

$$\frac{1}{\sqrt{2}}(\chi_{10} + \chi_{00}) = \alpha_+\beta_-, \quad \frac{1}{\sqrt{2}}(\chi_{10} - \chi_{00}) = \alpha_-\beta_+.$$

It is easy to see that the only non-zero matrix elements of the matrix $\langle S'S'_z|\mu_z|SS_z\rangle$, computed with the functions (84.4), are the elements

$$\langle 00|\mu_z|10\rangle = \langle 10|\mu_z|00\rangle = 2\mu_0. \quad (84.6)$$

In weak magnetic fields (when $\mu_0 H \ll \Delta$, where Δ is the difference between the energies of levels with $S = 0$ and $S = 1$) the initial approximation for computing the Zeeman splitting are the states with definite values of the total spin. In the first approximation this splitting is given by the average value of the energy of perturbation

$$\hat{V}_H = -\hat{\mu}_z H.$$

But all diagonal matrix elements of the operator $\hat{\mu}_z$, computed with the functions (84.4), are zero. Thus, in weak fields the linear Zeeman effect in positronium is absent.

In the limiting case of strong fields ($\mu_0 H \gg \Delta$) one can neglect the spin interaction, leading to the establishment of definite values of S . The components of the split level will in this case correspond to states with definite values of $\mu_z = \pm 2\mu_0$ (described by the functions (84.5)), and the magnitude of their shift is $\pm 2\mu_0 H$.

Problems

1. Determine the fine structure of the levels of parapositronium (*B. B. Berestetskii*, 1949)¹⁹.

Solution. The desired energy of level splitting is given by the average values of the correction terms in the Hamiltonian (84.1), computed with the wave functions of the unperturbed states with $j = l$ (for $S = 0$ the contribution that differs from zero arises only from $\hat{V}_1$ and the second member in $\hat{V}_3$).

The unperturbed wave functions (we denote them by ψ) satisfy the Schrödinger equation

$$\hat{\mathbf{p}}^2 \psi = -\Delta \psi = \left(E + \frac{1}{r}\right) \psi, \quad E = -\frac{1}{4n^2}.$$

Therefore

$$\begin{aligned} \hat{\mathbf{p}}^4 \psi &= \hat{\mathbf{p}}^2 \left(E + \frac{1}{r}\right) \psi = \left(E + \frac{1}{r}\right)^2 \psi - \psi \Delta \frac{1}{r} - 2 \left(\nabla \frac{1}{r}\right) (\nabla \psi) = \\ &= \left(E + \frac{1}{r}\right)^2 \psi + 4\pi \delta(\mathbf{r}) \psi + \frac{2}{r^2} \frac{\partial \psi}{\partial r} \end{aligned}$$

¹⁹The computation of the fine structure of orthopositronium see *Sokolov A. A., Tsitovich V. N.* JETP. 1953. V. 24. P. 253.

The average value

$$\overline{\mathbf{p}^4} = \overline{\left(E + \frac{1}{r}\right)^2} + 4\pi|\psi(0)|^2 + \int_0^\infty \int \frac{\partial|\psi|^2}{\partial r} dr d\Omega.$$

The last integral is non-zero only when $\psi(0) \neq 0$, i.e. only for $l = 0$, and the wave functions of S -states are spherically symmetric; the integral equals $-4\pi|\psi(0)|^2$ and cancels with the second term.

Introducing the operator of orbital angular momentum $\hat{\mathbf{l}} = [\mathbf{r}\hat{\mathbf{p}}]$ and writing

$$-\hat{\mathbf{p}}^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{\hat{\mathbf{l}}^2}{r^2}, \quad -\hat{\mathbf{p}}^2\psi = -\left(E + \frac{1}{r}\right)\psi.$$

From this, for the other mean value we need:

$$\begin{aligned} \int \psi^* \frac{r}{r^3} (\mathbf{r}\hat{\mathbf{p}})\hat{\mathbf{p}}\psi d^3x &= - \int \psi^* \frac{1}{r} \frac{\partial^2\psi}{\partial r^2} d^3x = \\ &= \frac{1}{r} \left(E + \frac{1}{r}\right) - 4\pi|\psi(0)|^2 - l(l+1)\overline{r^{-3}}. \end{aligned}$$

(for $l = 0$ the last member is absent).

According to the well-known formulas of the theory of the hydrogen atom (see III, (36.14), (36.16)) with the replacement of the electron mass m by $m/2$ we have

$$|\psi(0)|^2 = \frac{1}{8\pi n^3} \delta_{l0}, \quad \overline{r^{-2}} = \frac{1}{2n^3(2l+1)}, \quad \overline{r^{-3}} = \frac{1}{4n^3l(l+1)(2l+1)} \quad (l \neq 0).$$

With the help of these formulas we finally obtain for the desired energy levels of parapositronium

$$E_{nl} = -\frac{1}{4n^2} - \alpha^2 \frac{me^4}{\hbar^2} \frac{1}{2n^3} \left(\frac{1}{2l+1} - \frac{11}{32n} \right).$$

2. Determine the difference of energies of the ground states ($n = 1$, $l = 0$) of ortho- and parapositronium.

Solution. The dependence of energy on the total spin S at $l = 0$ is contained only in the average value of the second member in $\hat{V}_3$ (the first member vanishes on averaging in a spherically-symmetric S -state)²⁰. The ground level of orthopositronium (3S_1) lies above the ground level of parapositronium (1S_0) by the amount

$$E(^3S_1) - E(^1S_0) = \frac{7}{12} \alpha^2 \frac{me^4}{\hbar^2} = 8.2 \cdot 10^{-4} \text{ eV}.$$

²⁰The averaging over angles should be carried out before integration over r , as is seen from the manner of computation of the integral (83.14), leading to the first member in $\hat{V}_3$.

§ 85. Interaction of atoms at large distances

Between two neutral atoms, situated at large (in comparison with their own dimensions) distances r , there act forces of attraction. The ordinary quantum-mechanical calculation of these forces (see III, § 89) becomes, however, inapplicable at too large distances. The point is that in this calculation only the electrostatic interaction is considered, i.e. retardation effects are not taken into account. This consideration is valid only so long as the distance r remains small in comparison with the characteristic wavelengths λ_0 in the spectra of the interacting atoms. In this paragraph we shall carry out a calculation free from such a restriction.

We shall proceed in approximately the same way as in § 83, i.e. we shall compute in the first non-vanishing approximation the amplitude of elastic scattering (without change of internal state) of two different atoms. The obtained expression will be compared with the amplitude that would arise in describing the interaction of atoms by a potential energy $U(r)$.

In the latter case the first non-vanishing element of the S -matrix describing the given process would be the element of the first approximation of perturbation theory:

$$S_{fi} = -i \int \psi_1'^* \psi_2'^* U(r) \psi_1 \psi_2 d^3 x_1 d^3 x_2 \times \quad (\text{IX.7})$$

$$\times \int \exp\{-i(\varepsilon_1 + \varepsilon_2 - \varepsilon_1' - \varepsilon_2')t\} dt. \quad (85.1)$$

Here ψ_1, ψ_2 and ψ_1', ψ_2' are the time-independent parts of the wave functions (plane waves) describing the translational motion of the atoms with initial and final momenta; $\varepsilon_1, \varepsilon_2$ and $\varepsilon_1', \varepsilon_2'$ are the kinetic energies of this motion; the coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$ of the atoms as a whole may be understood as the coordinates of their nuclei, and the distance $r = |\mathbf{r}_1 - \mathbf{r}_2|$. The temporal integral gives, as usual, the δ -function expressing the law of conservation of energy.

The actual computation of the elastic scattering amplitude can be split, under the assumptions made, into two stages. First we average the S -operator over the wave functions of the unchanged (ground) states of both atoms (for given coordinates of their nuclei $\mathbf{r}_1$ and $\mathbf{r}_2$), and also over the photon vacuum—in the initial and final states of the process there are no photons. As a result we obtain a quantity which is a function of the distance between the nuclei; let us denote it by $\langle S(r) \rangle$ ²¹. To find the desired matrix element of transition, one must then compute the integral

$$S_{fi} = \int \psi_1'^* \psi_2'^* \langle S(r) \rangle \psi_1 \psi_2 d^3 x_1 d^3 x_2. \quad (85.3)$$

Comparing with (85.2), we see that if one obtains for $\langle S(r) \rangle$ the expression

$$\langle S(r) \rangle = -itU(r),$$

then $U(r)$ will be the desired energy of interaction of the atoms.

²¹Instead of a more cumbersome notation for the diagonal matrix element with indication of the states of the atom and photon field.

Since we have to deal in this case with collisions not of elementary particles but of complex systems (atoms, which in the intermediate states can be excited), the usual formal rules of diagram technique are here directly inapplicable, and we start from the initial expression for the S -operator in the form of the expansion (72.10).

For the interaction of atoms only the components of the fields with frequencies of the order of atomic (and smaller) are important. The corresponding wavelengths are large in comparison with atomic dimensions. Therefore the operator of electromagnetic interaction may be taken in the form

$$\hat{V} = -\hat{\mathbf{E}}(\mathbf{r}_1)\hat{\mathbf{d}}_1 - \hat{\mathbf{E}}(\mathbf{r}_2)\hat{\mathbf{d}}_2, \quad (85.4)$$

where $\hat{\mathbf{d}}_1$, $\hat{\mathbf{d}}_2$ are the operators of the dipole moments of the atoms (the Heisenberg operators, depending on time are meant), and $\hat{\mathbf{E}}(\mathbf{r})$ is the operator of the electric field, which is evaluated at the points of location of the corresponding atoms.

As is well known, the average values of the dipole moment of the atom in its stationary states are zero (see III, § 75). From this it follows that the non-zero scattering amplitude will appear only in the fourth approximation of perturbation theory, i.e. as the matrix element of the operator

$$\hat{S}^{(4)} = \frac{(-i)^4}{4!} \int dt_1 \dots \int dt_4 \cdot T\{\hat{V}(t_1)\hat{V}(t_2)\hat{V}(t_3)\hat{V}(t_4)\}. \quad (85.5)$$

Indeed, in lower orders each member in the products of the operators $\hat{V}$ will contain at most one of the operators $\hat{\mathbf{d}}_1$ or $\hat{\mathbf{d}}_2$ to the first power and upon averaging over the state of the corresponding atom will vanish.

We average the operator (85.5) over the photon vacuum. By Wick's theorem the average of the product of four operators of the field $\hat{\mathbf{E}}$ reduces to the sum of products of their pairwise averages (contractions). The splitting into pairs can be accomplished in three ways, which can be depicted by diagrams:

$$(85.6)$$

where the dashed lines depict the contractions, and the numbers correspond to the arguments t_1, t_2, t_3, t_4 . Furthermore, each point may correspond to the spatial coordinates $\mathbf{r}_1$ or $\mathbf{r}_2$ (since at each point one of the summands in the operator $\hat{V}$ of the operators $\hat{\mathbf{d}}_1$ or $\hat{\mathbf{d}}_2$ enters to the first power and upon averaging over the state of each atom will vanish). Obviously, in the points connected by lines there should stand different $\mathbf{r}_1$ and $\mathbf{r}_2$. In the opposite case the diagram (i.e. the corresponding member in the matrix element) would reduce to a product of independent functions of $\mathbf{r}_1$ and of $\mathbf{r}_2$, instead of being a function of the difference $\mathbf{r}_1 - \mathbf{r}_2$; such terms have no relation to scattering²². In accordance with these conditions one can assign to the four points of the diagrams the arguments $\mathbf{r}_1$ and $\mathbf{r}_2$ in four ways. Taking also the commutativity of the operators $\hat{\mathbf{d}}_1$ and $\hat{\mathbf{d}}_2$, and averaging over the state of each of the atoms, into account, we find that all the resulting $3 \cdot 4 = 12$ members are identical (they differ only in the labelling of the variables of integration).

²²They give corrections not of interest to us here to the self-energies of each of the atoms.

As a result we obtain

$$\begin{aligned} \langle S(r) \rangle = & \frac{1}{2} \int dt_1 \dots \int dt_4 \cdot \langle T(E_i(\mathbf{r}_1, t_1) E_k(\mathbf{r}_2, t_2)) \rangle \times \\ & \times \langle T(E_l(\mathbf{r}_2, t_3) E_m(\mathbf{r}_1, t_4)) \rangle \times \\ & \times \langle T(d_{1i}(t_1) d_{1m}(t_4)) \rangle \langle T(d_{2k}(t_2) d_{2l}(t_3)) \rangle, \end{aligned} \quad (85.7)$$

where $i, k, \dots$ are three-dimensional vector indices.

For computing the quantities

$$D_{ik}^E(x_1 - x_2) = \langle T(E_i(x_1) E_k(x_2)) \rangle \quad (85.8)$$

we use the gauge of the potentials in which the scalar potential $\Phi = 0$. Then $\hat{\mathbf{E}} = -\partial \hat{\mathbf{A}} / \partial t$, and we have

$$D_{ik}^E(x_1 - x_2) = \frac{\partial^2}{\partial t_1 \partial t_2} \langle T(A_i(x_1) A_k(x_2)) \rangle = i \frac{\partial^2}{\partial t^2} D_{ik}(x),$$

where $x = x_1 - x_2$, and $D_{ik}(x)$ is the photon propagator in the given gauge²³.

It will be more convenient below to use the propagator $D_{ik}(\omega, \mathbf{r})$ in the mixed ω , $\mathbf{r}$ -representation, which is related to $D_{ik}(t, \mathbf{r})$ by the relation

$$D_{ik}(t, \mathbf{r}) = \int D_{ik}(\omega, \mathbf{r}) e^{-i\omega t} \frac{d\omega}{2\pi}.$$

With this

$$D_{ik}^E(t, \mathbf{r}) = -i \int \omega^2 D_{ik}(\omega, \mathbf{r}) e^{-i\omega t} \frac{d\omega}{2\pi}. \quad (85.9)$$

The quantities

$$\alpha_{ik}(t_1 - t_2) = i \langle T(d_i(t_1) d_k(t_2)) \rangle \quad (85.10)$$

we expand into the Fourier integral

$$\alpha_{ik}(t) = \int e^{-i\omega t} \alpha_{ik}(\omega) \frac{d\omega}{2\pi}.$$

Setting for convenience $t_2 = 0$, $t_1 = t$, we write down the T-product by definition

$$\alpha_{ik}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} \alpha_{ik}(t) dt = \quad (IX.8)$$

$$= i \int_{-\infty}^0 e^{i\omega t} \langle d_k(0) d_i(t) \rangle dt + i \int_0^{\infty} e^{i\omega t} \langle d_i(t) d_k(0) \rangle dt. \quad (85.11)$$

The averages entering here (over the ground state of the atom) are expressed through the matrix elements of the dipole moment:

$$\langle d_k(0) d_i(t) \rangle = \sum_n (d_k)_{0n} (d_i)_{n0} e^{i\omega_{n0} t},$$

²³The first derivative $\frac{\partial}{\partial t} D_{ik}(t)$ has a finite jump at $t = 0$. Therefore the second derivative, i.e. the function $D_{ik}^E(t)$, contains also a δ -function term ($\sim \delta^{(4)}(x_2 - x_1)$). This term, however, is zero for all $\mathbf{r}_1 \neq \mathbf{r}_2$ and does not concern us here.

$$\langle d_i(t) d_k(o) \rangle = \sum_n (d_i)_{0n} (d_k)_{n0} e^{-i\omega_{n0}t}.$$

For the convergence of the integrals in (85.11) one must understand ω in the first of them as $\omega - i0$, and in the second as $\omega + i0$. Carrying out the integration, we obtain

$$\alpha_{ik}(\omega) = \sum_n \left\{ \frac{(d_i)_{0n} (d_k)_{n0}}{\omega_{n0} - \omega - i0} + \frac{(d_k)_{0n} (d_i)_{n0}}{\omega_{n0} + \omega + i0} \right\}. \quad (85.12)$$

If the ground state is an S -state, then this tensor reduces to a scalar: $\alpha_{ik} = \alpha \delta_{ik}$, where

$$\alpha(\omega) = \frac{1}{3} \sum_n |\mathbf{d}_{0n}|^2 \left(\frac{1}{\omega_{n0} - \omega - i0} + \frac{1}{\omega_{n0} + \omega + i0} \right). \quad (85.13)$$

If the atom possesses angular momentum, then the same result is obtained after averaging over its directions, which is to be understood (for our purposes, the interaction of atoms averaged over their mutual orientations).

Comparing (85.12) with expression (59.17), we see that $\alpha_{ik}(\omega)$ coincides with the tensor of coherent scattering of a photon of frequency ω on the atom. According to (59.23) $\alpha(\omega)$ at $\omega > 0$ coincides with the polarisability of the atom. The values of $\alpha(\omega)$ at $\omega < 0$ are expressed through the values at $\omega > 0$ with the help of the obvious from (85.13) relation $\alpha(-\omega) = \alpha(\omega)$.

Substituting the obtained expressions in (85.7), we find

$$\begin{aligned} \langle S(r) \rangle = & \frac{1}{2} \int dt_1 \dots dt_4 \frac{d\Omega_1}{2\pi} \frac{d\Omega_2}{2\pi} \frac{d\omega_1}{2\pi} \frac{d\omega_2}{2\pi} \times \\ & \times \alpha_1(\Omega_1) \alpha_2(\Omega_2) \omega_1^2 D_{ik}(\omega_1, \mathbf{r}) \omega_2^2 D_{ik}(\omega_2, \mathbf{r}) \times \\ & \times \exp\{-i\omega_1(t_1 - t_2) - i\omega_2(t_3 - t_4) - i\Omega_1(t_1 - t_4) - i\Omega_2(t_2 - t_3)\} \end{aligned} \quad (\text{IX.9})$$

($\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$; we have also taken into account the evenness of the function $D_{ik}(\omega, \mathbf{r})$ in $\mathbf{r}$).

Integration over three of the times gives δ -functions (by virtue of which $-\Omega_1 = \Omega_2 = \omega_2 = \omega_1$), and over the fourth—the factor t :

$$\langle S(r) \rangle = -itU(r),$$

where

$$U(r) = \frac{i}{2} \int_{-\infty}^{\infty} \omega^4 \alpha_1(\omega) \alpha_2(\omega) [D_{ik}(\omega, \mathbf{r})]^2 \frac{d\omega}{2\pi}. \quad (85.14)$$

This formula determines the energy of interaction of two atoms at any distances large in comparison with the atomic dimensions a . It remains to find and substitute into it the explicit expression for the function $D_{ik}(\omega, \mathbf{r})$.

Comparing expressions (76.14) and (76.8) with one another, we find that

$$D_{ik}(\omega, \mathbf{k}) = \left(\delta_{ik} - \frac{k_i k_k}{\omega^2} \right) D(\omega, \mathbf{k}),$$

where D is given by formula (76.8). In the $\omega, \mathbf{r}$ -representation this relation is expressed, evidently, by the equality

$$D_{ik}(\omega, \mathbf{r}) = - \left(\delta_{ik} + \frac{1}{\omega^2} \frac{\partial^2}{\partial x_i \partial x_k} \right) D(\omega, \mathbf{r}). \quad (85.15)$$

Substituting here $D(\omega, \mathbf{r})$ from (76.16) and performing the differentiation, we find

$$D_{ik}(\omega, \mathbf{r}) = \left[\delta_{ik} \left(1 + \frac{i}{|\omega|r} - \frac{1}{\omega^2 r^2} \right) + \frac{x_i x_k}{r^2} \left(\frac{3}{\omega^2 r^2} - \frac{3i}{|\omega|r} - 1 \right) \right] \frac{e^{i|\omega|r}}{r}. \quad (85.16)$$

Finally, substituting this expression in (85.14), after simple transformations taking into account the evenness of the function $\alpha(\omega)$ we obtain the following final expression for the energy of interaction of atoms:

$$U(r) = \frac{i}{\pi r^2} \int_0^\infty \omega^4 \alpha_1(\omega) \alpha_2(\omega) e^{2i\omega r} \left[1 + \frac{2i}{\omega r} - \frac{5}{(\omega r)^2} - \frac{6i}{(\omega r)^3} + \frac{3}{(\omega r)^4} \right] d\omega. \quad (85.17)$$

This general expression can be simplified in the limiting cases of “small” ($a \ll r \ll \lambda_0$) and “large” ($r \gg \lambda_0$) distances.

When $r \ll \lambda_0$, in the integral the values $\omega \sim \omega_0$ are important (see below), where $\omega_0 \sim c/\lambda_0$ are atomic frequencies; therefore $\omega r \ll 1$. In this case one can retain in the square brackets only the last member and replace the exponential by unity. Writing the integral from $-\infty$ to ∞ (with the aim of further transformation), we find

$$U(r) = \frac{3i}{2\pi r^6} \int_{-\infty}^\infty \alpha_1(\omega) \alpha_1(\omega) d\omega. \quad (85.18)$$

As expected, we have obtained for the interaction the law $1/r^6$. The integral in this formula is easily

computed, after substituting into it $\alpha(\omega)$ from (85.13), by closing the contour of integration by an infinitely distant semicircle in the lower half-plane of the complex variable ω ; at this the integral is determined by the residues of the subintegral expression in the poles $\omega = \omega_{n0} \sim \omega_0$. Assuming for simplicity of notation that both atoms are identical, we obtain (in ordinary units):

$$U(r) = -\frac{2}{3r^6} \sum_{n,n'} \frac{|\mathbf{d}_{0n}|^2 |\mathbf{d}_{0n'}|^2}{\hbar(\omega_{n0} + \omega_{n'0})}, \quad (85.19)$$

which coincides with the London formula (see III, § 89, Problem).

In the opposite limiting case of large distances, $r \gg \lambda_0$, in the integral the values $\omega \lesssim c/r \ll \omega_0$ are important; when $\omega \gtrsim \omega_0$ the integral is rapidly quenched by the oscillating factor $\exp(2i\omega r)$. Therefore one can replace $\alpha_1(\omega)$ and $\alpha_2(\omega)$ by their static values $\alpha_1(0)$ and $\alpha_2(0)$. After this the integration is carried out elementarily (for ensuring convergence one should replace $r \rightarrow r + i0$ in the exponent). As a result we finally obtain (in ordinary units):

$$U(r) = -\frac{23}{4\pi} \frac{\hbar c \alpha_1(0) \alpha_2(0)}{r^7} \quad (85.20)$$

(H. B. G. Casimir, D. Polder, 1948)²⁴.

²⁴In the outlined derivation we partially followed I. E. Dzyaloshinskii (1956).

Interaction of Electrons with Photons

§ 86. Scattering of a photon by an electron

Conservation of 4-momentum in the scattering of a photon by a free electron (*Compton effect*) is expressed by the equation

$$p + k = p' + k', \quad (\text{X.1})$$

where p and k are the 4-momenta of the electron and photon before the collision, and p' and k' are their 4-momenta after the collision. The kinematic invariants introduced in § 66 are:

$$\begin{aligned} s &= (p + k)^2 = (p' + k')^2 = m^2 + 2pk = m^2 + 2p'k', \\ t &= (p - p')^2 = (k' - k)^2 = 2(m^2 - pp') = -2kk', \\ u &= (p - k')^2 = (p' - k)^2 = m^2 - 2pk' = m^2 - 2p'k, \\ s + t + u &= 2m^2. \end{aligned} \quad (\text{X.2})$$

The process under consideration is represented by the two Feynman diagrams (74.14), and its amplitude is

$$M_{fi} = -4\pi e^2 e'^*_{\mu} e_{\nu} (\bar{u}' Q^{\mu\nu} u), \quad (\text{X.3})$$

where

$$Q^{\mu\nu} = \frac{1}{s - m^2} \gamma^{\mu} (\gamma p + \gamma k + m) \gamma^{\nu} - \frac{1}{u - m^2} \gamma^{\nu} (\gamma p - \gamma k' + m) \gamma^{\mu}. \quad (\text{X.4})$$

Here e, e' are the polarisation 4-vectors of the initial and final photons; u, u' are the bispinor amplitudes of the initial and final electrons.

According to the rules set out in § 65, for arbitrary polarisation states of the particles, $|M_{fi}|^2$ is replaced by

$$|M_{fi}|^2 \rightarrow 16\pi^2 e^4 \text{Sp} \left\{ \rho^{(e)'}_{\lambda\mu} Q^{(\gamma)'} Q^{\mu\rho} \rho^{(e)}_{\nu\sigma} Q^{\lambda\sigma} \right\}. \quad (\text{X.5})$$

Here $\rho^{(e)}, \rho^{(e)'}$ are the density matrices of the initial and final electrons, $\rho^{(\gamma)}, \rho^{(\gamma)'}$ are the same for the photons; the photonic (tensor) indices are written explicitly, while the electronic (bispinor) indices are implied; the Sp sign refers specifically to the latter indices. The same indices carry the hermitian conjugation sign in the definition $\bar{Q}_{\mu\nu} = \gamma^0 Q^+_{\mu\nu} \gamma^0$.

Let us consider the scattering of an unpolarised photon by an unpolarised electron, without concern for the polarisations after scattering. Averaging over the polarisations of all particles is achieved by means of the density matrices:

$$\rho^{(\gamma)}_{\lambda\mu} = \rho^{(\gamma)'}_{\lambda\mu} = -\frac{1}{2} g_{\lambda\mu}, \quad \rho^{(e)} = \frac{1}{2} (\gamma p + m), \quad \rho^{(e)'} = \frac{1}{2} (\gamma p' + m);$$

the transition to summation over polarisations of the final-state particles is accomplished by an additional factor of $2 \cdot 2 = 4$.

By formula (64.23) (in which one must now set $I^2 = \frac{1}{4}(s - m^2)^2$ —see (64.15a)) we obtain for the cross section

$$d\sigma = \frac{\pi e^4}{4} \frac{dt}{(s - m^2)^2} \text{Sp} \left\{ (\gamma p' + m) Q^{\lambda\mu} (\gamma p + m) \tilde{Q}_{\lambda\mu} \right\}.$$

With the aid of formula (65.2a) we find that $\tilde{Q}_{\mu\lambda} = Q_{\lambda\mu}$. Separating the terms that transform into one another under the substitution $k \leftrightarrow -k'$ (and correspondingly $s \leftrightarrow u$), we write the cross section in the form

$$d\sigma = dt \frac{\pi e^4}{(s - m^2)^2} [f(s, u) + g(s, u) + f(u, s) + g(u, s)],$$

where we have introduced the notation

$$\begin{aligned} f(s, u) &= \frac{1}{4(s - m^2)^2} \times \\ &\times \text{Sp} \{ (\gamma p' + m) \gamma^\mu (\gamma p + \gamma k + m) \gamma^\nu (\gamma p + m) \gamma_\mu (\gamma p + \gamma k + m) \gamma_\nu \}, \\ g(s, u) &= \frac{1}{4(s - m^2)(u - m^2)} \times \\ &\times \text{Sp} \{ (\gamma p' + m) \gamma^\mu (\gamma p + \gamma k + m) \gamma^\nu (\gamma p + m) \gamma_\mu (\gamma p - \gamma k' + m) \gamma_\nu \} \end{aligned}$$

(in these expressions we bear in mind in advance that the result will depend only on the invariant quantities).

Summation over μ and ν is carried out with the aid of formulas (22.6); discarding the terms with an odd number of γ factors, we obtain

$$\begin{aligned} f(s, u) &= \frac{1}{(s - m^2)^2} \text{Sp} \{ (\gamma p') (\gamma p + \gamma k) (\gamma p) (\gamma p + \gamma k) + \\ &+ 4m^2 (\gamma p + \gamma k) (\gamma k - \gamma p') + m^2 (\gamma p) (\gamma p') + 4m^4 \}. \end{aligned}$$

Computing the trace with the aid of formulas (22.13) and expressing all quantities through the invariants s, u , we find after straightforward manipulations

$$f(s, u) = \frac{2}{(s - m^2)^2} \{ 4m^2 - (s - m^2)(u - m^2) + 2m^2(s - m^2) \}.$$

In an analogous fashion, g is computed:

$$g(s, u) = \frac{2m^2}{(s - m^2)(u - m^2)} \{ 4m^2 + (s - m^2) + (u - m^2) \}.$$

As a result, for the cross section we obtain

$$\begin{aligned} d\sigma &= 8\pi r_e^2 \frac{m^2 dt}{(s - m^2)^2} \left\{ \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right)^2 + \right. \\ &\left. + \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right) - \frac{1}{4} \left(\frac{s - m^2}{u - m^2} + \frac{u - m^2}{s - m^2} \right) \right\}, \end{aligned} \quad (\text{X.6})$$

where $r_e = e^2/m$. This formula expresses the cross section through invariant quantities. With its help it is easy to express the cross section through the collision parameters in any particular reference frame.

Let us do this for the laboratory frame, in which the electron before the collision is at rest: $p = (m, \mathbf{0})$. Here

$$s - m^2 = 2m\omega, \quad u - m^2 = -2m\omega'. \quad (\text{X.7})$$

Writing the 4-momentum conservation equation in the form $p + k - k' = p'$ and squaring, we obtain

$$pk - pk' - kk' = 0,$$

whence (in the laboratory frame)

$$m(\omega - \omega') - \omega\omega'(1 - \cos \vartheta) = 0,$$

where ϑ is the photon scattering angle. This equation determines the relation between the change in the photon energy and the scattering angle:

$$\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1}{m}(1 - \cos \vartheta) = 0. \quad (\text{X.8})$$

Invariant t :

$$t = -2kk' = -2\omega\omega'(1 - \cos \vartheta) = 0.$$

At a given energy ω we find (using (86.8))

$$dt = 2\omega'^2 d \cos \vartheta = \frac{1}{\pi} \omega'^2 d\vartheta, \quad d\vartheta = 2\pi \sin \vartheta d\vartheta.$$

Substitution of the expressions written in (86.6) into the following formula for the scattering cross section in the laboratory frame gives:

$$d\sigma = \frac{r_e^2}{2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \vartheta \right) d\vartheta \quad (\text{X.9})$$

(*O. Klein, Y. Nishina, 1929; I. E. Tamm, 1930*).

Since the angle ϑ is uniquely related to ω' by the relation (86.8), the cross section can also be expressed through the energy of the scattered photon ω' :

$$d\sigma = \pi r_e^2 \frac{m d\omega'}{\omega^2} \left[\frac{\omega}{\omega'} + \frac{\omega'}{\omega} + \left(\frac{m}{\omega'} - \frac{m}{\omega} \right)^2 - 2m \left(\frac{1}{\omega'} - \frac{1}{\omega} \right) \right], \quad (\text{X.10})$$

where ω' varies within the limits

$$\frac{\omega}{1 + 2\omega/m} \leq \omega' \leq \omega. \quad (\text{X.11})$$

When $\omega \ll m$, one can set $\omega' \approx \omega$ in (86.9), and one obtains, as expected, the classical non-relativistic Thomson formula

$$d\sigma = \frac{1}{2} r_e^2 (1 + \cos^2 \vartheta) d\vartheta \quad (\text{X.12})$$

(cf. II, (78.7)).

For computing the total cross section let us return to formula (86.6) (the invariants s , t , and u entering it take values satisfying the inequalities

$$s \geq m^2, \quad t \leq 0, \quad us \leq m^4. \quad (\text{X.13})$$

These were already obtained in § 67 (corresponding to the physical region—I in Fig. 7, p. 299). It is easy to verify them directly, writing the expressions for the invariants in the centre-of-inertia frame. Here $\mathbf{p} + \mathbf{k} = 0$, and the energies of the electron ε and photon ω are related by $\varepsilon = \sqrt{\omega^2 + m^2}$. The invariants:

$$\begin{aligned} s &= (\varepsilon + \omega)^2 = m^2 + 2\omega(\omega + \varepsilon), \\ u &= m^2 - 2\omega(\varepsilon + \omega \cos \theta), \\ t &= -2\omega^2(1 - \cos \theta), \end{aligned} \quad (\text{X.14})$$

where θ is the scattering angle (the angle between $\mathbf{p}$ and $\mathbf{p}'$ or $\mathbf{k}$ and $\mathbf{k}'$). The three inequalities (86.13) follow from the conditions: $\omega \geq 0$, $-1 \leq \cos \theta \leq 1$.

At given s (given energy of the particles) the integration over t can be replaced by integration over $u = 2m^2 - s - t$ in the interval

$$m^4/s \geq u \geq 2m^2 - s.$$

Introducing, instead of s , u , the quantities

$$x = \frac{s - m^2}{m^2}, \quad y = \frac{m^2 - u}{m^2}, \quad (\text{X.15})$$

we obtain

$$\sigma = \frac{8\pi r_e^2}{x^2} \int_{x/(x+1)}^x \left[\left(\frac{1}{x} - \frac{1}{y} \right)^2 + \frac{1}{x} - \frac{1}{y} + \frac{1}{4} \left(\frac{x}{y} + \frac{y}{x} \right) \right] dy$$

and after elementary integration

$$\sigma = 2\pi r_e^2 \frac{1}{x} \left\{ \left(1 - \frac{4}{x} - \frac{8}{x^2} \right) \ln(1+x) + \frac{1}{2} + \frac{8}{x} - \frac{1}{2(1+x)^2} \right\}. \quad (\text{X.16})$$

The leading terms of the expansion for $x \ll 1$ (non-relativistic case) give

$$\sigma = \frac{8\pi r_e^2}{3} (1 - x) \quad (\text{n.r.}). \quad (\text{X.17})$$

The first term is the classical Thomson cross section. In the opposite, ultrarelativistic case $x \gg 1$, expansion of formula (86.16) gives

$$\sigma = 2\pi r_e^2 \frac{1}{x} \left(\ln x + \frac{1}{2} \right) \quad (\text{u.r.}). \quad (\text{X.18})$$

In the laboratory frame

$$x = 2\omega/m, \quad (\text{X.19})$$

so that formulas (86.16)–(86.18) directly give the dependence of the scattering cross section on the energy of the photon incident on a stationary electron as a function of ω/m .

We note that in the ultrarelativistic case the cross section decreases with increasing energy both in the laboratory frame ($\sigma \propto \omega^{-1} \ln \omega$) and in the centre-of-inertia frame ($x \approx 4\omega^2/m^2$, $\sigma \propto \omega^{-2} \ln \omega$). The angular distribution of the scattering in the ultrarelativistic case is entirely different in the two reference frames.

In the laboratory frame the differential cross section has a sharp maximum in the forward direction. In a narrow cone $\vartheta \lesssim \sqrt{m/\omega}$ we have $\omega' \sim \omega$ and the cross section $d\sigma/d\omega' \sim r_e^2$ (reaching the value r_e^2 at $\vartheta \rightarrow 0$). Outside this cone the cross section drops off, and in the region $\vartheta^2 \gg m/\omega$ (where $\omega' \approx m/(1 - \cos \vartheta)$) we have

$$\frac{d\sigma}{d\omega'} = \frac{r_e^2}{2} \frac{m}{\omega(1 - \cos \vartheta)},$$

i.e. the cross section decreases by a factor $\sim \omega/m$.

In the centre-of-inertia frame the differential cross section has a maximum in the backward direction. When $\pi - \theta \ll 1$ we have from (86.14)

$$\frac{s - m^2}{m^2} \approx \frac{4\omega^2}{m^2}, \quad \frac{m^2 - u}{m^2} \approx 1 + \frac{\omega^2}{m^2} (\pi - \theta)^2.$$

The leading term in the cross section (86.6) is

$$d\sigma \approx 8\pi r_e^2 \frac{m^2 dt}{4(s - m^2)(m^2 - u)};$$

whence

$$d\sigma = \frac{r_e^2}{2} \frac{d\omega'}{1 + (\pi - \theta)^2 \omega^2/m^2}. \quad (\text{X.20})$$

The cross section $d\sigma/d\omega' \sim r_e^2$ in a narrow cone $\pi - \theta \lesssim m/\omega$, and outside it the cross section decreases by a factor of order ω^2/m^2 .

§ 87. Scattering of a photon by an electron. Polarisation effects

Let us return to the original formulas of the preceding section and show how the calculations should be carried out taking into account the polarisations of the initial and final photons and electrons.

The photon density matrix is expressed, according to (8.17), with the aid of a pair of unit 4-vectors $e^{(1)}, e^{(2)}$, satisfying the conditions (8.16). In the present case one can choose these vectors in a unified manner for both photons. These are the 4-vectors introduced in § 70¹

$$e^{(1)} = \frac{N}{\sqrt{-N^2}}, \quad e^{(2)} = \frac{P}{\sqrt{-P^2}}, \quad (\text{X.21})$$

¹An alternative method of calculation consists in choosing from the outset a specific reference frame (e.g. the laboratory frame), and for each photon taking as $e^{(1)}, e^{(2)}$ two purely spatial-like ($\mathbf{e} = (0, \mathbf{e})$) vectors, orthogonal to the momentum of the photon and to each other. In that case, however, all the calculations will be carried out in three-dimensional form, and the result will not have an invariant appearance.

where

$$\begin{aligned} P^\lambda &= (p^\lambda + p'^\lambda) - K^\lambda \frac{pK + p'K}{K^2}, \quad N^\lambda = e^{\lambda\mu\nu\rho} P_\mu q_\nu K_\rho, \\ K^\lambda &= k^\lambda + k'^\lambda, \quad q^\lambda = k^\lambda - k'^\lambda = p^\lambda - p'^\lambda. \end{aligned} \quad (\text{X.22})$$

The quantities $Q^{\mu\nu}$ in (86.5) are given by formula (86.4). They can be regarded as components of a 4-tensor (in the sense that the 4-tensor formed after constructing the quantity $\bar{u}' Q^{\mu\nu} u$, as one says, “in bra-kets”). All the components of the 4-tensor can be exhausted by projecting it onto four mutually orthogonal 4-vectors, for example those defined above: P, N, q, K . Since the tensors $\rho_{\mu\nu}^{(\gamma)'}$, $\rho_{\mu\nu}^{(\gamma)}$ contain only components along P and N , in practice we need only the components of $Q_{\mu\nu}$ along these two 4-vectors.

In other words, it suffices to seek in $Q_{\mu\nu}$ only terms of the form

$$\begin{aligned} Q_{\mu\nu} &= Q_0 (e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + Q_1 (e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ &\quad - iQ_2 (e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + Q_3 (e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}); \end{aligned} \quad (\text{X.23})$$

the remaining terms upon substitution in (86.5) would vanish anyway. The quantities Q_0 and Q_3 are scalars—in the same sense as $Q_{\mu\nu}$ is a 4-tensor; they therefore contain γ matrices only in “invariant” combinations: γK and so on. In this same sense Q_1 and Q_2 are pseudoscalars (N is a pseudovector!) and must therefore contain the matrix γ^5 .

By direct projection of the tensor $Q_{\mu\nu}$ we find

$$Q_0 = \frac{1}{2} Q^{\mu\nu} (e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)})$$

and so on. For the calculations it is convenient first to express $Q_{\mu\nu}$ through the mutually orthogonal 4-vectors P, N, q, K :

$$Q^{\mu\nu} = \gamma^\mu \frac{\gamma P/2 + m}{s - m^2} \gamma^\nu + \gamma^\nu \frac{\gamma P/2 + m}{u - m^2} \gamma^\mu - \frac{1}{t} [\gamma^\mu (\gamma K) \gamma^\nu - \gamma^\nu (\gamma K) \gamma^\mu].$$

The further calculation reduces to purely algebraic manipulations with the aid of the formulas given in § 22. Furthermore, one can make in $Q^{\mu\nu}$ replacements that will not affect the result upon further formation of the product $\bar{u}' Q^{\mu\nu} u$. For example, since

$$\begin{aligned} \bar{u}' (\gamma p + \gamma p') u &= 2m \bar{u}' u, \\ \bar{u}' \gamma^5 (\gamma q) u &= \bar{u}' [\gamma^5 (\gamma p) + (\gamma p') \gamma^5] u = 2m \bar{u}' \gamma^5 u, \end{aligned}$$

one can replace in $Q^{\mu\nu}$ the terms

$$\gamma p + \gamma p' \rightarrow 2m, \quad \gamma^5 (\gamma q) \rightarrow 2m \gamma^5. \quad (\text{X.24})$$

Omitting the details of the calculations, we present the result²:

$$\begin{aligned} Q_0 &= -ma_+, \quad Q_1 = \frac{1}{2} a_+ \gamma^5 (\gamma K), \\ Q_2 &= -ma_+ \gamma^5, \quad Q_3 = ma_+ + \frac{1}{2} a_- (\gamma K), \end{aligned} \quad (\text{X.25})$$

²Expression (87.3) with the values (87.5) has the form of (70.11) (70.13), established in § 70 from general considerations. In addition to the equalities $f_1 = f_2 = 0$, which follow from T -invariance, here it also turns out that there is only one independent amplitude (f_+). This is a property of the Born approximation of perturbation theory under consideration, and it would vanish in higher approximations.

where

$$a_{\pm} = \frac{1}{s - m^2} \pm \frac{1}{u - m^2}.$$

For the further calculations it is convenient to apply to $Q_{\mu\nu}$ the same formal device that was described in § 8 for the photon density matrix: the four components of the tensor (87.3) in the directions $e^{(1)}, e^{(2)}$ are combined into a two-row matrix Q , which is then expanded in the Pauli matrices. Analogously to formula (8.18) we obtain

$$Q = Q_0 + \mathbf{Q}\sigma, \quad \mathbf{Q} = (Q_1, Q_2, Q_3). \quad (\text{X.26})$$

As regards the tensor $\tilde{Q}_{\mu\nu} = \gamma^0 Q_{\mu\nu}^+ \gamma^0$ appearing in (86.5), using (87.3) and (87.5) it is easy to verify (with the aid of the rules ((65.2a))) that its components are obtained from the components of $Q_{\mu\nu}$ by the replacement of the quantities $Q_0, Q_1, \dots$ by $\bar{Q}_0, \bar{Q}_1, \dots$, where

$$\bar{Q}_0 = Q_0, \quad \bar{Q}_1 = -Q_1, \quad \bar{Q}_2 = -Q_2, \quad \bar{Q}_3 = Q_3, \quad (\text{X.27})$$

and a simultaneous permutation of the indices $\mu\nu$ ³. In matrix form this means that

$$\bar{Q} = \bar{Q}_0 + \bar{\mathbf{Q}}\sigma. \quad (\text{X.28})$$

Let us now clarify the meaning of the 4-vectors $e^{(1)}, e^{(2)}$ in relation to the polarisation of the photons. For each photon the independent directions of polarisation will be determined by the transverse (with respect to the photon momentum $\mathbf{k}$) components of the 3-vectors $e^{(1)}, e^{(2)}$ ⁴. It is easy to see that, both in the centre-of-inertia frame and in the laboratory frame (the rest frame of the initial electron), the vector $\mathbf{P}$ lies in the plane $\mathbf{k}\mathbf{k}'$, and the vector $\mathbf{N}$ is perpendicular to this plane. Therefore the direction $e^{(1)}$ has the meaning of polarisation perpendicular to the scattering plane, and the direction $e^{(2)}$ is polarisation in the scattering plane. We note also that for the initial photon the x axis is the same for both photons and perpendicular to the scattering plane:

$$x \parallel [\mathbf{k}\mathbf{k}'],$$

while the y axes lie in the scattering plane:

$$y \parallel [\mathbf{k}[\mathbf{k}\mathbf{k}']], \quad y' \parallel [\mathbf{k}'[\mathbf{k}\mathbf{k}']].$$

For the initial photon the relevant vectors are $\mathbf{N}, -\mathbf{P}_{\perp}, \mathbf{k}$, and for the final photon—the vectors $\mathbf{N}, \mathbf{P}'_{\perp}$ ($\mathbf{P}_{\perp}, \mathbf{P}'_{\perp}$ are the components of $\mathbf{P}, \mathbf{P}'$ perpendicular to $\mathbf{k}$ and $\mathbf{k}'$ respectively). A change in the sign of $e^{(2)}$ in the photon density matrix (8.17) is equivalent to changing the sign of ξ_1 and ξ_2 . Therefore the photon density matrices of the initial and final photons, referred to the 4-orts $e^{(1)}, e^{(2)}$, will be

$$\begin{aligned} \rho^{(\gamma)} &= \frac{1}{2}(1 + \xi\sigma), \quad \xi = (\xi_1, \xi_2, \xi_3); \\ \rho^{(\gamma)'} &= \frac{1}{2}(1 + \xi'\sigma), \quad \xi' = (-\xi'_1, -\xi'_2, \xi'_3). \end{aligned} \quad (\text{X.29})$$

³For the matrix $Q_{\mu\nu}$ in the original form (86.4) we would simply have $\tilde{Q}_{\mu\nu} = Q_{\nu\mu}$. This property is, however, lost in the result of transformations including replacements of the type (87.4) and so on.

⁴The longitudinal components of e (as well as the time components of the 4-vectors e) can simply be ignored: their inessentiality is ensured by gauge invariance.

Now the tensor trace

$$\rho_{\lambda\mu}^{(\gamma)'} Q^{\mu\nu} \rho_{\nu\rho}^{(\gamma)} \bar{Q}^{\rho\lambda}$$

is computed as the trace of a matrix product of the matrices (87.6)–(87.9) with the aid of (33.5). The result is

$$\begin{aligned} |M_{fi}|^2 = & 8\pi^2 e^4 \text{Sp}_1 \{ (\rho^{(e)'} Q_0 \rho^{(e)} \bar{Q}_0 + \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{\mathbf{Q}}) + \\ & + (\xi + \xi') (\rho^{(e)'} Q_0 \rho^{(e)} \bar{\mathbf{Q}} + \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{Q}_0) - i(\xi - \xi') [\rho^{(e)'} Q_0 \rho^{(e)} \bar{\mathbf{Q}}] + \\ & + (\xi \xi') (\rho^{(e)'} Q_0 \bar{Q}_0 \rho^{(e)} - \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{\mathbf{Q}}) + \rho^{(e)'} (\xi' \mathbf{Q}) \rho^{(e)} (\xi \bar{\mathbf{Q}}) + \\ & + \rho^{(e)'} (\xi \mathbf{Q}) \rho^{(e)} (\xi' \bar{\mathbf{Q}}) - i[\xi \xi'] (\rho^{(e)'} Q_0 \rho^{(e)} \bar{\mathbf{Q}} - \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{Q}_0) \}. \end{aligned} \quad (\text{X.30})$$

Scattering on unpolarised electrons. Let us compute the complete cross section for scattering of polarised photons by an unpolarised electron, summed over the polarisations of the final electron. For this one must set in (87.10)

$$\rho^{(e)} = \frac{1}{2}(\gamma p + m), \quad \rho^{(e)'} = \frac{1}{2}(\gamma p' + m)$$

and double the result, which must be substituted in place of $|M_{fi}|^2$ in the formula for the cross section (64.22)

$$d\sigma = \frac{1}{32\pi^2} \frac{dt d\varphi}{(s - m^2)^2} |M_{fi}|^2$$

(φ is the azimuth in the centre-of-inertia frame or in the laboratory frame). The computation of the remaining traces leads to the following final result (in the notation of (86.15)):

$$\begin{aligned} d\sigma = & \frac{1}{2} d\bar{\sigma} + 2r_e^2 \frac{dy d\varphi}{x^2} \left\{ (\xi_3 + \xi'_3) \left[-\left(\frac{1}{x} - \frac{1}{y}\right)^2 - \left(\frac{1}{x} - \frac{1}{y}\right) \right] + \right. \\ & + \xi_1 \xi'_1 \left(\frac{1}{x} - \frac{1}{y} + \frac{1}{2} \right) + \xi_2 \xi'_2 \cdot \frac{1}{4} \left(\frac{x}{y} + \frac{y}{x} \right) \left(1 + \frac{2}{x} - \frac{2}{y} \right) + \\ & \left. + \xi_3 \xi'_3 \left[\left(\frac{1}{x} - \frac{1}{y} \right)^2 + \left(\frac{1}{x} - \frac{1}{y} \right) + \frac{1}{2} \right] \right\}, \end{aligned} \quad (\text{X.31})$$

where $d\bar{\sigma}$ is the cross section for scattering of unpolarised photons, given by formula (86.9); the factor $\frac{1}{2}$ is connected with the fact that in (87.11) there is no summation over polarisations of the final photon.

In the laboratory frame formula (87.11) takes the form

$$d\sigma = \frac{r_e^2}{4} \left(\frac{\omega'}{\omega} \right)^2 d\omega' \{ F_0 + F_3(\xi_3 + \xi'_3) + F_{11}\xi_1\xi'_1 + F_{22}\xi_2\xi'_2 + F_{33}\xi_3\xi'_3 \}, \quad (\text{X.32})$$

$$d\omega' = \sin \vartheta d\vartheta d\varphi,$$

where

$$\begin{aligned} F_0 = & \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2 \vartheta, \quad F_3 = \sin^2 \vartheta, \\ F_{11} = & 2 \cos \vartheta, \quad F_{22} = \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} \right) \cos \vartheta, \quad F_{33} = 1 + \cos^2 \vartheta \end{aligned} \quad (\text{X.33})$$

(*U. Fano*, 1949). We note that although expression (87.12) does not contain an explicit dependence on the azimuth of the scattering plane φ , there is an implicit dependence, since the Stokes parameters ξ_1, ξ_2, ξ_3 are defined relative to the x, y, z axes associated with the scattering plane. Recall that the x axis is the same for both photons and perpendicular to the scattering plane:

$$x \parallel [\mathbf{k}\mathbf{k}'],$$

while the y axes lie in the scattering plane:

$$y \parallel [\mathbf{k}[\mathbf{k}\mathbf{k}']], \quad y' \parallel [\mathbf{k}'[\mathbf{k}\mathbf{k}']].$$

Taking the sum of the cross sections differing by the sign of ξ' (i.e. setting $\xi' = 0$ and doubling the result), we obtain the complete (summed over polarisations of the final photon) cross section for scattering of a polarised photon on an unpolarised electron. Denoting it by $d\sigma(\xi)$, we have

$$d\sigma(\xi) = \frac{1}{2} r_e^2 \left(\frac{\omega'}{\omega} \right)^2 F d\omega', \quad (\text{X.34})$$

where

$$F = F_0 + \xi_3 F_3 = \frac{\omega'}{\omega} + \frac{\omega}{\omega'} - (1 - \xi_3) \sin^2 \vartheta. \quad (\text{X.35})$$

We see that the cross section for scattering of photons polarised perpendicular to the scattering plane ($\xi_3 = 1$) is greater than for photons polarised in the scattering plane ($\xi_3 = -1$). For circular polarisation the cross section does not depend on ξ_1 . Therefore the scattering cross section coincides with the cross section for unpolarised photons if there is no linear polarisation relative to the x or y axes ($\xi_3 = 0$) or even if there is polarisation relative to directions making 45° with these axes.

The cross section for scattering of unpolarised photons with detection of the polarised final

photon possesses analogous properties. This cross section (denote it by $d\sigma(\xi')$) is obtained from formula (87.12) by setting $\xi = 0$:

$$d\sigma(\xi') = \frac{1}{4} r_e^2 \left(\frac{\omega'}{\omega} \right)^2 F' d\omega', \quad F' = F_0 + \xi'_3 F_3. \quad (\text{X.36})$$

From formula (87.12) one can also find the polarisation of the secondary photon as such; the parameters of this polarisation are denoted $\xi^{(f)}$ in distinction from the detectable polarisation ξ' . According to the rules set out in § 65, the quantities $\xi_i^{(f)}$ are the ratios of the coefficients at ξ'_i to the term not containing ξ' :

$$\xi_1^{(f)} = \frac{F_{11}}{F} \xi_1, \quad \xi_2^{(f)} = \frac{F_{22}}{F} \xi_2, \quad \xi_3^{(f)} = \frac{F_3 + F_{33}\xi_3}{F}. \quad (\text{X.37})$$

In particular, when an unpolarised photon is scattered,

$$\xi_1^{(f)} = 0, \quad \xi_2^{(f)} = 0, \quad \xi_3^{(f)} = \frac{\sin^2 \vartheta}{\omega/\omega' + \omega'/\omega - \sin^2 \vartheta}. \quad (\text{X.38})$$

Here $\xi_3^{(f)} > 0$, i.e. the secondary photon is polarised perpendicular to the scattering plane. Circular polarisation of the secondary photon arises only if the primary photon is circularly polarised: $\xi_2^{(f)} \neq 0$ only when $\xi_2 \neq 0$.

Consider the case when the incident photon is completely linearly polarised ($\xi_2 = 0$, $\xi_1^2 + \xi_3^2 = 1$), and let us find the cross section for scattering in which a linearly polarised secondary photon is also detected. Expressing the Stokes parameters ξ_i and ξ_i' through the components of the polarisation vectors $\mathbf{e}$ and $\mathbf{e}'$, we obtain the following expression for the cross section of scattering:

$$d\sigma = \frac{r_e^2}{4} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - 2 + 4 \cos^2 \Theta \right) d\omega', \quad (\text{X.39})$$

where Θ is the angle between the polarisation directions of the incident and scattered photons⁵.

According to this formula, the cross section behaves substantially differently when the polarisations $\mathbf{e}$ and $\mathbf{e}'$ are mutually perpendicular and when they lie in one plane. Distinguishing these two cases by the subscripts $\perp$ and $\parallel$, we have in the non-relativistic limit ($\omega \ll m$, $\omega' \approx \omega$)

$$d\sigma_{\perp} = 0, \quad d\sigma_{\parallel} = r_e^2 \cos^2 \Theta d\omega' \quad (\text{X.40})$$

in agreement with the classical formulas. In the opposite, ultrarelativistic case $\omega \gg m$, $\omega' \approx m/(1 - \cos \vartheta)$. Here one must distinguish the regions of large and small angles (ω'/ω large or small):

$$\begin{aligned} d\sigma_{\perp} = d\sigma_{\parallel} &= \frac{1}{4} r_e^2 \frac{\omega'}{\omega} d\omega' = \frac{1}{4} r_e^2 \frac{m d\omega'}{\omega(1 - \cos \vartheta)}, \quad \vartheta^2 \gg \frac{m}{\omega}, \\ d\sigma_{\perp} &= 0, \quad d\sigma_{\parallel} = r_e^2 \cos^2 \Theta d\omega', \quad \vartheta^2 \ll \frac{m}{\omega}. \end{aligned} \quad (\text{X.41})$$

We see that in the region of very small scattering angles the cross section coincides with the classical one. The equality $d\sigma_{\perp} \approx d\sigma_{\parallel}$ at not too small angles does not mean that the scattered radiation in this region is unpolarised; we stress, however, that this conclusion refers specifically to a linearly polarised incident photon: from (87.17) it is seen that for a circularly polarised photon in the ultrarelativistic case $\xi_2^{(f)} \approx \cos \vartheta \cdot \xi_2$.

Scattering on polarised electrons. For polarised electrons the computation of the traces in formula (87.10) becomes very cumbersome, though it does not present any difficulties in principle. We give here some final results of such a calculation⁶.

In the general case the cross section depends both on the polarisation parameters of the initial and final photons ξ and ξ' , and on the polarisations of the initial and final

⁵Formula (87.19) could have been obtained more simply from the outset, by setting in the scattering amplitude (86.3) $e = (0, \mathbf{e})$, $e' = (0, \mathbf{e}')$ and computing the square of the amplitude in three-dimensional form (i.e. separating the time and spatial components of the 4-vectors). Averaging $\cos^2 \Theta = (\mathbf{e}\mathbf{e}')^2$ over the directions of $\mathbf{e}$ and $\mathbf{e}'$ (with the aid of (45.4a)) and doubling the cross section (the transition to summation over $\mathbf{e}'$), we return, of course, to (86.9).

⁶More detailed information can be found in the review articles: Tolhoek *H. A. Rev. Mod. Phys.* 1956. V. 28. P. 277; McMaster W. H. *Rev. Mod. Phys.* 1961. V. 33. P. 8.

electrons, characterised by the vectors ζ and ζ' . The dependence of the cross section on each of these parameters is linear. The cross section has the form

$$d\sigma = \frac{1}{2}d\sigma(\xi, \xi') + \frac{r_e^2}{8} \left(\frac{\omega'}{\omega} \right)^2 d\omega' \{ f\zeta\xi_2 + f'\zeta'\xi'_2 + g\zeta'\xi_2 + g'\zeta\xi'_2 + G_{ik}\zeta_i\zeta'_k + \dots \}. \quad (\text{X.42})$$

Here $d\sigma(\xi, \xi')$ is the cross section (87.12). Listed are all the terms containing products of two polarisation parameters; omitted are

the terms containing products of three or four parameters; these terms are inessential if we are interested only in correlations between the polarisations of two particles; they vanish when the polarisation parameters of the other two particles are set equal to zero. We give the values of some of the coefficients in the laboratory frame:

$$\begin{aligned} \mathbf{f} &= -\frac{1 - \cos \vartheta}{m} (\mathbf{k} \cos \vartheta + \mathbf{k}'), \\ \mathbf{f}' &= -\frac{1 - \cos \vartheta}{m} (\mathbf{k} + \mathbf{k}' \cos \vartheta), \\ \mathbf{g} &= -\frac{1 - \cos \vartheta}{m} \left[(\mathbf{k} \cos \vartheta + \mathbf{k}') - (1 + \cos \vartheta) \frac{\omega + \omega'}{\omega - \omega' + 2m} (\mathbf{k} - \mathbf{k}') \right], \\ \mathbf{g}' &= -\frac{1 - \cos \vartheta}{m} \left[(\mathbf{k} + \mathbf{k}' \cos \vartheta) - (1 + \cos \vartheta) \frac{\omega + \omega'}{\omega - \omega' + 2m} (\mathbf{k} - \mathbf{k}') \right]. \end{aligned}$$

In the cross section (87.22) there is no term of the form $\mathbf{G}\zeta$; this means that the polarisation of the electron does not affect the complete (summed over ξ' and ζ') cross section for scattering of polarised photons. There is also no term of the form $\mathbf{G}'\zeta'$; this means that upon scattering of unpolarised photons the recoil electron is not polarised.

We see also that the terms bilinear in the polarisations of the electron and photon contain only the parameters ξ_2, ξ'_2 corresponding to circular polarisation of the photon. The polarisation vectors of the electrons enter in the form of scalar products $f\zeta, \dots$, containing only the projections of these vectors onto the scattering plane. Therefore, for example, the cross section for scattering of a polarised photon by a polarised electron is

$$d\sigma(\xi, \zeta) = d\sigma(\xi) + \frac{1}{2}r_e^2 \left(\frac{\omega'}{\omega} \right)^2 \xi_2 \mathbf{f}\zeta d\omega' \quad (\text{X.24})$$

and differs from $d\sigma(\xi)$ only when the photon has circular polarisation and the electron has a non-zero projection of its mean spin on the scattering plane. For the same reason the recoil electron is polarised only when the photon possesses circular polarisation; the vector of the resulting electron polarisation lies in the scattering plane:

$$\zeta^{(f)} = \frac{1}{F} \xi_2 \mathbf{g}. \quad (\text{X.25})$$

Symmetry relations. In conclusion we point out that the qualitative properties of polarisation effects in the scattering of photons by electrons follow already from the general requirements of symmetry.

The parameter ξ_2 of circular polarisation is a pseudoscalar (see § 8). Therefore, by virtue of P -invariance, terms of the form $\propto \xi_2$ (or $\propto \xi_2'$) could appear in the cross section only as the product of ξ_2 with some other pseudoscalar that can be constructed from the vectors at our disposal $\mathbf{k}$ and $\mathbf{k}'$ ⁷. But from two polar vectors one cannot construct a pseudoscalar. It follows from this that the indicated terms cannot appear in the cross section.

The parameters of linear polarisation ξ_1 and ξ_2 are related to the components of the two-dimensional (in the plane perpendicular to $\mathbf{k}$) symmetric tensor

$$S_{\alpha\beta} = \frac{1}{2}(\rho_{\alpha\beta}^{(\gamma)} + \rho_{\beta\alpha}^{(\gamma)}) = \frac{1}{2} \begin{pmatrix} 1 + \xi_3 & \xi_1 \\ \xi_1 & 1 - \xi_3 \end{pmatrix}. \quad (\text{X.26})$$

In the present case one of the polarisation axes is chosen along the vector $\mathbf{v} = [\mathbf{k}\mathbf{k}']$, and the other lies in the plane $\mathbf{k}\mathbf{k}'$ (along the vector $[\mathbf{k}\mathbf{v}]$ or $[\mathbf{k}'\mathbf{v}]$ for one or the other photon). Terms $\propto \xi_1$ could appear in the cross section only as products of $S_{\alpha\beta}v_\alpha[\mathbf{k}'\mathbf{v}]_\beta$ (or, what is the same, $S_{\alpha\beta}v_\alpha k'_\beta$) and so on. But since $\mathbf{v}$ is an axial vector and $\mathbf{k}$ is a polar vector, and $S_{\alpha\beta}$ is a true tensor, such products are not invariant under inversion. Therefore terms $\propto \xi_1$ (or $\propto \xi_1'$) in the cross section are also excluded.

Terms in the cross section proportional to the electron polarisation ζ are not forbidden by parity, since such terms could be formed as products of two axial vectors: $\zeta\mathbf{v}$. However, they must be absent as a consequence of the hermiticity of the scattering matrix in the considered approximation (cf. § 71).

By virtue of this hermiticity, the square of the modulus of the scattering amplitude (and with it the cross section) does not change upon permutation of the initial and final states. At the same time the cross section must be invariant with respect to time reversal—permutation of the initial and final states with the simultaneous reversal of the sign of the momentum and angular momentum vectors of all particles (the Stokes parameters ξ_1, ξ_2, ξ_3 do not change—see § 8). Combining both requirements, we conclude that in the considered approximation the cross section must not change upon the simultaneous reversal of the sign of all momenta and angular momenta without the permutation of

initial and final states, i.e. under the transformation

$$\mathbf{k} \rightarrow -\mathbf{k}', \quad \mathbf{k}' \rightarrow -\mathbf{k}, \quad \zeta \rightarrow -\zeta', \quad \zeta' \rightarrow -\zeta \quad (\text{X.27})$$

and unchanged parameters ξ, ξ' .

Transformation (87.26) changes the sign of the products $\zeta\mathbf{v}$, and therefore such terms cannot appear in the cross section. We stress, however, that this prohibition is not a consequence of strict symmetry requirements and may be violated in the following approximations of perturbation theory.

We note also that parity forbids terms of the type $\xi_1\xi_3$ and $\xi_2\xi_3$ in the double correlation between the polarisations of the photons with one another, and imposes no restrictions on the form of the correlation of the photons with the electrons. However, all

⁷We consider the process in the laboratory frame, where $\mathbf{p} = 0$, $\mathbf{p}' = \mathbf{k} - \mathbf{k}'$. It is clear that the interesting consequences of the requirements of symmetry (the presence or absence of given terms in the cross section) do not depend on the choice of reference frame.

the terms of the type $\xi_1\xi_2$, $\xi_1\zeta$, $\xi_3\zeta$ are forbidden in the first Born approximation relative to the transformation (87.26). Thus, the terms of the type $\xi_1\xi_2$ and $\xi_1\zeta$ can be formed (from the point of view of obeying parity) as scalars, for example $\xi'_1(S_{\alpha\beta}k'_\alpha\nu_\beta)$ and $(S_{\alpha\beta}k'_\alpha\nu_\beta)(\zeta\mathbf{k})$; these combinations, however, change sign under the transformation (87.26).

The allowed correlation terms of the type $\xi_2\zeta$ can be formed only as products of the type $\xi_2(\zeta\mathbf{k})$; the electron polarisation vectors enter them only in the form of projections on the scattering plane.

Finally, a number of relations among the coefficients in the allowed terms arises from the requirements of cross-universality. The channels of the reaction, differing by the permutation of the initial and final photons, correspond to the same process—scattering of a photon by an electron. Therefore the square of the modulus of the amplitude, and with it the cross section, must be invariant with respect to the transformation expressing the transition from one of these channels to the other:

$$k \leftrightarrow -k', \quad e \leftrightarrow e'^* \quad (\text{X.28})$$

at unchanged momenta and polarisations of the electrons. In three-dimensional form this means the replacements:

$$\omega \leftrightarrow -\omega', \quad \mathbf{k} \leftrightarrow -\mathbf{k}', \quad \xi_1 \leftrightarrow \xi'_1, \quad \xi_2 \leftrightarrow -\xi'_2, \quad \xi_3 \leftrightarrow \xi'_3.$$

The change of sign of the parameter ξ_2 is obvious from the expression $\xi_2 = i[\mathbf{e}\mathbf{e}^*]\mathbf{n}$, in which the vector $[\mathbf{e}\mathbf{e}^*]$ changes sign under the substitution $\mathbf{e} \leftrightarrow -\mathbf{e}^*$, while the vector $\mathbf{n} = \mathbf{k}/\omega$ under $\mathbf{k} \leftrightarrow -\mathbf{k}$, $\omega \leftrightarrow -\omega$ does not change. The transformation (87.27), without affecting the momenta of the electrons, leaves the laboratory frame. Therefore the cross section (87.22) must not change its form under this transformation; formulas (87.12), (87.22), (87.23) indeed satisfy this condition.

§ 88. Two-photon annihilation of an electron-positron pair

The annihilation of an electron and a positron (4-momenta p_- and p_+) with the production of two photons (k_1 and k_2) corresponds to the two diagrams:

They differ from the diagrams for scattering of a photon by an electron by the substitutions

$$p \rightarrow p_-, \quad p' \rightarrow -p_+, \quad k \rightarrow -k_1, \quad k' \rightarrow k_2. \quad (\text{X.2})$$

Both processes are two crossed channels of one and the same (generalised) reaction. After the substitutions (88.2) the kinematic invariants (86.2) acquire the following meaning:

$$\begin{aligned} s &= (p_- - k_1)^2, \\ t &= (p_- + p_+)^2 = (k_1 + k_2)^2, \\ u &= (p_- - k_2)^2. \end{aligned} \quad (\text{X.3})$$

If the scattering of the photon was the s -channel, then annihilation is the t -channel.

The square $|M_{fi}|^2$ for annihilation (averaged over polarisations of the electrons and summed over polarisations of the photons) will, when expressed through the invariants s ,

u , coincide with the analogous quantity for scattering with a change only of the meaning of the invariants⁸. In the formula for the cross section (64.23), one must now replace $s \leftrightarrow t$, and for the quantity I we now have, according to (64.15a),

$$I^2 = \frac{1}{4}t(t - 4m^2).$$

Performing the corresponding changes in formula (86.6), we obtain the annihilation cross section:

$$d\sigma = 8\pi r_e^2 \frac{m^2 ds}{t(t - 4m^2)} \left\{ \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right)^2 + \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right) - \frac{1}{4} \left(\frac{s - m^2}{u - m^2} + \frac{u - m^2}{s - m^2} \right) \right\}. \quad (\text{X.4})$$

The physical region of the annihilation channel is region II in Fig. 7 (see p. 299). At given t (given energy in the centre

of inertia) the interval of variation of s is determined by the equation of the boundary $su = m^4$. Together with the relation $s + t + u = 2m^2$ this gives

$$\frac{t}{2} - \frac{1}{2}\sqrt{t(t - 4m^2)} \leq s - m^2 \leq -\frac{t}{2} + \frac{1}{2}\sqrt{t(t - 4m^2)}. \quad (\text{X.5})$$

Integration of expression (88.4) is elementary; the result must be divided by 2 as well, taking into account the identity of the two final-state particles (photons). Thus we obtain

$$\sigma = \frac{\pi r_e^2}{2\tau^2(\tau - 1)} \left[\left(\tau^2 + \tau - \frac{1}{2} \right) \ln \frac{\sqrt{\tau} + \sqrt{\tau - 1}}{\sqrt{\tau} - \sqrt{\tau - 1}} - (\tau + 1)\sqrt{\tau(\tau - 1)} \right], \quad (\text{X.6})$$

where $\tau = t/4m^2$ (P. A. M. Dirac, 1930).

In the non-relativistic limit ($\tau \rightarrow 1$) we find

$$\sigma = \frac{\pi r_e^2}{2\sqrt{\tau - 1}} \quad (\text{n.r.}). \quad (\text{X.7})$$

In the ultrarelativistic case ($\tau \rightarrow \infty$)

$$\sigma = \frac{\pi r_e^2}{2\tau} (\ln 4\tau - 1) \quad (\text{u.r.}). \quad (\text{X.8})$$

In the laboratory frame, in which one of the particles (say, the electron) before the collision is at rest, the invariant τ is

$$\tau = \frac{1}{2}(1 + \gamma), \quad \gamma = \frac{\varepsilon_+}{m}. \quad (\text{X.9})$$

Formulas (88.6)–(88.8) give the dependence of the total cross section on the energy of the positron in flight:

$$\sigma = \frac{\pi r_e^2}{4} \left[\frac{\gamma^2 + 4\gamma + 1}{\gamma^2 - 1} \ln \left(\gamma + \sqrt{\gamma^2 - 1} \right) - \frac{\gamma + 3}{\sqrt{\gamma^2 - 1}} \right]. \quad (\text{X.10})$$

⁸Here it is taken into account that the photons and electrons have the same number (two) of independent polarisations, and therefore the inessential factor by which $|M_{fi}|^2$ is averaged is the same.

In particular, in the non-relativistic limit⁹

$$\sigma = \pi r_e^2 / v_+ \quad (\text{n.r.}), \quad (\text{X.11})$$

where v_+ is the velocity of the positron.

In the centre-of-inertia frame, electron, positron, and both photons have equal energies $\varepsilon = \omega$. The invariants:

$$\begin{aligned} m^2 - s &= 2\varepsilon(\varepsilon - |\mathbf{p}| \cos \theta), & m^2 - u &= 2\varepsilon(\varepsilon + |\mathbf{p}| \cos \theta), \\ t &= 4\varepsilon^2, \end{aligned} \quad (\text{X.12})$$

where θ is the angle between the momenta of the electron and one of the photons.

Substituting (88.12) into (88.4), we obtain the angular distribution of the annihilation photons:

$$d\sigma = \frac{r_e^2 m^2}{4\varepsilon |\mathbf{p}|} \left[\frac{\varepsilon^2 + \mathbf{p}^2 (1 + \sin^2 \theta)}{\varepsilon^2 - \mathbf{p}^2 \cos^2 \theta} - \frac{2\mathbf{p}^4 \sin^4 \theta}{(\varepsilon^2 - \mathbf{p}^2 \cos^2 \theta)^2} \right] d\theta. \quad (\text{X.13})$$

In the ultrarelativistic case it has symmetric maxima in the directions $\theta = 0$ and $\theta = \pi$. Near $\theta = 0$

$$d\sigma \approx \frac{r_e^2 m^2 d\theta}{2\varepsilon^2 (\theta^2 + m^2/\varepsilon^2)} \quad (\text{u.r.}). \quad (\text{X.14})$$

The total cross section is obtained from (88.6):

$$\sigma = \pi r_e^2 \frac{1 - v^2}{4v} \left[\frac{3 - v^4}{v} \ln \frac{1 + v}{1 - v} - 2(2 - v^2) \right], \quad (\text{X.15})$$

where $v = |\mathbf{p}|/\varepsilon = \sqrt{\varepsilon^2 - m^2}/\varepsilon$ is the velocity of the colliding particles.

We shall not consider here the details of the polarisation effects in the annihilation. Let us note only a few qualitative properties for large and small velocities v of the colliding particles. We shall consider the process in the centre-of-inertia frame.

In the limit $v \rightarrow 0$ only the state with orbital angular momentum $l = 0$ contributes to the cross section. But the S -state of the system “electron + positron” has negative parity (see the problem to § 27). In odd states of the two-photon system their polarisations are mutually orthogonal (see § 9). The two-photon system must therefore possess the same property, and the annihilation photons must be polarised perpendicular to each other in the non-relativistic case.

If the electron and positron are polarised, and also in the non-relativistic case, one can assert that their annihilation is possible only for anti-parallel spins. Indeed, since the annihilation occurs from the S -state, the total angular momentum of the system coincides with the total spin, which equals 1 for parallel spins. But a system of two photons cannot have a total angular momentum equal to 1 (see § 9).

In the ultrarelativistic limit ($v \rightarrow 1$) the annihilation of a longitudinally polarised (helical) electron and positron is possible only for different signs of their helicities¹⁰. In

⁹This formula is, however, inapplicable when $v_+ \lesssim \alpha$, and one cannot neglect the Coulomb interaction of the pair components (cf. the end of § 94).

¹⁰Since the directions of the momenta of the particles in the centre-of-inertia frame are opposite, different signs of the helicities correspond to parallel spins.

this limit the helical particles behave like neutrinos (see the end of § 80), and therefore the annihilating electron and positron must be analogous to the neutrino and antineutrino, whence the statement made follows.

The annihilation of an electron and a positron with equal helicities arises in the ultrarelativistic case only upon taking into account terms containing m . In order of magnitude the amplitude of this process differs from the amplitude of annihilation of the pair with anti-parallel spins by a factor m/ε ; the cross section differs by the factor $(m/\varepsilon)^2$.

Problem

Find the cross section of electron-pair production upon collision of two photons (*G. Breit, J. A. Wheeler, 1934*).

Solution. This process is the reverse of two-photon annihilation of an electron-positron pair. The squared amplitudes of both processes are the same, and their relation to the cross section differs only in that now $I^2 = (k_1 k_2)^2 = t^2/4$. Therefore

$$d\sigma_{\text{pr}} = v^2 d\sigma_{\text{ann}}.$$

In the centre-of-inertia frame ($t = 4\varepsilon^2 = 4\omega^2$),

v = velocity of the pair components,

$$d\sigma_{\text{pr}} = v^2 d\sigma_{\text{ann}},$$

where the total cross section in the centre of inertia is

$$\sigma_{\text{pr}} = \frac{\pi r_e^2}{2} (1 - v^2) \left\{ (3 - v^4) \ln \frac{1 + v}{1 - v} - 2v(2 - v^2) \right\}. \quad (1)$$

In an arbitrary reference frame K , in which the two photons k_1 and k_2 are directed towards each other, we have (from the invariance of $k_1 k_2$)

$$\omega_1 \omega_2 = \omega^2.$$

Since the photon energies and the pair component energies coincide in the centre-of-inertia frame, $\omega = \varepsilon = m/\sqrt{1 - v^2}$. Therefore for the transition to the frame K one must put in (1)

$$v = \sqrt{1 - \frac{m^2}{\omega_1 \omega_2}}.$$

§ 89. Annihilation of positronium

By virtue of conservation of momentum the annihilation of the electron and positron in positronium must be accompanied by the emission of at least two photons. Such a decay is possible, however (principally) only for parapositronium; and for orthopositronium it

is three-photon annihilation. In § 9 it was shown that the total angular momentum of a system of two photons cannot equal 1. Therefore an orthopositronium atom, which is in a state 3S_1 , cannot decay into two photons. Moreover, since in the 3S_1 state positronium constitutes a charge-odd system (see the problem to § 27), by the Furry theorem (see § 79) its decay into any even number of photons is also impossible. On the contrary, in the 1S_0 state positronium is charge-even, and therefore the decay of parapositronium into any odd number of photons is forbidden.

The principal process determining the lifetime of positronium is, thus, two-photon annihilation in the case of parapositronium and three-photon annihilation in the case of orthopositronium (*I. Ya. Pomeranchuk*, 1948). The probability of decay can be related to the cross section of annihilation of a free pair.

The momenta of the electron and positron in positronium are $\sim me^2/\hbar$, i.e. small in comparison with mc . Therefore in computing the probability of annihilation one can go to the limit of two particles at rest in the origin of coordinates. Let $\bar{\sigma}_{2\gamma}$ be the cross section of two-photon annihilation of a free pair, averaged over spin directions of both particles. In the non-relativistic limit, according to (88.11)¹¹,

$$\bar{\sigma}_{2\gamma} = \pi \left(\frac{e^2}{mc^2} \right)^2 \frac{c}{v}, \quad (\text{X.16})$$

where v is the relative velocity of the particles. We obtain the probability of annihilation $\bar{w}_{2\gamma}$ by multiplying $\bar{\sigma}_{2\gamma}$ by the flux density $v|\psi(0)|^2$. Here $\psi(r)$ is the wave function of the ground state of positronium normalised to 1:

$$\psi = \frac{1}{\sqrt{\pi}a^3} e^{-r/a}, \quad a = \frac{2\hbar^2}{me^2} \quad (\text{X.17})$$

(the Bohr radius of positronium a is twice the Bohr radius of the hydrogen atom because the reduced mass is half that). This probability, however, corresponds to the averaging over spin states of the initial state. But among the four possible spin states of the two-particle system, only one (with total spin 0) is capable of two-photon annihilation. Therefore the average probability of decay $\bar{w}_{2\gamma}$ is related to the probability of decay of parapositronium w_0 by the relation $\bar{w}_{2\gamma} = w_0/4$. Thus,

$$w_0 = 4|\psi(0)|^2 (v\bar{\sigma}_{2\gamma})_{v \rightarrow 0}. \quad (\text{X.18})$$

Substituting the values of the quantities from (89.1) and (89.2), we obtain for the lifetime of parapositronium

$$\tau_0 = \frac{2\hbar}{mc^2\alpha^5} = 1.23 \cdot 10^{-10} \text{ s}. \quad (\text{X.19})$$

We draw attention to the fact that the level width $\Gamma_0 = \hbar/\tau_0$ is small compared with the energy:

$$|E_{\text{gr}}| = \frac{me^4}{4\hbar^2} = mc^2 \frac{\alpha^2}{4}.$$

¹¹Formulas (89.1) and (89.7) are written in ordinary units.

It is precisely this circumstance that allows one to treat positronium as a quasi-stationary system.

In an analogous manner we find that the probability of decay of orthopositronium is related to the spin-averaged cross section of three-photon annihilation of a free pair by the relation

$$w_1 = \frac{4}{3} \bar{w}_{3\gamma} = \frac{4}{3} |\psi(0)|^2 (v \bar{\sigma}_{3\gamma})_{v \rightarrow 0} \quad (\text{X.20})$$

($\frac{3}{4}$ is the relative statistical weight of the state with spin 1). Looking ahead, we note that

$$\bar{\sigma}_{3\gamma} = \frac{4(\pi^2 - 9)}{3v} \alpha \left(\frac{e^2}{mc^2} \right)^2. \quad (\text{X.21})$$

Therefore the lifetime of orthopositronium is

$$\tau_1 = \frac{9\pi}{2(\pi^2 - 9)} \frac{\hbar}{mc^2 \alpha^6} = 1.4 \cdot 10^{-7} \text{ s}. \quad (\text{X.22})$$

The inequality $\Gamma_1 \ll |E_{\text{gr}}|$ is in this case satisfied, of course, even better than for parapositronium.

Let us compute the cross section of three-photon annihilation of a free pair (A. Ore, J. L. Powell, 1949).

According to (64.18) the cross section of the process under consideration in the centre-of-inertia frame is expressed through the squared amplitude by the formula

$$\begin{aligned} d\sigma_{3\gamma} &= \frac{(2\pi)^4 |M_{fi}|^2}{4I} \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) \delta(\omega_1 + \omega_2 + \omega_3 - 2m) \times \\ &\times \frac{d^3 k_1 d^3 k_2 d^3 k_3}{(2\pi)^9 2\omega_1 \cdot 2\omega_2 \cdot 2\omega_3}, \end{aligned} \quad (\text{X.23})$$

where, according to (64.16), $I = 2m \frac{m}{2} v = m^2 v$, where v is the relative velocity of the positron and electron (which we assume to be small); $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3$ and $\omega_1, \omega_2, \omega_3$ are the wave vectors and frequencies of the emitted photons; the δ -functions express the laws of conservation of energy and momentum. By virtue of these laws the three frequencies $\omega_1, \omega_2, \omega_3$ must be representable as the side lengths of a triangle with perimeter $2m$.

Three-photon annihilation corresponds to the diagram

and five more diagrams obtained by permutation of the photons k_1, k_2, k_3 . The corresponding amplitude is written in the form

$$M_{fi} = (4\pi)^{3/2} e_{\lambda}^{(3)*} e_{\mu}^{(2)*} e_{\nu}^{(1)*} \bar{u}(-p_+) Q^{\lambda\mu\nu} u(p_-), \quad (\text{X.24})$$

where

$$Q^{\lambda\mu\nu} = \sum_{\text{perm}} \gamma^{\lambda} G(k_3 - p_+) \gamma^{\mu} G(p_- - k_1) \gamma^{\nu}, \quad (\text{X.25})$$

where the sum is taken over all permutations of the photon numbers 1, 2, 3 together with the simultaneously corresponding permutations of the tensor indices $\lambda\mu\nu$. The squared

amplitude, averaged over the polarisations of the electron and positron and summed over the polarisations of the photons:

$$\frac{1}{4} \sum_{\text{polar}} |M_{fi}|^2 = (4\pi)^3 \text{Sp} \left\{ \rho_+ Q^{\lambda\mu\nu} \rho_- \tilde{Q}_{\lambda\mu\nu} \right\}, \quad (\text{X.26})$$

where

$$\rho_- = \frac{1}{2}(\gamma p_- + m), \quad \rho_+ = \frac{1}{2}(\gamma p_+ - m).$$

The matrices $\tilde{Q}^{\lambda\mu\nu}$ differ from the matrices $Q^{\lambda\mu\nu}$ by the reversal of the order of factors in each term of the sum. In the case of interest to us, where the velocities of the electron and positron are small, one can set their 3-momenta $\mathbf{p}_-$ and $\mathbf{p}_+$ equal to zero, i.e. set $p_- = p_+ = (m, 0)$. Then the electron Green's functions become

$$G(p_- - k_1) = \frac{\gamma p_- - \gamma k_1 + m}{(p_- - k_1)^2 - m^2} \approx \frac{-\gamma k_1 + m(\gamma^0 + 1)}{-2m\omega_1}$$

and so on, and the density matrices reduce to

$$\rho_{\pm} = \frac{m}{2}(\gamma^0 \pm 1).$$

Upon multiplying out the terms in (89.11) there arises a large number of terms. However, the number of terms requiring computation can be considerably reduced by making full use of the symmetry with respect to the permutations of the photons. It suffices to multiply out the six terms in $Q^{\lambda\mu\nu}$ (89.10) with any one term in $\tilde{Q}^{\lambda\mu\nu}$. In the remaining five traces one can also identify certain parts that transform into one another under different permutations of the photons. The products of traces of 4-vectors p , k_1 , k_2 , k_3 arising upon expanding the traces are expressed through the frequencies ω_1 , ω_2 , ω_3 . Since $p = (m, 0)$, we have $pk_1 = m\omega_1, \dots$. The products $k_1k_2, \dots$ are determined from the equation of 4-momentum conservation: $2p = k_1 + k_2 + k_3$; rewriting this equation in the form $2p - k_3 = k_1 + k_2$ and squaring, we obtain

$$k_1k_2 = 2m(m - \omega_3), \dots \quad (\text{X.27})$$

As a result, after a rather lengthy calculation one obtains

$$\frac{1}{4} \sum_{\text{polar}} |M_{fi}|^2 = (4\pi)^3 e^6 \cdot 16 \left[\left(\frac{m - \omega_1}{\omega_2\omega_3} \right)^2 + \left(\frac{m - \omega_2}{\omega_1\omega_3} \right)^2 + \left(\frac{m - \omega_3}{\omega_1\omega_2} \right)^2 \right].$$

Substituting this expression into (89.8), we find the differential cross section of three-photon annihilation:

$$\begin{aligned} d\bar{\sigma}_{3\gamma} = & \frac{e^6}{\pi^2 m^2 v} \left[\left(\frac{m - \omega_1}{\omega_2\omega_3} \right)^2 + \left(\frac{m - \omega_2}{\omega_1\omega_3} \right)^2 + \left(\frac{m - \omega_3}{\omega_1\omega_2} \right)^2 \right] \times \\ & \times \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) \delta(\omega_1 + \omega_2 + \omega_3 - 2m) \frac{d^3k_1 d^3k_2 d^3k_3}{\omega_1\omega_2\omega_3}. \end{aligned} \quad (\text{X.28})$$

Here we must still eliminate the δ -functions. The first of them is removed by integration over d^3k_3 , after which we replace the remaining differentials:

$$d^3k_1 d^3k_2 \rightarrow 4\pi\omega_1^2 d\omega_1 \cdot 2\pi\omega_2^2 d(\cos\theta_{12}) d\omega_2,$$

where θ_{12} is the angle between $\mathbf{k}_1$ and $\mathbf{k}_2$; it is understood that the integration over the directions of $\mathbf{k}_1$ and the azimuth of $\mathbf{k}_2$ relative to $\mathbf{k}_1$ has already been performed. Differentiating the equation

$$\omega_3 = \sqrt{\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2 \cos\theta_{12}},$$

we find

$$d \cos \theta_{12} = \frac{\omega_3}{\omega_1 \omega_2} d\omega_3.$$

By integrating over ω_3 we eliminate the second δ -function. As a result we obtain the cross section of annihilation with the production of photons having given energies in the form

$$d\bar{\sigma}_{3\gamma} = \frac{1}{6} \frac{8e^6}{vm^2} \left\{ \left(\frac{m - \omega_1}{\omega_2 \omega_3} \right)^2 + \left(\frac{m - \omega_2}{\omega_1 \omega_3} \right)^2 + \left(\frac{m - \omega_3}{\omega_1 \omega_2} \right)^2 \right\} d\omega_1 d\omega_2 \quad (\text{X.29})$$

(with a view to subsequent integration over the frequencies, we have introduced the factor $\frac{1}{6}$ taking into account the identity of the photons (cf. the footnote on p. 286)).

Each of the frequencies $\omega_1, \omega_2, \omega_3$ can take values between 0 and m (the value m is attained by two of the frequencies when the third equals zero). At given ω_1 the frequency ω_2 varies between $m - \omega_1$ and m . Integrating (89.14) over ω_2 in these limits, we obtain the spectral distribution of the photons of the decay:

$$d\bar{\sigma}_{3\gamma} = \frac{8e^6}{3vm^3} F(\omega_1) d\omega_1,$$

$$F(\omega_1) = \frac{\omega_1(m - \omega_1)}{(2m - \omega_1)^2} + \frac{2m - \omega_1}{\omega_1} + \left[\frac{2m(m - \omega_1)}{\omega_1^2} - \frac{2m(m - \omega_1)^2}{(2m - \omega_1)^3} \right] \ln \frac{m - \omega_1}{m}.$$

The function F monotonically increases from zero at $\omega_1 = 0$ to 1 at $\omega_1 = m$; in Fig. 14 its graph is shown.

The total annihilation cross section is obtained by integrating (89.14) over both frequencies:

$$\bar{\sigma}_{3\gamma} = \frac{4e^6}{3vm^2} \cdot 3 \int_0^m \int_{m-\omega_1}^m \frac{(\omega_1 + \omega_2 - m)^2}{\omega_1^2 \omega_2^2} d\omega_1 d\omega_2.$$

The double integral here equals $(\pi^2 - 9)/3$, and we arrive at the formula (89.6) given above.

§ 90. Magnetobremssstrahlung

According to classical theory (see II, § 74) an ultrarelativistic electron moving in a constant magnetic field H emits a quasi-continuous spectrum with maximum at a frequency of order

$$\omega \sim \omega_0 \left(\frac{\varepsilon}{m} \right)^3, \quad (\text{X.30})$$

where

$$\omega_0 = \frac{v|e|H}{|\mathbf{p}|} \approx \frac{|e|H}{\varepsilon} \quad (\text{X.31})$$

is the frequency of revolution of an electron with energy ε in a circular orbit (in the plane perpendicular to the field)¹². We shall assume that the longitudinal (along $\mathbf{H}$) component of the electron velocity is zero; this can always be achieved by choosing the appropriate reference frame.

Quantum effects in magnetobremssstrahlung radiation have a twofold origin: quantisation of the motion of the electron and quantum recoil upon emission of a photon. The latter is determined by the ratio $\hbar\omega_0/\varepsilon$, and the condition for applicability of classical theory requires its smallness. Since

$$\frac{\hbar\omega_0}{\varepsilon} = \frac{H}{H_0} \left(\frac{m}{\varepsilon} \right)^2, \quad (\text{X.32})$$

where $H_0 = m^2/(|e|\hbar)$ ($= m^2 c^3/(|e|\hbar) = 4.4 \cdot 10^{13}$ G). In the classical region $\chi \sim \hbar\omega/\varepsilon \ll 1$. When $\chi \gtrsim 1$ the energy of the emitted photon $\hbar\omega \sim \varepsilon$, and moreover when $\chi \gg 1$ (as we shall see below) the essential region of the spectrum extends to frequencies at which the energy of the electron after emission is

$$\varepsilon' \sim m \frac{H_0}{H} \ll \varepsilon. \quad (\text{X.33})$$

In order for the electron to remain ultrarelativistic, the field must satisfy the condition

$$\frac{H}{H_0} \ll 1. \quad (\text{X.34})$$

As regards the quantisation of the motion itself, it is characterised by the ratio $\hbar\omega_0/\varepsilon$. Since

$$\frac{\hbar\omega_0}{\varepsilon} = \frac{H}{H_0} \left(\frac{m}{\varepsilon} \right)^2,$$

by virtue of (90.5) $\hbar\omega_0 \ll \varepsilon$, i.e. the motion of the electron is quasi-classical irrespective of the value of χ . In other words, one can neglect the non-commutativity of the operators of dynamic variables of the electron with one another (quantities $\approx \hbar\omega_0/\varepsilon$), while at the same time taking into account their non-commutativity with the operators of the photon field (quantities $\sim \hbar\omega/\varepsilon$)¹³.

¹²In this section we set $c = 1$, but retain the factors $\hbar$.

¹³The complete solution of the quantum problem of magnetobremssstrahlung radiation was given by *N. P. Klepikov* (1954), and the first quantum correction to the classical formula by *A. A. Sokolov, N. P. Klepikov* and *I. M. Ternov* (1952). The derivation presented in this section, which makes explicit use of quasi-classicality of the motion, is due to *V. N. Baier* and *V. M. Katkov* (1967). An analogous method was used earlier by *Schwinger* (*J. Schwinger*, 1967) to obtain the first quantum correction to the intensity of radiation.

The quasi-classical wave functions of the stationary states of an electron in the external field can be symbolically represented in the form

$$\psi = (2\hat{H})^{-1/2} u(\hat{p}) \exp\left(-\frac{i}{\hbar}\hat{H}t\right) \varphi(\mathbf{r}), \quad (\text{X.35})$$

where $\varphi(\mathbf{r}) \sim \exp(iS/\hbar)$ are the quasi-classical wave functions of a spinless particle ($S(\mathbf{r})$ is its classical action); $u(\hat{p})$ is an operator bispinor

$$u(\hat{p}) = \begin{pmatrix} (\hat{H} + m)^{1/2} w \\ (\hat{H} + m)^{-1/2} (\boldsymbol{\sigma} \hat{\mathbf{p}}) w \end{pmatrix},$$

obtained from the bispinor amplitude of the plane wave $u(p)$ (23.9) by the replacement of $\mathbf{p}$ and ε by operators¹⁴

$$\hat{\mathbf{p}} = \hat{\mathbf{P}} - e\mathbf{A} = -i\hbar\nabla - e\mathbf{A}, \quad \hat{H} = (\hat{\mathbf{p}}^2 + m^2)^{1/2},$$

$\mathbf{P}$ being the generalised momentum of the particle in the field with vector potential $\mathbf{A}(\mathbf{r})$; the ordering of operator factors in ψ is inessential, since their non-commutativity is neglected; the spin state of the electron is determined by the 3-spinor w .

For computing the probability of emission of a photon it is convenient in the quasi-classical case to start not from the final formula of perturbation theory (44.3), but from the formula in which the integration over time has not yet been performed. For the complete (over all time) differential probability we have¹⁵

$$dw = \sum_f |a_{fi}|^2 \frac{d^3k}{(2\pi)^3}, \quad a_{fi} = \int_{-\infty}^{\infty} V_{fi}(t) dt \quad (\text{X.36})$$

(cf. III, (41.2)); the summation is over the final states of the electron.

Using (90.6), we write the matrix element for the emission of a photon ω , $\mathbf{k}$ in the form

$$\begin{aligned} V_{fi}(t) = & -\frac{e\sqrt{4\pi}}{\sqrt{2\hbar\omega}} \int \varphi_f^* \exp\left(\frac{i}{\hbar}\hat{H}t\right) \frac{u^\dagger(\hat{p}')}{(2\hat{H}')^{1/2}} (\mathbf{e}^* \boldsymbol{\alpha}) \frac{u(\hat{p})}{(2\hat{H})^{1/2}} \times \\ & \times e^{i\omega t - i\mathbf{k}\mathbf{r}} \exp\left(-\frac{i}{\hbar}\hat{H}t\right) \varphi_i d^3x, \end{aligned}$$

where the operators in the square brackets act to the left; the field of the photon is chosen in three-dimensional transverse gauge. The factors $\exp(\pm i\hat{H}t/\hbar)$ convert the operators standing between them into operators that are explicitly time-dependent, in the Heisenberg representation. We write $V_{fi}(t)$ in the form

$$V_{fi}(t) = e\sqrt{\frac{2\pi}{\hbar\omega}} \langle f | Q(t) | i \rangle e^{i\omega t}, \quad (\text{X.37})$$

¹⁴In this section (in contrast to Ch. IV) the generalised momentum is denoted by the capital letter $\mathbf{P}$; the symbol $\mathbf{p}$ is used for the ordinary (kinetic) momentum.

¹⁵Substituting $V_{fi}(t) = V_{fi} \exp(i\omega_{fi}t)$, we obtain $a_{fi} = 2\pi V_{fi} \delta(\omega_{fi})$. Bearing in mind that the square of the δ -function must be understood as $|\delta(\omega)|^2 \rightarrow (t/2\pi) \delta(\omega)$, where t is the total observation time (cf. the derivation of formula (64.5)), we obtain from (90.7) the probability per unit time (44.3).

where $Q(t)$ denotes the Heisenberg operator

$$\hat{Q}(t) = \frac{u_f^\dagger(\hat{p}')}{(2H')^{1/2}} (\alpha \mathbf{e}^*) e^{-i\mathbf{k}\hat{\mathbf{r}}(t)} \frac{u_i(\hat{p})}{(2H)^{1/2}}, \quad (\text{X.38})$$

and the matrix element is taken with respect to the functions φ_i, φ_f .

The summation in (90.7) is over all final wave functions φ_f ; it is carried out with the aid of the completeness relation

$$\sum_f \varphi_f^*(\mathbf{r}') \varphi_f(\mathbf{r}) = \delta(\mathbf{r}' - \mathbf{r}),$$

expressing the completeness of the system of functions φ_f . As a result,

$$dw = \frac{e^2}{\hbar\omega} \frac{d^3k}{4\pi^2} \iint dt_1 dt_2 \cdot e^{i\omega(t_1-t_2)} \langle i|Q^+(t_2)Q(t_1)|i\rangle. \quad (\text{X.39})$$

If the integration is over a sufficiently large time interval, one can introduce instead of t_1, t_2 the new variables

$$\tau = t_2 - t_1, \quad t = \frac{t_1 + t_2}{2}$$

and in the integral over dt regard the sub-integral expression as the probability of emission per unit time. Multiplying it by $\hbar\omega$, we obtain the intensity

$$dI = \frac{e^2}{4\pi^2} d^3k \int e^{-i\omega\tau} \langle i|Q^+\left(t + \frac{\tau}{2}\right)Q\left(t - \frac{\tau}{2}\right)|i\rangle d\tau. \quad (\text{X.40})$$

An ultrarelativistic electron radiates into a narrow cone at angles $\vartheta \sim m/\varepsilon$ about the direction of its velocity $\mathbf{v}$. Therefore radiation in a given direction $\mathbf{n} = \mathbf{k}/\omega$ is formed on that segment of the trajectory on which $\mathbf{v}$ turns through an angle $\sim m/\varepsilon$. This segment is traversed in a time τ such that $\tau|\dot{\mathbf{v}}| \approx \tau\omega_0 \sim m/\varepsilon \ll 1$. It is precisely this region that gives the principal contribution to the integral over τ . Therefore in the further calculations we shall systematically expand all quantities in powers of $\omega_0\tau$. In doing this, however, it may be necessary to retain more than just the leading term of the expansion because of cancellations resulting from the fact that $1 - \mathbf{n}\mathbf{v} \sim \vartheta^2 \sim (m/\varepsilon)^2$.

If we bring the operator $\hat{Q}^+\hat{Q}$ to the form of a product of commutative (to the required accuracy) operators, then taking the diagonal matrix element $\langle i|\dots|i\rangle$ reduces to replacing these operators by the classical values (functions of time) of the corresponding quantities. This goal is achieved as follows.

In the expression for $\hat{Q}(t)$ (90.8) we factor out the operator $\exp(-i\mathbf{k}\hat{\mathbf{r}}(t))$ to the left and rewrite $\hat{Q}(t)$ in the form

$$\hat{Q}(t) = \exp[-i\mathbf{k}\hat{\mathbf{r}}(t)] \hat{R}(t), \quad \hat{R}(t) = \frac{u_f^\dagger(\hat{p}')}{(2H')^{1/2}} (\alpha \mathbf{e}^*) \frac{u_i(\hat{p})}{(2H)^{1/2}}, \quad (\text{X.41})$$

where $\hat{H}' = \hat{H} - \hbar\omega$, $\hat{\mathbf{p}}' = \hat{\mathbf{p}} - \hbar\mathbf{k}$.

Now,

$$\hat{Q}_2^+ \hat{Q}_1 = \hat{R}_2 \exp(i\mathbf{k}\hat{\mathbf{r}}_2) \exp(-i\mathbf{k}\hat{\mathbf{r}}_1) \hat{R}_1 \quad (\text{X.42})$$

(here and below the subscripts 1 and 2 label values of the quantities at the times $t_1 = t - \tau/2$ and $t_2 = t + \tau/2$). It remains to compute the product of the two non-commutative operators $\exp(i\mathbf{k}\hat{\mathbf{r}}_2)$ and $\exp(-i\mathbf{k}\hat{\mathbf{r}}_1)$. This product can itself already be considered commutative with the remaining factors. Denoting

$$\hat{L}(\tau) = \exp(-i\omega\tau) \exp(i\mathbf{k}\hat{\mathbf{r}}_2) \exp(-i\mathbf{k}\hat{\mathbf{r}}_1); \quad (\text{X.43})$$

this is precisely the combination of operators that enters (90.10). By the meaning of the operator $\exp(i\hat{H}\tau/\hbar)$ as the time-shift operator we have

$$\exp(i\mathbf{k}\hat{\mathbf{r}}_2) = \exp\left(i\hat{H}\frac{\tau}{\hbar}\right) \exp(i\mathbf{k}\hat{\mathbf{r}}_1) \exp\left(-i\hat{H}\frac{\tau}{\hbar}\right).$$

Substituting this expression into (90.14) and noting that $\exp(i\mathbf{k}\hat{\mathbf{r}}_1)$ is the operator of momentum-space shift, we transform L to the form

$$\hat{L}(\tau) = \exp\left\{i\left[\hat{H} - \hbar\omega\right]\frac{\tau}{\hbar}\right\} \exp\left\{-i\hat{H}(\hat{\mathbf{p}}_1 - \hbar\mathbf{k})\frac{\tau}{\hbar}\right\}. \quad (\text{X.44})$$

Differentiating (90.15) with respect to τ and noting that $\exp(i\mathbf{k}\hat{\mathbf{r}}_1)$ is the momentum-space shift operator, we transform L to the form¹⁶

$$i\hbar \frac{d\hat{L}}{d\tau} = \frac{\varepsilon}{\varepsilon'} \hbar(\omega - \mathbf{v}_2\mathbf{k})L. \quad (\text{X.45})$$

This equation must be solved with the obvious initial condition $L(0) = 1$. Noting that

$$\int_0^\tau \mathbf{v}_2 d\tau = \mathbf{r}_2 - \mathbf{r}_1,$$

we obtain

$$L(\tau) = \exp\left\{i \frac{\varepsilon}{\varepsilon'} (\mathbf{k}\mathbf{r}_2 - \mathbf{k}\mathbf{r}_1 - \omega\tau)\right\}. \quad (\text{X.46})$$

Up to this point we have not used the specific form of the electron trajectory. Let us now express $\mathbf{r}_2 - \mathbf{r}_1$ through $\mathbf{p}_1$ with the aid of the equations of motion of the electron in the plane perpendicular to the magnetic field $\mathbf{H}$ (see II, § 21):

$$\mathbf{r}_2 - \mathbf{r}_1 = \frac{\mathbf{p}_1}{eH} \sin \frac{eH\tau}{\varepsilon} + \frac{[\mathbf{p}_1, \mathbf{H}]}{|eH|} \left(1 - \cos \frac{eH\tau}{\varepsilon}\right).$$

Expanding in powers of $\omega_0\tau$ and retaining terms up to the third order, and noting that $1 - \mathbf{nv} \sim \vartheta^2 \sim (m/\varepsilon)^2$, we obtain

$$\mathbf{k}(\mathbf{r}_2 - \mathbf{r}_1) - \omega\tau \approx \omega\tau \left\{ -\frac{i\omega\tau\varepsilon}{\varepsilon'} \left(1 - \mathbf{vn} + \frac{\tau^2}{24}\omega_0^2\right) \right\}. \quad (\text{X.47})$$

(In the last term we have set $\mathbf{nv}_1 \approx 1$.)

¹⁶By virtue of energy conservation the Heisenberg energy operators $\hat{H}(\hat{\mathbf{p}}_1)$ and $\hat{H}(\hat{\mathbf{p}}_2)$ coincide, and therefore in such cases we write the argument of $\hat{H}$ without a subscript. But, of course, $\hat{H}(\hat{\mathbf{p}}_1 - \hbar\mathbf{k})$ does not coincide with $\hat{H}(\hat{\mathbf{p}}_2 - \hbar\mathbf{k})$.

Let us now transform the remaining factors in (90.13). By direct expansion of the products in $R(t)$ (with the matrix α from (21.20)) we find

$$R(t) = w_f^\dagger \mathbf{e}^* (A + i[\mathbf{B}\sigma]) w_i,$$

$$\mathbf{A} = \frac{\mathbf{p}}{2} \left(\frac{1}{\varepsilon} + \frac{1}{\varepsilon'} \right) = \frac{\varepsilon + \varepsilon'}{2\varepsilon'} \mathbf{v}, \quad (\text{X.48})$$

$$\mathbf{B} = \frac{1}{2} \left(\frac{\mathbf{p}}{\varepsilon + m} - \frac{\mathbf{p}'}{\varepsilon' + m} \right) \approx \frac{\hbar\omega}{2\varepsilon'} \left(\mathbf{n} - \mathbf{v} + \mathbf{v} \frac{m}{\varepsilon} \right),$$

where $p'(t) = p(t) - \hbar\mathbf{k}$; terms of higher orders in m/ε are omitted. Thus, finally we have

$$\exp(-i\omega\tau) \langle i | Q_2^\dagger Q_1 | i \rangle = R_2^* R_1 L(\tau), \quad (\text{X.49})$$

$$R_2^* R_1 = \text{Sp} \frac{1 + \zeta\sigma}{2} ((\mathbf{A}_2 - i[\mathbf{B}_2\sigma])\mathbf{e}) \frac{1 + \zeta'\sigma}{2} ((\mathbf{A}_1 - i[\mathbf{B}_1\sigma])\mathbf{e}^*).$$

The factors $(1 + \zeta\sigma)/2$ are the two-row polarisation density matrices of the initial and final electrons.

Let us consider the intensity of the radiation, summed over polarisations of the photon and the final electron and averaged over polarisations of the initial electron. As a result of the indicated operations we obtain after a simple calculation¹⁷

$$\frac{1}{2} \sum_{\text{polar}} R_2^* R_1 = \frac{\varepsilon^2 + \varepsilon'^2}{2\varepsilon'^2} (\mathbf{v}_1 \mathbf{v}_2 - 1) + \frac{1}{2} \left(\frac{\hbar\omega}{\varepsilon'} \right)^2 \left(\frac{m}{\varepsilon} \right)^2.$$

To the required accuracy

$$\mathbf{v}_1 \mathbf{v}_2 = \mathbf{v}^2 - \frac{\tau^2}{4} \dot{\mathbf{v}}^2 + \frac{\tau}{4} \mathbf{v} \ddot{\mathbf{v}} = 1 - \frac{m^2}{\varepsilon^2} - \frac{1}{2} \omega_0^2 \tau^2.$$

Substituting these expressions into (90.21), and then into (90.10), we obtain

$$dI = -\frac{e^2}{4\pi^2} \omega^2 d\omega d\mathbf{o}_{\mathbf{n}} \times \int_{-\infty}^{\infty} \left(\frac{m^2}{\varepsilon\varepsilon'} + \frac{\varepsilon^2 + \varepsilon'^2}{4\varepsilon'^2} \omega_0^2 \tau^2 \right) \exp \left\{ -\frac{i\omega\tau\varepsilon}{\varepsilon'} \left(1 - \mathbf{v}\mathbf{n} + \frac{\tau^2}{24} \omega_0^2 \right) \right\} d\tau. \quad (\text{X.50})$$

This formula gives the spectral and angular distribution of the intensity of the radiation. For finding the spectral distribution of the radiation the product is integrated over $d\mathbf{o}_{\mathbf{n}}$. Choosing $\mathbf{v}$ as the polar axis with angle ϑ between $\mathbf{n}$ and $\mathbf{v}$, we have

$$\mathbf{n}\mathbf{v} = v \cos \vartheta, \quad d\mathbf{o}_{\mathbf{n}} = \sin \vartheta d\vartheta d\varphi,$$

¹⁷Here we have also used the following circumstance. Upon summation over $\mathbf{e}$: $\sum_{\mathbf{e}} = (\mathbf{v}_1 \mathbf{e})(\mathbf{v}_2 \mathbf{e}^*) = \mathbf{v}_1 \mathbf{v}_2 - (\mathbf{v}_1 \mathbf{n})(\mathbf{v}_2 \mathbf{n})$. But upon substitution of (90.21) into (90.10) one can integrate by parts, noting that $(\mathbf{v}_1 \mathbf{n}) \exp(-\frac{i\varepsilon}{\varepsilon'} \mathbf{k}\mathbf{r}_1) = \frac{i\varepsilon'}{\varepsilon\omega} \frac{d}{dt_1} \exp(-\frac{i\varepsilon}{\varepsilon'} \mathbf{k}\mathbf{r}_1)$ and analogously for $\mathbf{v}_2 \mathbf{n}$. As a result, we find that for the purpose of further integration one can replace $\mathbf{v}_1 \mathbf{n}$ and $\mathbf{v}_2 \mathbf{n}$ by unity.

and the integral

$$\int \exp\left(\frac{i\omega\tau\varepsilon}{\varepsilon'} \mathbf{n}\mathbf{v}\right) d\mathbf{o}_n = \frac{2\pi\varepsilon'}{i\omega\tau\varepsilon v} \left\{ \exp\left(\frac{i\omega\tau\varepsilon v}{\varepsilon'}\right) - \exp\left(-\frac{i\omega\tau\varepsilon v}{\varepsilon'}\right) \right\}.$$

Upon substitution of this expression into (90.22) we obtain two terms with exponents of different order of magnitude. The exponent of the second term turns out to be much larger, since it contains the factor $1 + v \approx 2$ instead of the small factor $1 - v \approx m^2/(2\varepsilon^2)$ in the first term. Shifting the contour of integration over τ to the lower half-plane of the complex variable τ , one can make the second term small and neglect it. After this one can again shift the contour of integration back to the real axis. From the conclusion it is clear, however, that the pole appearing in the sub-integral expression at $\tau = 0$ must be bypassed from below. Thus,

$$\frac{dI}{d\omega} = \frac{ie^2\omega}{2\pi} \int_{-\infty}^{\infty} \left(\frac{m^2}{\varepsilon^2\tau} + \frac{\varepsilon^2 + \varepsilon'^2}{4\varepsilon'^2} \omega_0^2 \tau^2 \right) \exp\left\{ -\frac{i\omega\tau\varepsilon}{\varepsilon'} \left(1 - v + \frac{\tau^2}{24} \omega_0^2 \right) \right\} d\tau, \quad (\text{X.51})$$

where the contour of integration is chosen as indicated above. Using the integral representation of the Airy function Φ (see III, § b), it is not difficult to show that the first term reduces to an integral of the Airy function, and the second to its derivative. We finally obtain

$$\frac{dI}{d\omega} = -\frac{e^2 m^2 \omega}{\sqrt{\pi} \varepsilon^2} \left[\int_x^{\infty} \Phi(\xi) d\xi + \left(\frac{2}{x} + \frac{\hbar\omega}{\varepsilon} \chi x^{1/2} \right) \Phi'(x) \right], \quad (\text{X.52})$$

$$x = \left(\frac{\hbar\omega}{\varepsilon' \chi} \right)^{2/3} = \frac{m^2}{\varepsilon^2} \left(\frac{\varepsilon\omega}{\varepsilon' \omega_0} \right)^{2/3}, \quad (\text{X.53})$$

(A. I. Nikishov, V. I. Ritus, 1967). The maximum in the frequency distribution lies at $x \sim 1$; when $\chi \ll 1$ it follows from this that (90.1), and when $\chi \gg 1$ —(90.4). In the classical limiting case we have $\hbar\omega \ll \varepsilon$, so that $\varepsilon' \approx \varepsilon$, $x \approx (\omega/\omega_0)^{2/3} (m/\varepsilon)^2$; the second term in the square brackets is small and (90.23) goes over into the classical formula (74.13) (see II).

The quantity I_{cl} is the classical total intensity of the radiation (cf. II, (74.2)).

For computing the total intensity of the radiation, expression (90.23) must be integrated over ω from 0 to ε . Passing to integration over x , noting that

$$\hbar\omega = \varepsilon \left(1 - \frac{1}{1 + \chi x^{3/2}} \right),$$

and therefore x varies from 0 to ∞ . Carrying out in the first term of (90.23) two integrations by parts, we obtain

$$I = -\frac{e^2 m^2 \chi^2}{2\sqrt{\pi} \hbar^2} \int_0^{\infty} \frac{4 + 5\chi x^{3/2} + 4\chi^2 x^3}{(1 + \chi x^{3/2})^4} \Phi'(x) x dx. \quad (\text{X.54})$$

In Fig. 16 the graph of the function $I(\chi)/I_{cl}$ is shown.

When $\chi \ll 1$, in the integral the essential region is $x \sim 1$. Expanding the sub-integral expression in χ and integrating this expansion with the aid of the formula

$$\int_0^\infty x^\nu \Phi'(x) dx = -\frac{1}{2\sqrt{\pi}} 3^{(\nu-1)/6} \Gamma\left(\frac{\nu}{3} + 1\right) \Gamma\left(\frac{\nu}{3} + \frac{1}{3}\right),$$

we obtain

$$I = I_{cl} \left(1 - \frac{55\sqrt{3}}{16} \chi + 48\chi^2 - \dots \right). \quad (\text{X.55})$$

When $\chi \gg 1$, in the integral the essential region is that in which $\chi x^{3/2} \sim 1$, i.e. $x \ll 1$. In the first approximation one can therefore replace $\Phi'(x)$ by $\Phi'(0) = -3^{1/6} \Gamma(2/3)/(2\sqrt{\pi})$, after which integration gives

$$I \approx \frac{32\Gamma(2/3)e^2 m^2}{243\hbar^2} (3\chi)^{2/3} = 0.37 \frac{e^2 m^2}{\hbar^2} \left(\frac{H\varepsilon}{H_0 m} \right)^{2/3}. \quad (\text{X.56})$$

Magnetobremssstrahlung radiation leads to the appearance of polarisation of the moving electrons (*A. A. Sokolov, I. M. Ternov,*

1963). To examine this question one must find the probability of a radiative transition with reversal of the spin direction. Setting in (90.21) $\zeta_i = -\zeta_f \equiv \zeta$, $|\zeta| = 1$, we obtain

$$\begin{aligned} R_2^* R_1 &= (\mathbf{B}_1 \mathbf{B}_2) - (\mathbf{e}^* \mathbf{B}_2)(\mathbf{e} \mathbf{B}_2) - \\ &- (\mathbf{e}^* [\mathbf{B}_1 \zeta])(\mathbf{e} [\mathbf{B}_2 \zeta]) - i(\zeta \mathbf{e}^*)(\mathbf{e} [\mathbf{B}_1 \mathbf{B}_2]). \end{aligned}$$

Summation over the polarisations of the photon gives after straightforward transformations

$$\sum_{\mathbf{e}} R_2^* R_1 = (\mathbf{B}_1 \mathbf{B}_2)(1 - (\zeta \mathbf{n})^2) + (\zeta \mathbf{n})(\mathbf{n} \mathbf{B}_2)(\zeta \mathbf{B}_1) - i(\zeta - \mathbf{n}(\mathbf{n} \zeta))[\mathbf{B}_1 \mathbf{B}_2]. \quad (\text{X.57})$$

We shall assume that $\chi \ll 1$, and seek only the leading term of the expansion of the probability in powers of $\hbar$. Since $\mathbf{B}$ (from (90.20)) already contains $\hbar^2$, all remaining quantities (including those in the exponent in (90.18)) can be replaced by ε .

Expanding

$$\begin{aligned} \mathbf{B}_1 &= \frac{\omega}{2\varepsilon} \left(\mathbf{n} - \mathbf{v} + \frac{\tau}{2} \dot{\mathbf{v}} + \mathbf{v} \frac{m}{\varepsilon} \right), \\ \mathbf{B}_2 &= \frac{\omega}{2\varepsilon} \left(\mathbf{n} - \mathbf{v} - \frac{\tau}{2} \dot{\mathbf{v}} + \mathbf{v} \frac{m}{\varepsilon} \right), \\ \mathbf{r}_2 - \mathbf{r}_1 &= \tau \mathbf{v} + \frac{\tau^3}{24} \ddot{\mathbf{v}} \end{aligned}$$

and substituting (90.28) into (90.21) and then into (90.10), we find the differential probability of the transition per unit time ($d\omega = dI/\hbar\omega$). It is integrated with the aid of formula

$$\int f(k_\mu) e^{-ikx} \frac{d^3 k}{\omega} = -f(i\partial_\mu) \frac{4\pi}{x_0^2 - \mathbf{x}^2}, \quad (\text{X.58})$$

where in the present case

$$x_0 = \tau, \quad \mathbf{x} = \mathbf{r}_2 - \mathbf{r}_1, \quad x^2 = x_0^2 - \mathbf{x}^2 = \tau^2 \left(\frac{m^2}{\varepsilon^2} + \frac{\tau^2 \omega_0^2}{12} \right).$$

The computation leads to the result

$$w = \frac{5\sqrt{3}\alpha}{16} \frac{\hbar^2}{m^2} \left(\frac{\varepsilon}{m} \right)^5 \omega_0^3 \left(1 - \frac{2}{9} \zeta_{\parallel}^2 - \frac{8\sqrt{3}}{15} \frac{\varepsilon}{|e|} \zeta_{\perp} \right), \quad (\text{X.59})$$

where $\zeta_{\parallel} = \boldsymbol{\zeta} \mathbf{v}$, $\zeta_{\perp} = \boldsymbol{\zeta} \mathbf{H}/H$. This formula is valid both for electrons ($e < 0$) and for positrons ($e > 0$).

The probability (90.30) does not depend on the sign of the longitudinal polarisation $\zeta_{\parallel}$, but does depend on the sign of $\zeta_{\perp}$. Therefore the polarisation arising as a result of the radiation is transverse. For electrons the probability of transition from the state with spin “along the field” ($\zeta_{\perp} = 1$) to the state with spin “against the field” is greater than the probability of the reverse transition. Therefore the radiative polarisation of electrons is directed against the field, and its degree in the stationary state (at $\zeta_{\parallel} = 0$) is

$$\frac{w(\zeta_{\perp} = -1) - w(\zeta_{\perp} = 1)}{w(\zeta_{\perp} = -1) + w(\zeta_{\perp} = 1)} = \frac{8\sqrt{3}}{15} = 0.92.$$

Positrons are polarised (to the same degree) in the direction of the field.

§ 91. Pair production by a photon in a magnetic field

The production of an electron-positron pair by a photon in a magnetic field and magnetobremstrahlung radiation are two crossed channels of one and the same reaction. Therefore the amplitude M_{fi} of the process of pair production is obtained from the amplitude of bremsstrahlung radiation simply by the substitutions

$$\varepsilon, \mathbf{p} \rightarrow -\varepsilon_+, -\mathbf{p}_+; \quad \varepsilon', \mathbf{p}' \rightarrow \varepsilon_-, -\mathbf{p}_-; \quad \omega, \mathbf{k} \rightarrow -\omega, -\mathbf{k} \quad (\text{X.60})$$

(here $\varepsilon_-, \mathbf{p}_-$ and $\varepsilon_+, \mathbf{p}_+$ are the energies and momenta of the electron and positron in the pair; $\varepsilon, \mathbf{p}$ and $\varepsilon', \mathbf{p}'$ are the initial and final energies and momenta of the electron in bremsstrahlung radiation)¹⁸. In terms of angles and absolute values the transformation of momenta is

$$|\mathbf{p}| \rightarrow |\mathbf{p}_+|, \quad |\mathbf{p}'| \rightarrow |\mathbf{p}_-|, \quad \theta \rightarrow \pi - \theta_+, \quad \theta' \rightarrow \theta_-, \quad \varphi \rightarrow \varphi - \pi, \quad (\text{X.61})$$

where $\theta_{\pm}$ are the angles between the momenta of the electron and positron and $\mathbf{k}$, and φ is the angle between the planes $\mathbf{k}, \mathbf{p}_+$ and $\mathbf{k}, \mathbf{p}_-$. In the case of bremsstrahlung radiation the cross section is expressed

through the amplitude by the formula¹⁹

$$d\sigma = |M_{fi}|^2 \frac{1}{8|\mathbf{p}|\varepsilon'\omega} \delta(\varepsilon - \varepsilon' - \omega) \frac{d^3 p' d^3 k}{(2\pi)^5} \quad (\text{X.62})$$

¹⁸This circumstance is, however, clear in advance: the axial vector of the arising polarisation can only be directed along the sole axial vector $\mathbf{H}$ figuring in the problem.

¹⁹In this section we again set not only $c = 1$, but also $\hbar = 1$.

(see (64.25)); the δ -function is removed by integration over ε' . Noting that

$$d^3 p' = |\mathbf{p}'| \varepsilon' d\varepsilon' d\mathbf{o}', \quad d^3 k = \omega^2 d\omega d\mathbf{o}_k,$$

we must simply replace

$$\delta(\varepsilon - \varepsilon' - \omega) d^3 p' d^3 k \rightarrow \omega^2 |\mathbf{p}'| \varepsilon' d\mathbf{o}_k d\mathbf{o}' d\omega.$$

Then

$$d\sigma = |M_{fi}|^2 \frac{\omega |\mathbf{p}'|}{8(2\pi)^5 |\mathbf{p}|} d\mathbf{o}_k d\mathbf{o}' d\omega. \quad (\text{X.63})$$

In the case of pair production by a photon the cross section is expressed through the amplitude as

$$d\sigma = |M_{fi}|^2 \frac{1}{8\omega\varepsilon_+\varepsilon_-} \delta(\omega - \varepsilon_+ - \varepsilon_-) \frac{d^3 p_+ d^3 p_-}{(2\pi)^5}$$

or, after elimination of the δ -function:

$$d\sigma = |M_{fi}|^2 \frac{|\mathbf{p}_+||\mathbf{p}_-|}{8(2\pi)^5 \omega} d\mathbf{o}_+ d\mathbf{o}_- d\varepsilon_+. \quad (\text{X.64})$$

Comparing with (91.4), we see that to obtain the cross section of pair production from the cross section of bremsstrahlung radiation one must carry out the substitution (91.1) and multiply the latter by

$$\frac{\mathbf{p}_+^2 d\varepsilon_+}{\omega^2 d\omega} \quad (\text{X.65})$$

and replace $d\mathbf{o}' d\mathbf{o}_k$ by $d\mathbf{o}_+ d\mathbf{o}_-$.

In the ultrarelativistic case ($\omega \gg m$)²⁰ this can be done in the formulas obtained in the preceding section. It is assumed that both particles of the pair are ultrarelativistic; it is easy to verify that in such a case all the approximations used in § 90 remain valid.

In particular, the probability of pair production by an unpolarised photon, summed over the spin projections of the electron and positron and integrated over the emission directions of the electron, is obtained by performing the substitution (91.1) in the formula (90.22) (more precisely, in the expression for dI/ω), where $d^3 k = \omega^2 d\omega d\mathbf{o}_n$ is replaced by $d^3 p_+$:

$$\begin{aligned} d\omega = & \frac{e^2}{4\pi^2} \frac{d^3 p_+}{\omega} \int_{-\infty}^{\infty} \left(\frac{m^2}{\varepsilon_+ \varepsilon_-} - \frac{\varepsilon_+^2 + \varepsilon_-^2}{4\varepsilon_-^2} \omega_{0+}^2 \tau^2 \right) \times \\ & \times \exp \left\{ i \frac{\omega \tau \varepsilon_+}{\varepsilon_-} \left(1 - \mathbf{n} \mathbf{v}_+ + \frac{\tau^2}{24} \omega_{0+}^2 \right) \right\} d\tau, \end{aligned} \quad (\text{X.66})$$

where $\omega_{0+} = |e|H/\varepsilon_+$; $\mathbf{n}$ is the unit vector in the direction of the photon momentum, lying in the plane perpendicular to the magnetic field. The integration is performed in the same way as in § 90, since (expression (91.7)) depends only on the angle between $\mathbf{n}$

²⁰More precisely, one should have $\omega \sin \vartheta \gg m$, where ϑ is the angle between $\mathbf{k}$ and $\mathbf{H}$; when $\vartheta = 0$ pairs are not produced at all. Below we set $\vartheta = \pi/2$.

and $\mathbf{v}_+$ —one can integrate indifferently over do_+ or over $do_{\mathbf{n}}$. The result can therefore be obtained directly by analogy with (90.23):

$$dw = \frac{m^2 e^2 d\varepsilon_+}{\sqrt{\pi} \omega^2} \left[\int_x^\infty \Phi(\xi) d\xi + \left(\frac{2}{x} - \kappa x^{1/2} \right) \Phi'(x) \right], \quad (\text{X.67})$$

where now

$$x = \left(\frac{m^3 \omega}{|e| H \varepsilon_+ \varepsilon_-} \right)^{2/3}, \quad \kappa = \frac{|e| H \omega}{m^3} \left(= \frac{\hbar^2 |e| H \omega}{m^3 c^5} \right). \quad (\text{X.68})$$

The total probability of pair production per unit time is obtained by integrating (91.8) over ε_+ (and, in view of the obvious symmetry with respect to ε_+ and $\varepsilon_- = \omega - \varepsilon_+$, it suffices to integrate from 0 to $\omega/2$ and then double the result). Performing the change of the integration variable from ε_+ to x and integrating the first term in (91.8) twice by parts, we obtain

$$w = -\frac{|e|^3 H}{m \kappa \sqrt{\pi}} \int_{(4/\kappa)^{2/3}}^\infty \frac{2(x^{3/2} + 1/\kappa)}{x^{11/4}(x^{3/2} - 4/\kappa)^{1/2}} \Phi'(x) dx \quad (\text{X.69})$$

(A. I. Nikishov, V. I. Ritus, 1967).

In the limiting case of weak fields ($\kappa \ll 1$) the essential values of x in the integral (91.10) are those near the lower limit. Since these values are large, one can use the asymptotic expression for the Airy function

$$\Phi(x) \approx \frac{1}{2x^{1/4}} \exp\left(-\frac{2}{3}x^{3/2}\right)$$

(see III, § b). Introducing the variable $y = x^{3/2} - 4/\kappa$ and setting $y = 0$ everywhere possible, we obtain in the result

$$w = \frac{3^{3/2} |e|^3 H}{2^{5/2} m} \exp\left(-\frac{8}{3\kappa}\right), \quad \kappa \ll 1. \quad (\text{X.70})$$

The exponential decrease of the probability when $\kappa \rightarrow 0$ corresponds to the impossibility of pair production in the classical limit.

In the opposite limiting case of strong fields ($\kappa \gg 1$) only the second term is essential in the integral (91.10), and it is determined by the region of values of x in which $x^{3/2} \sim 1/\kappa \ll 1$. In this region one can replace $\Phi'(x)$ by its value

$$\Phi'(0) = -\frac{3^{1/6} \Gamma(2/3)}{2\pi^{1/2}}.$$

Using the value of the integral

$$\int_1^\infty y^{-\nu} (y-1)^{\mu-1} dy = \frac{\Gamma(\nu-\mu) \Gamma(\mu)}{\Gamma(\nu)},$$

we obtain

$$w = \frac{3^{1/6} \cdot 5 \Gamma^2(2/3)}{2^{4/3} \cdot 7 \pi^{1/2} \Gamma(7/6)} \frac{|e|^3 H}{m \kappa^{1/3}} = 0.38 \frac{|e|^3 H}{m \kappa^{1/3}}, \quad \kappa \gg 1. \quad (\text{X.71})$$

The function $mw(x)/|e|^3 H$ has a maximum of 0.11 at $\kappa \approx 11$.

§ 92. Bremsstrahlung of an electron on a nucleus. Non-relativistic case

This and several of the following sections are devoted to the important phenomenon of *bremsstrahlung*, the radiation accompanying the collision of particles. We begin with the non-relativistic case, i.e. we consider the radiation produced during scattering of an electron in the Coulomb field of a stationary centre (*A. Sommerfeld*, 1931).

We start from formula (45.5) for the probability of dipole radiation

$$dw = \frac{\omega^3}{2\pi} |\mathbf{e}^* \mathbf{d}_{fi}|^2 d\mathbf{k}. \quad (\text{X.72})$$

In the present case the initial and final states of the electron belong to the continuous spectrum, and the photon frequency is

$$\omega = \frac{1}{2m} (\mathbf{p}^2 - \mathbf{p}'^2), \quad (\text{X.73})$$

where $\mathbf{p} = m\mathbf{v}$ and $\mathbf{p}' = m\mathbf{v}'$ are the initial and final momenta of the electron. If the initial and final electron wave functions are normalised to one particle in a volume $V = 1$, then expression

(92.1), multiplied by $d^3 p' / (2\pi)^3$ and divided by the density of the incident flux $v/V = v$, gives the cross section $d\sigma_{\mathbf{k}\mathbf{p}'}$ of the radiation of a photon $\mathbf{k}$ in the solid angle $d\mathbf{k}$ with the scattering of the electron into the interval of final-state momenta $d^3 p'$. Replacing the matrix element of the dipole moment $\mathbf{d} = e\mathbf{r}$ by the matrix element of momentum according to

$$\mathbf{d}_{fi} = -\frac{1}{i\omega} \frac{e}{m} \mathbf{p}_{fi},$$

we write the expression for the cross section in the form

$$d\sigma_{\mathbf{k}\mathbf{p}'} = \frac{\omega e^2}{(2\pi)^4 m p} |\mathbf{e}^* \mathbf{p}_{fi}|^2 d\mathbf{k} d^3 p', \quad (\text{X.74})$$

where

$$\mathbf{p}_{fi} = \int \psi_f^* \hat{\mathbf{p}} \psi_i d^3 x = -i \int \psi_f^* \nabla \psi_i d^3 x. \quad (\text{X.75})$$

The functions ψ_i and ψ_f must be taken as exact wave functions in the Coulomb field of attraction, those functions which asymptotically contain a plane wave and a spherical wave; in ψ_f the spherical wave must be converging, and in ψ_i —diverging (see III, § 136). These functions have the form

$$\begin{aligned} \psi_i &= A_i e^{i\mathbf{p}\mathbf{r}} F(i\nu, 1, i(pr - \mathbf{p}\mathbf{r})), & \nu &= \frac{Ze^2 m}{p}, \\ \psi_f &= A_f e^{i\mathbf{p}'\mathbf{r}} F(-i\nu', 1, -i(p'r + \mathbf{p}'\mathbf{r})), & \nu' &= \frac{Ze^2 m}{p'}, \end{aligned} \quad (\text{X.76})$$

with the normalisation coefficients

$$A_i = e^{\pi\nu/2} \Gamma(1 - i\nu), \quad A_f = e^{\pi\nu'/2} \Gamma(1 + i\nu'). \quad (\text{X.77})$$

Noting that

$$\nabla F(iv, 1, i(pr - \mathbf{p}\mathbf{r})) = i \left(p \frac{\mathbf{r}}{r} - \mathbf{p} \right) F' = - \left(\frac{\partial F}{\partial \mathbf{p}} \right),$$

we write $\nabla \psi_i$ in the form

$$\nabla \psi_i = i \mathbf{p} \psi_i - A_i e^{i \mathbf{p} \mathbf{r}} \frac{p}{r} \left(\frac{\partial F}{\partial \mathbf{p}} \right).$$

Upon multiplying by ψ_f^* and integrating, the first term vanishes by virtue of the orthogonality of ψ_i and ψ_f . Therefore for the matrix element of momentum we have

$$\mathbf{p}_{fi} = i A_i A_f p \frac{\partial J}{\partial \mathbf{p}}, \quad (\text{X.78})$$

where

$$J = \int \frac{e^{-i \mathbf{q} \mathbf{r}}}{r} F(iv', 1, i(p'r + \mathbf{p}'\mathbf{r})) F(iv, 1, i(pr - \mathbf{p}\mathbf{r})) d^3x, \quad \mathbf{q} = \mathbf{p}' - \mathbf{p}. \quad (\text{X.79})$$

We have taken $\partial/\partial \mathbf{p}$ out from under the integral sign, bearing in mind that upon differentiating J the quantities $\nu, \nu', \mathbf{q}$ must be treated as independent parameters and only after the differentiation should one express ν and $\mathbf{q}$ through $\mathbf{p}$.

The integral is computed by replacing each of the degenerate hypergeometric functions by its expression in the form of a contour integral. We give here only the result:

$$\begin{aligned} J &= BF(iv', iv, 1, z), \\ B &= 4\pi e^{-\pi\nu} (-\mathbf{q}^2 - 2\mathbf{q}\mathbf{p})^{-i\nu} (\mathbf{q}^2 - 2\mathbf{q}\mathbf{p}')^{-iv'} (\mathbf{q}^2)^{i\nu+iv'-1}, \\ z &= 2 \frac{\mathbf{q}^2(p p' + \mathbf{p}\mathbf{p}') - 2(\mathbf{q}\mathbf{p})(\mathbf{q}\mathbf{p}')}{(\mathbf{q}^2 - 2\mathbf{q}\mathbf{p}')(\mathbf{q}^2 + 2\mathbf{q}\mathbf{p})}. \end{aligned} \quad (\text{X.80})$$

Here $F(iv', iv, 1, z)$ is the complete hypergeometric function.

After the differentiation in (92.7) one can set $\mathbf{q} = \mathbf{p}' - \mathbf{p}$; then

$$z = -2 \frac{p p' - \mathbf{p}\mathbf{p}'}{(p - p')^2}, \quad \mathbf{q}^2 = (p - p')^2(1 - z) \quad (\text{X.81})$$

($z < 0$). We note also that

$$-\mathbf{q}^2 - 2\mathbf{q}\mathbf{p} = \mathbf{q}^2 - 2\mathbf{q}\mathbf{p}' = \mathbf{p}^2 - \mathbf{p}'^2 > 0.$$

As a result we find for the matrix element the following final expression:

$$\begin{aligned} \mathbf{p}_{fi} &= A_i A_f \frac{8\pi i e^{-\pi\nu}}{(p - p')^3(p + p')} \left(\frac{p + p'}{p - p'} \right)^{-i(\nu+\nu')} \times \\ &\times (1 - z)^{i(\nu+\nu')-1} [i\nu p \mathbf{q} F'(z) + (1 - z) F'(z)(p'\mathbf{p} - p\mathbf{p}')], \end{aligned} \quad (\text{X.82})$$

where for brevity we have denoted

$$F(z) = F(iv', iv, 1, z). \quad (\text{X.83})$$

The cross section is obtained by substitution of (92.10) into (92.3), but the general formula is very cumbersome and difficult to survey. Therefore we proceed directly to computing the spectral distribution of the radiation, i.e. we integrate the cross section over the directions of the photon and the final electron.

Integration over $do_{\mathbf{k}}$ and summation over the polarisations of the photon reduces to averaging over all directions of $\mathbf{e}$ and multiplication by $2 \cdot 4\pi$, i.e. to the replacement

$$e_i e_k^* do_{\mathbf{k}} \rightarrow \frac{8\pi}{3} \delta_{ik}.$$

After this the cross section becomes

$$d\sigma_{\mathbf{p}'} = \frac{\omega e^2 p'}{6\pi^3 p} |\mathbf{p}_{fi}|^2 d\omega do_{\mathbf{p}'}. \quad (\text{X.84})$$

The square $|\mathbf{p}_{fi}|^2$ is computed using (92.9)–(92.11), and taking into account that

$$|\Gamma(1 - i\nu)|^2 = \frac{\pi\nu}{\sinh \pi\nu}.$$

We obtain

$$|\mathbf{p}_{fi}|^2 = \frac{2^8 \pi^4 (Ze^2)^2 m^2}{(p + p')^2 (p - p')^4 (1 - e^{-2\pi\nu})(e^{2\pi\nu} - 1)} \times \\ \times \left\{ \frac{\nu\nu'}{1 - z} |F|^2 - z|F'|^2 + i \frac{\nu + \nu'}{2} \frac{z}{1 - z} (FF'^* - F^*F') \right\}. \quad (\text{X.85})$$

For integrating the cross section (92.12), $do_{\mathbf{p}'} = 2\pi \sin \vartheta d\vartheta$ where ϑ is the scattering angle, we pass from ϑ to the variable

$$z = -\frac{2pp'}{(p - p')^2} (1 - \cos \vartheta), \quad do_{\mathbf{p}'} \rightarrow \frac{\pi(p - p')^2}{pp'} dz.$$

To take the integral over dz , we transform the expression in curly brackets in (92.13). According to the differential equation for hypergeometric functions (see III, (e. 2)) we have

$$z(1 - z)F'' + [1 - (1 + i\nu + i\nu')z]F' + \nu\nu'F = 0, \\ z(1 - z)F''^* + [1 - (1 - i\nu - i\nu')z]F'^* + \nu\nu'F^* = 0.$$

Multiplying these two equations by F^* and F respectively and adding, we obtain

$$(1 - z) \left[\frac{d}{dz} z(F'F^* + F'^*F) - 2z|F'|^2 + \frac{i(\nu + \nu')z}{1 - z} (F'^*F + F^*F') + \frac{2\nu\nu'}{1 - z} |F|^2 \right] = 0.$$

From this one sees that the expression in curly brackets in (92.13) equals

$$\{\dots\} = -\frac{1}{2} \frac{d}{dz} z(F'F^* + F'^*F) \quad (\text{X.86})$$

and integrates directly.

Collecting the obtained formulas, we find the final expression for the cross section of bremsstrahlung radiation in the frequency interval $d\omega$ ²¹:

$$d\sigma_\omega = \frac{64\pi^2}{3} Z^2 \alpha r_e^2 \frac{m^2 c^2}{(p - p')^2} \frac{p'}{p} \frac{1}{(1 - e^{-2\pi\nu'})(e^{2\pi\nu} - 1)} \times \left(-\frac{d}{d\xi} |F(\xi)|^2 \right) \frac{d\omega}{\omega}, \quad (\text{X.87})$$

where

$$\nu = \frac{Z\alpha mc}{p} = \frac{Ze^2}{\hbar v}, \quad \nu' = \frac{Ze^2}{\hbar v'}, \quad p' = \sqrt{p^2 - 2m\hbar\omega},$$

$$F(\xi) = F(i\nu', i\nu, 1, \xi), \quad \xi = -\frac{4pp'}{(p - p')^2}.$$

Let us consider the limiting case when both velocities ν and ν' are so large that $\nu \ll 1$, $\nu' \ll 1$ (but still, of course, $\nu \ll 1$, so that $Z\alpha \ll \nu \ll 1$; this is possible only for small Z). For computing the derivative $F'(\xi)$ in this case we use the formula

$$\frac{d}{dz} F(\alpha, \beta, \gamma, z) = \frac{\alpha\beta}{\gamma} F(\alpha + 1, \beta + 1, \gamma + 1, z),$$

which is easily obtained by direct differentiation of the hypergeometric series. We have

$$F'(\xi) = i\nu \cdot i\nu' F(1, 1, 2, \xi) = \frac{i\nu\nu'}{\xi} \ln(1 - \xi)$$

(the last equality follows from direct comparison of the corresponding series). For the function $F(\xi)$ itself we have simply

$$F(\xi) \approx F(0, 0, 1, \xi) = 1.$$

As a result, from (92.15) we find

$$d\sigma_\omega = \frac{16}{3} Z^2 \alpha r_e^2 \frac{c^2}{v^2} \ln \frac{v + v'}{v - v'} \frac{d\omega}{\omega}, \quad \frac{Ze^2}{\hbar v} \ll 1, \quad \frac{Ze^2}{\hbar v'} \ll 1. \quad (\text{X.88})$$

The smallness of ν and ν' is precisely the condition for applicability of the Born approximation in the case of Coulomb interaction. Therefore formula (92.16) itself can be more simply obtained directly with the aid of perturbation theory (see Problem 1).

Now let the fast ($\nu \ll 1$) electron lose a considerable fraction of its energy to radiation, so that $\nu' \ll \nu$ and ν' may not be small. Then

$$-\xi \approx 4p'/p = 4\nu/\nu' \ll 1, \quad F(\xi) \approx F(i\nu', 0, 1, \xi) = 1,$$

$$F'(\xi) = -\nu\nu' F(1 + i\nu', 1, 2, \xi) \approx -\nu\nu',$$

and the cross section

$$d\sigma_\omega = \frac{64\pi^3}{3} Z^3 \alpha^2 r_e^2 \left(\frac{c}{v} \right)^3 \frac{1}{1 - \exp(-2\pi Ze^2/\hbar v')} \frac{d\omega}{\omega}, \quad (\text{X.89})$$

²¹Formulas (92.15) and (92.25) are written in ordinary units.

$$\frac{Ze^2}{\hbar v} \ll 1, \quad \frac{Ze^2}{\hbar v'} \gtrsim 1.$$

When $v' \ll 1$ this formula gives the same limiting expression

$$d\sigma_\omega = \frac{32}{3} Z^2 \alpha r_e^2 \frac{c^2 v'}{v^3} \frac{d\omega}{\omega} \quad (\text{X.90})$$

as formula (92.16) when $v' \ll v$. Therefore formulas (92.16) and (92.17) together cover (for $v \ll 1$) the entire range of values of v' .

When $\omega \rightarrow \omega_0$ (where $\hbar\omega_0 = mv^2/2$) the velocity $v' \rightarrow 0$ and $v' \rightarrow \infty$. In this limit (92.17) gives

$$d\sigma_\omega = \frac{128\pi}{3} Z^3 \alpha^2 r_e^2 \left(\frac{c}{v}\right)^3 \frac{\hbar d\omega}{mv^2}. \quad (\text{X.91})$$

Thus, $d\sigma_\omega/d\omega$ tends to a finite limit as $\omega \rightarrow \omega_0$. This circumstance can be justified in general terms by arguments analogous to those set out in vol. III, § 147. Physically it is related to the fact that the frequency $\omega = \omega_0$ is the boundary of a continuous bremsstrahlung spectrum only. The electron can also radiate at the frequency $\omega > \omega_0$, passing into a bound state. But strongly excited bound states in the Coulomb field are so similar in their properties to states near the boundary of the free spectrum that the continuous spectrum boundary is not a physically distinguished point.

Let us pass to the case when both parameters $v, v' \gg 1$. In this case the motion of both the initial and the final electrons is quasi-classical. We shall assume that $p^2/(2m) \sim \hbar\omega$; then we shall need the asymptotic expression for the function $F(\xi)$ when $v, v' \rightarrow \infty$ and $\xi \sim 1$ (the more precise condition will be formulated below, see (92.24)).

For obtaining this expression we start from the integral representation of the hypergeometric function (e. 3) (see III), which we write in the form

$$F(i\eta v', iv', 1, \xi) = \frac{e^{-\pi\eta v'}}{2\pi i} \oint e^{v' f(t)} \frac{dt}{t}, \quad (\text{X.92})$$

where we have introduced the notation

$$\eta = v/v', \quad 0 < \eta < 1, \quad (\text{X.93})$$

so that

$$\xi = -\frac{4\eta}{(1-\eta)^2}. \quad (\text{X.94})$$

As integration contour we choose the path shown in Fig. 17, passing along a segment of the real axis and going around the points $t = 0$ and $t = 1$ ²².

When $v, v' \gg 1$ the value of the sub-integral expression on the lower part of this contour is small and can be neglected: upon going around the point $t = 0$ from above downwards the sub-integral expression is multiplied by the small factor $\exp(-2\pi\eta v')$,

²²For the hypergeometric function $F(\alpha, \beta, \gamma, \xi)$ the contour must be chosen so that upon going around it the function $V(t) = e^t t^{\gamma-\beta-1} (t-1)^{1-\alpha}$ returns to its original value. When γ is integer (in the present case $\gamma = 1$) the indicated contour satisfies this condition.

while upon going around the point $t = 1$ from below upwards it is multiplied by $\exp(2\pi\eta\nu')$. The integral

$$E = \frac{e^{-\pi\eta\nu'}}{2\pi i} \int e^{\nu' f(t)} \frac{dt}{t}, \quad f(t) = i \ln \frac{t^\eta}{(1-t)^\eta (1-t\xi)^{i\nu'}},$$

is evaluated by the saddle-point method. The saddle point t_0 is determined by the condition $f'(t_0) = 0$, whence $t_0 = (1-\eta)/2$. At this point, however, f'' also vanishes, as does the derivative $f'''(t_0)$, so that one must write

$$f(t) \approx f(t_0) + \frac{ia}{3} \tau^3, \quad \tau = t - t_0,$$

where

$$f(t_0) = 2\pi\eta i + i(1+\eta) \ln \frac{1-\eta}{1+\eta}, \quad a = \frac{1}{2} f'''(t_0) = \frac{16\eta}{(1-\eta^2)^2}.$$

The pre-exponential factor $1/t$ in the sub-integral expression is written as

$$\frac{1}{t} \approx \frac{1}{t_0} - \frac{\tau}{t_0^2}$$

(one cannot restrict oneself to the term $1/t_0$ alone, as this would lead to the vanishing of the derivative $d|F(\xi)|^2/d\xi$ appearing in (92.15)). Thus, we find, after an obvious substitution, in the integrals,

$$F \approx \frac{1}{2\pi i t_0 (\alpha\nu')^{1/3}} \exp\{-\pi\eta\nu' + \nu' f(t_0)\} \times \\ \times \left[- \int_{-\infty}^{\infty} e^{ix^3/3} dx + \frac{i}{t_0 (\alpha\nu')^{1/3}} \int_{-\infty}^{\infty} x e^{ix^3/3} dx \right]. \quad (\text{X.95})$$

The integrals here are respectively

$$2 \int_0^{\infty} \cos \frac{x^3}{3} dx = \frac{2\pi}{3^{2/3} \Gamma(2/3)}, \quad 2 \int_0^{\infty} x \sin \frac{x^3}{3} dx = 3^{1/6} \Gamma(2/3).$$

In an analogous fashion the derivative $F'(\xi)$ is computed; according to (92.19) it is given by an integral differing from (92.21) only by the

replacement of the pre-exponential factor $1/it$ by $\nu'/(l-\xi t)$. After this straightforward calculation we arrive at the result

$$-\frac{d}{d\xi} |F(\xi)|^2 = \frac{(1-\eta)^2 e^{2\pi\nu}}{4\sqrt{3}\pi\nu}.$$

Finally, substituting this expression into formula (92.15), we find, to the required accuracy, the following simple result:

$$d\sigma_\omega = \frac{16\pi}{3^{3/2}} Z^2 \alpha r_e^2 \frac{m^2 c^2}{p^2} \frac{d\omega}{\omega}. \quad (\text{X.96})$$

The condition for applicability of this formula, i.e. the condition for applicability of the asymptotic expression (92.23), consists in requiring the smallness of the second term

in comparison with the first: $(1 - \eta)v \gg 1$, or, after expressing the parameters of the hypergeometric function through the physical quantities:

$$\omega \gg \frac{v}{Ze^2} \frac{mv^2}{2}. \quad (\text{X.97})$$

Condition (92.24) coincides with that defining the “high-frequency limit” in the classical radiation in the Coulomb field of attraction, and the quantity $\hbar\omega d\sigma_\omega$ from (92.23) coincides with the expression (70.22) (see II) for the “effective retardation” in this limit. This result needs some discussion. It might seem that for the applicability of the classical radiation formula one needs, besides the quasi-classicality of the motion, also the smallness of the energy of the quantum compared to the energy of the electron, i.e. the condition $\hbar\omega \ll mv^2/2$, which was not assumed in the derivation of (92.23). In reality, however, the value $\hbar\omega$ need not be small compared with the energy of the electron at infinity, but rather compared with its kinetic energy on that segment of the trajectory where the radiation occurs in the main. This energy is much larger than the initial one, because of the acceleration of the electron by the field of the ion.

Indeed, the radiation of high frequencies occurs mainly at small distances from the ion, where

$$v(r)/r \sim \omega. \quad (\text{X.98})$$

(We have denoted by $v(r)$ the velocity of the electron at distance r from the ion, in contrast to the velocity v at infinity.) Taking into account that here $Ze^2/r \sim mv^2(r)$, we find that the kinetic energy on the segment where the radiation is produced is

$$\frac{mv^2(r)}{2} \sim \frac{m}{2} \left(\frac{\omega Ze^2}{m} \right)^{2/3} \gg \frac{mv^2}{2}.$$

Therefore the radiation of a quantum with energy of order $mv^2/2$ does not change the motion on the radiation segment, and no additional condition of smallness of $\hbar\omega$ is required.

We note also that the motion on the segment (92.25) for given angular momentum $\hbar l$ does not depend on the initial energy. Correspondingly, the energy and $d\sigma_\omega$ from (92.23) can be obtained by multiplying the probability of radiation $d\mathcal{E}_\omega/\hbar\omega$ by $2\pi\rho d\rho$ (ρ is the impact parameter) and integrating over all ρ . Since in the quasi-classical case

$$\rho d\rho = \hbar^2 l dl / (m^2 v^2),$$

this leads to the dependence $d\sigma_\omega \sim 1/v^2$, i.e. to (92.23). The above argument clarifies why it is the initial (and not the final) velocity of the electron that enters this formula.

In order to go over to the classical formulas in the entire region $(1 - \eta)v \sim 1$, $v \gg 1$, one would have to find the asymptotics of the hypergeometric function near the saddle point t_0 at closer range; we shall not pursue this here in view of the evident nature of the final result.

All the formulas written above refer to the Coulomb field of attraction. The cross section for radiation in the field of repulsion is obtained from (92.15) by the substitution:

$\nu \rightarrow -\nu$, $\nu' \rightarrow -\nu'$. In this case, in particular, the limiting Born formula (92.16) does not change at all. In the limit $\nu \ll 1$, $\nu' \rightarrow \infty$ we obtain instead of (92.18)

$$d\sigma_\omega = \frac{128\pi}{3} Z^3 \alpha^2 r_e^2 \left(\frac{c}{v}\right)^3 \exp\left(-\frac{\sqrt{2mc^2} \pi Z \alpha}{\sqrt{\hbar(\omega_0 - \omega)}}\right) \frac{\hbar d\omega}{mv^2}, \quad (\text{X.99})$$

i.e. the differential cross section tends to zero exponentially as $\omega \rightarrow \omega_0$. This result is again natural, since in the field of repulsion there are no bound states and the frequency $\omega = \omega_0$ is the true boundary of the radiation spectrum.

Problems

1. Find in the Born approximation the cross section of bremsstrahlung radiation in the non-relativistic collision of two particles with different ratios e/m .

Solution. The dipole moment of two particles with charges e_1 , e_2 and masses m_1 , m_2 in the system of their centre of inertia is

$$\mathbf{d} = \mu \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right) \mathbf{r},$$

where $\mu = m_1 m_2 / (m_1 + m_2)$, $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. Hence

$$\dot{\mathbf{d}} = \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right) \mu \dot{\mathbf{r}} = - \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right) \frac{e_1 e_2}{r} \nabla \frac{1}{r}.$$

The matrix element

$$\mathbf{d}_{\mathbf{p}'\mathbf{p}} = -\frac{1}{i\omega^2} (\dot{\mathbf{d}})_{\mathbf{p}'\mathbf{p}}, \quad \omega = \frac{p^2 - p'^2}{2\mu}$$

($\mathbf{p} = \mu \mathbf{v}$, $\mathbf{p}' = \mu \mathbf{v}'$ are the momenta of relative motion) is computed using plane waves²³

$$\psi_{\mathbf{p}} = e^{i\mathbf{p}\mathbf{r}}, \quad \psi_{\mathbf{p}'} = e^{i\mathbf{p}'\mathbf{r}}$$

with the aid of the formula

$$\left(\nabla \frac{1}{r} \right)_{\mathbf{p}'\mathbf{p}} = \frac{4\pi i \mathbf{q}}{q^2}, \quad \mathbf{q} = \mathbf{p}' - \mathbf{p}.$$

As a result we find

$$d\sigma_{\mathbf{k}\mathbf{p}'} = \frac{e_1^2 e_2^2}{\pi^2} \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right)^2 \frac{v'}{v} \frac{\mu^2}{q^4} (\mathbf{e}\mathbf{q})(\mathbf{e}^*\mathbf{q}) \frac{d\omega}{\omega} d\mathbf{o}_{\mathbf{k}} d\mathbf{o}_{\mathbf{p}'}.$$

After summation over the polarisations, the angular distribution of the radiation is described by a factor $\sin^2 \Theta$, where Θ is the angle between the photon $\mathbf{k}$ and the vector $\mathbf{q}$, lying in the scattering plane (see (45.4a)). After integration over the photon directions

$$d\sigma_{\vartheta\omega} = \frac{16}{3} e_1^2 e_2^2 \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right)^2 \frac{v'}{v} \frac{d\omega}{\omega} \frac{\sin \theta d\theta}{v^2 + v'^2 - 2vv' \cos \theta'},$$

²³The replacement of two particles with the reduced mass is admissible, of course, only in the non-relativistic case.

where θ is the scattering angle. Finally, integrating over θ ,

$$d\sigma_{\omega} = \frac{16}{3} e_1^2 e_2^2 \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right)^2 \frac{1}{v^2} \ln \frac{v + v'}{v - v'} \frac{d\omega}{\omega}.$$

For radiation in the field of a stationary Coulomb centre this formula coincides with (92.16).

2. Find in the Born approximation the cross section of bremsstrahlung radiation in the non-relativistic collision of two electrons²⁴.

Solution. Dipole radiation in this case is absent, and one must consider quadrupole radiation. In classical theory the spectral distribution of the total intensity of quadrupole radiation is given by the formula

$$I_{\omega} = (1/90) |\hat{D}_{ik})_{\omega}|^2,$$

where $D_{ik} = \sum e(3x_i x_k - r^2 \delta_{ik})$ is the tensor of the quadrupole moment of the system of charges²⁵. For two electrons in the system of their centre of inertia

$$D_{ik} = e(3x_i x_k - r^2 \delta_{ik}), \quad \mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2.$$

In the transition to quantum theory the Fourier components must be replaced by matrix elements (cf. what was said in §45 on dipole radiation), and with the appropriate normalisation of the wave functions (plane waves) one obtains, after division by the photon energy ω , the cross section of radiation with the scattering of the electrons into the interval of final states $d^3 p'$:

$$d\sigma_{\mathbf{p}'} = \frac{1}{90\omega} |(\hat{D}_{ik})_{\mathbf{p}\mathbf{p}'}|^2 \frac{d^3 p'}{v(2\pi)^3}.$$

$v = 2p/m$ is the initial velocity of relative motion; the emitted frequency $\omega = (p^2 - p'^2)/m$.

The operator $\hat{D}_{ik}$ is computed by threefold commutation of D_{ik} with the Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{m} + \frac{e^2}{r}$$

and is²⁶

$$\hat{\hat{D}}_{ik} = \frac{2e^3}{m} \left[6 \left(\frac{x_i}{r^3} \hat{p}_k + \hat{p}_k \frac{x_i}{r^3} \right) + 6 \left(\frac{x_k}{r^3} \hat{p}_i + \hat{p}_i \frac{x_k}{r^3} \right) - 9 \left(\frac{x_i x_k x_l}{r^5} \hat{p}_l + \hat{p}_l \frac{x_i x_k x_l}{r^5} \right) - \delta_{ik} \left(\frac{x_l}{r^3} \hat{p}_l + \hat{p}_l \frac{x_l}{r^3} \right) \right]$$

Taking into account the identity of both particles (electrons) the matrix elements are computed using the wave functions

$$\psi_{\mathbf{p}} = \frac{1}{\sqrt{2}} (e^{i\mathbf{p}\mathbf{r}} \pm e^{-i\mathbf{p}\mathbf{r}}), \quad \psi_{\mathbf{p}'} = \frac{1}{\sqrt{2}} (e^{i\mathbf{p}'\mathbf{r}} \pm e^{-i\mathbf{p}'\mathbf{r}}),$$

²⁴The velocity of the collision satisfies the condition $\alpha \ll e^2/(\hbar v) \ll 1$.

²⁵This formula is obtained from (71.5) (see II), just as (67.11) (see II) is obtained from (67.8) (see II).

²⁶This expression is analogous to the classical formula $\ddot{D}_{ik} = \frac{4e^3}{m} \left[6 \frac{x_i x_k}{r^3} \hat{p}_k + 6 \frac{x_k x_i}{r^3} \hat{p}_i - 9 \frac{x_i x_k x_l}{r^5} \mathbf{p} \cdot \mathbf{r} - \frac{1}{r^3} \delta_{ik} \mathbf{p} \cdot \mathbf{r} \right]$, which would be obtained by differentiating D_{ik} using the classical equation of motion $\frac{m}{2} \ddot{\mathbf{r}} = \frac{e^2}{r^3} \mathbf{r}$.

where the signs “+” and “-” correspond respectively to total spins of the electrons 0 and 1 (the permutation of the electrons corresponds to the substitution $\mathbf{r} \rightarrow -\mathbf{r}$).

The cumbersome calculation leads to the following formula for the spectral distribution of the radiation:

$$d\sigma_\omega = \frac{4}{10} \alpha r_e^2 \left\{ 17 - \frac{3x^2}{(2-x)^2} + \frac{12(2-x)^4 - 7(2-x)^2 x^2 - 3x^4}{(2-x)^3 \sqrt{1-x}} \operatorname{Arcth} \frac{1}{\sqrt{x}} \right\} \sqrt{1-x} dx,$$

where $x = \omega/\varepsilon$, and $\varepsilon = p^2/m$ is the initial energy of relative motion of the electrons; the cross section is averaged over the values of the total spin. The effective retardation is

$$\kappa_{\text{eff}} = \hbar \int_0^\varepsilon \omega d\sigma_\omega = 8.1 \alpha r_e^2 \varepsilon$$

(B. K. Fedyushin, 1952).

3. Determine the energy of radiation arising upon emission of a non-relativistic electron by a nucleus in the s -state.

Solution. The wave function of the emitted electron—a diverging spherical s -wave, normalised to unit full flux:

$$\psi_i = \frac{1}{\sqrt{4\pi v}} \frac{e^{ipr}}{r}$$

(see III, (33.14)). As the wave function of the final (after emission of the photon) state of the electron we take a plane wave

$$\psi_f = e^{i\mathbf{p}'\mathbf{r}}.$$

The matrix element of the transition

$$\begin{aligned} \mathbf{p}_{fi} &= (\mathbf{p}_{fi})^* = \left(\int \psi_f^* \hat{\mathbf{p}} \psi_i d^3x \right)^* = \frac{\mathbf{p}'}{\sqrt{4\pi v}} \int e^{-i\mathbf{p}'\mathbf{r} + ipr} \frac{d^3x}{r} \\ &= \sqrt{\frac{\pi}{v}} \frac{\mathbf{p}'}{p^2 - p'^2} = \sqrt{\frac{\pi}{v}} \frac{v'}{v \omega} \end{aligned}$$

(the integral is evaluated according to (57.6a)). The energy of the radiation is obtained from formula (45.8), multiplied by $d^3p'/(2\pi)^3$ and integrated over the directions of $\mathbf{p}'$. As a result we obtain the spectral distribution of the radiated energy

$$dE_\omega = \frac{2e^2 v'^3}{3\pi v} d\omega.$$

When $\omega \rightarrow 0$ the final velocity of the electron $v' \rightarrow v$, and this formula coincides, as it should, with the non-relativistic limit of the classical result (see the problem in II, § 69). The total radiated energy (in ordinary units)

$$E = \frac{4}{15\pi} \alpha \left(\frac{v}{c} \right)^2 \varepsilon,$$

where $\varepsilon = mv^2/2$ is the initial energy of the electron.

4. Determine the energy of radiation arising upon the reflection of a non-relativistic electron from an infinitely high “potential wall”.

Solution. Let the electron move normally to the wall. Although the photon may be emitted in any direction, in the non-relativistic case the photon impulse is small in comparison with the electron impulse, and one can consider that the reflected electron also moves normally to the plane.

Let the wall be at $x = 0$, and the electron be on the side $x > 0$. The wave functions of the stationary states of one-dimensional motion, normalised to $\delta(p/2\pi)$ (see III, § 21):

$$\psi_i = 2 \sin px, \quad \psi_f = 2 \sin p'x.$$

The matrix element of the operator $\hat{p} = \hat{p}_x$:

$$p_{fi} = -4i \int_0^\infty \sin p'x \frac{d}{dx} \sin px \, dx = -\frac{4ipp'}{p^2 - p'^2},$$

(the integrals of this kind are to be understood as the limit when $\delta \rightarrow +0$ of the value obtained by introducing in the sub-integral expression the factor $e^{-\delta x}$).

The energy radiated upon single reflection of the electron is obtained from (45.8) by multiplication by $d\omega/v'$ and division by $v/2$ (the flux density of the running wave in the initial wave function ψ_i):

$$dE_\omega = \frac{4\omega^2 e^2}{3m^2} |p_{fi}|^2 \frac{2\pi d\omega}{vv'} = \frac{8}{3\pi} e^2 v v' d\omega. \quad (1)$$

At small frequencies ($\omega \ll \varepsilon = mv^2/2$) we have $v' \approx v$ and (1) goes over into the classical formula (69.5) (see II), which must be integrated over angles and divided by the change in velocity $\Delta v = 2v$ upon reflection; and it should be, since from the reflection from the wall the condition of smallness of the collision time (69.1) (see II) is always satisfied.

The quantum formula (1) allows, however, also to find the total radiated energy:

$$E = \int \frac{dE_\omega}{d\omega} d\omega = \frac{16}{9\pi} \alpha \varepsilon \frac{v^2}{c^2}$$

(in ordinary units).

5. Determine the energy of bremsstrahlung radiation arising upon scattering of a slow electron on an atom.

Solution. When the condition $pa \ll 1$ (where a is the atomic dimensions) is satisfied, the scattering on the atom is isotropic and does not depend on the energy of the electron (see III, § 132). The wave functions of the initial and final states of the electron are written in the form

$$\psi_i = e^{i\mathbf{p}\mathbf{r}} + f \frac{e^{ipr}}{r}, \quad \psi_f = e^{i\mathbf{p}'\mathbf{r}} + f \frac{e^{-ip'r}}{r},$$

where f is a real constant scattering amplitude. These expressions refer to the asymptotic region of distances $r \gg a$, which in the present case are just the essential ones:

$r \sim 1/p \gg a$. The computed matrix element from these functions is

$$\mathbf{p}_{fi} = \frac{2\pi f}{\omega}(\mathbf{v} - \mathbf{v}')$$

(the integrals are evaluated as in Problem 3). Substituting this expression into (92.12), we obtain the cross section of radiation with scattering of the electron in the direction $\mathbf{p}'$ (ordinary units):

$$d\sigma_{\omega\mathbf{p}'} = \frac{2\alpha p'}{3\pi pc^2}(\mathbf{v} - \mathbf{v}')^2 d\sigma_{\text{el}} \frac{d\omega}{\omega}, \quad (1)$$

where $d\sigma_{\text{el}} = f^2 d\sigma_{\mathbf{p}'}$ is the differential cross section of elastic scattering. When $\hbar\omega \ll p^2/(2m)$ one can set $p \approx p'$, and then this formula goes over, in the non-relativistic limit, into the formula for radiation of soft photons (see § 98)²⁷.

Integrating (1) over the directions of $\mathbf{p}'$, we get

$$d\sigma_{\omega} = \frac{2\alpha p'}{3\pi pc^2}(v^2 + v'^2)\sigma_{\text{el}} \frac{d\omega}{\omega}. \quad (2)$$

where $\sigma_{\text{el}} = 4\pi f^2$ is the total elastic scattering cross section. Finally, multiplying by $\hbar\omega$ and integrating over ω from 0 to $p^2/(2m) = \varepsilon$, we obtain the “effective retardation”

$$\kappa_{\text{eff}} = \int \hbar\omega d\sigma_{\omega} = \frac{32}{45\pi}\alpha\sigma_{\text{el}}\varepsilon \left(\frac{v}{c}\right)^2. \quad (3)$$

§ 93. Bremsstrahlung of an electron on a nucleus. Relativistic case

Let us turn to the bremsstrahlung radiation of an electron on a nucleus in the case of relativistic electron velocities²⁸. Here we shall again assume the Born approximation condition to be satisfied, i.e. for both the initial (v) and the final (v') velocities of the electron: $Ze^2/\hbar v \ll 1$, $Ze^2/\hbar v' \ll 1$. At the same time the charge of the nucleus must not be too large: $Z\alpha \ll 1$.

As in the preceding section, we neglect the recoil of the nucleus, so that it serves merely as the source of the external field (for the justification of such a neglect see § 97).

According to (91.4) the bremsstrahlung cross section is expressed through the amplitude by the formula

$$d\sigma = |M_{fi}|^2 \frac{\omega|\mathbf{p}'|}{8(2\pi)^5|\mathbf{p}|} d\mathbf{k} d\mathbf{o}' d\omega. \quad (\text{X.100})$$

In the first non-vanishing approximation the matrix element M_{fi} corresponds to two diagrams:

The free end q corresponds to the external field, so that $q = p' - p + k$ is the 4-vector of the momentum transfer to the nucleus. The neglect of the recoil means that the time component $q^0 = 0$.

²⁷The fact that “factorisation” of the cross section (i.e. the extraction of the factor σ_{el}) has occurred in the present case for arbitrary ω is, in a certain sense, accidental and is related to the independence of the scattering amplitude from energy.

²⁸The greater part of the results presented below was obtained by *Bethe* and *Heitler* (*H. A. Bethe, W. Heitler*, 1934) and independently by *Sauter* (*F. Sauter*, 1934).

According to the diagrams (93.2) we have

$$M_{fi} = -e^2 A_0^{(e)}(\mathbf{q}) \sqrt{4\pi} e_\nu^* \bar{u}' \left(\gamma^\mu \frac{\gamma f' + m}{f'^2 - m^2} \gamma^\nu + \gamma^\nu \frac{\gamma f + m}{f^2 - m^2} \gamma^\mu \right) u. \quad (\text{X.3})$$

The intermediate 4-momenta $f = p - k$, $f' = p' + k$; we introduce the notation:

$$f^2 - m^2 = -2kp \equiv -2\kappa\omega, \quad f'^2 - m^2 = -2kp' \equiv -2\kappa'\omega. \quad (\text{X.4})$$

$A_0^{(e)}$ is the scalar potential of the external field; for a purely Coulomb field

$$A_0^{(e)}(\mathbf{q}) = \frac{4\pi Ze}{\mathbf{q}^2}. \quad (\text{X.5})$$

Substituting into (93.1), we have for the cross section

$$d\sigma = \frac{Z^2 e^6}{4\pi^2} \frac{|\mathbf{p}'| \omega}{|\mathbf{p}| \mathbf{q}^4} e_\nu^* e_\mu (\bar{u}' Q^\mu u) (\bar{u} \tilde{Q}^\nu u') d\mathbf{o}_\mathbf{k} d\mathbf{o}' d\omega, \quad (\text{X.6})$$

where

$$Q^\mu = \gamma^\mu \frac{\gamma f' + m}{2\omega\kappa'} \gamma^0 - \gamma^0 \frac{\gamma f + m}{2\omega\kappa} \gamma^\mu, \\ \tilde{Q}^\nu = \gamma^0 \frac{\gamma f' + m}{2\omega\kappa'} \gamma^\nu - \gamma^\nu \frac{\gamma f + m}{2\omega\kappa} \gamma^0.$$

Without considering polarisation effects, we average the cross section over the spin directions of the initial electron and sum over the polarisations of the final electron and the photon. This reduces to the replacement

$$e_\nu^* e_\mu (\bar{u}' Q^\mu u) (\bar{u} \tilde{Q}^\nu u') \rightarrow -\frac{1}{2} \text{Sp } Q_\mu (\gamma p + m) \tilde{Q}^\mu (\gamma p' + m).$$

Computation of the trace is carried out by the standard formulas (see § 22). A certain simplification of the algebra is achieved by using the identity

$$\gamma^0 (\gamma p) \gamma^0 = \gamma \tilde{p},$$

where $\tilde{p} = (\varepsilon, -\mathbf{p})$, if $p = (\varepsilon, \mathbf{p})$. Furthermore, the number of terms requiring computation can be reduced by using the symmetry with respect to the substitution $p \leftrightarrow p'$, $k \rightarrow -k$, $q \rightarrow -q$ (this substitution leads only to a cyclic permutation of the factors in the products of matrices and therefore does not change the trace).

As a result one obtains the following expression for the differential cross section of bremsstrahlung with the emission of a photon of given frequency and direction, and with the emission of the secondary electron in a given direction²⁹:

$$d\sigma = \frac{Z^2 \alpha r_e^2}{4\pi^2} \frac{p' m^4}{p q^4} \frac{d\omega}{\omega} d\mathbf{o}_\mathbf{k} d\mathbf{o}' \times \\ \times \left\{ \frac{q^2}{\kappa^2 m^2} (2\varepsilon^2 + 2\varepsilon'^2 - q^2) + \left(q^2 \left(\frac{1}{\kappa} - \frac{1}{\kappa'} \right)^2 - \left(\frac{\varepsilon}{\kappa} - \frac{\varepsilon'}{\kappa'} \right)^2 \right) + \right. \\ \left. + \frac{2\omega q^2}{m^2} \left(\frac{1}{\kappa} - \frac{1}{\kappa'} \right) - \frac{2\omega^2}{m^2} \left(\frac{\varepsilon^2}{\kappa'^2} - \frac{\varepsilon'^2}{\kappa^2} \right) \right\}, \quad (\text{X.7})$$

²⁹Here and below in this section p , p' , q denote the absolute values of the three-dimensional vectors: $p = |\mathbf{p}|$, $p' = |\mathbf{p}'|$, $q = |\mathbf{q}|$.

where

$$\kappa = \varepsilon - \mathbf{n}\mathbf{p}, \quad \kappa' = \varepsilon' - \mathbf{n}\mathbf{p}' \quad (\mathbf{n} = \mathbf{k}/\omega), \quad \mathbf{q} = \mathbf{p}' + \mathbf{k} - \mathbf{p}.$$

By straightforward transformations this formula can be given a somewhat more convenient form for investigation:

$$\begin{aligned} d\sigma = & \frac{Z^2 \alpha r_e^2}{2\pi^2} \frac{d\omega}{\omega} \frac{p'}{p} \frac{m^2}{q^4} \sin \theta d\theta \sin \theta' d\theta' d\varphi \times \\ & \times \left\{ \frac{p'^2}{\kappa'^2} (4\varepsilon^2 - q^2) \sin^2 \theta' + \frac{p^2}{\kappa^2} (4\varepsilon'^2 - q^2) \sin^2 \theta + \right. \\ & + \frac{2\omega^2}{\kappa\kappa'} (p^2 \sin^2 \theta + p'^2 \sin^2 \theta') - \\ & \left. - \frac{2pp'}{\kappa\kappa'} (2\varepsilon^2 + 2\varepsilon'^2 - q^2) \sin \theta \sin \theta' \cos \varphi \right\}, \end{aligned} \quad (\text{X.8})$$

where

$$\kappa = \varepsilon - p \cos \theta, \quad \kappa' = \varepsilon' - p' \cos \theta',$$

$$q^2 = p^2 + p'^2 + \omega^2 - 2p\omega \cos \theta + 2p'\omega \cos \theta' - 2pp'(\cos \theta \cos \theta' + \sin \theta \sin \theta' \cos \varphi),$$

θ and θ' are the angles between $\mathbf{k}$ and, respectively, $\mathbf{p}$ and $\mathbf{p}'$, φ the angle between the planes $\mathbf{k}, \mathbf{p}$ and $\mathbf{k}, \mathbf{p}'$.

Integration of (93.8) over the directions of the photon and secondary electron is quite cumbersome. It leads to the following formula for the spectral distribution of the radiation³⁰:

$$\begin{aligned} d\sigma_\omega = & Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{p'}{p} \left\{ \frac{4}{3} - 2\varepsilon\varepsilon' \frac{p^2 + p'^2}{p^2 p'^2} + \right. \\ & + m^2 \left(l \frac{\varepsilon'}{p^3} + l' \frac{\varepsilon}{p'^3} - \frac{ll'}{pp'} \right) + L \left[\frac{8\varepsilon\varepsilon'}{3pp'} + \frac{\omega^2}{p^3 p'^3} (\varepsilon^2 \varepsilon'^2 + p^2 p'^2 + m^2 \varepsilon\varepsilon') \right] + \\ & \left. + \frac{m^2 \omega}{2pp'} \left(l \frac{\varepsilon\varepsilon' + p^2}{p^3} - l' \frac{\varepsilon\varepsilon' + p'^2}{p'^3} + \frac{2\omega\varepsilon\varepsilon'}{p^2 p'^2} \right) \right\}, \end{aligned} \quad (\text{X.9})$$

where

$$L = \ln \frac{\varepsilon\varepsilon' + pp' - m^2}{\varepsilon\varepsilon' - pp' - m^2}, \quad l = \ln \frac{\varepsilon + p}{m}, \quad l' = \ln \frac{\varepsilon' + p'}{m}.$$

We recall that the permissible values of frequencies in the obtained formulas are restricted only by the condition imposed on the final velocity of the electron ($Ze^2/v' \ll 1$): the electron must not lose almost all its energy. When $\omega \rightarrow 0$ the cross section of radiation diverges as $d\omega/\omega$; this is a manifestation of the general rule discussed in § 98.

In the non-relativistic limit ($p \ll m$) the photon impulse is small compared to the electron impulse, so that $q^2 \approx (\mathbf{p}' - \mathbf{p})^2$. Setting in (93.8) $\varepsilon = \varepsilon' = m$, and neglecting all p, p', ω compared to m , we obtain

$$d\sigma = \frac{2}{\pi} Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{p'}{p} \frac{m}{q^4} \sin \theta d\theta \sin \theta' d\theta' d\varphi \times (p^2 \sin^2 \theta + p'^2 \sin^2 \theta' - 2pp' \sin \theta \sin \theta' \cos \varphi),$$

³⁰Integration over the directions of the secondary electron alone can also be performed in analytic form—see *Gluckstern R. L., Hull M. H.* Phys. Rev. 1953. V. 90. P. 1030.

or

$$d\sigma = \frac{Z^2 \alpha^3}{\pi^2} \frac{p'}{p} [\mathbf{nq}]^2 \frac{d\mathbf{q} d\mathbf{q}' d\omega}{q^4 \omega}, \quad (\text{X.10})$$

in agreement with the formula obtained in the Born approximation in Problem 1 of § 92. Correspondingly, for the spectral distribution of the radiation one also obtains the already known result (92.16).

In the ultrarelativistic case, when both the initial and final energies of the electron are large ($\varepsilon, \varepsilon' \gg m$), the angular distribution of the photon and secondary electrons has a very specific character. At small angles θ, θ' the quantities $\varkappa, \varkappa'$ appearing in the denominators of formula (93.8) are

$$\varkappa \approx \frac{\varepsilon}{2} \left(\frac{m^2}{\varepsilon^2} + \theta^2 \right), \quad \varkappa' \approx \frac{\varepsilon'}{2} \left(\frac{m^2}{\varepsilon'^2} + \theta'^2 \right) \quad (\text{X.11})$$

and in the region $\theta \lesssim m/\varepsilon$ become very small. In this region the magnitude of the vector $\mathbf{q}$ ($q \sim m$) is also small. Thus, in the ultrarelativistic case the photon and the secondary electron fly forward in a narrow cone with opening angle $\sim m/\varepsilon$.

A quantitative formula for the angular distribution in the ultrarelativistic case is easily obtained from (93.8) by substituting $\varkappa, \varkappa'$ from (93.11), replacing everywhere else p, p' by $\varepsilon, \varepsilon'$ and neglecting q^2 compared to ε^2 . Introducing convenient notation:

$$\delta = \frac{\varepsilon}{m} \theta, \quad \delta' = \frac{\varepsilon'}{m} \theta', \quad (\text{X.12})$$

we write this formula in the form

$$\begin{aligned} d\sigma = & \frac{8}{\pi} Z^2 \alpha r_e^2 \frac{\varepsilon' m^4 d\omega}{\varepsilon q^4 \omega} \delta d\delta \cdot \delta' d\delta' d\varphi \times \\ & \times \left\{ \frac{\delta^2}{(1 + \delta^2)^2} + \frac{\delta'^2}{(1 + \delta'^2)^2} + \frac{\omega^2}{2\varepsilon\varepsilon'} \frac{\delta^2 + \delta'^2}{(1 + \delta^2)(1 + \delta'^2)} - \right. \\ & \left. - \left(\frac{\varepsilon'}{\varepsilon} + \frac{\varepsilon}{\varepsilon'} \right) \frac{\delta\delta' \cos \varphi}{(1 + \delta^2)(1 + \delta'^2)} \right\}. \end{aligned} \quad (\text{X.13})$$

Writing $q^2 = [\mathbf{nq}]^2 + (\mathbf{nq})^2$ ($\mathbf{n} = \mathbf{k}/\omega$), it is easy to find that for small angles

$$\frac{q^2}{m^2} = (\delta^2 + \delta'^2 - 2\delta\delta' \cos \varphi) + m^2 \left(\frac{1 + \delta^2}{2\varepsilon} - \frac{1 + \delta'^2}{2\varepsilon'} \right)^2. \quad (\text{X.14})$$

When $\delta \sim \delta' \sim 1$ the second term here is small in comparison with the first. These terms become comparable in the region of even smaller angles, where $\delta \sim m/\varepsilon$. Although q here becomes especially small ($q \sim m^2/\varepsilon \ll m$), the integral contribution of this region to the cross section is still small by the factor m^2/ε^2 (as it is easy to see) compared to the contribution of the whole region $\delta \lesssim 1$. However, q can also reach values $\sim m^2/\varepsilon$ when $\delta \sim \delta' \sim 1$, if in addition

$$|\delta - \delta'| \lesssim \frac{m}{\varepsilon}, \quad \varphi \lesssim \frac{m}{\varepsilon}. \quad (\text{X.15})$$

The contribution of this region is of the same order as the total integral cross section (or even constitutes the main part of it—see below).

Integration of formula (93.13) over φ and over δ' gives the angular distribution of the photon (of given frequency) independently of the directions of the secondary electrons³¹:

$$d\sigma = 8Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{\varepsilon'}{\varepsilon} \frac{\delta \cdot d\delta}{(1 + \delta^2)^2} \times \\ \times \left\{ \left[\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} - \frac{4\delta^2}{(1 + \delta^2)^2} \right] \ln \frac{2\varepsilon\varepsilon'}{m\omega} - \right. \\ \left. - \frac{1}{2} \left[\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} + 2 - \frac{16\delta^2}{(1 + \delta^2)^2} \right] \right\}. \quad (\text{X.16})$$

Integrating over δ , we find the spectral distribution of the radiation in the ultrarelativistic case:

$$d\sigma_\omega = 4Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{\varepsilon'}{\varepsilon} \left[\left(\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} - \frac{2}{3} \right) \left(\ln \frac{2\varepsilon\varepsilon'}{m\omega} - \frac{1}{2} \right) \right]. \quad (\text{X.17})$$

We draw attention to the presence in (93.16) and (93.17) of a logarithm of a large quantity (even at $\omega \sim \varepsilon$ the ratio $\varepsilon\varepsilon'/(m\omega) \sim \varepsilon/m \gg 1$). If it is so large that its logarithm is also large, then in the formulas named, the terms containing the logarithm become dominant. We note that this logarithm arises from integration over the region (93.15)³². Thus, in the logarithmic approximation (i.e. neglecting terms not containing the large logarithm), the secondary electron flies at an angle $\sim (m/\varepsilon)^2$ to the direction of incidence.

Finally, we give the limiting formula for the region near the hard boundary of the spectrum, when the ultrarelativistic electron radiates almost all its energy: $\omega \approx \varepsilon \gg \varepsilon'$. From (93.9) one easily finds

$$d\sigma_\omega = 2Z^2 \alpha r_e^2 \frac{d\omega}{\varepsilon} \left\{ \frac{\varepsilon'^2}{p'^2} \ln \frac{\varepsilon' + p'}{m} - \frac{m^2 \varepsilon'^2}{4p'^3} \ln^2 \frac{\varepsilon' + p'}{m} + \frac{\varepsilon' + p'}{3pp'} \right\}. \quad (\text{X.18})$$

Formulas (93.17) and (93.18) cover the entire range of values of ω for an ultrarelativistic initial electron; when $\omega \approx \varepsilon \gg \varepsilon' \gg m$ these formulas coincide. If the secondary electron is non-relativistic ($p' \ll m$), then

$$d\sigma_\omega = 2Z^2 \alpha r_e^2 \frac{\sqrt{2m(\varepsilon - \omega)}}{m} \frac{d\omega}{\varepsilon}. \quad (\text{X.19})$$

Polarisation effects. Polarisation effects in bremsstrahlung can be investigated by the same general method as was done in § 87 for the Compton effect. The question of

³¹First one integrates over φ (in the limits from 0 to 2π). The integration over δ' is conveniently replaced by integration over the difference $|\Delta| = |\delta' - \delta|$, splitting its range into two parts: from 0 to some Δ_0 and from Δ_0 to ∞ , where Δ_0 satisfies the inequalities $m/\varepsilon \ll \Delta_0 \ll 1$. In each domain corresponding simplifications in the sub-integral expression of (93.13) are possible.

³²This is easy to verify by considering the domain of integration in which φ and $\Delta = \delta' - \delta$ satisfy conditions: $m/\varepsilon \ll \Delta_0 \ll 1$, $\varphi \ll 1$. In this domain $q^2/m^2 \approx \Delta^2 + \varphi^2 \delta^2$, and the expression in curly brackets in (93.13) becomes the expression proportional to φ^2 or Δ^2 (when $\varphi = 0$ and $\Delta = 0$ it vanishes). The integrals $\int \frac{\varphi^2 d\varphi d\Delta}{(\Delta^2 + \delta^2 \varphi^2)^2}$ or $\int \frac{\Delta^2 d\varphi d\Delta}{(\Delta^2 + \delta^2 \varphi^2)^2}$ diverge logarithmically; their "cutting off" occurs at the boundaries of the indicated domain of values of the variables.

the choice of 4-vectors $e^{(1)}, e^{(2)}$ in this case is particularly simple. Since it suffices in practice to consider the process in one specific reference frame (the rest frame of the nucleus), it is sufficient to set

$$e^{(1)} = (0, \mathbf{e}^{(1)}), \quad e^{(2)} = (0, \mathbf{e}^{(2)}),$$

where $\mathbf{e}^{(1)}, \mathbf{e}^{(2)}$ are unit vectors perpendicular to $\mathbf{k}$, one of which lies in the plane $\mathbf{k}\mathbf{p}$ and the other is perpendicular to it.

We shall not give here the rather cumbersome calculations themselves, but merely indicate some qualitative properties of the polarisation effects.

These properties can be obtained with the aid of various symmetry relations, similarly to what was done in § 87 for the Compton effect.

The theory expounded corresponds to the first Born approximation. In this approximation the cross section cannot contain a term proportional only to the vector of polarisation of one of the initial (ζ) or the final (ζ') electron. The absence of a term $\propto \zeta$ means that the complete (summed over photon polarisations and the polarisation of the secondary electron) bremsstrahlung cross section does not depend on the polarisation of the incident electron.

Of the terms proportional to only the photon polarisation parameters (ξ_1, ξ_2, ξ_3), the one $\propto \xi_2$ is absent. This means that upon radiation by an unpolarised electron the photon does not possess circular polarisation. Here there is, however, a difference from the analogous result for the Compton effect. In the latter case such terms were forbidden by the impossibility of constructing a pseudoscalar from the only two independent vectors at our disposal $\mathbf{k}$ and $\mathbf{k}'$. In the case of bremsstrahlung there are three independent momenta ($\mathbf{p}, \mathbf{p}', \mathbf{k}$), which suffice for the construction of a pseudoscalar ($\mathbf{k}[\mathbf{p}\mathbf{p}']$). The term of the form $\xi_2'(\mathbf{k}[\mathbf{p}\mathbf{p}'])$ does not contradict spatial parity and, strictly speaking, differs from zero. However, it is not invariant under the transformation (87.26) and therefore is absent in the first Born approximation.

The existence of the pseudoscalar ($\mathbf{k}[\mathbf{p}\mathbf{p}']$) leads also to the fact that along with the term $\propto \xi_3$ there also appears in the cross section a term $\propto \xi_1$ (in contrast to the situation in the Compton effect). The term of the form

$$S_{\alpha\beta} \nu_{\alpha}(\mathbf{k}[\mathbf{p}\mathbf{p}'])$$

(where $\nu = [\mathbf{k}\mathbf{p}]$), invariant under spatial inversion as well as under the transformation (87.26), is non-zero.

When also the direction of the secondary electron is detected and recorded, the cross section contains the terms $\propto \xi_1$ (allowed in the cross section also for the Compton effect) and also $\propto \xi_1'$, disappearing upon integration over all directions of $\mathbf{p}'$.

The degree of linear polarisation does not depend on the polarisation state of the incident electron: the correlation terms of the form $\xi_1'\zeta$ and $\xi_3\zeta$ are forbidden in the first Born approximation. However, the term $\xi_2'\zeta$ is allowed, so that upon radiation by a polarised electron the photon possesses circular polarisation (*Ya. B. Zeldovich, 1952*).

Screening. The formulas obtained above are derived for a purely Coulomb field. If, however, one is dealing with radiation not from a "bare" nucleus but from an atom, then the screening of the nuclear charge by the atomic electrons, leading to a reduction of the

cross section, must be taken into account. For this one must introduce into the potential of the external field $A_0^{(e)}(\mathbf{q})$ the atomic form factor $F(q)$ (see III, § 139). According to (139.2) (see III) this is achieved by the replacement of Z by $Z - F(q)$. We determine under what conditions the screening is substantial.

A definite value of q in the form factor corresponds to distances $r \sim 1/q$ in the spatial distribution of the electronic charge in the atom. The form factor approaches the value Z (complete screening) when $q \lesssim 1/a$, where a is the dimensions of the atom.

On the other hand, in the ultrarelativistic case a substantial contribution to the bremsstrahlung cross section arises, as we have seen above, already from the region of values of q close to the smallest value which q can have at given initial and final energies of the electron. In the ultrarelativistic case

$$q_{\min} = p - p' - \omega = \sqrt{\varepsilon^2 - m^2} - \sqrt{\varepsilon'^2 - m^2} - (\varepsilon - \varepsilon') \approx \frac{m^2 \omega}{2\varepsilon \varepsilon'}. \quad (\text{X.20})$$

Screening is substantial if $q_{\min} \lesssim 1/a$ or

$$\frac{\varepsilon \varepsilon'}{m \omega} \gtrsim \frac{a}{1/m} = \frac{a m}{1}. \quad (\text{X.21})$$

This condition is always satisfied at sufficiently large energies of the incident electron.

If $q_{\min} \ll 1/a$ ("complete screening"), then with logarithmic accuracy one can immediately write the answer for the spectral distribution of the radiation. Indeed, under the sign of the logarithm in (93.17) just the left-hand side of inequality (93.21) stands, the logarithm that leads to it. Upon observing the inequality the integral over dq , giving rise to this logarithm, is cut off at the value of order of the right-hand side of inequality. According to the Thomas–Fermi model $a \sim a_0 Z^{-1/3}$, where $a_0 \sim 1/(me^2)$ is the Bohr radius (see III, § 70); then $am \sim 1/(\alpha Z^{1/3})$. Thus, under complete screening the logarithm in (93.17) should be replaced by $\ln(1/(\alpha Z^{1/3}))$.

The ratio $\kappa_{\text{rad}}/\varepsilon$ is also called the cross section of energy loss to radiation. For large ε it grows logarithmically. This growth is eliminated, however, upon inclusion of screening. Under complete screening $\kappa_{\text{rad}}/\varepsilon$ tends to a constant limit $\approx 4Z^2 \alpha r_e^2 \ln(1/(\alpha Z^{1/3}))$.

When an atom is bombarded, some radiation is produced not only on the nucleus but also on the electrons. We shall see below (see § 97) that in the ultrarelativistic case the cross section of the radiation of an electron on an electron differs from the cross section of radiation on the nucleus only by the absence of the factor Z^2 . Therefore the presence of Z atomic electrons can approximately be taken into account by the replacement of Z^2 by $Z(Z+1)$.

Energy loss. The energy loss of the electron to radiation is characterised by the "effective retardation":

$$\kappa_{\text{rad}} = \int_0^{\varepsilon-m} \omega d\sigma_\omega. \quad (\text{X.22})$$

Computation of the integral of $d\sigma_\omega$ from (93.17) leads to the following result³³:

$$\kappa_{\text{rad}} = Z^2 \alpha r_e^2 \varepsilon \left\{ \frac{12\varepsilon^2 + 4m^2}{3\varepsilon p} \ln \frac{\varepsilon + p}{m} - \frac{(8\varepsilon + 6p)m^2}{3\varepsilon p^2} \ln^2 \frac{\varepsilon + p}{m} - \frac{4}{3} + \frac{2m^2}{\varepsilon p} F\left(\frac{2p(\varepsilon + p)}{m^2}\right) \right\}, \quad (\text{X.23})$$

where $F(\xi)$ is the Spence function, defined according to (131.19). In the non-relativistic case formula (93.23) goes over into

$$\kappa_{\text{rad}} = \frac{16}{3} Z^2 \alpha r_e^2 m \quad (\text{n.r.}) \quad (\text{X.24})$$

(one uses $F(\xi) \approx \xi$ at $\xi \ll 1$ —see (131.23)). This expression can, of course, be obtained also by direct integration of the non-relativistic Born formula (92.16). In the ultrarelativistic case

$$\kappa_{\text{rad}} = 4Z^2 \alpha r_e^2 \varepsilon \left(\ln \frac{2\varepsilon}{m} - \frac{1}{3} \right) \quad (\text{u.r.}) \quad (\text{X.25})$$

(when $\xi \gg 1$ we have $F(\xi) \approx \frac{1}{2} \ln^2 \xi$ —see (131.20); both terms with the square of the logarithm in (93.23) can then be omitted).

When passing through a medium containing N atoms per unit volume, a fast electron loses on average its energy to bremsstrahlung at a rate $-d\varepsilon/dx = N\kappa_{\text{rad}}$. Hence the energy decreases exponentially along the path, $\varepsilon = \varepsilon_0 e^{-x/l_{\text{rad}}}$, at distances of order

$$l_{\text{rad}} \sim \frac{\varepsilon}{N\kappa_{\text{rad}}} \sim \left[Z^2 \alpha N r_e^2 \ln \frac{1}{\alpha Z^{1/3}} \right]^{-1}; \quad (\text{X.26})$$

this length is called the *radiation length*.

Coherence length. Formula (93.20) admits also another, more general interpretation: for the applicability of the obtained formulas it is necessary that the external field in which the electron moves change little (in the direction of motion) over distances

$$l_{\text{coh}} \sim \frac{1}{q_{\text{min}}} \sim \frac{\varepsilon \varepsilon'}{m^2 \omega} \left(= \frac{\varepsilon \varepsilon'}{c^2 m^2 \omega} \right); \quad (\text{X.27})$$

this length is called the *formation length* of the radiation or the *coherence length*³⁴. The value (93.27), obtained in the Born approximation, has in fact (for ultrarelativistic particles) a completely general character—it is easy to obtain it also in the opposite limiting case of quasi-classical motion. Indeed, from formula (90.22)³⁵ it is immediately clear that for radiation at small angles to the direction of motion the essential times are

$$\tau \sim \frac{\varepsilon'}{\varepsilon \omega (1 - v)} \sim \frac{\varepsilon \varepsilon'}{m^2 \omega},$$

³³Though formula (93.17) is inapplicable close to the upper limit, this is inessential because of the convergence of the integral.

³⁴The considerations presented are due to *M. L. Ter-Mikaelyan* (1953).

³⁵The derivation of formula (90.22) is based only on the smallness of the curvature of the trajectory and in this sense is not connected with the fact that in § 90 a specifically magnetic field was considered.

i.e. a segment of trajectory of length $c\tau \sim l_{\text{coh}}$.

At a given frequency ω the coherence length grows with increase of the electron energy. Meanwhile the formulas obtained for bremsstrahlung on a single isolated atom can be valid for radiation during passage through a medium only under the condition that on the coherence length no repeated scattering of the photon or scattering of the electron occurs. The first condition means that one must have $l_{\text{coh}} \ll l_{\text{rad}}$. But already significantly earlier the second condition is violated—over a path $\sim l_{\text{rad}}$ there arises multiple scattering of the electron on nuclei of atoms of the medium.

For a quantitative formulation of the condition let us return to formula (90.22) and the expression in the exponent of the exponential; carrying out the integration over time, we write it in the form

$$-i \frac{\omega \varepsilon}{\varepsilon'} \int_{t_1}^{t_1+\tau} (1 - \mathbf{n} \mathbf{v}) dt \approx -\frac{i\omega \varepsilon}{\varepsilon'} (1 - \nu) \tau + \frac{1}{2} \int_{t_1}^{t_1+\tau} \theta^2 dt, \quad (\text{X.28})$$

where θ is the small angle between $\mathbf{v}$ and $\mathbf{n}$, associated with scattering on nuclei. In Coulomb scattering the angle θ changes by small increments, so that the variation of θ with time has the character of a slow “diffusion in angles”. The mean square deviation of the electron over a path $t - t_1 (= c(t - t_1))$

$$\overline{\theta^2} \sim (t - t_1) / l_{\text{Coul}},$$

where l_{Coul} is the mean free path with respect to Coulomb collisions. For this mean free path we have

$$\frac{1}{l_{\text{Coul}}} \sim \frac{NZ^2 e^4}{\varepsilon^2} \ln \frac{\chi_{\text{max}}}{\chi_{\text{min}}},$$

where χ_{min} and χ_{max} are the minimum and maximum scattering angles in a single collision for which the scattering can still be considered Rutherford (cf. X, § 41)³⁶. The first of them is determined by the atomic dimensions a , at which the field of the nucleus is screened: $\chi_{\text{min}} \sim 1/(pa)$. Large scattering angles are bounded (for an ultrarelativistic electron) by distances of order the nuclear radius R : $\chi_{\text{max}} \approx 1/(pR)$. Putting $R \approx 1.5 \cdot 10^{-13} Z^{1/3} \text{ cm} \sim r_e Z^{1/3}$, we obtain

$$l_{\text{Coul}} \sim \frac{\varepsilon^2}{NZ^2 e^4 \ln(1/(\alpha Z^{1/3}))} \sim \frac{\alpha \varepsilon^2}{m^2} l_{\text{rad}}. \quad (\text{X.29})$$

The second term in (93.28), accumulated over the time $\tau \sim l_{\text{coh}}$, is now estimated as

$$\overline{\theta^2} \tau \sim \frac{\omega \varepsilon l_{\text{coh}}^2}{\varepsilon' l_{\text{Coul}}} \sim \frac{l_{\text{coh}}}{\alpha l_{\text{rad}}}.$$

For the applicability of the bremsstrahlung formulas obtained without account of multiple scattering, this term must be small compared with unity. From this we find the condition

$$l_{\text{coh}} \ll \alpha l_{\text{rad}}, \quad (\text{X.30})$$

stronger than the condition $l_{\text{coh}} \ll l_{\text{rad}}$ (*L. D. Landau, I. Ya. Pomeranchuk, 1953*).

³⁶Recall that the mean free path is defined by the transport cross section $\sigma_{\text{tr}} = \int (1 - \cos \chi) d\sigma(\chi)$. For scattering of ultrarelativistic electrons on a Coulomb centre the cross section $d\sigma(\chi)$ is given by formula (80.10).

§ 94. Pair production by a photon in the field of a nucleus

The production of an electron-positron pair in the collision of a photon with a nucleus ($Z + \gamma \rightarrow Z + e^- + e^+$) and bremsstrahlung in the collision of an electron with a nucleus ($Z + e^- \rightarrow Z + e^+ \gamma$) are two crossed channels of one and the same reaction. In § 91 the rules were already formulated by which the formulas are transformed in passing from the second of these cases to the first. In the present case, applying these rules to formula (93.8), we obtain the following expression for the differential cross section of pair production by an unpolarised photon, averaged over the polarisations of the pair components³⁷:

$$\begin{aligned}
 d\sigma = & \frac{Z^2 \alpha r_e^2 m^2 p_+ p_-}{2\pi \omega^3 q^4} \sin \theta_+ d\theta_+ \sin \theta_- d\theta_- d\varphi \times \\
 & \times \left\{ \frac{p_+^2}{\kappa_+^2} (4\varepsilon_-^2 - q^2) \sin^2 \theta_+ + \frac{p_-^2}{\kappa_-^2} (4\varepsilon_+^2 - q^2) \sin^2 \theta_- - \right. \\
 & - \frac{2\omega^2}{\kappa_+ \kappa_-} (p_+^2 \sin^2 \theta_+ + p_-^2 \sin^2 \theta_-) - \\
 & \left. - \frac{2p_+ p_-}{\kappa_+ \kappa_-} (2\varepsilon_+^2 + 2\varepsilon_-^2 - q^2) \sin \theta_+ \sin \theta_- \cos \varphi \right\},
 \end{aligned} \tag{X.31}$$

$$\kappa_{\pm} = \varepsilon_{\pm} - p_{\pm} \cos \theta_{\pm}, \quad q^2 = (\mathbf{p}_+ + \mathbf{p}_- - \mathbf{k})^2, \quad \varepsilon_+ + \varepsilon_- = \omega$$

(H. A. Bethe, W. Heitler, 1934).

By the same transformation we obtain from (93.9) the distribution of the pair components in energy:

$$\begin{aligned}
 d\sigma_{\varepsilon_+} = & Z^2 \alpha r_e^2 \frac{p_+ p_-}{\omega^3} d\varepsilon_+ \left\{ -\frac{4}{3} - 2\varepsilon_+ \varepsilon_- \frac{p_+^2 + p_-^2}{p_+^2 p_-^2} + \right. \\
 & + m^2 \left(l_- \frac{\varepsilon_+}{p_-^2} + l_+ \frac{\varepsilon_-}{p_+^2} - \frac{l_+ l_-}{p_+ p_-} \right) + \\
 & + L \left[-\frac{8\varepsilon_+ \varepsilon_-}{3p_+ p_-} + \frac{\omega^3}{p_+^3 p_-^3} (\varepsilon_+^2 \varepsilon_-^2 + p_+^2 p_-^2 - m^2 \varepsilon_+ \varepsilon_-) - \right. \\
 & \left. \left. - \frac{m^2 \omega}{2p_+ p_-} \left(l_+ \frac{\varepsilon_+ \varepsilon_- - p_+^2}{p_+^3} + l_- \frac{\varepsilon_+ \varepsilon_- - p_-^2}{p_-^3} \right) \right] \right\},
 \end{aligned} \tag{X.32}$$

$$L = \ln \frac{\varepsilon_+ \varepsilon_- + p_+ p_- + m^2}{\varepsilon_+ \varepsilon_- - p_+ p_- + m^2}, \quad l_{\pm} = \ln \frac{\varepsilon_{\pm} + p_{\pm}}{\varepsilon_{\pm} - p_{\pm}}.$$

In the ultrarelativistic case ($\varepsilon_{\pm} \gg m$) the electron and positron emerge at angles $\theta_{\pm} \sim m/\varepsilon_{\pm}$ to the direction of the incident photon. The angular distribution is given by a

³⁷On polarisation effects in pair production by photons see the same literature as indicated in § 93 for bremsstrahlung.

formula analogous to (93.13):

$$d\sigma = \frac{8}{\pi} Z^2 \alpha r_e^2 \frac{m^2 \varepsilon_+ \varepsilon_-}{\omega^3 q^4} d\varepsilon_+ \times \\ \times \left\{ -\frac{\delta_+^2}{(1 + \delta_+^2)^2} - \frac{\delta_-^2}{(1 + \delta_-^2)^2} + \frac{\omega^2}{2\varepsilon_+ \varepsilon_-} \frac{\delta_+^2 + \delta_-^2}{(1 + \delta_+^2)(1 + \delta_-^2)} + \right. \\ \left. + \left(\frac{\varepsilon_+}{\varepsilon_-} + \frac{\varepsilon_-}{\varepsilon_+} \right) \frac{\delta_+ \delta_- \cos \varphi}{(1 + \delta_+^2)(1 + \delta_-^2)} \right\} \delta_+ \delta_- d\delta_+ d\delta_- d\varphi, \quad (\text{X.33})$$

where

$$\frac{q^2}{m^2} = \delta_+^2 + \delta_-^2 - 2\delta_+ \delta_- \cos \varphi + m^2 \left(\frac{1 + \delta_+^2}{2\varepsilon_+} + \frac{1 + \delta_-^2}{2\varepsilon_-} \right)^2. \quad (\text{X.34})$$

The energy distribution in this case:

$$d\sigma = 4Z^2 \alpha r_e^2 \frac{d\varepsilon_+}{\omega^3} \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \left(\ln \frac{2\varepsilon_+ \varepsilon_-}{m\omega} - \frac{1}{2} \right) \quad (\text{u.r.}), \quad (\text{X.35})$$

Integration of this expression over ε_+ (in the limits from m to ω) gives the total cross section of pair production by a photon of given energy³⁸:

$$\sigma = \frac{28}{9} Z^2 \alpha r_e^2 \left(\ln \frac{2\omega}{m} - \frac{109}{42} \right), \quad \omega \gg m. \quad (\text{X.36})$$

As for bremsstrahlung, the logarithmic term in the cross section in the ultrarelativistic case originates from the region of values $q \sim m^2/\varepsilon$. To this there now correspond angles for which

$$|\delta_+ - \delta_-| \lesssim \frac{m}{\varepsilon}, \quad |\pi - \varphi| \lesssim \frac{m}{\varepsilon}$$

(instead of $\varphi \lesssim m/\varepsilon$ in (93.15)). Thus, in the logarithmic approximation the directions of the electron and positron make small angles inversely proportional to the particle energies with the direction of the photon, and lie almost in one plane with the latter but on opposite sides of it.

Near the reaction threshold ($\omega \rightarrow 2m$) the Born approximation is inapplicable. The derivation of a quantitative formula in this case would require exact account of the Coulomb interaction of the three charged particles present in the final state (the nucleus and the pair). The symmetry with respect to the electron (attracted

to the nucleus) and the positron (repelled from the nucleus) in this case, of course, disappears.

If

$$Z\alpha \sqrt{\frac{\omega - 2m}{\omega}} \ll 1, \quad (\text{X.37})$$

then the Born approximation is still applicable. At non-relativistic energies of the pair $\omega \approx 2m \gg p_{\pm}$, and therefore $q \approx \omega$. In (94.1) one can put everywhere $\varepsilon_{\pm} = \kappa_{\pm} = m$, $\omega = 2m$, after which this formula reduces to the expression

$$d\sigma = \frac{Z^2 \alpha r_e^2 p_+ p_-}{64\pi^2 m^6} (p_+^2 \sin^2 \theta_+ + p_-^2 \sin^2 \theta_-) d\theta_+ d\theta_- d\varepsilon_+. \quad (\text{X.38})$$

³⁸Because of the convergence of the integral at both limits, the inapplicability of formula (94.5) when $\varepsilon_{\pm} - m$ is small is inessential.

After integration over angles

$$\begin{aligned} d\sigma &= \frac{1}{6} Z^2 \alpha r_e^2 \frac{p_+ p_- (p_+^2 + p_-^2)}{m^5} d\varepsilon_+ = \\ &= \frac{2Z^2 \alpha r_e^2}{3m^3} (\omega - 2m) \sqrt{(\varepsilon_+ - m)(\varepsilon_- - m)} d\varepsilon_+. \end{aligned} \quad (\text{X.39})$$

Finally, integrating over ε_+ (in the limits from m to $\omega - m$), we obtain the total cross section

$$\sigma = \frac{\pi}{12} Z^2 \alpha r_e^2 \left(\frac{\omega - 2m}{m} \right)^3. \quad (\text{X.40})$$

If the relative velocity v_0 of the components of the produced pair is small, it is necessary to take into account their Coulomb interaction with each other (*A. D. Sakharov*, 1948). It becomes substantial when v_0 is of the order of (or less than) the velocity of a particle in the bound state of the electron and positron (positronium):

$$v_0 \lesssim \alpha. \quad (\text{X.41})$$

Let us consider the process in the centre-of-mass frame of the pair. In the diagrams depicting the process in this frame, the essential virtual momenta are $\sim m$. In other words, the essential distances between the electron and positron are $\sim 1/m$. Meanwhile the wave function of their relative motion $\psi(r)$ changes substantially only at distances $r \sim 1/(mv_0) \sim 1/(m\alpha)$, i.e. large compared with $1/m$. Therefore the account of the interaction of the particles reduces to the appearance in the transition matrix element of a factor $\psi^*(0)$. Correspondingly, the differential cross section is multiplied by $|\psi(0)|^2$, i.e. by

$$\frac{2\pi\alpha/v_0}{1 - e^{-2\pi\alpha/v_0}} \quad (\text{X.42})$$

(see III, (136.11)). The relative velocity of two particles is the velocity of one of them in the rest frame of the other. Comparing the values of the invariant $p_- \mu p_+^\mu$ in this frame and in the laboratory frame (the rest frame of the nucleus), we obtain

$$\frac{m^2}{\sqrt{1 - v_0^2}} = \varepsilon_+ \varepsilon_- - \mathbf{p}_+ \mathbf{p}_-,$$

from which v_0 can be found. If $\mathbf{p}_+$ and $\mathbf{p}_-$ are close to each other in magnitude and direction, then for v_0 we obtain the approximate formula

$$v_0^2 = \frac{p^2}{m^2} \vartheta^2 + \frac{(p_+ - p_-)^2}{\varepsilon^2}, \quad (\text{X.43})$$

applicable when $v_0 \ll 1$; $p = (p_+ + p_-)/2$, $\varepsilon = (\varepsilon_+ + \varepsilon_-)/2$, ϑ is the angle between $\mathbf{p}_+$ and $\mathbf{p}_-$.

The correction to the cross section, determined by formulas (94.12), (94.13), leads to the appearance of an anomaly in the correlation between the momenta of the produced electron and positron: a narrow maximum at $\mathbf{p}_+ \approx \mathbf{p}_-$.

§ 95. Exact theory of pair production in the ultrarelativistic case

In the two preceding sections bremsstrahlung and pair production by a photon in the relativistic region were studied on the basis of the Born approximation, for which in any case the fulfilment of the condition $Z\alpha \ll 1$ was required. In §§ 95, 96 there is presented the theory of these processes, free from this restriction, i.e. valid also when $Z\alpha \sim 1$ (H. A. Bethe, L. Maximon, 1954). It is assumed that both particles (the initial and final electron, or the components of the pair) are ultrarelativistic; their energy $\varepsilon \gg m$.

We have seen that in the ultrarelativistic case both particles fly at small angles (θ, θ' or θ_+, θ_-) to the direction of the photon: $\theta \lesssim m/\varepsilon$. This property is preserved also in the exact (in $Z\alpha$) theory, and we shall consider precisely this region of angles.

The momentum transfer to the nucleus in this region is $q \sim m$. This means that in the wave functions the essential impact parameters are $\rho \sim 1/q \sim 1/m$, i.e. “large” distances. At such distances one can use the wave function obtained in § 39. Let us present the corresponding calculations for pair production.

The cross section of pair production has a form analogous to the cross section of the photoeffect (cf. (56.1), (56.2)):

$$d\sigma = 2\pi \left| e\sqrt{4\pi} \frac{1}{\sqrt{2\omega}} M_{fi} \right|^2 \delta(\omega - \varepsilon_+ - \varepsilon_-) \frac{d^3p_+ d^3p_-}{(2\pi)^3}, \quad (\text{X.44})$$

where

$$M_{fi} = \int \psi_{\varepsilon_-, \mathbf{p}_-}^{(-)*}(\alpha \mathbf{e}) e^{i\mathbf{k}\mathbf{r}} \psi_{-\varepsilon_+, -\mathbf{p}_+}^{(+)} d^3x. \quad (\text{X.45})$$

Here $\psi_{\varepsilon_-, \mathbf{p}_-}^{(-)}$ is the wave function of the electron, and $\psi_{-\varepsilon_+, -\mathbf{p}_+}^{(+)}$ is the wave function with negative energy ($-\varepsilon_+$) and momentum $-\mathbf{p}_+$.

The function $\psi_{\varepsilon_-, \mathbf{p}_-}^{(-)}$, pertaining to a particle in the final state, must contain in its asymptotics (alongside the plane wave) a converging spherical wave; this circumstance is indicated by the upper index $(-)$ on the function. According to (39.10) such a wave function is³⁹

$$\psi_{\varepsilon_-, \mathbf{p}_-}^{(-)} = \frac{C^{(-)}}{\sqrt{2\varepsilon_-}} e^{i\mathbf{p}_- \mathbf{r}} \left(1 - \frac{i\alpha \nabla}{2\varepsilon_-} \right) F(-i\nu, 1, -i(p_- r + \mathbf{p}_- \mathbf{r})) u(\mathbf{p}_-), \quad (\text{X.46})$$

$$C^{(-)} = e^{\pi\nu/2} \Gamma(1 + i\nu), \quad \nu = Z\alpha.$$

The function $\psi_{-\varepsilon_+, -\mathbf{p}_+}^{(+)}$ must contain in its asymptotics a diverging spherical wave (index $(+)$ above), since by its meaning it is the wave function of an “initial state with negative energy”. The wave function of the positron (formed from $\psi_{\varepsilon_+, \mathbf{p}_+}^{(+)*}$) will then have a converging wave in its asymptotics, as required for a final-state particle. According to (39.11) such a function is

$$\psi_{-\varepsilon_+, -\mathbf{p}_+}^{(+)} = \frac{C^{(+)}}{\sqrt{2\varepsilon_+}} e^{-i\mathbf{p}_+ \mathbf{r}} \left(1 + \frac{i\alpha \nabla}{2\varepsilon_+} \right) F(-i\nu, 1, i(p_+ r + \mathbf{p}_+ \mathbf{r})) u(-\mathbf{p}_+), \quad (\text{X.47})$$

$$C^{(+)} = e^{-\pi\nu/2} \Gamma(1 + i\nu).$$

³⁹In this section $p_{\pm} = |\mathbf{p}_{\pm}|$, $q = |\mathbf{q}|$.

Note that the necessity of including the terms $\sim 1/\varepsilon$ in (95.3), (95.4) is connected with the matrix structure of M_{fi} (95.2). The matrix element $(\alpha)_{fi}$ is a vector with a direction close to $\mathbf{k}$. Therefore the principal term in $(\alpha\mathbf{e})_{fi}$ turns out to be small, while the correction terms are of the same order of magnitude as it.

Substituting (95.3), (95.4) in (95.2) and neglecting terms $\sim 1/(\varepsilon_+\varepsilon_-)$, we find

$$M_{fi} = \frac{N}{2\sqrt{\varepsilon_+\varepsilon_-}} u^*(\mathbf{p}_-)\{(\mathbf{e}\alpha)I + (\mathbf{e}\alpha)(\alpha\mathbf{I}_+) + (\alpha\mathbf{I}_-)(\mathbf{e}\alpha)\}u(-\mathbf{p}_+), \quad (\text{X.48})$$

where

$$N = C^{(+)}C^{(-)} = \frac{\pi\nu}{\sinh \pi\nu}, \quad (\text{X.49})$$

$$I = \int e^{-i\mathbf{q}\mathbf{r}} F_-^* F_+ d^3x, \quad (\text{X.50})$$

$$\mathbf{I}_+ = \frac{i}{2\varepsilon_+} \int e^{-i\mathbf{q}\mathbf{r}} F_-^* \nabla F_+ d^3x, \quad \mathbf{I}_- = \frac{i}{2\varepsilon_-} \int e^{-i\mathbf{q}\mathbf{r}} (\nabla F_-)^* F_+ d^3x,$$

$$\mathbf{q} = \mathbf{p}_+ + \mathbf{p}_- - \mathbf{k}$$

(F_- and F_+ denote for brevity the hypergeometric functions appearing in (95.3) and (95.4)). Let us immediately note that the integrals $I, \mathbf{I}_+, \mathbf{I}_-$ are connected by one identity: from

$$\int \nabla(e^{-i\mathbf{q}\mathbf{r}} F_-^* F_+) d^3x = 0$$

we have

$$\mathbf{q}I + 2\varepsilon_+\mathbf{I}_+ + 2\varepsilon_-\mathbf{I}_- = 0. \quad (\text{X.51})$$

The square $|M_{fi}|^2$ is averaged over the polarisations of the incident photon and summed over the spin directions of the electron and positron⁴⁰. This is accomplished by the replacement of the tensor:

$$e_i e_k^* \rightarrow \frac{1}{2}(\delta_{ik} - n_i n_k), \quad \mathbf{n} = \frac{\mathbf{k}}{\omega},$$

and of the bispinor products:

$$u_\pm \bar{u}_\pm \rightarrow 2\rho_\pm = (\varepsilon_\pm \gamma^0 - \mathbf{p}_\pm \boldsymbol{\gamma} \mp m).$$

Replacing also $\alpha = \gamma^0 \boldsymbol{\gamma}$, we find

$$|M_{fi}|^2 \rightarrow \frac{N^2}{2\varepsilon_+\varepsilon_-} \{\text{Sp } \rho_- Q \rho_+ \bar{Q} - \text{Sp } \rho_- (\mathbf{n}Q) \rho_+ (\mathbf{n}\bar{Q})\},$$

$$Q = \gamma I - \gamma^0 \boldsymbol{\gamma} (\gamma \mathbf{I}_+) - \gamma^0 (\gamma \mathbf{I}_-) \boldsymbol{\gamma},$$

$$\bar{Q} = \gamma I^* - \gamma^0 \boldsymbol{\gamma} (\gamma \mathbf{I}_+^*) - \gamma^0 (\gamma \mathbf{I}_-^*) \boldsymbol{\gamma}.$$

⁴⁰For calculations with account of the polarisation of all particles see *Olsen H., Maximon L.* Phys. Rev. 1959. V. 114. P. 887, and also the book by V. Ya. Baier, V. M. Katkov and V. S. Fadin indicated on p. 454.

Let us write out immediately the result, obtained after the appropriate approximations, for the ultrarelativistic case of interest in the region of small angles

$$\theta_{\pm} \sim m/\varepsilon \ll 1. \quad (\text{X.52})$$

We introduce auxiliary vectors:

$$\delta_{\pm} = \frac{1}{m} (\mathbf{p}_{\pm})_{\perp}, \quad \delta_{\pm} = \frac{\varepsilon_{\pm}}{m} \theta_{\pm} \quad (\text{X.53})$$

(the subscript $\perp$ denotes the component perpendicular to the direction of $\mathbf{k}$). With their help the result is written in the form

$$|M_{fi}|^2 \rightarrow \frac{N^2}{4} \left\{ \frac{m^2 \omega^2}{\varepsilon_+^2 \varepsilon_-^2} |I|^2 + 2 \left| I \frac{m \delta_+}{2 \varepsilon_+} + \mathbf{I}_+ \right|^2 + 2 \left| I \frac{m \delta_-}{2 \varepsilon_-} + \mathbf{I}_- \right|^2 \right\}. \quad (\text{X.54})$$

Here it is taken into account that $I \sim \varepsilon/(q) \cdot I_{\pm}$ (as is evident from (95.8)), and terms of higher order in m/ε are omitted.

The integrals $\mathbf{I}_{\pm}$ can be represented in the form

$$\mathbf{I}_{\pm} = i \frac{p_{\pm}}{2 \varepsilon_{\pm}} \frac{\partial J}{\partial \mathbf{p}_{\pm}},$$

$$J = \int \frac{e^{-i\mathbf{q}\mathbf{r}}}{r} F(-i\nu, 1, i(p_+r + \mathbf{p}_+\mathbf{r})) F(i\nu, 1, i(p_-r + \mathbf{p}_-\mathbf{r})) d^3x. \quad (\text{X.55})$$

The integral J is expressed through the complete hypergeometric function⁴¹:

$$J = \frac{4\pi}{q^2} \left(\frac{q^2 - 2\mathbf{p}_+\mathbf{q}}{q^2 - 2\mathbf{p}_-\mathbf{q}} \right)^{i\nu} F(-i\nu, i\nu, 1, z), \quad (\text{X.56})$$

$$z = 2 \frac{q^2(p_+p_- - \mathbf{p}_+\mathbf{p}_-) + 2(\mathbf{p}_+\mathbf{q})(\mathbf{p}_-\mathbf{q})}{(q^2 - 2\mathbf{p}_+\mathbf{q})(q^2 - 2\mathbf{p}_-\mathbf{q})}.$$

Differentiation with respect to $\mathbf{p}_{\pm}$ is to be carried out at a given value of the parameter $\mathbf{q}$, and only afterwards can one set $\mathbf{q} = \mathbf{p}_+ + \mathbf{p}_- - \mathbf{k}$. We give the result in a form in which the approximations corresponding to the ultrarelativistic case and the conditions (95.9) have already been made:

$$\mathbf{I}_{\pm} = \frac{4\pi}{q^2} \frac{\varepsilon_{\mp}}{m^2 \omega} \left(\frac{\varepsilon_{\pm} \xi_{\pm}}{\varepsilon_{\mp} \xi_{\mp}} \right)^{i\nu} \left\{ \pm \nu \mathbf{q} \xi_{\mp} F(z) + i \frac{q^2}{m^2} F'(z) (\mathbf{q} \xi_{\mp} - m \delta_{\pm}) \right\}. \quad (\text{X.57})$$

Here the following notation has been introduced:

$$\xi_{\pm} = \frac{1}{1 + \delta_{\pm}^2}, \quad z = 1 - \frac{q^2}{m^2} \xi_+ \xi_-, \quad F(z) = F(-i\nu, i\nu, 1, z) \quad (\text{X.58})$$

($F(z)$ is a real function). The integral I is then computed directly from (95.8).

⁴¹For the computation see the paper by Nordsieck indicated on p. 438.

Substituting the values of the integrals in (95.11), and then in (95.1), we obtain the desired differential cross section. The final formula:

$$\begin{aligned}
 d\sigma = & \frac{4}{\pi} \left(\frac{\pi\nu}{\sinh \pi\nu} \right)^2 Z^2 \alpha r_e^2 \frac{m^2}{q^4 \omega^3} \delta_+ d\delta_+ \cdot \delta_- d\delta_- \cdot d\varphi d\varepsilon_+ \left\{ F^2(z) \times \right. \\
 & \times [-2\varepsilon_+ \varepsilon_- (\delta_+^2 \xi_+^2 + \delta_-^2 \xi_-^2) + \omega^2 (\delta_+^2 + \delta_-^2) \xi_+ \xi_- + 2(\varepsilon_+^2 + \varepsilon_-^2) \times \\
 & \times \delta_+ \delta_- \xi_+ \xi_- \cos \varphi] + \frac{q^4}{m^4} \frac{\xi_+^2 \xi_-^2}{\nu^2} F'^2(z) [-2\varepsilon_+ \varepsilon_- (\delta_+^2 \xi_+^2 + \delta_-^2 \xi_-^2) + \\
 & \left. + \omega^2 (1 + \delta_+^2 \delta_-^2) \xi_+ \xi_- - 2(\varepsilon_+^2 + \varepsilon_-^2) \delta_+ \delta_- \xi_+ \xi_- \cos \varphi] \right\}. \quad (\text{X.59})
 \end{aligned}$$

When $\nu \rightarrow 0$

$$\pi\nu/(\sinh \pi\nu) \rightarrow 1, \quad F(z) \rightarrow 1, \quad F'(z) \approx \nu^2 \rightarrow 0.$$

Expression (95.16) reduces in this case, as it should, to the Bethe–Heitler formula (94.3), corresponding to the Born approximation. It reduces to the same formula also for arbitrary ν , if the emergence angles of the pair satisfy the conditions

$$|\delta_+ - \delta_-| \ll 1, \quad |\pi - \varphi| \ll 1.$$

Indeed, in this case $q \ll m$, so that the second term in the curly brackets in (95.16) can be dropped in view of the presence in it of the extra (compared with the first term) factor $(q/m)^4$. In the first term, however, we have (noting that $(1 - z) \sim q^2/m^2 \ll 1$)

$$F(z) \rightarrow F(1) \equiv F(-i\nu, i\nu, 1, 1) = \frac{1}{\Gamma(1 - i\nu)\Gamma(1 + i\nu)} = \frac{\sinh \pi\nu}{\pi\nu}, \quad (\text{X.60})$$

as a result of which the analogous factor before the curly brackets cancels⁴².

Let us now turn to the integration of the cross section over the directions of emergence of the pair.

We divide the integration over angles into two regions, I and II, in which respectively

$$\text{I) } 1 - z > 1 - z_1, \quad \text{II) } 1 - z < 1 - z_1,$$

where z_1 is some value for which $1 \gg 1 - z_1 \gg (m/\varepsilon)^2$.

Since in region II $1 - z \ll 1$, $q \ll m$, then, according to what was said above, $d\sigma \approx d\sigma_B \equiv d\sigma|_{\nu \rightarrow 0}$, where $d\sigma_B$ is the cross section in the Born approximation. Therefore the integral over angles:

$$d\sigma_{\varepsilon_+} \equiv \int d\sigma = \int_I d\sigma + \int_{II} d\sigma|_{\nu \rightarrow 0} = (d\sigma_{\varepsilon_+})_B + \int_I (d\sigma - d\sigma|_{\nu \rightarrow 0}), \quad (\text{X.61})$$

where $(d\sigma_{\varepsilon_+})_B$ is the Born cross section (94.5) integrated over angles.

In region I we have

$$q^2/m^2 \approx \delta_+^2 + \delta_-^2 + 2\delta_+ \delta_- \cos \varphi.$$

⁴²This value of the function can be obtained from formula (e. 7) (see II), connecting hypergeometric functions of the arguments z and $1 - z$.

From the variables δ_+ , δ_- , φ let us go over to the variables ξ_+ , ξ_- , z . By direct computation of the Jacobian of the transformation we find

$$\delta_+ d\delta_+ \cdot \delta_- d\delta_- \cdot d\varphi \rightarrow \frac{\varepsilon_+ \varepsilon_-}{8m^2} \frac{d\xi_+ d\xi_- d\varphi}{(\xi_+ \xi_-)^3 \sin \varphi},$$

where

$$1 - z = \frac{q^2}{m^2} \xi_+ \xi_- = \xi_+ + \xi_- - 2\xi_+ \xi_- + 2\sqrt{\xi_+ \xi_- (1 - \xi_+)(1 - \xi_-)} \cos \varphi.$$

Expressing from this $\cos \varphi$ and $\sin \varphi$ and substituting in (95.16), after simple algebraic transformations we obtain

$$\begin{aligned} d\sigma = A d\varepsilon_+ & \frac{2 d\xi_+ d\xi_- dz}{[z(1-z) - (1-z)(\xi_+ + \xi_- - 1)^2 - z(\xi_+ - \xi_-)^2]^{1/2}} \times \\ & \times \left\{ \frac{F^2(z)}{1-z} [(\varepsilon_+^2 + \varepsilon_-^2)z + 2\varepsilon_+ \varepsilon_- (\xi_+ - \xi_- - 1)^2] + \right. \\ & \left. + \frac{F'^2(z)}{\nu^2} [(\varepsilon_+^2 + \varepsilon_-^2)z + 2\varepsilon_+ \varepsilon_- (\xi_+ + \xi_- - 1)^2] \right\}, \\ A = & \left(\frac{\pi \nu}{\sinh \pi \nu} \right)^2 \frac{Z^2 \alpha r_e^2}{2\pi \omega^3}. \end{aligned}$$

Finally, we introduce instead of ξ_+ , ξ_- new "spherical" variables χ , ψ :

$$\begin{aligned} \xi_+ + \xi_- - 1 &= \sqrt{z} \sin \chi \cos \psi; \\ \xi_+ - \xi_- &= \sqrt{1-z} \sin \chi \sin \psi; \\ 0 &\leq \chi \leq \pi/2, \quad 0 \leq \psi \leq 2\pi; \\ 2 d\xi_+ d\xi_- &\rightarrow \sqrt{z(1-z)} \sin \chi \cos \chi d\chi d\psi. \end{aligned}$$

The indicated intervals of variation of χ and ψ correspond to the variation of ξ_+ , ξ_- from 0 to 1, i.e. δ_+ , δ_- (or, what is the same, θ_+ , θ_-) from 0 to ∞ ; the fast convergence of the integral permits such an extension of the region of variation of angles. After the transformation, the square root in the denominator reduces to $\sqrt{z(1-z)} \cos \chi$; the integration over $d\chi d\psi$ is elementary and gives

$$d\sigma = 2A \cdot 2\pi dz \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \left[\frac{F^2(z)}{1-z} + \frac{z}{\nu^2} F'^2(z) \right] d\varepsilon_+.$$

Here an extra factor of 2 has been introduced, taking into account the fact that the integration over z will be carried out from 0 to z_1 , whereas as the azimuth φ varies from 0 to π and from π to 2π each value of z is traversed twice.

The integration over dz is performed with the aid of formula (92.14), which for $\nu' = -\nu$ (and correspondingly real $F(z)$) takes the form

$$\frac{F^2}{1-z} + \frac{z}{\nu^2} F'^2 = \frac{1}{\nu^2} \frac{d}{dz} (z F F').$$

The integral of this expression is $z_1 F(z_1) F'(z_1) / \nu^2$. The value $z_1 F(z_1) \approx F(1)$, while the limiting expression for $F'(z_1 \rightarrow 1)$ is given by the formula⁴³

$$\frac{1}{\nu^2} F'(z) = F(1 - i\nu, 1 + i\nu, 2, z) \approx -[\ln(1 - z) + 2f(\nu)] \frac{\sinh \pi \nu}{\pi \nu},$$

where

$$f(\nu) = \frac{1}{2} [\Psi(1 + i\nu) + \Psi(1 - i\nu) - 2\Psi(1)] = \nu^2 \sum_{n=1}^{\infty} \frac{1}{n(n^2 + \nu^2)}, \quad (\text{X.62})$$

$$\Psi(z) = \Gamma'(z) / \Gamma(z).$$

Substituting all the found expressions in (95.18), we obtain the following final formula:

$$d\sigma_{\varepsilon_+} = 4Z^2 \alpha r_e^2 \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \left[\ln \frac{2\varepsilon_+ \varepsilon_-}{m\omega} - \frac{1}{2} - f(\alpha Z) \right] \frac{d\varepsilon_+}{\omega^3}. \quad (\text{X.63})$$

The total cross section of pair production by a photon with energy ω :

$$\sigma = \frac{28}{9} Z^2 \alpha r_e^2 \left[\ln \frac{2\omega}{m} - \frac{109}{42} - f(\alpha Z) \right]. \quad (\text{X.64})$$

We see that in these formulas the changes reduce to the subtraction from the logarithm of a universal function of the atomic number $f(\alpha Z)$. In Fig. 18 a graph of this function is given. When $\nu \ll 1$ the function $f(\nu) \approx 1.2\nu^2$.

§96. Exact theory of bremsstrahlung in the ultrarelativistic case

The matrix element for bremsstrahlung

$$M_{fi} = \int \psi_{\varepsilon' \mathbf{p}'}^{(-)*} (\alpha \mathbf{e}^*) e^{-i\mathbf{k}\mathbf{r}} \psi_{\varepsilon \mathbf{p}}^{(+)} d^3x; \quad (\text{X.65})$$

the wave functions of the initial $(\varepsilon, \mathbf{p})$ and final $(\varepsilon', \mathbf{p}')$ electrons contain in their asymptotics respectively an incoming and outgoing spherical wave. The computation of this integral is analogous to the computation of the matrix element (95.2). We shall, however, present here a different method of computing the bremsstrahlung cross section, based on the quasi-classicality of the process and not utilising the explicit form of the wave functions of the electron in the field of the nucleus; in this sense the method is not connected with the specific form of the potential of the field (*V. N. Baier, V. M. Katkov, 1968*).

In the process of bremsstrahlung the nucleus transfers to the electron and photon a momentum $\mathbf{q} = \mathbf{p}' + \mathbf{k} - \mathbf{p}$. As in the problem of pair production, one must distinguish two regions of values of the momentum transfer $q_{\perp}$, transverse with respect to $\mathbf{p}$:

$$\text{I) } m \gtrsim q_{\perp} \gg \omega m^2 / \varepsilon^2, \quad \text{II) } q_{\perp} \sim \omega m^2 / \varepsilon^2 \ll m. \quad (\text{X.66})$$

⁴³The derivation of these formulas can be found in the appendix to the paper by *Davies H., Bethe H. A., Maximon L. C. Phys. Rev. 1954. V. 93. P. 788.*

It is evident that in region I the cross section of photon emission is given by its Born value: for such $q_{\perp}$ the change in the recoil momentum of the nucleus upon radiation is inessential, as will be shown in § 98 (see the derivation of condition (98.10)). Therefore in region I the cross section of the process equals the product of the exact cross section of scattering of the electron in the field of a fixed nucleus and the probability of photon emission, which is independent of the form of the field. But according to (80.10) the scattering cross section in a Coulomb field for small angles coincides with its Born value. The same therefore applies to the cross section of the whole process in region I.

Thus, only region II requires special consideration. To small momentum transfers there corresponds the passage of the electron past the nucleus at large impact parameters: $\rho \sim 1/q_{\perp} \gtrsim \varepsilon/m^2$

$\gtrsim \varepsilon/m^2$. But at such distances the motion of the electron is certainly quasi-classical, as one can easily verify by a simple application of the usual quasi-classicality condition (46.7) (see III) to the ultrarelativistic equation (39.5).

The quasi-classicality of the motion permits applying the method already used in § 90 for magnetobremstrahlung radiation. In this case the expression (90.7) represents the probability of emission during a single passage of the electron past the nucleus.

For the function L that appeared in § 90 the formula (90.18) remains in force; the sole distinction consists in the form of the quasi-classical trajectory of the electron $\mathbf{r} = \mathbf{r}(t)$, along which the difference $\mathbf{r}_2 - \mathbf{r}_1$ is computed.

At large impact parameters the field of the nucleus can be considered weak. In the zeroth approximation the trajectory is a straight line passing at a distance ρ from the centre. In the next approximation we have the equation of motion (cf. I, § 20)

$$\frac{d\mathbf{p}}{dt} = -\frac{\rho}{r} \frac{dU}{dr},$$

where ρ is a vector in the xy plane, perpendicular to the initial momentum of the electron, and for r on the right-hand side of the equation one should take the zeroth-approximation function:

$$r \approx \sqrt{\rho^2 + v^2 t^2} \approx \sqrt{\rho^2 + t^2}.$$

Consequently,

$$\mathbf{p}(t) - \mathbf{p}_1 = -\rho \int_{t_1}^t \frac{dU}{dr} \frac{dt}{r}. \quad (\text{X.67})$$

With sufficient accuracy the velocity $\mathbf{v}(t) = \mathbf{p}(t)/\varepsilon$ (where the energy ε depends only on the magnitude, but not on the direction of $\mathbf{p}$) can be considered constant. One more integration then gives

$$\mathbf{r}(t) - \mathbf{r}_1 = \mathbf{v}_1(t - t_1) - \frac{1}{\varepsilon} \int_{t_1}^t [\mathbf{p}(t') - \mathbf{p}_1] dt'. \quad (\text{X.68})$$

Setting $t_1 = -\infty$, the quantities $\mathbf{p}_1 = \mathbf{p}(-\infty) \equiv \mathbf{p}$ and $\mathbf{v} = \mathbf{p}/\varepsilon$ will be the initial momentum and velocity of the electron.

Let us represent the probability (90.7) in the form

$$dw = |a(\rho)|^2 \frac{d^3k}{(2\pi)^3}, \quad (\text{X.69})$$

where

$$a(\boldsymbol{\rho}) = e \sqrt{\frac{2\pi}{\omega}} \int_{-\infty}^{\infty} R(t) \exp \left\{ i \frac{\varepsilon}{\varepsilon'} [\omega t - \mathbf{k} \mathbf{r}(t)] \right\} dt, \quad (\text{X.70})$$

$$R(t) = \frac{u_{\varepsilon'}^* \mathbf{p}'}{\sqrt{2\varepsilon'}} (\boldsymbol{\alpha} \mathbf{e}^*) \frac{u_{\varepsilon} \mathbf{p}}{\sqrt{2\varepsilon}},$$

where $\varepsilon' = \varepsilon - \omega$, $\mathbf{p}'(t) = \mathbf{p}(t) - \mathbf{k}$. The classical function $\mathbf{p}(t)$ is given by formula (96.3). If $\mathbf{p}$ is the initial momentum of the electron, then for a Coulomb field ($U = -\nu/r$, $\nu = Z\alpha$) we have

$$\mathbf{p}(t) = \mathbf{p} - \frac{\nu \boldsymbol{\rho}}{\rho^2} \left[\frac{t}{\sqrt{\rho^2 + t^2}} + 1 \right],$$

$$\mathbf{r}(t) = \frac{\mathbf{p}}{\varepsilon} t - \frac{\boldsymbol{\rho}}{\rho^2} \frac{\nu}{\varepsilon} \left[\sqrt{\rho^2 + t^2} + t \right].$$

Introducing the momentum transfer in classical scattering

$$\Delta = \mathbf{p}(\infty) - \mathbf{p}(-\infty) = -\frac{2\rho\nu}{\rho^2}, \quad (\text{X.71})$$

we can rewrite these formulas as

$$\mathbf{p}(t) = \mathbf{p} + \frac{1}{2} \Delta \left[\frac{t}{\sqrt{\rho^2 + t^2}} + 1 \right], \quad (\text{X.72})$$

$$\mathbf{r}(t) = \left(\mathbf{p} + \frac{1}{2} \Delta \right) \frac{t}{\varepsilon} + \frac{\Delta}{2\varepsilon} \sqrt{t^2 + \rho^2}.$$

Using now formula (90.20) for $R(t)$ and the expressions (96.8) for $\mathbf{p}(t)$ and $\mathbf{r}(t)$, one can carry out the integration over time in (96.6). It is accomplished by the introduction of the variable

$$\xi = -\frac{\varepsilon}{\varepsilon'} [\omega t - \mathbf{k} \mathbf{r}(t)]$$

instead of t and the use of the formula

$$\int_{-\infty}^{\infty} \frac{\xi e^{-i\xi} d\xi}{\sqrt{\chi^2 + \xi^2}} = 2i\chi K_1(\chi),$$

where K_1 is the MacDonald function. In the full execution of this computation, however, there is no necessity, since we require the expression $a(\boldsymbol{\rho})$ only for small values of the independent parameter Δ ($\Delta \ll m$). In this case we find

$$a(\boldsymbol{\rho}) = w_f^* \mathbf{D} w_i \cdot \Delta \chi K_1(\chi), \quad (\text{X.73})$$

where

$$\chi = \rho \frac{\omega \varepsilon}{\varepsilon'} (1 - \mathbf{n} \mathbf{v}),$$

$\mathbf{n} = \mathbf{k}/\omega$, and $\mathbf{D}$ is some function of $\mathbf{p}$, ε and $\mathbf{k}$ (but not $\boldsymbol{\rho}$), the precise form of which is inessential⁴⁴.

⁴⁴The spinors w_i and w_f can be considered constant during the integration, i.e. one can neglect the change of polarisation of the electron during its classical ultrarelativistic motion. This follows from the equations obtained in § 41.

Since in the ultrarelativistic case the photon is emitted at a small angle θ to the direction of velocity of the electron, we have

$$\chi \approx \rho \frac{\varepsilon}{\varepsilon'} \omega \left(1 - v + \frac{\theta^2}{2} \right),$$

or

$$\chi = \rho \frac{\omega m^2}{2\varepsilon\varepsilon'} (1 + \delta^2), \quad \delta = \frac{\theta\varepsilon}{m}. \quad (\text{X.74})$$

It was already mentioned that (96.5) is the probability of photon emission during a single passage of the electron past the nucleus at impact parameter ρ . The cross section of emission of a photon with given frequency and direction is obtained by multiplying this probability by $dp_x dp_y / v \approx dp_x dp_y \equiv d^2\rho$ and integrating over the impact parameters:

$$d\sigma = \frac{d^3k}{(2\pi)^3} \int |a(\rho)|^2 d^2\rho. \quad (\text{X.75})$$

One should not, however, think that this formula without the integration over $d^2\rho$ would also give the distribution of the final electrons in direction. The deflection of the electron during its motion along the classical orbit is determined uniquely by the external field and certainly does not coincide with the uncertain quantum-mechanical deflection (and the limiting value $\mathbf{p}'(\infty)$ of the classical function $\mathbf{p}'(t)$ does not therefore coincide with the real values of the momentum of the electron). To find this distribution it is necessary, therefore, to expand the wave function of the electrons in plane waves.

As is evident from (96.11), $a(\rho)$ is the amplitude of photon emission in a collision at impact parameter ρ . From the expressions (96.5), (96.6) this amplitude is determined only up to a phase factor. The latter is, obviously, $e^{-ik\rho}$ —by virtue of the presence of the time-independent term $r_\perp(\infty) = \rho$ in $\mathbf{r}(t)$, this constant factor must appear in $V_{fi}(t)$ and can be taken outside the integral sign. Since it is not an operator, it is not affected by commutation operations and, thus, the amplitude of the emission process is

$$e^{-ik\rho} a(\rho), \quad (\text{X.76})$$

where $a(\rho)$ is given by expression (96.9).

Let the electron be described as $z \rightarrow -\infty$ by a plane wave with momentum $\mathbf{p}$, directed along the z axis. This means that the wave function of the electron as $z \rightarrow -\infty$ does not depend on x and y and reduces to a constant, which can be set equal to 1.

Then the wave function of the electron that has passed through the field, as $z \rightarrow \infty$, equals⁴⁵

$$\psi(\infty) = S(\rho) = \exp \left\{ -i \int_{-\infty}^{\infty} U(x, y, z) dz \right\}. \quad (\text{X.77})$$

⁴⁵Cf. III, (131.4). We have in mind here the analogy between equation (39.5) (in which we set $p^2 \approx \varepsilon^2$) and the non-relativistic Schrödinger equation. Taking into account the difference of coefficients in these equations, it is easy to see that in our case the condition (131.1) (see III) of applicability of equation (131.4) (see III) is indeed satisfied. The fact that this formula does not apply to the region of arbitrarily large z is inessential for the same reasons as in III, § 131.

On the other hand, by the meaning of the transition amplitude (96.12) the wave function of the electron that has passed through the field and emitted a photon is

$$e^{-i\mathbf{k}\rho} a(\rho) S(\rho). \quad (\text{X.78})$$

The amplitude of the process of photon emission, in which the electron remains in a state with definite momentum $\mathbf{p}'$, is given by the corresponding Fourier component of the function (96.14), i.e.

$$a(\mathbf{q}_\perp) = \int e^{-i\mathbf{p}'\rho} e^{-i\mathbf{k}\rho} a(\rho) S(\rho) d^2\rho = \int e^{-i\mathbf{q}_\perp\rho} a(\rho) S(\rho) d^2\rho, \quad (\text{X.79})$$

where $\mathbf{q}_\perp$ is the transverse component of the momentum transfer vector to the nucleus (cf. III, (131.7)). The cross section of scattering with given value of $\mathbf{q}_\perp$ is then

$$d\sigma = |a(\mathbf{q}_\perp)|^2 \frac{d^3k}{(2\pi)^3} \frac{d^2q_\perp}{(2\pi)^2}. \quad (\text{X.80})$$

Let us now compute $S(\rho)$. In the case under consideration of a Coulomb field the integral in the exponent diverges, in correspondence with the divergence of the phase in Coulomb scattering. Therefore the integral must be taken between finite limits:

$$\int_{-R}^R U dz = -2\nu \int_0^R \frac{dz}{\sqrt{\rho^2 + z^2}} = -2\nu [\ln(R + \sqrt{R^2 + \rho^2}) - \ln \rho] \approx -2\nu \ln R + 2\nu \ln \rho$$

($R \gg \rho$). The first, constant, term is inessential, so that

$$S(\rho) = \exp(-2i\nu \ln \rho) = \rho^{-2i\nu}. \quad (\text{X.81})$$

Substituting (96.9), (96.17) in (96.15) and integrating over the directions of the vector ρ in the xy plane, we find

$$a(\mathbf{q}_\perp) \propto \nu \int_0^\infty \rho^{-2i\nu} K_1(\chi) J_1(q_\perp \rho) \rho d\rho, \quad (\text{X.82})$$

where J_1 is the Bessel function. The factors not containing $\nu = Z\alpha$ are not written out here.

We see that the dependence of the amplitude $a(\mathbf{q}_\perp)$ (and consequently also of the cross section (96.16)) on ν is contained in a separate factor. On the other hand, when $\nu \rightarrow 0$ the cross section must tend to its Born value. Therefore it is clear that the cross section will differ from the Born one only by a factor that is independent of the polarisation of the electron and does not affect the polarisation effects.

The integral (96.18) can be expressed through a hypergeometric function with the aid of the formula

$$\int_0^\infty x^{-\lambda} K_1(ax) J_1(bx) x dx = \frac{b \Gamma(2 - \lambda/2) \Gamma(1 - \lambda/2)}{2^\lambda a^{3-\lambda}} \left(1 + \frac{b^2}{a^2}\right)^{-1+\lambda/2} F\left(\frac{\lambda}{2}, 1 - \frac{\lambda}{2}, 2, \frac{b^2}{a^2 + b^2}\right)$$

This gives

$$a(\mathbf{q}_\perp) \propto \nu(1 - i\nu) \left(\frac{q}{2}\right)^{2i\nu} \Gamma^2(1 - i\nu) F(i\nu, 1 - i\nu, 2, z), \quad (\text{X.83})$$

where

$$z = 1 - \frac{m^4 \omega^2}{4q^2 \varepsilon^2 \varepsilon'^2} (1 + \delta^2)^2, \quad \delta = \frac{\varepsilon \theta}{m}; \quad (\text{X.84})$$

here it is used that in region II (see (96.2)) the component of the vector $\mathbf{q}$ parallel to $\mathbf{p}$ is

$$q_{\parallel}^2 = q^2 - q_{\perp}^2 \approx \frac{m^4 \omega^2}{4\varepsilon^2 \varepsilon'^2} (1 + \delta^2)^2. \quad (\text{X.85})$$

In this one can easily convince oneself, if one takes into account that in the indicated region the angles between the momenta $\mathbf{p}$, $\mathbf{p}'$ and $\mathbf{k}$ satisfy conditions (93.15).

The hypergeometric function in (96.19) can be reduced to the function $F(z)$ (95.15) with the aid of the formula

$$F(a, b+1, c+1, z) = \frac{c}{c-a} F(a, b, c, z) + \frac{c(1-z)}{b(a-c)} F'(a, b, c, z).$$

The final result takes then the form

$$d\sigma = d\sigma_B \frac{1}{F^2(1)} \left[F^2(z) + \frac{(1-z)^2}{v^2} F'^2(z) \right], \quad (\text{X.86})$$

where $d\sigma_B$ is the Born cross section (93.13) (*H. A. Bethe, L. Maximon, 1954*). When $q \gg m^2/\varepsilon$ we have $z \approx 1$, so that the entire coefficient

at $d\sigma_B$ tends to unity. In this sense formula (96.22), derived for region II, is automatically valid for all $q \lesssim m$. When $q \lesssim m^2/\varepsilon$ and the correction factor in (96.22) differs from unity, the vectors $\mathbf{p}$, $\mathbf{p}'$, $\mathbf{k}$ are almost coplanar and the quantities δ and δ' are almost equal to each other; this has already been taken into account in (96.22). Thus, q in expression (96.20) for z can be rewritten as

$$\frac{q^2}{m^2} = \delta^2 + \delta'^2 - 2\delta\delta' \cos \varphi + \frac{m^2 \omega^2}{4\varepsilon^2 \varepsilon'^2} (1 + \delta^2)^2, \quad (\text{X.87})$$

i.e. one can set $\delta = \delta'$ in the second term in (93.14), but not in the first term, which does not contain the small coefficient ($\sim m^2/\varepsilon^2$).

For finding the integral (over angles) cross section of radiation there is no need to carry out the integration anew, as is clear from the following considerations (*H. Olsen, 1955*). The different directions of $\mathbf{p}'$ (at given energy ε') correspond to the degeneracy of the final state of the electron. It is evident that the result of summation over states belonging to one degenerate level does not depend on the manner in which the complete set of these states is chosen. We can therefore use for the purposes of summation over directions of $\mathbf{p}'$ the system of functions $\psi_{\varepsilon' \mathbf{p}'}^{(-)}$ instead of the system $\psi_{\varepsilon' \mathbf{p}'}^{(+)}$ (necessary for computing the differential cross section), i.e. define the matrix element of bremsstrahlung as

$$M_{fi}^{\text{brem}} = \int \psi_{\varepsilon' \mathbf{p}'}^{(+)*} (\alpha \mathbf{e}^*) e^{-i\mathbf{k}\mathbf{r}} \psi_{\varepsilon \mathbf{p}}^{(+)} d^3x.$$

It is easy to verify that this integral coincides with the integral $(M_{fi}^{\text{pair}})^*$, if in the latter one replaces the parameters of the wave functions according to

$$\mathbf{p}_+, p_+, \varepsilon_+ \rightarrow -\mathbf{p}_-, -p_-, -\varepsilon_-; \quad \mathbf{p}_-, p_-, \varepsilon_- \rightarrow \mathbf{p}', p', \varepsilon'; \quad \mathbf{k} \rightarrow -\mathbf{k}$$

(and also replaces the variables of integration: $\mathbf{r} \rightarrow -\mathbf{r}$).

From this it is clear that the integral (over angles) bremsstrahlung cross section can be obtained from the integral pair-production cross section (95.20) by multiplying the latter by

$$\frac{\omega^2 d\omega}{p_+^2} \frac{d\omega}{\omega} \approx \frac{\omega^2 d\omega}{\varepsilon_+^2 \varepsilon_-}$$

(cf. (91.6)) and replacing $\varepsilon_+ \rightarrow -\varepsilon$, $\varepsilon_- \rightarrow \varepsilon'$. Thus, we find

$$d\sigma = 4Z^2 \alpha r_e^2 \frac{\varepsilon'}{\varepsilon} \left[\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} - \frac{2}{3} \right] \left(\ln \frac{2\varepsilon\varepsilon'}{m\omega} - \frac{1}{2} - f(\alpha Z) \right) \frac{d\omega}{\omega}. \quad (\text{X.88})$$

We see that the corrections to the Born formulas for the integral cross sections of bremsstrahlung and pair production are given by one and the same function $f(\alpha Z)$.

Formula (96.24), not connected with any restrictions on the magnitude of $Z\alpha$, permits passage to the classical limit $\hbar \rightarrow 0$, $Z\alpha \rightarrow \infty$. In this limit one must also set $\varepsilon \approx \varepsilon'$. Bearing in mind the asymptotic expression $\Psi(z) \approx \ln z$ for $|z| \rightarrow \infty$ and the value $\Psi(1) = -C$ (C is the Euler constant), we find for the effective retardation

$$\hbar\omega d\sigma = \frac{16Z^2 r_e^2 e^2}{3c} \left[\ln \frac{2\varepsilon^2}{mc\omega Ze^2} - \frac{1}{2} - C \right] d\omega. \quad (\text{X.89})$$

This expression, not containing $\hbar$, is the classical spectral distribution of the intensity of bremsstrahlung.

§97. Bremsstrahlung of an electron on an electron in the ultrarelativistic case

The bremsstrahlung of an electron on an electron is depicted by eight Feynman diagrams: four diagrams

and four “exchange” diagrams, obtained from the depicted ones by the interchange of p'_1 and p'_2 . We present here the results of the calculations for the ultrarelativistic case (*G. Altarelli, F. Buccella, 1964; V. N. Baier, V. S. Fadin, V. A. Khoze, 1966*)⁴⁶.

In the laboratory frame (the rest frame of one of the initial electrons—say, the second), the cross section of radiation integrated over the photon directions can be represented as a sum $d\sigma = d\sigma^{(1)} + d\sigma^{(2)}$, where

$$d\sigma^{(1)} = 4\alpha r_e^2 \frac{d\omega}{\omega} \frac{\varepsilon - \omega}{\varepsilon} \left[\frac{\varepsilon}{\varepsilon - \omega} + \frac{\varepsilon - \omega}{\varepsilon} - \frac{2}{3} \right] \left(\ln \frac{2\varepsilon(\varepsilon - \omega)}{m\omega} - \frac{1}{2} \right); \quad (\text{X.90})$$

$$d\sigma^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{m d\omega}{\omega^2} \left\{ \left(4 - \frac{m}{\omega} + \frac{m^2}{4\omega^2} \right) \ln \frac{2\varepsilon}{m} - 2 + \frac{2m}{\omega} - \frac{5m^2}{8\omega^2} \right\} \omega \gg \frac{m}{2}; \quad (\text{X.91})$$

$$d\sigma^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{d\omega}{\omega} \left\{ 8 \left(1 - \frac{m}{\omega} + \frac{\omega^2}{m^2} \right) \ln \frac{\varepsilon}{m} - \left(1 - \frac{2\omega}{m} \right) \ln \left(1 - \frac{2\omega}{m} \right) \times \right. \\ \left. \times \left[\frac{m^3}{4\omega^3} - \frac{m^2}{2\omega^2} + \frac{3m}{\omega} - \frac{4\omega}{m} \right] - \frac{m^2}{2\omega^2} + \frac{3m}{\omega} - 2 + \frac{m^2}{2\omega^2} - \frac{4\omega^2}{m^2} \right\} \omega \leq \frac{m}{2}; \quad (\text{X.92})$$

⁴⁶The corresponding calculations can be found in the book by Baier, Katkov and Fadin indicated on p. 454.

(ε is the initial energy of the first electron).

The accuracy of these formulas is up to terms of relative order m/ε . With this accuracy it turns out that the contributions to the cross section from the various diagrams do not interfere with each other, and in this sense $d\sigma^{(1)}$ and $d\sigma^{(2)}$ correspond to radiation by each of the two electrons—respectively the fast electron and the recoil electron (diagrams (97.1a) and (97.1b)).

The exchange-type diagrams give the same contribution to the cross section as the “direct” diagrams. By virtue of the identity of the electrons the total contribution of the direct and exchange diagrams must be divided by 2; therefore one can consider only the contribution of the direct diagrams and not take into account the identity of the particles. For the collision of an electron with a positron, in place of exchange diagrams there appear annihilation diagrams. Their contribution, however, turns out to be of relative order m/ε , i.e. negligible. Therefore, with the indicated accuracy, the bremsstrahlung cross sections in electron-electron and electron-positron collisions are the same.

When $\omega \gg m$ the ratio

$$\frac{d\sigma^{(2)}}{d\sigma^{(1)}} \sim \frac{m}{\omega} \ll 1,$$

i.e. the radiation by the recoil electron is small compared with the radiation by the fast electron (when this ratio reaches order m/ε , formula (97.3), of course, loses its meaning). On the contrary, when $\omega \ll m$ both parts of the cross section are nearly equal:

$$d\sigma^{(1)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega}{\omega} \ln \frac{2\varepsilon^2}{m\omega}, \quad d\sigma^{(2)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega}{\omega} \ln \frac{\varepsilon}{\omega}, \quad \omega \ll m. \quad (\text{X.93})$$

For the validity of formulas (97.2)–(97.5) it is necessary that at least one of the electrons after the radiation remain ultrarelativistic. In other words, the photon frequency must be sufficiently far from the hard boundary of the spectrum, i.e. from the maximum frequency $\omega_{\max}$ that can be radiated. The final energy of the electrons will be minimal, and the photon energy maximal, when both electrons move after the radiation in the direction of the photon and have equal velocities. Then from the conservation laws we have

$$\varepsilon + m = \omega_{\max} + 2\varepsilon', \quad |\mathbf{p}| = \omega_{\max} + 2|\mathbf{p}'|.$$

Eliminating from this ε' and $\mathbf{p}'$, we obtain

$$(\varepsilon + m - \omega_{\max})^2 - (|\mathbf{p}| - \omega_{\max})^2 = 4m^2,$$

whence

$$\omega_{\max} = \frac{m(\varepsilon - m)}{m + \varepsilon - |\mathbf{p}|}. \quad (\text{X.94})$$

When $\varepsilon \gg m$ we have $\omega_{\max} \approx \varepsilon$. Thus, formulas (97.2)–(97.4) are valid under the condition

$$\omega_{\max} - \omega \sim \varepsilon - \omega \gg m. \quad (\text{X.95})$$

The cross section of radiation by the fast electron (97.2) coincides exactly with the cross section of radiation of an electron on a nucleus with $Z = 1$ (formula (93.17)). This coincidence is not accidental, and its causes are elucidated by analysis of the role of recoil in the radiation process.

In deriving formula (93.17) we neglected the recoil of the stationary particle (the nucleus)—replaced it by a constant external field. This reduces to the neglect of the time component of the 4-vector of momentum transfer $q = p' - p + k$ (the recoil energy). Let us show that in the ultrarelativistic case such neglect is permissible for radiation by an electron not only on a nucleus but also on an electron.

Let us write q^2 in the form

$$-q^2 = -(\varepsilon' + \omega - \varepsilon)^2 + (p'_{\parallel} + \omega - p_{\parallel})^2 + (\mathbf{p}'_{\perp} - \mathbf{p}_{\perp})^2, \quad (\text{X.96})$$

where the subscripts denote the components of the vectors $\mathbf{p}'$ and $\mathbf{p}$ (the initial and final momenta of the electron), parallel and perpendicular to the direction of the photon $\mathbf{k}$. In the ultrarelativistic case the angles θ and θ' (between $\mathbf{k}$ and respectively $\mathbf{p}$ and $\mathbf{p}'$) are small: $\theta \lesssim m/\varepsilon$, $\theta' \lesssim m/\varepsilon'$. Therefore

$$|\mathbf{p}_{\perp}| \sim |\mathbf{p}|\theta \sim m, \quad p_{\parallel} \approx |\mathbf{p}| - \frac{p_{\perp}^2}{2|\mathbf{p}|} \approx \varepsilon - \frac{m^2}{2\varepsilon} - \frac{\mathbf{p}_{\perp}^2}{2\varepsilon}, \quad (\text{X.97})$$

and analogously for $\mathbf{p}'_{\perp}$, $p'_{\parallel}$.

Without account of recoil we have $\varepsilon' + \omega - \varepsilon = 0$, the difference $p'_{\parallel} + \omega - p_{\parallel} \sim m^2/\varepsilon$, so that

$$-q^2 \approx (\mathbf{p}'_{\perp} - \mathbf{p}_{\perp})^2 \sim m^2. \quad (\text{X.98})$$

The recoil energy (on an electron):

$$q_0 = \varepsilon' + \omega - \varepsilon \sim \mathbf{q}^2/(2m) \sim m. \quad (\text{X.99})$$

The change of $\mathbf{p}'_{\perp}$ due to the change of ε' can be neglected. Therefore the first two terms in (97.8) give the change of q^2 upon account of

recoil; let us denote it by Δq^2 . Using (97.9), we obtain

$$\Delta q^2 \approx (\varepsilon' + \omega - \varepsilon) \left(-\frac{m^2}{\varepsilon'} - \frac{\mathbf{p}'_{\perp}^2}{\varepsilon'} + \frac{m^2}{\varepsilon} + \frac{\mathbf{p}_{\perp}^2}{\varepsilon} \right) \sim m^2 \cdot \frac{m}{\varepsilon}.$$

Comparing with (97.10), we see that $\Delta q^2 \ll |q^2|$, which justifies the neglect of recoil⁴⁷.

The fact that the fast particle radiates into a narrow cone (with opening angle $\sim m/\varepsilon$) in the direction of its motion permits obtaining the radiation cross section in the centre-of-mass frame by a simple recalculation of the cross section (97.2) from the laboratory frame⁴⁸.

In the centre-of-mass frame both electrons radiate equally, each in the direction of its motion (this circumstance visually explains the reason for the absence of interference between the radiation of both particles). The energy of the ultrarelativistic electron in the centre-of-mass frame is related to its energy ε in the laboratory frame by $2E^2 = m\varepsilon$, and the frequencies Ω and ω of the photon in these frames by $\omega/\varepsilon = \Omega/E$ (these equalities

⁴⁷This conclusion is, of course, all the more valid for radiation of an electron on a nucleus, for which the recoil energy $q_0 \approx \mathbf{q}^2/(2M) \sim m^2/M$, where M is the mass of the nucleus.

⁴⁸In the general case such a recalculation is impossible, since the contribution to the spectrum in a given frequency interval $d\omega$ arises from photons emitted in essentially different directions.

are easy to obtain by comparing the values of the invariants $(p_1 p_2)$ and $(p_1 k)$ in both frames). Therefore for the cross section of radiation by each of the electrons in the centre-of-mass frame we find

$$d\sigma^{(1)} = d\sigma^{(2)} = 4\alpha r_e^2 \frac{d\Omega}{\Omega} \frac{E - \Omega}{E} \left[\frac{E}{E - \Omega} + \frac{E - \Omega}{E} - \frac{2}{3} \right] \times \\ \times \left(\ln \frac{4E^2(E - \Omega)}{m^2 \Omega} - \frac{1}{2} \right). \quad (\text{X.100})$$

For the applicability of (97.12) it is also necessary that the photon frequency not be close to the boundary of the spectrum. For an ultrarelativistic particle the transformation indicated above directly gives from $\omega_{\max} \approx \varepsilon$

$$\Omega_{\max} \sim \omega_{\max} E / \varepsilon \approx E. \quad (\text{X.101})$$

Thus, in the centre-of-mass frame the electrons can radiate only half their total energy $2E$. A direct computation of $\Omega_{\max}$ is easily carried out by noting that after radiation of such a photon the electrons will move (in the same frame) with equal velocities in the direction opposite to the direction of the photon. We have

$$2E = 2E' + \Omega_{\max}, \quad 2|\mathbf{p}'| = \Omega_{\max},$$

whence

$$\Omega_{\max} = \frac{\mathbf{p}^2}{m} = E - \frac{m^2}{E}, \quad (\text{X.102})$$

and in the ultrarelativistic case we again obtain (97.13). Thus, formula (97.12) is applicable under the condition

$$\Omega_{\max} - \Omega \sim E - \Omega \gg m. \quad (\text{X.103})$$

Let us now give formulas for radiation in the centre-of-mass frame in the opposite limiting case, near the boundary of the spectrum, when⁴⁹

$$\Omega_{\max} - \Omega \ll m. \quad (\text{X.104})$$

Since in this case the recoil is very substantial, the results differ from the case of scattering on a fixed centre and turn out to be different for electron-electron and electron-positron scattering (*V. N. Baier, V. S. Fadin, V. A. Khoze, 1967*).

In the case of electron scattering on an electron, besides the squares of diagrams (97.1), a contribution to the radiation cross section near the boundary of the spectrum is also given by the products (interference terms) of direct and exchange diagrams in which one and the same initial particle radiates.

This is connected with the fact that near the boundary the final particles have close momenta and there are no reasons for the smallness of the exchange terms. The final answer for the cross section:

$$d\sigma = 2\alpha r_e^2 \frac{[E(\Omega_{\max} - \Omega)]^{1/2}}{m} \frac{d\Omega}{\Omega_{\max}}. \quad (\text{X.105})$$

⁴⁹The result obtained in the Born approximation is, of course, valid only so long as the relative velocity of the final electrons is large compared with α . In the opposite case one should take into account the interaction of the particles in the final state.

In the scattering of an electron on a positron a logarithmically large contribution to the radiation cross section comes from the squares of annihilation diagrams in which the initial particles radiate:

With non-logarithmic accuracy the squares of other diagrams are also substantial. The interference terms, however, are small. The final answer:

$$d\sigma = 2\alpha r_e^2 \frac{[E(\Omega_{\max} - \Omega)]^{1/2}}{m} \left(\ln \frac{2E}{m} + 1 \right) \frac{d\Omega}{\Omega_{\max}}. \quad (\text{X.106})$$

Thus, radiation in electron-positron scattering is logarithmically large compared with radiation in electron-electron scattering.

§ 98. Emission of soft photons in collisions

Let $d\sigma_0$ be the cross section of some process of scattering of charged particles, which may be accompanied by the emission of a definite number of photons. Alongside this process let us consider also another process, differing from the first only by the emission of one additional photon. If the frequency ω of this additional photon is sufficiently small (the corresponding conditions will be formulated below), then the cross section $d\sigma$ is in a simple way related to $d\sigma_0$.

Indeed, at small $d\sigma_0$ one can neglect the reverse influence of the emission of this quantum on the scattering process. In other words, the cross section $d\sigma$ can be represented in the form of a product of two independent factors: the cross section $d\sigma_0$ and the probability $d\omega$ of emission of one photon in the collision. The emission of a soft photon is a quasi-classical process; therefore its probability coincides with the classically computed number of emitted quanta, i.e. with the classical intensity (total energy) of the radiation dI , divided by ω ($= \hbar\omega$). Thus,

$$d\sigma = d\sigma_0 \frac{dI}{\omega}. \quad (\text{X.107})$$

Let us show that this formula can be obtained by the general rules of the diagram technique (*J. M. Jauch, F. Rohrlich*, 1954). The diagrams of the process with an additional photon are obtained from the diagrams of the basic process by adding an external photon line, “branching off” from some (external or internal) electron line, i.e. by the replacement

It is easy to see that the main role will be played by diagrams obtained by such a replacement in the external electron lines. Indeed, if p is the momentum of the external line ($p^2 = m^2$), then for small k also $(p - k)^2 \approx m^2$, i.e. the propagator $G(p - k)$ added in the diagram will be near its pole.

For the line of the initial electron p the replacement (98.2) reduces to a replacement in the amplitude of the reaction:

$$\begin{aligned} u(p) &\rightarrow e\sqrt{4\pi} G(p - k)(\gamma e^*) u(p) = \\ &= e\sqrt{4\pi} \frac{\gamma p - \gamma k + m}{(p - k)^2 - m^2} (\gamma e^*) u(p) \approx -e\sqrt{4\pi} \frac{\gamma p + m}{2(pk)} (\gamma e^*) u(p). \end{aligned}$$

Noting that

$$(\gamma p)(\gamma e^*) = 2pe^* - (\gamma e^*)(\gamma p), \quad \gamma p u(p) = mu(p),$$

we obtain the replacement rule in the form

$$u(p) \rightarrow -e\sqrt{4\pi} \frac{(pe^*)}{(pk)} u(p). \quad (\text{X.108})$$

Analogously, for the line of the final electron p' the replacement in the diagram means a replacement in the amplitude:

$$u(p') \rightarrow e\sqrt{4\pi} \bar{u}(p') \frac{(p'e^*)}{(p'k)}. \quad (\text{X.109})$$

In all remaining parts of the diagrams one can in general neglect the changes in the momenta of lines associated with the emission of the photon k . It is assumed here that the photon energy ω is in all cases small compared with the energies of all particles participating in the reaction (including compared with the energies of hard photons radiated, if there are such).

For definiteness let the cross section $d\sigma_0$ refer to the scattering of an electron on a fixed nucleus (with possible emission of hard photons). The amplitude of this process, which we shall conventionally call elastic, has the form

$$M_{fi}^{\text{el}} = \bar{u}(p') M u(p).$$

Carrying out in it the replacement (98.3) once and (98.4) another time, and combining the results, we obtain the amplitude of bremsstrahlung of the same hard photons and a soft photon k ⁵⁰:

$$M_{fi} = M_{fi}^{\text{el}} e\sqrt{4\pi} \left(\frac{(p'e^*)}{(p'k)} - \frac{(pe^*)}{(pk)} \right). \quad (\text{X.110})$$

The corresponding cross section

$$d\sigma = d\sigma_{\text{el}} \cdot 4\pi e^2 \left| \frac{(p'e)}{(p'k)} - \frac{(pe)}{(pk)} \right|^2 \frac{d^3k}{(2\pi)^3\omega}. \quad (\text{X.111})$$

Summing over the polarisations of the photon k , we obtain

$$d\sigma = -e^2 \left| \frac{p'}{(p'k)} - \frac{p}{(pk)} \right|^2 \frac{d^3k}{4\pi^2\omega} d\sigma_{\text{el}}. \quad (\text{X.112})$$

Expressed through three-dimensional quantities, this formula has the form⁵¹

$$d\sigma = \alpha \left(\frac{[\mathbf{v}'\mathbf{n}]}{1 - \mathbf{v}'\mathbf{n}} - \frac{[\mathbf{v}\mathbf{n}]}{1 - \mathbf{v}\mathbf{n}} \right)^2 \frac{d\omega d\mathbf{o}_k}{4\pi^2\omega} d\sigma_{\text{el}}, \quad (\text{X.113})$$

⁵⁰Let us draw attention to the fact that the appearance of the difference in this formula is a natural result of gauge invariance: the amplitude of the reaction must not change under the replacement of the 4-vector of polarisation $e \rightarrow e + \text{const} \cdot k$.

⁵¹For the derivation it is convenient to return to (98.6), setting $p = (\varepsilon, \varepsilon\mathbf{v})$, $pk = \varepsilon\omega(1 - \mathbf{v}\mathbf{n})$, ..., $e = (0, \mathbf{e})$ and carrying out the summation over polarisations afresh with the aid of (45.4a).

where $\mathbf{n} = \mathbf{k}/\omega$, and $\mathbf{v}$ and $\mathbf{v}'$ are the initial and final velocities of the electron. We see that the expression standing before $d\sigma_{\text{el}}$ indeed coincides (divided by ω) with the classical intensity of radiation (cf. II, (69.4)), as was asserted in formula (98.1).

The condition of applicability of the obtained formulas requires also, besides the smallness of ω compared with ε , that the momentum transfer to the nucleus q be large compared with the change $\delta\mathbf{q}$ of this quantity associated with the emission of a soft photon. We have

$$\delta\mathbf{q} = (\mathbf{p}' - \mathbf{p} - \mathbf{k}) - (\mathbf{p}' - \mathbf{p})_{\omega=0} = \delta\mathbf{p}' - \mathbf{k},$$

where

$$|\delta\mathbf{p}'| \sim \frac{\partial|\mathbf{p}'|}{\partial\varepsilon} \omega \sim \frac{\omega}{v},$$

and $|\mathbf{k}| = \omega$. In the non-relativistic case ($v \ll 1$) we therefore obtain the condition

$$\omega/|\mathbf{q}|v \ll 1. \quad (\text{X.114})$$

In scattering on a Coulomb (and in general on a slowly decreasing with distance) potential $|\mathbf{q}| \sim 1/\rho$ (ρ is the impact parameter), so that this condition can also be represented in the form $\omega\tau \ll 1$, where $\tau \sim \rho/v$ is the characteristic collision time.

In the ultrarelativistic case the photons are radiated mainly in directions close to $\mathbf{v}$ or $\mathbf{v}'$ (as is evident from one of the denominators in (98.8)). If the scattering angle θ of the electron is small, then the directions of all three vectors $\mathbf{p}$, $\mathbf{p}'$, $\mathbf{n}$ are close to each other. Then

$$|\delta\mathbf{q}| = |\delta\mathbf{p}'| - |\mathbf{k}| = \omega \left(\frac{1}{v} - 1 \right) \sim \frac{\omega m^2}{\varepsilon^2},$$

and since $|\mathbf{q}| \sim \varepsilon\theta$, we obtain the condition

$$\theta \gg \frac{\omega}{\varepsilon} \frac{m^2}{\varepsilon^2}. \quad (\text{X.115})$$

By virtue of the quasi-classical character of formulas (98.5)–(98.8) they are valid for radiation by any charged particles (not necessarily electrons, for which the derivation was carried out). In the general case, when several particles participate in the reaction, formula (98.5) must be written in the form

$$M_{fi} = M_{fi}^{\text{el}} e\sqrt{4\pi} \sum Z \left(\frac{(p'e^*)}{(p'k)} - \frac{(pe^*)}{(pk)} \right), \quad (\text{X.116})$$

where the summation is carried out over all particles (with charges Ze); correspondingly formulas (98.6)–(98.8) also change. In particular, in the non-relativistic case

$$M_{fi} = M_{fi}^{\text{el}} \frac{e\sqrt{4\pi}}{\omega} \sum Z(\mathbf{v}' - \mathbf{v})\mathbf{e}^*. \quad (\text{X.117})$$

For two particles this formula takes the form

$$M_{fi} = M_{fi}^{\text{el}} \frac{\sqrt{4\pi}}{\omega} \left(\frac{Z_1 e}{m_1} - \frac{Z_2 e}{m_2} \right) \mathbf{q}\mathbf{e}^*, \quad (\text{X.118})$$

$$\mathbf{q} = m(\mathbf{v}' - \mathbf{v}), \quad m = \frac{m_1 m_2}{m_1 + m_2},$$

where $\mathbf{v}$ and $\mathbf{v}'$ are the relative velocity of the particles before and after the collision. Integrating the square $|M_{fi}|^2$ over the photon emission directions and summing over its polarisations, we obtain from this the non-relativistic spectral distribution of the radiation in the form

$$d\sigma = d\sigma_{\text{el}} \frac{2e^2}{3\pi} \left(\frac{Z_1}{m_1} - \frac{Z_2}{m_2} \right)^2 \mathbf{q}^2 \frac{d\omega}{\omega}.$$

The obtained results generalise to the case of simultaneous emission of several soft photons. For each of the photons one adds to the amplitude M_{fi} its own factor of the type standing at M_{fi}^{el} in (98.5). In this one can easily convince oneself directly, say, on the example of two photons. The lines of both emitted photons must be added to the external electron lines, in two different sequences, i.e. the diagram with external line p is replaced by two diagrams with lines

Each of them contains the factor (the denominators of the electron propagators)

$$\frac{1}{2((pk_1) + (pk_2))} \frac{1}{2(pk_1)} \quad \text{or} \quad \frac{1}{2((pk_1) + (pk_2))} \frac{1}{2(pk_2)}.$$

Their sum is

$$\frac{1}{2(pk_1) \cdot 2(pk_2)},$$

i.e. it contains the product of two independent factors, corresponding to the first and second photon. After this the sum of all the diagrams is assembled (by virtue of gauge invariance) into the product of differences

$$\left(\frac{(p'e_1^*)}{(p'k_1)} - \frac{(pe_1^*)}{(pk_1)} \right) \left(\frac{(p'e_2^*)}{(p'k_2)} - \frac{(pe_2^*)}{(pk_2)} \right).$$

Correspondingly the factorisation of the amplitude extends to the factor and also to the cross section of the process. Thus, soft photons are emitted independently. The cross section of the process with emission of n soft photons can be represented in the form

$$d\sigma = d\sigma_{\text{el}} dw_1 dw_2 \dots dw_n, \quad (\text{X.119})$$

where $dw_1, dw_2, \dots$ are the probabilities of emission of each individual photon $dk_1, dk_2, \dots$. Upon integration of this formula over a finite interval of the variables (frequencies and directions), equal for all the quanta, a factor $1/n!$ must be introduced, taking into account the identity of the photons.

If one integrates the radiation cross section (98.1) over frequencies in some finite interval from ω_1 to ω_2 , we obtain an expression of the form

$$d\sigma \sim \alpha \ln \frac{\omega_2}{\omega_1} d\sigma_{\text{el}} \quad (\text{X.120})$$

(cf. (98.8)). It is assumed here that both frequencies are soft, so that their possible values are limited by the condition of applicability of the method. With logarithmic accuracy, however, one can set $\omega_2 \sim \varepsilon$, where ε is the initial energy of the radiating particle. The

values of ω_1 are in no way limited from below. Sending ω_1 to zero, we see that the cross section of emission of all possible soft quanta turns to infinity. Let us elucidate the meaning of this situation—the so-called *infrared catastrophe* (F. Bloch, A. Nordsieck, 1937). When

$$\alpha \ln \frac{\varepsilon}{\omega_1} \gtrsim 1 \quad (\text{X.121})$$

$d\sigma$ will be $d\sigma \gtrsim d\sigma_{\text{el}}$. But this means the inapplicability of perturbation theory—the impossibility of computing $d\sigma$ as a quantity of higher order of smallness than $d\sigma_{\text{el}}$. In other words, the parameter of smallness should in the given case be not α , but the product $\alpha \ln(\varepsilon/\omega_1)$:

Thus, the derivation of formulas (98.5), (98.6) on the basis of perturbation theory turns out to be incorrect at sufficiently small frequencies. On the other hand, the classical formula for the intensity dI (see II, (69.4)) is applicable to a greater extent, the smaller is ω . Therefore formula (98.1) remains correct, if one somewhat modifies its meaning toward greater classicality. Namely, in (98.1) it was assumed that one photon is radiated; then the energy lost by the particle to radiation coincides with ω and the “cross section of the relative energy loss” is given by the expression $\omega d\sigma/\varepsilon$, i.e.

$$d\sigma_{\text{el}} \frac{dI}{\varepsilon}. \quad (\text{X.122})$$

In reality, however, at sufficiently small ω the probability of radiation is not small, and the probability of radiation of two or more photons is not less, but greater than the probability of radiation of one photon. Under these conditions expression (98.17) remains valid, but the classical intensity dI will determine not the probability of radiation of one photon, but the mean number of radiated photons, and

$$d\bar{n} = \frac{dI}{\omega}, \quad (\text{X.123})$$

or in a finite interval of frequencies

$$\bar{n} = \int_{\omega=\omega_1}^{\omega_2} \frac{dI}{\omega}. \quad (\text{X.124})$$

Since soft photons are radiated statistically independently (this is valid in all approximations of perturbation theory), the Poisson formula can be applied to the process of multiple radiation: the probability $w(n)$ of emission of n photons is expressed through the mean number $\bar{n}$ by the formula

$$w(n) = \frac{\bar{n}^n}{n!} \exp(-\bar{n}). \quad (\text{X.125})$$

Let us represent the cross section of the scattering process with radiation of photons in the form

$$d\sigma = d\sigma_{\text{el}} \cdot w(n). \quad (\text{X.126})$$

Since $\sum w(n) = 1$, $d\sigma_{\text{el}}$ represents the total cross section of scattering accompanied by any soft radiation. This circumstance is obvious from classical considerations; in the

theory of perturbations, however, $d\sigma_{\text{el}}$ is the cross section of purely elastic scattering. But the theory of perturbations here is inapplicable. It turns out that

$d\sigma_{\text{el}}$, computed by perturbation theory as the cross section of elastic scattering, in fact takes into account the emission of any soft photons. As regards the cross section of purely elastic scattering, it is in fact equal to zero: when $\omega_1 \rightarrow 0$ the mean number $\bar{n} \rightarrow \infty$, and according to (98.20) the probability of emission of any finite number of photons goes to zero⁵².

Problems

1. Find the spectral distribution of the bremsstrahlung of soft photons in scattering of an ultrarelativistic electron on a nucleus⁵³.

Solution. Integration of formula (98.8) over $d\mathbf{o}_k$ gives

$$d\sigma = \alpha F(\xi) \frac{d\omega}{\omega} d\sigma_{\text{el}}, \quad (1)$$

where

$$F(\xi) = \frac{2}{\pi} \left[\frac{2\xi^2 + 1}{\sqrt{\xi^2 + 1}} \ln(\xi + \sqrt{\xi^2 + 1}) - 1 \right], \quad \xi = \frac{|\mathbf{p}|}{m} \sin \frac{\theta}{2} \quad (2)$$

($\mathbf{p}$ is the momentum, θ the scattering angle of the electron). In the ultrarelativistic case the essential role is played by the angular region

$$\frac{m^2\omega}{\varepsilon^2} \ll \theta \ll \frac{m}{\varepsilon} \quad (3)$$

(the lower bound is condition (98.10), the upper bound from above). At this $\xi \approx \varepsilon\theta/(2m) \ll 1$, so that $F(\xi) \approx (8/(3\pi))\xi^2$, and the elastic scattering cross section on a nucleus (see (80.10))

$$d\sigma_{\text{el}} \approx 4Z^2 r_e^2 \frac{m^2 d\omega}{\varepsilon^2 \theta^4}. \quad (4)$$

The integral

$$d\sigma_\omega = \frac{16}{3} Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \int \frac{d\theta}{\theta} \quad (5)$$

diverges logarithmically; it is cut off from below at $\theta \sim m^2\omega/\varepsilon^2$, from above at $\theta \sim m/\varepsilon$, so that

$$d\sigma_\omega = \frac{16}{3} Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \ln \frac{\varepsilon^2}{m\omega}$$

in agreement with the logarithmic part of formula (93.17) (in which one should set $\varepsilon \approx \varepsilon'$). Achieving non-logarithmic accuracy is possible only by going beyond the quasi-classical region.

⁵²We shall return to the discussion of this situation in § 130 in connection with the study of radiative corrections.

⁵³The formulas given below in the application of formula (98.7) are due to V. N. Baier and V. M. Galitsky (1964).

2. For the collision of two ultrarelativistic electrons, determine (in the centre-of-mass frame) the cross section of simultaneous emission of two soft photons in opposite directions at small angles to the momenta of the electrons.

Solution. Photons flying in opposite directions are emitted by different electrons. The cross section of simultaneous emission:

$$d\sigma = d\sigma_{\text{el}} \cdot \alpha F(\xi) \frac{d\omega_1}{\omega_1} \cdot \alpha F(\xi) \frac{d\omega_2}{\omega_2}, \quad \xi = \frac{\varepsilon}{m} \sin \frac{\theta}{2} \quad (6)$$

where ε is the energy of each electron, θ the scattering angle in the centre-of-mass frame. In the ultrarelativistic case the elastic scattering cross section for small angles in the centre of mass coincides with (4) (cf. (81.11)). Unlike σ (1) the cross section (6) behaves as $\theta d\theta$ when $\theta \rightarrow 0$, so that the integration over angles is convergent down to $\theta = 0$, and one should use the exact expression (2). On the other hand, the main contribution to the integral cross section comes from $\theta \sim m/\varepsilon$, so that $F(\xi)$ should also be used exactly. The result of integration of the cross section over the scattering angles:

$$d\sigma_{\omega_1\omega_2} = \frac{2}{\pi} \left[5 + \frac{7}{2} \zeta(3) \right] r_e^2 \alpha^2 \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2} = 5.9 r_e^2 \alpha^2 \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}.$$

(ζ is the Riemann zeta function; $\zeta(3) = 1.202$).

§ 99. The method of equivalent photons

Let us compare two processes, described by the diagrams:

The circles depict conventionally the entire internal part of the diagrams. Diagram *a*) depicts the collision of a photon k ($k^2 = 0$) with some particle with 4-momentum q (and mass m ; $q^2 = m^2$). As a result of the collision a system (particle or group of particles) with total 4-momentum Q is formed. Diagram *b*) depicts the collision of the same particle q with another particle, whose 4-momentum is p , and mass M ($p^2 = M^2$). As a result of the collision this latter particle acquires 4-momentum p' and there is formed the same system Q . The second process can be considered as the collision of the particle q with a virtual photon emitted by the particle p , the momentum of which $k = p - p'$ ($k^2 < 0$). If $|k^2|$ is small,

the virtual photon differs little from a real one. It is obvious that such a situation can be encountered in collisions of very fast particles: the electromagnetic field of a charged particle moving with velocity $v \approx 1$ is almost transverse and therefore close in its properties to the field of a light wave. Under these conditions the cross section of process *b*) can be expressed through the cross section of process *a*)⁵⁴.

Thus, let us consider particle M as ultrarelativistic: its energy (in the rest frame of particle m) $\varepsilon \gg M$. If the masses of the colliding particles m and M are different, then for definiteness we shall assume that $m < M$.

⁵⁴The method described below was developed by Weizsäcker and Williams (*K. Weizsäcker, E. J. Williams, 1934*); the basic idea of this method was already put forward earlier by Fermi (*E. Fermi, 1924*).

The amplitude of process *a*) (with participation of a real photon) can be represented in the form

$$M_{fi}^{(r)} = -e\sqrt{4\pi}(e_\mu J^\mu), \quad (\text{X.127})$$

where e_μ is the 4-vector of polarisation of the photon, and J^μ the transition current, corresponding to the upper vertex (circle) of the diagram. The amplitude of process *b*)

$$M_{fi} = Ze^2 \frac{4\pi}{k^2} (j_\mu J^\mu), \quad (\text{X.128})$$

where j_μ is the transition current of particle *m* (the lower vertex of the diagram); Ze is the charge of this particle. The current J is a function of $k = Q - q$ and therefore in the two cases is different: $k^2 = 0$ in (99.2) and $k^2 \neq 0$ in (99.3). But if in the second case

$$|k^2| \ll m^2, \quad (\text{X.129})$$

then also here one can take J at $k^2 = 0$.

The change of the momentum of particle *M* upon emission of a virtual photon, $\mathbf{p} - \mathbf{p}' = \mathbf{k}$, is small compared with its original momentum $|\mathbf{p}| \approx \varepsilon$; therefore in the current *j* one can set $\mathbf{p} = \mathbf{p}'$. In other words, we consider the motion of particle *M* as rectilinear and uniform. Since such motion is quasi-classical, the corresponding current does not depend on the spin of the particle⁵⁵:

$$j_\mu = 2p_\mu. \quad (\text{X.130})$$

The condition of transversality of the current ($jk = 0$) gives now $\varepsilon\omega - p_x k_x = 0$, where the *x* axis is chosen in the direction of $\mathbf{p}$. From this

$$\omega = vk_x, \quad (\text{X.131})$$

where $v = p_x/\varepsilon$ is the velocity of particle *M*. Since

$$-\mathbf{k}^2 = -\omega^2 + k_x^2 + \mathbf{k}_\perp^2 \approx \omega^2(1 - v^2) + \mathbf{k}_\perp^2 \quad (\text{X.132})$$

($\mathbf{k}_\perp$ is the component of the vector $\mathbf{k}$ transverse to the *x* axis), condition (99.4) is equivalent to the inequality $|\mathbf{k}_\perp| \ll m$ and a considerably stronger inequality for ω : $\omega \ll m\sqrt{1 - v^2}$.

Further, from the condition of transversality of the current J ($Jk = 0$) it follows, using (99.6), that

$$J_0 = \frac{J_x}{v} + \frac{\mathbf{J}_\perp \mathbf{k}_\perp}{\omega}.$$

Therefore for the scalar product jJ we obtain

$$jJ = 2(J_0\varepsilon - J_x p_x) \approx 2 \frac{\varepsilon}{\omega} \left(\mathbf{J}_\perp \mathbf{k}_\perp + \frac{\omega M^2}{\varepsilon^2} J_x \right). \quad (\text{X.133})$$

⁵⁵In the normalisation of wave functions to one particle in unit volume the current is $j^\mu = (1, \mathbf{v})$, where v is the velocity. But we have agreed (see § 6) to omit in the wave functions the normalisation factor $1/\sqrt{2\varepsilon}$, and we arrive at expression (99.5).

The product Je in (99.2) we expand, choosing the 4-vector of polarisation of the real photon in the three-dimensional transverse gauge: $ek = -\mathbf{ek} = 0$, whence $e_x \approx -\mathbf{e}_\perp \mathbf{k}_\perp / \omega$. Then

$$je = -\mathbf{e}_\perp \left(\mathbf{J}_\perp - \frac{\mathbf{k}_\perp}{\omega} J_x \right). \quad (\text{X.134})$$

Comparing expressions (99.8) and (99.9), they will turn out to be proportional to each other if one can neglect the second terms in the brackets. Since the current J pertains to the upper vertex of the diagrams (99.1), it is not connected with the direction of $\mathbf{p}$; therefore J_x and $\mathbf{J}_\perp$ should be considered as quantities of the same order. The permissibility of the indicated neglect requires, therefore, the fulfilment of the conditions $|\mathbf{k}_\perp| \ll \omega$ and $\omega \ll \varepsilon^2 |\mathbf{k}_\perp| / M^2$; they do not contradict the conditions already imposed on $\mathbf{k}_\perp$ and ω .

Accepting that in (99.9) the photon is polarised in the plane $x\mathbf{k}$ (so that $\mathbf{e}_\perp \parallel \mathbf{k}_\perp$), and noting that by virtue of the established conditions $\mathbf{e}_\perp^2 \approx \mathbf{e}^2 = 1$, we obtain now

$$M_{fi} = M_{fi}^{(r)} \frac{Ze\sqrt{4\pi}}{-k^2} \frac{2\varepsilon}{\omega} |\mathbf{k}_\perp|. \quad (\text{X.135})$$

According to what was said above, it is assumed here that the following conditions are fulfilled

$$|\mathbf{k}_\perp| \ll \omega \ll m\gamma, \quad (\text{X.136})$$

$$\frac{\omega}{\gamma^2} \ll |\mathbf{k}_\perp| \ll m, \quad (\text{X.137})$$

where for brevity

$$\gamma = \frac{\varepsilon}{M} = \frac{1}{\sqrt{1-v^2}}.$$

From this one can find the relation between the corresponding cross sections. According to the general formula (64.18) we have (in the rest frame of particle m)

$$\begin{aligned} d\sigma_r &= |M_{fi}^{(r)}|^2 (2\pi)^4 \delta^{(4)}(P_f - P_i) \frac{1}{4m\omega} d\rho_Q, \\ d\sigma &= |M_{fi}|^2 (2\pi)^4 \delta^{(4)}(P_f - P_i) \frac{1}{4m\omega} \frac{d^3 p'}{2\varepsilon(2\pi)^3} d\rho_Q, \end{aligned}$$

where $d\rho_Q$ are the statistical weights of the system Q . Using (99.10) and (99.7), we obtain

$$d\sigma = d\sigma_r \cdot n(\mathbf{k}) d^3 p', \quad (\text{X.138})$$

where

$$n(\mathbf{k}) = \frac{Z^2 e^2}{\pi^2} \frac{\mathbf{k}_\perp^2}{\omega(\mathbf{k}_\perp^2 + \omega^2/\gamma^2)^2}. \quad (\text{X.139})$$

Let us recall that $d\sigma_r$ is the cross section of process a), caused by a collision with a real photon, with the resting particle, and there is formed a system of particles Q in definite intervals of their momenta. The cross section $d\sigma$ refers to process b)—the formation of the same system Q in the collision of a fast particle (mass M) with the same

resting particle, with the fast particle remaining in the interval $d^3 p'$ of values of $\mathbf{p}'$. The factor $n(\mathbf{k})$ in (99.13) can be interpreted as a density of photons (in $\mathbf{k}$ -space) equivalent to the electromagnetic field of the fast particle.

Integration over $d^3 p'$ is equivalent to integration over $d^3 k = d\omega d^2 k_\perp$. Carrying out the integration over $d^2 k_\perp$, we obtain the cross section of the process in which the total energy E of the system Q lies in a given interval $dE = d\omega$ ($E - m = \varepsilon - \varepsilon' = \omega$, where ε and ε' are the initial and final energies of particle M). The integration over directions of $\mathbf{k}_\perp$ means averaging over the polarisation directions of the incident photon (together with multiplication by 2π). After this we obtain

$$d\sigma = n(\omega) d\sigma_r d\omega, \\ n(\omega) = \int n(\mathbf{k}) \cdot 2\pi k_\perp dk_\perp = \frac{2Z^2 e^2}{\pi\omega} \int \frac{k_\perp^3 dk_\perp}{(k_\perp^2 + \omega^2/\gamma^2)^2}. \quad (\text{X.140})$$

The integral over $k_\perp$ diverges for large $k_\perp$. The divergence, however, is only logarithmic. This circumstance permits (within the limits of applicability of the method described) obtaining

the answer in the logarithmic approximation: it is assumed that not only the argument of the logarithm, but the logarithm itself, is large. With such accuracy it suffices to put for the upper limit of integration $k_{\perp\text{max}} \sim m$ —the upper limit of inequality (99.12). Carrying out the integration, we obtain for the spectral distribution of equivalent photons (in ordinary units)

$$n(\omega) d\omega = \frac{2}{\pi} Z\alpha \ln \frac{\gamma mc^2}{\hbar\omega} \frac{d\omega}{\omega}. \quad (\text{X.141})$$

The adopted approximation means that the numerical coefficient in the argument of the logarithm remains undefined: the introduction of such a coefficient would mean the addition to the large logarithm of a relatively small quantity (~ 1) and would represent an excess of the permissible accuracy.

Problems

1. Find the bremsstrahlung cross section in the collision of a fast electron with a nucleus, starting from the scattering of photons on the electron.

Solution. In the system of reference K_1 in which the electron is at rest before the collision, the process can be considered as scattering on the electron of equivalent photons of the field of the nucleus⁵⁶. According to (86.10) the cross section of photon scattering by the electron in the frame K_1

$$d\sigma_{\text{scatt}}(\omega_1, \omega'_1) = \pi r_e^2 \frac{m d\omega'_1}{\varepsilon\omega_1} \left[\frac{\omega_1}{\omega'_1} + \frac{\omega'_1}{\omega_1} + \left(\frac{m}{\omega'_1} - \frac{m}{\omega_1} \right)^2 - 2m \left(\frac{1}{\omega'_1} - \frac{1}{\omega_1} \right) \right], \quad (1)$$

⁵⁶The scattering of virtual photons on the nucleus (in the rest frame of the nucleus) is excluded by the large mass of the latter: the scattering cross section tends to zero with increasing mass of the scattering particle.

where ω_1 and ω'_1 are the initial and final energies of the photon in this frame. The bremsstrahlung cross section in the frame K_1

$$d\sigma_{\text{rad}}(\omega'_1) = \int d\omega_1 \cdot n(\omega_1) d\sigma_{\text{scatt}}(\omega_1, \omega'_1), \quad (2)$$

where $n(\omega_1)$ is function (99.16). By virtue of the invariance of the cross section the transition to the frame K in which the nucleus is at rest reduces to the change of the designated variables. For the region of values probed by ω_1 at given ω'_1 , we have (cf. (4))

$$\begin{aligned} \omega \leq \omega \leq \infty & \quad \text{when} \quad \omega' > \frac{m}{2}; \\ \omega \leq \omega \leq \frac{\omega'}{m - 2\omega'} & \quad \text{when} \quad \omega' < \frac{m}{2}. \end{aligned}$$

When $\omega' < m/2$ the integration over ω gives

$$d\sigma_{\text{rad}}^{(2)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega'}{\omega'} \left(1 - \frac{\omega'}{m} + \frac{\omega'^2}{m^2} \right) \ln \frac{\varepsilon}{\omega'},$$

in agreement with (97.4). If, however, $\omega' > m/2$, one must distinguish the case $\omega' \sim m$ and

$\omega' \sim \varepsilon \gg m$. In the first case we obtain

$$d\sigma_{\text{rad}}^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{m d\omega'}{\omega'^2} \left(4 - \frac{m}{\omega'} + \frac{m^2}{4\omega'^2} \right) \ln \frac{\varepsilon}{m}$$

in agreement with (97.3) (in the argument of the logarithm the replacement of ε/ω' by ε/m is made with the required accuracy). In the case $\omega' \sim \varepsilon$ the method of equivalent photons is generally inapplicable to computing $d\sigma_{\text{rad}}^{(2)}$. The frequency of the virtual photon ω ranges over values starting from ω' , and at $\omega = \omega' \sim \varepsilon$, i.e. consequently, condition (99.11) is not fulfilled.

2. The same for bremsstrahlung of the electron on an electron.

Solution. In this case the virtual photon can scatter either on the fast electron, or on the recoil electron, and conversely. The scattering of virtual photons on the fast electron gives a cross section $d\sigma_{\text{rad}}^{(1)}$ coinciding with the cross section of radiation of the electron on the nucleus with $Z = 1$.

The scattering of virtual photons on the recoil electron gives a cross section of radiation

$$d\sigma_{\text{rad}}^{(2)} = \int d\omega \cdot n(\omega) d\sigma_{\text{scatt}}(\omega, \omega'),$$

with $d\sigma_{\text{scatt}}(\omega, \omega')$ from (1) (with the correspondingly changed notation of variables). For the region of values probed by ω at given ω' , we have (cf. (4))

$$\begin{aligned} \omega \leq \omega \leq \infty & \quad \text{when} \quad \omega' > \frac{m}{2}; \\ \omega \leq \omega \leq \frac{\omega'}{m - 2\omega'} & \quad \text{when} \quad \omega' < \frac{m}{2}. \end{aligned}$$

When $\omega' < m/2$ the integration over ω gives

$$d\sigma_{\text{rad}}^{(2)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega'}{\omega'} \left(1 - \frac{\omega'}{m} + \frac{\omega'^2}{m^2} \right) \ln \frac{\varepsilon}{\omega'},$$

in agreement with (97.4). If $\omega' > m/2$, one must distinguish $\omega' \sim m$ and $\omega' \sim \varepsilon$. In the first case:

$$d\sigma_{\text{rad}}^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{m d\omega'}{\omega'^2} \left(4 - \frac{m}{\omega'} + \frac{m^2}{4\omega'^2} \right) \ln \frac{\varepsilon}{m}$$

in agreement with (97.3). In the case $\omega' \sim \varepsilon$ the method of equivalent photons is generally inapplicable.

3. Determine the total cross section of pair production by a photon in the field of a nucleus, starting from the cross section of pair production by two photons.

Solution. The energy of the photon in the rest frame of the nucleus (system K): $\omega \gg m$. We go to the frame K_0 in which the nucleus moves towards the photon with velocity v_0 such that

$$\frac{1}{\sqrt{1-v_0^2}} = \frac{\omega}{2m}.$$

In this frame the photon energy is

$$\omega_0 = \omega \frac{1-v_0}{\sqrt{1-v_0^2}} \approx \frac{\omega}{2} \sqrt{1-v_0^2} = m.$$

The desired cross section σ in the frame K_0 is computed as the cross section of pair production at the collision of the incident photon ω_0 with equivalent photons of the nuclear field, whose energies we denote by ω' :

$$\sigma = \int \sigma_{\gamma\gamma} n(\omega') d\omega',$$

where $\sigma_{\gamma\gamma}$ is the cross section of pair production by two photons; it is given by the solution of the Problem to § 88, in which one must set

$$v = \sqrt{1 - \frac{m^2}{\omega_0 \omega'}} = \sqrt{1 - \frac{m}{\omega'}}.$$

Changing to the integration variable v instead of ω' , we obtain

$$\sigma = 2r_e^2 \alpha Z \int v \ln \left[\frac{\omega}{m} (1-v^2) \right] \left[(3-v^4) \ln \frac{1+v}{1-v} - 2v(2-v^2) \right] dv.$$

Taking into account the convergence of the integral at the upper limit, the integration extends over the whole region from the reaction threshold $\omega' = m$ ($v = 0$) to $\omega' = \infty$ ($v = 1$) and is carried out with logarithmic accuracy (i.e. the logarithm

$\ln[\omega(1 - v^2)/m]$ is replaced by its value at $v = 0$ and taken outside the integral sign). In the result we obtain

$$\sigma = \frac{28}{9} \alpha Z^2 r_e^2 \ln \frac{\omega}{m}$$

in agreement with (94.6); the formula is valid when $\ln(\omega/m) \gg 1$.

§ 100. Pair production in particle collisions

The production of an electron pair in the collision of two charged particles is described by diagrams of two types:

The two upper solid lines correspond to the colliding particles, the lower one to the pair being produced.

Let us consider in the ultrarelativistic case the collision of two heavy particles (nuclei). The change in the state of motion of the particles themselves in such a collision can be neglected, i.e. they can be considered as sources of an external field⁵⁷. To this there correspond two diagrams of the first type:

where $q^{(1)}, q^{(2)}$ are the “momenta” of the Fourier components of the fields of the two particles.

The potential $A^\mu = (A_0, \mathbf{A})$, created by a uniformly moving with velocity $\mathbf{v}$ classical particle, satisfies the equations

$$\square A_0 = -4\pi Ze \delta(\mathbf{r} - \mathbf{v}t - \mathbf{r}_0), \quad \square \mathbf{A} = -4\pi Ze \mathbf{v} \delta(\mathbf{r} - \mathbf{v}t - \mathbf{r}_0).$$

Its Fourier components:

$$A_0(\omega, \mathbf{k}) = -\frac{8\pi^2 Ze}{\omega^2 - \mathbf{k}^2} e^{-i\mathbf{k}\mathbf{r}_0} \delta(\omega - \mathbf{k}\mathbf{v})$$

and analogously for $\mathbf{A}(\omega, \mathbf{k})$. In four-dimensional form

$$A^\mu(q) = -\frac{8\pi^2 Ze}{q^2} e^{iqx_0} U^\mu \delta(Uq),$$

where U is the 4-velocity of the particle, and the 4-vector $x_0 = (0, \mathbf{r}_0)$. If nucleus 1 is at rest at the origin ($\mathbf{r}_0^{(1)} = 0$), then $\boldsymbol{\rho} \equiv \mathbf{r}_0^{(2)}$ is the impact parameter vector (in the plane perpendicular to the direction of motion of nucleus 2). This expression for $A^\mu(q)$ should be used in the analytic representation of diagrams (100.2).

The pair production cross section can be determined by the method of equivalent photons from the known cross section of pair production by a photon on a nucleus. Replacing in diagrams (100.2) the field lines $q^{(1)}$ of one of the particles by real photon lines, the set of these two diagrams becomes identical to the set of diagrams for pair production by a photon on nucleus 2. For $\varepsilon_+, \varepsilon_- \gg m$ this cross section is given by

⁵⁷The case of the collision of two light particles (electrons), the change in the motion of which cannot be neglected, is considerably more complicated. See about this the book by V. Ya. Baier, V. M. Katkov and V. S. Fadin indicated on p. 454.

formula (94.5). Multiplying it by the equivalent photon spectrum (99.16) of the first nucleus, we obtain the differential cross section for pair production in collisions:

$$d\sigma = \frac{8}{\pi} r_e^2 (Z_1 Z_2 \alpha)^2 \frac{d\varepsilon_+ d\varepsilon_-}{(\varepsilon_+ + \varepsilon_-)^4} \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \ln \frac{2\varepsilon_+ \varepsilon_-}{m(\varepsilon_+ + \varepsilon_-)} \ln \frac{m\gamma}{\varepsilon_+ + \varepsilon_-}, \quad (\text{X.142})$$

where $\gamma = 1/\sqrt{1 - v^2} \gg 1$. This formula applies in the region

$$m \ll \varepsilon_+, \varepsilon_- \ll m\gamma. \quad (\text{X.143})$$

The upper inequality is the applicability condition for the equivalent photon method. The region (100.4) coincides with the region of energies of the electron and positron that are essential in integrating expression (100.3).

Integrating over ε_+ or ε_- at fixed sum $\varepsilon \equiv \varepsilon_+ + \varepsilon_- (\gg m)$, the essential region lies near the upper limit; discarding terms that are non-leading in the logarithm:

$$d\sigma = \frac{56}{9\pi} r_e^2 (Z_1 Z_2 \alpha)^2 \ln \frac{\varepsilon}{m} \ln \frac{m\gamma}{\varepsilon} \frac{d\varepsilon}{\varepsilon}.$$

The integral over ε taken over region (100.4) diverges as the cube of the logarithm, while at its boundaries only as the square of the logarithm. In the logarithmic approximation ($\ln \gamma \gg 1$), region (100.4) is the principal one, and the integral can be taken from m to $m\gamma$. Using

$$\int_1^\gamma \ln \xi (\ln \gamma - \ln \xi) \frac{d\xi}{\xi} = \frac{1}{6} \ln^3 \gamma,$$

we obtain the total cross section for pair production:

$$\sigma = \frac{28}{27\pi} r_e^2 (Z_1 Z_2 \alpha)^2 \ln^3 \frac{1}{\sqrt{1 - v^2}} \quad (\text{X.144})$$

(*L. D. Landau, E. M. Lifshitz, 1934*).

Let us now consider the non-relativistic case of colliding nuclei. The change in the state of motion of the nuclei becomes essential, and the main contribution comes from diagrams of the second type in (100.1). There are four such diagrams:

and two analogous diagrams in which the virtual photon k (producing the pair) is emitted by the first rather than the second nucleus⁵⁸.

We assume that the pair energy is small compared with the kinetic energy of relative motion of the nuclei in their centre-of-mass system:

$$\varepsilon_+ + \varepsilon_- \ll \frac{1}{2} M v^2 \quad (\text{X.145})$$

(v is the initial relative velocity, $M = M_1 M_2 / (M_1 + M_2)$ is the reduced mass). Then one can neglect the back-reaction of pair production on the nuclear motion. If one removes the electron–positron line from diagrams (100.6), the remaining parts depict emission of a virtual photon of small frequency ($\omega = \varepsilon_+ + \varepsilon_-$) by colliding particles. We thus return

⁵⁸Note that pair production in the collision of two electrons corresponds to 36 diagrams: $2! \cdot 3! = 12$ diagrams of type *a*), obtained by permutations among two initial and three final electrons, plus $2 \cdot 2! \cdot 3! = 24$ diagrams of type *b*), obtained similarly from the two diagrams (100.6).

to the situation considered in § 98 for emission of a real soft photon, and can use formula (98.13) obtained there for the non-relativistic case (with the difference that in place of the amplitude $\sqrt{4\pi} e^*$ of the real photon, the propagator of the virtual photon will stand)⁵⁹.

Thus the amplitude of the entire pair production process is written as:

$$M_{fi} = M_{fi}^{\text{el}} \frac{1}{\omega} \left(\frac{Z_1 e}{M_1} - \frac{Z_2 e}{M_2} \right) q^\lambda D_{\lambda\mu}(k) [-ie(\bar{u}_- \gamma^\mu u_+)], \quad (\text{X.146})$$

where $\mathbf{q} = M(\mathbf{v}' - \mathbf{v})$, $q = (0, \mathbf{q})$.

As usual, in the non-relativistic case the photon propagator should be chosen in the Coulomb gauge (76.14). From the amplitude (100.8) we find the cross section of the process:

$$d\sigma = d\sigma_{\text{sc}} \cdot e^4 \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 \frac{d^3 p_+ d^3 p_-}{2\varepsilon_+ 2\varepsilon_- (2\pi)^6} \frac{(4\pi)^2}{\omega^2 (\omega^2 - \mathbf{k}^2)^2} |\bar{u}_- \gamma \mathbf{Q} u_+|^2, \quad (\text{X.147})$$

where

$$\omega = \varepsilon_+ + \varepsilon_-, \quad \mathbf{k} = \mathbf{p}_+ + \mathbf{p}_-, \quad \mathbf{Q} = \mathbf{q} - \frac{1}{\omega^2} \mathbf{k}(\mathbf{q} \cdot \mathbf{k});$$

$d\sigma_{\text{sc}}$ is the cross section of elastic scattering of nuclei off each other (in their centre-of-mass system). It is given by the Rutherford formula:

$$d\sigma_{\text{sc}} = 4(Z_1 Z_2 e^2)^2 \frac{M^2 d\sigma}{q^4} \approx \frac{4(Z_1 Z_2 e^2)^2 dq_y dq_z}{v^2 q^4} \quad (\text{X.148})$$

(the approximate equality assumes small deflection of the nuclei from their initial direction of motion—the x -axis)⁶⁰. Substituting (100.10) into (100.9) and performing the usual summation over polarisations of the pair:

$$d\sigma = \frac{(Z_1 Z_2 e^2)^2 e^4}{v^2} \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 \frac{d^3 p_+ d^3 p_-}{4\pi^2 \varepsilon_+ \varepsilon_-} \frac{dq_y dq_z}{q^4 (\omega^2 - \mathbf{k}^2)^2 \omega^2} \times \text{Sp}\{(\gamma p_- + m)(\gamma \mathbf{Q})(\gamma p_+ - m)(\gamma \mathbf{Q})\}. \quad (\text{X.149})$$

Further computation is performed in the approximation where all logarithms arising during integration are considered large quantities. We shall see that with this accuracy the main role is played by pair energies ε_+ , $\varepsilon_- \gg m$ and angles θ between $\mathbf{p}_+$ and $\mathbf{p}_-$ in the region

$$m/\varepsilon \ll \theta \ll 1. \quad (\text{X.150})$$

⁵⁹In the non-relativistic case the photon momentum is small compared with the momentum of the radiating particles ($|\delta \mathbf{p}| \sim \omega/v$), and therefore can be neglected (compared with $\delta \mathbf{p}$) even when it is not negligible compared with the photon energy. This applies all the more in the present case to the virtual photon, for which $k^2 = (p_+ + p_-)^2 > 0$, so that $|\mathbf{k}| < \omega$. Under these conditions the difference between a real and a virtual photon vanishes, which justifies the use of formula (98.13).

⁶⁰Diagrams (100.6) correspond to the Born approximation for nuclear scattering. However, since the Rutherford formula (for Coulomb interaction) is exact, the validity of the obtained results does not actually require the applicability condition of the Born approximation.

With the corresponding approximations, the trace computation in (100.11) gives:

$$\text{Sp}\{\dots\} = 4 \left[(\varepsilon_+ \varepsilon_- - \mathbf{p}_+ \cdot \mathbf{p}_-) \left(q^2 - \frac{(\mathbf{q} \cdot \mathbf{k})^2}{\omega^2} \right) + 2(\mathbf{p}_+ \cdot \mathbf{q})(\mathbf{p}_- \cdot \mathbf{q}) \right. \\ \left. + \frac{2\varepsilon_+ \varepsilon_-}{\omega^2} (\mathbf{q} \cdot \mathbf{k})^2 - \frac{2(\mathbf{q} \cdot \mathbf{k})}{\omega} (\varepsilon_+ (\mathbf{q} \cdot \mathbf{p}_-) + \varepsilon_- (\mathbf{q} \cdot \mathbf{p}_+)) \right],$$

where one can set $|\mathbf{p}_+| = \varepsilon_+$, $|\mathbf{p}_-| = \varepsilon_-$. In the denominator:

$$\omega^2 - \mathbf{k}^2 \approx \varepsilon_+ \varepsilon_- \theta^2 + \frac{m^2 (\varepsilon_+ + \varepsilon_-)^2}{\varepsilon_+ \varepsilon_-}.$$

Integrating over the directions of $\mathbf{p}_+$ and $\mathbf{p}_-$ at fixed angle θ between them:

$$d\sigma = \frac{8}{3\pi^2} (Z_1 Z_2 e^2)^2 \frac{e^4}{v^2} \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 (\varepsilon_+^2 + \varepsilon_-^2) d\varepsilon_+ d\varepsilon_- \\ \times \frac{\theta^3 d\theta}{[\theta^2 + m^2 (\varepsilon_+ + \varepsilon_-)^2 / (\varepsilon_+^2 \varepsilon_-^2)]^2} \frac{dq_y dq_z}{q^2}. \quad (\text{X.151})$$

The form of the θ -dependence confirms the assumption (100.12). Integration over the last factor in (100.13) is carried out from $q_y = q_z = 0$ to $\sqrt{q_y^2 + q_z^2} \sim 1/R$, where R is a quantity of order of the nuclear radius (this value corresponds to the minimum impact parameter—see below); this integration gives:

$$\pi \ln(q_x^2 + q_y^2 + q_z^2) \Big|_{q_y=q_z=0}^{q_y=q_z=1/R} \approx 2\pi \ln \frac{1}{Rq_x}.$$

On the other hand, the total pair energy, equal to the change in the kinetic energy of the nuclei, is:

$$\varepsilon \equiv (\varepsilon_+ + \varepsilon_-) = \frac{1}{2} M(v'^2 - v^2) \approx Mv(v'_x - v_x) = vq_x,$$

so that $q_x = \varepsilon/v$. Thus we find:

$$d\sigma = \frac{16}{3\pi} (Z_1 Z_2 e^2)^2 \frac{e^4 m^2}{v^2} \left(\frac{Z_2}{M_2} - \frac{Z_1}{M_1} \right)^2 \frac{\varepsilon_+^2 + \varepsilon_-^2}{\varepsilon^4} \\ \times \ln \frac{v}{R\varepsilon} \ln \frac{\varepsilon_+ \varepsilon_-}{m(\varepsilon_+ + \varepsilon_-)} d\varepsilon_+ d\varepsilon_-,$$

and after integration over ε_+ or ε_- at fixed sum ε :

$$d\sigma = \frac{32}{9\pi} (Z_1 Z_2 e^2)^2 \frac{e^4 m^2}{v^2} \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 \ln \frac{v}{R\varepsilon} \ln \frac{\varepsilon}{m} \frac{d\varepsilon}{\varepsilon}. \quad (\text{X.152})$$

The energy ε can be related to the impact parameter: $\rho \sim v/\varepsilon$ (the pair energy is of order of the frequency corresponding to the collision time). Therefore the logarithmic divergence in ε in (100.14) signifies the same divergence in impact parameters. This means that the essential ρ are large (which, incidentally, justifies the use of the

Rutherford scattering cross section (100.10) in the pure Coulomb field of the nucleus). Correspondingly, the essential energy region is: $m \ll \varepsilon \ll \nu/R$. Integrating (100.14) gives the total pair production cross section; finally (in ordinary units)⁶¹:

$$\sigma = \frac{16}{27\pi} (Z_1 Z_2 \alpha)^2 r_e^2 \frac{c^2}{v^2} \left(\frac{Z_2 m}{M_2} - \frac{Z_1 m}{M_1} \right)^2 \ln^3 \frac{\hbar \nu}{mc^2 R} \quad (\text{X.153})$$

(*E. M. Lifshitz*, 1935).

§ 101. Emission of a photon by an electron in the field of an intense electromagnetic wave

The applicability of perturbation theory to processes of interaction of an electron with the radiation field presupposes (besides the smallness of the coupling constant α) also sufficient weakness of this field. If a is the amplitude of the classical 4-potential of the electromagnetic wave field, the characteristic quantity in this sense is the dimensionless invariant ratio

$$\xi = \frac{e\sqrt{-a^2}}{m}. \quad (\text{X.154})$$

In this section we consider radiation processes arising in the interaction of an electron with the field of a strong electromagnetic wave, for which ξ can have any value. The method used is based on exact account of this interaction; the interaction of the electron with the newly emitted photons can still be treated as a small perturbation (*A. I. Nikishov*, *V. I. Ritus*, 1964).

Consider a monochromatic plane wave, for definiteness circularly polarised. Its 4-potential:

$$A = a_1 \cos \varphi + a_2 \sin \varphi, \quad \varphi = kx, \quad (\text{X.155})$$

where $k^\mu = (\omega, \mathbf{k})$ is the wave 4-vector ($k^2 = 0$), and the 4-amplitudes a_1 and a_2 are equal in magnitude and mutually orthogonal:

$$a_1^2 = a_2^2 \equiv a^2, \quad a_1 a_2 = 0.$$

We assume the potential is calibrated by the Lorentz condition, so $a_1 k = a_2 k = 0$.

The exact wave function for an electron in the field of an arbitrary plane electromagnetic wave was found in § 40 (formulae (40.7), (40.8)). We change, however, the normalisation: we require that ψ_p correspond to unit average spatial particle density — similar to how we normalise the wave functions of free particles to one particle per unit volume. Since for function (40.7) the average density is $\bar{j}_0 = q_0/p_0$, to obtain the required normalisation we multiply it by $\sqrt{p_0/q_0}$, i.e. replace in (40.7) the factor $1/\sqrt{2p_0}$ by $1/\sqrt{2q_0}$. For the wave with 4-potential (X.155) we obtain:

$$\psi_p = \left\{ 1 + \frac{e}{2(kp)} [(\gamma k)(\gamma a_1) \cos \varphi + (\gamma k)(\gamma a_2) \sin \varphi] \right\} \frac{u(p)}{\sqrt{2q_0}} \exp \left\{ -\frac{ie(a_1 p)}{(kp)} \sin \varphi + \frac{ie(a_2 p)}{(kp)} \cos \varphi \right\} \quad (\text{X.156})$$

⁶¹A numerical error was corrected by *L. B. Okun'* (1953).

where

$$q^\mu = p^\mu - \frac{e^2 a^2}{2(kp)} k^\mu. \quad (\text{X.157})$$

According to (40.14) the 4-vector q is the average kinetic 4-momentum of the electron; we shall call it the *quasi-momentum*.

The S -matrix element for transition of the electron from state ψ_p to state $\psi_{p'}$ with emission of a photon with 4-momentum $k'^\mu = (\omega', \mathbf{k}')$ and polarisation 4-vector e' :

$$S_{fi} = -ie \int \bar{\psi}_{p'}(\gamma e'^*) \psi_p \frac{e^{ik'x}}{\sqrt{2\omega'}} d^4x. \quad (\text{X.158})$$

The integrand in (X.158) is a linear combination of the quantities

$$\exp(-i\alpha_1 \sin \varphi + i\alpha_2 \cos \varphi) \cdot \{1, \cos \varphi, \sin \varphi\},$$

where

$$\alpha_1 = e \left(\frac{(a_1 p)}{(kp)} - \frac{(a_1 p')}{(kp')} \right), \quad \alpha_2 = e \left(\frac{(a_2 p)}{(kp)} - \frac{(a_2 p')}{(kp')} \right). \quad (\text{X.159})$$

Together with the factor $\exp[i(k' + q' - q)x]$ these quantities exhaust all the dependence of the integrand on x .

Expand them in Fourier series, denoting the expansion coefficients B_s, B_{1s}, B_{2s} ; for example:

$$\exp(-i\alpha_1 \sin \varphi + i\alpha_2 \cos \varphi) = \exp[-i\sqrt{\alpha_1^2 + \alpha_2^2} \sin(\varphi - \varphi_0)] = \sum_{s=-\infty}^{\infty} B_s e^{-is\varphi}.$$

These coefficients are expressed through Bessel functions:

$$\begin{aligned} B_s &= J_s(z) e^{is\varphi_0}, \\ B_{1s} &= \frac{1}{2} [J_{s+1}(z) e^{i(s+1)\varphi_0} + J_{s-1}(z) e^{i(s-1)\varphi_0}], \\ B_{2s} &= \frac{1}{2i} [J_{s+1}(z) e^{i(s+1)\varphi_0} - J_{s-1}(z) e^{i(s-1)\varphi_0}], \end{aligned} \quad (\text{X.160})$$

where $z = \sqrt{\alpha_1^2 + \alpha_2^2}$, $\cos \varphi_0 = \alpha_2/z$, $\sin \varphi_0 = \alpha_1/z$. The functions B_s, B_{1s}, B_{2s} are related by

$$\alpha_1 B_{1s} + \alpha_2 B_{2s} = s B_s, \quad (\text{X.161})$$

which is a consequence of the known recurrence relation for Bessel functions:

$$J_{s-1}(z) + J_{s+1}(z) = \frac{2s}{z} J_s(z).$$

The matrix element (X.158) thus takes the form

$$S_{fi} = \frac{1}{\sqrt{2\omega' \cdot 2q_0 \cdot 2q'_0}} \sum_s M_{fi}^{(s)} (2\pi)^4 i \delta^{(4)}(sk + q - q' - k'); \quad (\text{X.162})$$

the rather cumbersome expressions for the amplitudes $M_{fi}^{(s)}$ we shall not write out here. Thus S_{fi} is an infinite sum of terms, each of which corresponds to a conservation law:

$$sk + q = q' + k'. \quad (\text{X.163})$$

Since

$$q^2 = q'^2 = m^2(1 + \xi^2) \equiv m_*^2 \quad (\text{X.164})$$

(cf. (40.15)), and $k^2 = k'^2 = 0$, the equality (X.163) is possible only for $s \geq 1$. The s -th term of the sum describes emission of a photon k' due to absorption of s photons with 4-momenta k from the wave. From the form of the equality (X.163) it is clear that all kinematic relations for the Compton effect will apply to the processes under consideration, if we replace the electron momenta by quasi-momenta q and the incident photon momentum by the 4-vector

sk . In particular, for the frequency of the emitted photon in the system where the electron is on average at rest ($\mathbf{q} = 0$, $q_0 = m_*$), we have

$$\omega' = \frac{s\omega}{1 + (s\omega/m_*)(1 - \cos \theta)}, \quad (\text{X.165})$$

where θ is the angle between $\mathbf{k}$ and $\mathbf{k}'$ (cf. (86.8)). One can say that the frequencies ω' are harmonics of the frequency ω . In our adopted notation (§ 64) the amplitude of the s -th harmonic radiation process coincides with $M_{fi}^{(s)}$, and the expression

$$dW_s = \frac{|M_{fi}^{(s)}|^2 d^3k' d^3q'}{(2\pi)^6 \cdot 2\omega' \cdot 2q_0 \cdot 2q'_0} (2\pi)^4 \delta^{(4)}(sk + q - q' - k') \quad (\text{X.166})$$

gives the corresponding differential probability (per unit time)⁶².

The structure of the amplitudes $M_{fi}^{(s)}$ is similar to the structure of the scattering amplitude with plane waves: $\bar{u}(p') \dots u(p)$. Therefore the operations of summation over polarisations of the particles are carried out in the usual way. After summation over polarisations of the final electron and photon and averaging over polarisations of the initial electron:

$$\begin{aligned} dW_s &= \frac{e^2 m^2}{4\pi} \frac{d^3k' d^3q'}{q_0 q'_0 \omega'} \delta^{(4)}(sk + q - q' - k') \times \\ &\times \left\{ -2J_s^2(z) + \xi^2 \left(1 + \frac{(kk')^2}{2(kp)(kp')} \right) (J_{s+1}^2 + J_{s-1}^2 - 2J_s^2) \right\}. \end{aligned} \quad (\text{X.167})$$

For integration of this expression we note that due to the axial symmetry of the circularly polarised wave field, the differential probability does not depend on the overall azimuthal angle ϕ around the direction $\mathbf{k}$. Together with the δ -function, this allows integration over all variables except one; as the latter we choose the invariant quantity $u = (kk')/(kp)$. Then after integration over $d^3k' d\phi d(q'_0 + \omega')$ we have

$$\delta^{(4)}(sk + q - q' - k') \frac{d^3q' d^3k'}{q'_0 \omega'} \rightarrow \frac{2\pi du}{(1+u)^2}.$$

⁶²Note that the normalisation of the functions ψ_p to unit density corresponds to normalisation of the δ -function "on the scale $q/(2\pi)$ " (cf. (40.17), where the factor q_0/p_0 on the right side of the equality will now be absent). It is precisely for this reason that the number of final states of the electron must be measured by the element $d^3q'/(2\pi)^3$.

Indeed, in the centre-of-inertia system (the system where $s\mathbf{k} + \mathbf{q} = \mathbf{q}' + \mathbf{k}' = 0$), the indicated integration gives $2\pi|\mathbf{q}'| d\cos\theta/E_s$, where $E_s = s\omega + q_0 = \omega' + q'_0$ and θ is the angle between $\mathbf{k}$ and $\mathbf{q}'$. On the other hand, in this same system,

$$u = \frac{E_s}{q'_0 - |\mathbf{q}'| \cos\theta} - 1, \quad d\cos\theta = \frac{E_s du}{|\mathbf{q}'|(1+u)^2}.$$

The interval $-1 \leq \cos\theta \leq 1$ corresponds to the interval

$$0 \leq u \leq u_s \equiv \frac{E_s^2}{m_*^2} - 1 = \frac{2s(kp)}{m_*^2}$$

(in the transformations one should remember that $(kp) = (kq)$).

Thus the total probability of radiation per unit time:

$$W = \sum_{s=1}^{\infty} W_s = \frac{e^2 m^2}{4q_0} \sum_{s=1}^{\infty} \int_0^{u_s} \frac{du}{(1+u)^2} \left\{ -4J_s^2(z) + \right. \\ \left. + \xi^2 \left(2 + \frac{u^2}{1+u} \right) (J_{s+1}^2 + J_{s-1}^2 - 2J_s^2) \right\}, \quad (\text{X.168})$$

where⁶³

$$u = \frac{(kk')}{(kp')}, \quad u_s = \frac{2s(kp)}{m_*^2}, \quad z = \frac{2sm^2\xi}{m_*\sqrt{1+\xi^2}} \sqrt{\frac{u}{u_s} \left(1 - \frac{u}{u_s} \right)}. \quad (\text{X.169})$$

For $\xi \ll 1$ (the condition for applicability of perturbation theory) the integrand expressions in (X.168) can be expanded in powers of ξ . Thus, for the first term of the expansion W_1 (at $\xi \ll 1$):

$$W_1 = \frac{e^2 m^2 \xi^2}{4p_0} \int_0^{u_1} \left[2 + \frac{u^2}{1+u} - \frac{4u}{u_1} \left(1 - \frac{u}{u_1} \right) \right] \frac{du}{(1+u)^2} \\ = \frac{e^2 m^2 \xi^2}{4p_0} \left[\left(1 - \frac{4}{u_1} - \frac{8}{u_1^2} \right) \ln(1+u_1) + \frac{1}{2} + \frac{8}{u_1} - \frac{1}{2(1+u_1)^2} \right], \quad (\text{X.170})$$

with $u_1 \approx 2(kp)/m^2$. As expected, this result coincides with the Klein–Nishina formula for photon scattering on an electron: substituting in (X.170) $-a^2 = 4\pi/\omega$, $\xi^2 = 4\pi e^2/(m^2\omega)$ and dividing by the flux density (64.14), we return to (86.16) (the total scattering cross section does not depend on the initial polarisation of the photon).

We also give the expression for the probability of emission of the second harmonic (first term of the expansion of W_2 at $\xi \ll 1$):

$$W_2 = \frac{e^2 m^2 \xi^4}{p_0} \int_0^{u_2} \frac{du}{(1+u)^2} \frac{u}{u_2} \left(1 - \frac{u}{u_2} \right) \left[2 + \frac{u^2}{1+u} - \frac{4u}{u_2} \left(1 - \frac{u}{u_2} \right) \right]. \quad (\text{X.171})$$

In general, the main term in W_s (for not too large s) is proportional to ξ^{2s} .

⁶³The indicated value of a^2 corresponds to normalisation of the 4-potential to one photon per unit volume. For its determination one must equate ω times the energy of the classical field with the (real) 4-potential (X.155).

Now consider the opposite case: $\xi \gg 1$. The parameter ξ can be made large, for example, by decreasing the frequency ω at fixed field intensity (obviously, $\xi = eF/(m\omega)$, where F is the field amplitude). Therefore it is clear that the case $\xi \gg 1$ essentially reduces to processes in a constant uniform field, the intensities $\mathbf{E}$ and $\mathbf{H}$ of which are mutually perpendicular and equal in magnitude (such a field is called a *crossed field*). The radiation probability can be obtained by the limiting transition $\xi \rightarrow \infty$, but it is simpler to carry out the calculations directly for a constant field, taking the 4-potential in the form

$$A^\mu = a^\mu \varphi, \quad \varphi = kx, \quad ak = 0 \quad (\text{X.172})$$

(so that $F_{\mu\nu} = k_\mu a_\nu - k_\nu a_\mu = \text{const}$). The exact wave function of the electron in this field is obtained by substitution of (X.172) into (40.7), (40.8):

$$\psi_p = \left[1 + \frac{e(\gamma k)(\gamma a)}{2(kp)} \varphi \right] \frac{u(p)}{\sqrt{2p_0}} \exp \left\{ -\frac{ie(ap)}{2(kp)} \varphi^2 + \frac{ie^2 a^2}{6(kp)} \varphi^3 - ipx \right\}. \quad (\text{X.173})$$

The result obtained with this function is exact for radiation of an electron in a crossed field at any electron energy. But in the ultrarelativistic case this result applies to radiation of an electron not only in a crossed field, but in any constant uniform electromagnetic field, including a constant magnetic field (which was considered in § 90).

To formulate this assertion we note that the state of a particle in an arbitrary constant uniform field is determined by the same quantum numbers as the state of a free particle, and these quantum numbers can always be chosen so that when the field is switched off they go over into the quantum numbers of the free particle, i.e. its 4-momentum p^μ ($p^2 = m^2$). Thus the state of the particle in the field will be described by a constant 4-vector p .

The total radiation intensity, being an invariant quantity, depends only on invariants that can be formed from the constant 4-tensor $F_{\mu\nu}$ and the 4-vector p . Taking into account also that $F_{\mu\nu}$ must enter the intensity only together with the charge e , we obtain three dimensionless invariants⁶⁴:

$$\chi^2 = -\frac{e^2}{m^6} (F_{\mu\nu} p^\nu)^2, \quad f = \frac{e^2 (F_{\mu\nu})^2}{m^4}, \quad g = \frac{e^2}{m^4} e_{\lambda\mu\nu\rho} F^{\lambda\mu} F^{\nu\rho}. \quad (\text{X.174})$$

In a crossed field $f = g \equiv 0$, while in the general case all three invariants are non-zero. But if the electron is ultrarelativistic ($p_0 \gg m$), and its momentum $\mathbf{p}$ makes with the fields $\mathbf{E}$, $\mathbf{H}$ angles $\theta \gg m/p_0$, then $\chi^2 \gg f$, g (in other words, for an ultrarelativistic particle almost for all directions of $\mathbf{p}$ any constant field looks like a crossed one). If, moreover, the field intensities $|\mathbf{E}|$, $|\mathbf{H}| \ll m^2/e (= m^2 c^3/(e\hbar))$, then $|f|$, $|g| \ll 1$. Under these conditions the radiation intensity, computed for a crossed field, also applies to radiation in any constant field, expressed through the invariant χ computed for a crossed field.

The invariant χ is expressed through the field intensities $\mathbf{E}$, $\mathbf{H}$ as

$$\chi^2 = \frac{e^2}{m^6} \{ ([\mathbf{p}\mathbf{H}] + p_0 \mathbf{E})^2 - (\mathbf{p} \cdot \mathbf{E})^2 \}.$$

⁶⁴In the expression for χ one can with the same accuracy consider p as the ordinary kinematic 4-momentum of the particle.

For a constant magnetic field, χ coincides with the quantity (90.3) introduced in § 90, so that the considerations presented here give another way to obtain the results of § 90⁶⁵.

⁶⁵Detailed treatment of the theory of various processes in strong fields is given in the reviews by *A. I. Nikishov* and *V. I. Ritus* in the collection “Kvantovaya elektrodinamika yavlenii v intensivnom pole” (Trudy FIAN, M.: Nauka, 1979, T. 111).

Exact Propagators and Vertex Parts

§ 102. Field operators in the Heisenberg representation

Up to now when considering various concrete electrodynamical processes we limited ourselves to the first non-vanishing approximation of perturbation theory. We now turn to the study of effects arising when higher-order approximations are taken into account. These effects are called *radiative corrections*.

A deeper understanding of the structure of higher approximations can be achieved on the basis of a preliminary investigation of general properties possessed by the exact (i.e. not expanded in powers of e^2) scattering amplitudes. We saw (see § 72) that successive terms of the perturbation theory series are expressed through field operators in the interaction representation—operators whose time dependence is determined by the Hamiltonian of the system of free particles $\hat{H}_0$. The exact scattering amplitudes are more conveniently expressed through operators of the field not in this, but in the Heisenberg representation, in which the time dependence is determined directly by the exact Hamiltonian of the system of interacting particles $\hat{H} = \hat{H}_0 + \hat{V}$.

By the general rule for constructing Heisenberg operators we have:¹

$$\hat{\psi}(x) \equiv \hat{\psi}(t, \mathbf{r}) = e^{i\hat{H}t} \hat{\psi}(\mathbf{r}) e^{-i\hat{H}t} \quad (\text{XI.1})$$

and similarly for $\hat{\bar{\psi}}(x)$ and $\hat{A}(x)$, where $\hat{\psi}(\mathbf{r}), \dots$ are time-independent (Schrödinger, or pre-interaction) operators.

We immediately note that Heisenberg operators taken at equal times satisfy the same commutation rules as operators in the Schrödinger representation or in the interaction representation. Indeed, we have, for example,

$$\{\hat{\psi}_i(t, \mathbf{r}), \hat{\psi}_k^\dagger(t, \mathbf{r}')\}_+ = e^{i\hat{H}t} \{\hat{\psi}_i(\mathbf{r}), \hat{\psi}_k^\dagger(\mathbf{r}')\}_+ e^{-i\hat{H}t} = \gamma_{ik}^0 \delta(\mathbf{r} - \mathbf{r}') \quad (\text{XI.2})$$

(cf. (75.6)). Analogously the operators $\hat{\psi}(t, \mathbf{r})$ and $\hat{A}(t, \mathbf{r}')$ are commutative:

$$\{\hat{\psi}_i(t, \mathbf{r}), \hat{A}(t, \mathbf{r}')\}_- = 0$$

(at different times this is no longer so!).

The “equation of motion” satisfied by the Heisenberg ψ -operator can be obtained from the general formula (13.7) (see III):

$$-i \frac{\partial \hat{\psi}(x)}{\partial t} = \hat{H} \hat{\psi}(x) - \hat{\psi}(x) \hat{H}. \quad (\text{XI.3})$$

For the Hamiltonians the Schrödinger and Heisenberg representations are identical, and the Hamiltonian is expressed identically through field operators in both representations.

¹In this chapter operators with a time argument will refer to the Heisenberg representation, and operators in the interaction representation will be marked with the additional index “int”.

In this case, when computing the right side of (XI.3), one can omit the part depending only on the operator $\hat{A}(x)$ (the Hamiltonian of the free electromagnetic field), since this part is commutative with $\hat{\psi}(x)$. According to (21.13) and (43.3) we have:

$$\begin{aligned}\hat{H} &= \int \hat{\psi}^*(t, \mathbf{r}) (\alpha \hat{\mathbf{p}} + \beta m) \hat{\psi}(t, \mathbf{r}) d^3x \\ &\quad + e \int \hat{\bar{\psi}}(t, \mathbf{r}) (\gamma \hat{A}(t, \mathbf{r})) \hat{\psi}(t, \mathbf{r}) d^3x \\ &= \int \hat{\bar{\psi}}(t, \mathbf{r}) \{ \gamma \hat{p} + m + e(\gamma \hat{A}(t, \mathbf{r})) \} \hat{\psi}(t, \mathbf{r}) d^3x.\end{aligned}\quad (\text{XI.4})$$

Computing the commutator $\{\hat{H}, \hat{\psi}(t, \mathbf{r})\}_-$ with the help of (XI.2) and eliminating the δ -function by integration over d^3x , we obtain:

$$(\gamma \hat{p} - e \gamma \hat{A} - m) \hat{\psi}(t, \mathbf{r}) = 0. \quad (\text{XI.5})$$

As expected, the operator $\hat{\psi}(t, \mathbf{r})$ satisfies the equation formally coinciding with the Dirac equation.

The equation for the operator of the electromagnetic field $\hat{A}(t, \mathbf{r})$ is obviously obtained from correspondence with the classical case. In this case (large occupation numbers—see § 5) after averaging over the field state the operator equation should go over into the classical Maxwell equation for potentials (30.2) (see II). Therefore it is clear that the equation for the operator simply coincides in form with the Maxwell equation, i.e. (in arbitrary gauge):

$$\partial_\nu \partial^\nu \hat{A}^\mu(x) - \partial^\mu \partial_\nu \hat{A}^\nu(x) = -4\pi e \hat{j}^\mu(x), \quad (\text{XI.6})$$

where $\hat{j}^\mu = \hat{\bar{\psi}}(x) \gamma^\mu \hat{\psi}(x)$ is the current operator, identically satisfying the continuity equation:

$$\partial_\mu \hat{j}^\mu(x) = 0. \quad (\text{XI.7})$$

It is essential that equations (XI.6) are linear in $\hat{A}^\mu$ and $\hat{j}^\mu$, and therefore no question arises about the ordering of these operators.

Like the equations for wave functions, the system of operator equations (XI.6), (XI.7) is invariant with respect to gauge transformation:²

$$\hat{A}^\mu(x) \rightarrow \hat{A}^\mu(x) - \partial^\mu \hat{\chi}(x), \quad \hat{\psi}(x) \rightarrow \hat{\psi}(x) e^{ie\hat{\chi}}, \quad \hat{\bar{\psi}}(x) \rightarrow e^{-ie\hat{\chi}} \hat{\bar{\psi}}(x), \quad (\text{XI.8})$$

where $\hat{\chi}(x)$ is an arbitrary Hermitian operator, commuting (at one and the same moment of time) with $\hat{\psi}$.

Now let us establish the connection between operators in the Heisenberg representation and in the interaction representation. To simplify the reasoning it is convenient to make the formal assumption (not affecting the final result) that the interaction $\hat{V}(t)$ is adiabatically “switched on” from $t = -\infty$ to finite times. Then when $t \rightarrow -\infty$ both representations—the

²We emphasise that here it is precisely about Heisenberg ψ -operators. In the interaction representation the gauge transformation of electromagnetic potentials does not at all affect the ψ -operators.

Heisenberg representation and the interaction representation—simply coincide. The corresponding wave functions of the system Φ and Φ_{int} also coincide:

$$\Phi_{\text{int}}(t = -\infty) = \Phi. \quad (\text{XI.9})$$

On the other hand, the wave function in the Heisenberg representation does not depend on time at all (all time dependence is transferred to the operators), while in the interaction representation the dependence of the wave function on time is given, according to (72.7):

$$\Phi_{\text{int}}(t) = \hat{S}(t, -\infty) \Phi_{\text{int}}(-\infty), \quad (\text{XI.10})$$

where the operator is introduced:

$$\hat{S}(t_2, t_1) = T \exp \left\{ -i \int_{t_1}^{t_2} \hat{V}(t') dt' \right\} \quad (\text{XI.11})$$

with the obvious properties:

$$\hat{S}(t, t_1) \hat{S}(t_1, t_0) = \hat{S}(t, t_0), \quad \hat{S}^{-1}(t, t_1) = \hat{S}(t_1, t). \quad (\text{XI.12})$$

Comparing formulas (XI.10) and (XI.9), we find the relation:

$$\Phi_{\text{int}}(t) = \hat{S}(t, -\infty) \Phi, \quad (\text{XI.13})$$

establishing the connection between the wave functions in both representations. Correspondingly the operator transformation formula:

$$\hat{\psi}(t, \mathbf{r}) = \hat{S}^{-1}(t, -\infty) \hat{\psi}_{\text{int}}(t, \mathbf{r}) \hat{S}(t, -\infty) = \hat{S}(-\infty, t) \hat{\psi}_{\text{int}}(t, \mathbf{r}) \hat{S}(t, -\infty) \quad (\text{XI.14})$$

(the same for $\hat{\bar{\psi}}$ and $\hat{A}$).

In conclusion let us make one more general remark. We have already repeatedly indicated that in relativistic quantum field theory the physical meaning of field operators is very limited due to the infinity of zero-point fluctuations. This applies even more to operators in the Heisenberg representation, which actually contain in themselves also divergences associated with the interaction. In this chapter, §§ 102, 109 are devoted to the exposition of formal theory, in which questions of removal of these infinities are not discussed and operations with all quantities are carried out as if they were finite. The results obtained in this way have primarily heuristic value: they allow one to more deeply understand the meaning of the expansion of perturbation theory; it is also possible that they are preserved in some form in a future theory free of the present difficulties.

§ 103. The exact photon propagator

The main role in the exact theory apparatus (without expansion in powers of e^2) is played by the concepts of exact propagators.

The exact photon propagator (which we shall denote by the script letter $\mathcal{D}$) is defined by the formula:

$$\mathcal{D}_{\mu\nu}(x - x') = i \langle 0 | T \hat{A}_\mu(x) \hat{A}_\nu(x') | 0 \rangle, \quad (\text{XI.15})$$

where $\hat{A}_\mu(x)$ are Heisenberg operators, in contrast to the definition (??):

$$D_{\mu\nu}(x - x') = i\langle 0 | T \hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') | 0 \rangle, \quad (\text{XI.16})$$

in which operators in the interaction representation figured. In contrast to the exact propagator (XI.15), function (XI.16) can be called the propagator of free photons.

In view of the impossibility of exact computation of the vacuum mean value (XI.15), the definition allows one to establish some general properties of this function. This will be the topic of § 111, and for now we shall concern ourselves with the computation of $\mathcal{D}_{\mu\nu}$ by perturbation theory with the help of diagrammatic technique. For this it is necessary to express $\mathcal{D}_{\mu\nu}$ through operators in the interaction representation.

Let first $t > t'$. Using the connection between $\hat{A}(x)$ and $\hat{A}_{\text{int}}(x)$ (cf. (XI.14)), we write:

$$\begin{aligned} \mathcal{D}_{\mu\nu}(x - x') &= i\langle 0 | \hat{A}_\mu(x) \hat{A}_\nu(x') | 0 \rangle \\ &= i\langle 0 | \hat{S}(-\infty, t) \hat{A}_\mu^{\text{int}}(x) \hat{S}(t, t') \hat{A}_\nu^{\text{int}}(x') \hat{S}(t', -\infty) | 0 \rangle. \end{aligned} \quad (\text{XI.17})$$

where for brevity it is denoted:

$$\hat{S} \equiv \hat{S}(+\infty, -\infty). \quad (\text{XI.18})$$

According to (XI.12):

$$\hat{S}(t, -\infty) \hat{S}(-\infty, t') = \hat{S}(t, t'), \quad \hat{S}(-\infty, t) = \hat{S}(-\infty, +\infty) \hat{S}(+\infty, t) = \hat{S}^{-1} \hat{S}(\infty, t).$$

Then:

$$\mathcal{D}_{\mu\nu} = i\langle 0 | \hat{S}^{-1} [\hat{S}(\infty, t) \hat{A}_\mu^{\text{int}}(x) \hat{S}(t, t') \hat{A}_\nu^{\text{int}}(x') \hat{S}(t', -\infty)] | 0 \rangle.$$

Since according to the definition (XI.11) $\hat{S}(t_2, t_1)$ contains only operators at times between t_1 and t_2 , arranged in chronological order, it is obvious that altogether all operator factors in square brackets are arranged in order of decreasing times from left to right. Placing before the bracket the sign of chronologisation T , we can then arbitrarily rearrange the order of factors, since the operator T automatically establishes them in the needed order. Using this, we rewrite the expression in brackets in the form:

$$[\dots] = T [\hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') \hat{S}(\infty, t) \hat{S}(t, t') \hat{S}(t', -\infty)] = T [\hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') \hat{S}].$$

Thus:

$$\mathcal{D}_{\mu\nu}(x - x') = i\langle 0 | \hat{S}^{-1} T [\hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') \hat{S}] | 0 \rangle, \quad (\text{XI.19})$$

and it is easy to verify analogously that this formula is valid also for $t < t'$.

We now show that the factor $\hat{S}^{-1}$ can be taken out from under the vacuum averaging sign in the form of some phase factor. For this we recall that the Heisenberg wave function of the vacuum Φ coincides with the value $\Phi_{\text{int}}(-\infty)$ of the wave function of this same state in the interaction representation (see (XI.9)). But the vacuum is a strictly stationary state; in it no spontaneous processes of particle production are possible; with

the passage of time the vacuum remains vacuum; this means that $\Phi_{\text{int}}(+\infty)$ can differ from $\Phi_{\text{int}}(-\infty)$ only by some phase factor $e^{i\alpha}$. According to (??) we have:

$$\hat{S} \Phi_{\text{int}}(-\infty) = e^{i\alpha} \Phi_{\text{int}}(-\infty) = \langle 0|S|0 \rangle \Phi_{\text{int}}(-\infty). \quad (\text{XI.20})$$

Taking the complex conjugate and using the unitarity of operator $\hat{S}$, we obtain:

$$\Phi_{\text{int}}^*(-\infty) \hat{S}^{-1} = \langle 0|S|0 \rangle^{-1} \Phi_{\text{int}}^*(-\infty).$$

From this it is clear that expression (XI.19) can be rewritten in the form:

$$\mathcal{D}_{\mu\nu}(x-x') = \frac{i \langle 0|T \hat{A}_{\mu}^{\text{int}}(x) \hat{A}_{\nu}^{\text{int}}(x') S|0 \rangle}{\langle 0|S|0 \rangle}. \quad (\text{XI.21})$$

Substituting here (in both numerator and denominator) the expansion (??) for $\hat{S}$ and performing the averaging with the help of Wick's theorem (see § 77), we obtain an expansion of $\mathcal{D}_{\mu\nu}$ in powers of e^2 .

In the numerator of (XI.21) the averaged expressions differ from the matrix elements of the type (??) considered in § 77, only in that instead of “external” operators of creation or annihilation of photons, there stand the operators $\hat{A}_{\mu}^{\text{int}}(x)$ and $\hat{A}_{\nu}^{\text{int}}(x')$. Since all factors in the averaged products stand under the chronologisation sign, the pairwise contractions of these operators with “internal” operators of creation or annihilation of photons will give photon propagators $D_{\mu\nu}$. Thus the results of the averaging will be expressed as sets of diagrams with two external ends, constructed according to the rules described in § 77, with the only difference that external photon lines, like internal ones, will now correspond to propagators $D_{\mu\nu}$ (instead of amplitudes e of real photons). In the zeroth approximation, setting $\hat{S} = 1$, the numerator of (XI.21) coincides simply with $D_{\mu\nu}(x-x')$. The next non-vanishing terms will be of order $\sim e^2$. They are represented by a set of diagrams containing two external ends and two vertices:

$$(\text{XI.22})$$

The second of these diagrams consists of two unconnected parts: a dashed line (which corresponds to $-iD_{\mu\nu}$) and

a closed loop. Such a decomposition of the diagram means decomposition of the corresponding analytic expression into two independent factors. Adding to diagrams (XI.22) the diagram (dashed line) of the zeroth approximation and “taking it outside the bracket”, we find that as a result, to the accuracy of terms of second order, the numerator in (XI.21) equals:

The expression $\langle 0|S|0 \rangle$ in the denominator of (XI.21) represents the amplitude of the “transition” from vacuum to vacuum. Its expansion therefore contains only diagrams without external ends. In the zeroth approximation $\langle 0|S|0 \rangle = 1$, and to accuracy of terms of second order we get:

Dividing numerator by denominator to this accuracy, we find that the curly bracket cancels and there remains:

Thus the diagram with the disconnected loop drops out of the answer. This result has a general character. Thinking about the way diagrams are constructed for the

numerator and denominator in (XI.21), it is not difficult to understand that the role of the denominator $\langle 0|S|0\rangle$ reduces to the fact that in any order of perturbation theory the exact propagator $\mathcal{D}_{\mu\nu}$ will be represented only by diagrams not containing disconnected parts.

We note that diagrams without external ends (closed loops) have no physical significance whatsoever and should not be taken into account even independently of the fact that they drop out in the formation of propagator $\mathcal{D}$. Indeed, such loops represent radiative corrections to a diagonal matrix element of the S -matrix for the vacuum-to-vacuum transition. But according to (XI.20) the sum of all these loops (together with unity of the zeroth approximation) gives only an insignificant phase factor, which cannot affect any physical results.

The transition from coordinate to momentum representation occurs in the usual way. Thus in the second approximation of perturbation theory the propagator $-i\mathcal{D}_{\mu\nu}(k)$ (which we will depict by a bold dashed line) is given by the sum:

$$(XI.23)$$

where all diagrams are computed by the usual rules (enumerated in § 77), with the only difference that external photon lines, like internal ones, are also assigned factors $-iD_{\mu\nu}(k)$.

The analytic expression of this formula gives therefore:

$$\mathcal{D}_{\mu\nu}(k) = D_{\mu\nu}(k) + ie^2 D_{\mu\lambda}(k) \int \text{Sp } \gamma^\lambda G(p+k) \gamma^\rho G(p) \frac{d^4 p}{(2\pi)^4} D_{\rho\nu}(k) \quad (XI.24)$$

(bispinor indices on matrices γ and G , as usual, are not written out).

The terms of the following approximations are constructed analogously; they are represented by sets of diagrams with two external photon ends and the needed number of vertices. Thus, terms $\sim e^4$ correspond to the following diagrams with four vertices:

$$(XI.25)$$

Four vertices also has the diagram

the upper part of which constitutes a loop formed by a single “closed on itself” electron line. Such a loop corresponds to the convolution $\hat{\psi}(x)\gamma\hat{\psi}(x)$, i.e. simply to the vacuum-average value of the current: $\langle 0|j(x)|0\rangle$. But by the very definition of the vacuum, this quantity must identically vanish, and this identity cannot, of course, be altered by any further radiative corrections to such a loop.³

Therefore altogether no diagrams with “closed on themselves” electron lines should be taken into account in any approximation.⁴

The part of the diagram (“block”) enclosed between two photon lines (external or internal) is called the *photon self-energy part*. In the general case such a block itself can be further subdivided into parts connected pairwise by one photon line, i.e. has the structure of the form:

³Although direct computation of the diagrams would lead to divergent integrals.

⁴In determining the signs one should not forget about the factor -1 contributed by a closed electron loop!

where circles denote blocks that can no longer be further subdivided in such a way; these parts are called *compact* (for example, of the four self-energy parts of fourth order (XI.25) the first three are compact).

Denote by the symbol $i\mathcal{P}_{\mu\nu}/4\pi$ the sum of all (infinite number of) compact self-energy parts; the function $\mathcal{P}_{\mu\nu}(k)$ is called the *polarisation operator*. Classifying the diagrams by the number of compact parts they contain, one can represent the exact propagator $\mathcal{D}_{\mu\nu}$ in the form of a series:

Analytically this series is written as:

$$\begin{aligned}\mathcal{D} &= D + D \frac{\mathcal{P}}{4\pi} D + D \frac{\mathcal{P}}{4\pi} D \frac{\mathcal{P}}{4\pi} D + \dots \\ &= D \left\{ 1 + \frac{\mathcal{P}}{4\pi} \left[D + D \frac{\mathcal{P}}{4\pi} D + \dots \right] \right\}\end{aligned}\quad (\text{XI.26})$$

(tensor indices omitted for brevity). But the series in square brackets again coincides with the series for $\mathcal{D}$. Therefore we have:

$$\mathcal{D}_{\mu\nu}(k) = D_{\mu\nu}(k) + D_{\mu\lambda}(k) \frac{\mathcal{P}^{\lambda\rho}(k)}{4\pi} \mathcal{D}_{\rho\nu}(k). \quad (\text{XI.27})$$

Multiplying this equality on the left by the inverse tensor $(D^{-1})^{\tau\mu}$ and on the right by $(\mathcal{D}^{-1})^{\nu\sigma}$ (and changing the notation of indices), we obtain its equivalent form:

$$\mathcal{D}_{\mu\nu}^{-1} = D_{\mu\nu}^{-1} - \frac{\mathcal{P}_{\mu\nu}}{4\pi}. \quad (\text{XI.28})$$

We emphasise that the representation of $\mathcal{D}$ in the form (XI.26) implies that from diagrams one can extract simpler blocks, which are computed by the general rules of diagrammatic technique. Combining such blocks with each other, we obtain correct expressions for the diagrams as a whole. The admissibility of such a decomposition constitutes an important (and by no means trivial) feature of diagrammatic technique. It is connected with the fact that the total numerical coefficient in a diagram does not depend on its order.

This same property allows one to use the function $\mathcal{D}$ (if it is known) for simplification of computations of radiative corrections to amplitudes of various scattering processes: instead of each time considering diagrams with corrections to internal photon lines, we can simply replace these lines by bold ones, i.e. assign to them propagators $\mathcal{D}$ (instead of D).

If the photon line corresponds to a real (not virtual) photon, i.e. if it is an external end of the diagram as a whole, then after including in it all self-energy corrections we get what is called an *effective external line*. It corresponds, in distinction from (XI.27), to replacing the multiplier D by the polarisation amplitude of the real photon:

$$e_\mu + D_{\mu\rho}(k) \frac{\mathcal{P}^{\rho\lambda}(k)}{4\pi} e_\lambda. \quad (\text{XI.29})$$

If we are talking about a line of an external field, then instead of e_μ here one should write $A_\mu^{(e)}$.

Everything said in § 76 about the tensor structure and gauge ambiguity of the approximate propagator $D_{\mu\nu}$ also applies to the exact function $\mathcal{D}_{\mu\nu}$. Remaining within the framework of relativistically invariant representations of this function, let us write its general form:

$$\mathcal{D}_{\mu\nu}(k) = \mathcal{D}(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + \mathcal{D}^{(l)}(k^2) \frac{k_\mu k_\nu}{k^2}; \quad (\text{XI.30})$$

the first term corresponds to the Landau gauge, and in the second term $\mathcal{D}^{(l)}$ is a gauge-arbitrary function. An analogous representation for the approximate propagator:

$$D_{\mu\nu}(k) = D(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + D^{(l)}(k^2) \frac{k_\mu k_\nu}{k^2}. \quad (\text{XI.31})$$

The polarisation operator, in accordance with its definition, is also a transverse tensor:⁵

$$\mathcal{P}_{\mu\nu} = \mathcal{P}(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right). \quad (\text{XI.32})$$

For the direct tensors (XI.30) and (XI.31) the inverse tensors have the form:

$$\mathcal{D}_{\mu\nu}^{-1}(k) = \frac{1}{\mathcal{D}} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + \frac{1}{\mathcal{D}^{(l)}} \frac{k_\mu k_\nu}{k^2}, \quad D_{\mu\nu}^{-1}(k) = \frac{1}{D} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + \frac{1}{D^{(l)}} \frac{k_\mu k_\nu}{k^2}. \quad (\text{XI.33})$$

Substituting (XI.32) and (XI.33) into (XI.28) and comparing the transverse parts, we find:

$$\frac{1}{\mathcal{D}} = \frac{1}{D} - \frac{\mathcal{P}}{4\pi}, \quad (\text{XI.34})$$

where $\mathcal{P} = \mathcal{P}(k^2)$, $D = D(k^2) = 4\pi/k^2$, or:

$$\mathcal{D}(k^2) = \frac{4\pi}{k^2(1 - \mathcal{P}(k^2)/k^2)}. \quad (\text{XI.35})$$

Thus the polarisation operator is (in contrast to the photon propagator itself) a gauge-invariant quantity.

§ 104. The photon self-energy function

For further investigation of the analytic properties of the photon propagator it will be useful to introduce, alongside the polarisation operator, one more auxiliary quantity $\Pi_{\mu\nu}(k)$, which is called the *photon self-energy function*. Namely, $i\Pi_{\mu\nu}/(4\pi)$ is defined as the sum of all (not only compact) self-energy photon parts. Depicting this sum as a square on the diagrams, we represent the exact propagator as the sum:
i.e.:

$$\mathcal{D}_{\mu\nu} = D_{\mu\nu} + D_{\mu\lambda} \frac{\Pi^{\lambda\rho}}{4\pi} D_{\rho\nu}. \quad (\text{XI.36})$$

⁵The definition of $D^{(l)}$ in this formula does not coincide with the definition in (??).

Expressing from this $\Pi_{\mu\nu}$ in the form:

$$\frac{1}{4\pi} \Pi_{\mu\nu} = D_{\mu\lambda}^{-1} \mathcal{D}^{\lambda\rho} D_{\rho\nu}^{-1} - D_{\mu\nu}^{-1}$$

and substituting into this equation (103.16), (103.19), and then (103.21), we get:

$$\Pi_{\mu\nu} = \Pi(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right), \quad \Pi = \frac{\mathcal{P}}{1 - \mathcal{P}/k^2}. \quad (\text{XI.37})$$

We see that $\Pi_{\mu\nu}$ (like $\mathcal{P}_{\mu\nu}$) is a gauge-invariant tensor.

The usefulness of the quantity $\Pi_{\mu\nu}$ is connected with its expression in the coordinate representation. It is easy to find, noting that the equality

$$\frac{1}{4\pi} \Pi_{\mu\nu}(k) = D_{\mu\lambda}^{-1} \mathcal{D}^{\lambda\rho} D_{\rho\nu}^{-1},$$

taking into account from (103.18) the transverse tensor $\mathcal{D}^{\lambda\rho} - D^{\lambda\rho}$, in the coordinate representation can be written in the form:

$$\Pi_{\mu\nu}(x-x') = \frac{1}{4\pi} (\partial_\mu \partial_\lambda - g_{\mu\lambda} \partial_\sigma \partial^\sigma) (\partial'_\nu \partial'_\rho - g_{\nu\rho} \partial'_\sigma \partial'^\sigma) \{ \mathcal{D}^{\lambda\rho}(x-x') - D^{\lambda\rho}(x-x') \}.$$

For carrying out differentiation it is necessary to substitute here:

$$\mathcal{D}^{\lambda\rho}(x-x') - D^{\lambda\rho}(x-x') = i \langle 0 | T \hat{A}^\lambda(x) \hat{A}^\rho(x') - T \hat{A}^{\text{int},\lambda}(x) \hat{A}^{\text{int},\rho}(x') | 0 \rangle. \quad (\text{XI.38})$$

We saw in § 75 that differentiation of T -products requires, generally speaking, caution due to their discontinuous character. But the averaged expression in (XI.38) is continuous, since the commutation rules for components of operators $\hat{A}^\lambda(x)$ and $\hat{A}^{\text{int}}(x)$ (taken at one and the same moment of time) are identical and the corresponding jumps cancel (cf. § 75). Therefore differentiation of the difference (XI.38) can be carried out under the T sign. According to (XI.6) (and similarly to the equation without the right-hand part for operators of the free electromagnetic field $\hat{A}_\mu^{\text{int}}(x)$) we get in the result the expression:

$$\Pi_{\mu\nu}(x-x') = 4\pi i e^2 \langle 0 | T \hat{j}_\mu(x) \hat{j}_\nu(x') | 0 \rangle. \quad (\text{XI.39})$$

This manifestly reveals the gauge invariance of $\Pi_{\mu\nu}$, since such are the current operators.

From (XI.39) one can obtain an important integral representation of this function.

In view of (XI.37) it is sufficient to consider the scalar function $\Pi = \Pi_\mu^\mu/3$. In the coordinate representation:

$$\Pi(x-x') = \frac{4\pi}{3} i e^2 \langle 0 | T \hat{j}_\mu(x) \hat{j}^\mu(x') | 0 \rangle = \frac{4\pi}{3} i e^2 \left\{ \sum_n \langle 0 | \hat{j}_\mu(x) | n \rangle \langle n | \hat{j}^\mu(x') | 0 \rangle, \quad t > t', \right. \\ \left. \sum_n \langle 0 | \hat{j}_\mu(x') | n \rangle \langle n | \hat{j}^\mu(x) | 0 \rangle, \quad t < t', \right. \quad (\text{XI.40})$$

where the symbol n numbers the states of the system “electromagnetic + electron-positron field”.⁶

⁶The current operator conserves charge, therefore the states $|n\rangle$ in (XI.40) can contain only equal numbers of electrons and positrons.

Since the current operator $\hat{j}(x)$ depends on $x^\mu = (t, \mathbf{r})$, also the matrix elements depend on x . This dependence can be established explicitly, if one chooses as states $|n\rangle$ the states with definite values of total 4-momentum of the system (see III, (15.13)). The operator of such a translation is $e^{i\mathbf{r}\hat{\mathbf{P}}}$, where $\hat{\mathbf{P}}$ is the operator of total momentum of the system (see III, (15.13)). Bearing in mind the general rule for transformation of matrix elements (see III, (12.7)), we find:

$$\langle n|\hat{j}^\mu(t, \mathbf{r})|m\rangle = \langle n|\hat{j}^\mu(0)|m\rangle e^{-i(P_m - P_n) \cdot x}. \quad (\text{XI.41})$$

We also note that the matrix $\langle n|\hat{j}_\mu(0)|m\rangle$ is Hermitian (like the matrix (XI.41) of the operator $\hat{j}^\mu(t, \mathbf{r})$ as a whole), and by virtue of the continuity equation (XI.7) satisfies the transversality condition:

$$(P_n - P_m)^\mu \langle n|\hat{j}_\mu(0)|m\rangle = 0. \quad (\text{XI.42})$$

Let us return to the computation of the function $\Pi(x - x')$. Substituting (XI.41) into (XI.40), we get:

$$\Pi(\xi) = \frac{4\pi i e^2}{3} \sum_n \langle 0|\hat{j}_\mu(0)|n\rangle \langle n|\hat{j}^\mu(0)|0\rangle e^{\mp i P_n \cdot \xi}, \quad \tau \gtrless 0, \quad (\text{XI.43})$$

where $x - x' = \xi = (\tau, \boldsymbol{\xi})$. Denote:

$$\rho(k^2) = -\frac{4\pi e^2}{3} (2\pi)^3 \sum_n \langle 0|\hat{j}_\mu(0)|n\rangle \langle 0|\hat{j}^\mu(0)|n\rangle^* \delta^{(4)}(k - P_n). \quad (\text{XI.44})$$

The summation is over all systems of real electron pairs and photons that can be produced by a virtual photon with 4-momentum $k = (\omega, \mathbf{k})$ ($k^2 > 0$), and for each of such systems—also over its internal variables (polarisations and momenta of particles in the centre-of-inertia system).⁷

As a result of such summation the function ρ can depend only on k , and by virtue of its scalar nature—only on k^2 . In particular, ρ does not depend on the direction of $\mathbf{k}$. Having in mind these properties of the function ρ , we rewrite (XI.43) in the form:

$$\Pi(\xi) = -i \int_0^\infty d\omega \int \frac{d^3 k}{(2\pi)^3} \rho(k^2) e^{i\mathbf{k} \cdot \boldsymbol{\xi} - i\omega|\tau|} = -i \int \frac{d^3 k}{(2\pi)^3} \int_0^\infty d\omega d(\mu^2) \delta(\mu^2 - k^2) \rho(\mu^2) e^{i\mathbf{k} \cdot \boldsymbol{\xi} - i\omega|\tau|}$$

The transition to the momentum representation is effected by substitution of the formula:

$$e^{-i\omega|\tau|} = 2i\omega \int_{-\infty}^\infty \frac{dk_0}{2\pi} \frac{e^{-ik_0\tau}}{k_0^2 - \omega^2 + i0} \quad (\text{XI.45})$$

(already used in § 76) and gives:

$$\Pi(k^2) = \int d(\mu^2) \int d(\omega^2) \delta(\mu^2 - k^2 - \omega^2) \frac{\rho(\mu^2)}{k_0^2 - \omega^2 + i0},$$

⁷This definition of states $|n\rangle$ is obviously identical with identifying them as states for which the matrix elements $\langle 0|\hat{j}|n\rangle$ of the charge-odd operator are non-zero.

or finally:⁸

$$\Pi(k^2) = \int_0^\infty \frac{\rho(\mu^2) d(\mu^2)}{k^2 - \mu^2 + i0}. \quad (\text{XI.46})$$

The coefficient ρ in this integral representation is called the *spectral density* of the function $\Pi(k^2)$. It possesses the properties:

$$\rho(k^2) = 0 \quad \text{for } k^2 < 0, \quad \rho(k^2) > 0 \quad \text{for } k^2 > 0. \quad (\text{XI.47})$$

Indeed, the 4-momentum k of a virtual photon that can produce a system of real particles is necessarily timelike (k^2 coincides with the square of the total energy of the particles in their centre-of-inertia system). By virtue of the transversality condition (XI.42) we have:

$$P_n^\mu \langle 0 | \hat{j}_\mu | n \rangle = 0.$$

But the 4-vector $\langle 0 | \hat{j} | n \rangle$, orthogonal to a timelike 4-vector (P_n), is spacelike, i.e.:

$$\langle 0 | \hat{j}_\mu | n \rangle \langle 0 | \hat{j}^\mu | n \rangle^* < 0,$$

and therefore, according to definition (XI.44), $\rho > 0$.

§ 105. The exact electron propagator

Similarly to the photon, the exact electron propagator is defined by the formula:

$$\mathcal{G}_{ik}(x - x') = -i \langle 0 | T \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x') | 0 \rangle \quad (\text{XI.48})$$

(i, k are bispinor indices), differing from the definition (??) of the propagator of free particles:

$$G_{ik}(x - x') = -i \langle 0 | T \hat{\psi}_i^{\text{int}}(x) \hat{\bar{\psi}}_k^{\text{int}}(x') | 0 \rangle, \quad (\text{XI.49})$$

by the replacement of $\hat{\psi}$ -operators in the interaction representation by Heisenberg ones.

The same reasoning as in deriving (XI.21) allows one to transform the propagator $\mathcal{G}$ to the form:

$$\mathcal{G}_{ik}(x - x') = \frac{\langle 0 | T \hat{\psi}_i^{\text{int}}(x) \hat{\bar{\psi}}_k^{\text{int}}(x') S | 0 \rangle}{\langle 0 | S | 0 \rangle}. \quad (\text{XI.50})$$

Its expansion in powers of e^2 leads to the representation of the $\mathcal{G}$ -function as a set of diagrams with two external electron lines and various numbers of vertices. Here the role of the denominator in (XI.50) again reduces to the necessity of including only diagrams without isolated “vacuum loops”. Thus, to terms $\sim e^2$ we have:⁹

$$(\text{XI.51})$$

⁸Formal calculations, analogous to those carried out above, require caution in view of the presence of the already-mentioned divergences. This leads, in particular, to the appearance on the right-hand side of (XI.46) of additional divergent terms that do not have a relativistically invariant form (so-called Schwinger terms). We do not write them out, since they are eliminated in renormalisation (see § 110) and do not affect the further results.

⁹As was already explained in § 103, one should also not include here diagrams with electron lines “closed on themselves”, which would appear already in second order.

The bold solid line corresponds (in the momentum representation) to the function $i\mathcal{G}(p)$, and all thin solid and dashed lines in the diagrams on the right side of the equality correspond to the propagators of free particles iG and $-iD$ respectively.

The block enclosed between two electron lines is called the *electron self-energy part*. As in the photon case, a self-energy part is called *compact* if it cannot be further subdivided into two self-energy parts by cutting one electron line. The sum of all possible compact parts we denote by $-i\mathcal{M}_{ik}$; the function $\mathcal{M}_{ik}(p)$ is called the *mass operator*. Thus, to accuracy $\sim e^4$ we have:

$$(XI.52)$$

By summation, in exact analogy with the derivation of (XI.27), we get:

$$\mathcal{G}(p) = G(p) + G(p) \mathcal{M}(p) \mathcal{G}(p) \quad (XI.53)$$

(bispinor indices omitted), or for inverse matrices:

$$\mathcal{G}^{-1}(p) = G^{-1}(p) - \mathcal{M}(p) = \gamma p - m - \mathcal{M}(p). \quad (XI.54)$$

In § 102 it was already noted that the Heisenberg $\hat{\psi}$ -operators (in contradistinction to $\hat{\psi}$ -operators in the interaction representation) change under gauge transformation of electromagnetic potentials. Together with them also turns out to be gauge-non-invariant the exact electron propagator $\mathcal{G}$. Let us clarify the law of its gauge transformation (L. D. Landau, I. M. Khalatnikov, 1952).

It is clear in advance that the change of $\mathcal{G}$ under gauge transformation must be expressible through the very same quantity $\mathcal{D}^{(l)}$ that is added to the photon propagator under this transformation. This will become obvious if one notes that in computing $\mathcal{G}$ from the diagrams of perturbation theory each term of the series is expressed through the functions D and no other electromagnetic quantities appear in it. This circumstance can be used to simplify the derivation: one can make any particular assumptions about the properties of the arbitrary operator $\hat{\chi}$ in the transformation (XI.8), as long as the answer is expressed through $\mathcal{D}^{(l)}$.

As a result of the transformation (XI.8) the propagators $\mathcal{D}$ (XI.15) and $\mathcal{G}$ (XI.48) go over to:

$$\mathcal{D}_{\mu\nu} \rightarrow i\langle 0|T[\hat{A}_\mu(x) - \partial_\mu\hat{\chi}(x)][\hat{A}_\nu(x') - \partial'_\nu\hat{\chi}(x')]|0\rangle, \quad (XI.55)$$

$$\mathcal{G}_{ik} \rightarrow -i\langle 0|T\hat{\psi}_i(x)e^{ie\hat{\chi}(x)}e^{-ie\hat{\chi}(x')}\hat{\psi}_k(x')|0\rangle.$$

We assume now that the operators $\hat{\chi}$ are averaged independently of all remaining operator factors in the T -product; this assumption is quite natural, since by virtue of gauge invariance the “field” $\hat{\chi}$ takes no part in the interaction. We also assume that the vacuum average of $\hat{\chi}$ itself vanishes: $\langle 0|\hat{\chi}|0\rangle = 0$. Then in (XI.55) the terms containing $\hat{\chi}$ separate out and we get:

$$\mathcal{D}_{\mu\nu} \rightarrow \mathcal{D}_{\mu\nu} + i\langle 0|T\partial_\mu\hat{\chi}(x) \cdot \partial'_\nu\hat{\chi}(x')|0\rangle, \quad (XI.56)$$

$$\mathcal{G}_{ik} \rightarrow \mathcal{G}_{ik} \langle 0|T e^{ie\hat{\chi}(x)} e^{-ie\hat{\chi}(x')} |0\rangle. \quad (XI.57)$$

The further derivation is carried out for an infinitesimally small transformation; to emphasise this smallness we shall write $\delta\hat{\chi}$ instead of $\hat{\chi}$.

The transformation (XI.56) can (independently of the smallness of $\delta\hat{\chi}$) be written as:

$$\mathcal{D}_{\mu\nu} \rightarrow \mathcal{D}_{\mu\nu} + \delta\mathcal{D}_{\mu\nu}, \quad \delta\mathcal{D}_{\mu\nu} = \partial_\mu \partial'_\nu d^{(l)}(x - x'), \quad (\text{XI.58})$$

where

$$d^{(l)}(x - x') = i\langle 0 | T \delta\hat{\chi}(x) \delta\hat{\chi}(x') | 0 \rangle. \quad (\text{XI.59})$$

From this it is clear that the function $d^{(l)}$ determines the change under gauge transformation of the longitudinal part of the photon propagator $\mathcal{D}^{(l)}$.¹⁰

The assumption about the dependence of this function only on the difference $x - x'$ means, of course, a definite restriction on the properties of the operator $\delta\hat{\chi}$; in the general case a fully arbitrary gauge transformation can violate the space-time homogeneity of the propagator.

In (XI.57) we expand the exponential factors in powers of $\delta\hat{\chi}$ to the accuracy of quadratic terms:¹¹

$$\langle 0 | T e^{ie\delta\hat{\chi}(x)} e^{-ie\delta\hat{\chi}(x')} | 0 \rangle \approx 1 - \frac{1}{2}e^2 \langle 0 | \delta\hat{\chi}^2(x) + \delta\hat{\chi}^2(x') - 2T \delta\hat{\chi}(x) \delta\hat{\chi}(x') | 0 \rangle.$$

Taking into account definition (XI.59), we find therefore the following law of transformation of the electron propagator:

$$\mathcal{G} \rightarrow \mathcal{G} + \delta\mathcal{G}, \quad \delta\mathcal{G} = ie^2 \mathcal{G}(x - x') [d^{(l)}(0) - d^{(l)}(x - x')]. \quad (\text{XI.60})$$

In the momentum representation:

$$\delta\mathcal{G}(p) = ie^2 \int d^{(l)}(q) [\mathcal{G}(p) - \mathcal{G}(p - q)] \frac{d^4 q}{(2\pi)^4}. \quad (\text{XI.61})$$

Moreover the function $d^{(l)}(q)$ is connected with the change of the function $\mathcal{D}^{(l)}$ by:

$$\delta\mathcal{D}^{(l)}(q) = q^2 d^{(l)}(q). \quad (\text{XI.62})$$

For the electron propagator one could also obtain an integral representation analogous to formula (104.11). Its derivation is based on the expressions

$$\psi_{nm}(x) = \psi_{nm}(0) e^{-i(P_m - P_n)x} \quad (\text{XI.63})$$

for matrix elements of the $\hat{\psi}$ -operator, similar to those used in § 104 expressions (104.6) for matrix elements of the current. In contradistinction to the current, however, the

¹⁰The transition from (XI.56) to (XI.58) is possible if the function $d^{(l)}$ and its derivative with respect to t are continuous at $t = t'$. In the opposite case the right-hand sides of these expressions would differ by δ -functional terms (cf. the derivation of formula (75.2)). In the momentum representation this condition is equivalent to the assumption that $d^{(l)}(q)$ decreases at $|q^2| \rightarrow \infty$ faster than $1/q^2$.

¹¹If the function $f(x) = f_1(x) f_2(x)$, then its Fourier components $f(p) = \int f(x) e^{ipx} d^4 x = \iiint d^4 x \frac{d^4 q_1 d^4 q_2}{(2\pi)^8} e^{ix(p - q_1 - q_2)} f_1(q_1) f_2(q_2) = \iint \frac{d^4 q_1 d^4 q_2}{(2\pi)^4} \delta^{(4)}(p - q_1 - q_2) f_1(q_1) f_2(q_2) = \int \frac{d^4 q}{(2\pi)^4} f_1(q) f_2(p - q)$. In the transition from (XI.60) to (XI.61) it was also taken into account that $f(x = 0) = \int f(q) d^4 q / (2\pi)^4$.

$\hat{\psi}$ -operators themselves are gauge-non-invariant. Therefore the coordinate dependence of the form

(XI.63) does not have a general character, and relates only to a particular gauge, and the integral representation of the propagator based on (XI.63) has no general character. The deep physical reason for this situation is the equality to zero of the photon mass, and the resulting infrared catastrophe (see § 98). Because of this an electron in the process of interaction emits an infinite number of soft quanta, which to a considerable extent deprives the “single-particle” propagator (XI.48) of direct physical meaning.

§ 106. The vertex operator

In complex diagrams one can distinguish certain standard structural elements (blocks), of which these diagrams are composed. In §§ 103 and 105 two of such elements were in essence already considered: the photon and electron self-energy parts. Below we shall encounter yet another type — the *vertex part*.

For this purpose let us define the *vertex function*

$$K_{ik}^{\mu}(x_1, x_2, x_3) = \langle 0 | T \hat{A}^{\mu}(x_1) \hat{\psi}_i(x_2) \hat{\bar{\psi}}_k(x_3) | 0 \rangle \quad (\text{XI.64})$$

— the vacuum average of the T -product of Heisenberg operators of the electromagnetic and electron-positron fields (one factor of each type). In the interaction representation:

$$K_{ik}^{\mu}(x_1, x_2, x_3) = \frac{\langle 0 | T \hat{A}^{\text{int}, \mu}(x_1) \hat{\psi}_i^{\text{int}}(x_2) \hat{\bar{\psi}}_k^{\text{int}}(x_3) S | 0 \rangle}{\langle 0 | S | 0 \rangle}. \quad (\text{XI.65})$$

The transition to the momentum representation is carried out by the formula

$$\begin{aligned} (2\pi)^4 \delta^{(4)}(p_1 + k - p_2) K_{ik}^{\mu}(p_2, p_1; k) = \\ = \iiint K_{ik}^{\mu}(x_1, x_2, x_3) e^{-ikx_1 + ip_2x_2 - ip_1x_3} d^4x_1 d^4x_2 d^4x_3. \end{aligned} \quad (\text{XI.66})$$

In diagrammatic technique the functions K_{ik}^{μ} correspond to blocks (three-tails) of the type

$$(\text{XI.67})$$

with three (one photon and two electron) ends, whose momenta are related by the conservation law

$$p_1 + k = p_2. \quad (\text{XI.68})$$

The zeroth-order term in the expansion of (XI.65) vanishes. In first order in the coordinate representation:

$$K^{\mu} = e \int G(x_2 - x) \gamma_{\nu} G(x - x_3) \cdot D^{\nu\mu}(x_1 - x) d^4x,$$

and in the momentum representation:

$$K^{\mu}(p_2, p_1; k) = e G(p_2) \gamma_{\nu} G(p_1) \cdot D^{\nu\mu}(k), \quad (\text{XI.69})$$

corresponding to the diagram:

(XI.70)

In higher approximations new vertices arise. In third order there are the following diagrams:

(XI.71)

The first three of these can be cut into two parts, one of which is the simple vertex (XI.70), and the other a second-order self-energy part of one of the external lines.

Corrections of this kind to the function K^μ simply amount to the replacement of the free propagators G and D of the external lines by the exact propagators $\mathcal{G}$ and $\mathcal{D}$. But the fourth diagram cannot be decomposed in this manner.

Separating from the sum of all diagrams for K^μ the propagators of the three external lines, we write by definition:

$$K^\mu(p_2, p_1; k) = \{i\mathcal{G}(p_2) [-ie\Gamma_\nu(p_2, p_1; k)] i\mathcal{G}(p_1)\} \{-i\mathcal{D}^{\nu\mu}(k)\}. \quad (\text{XI.72})$$

The block that cannot be divided into parts by cutting a single (electron or photon) line of an internal diagram is called the *vertex part*. The quantity Γ^μ , equal to the sum of all vertex parts including the simple vertex γ^μ , is called the *vertex operator* (or *vertex function*).

All diagrams of the vertex operator up to fifth order are:

(XI.73)

The operator Γ has two bispinor indices and one four-vector index; it is a function of p_1, p_2 (or equivalently of p_1, k , since $p_2 = p_1 + k$). We note that not all three four-momenta p_1, p_2, k can correspond simultaneously to real particles, since the conservation law (XI.68) cannot be satisfied when $p_1^2 = p_2^2 = m^2, k^2 = 0$.

The vertex parts can be split into *irreducible* and *reducible* ones. An irreducible vertex part is one that cannot be split into simpler parts by removing a single electron or photon line; in the terminology previously introduced these are vertex parts that do not contain self-energy insertions in internal lines and do not contain vertex corrections within themselves. From the diagrams (XI.73): only *b*) and *d*) are irreducible (not counting the bare vertex *a*)); diagrams *g*), *h*), *i*) contain self-energy parts; in diagram *c*) the upper horizontal dashed line can be regarded as a correction to the upper vertex, and the lateral dashed lines in diagrams *e*) and *f*) as corrections to the lateral vertices.

In irreducible diagrams, if one replaces the internal lines by bold (exact propagator) lines and the vertices by bold points, i.e. replaces D, G by $\mathcal{D}, \mathcal{G}$ and γ by Γ , one obtains all vertex parts whatsoever. The resulting diagrams are called *skeleton diagrams*.

This decomposition of the vertex operator can be written symbolically in the form:

(XI.74)

This is an integral equation for Γ , with an infinite number of terms on the right-hand side.

Let us state the general principle of constructing exact expressions corresponding to diagrammatic blocks with an arbitrary number of external lines. Such an expression is constructed as the vacuum average of a T -product of Heisenberg operators: one factor of $\hat{\psi}(x)$ for each final electron line, one factor of $\hat{\bar{\psi}}(x)$ for each initial electron line, and one factor of $\hat{A}(x)$ for each photon line.

As an example let us consider four-electron diagrams — diagrams with four electron external lines:

(XI.75)

The corresponding function is:

$$K_{ik,lm}(x_1, x_2; x_3, x_4) = \langle 0 | T \hat{\psi}_i(x_1) \hat{\bar{\psi}}_k(x_2) \hat{\psi}_l(x_3) \hat{\bar{\psi}}_m(x_4) | 0 \rangle. \quad (\text{XI.76})$$

Its Fourier transform contains a δ -function expressing the overall conservation of four-momentum:

$$K_{ik,lm}(p_3, p_4; p_1, p_2) \sim (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4). \quad (\text{XI.77})$$

We write the function K in the form:

$$\begin{aligned} K_{ik,lm}(p_3, p_4; p_1, p_2) = & (2\pi)^4 \left[\delta^{(4)}(p_1 - p_3) \mathcal{G}_{il}(p_1) \mathcal{G}_{km}(p_2) \right. \\ & \left. - \delta^{(4)}(p_2 - p_3) \mathcal{G}_{im}(p_1) \mathcal{G}_{kl}(p_2) \right] \\ & + \mathcal{G}_{in}(p_3) \mathcal{G}_{kr}(p_4) \left[-i\Gamma_{nr,st}(p_3, p_4; p_1, p_2) \right] \mathcal{G}_{sl}(p_1) \mathcal{G}_{tm}(p_2). \end{aligned} \quad (\text{XI.78})$$

The first two terms in (XI.78) exclude from Γ diagrams that split into two disconnected parts, each having two external lines:

The third term: the $\mathcal{G}$ -factors on all four external lines exclude corrections to the external electron lines (self-energy insertions on external legs).

The antisymmetry of Γ (following from the properties of the T -product of fermionic $\hat{\psi}$ -operators) is expressed by:

$$\Gamma_{ik,lm}(p_3, p_4; p_1, p_2) = -\Gamma_{ki,lm}(p_4, p_3; p_1, p_2) = -\Gamma_{ik,ml}(p_3, p_4; p_2, p_1). \quad (\text{XI.79})$$

If p_1, p_2, p_3, p_4 correspond to real particles, the non-decomposing diagrams (XI.75) describe electron-electron scattering. The scattering amplitude (with the factors $\mathcal{G}$ replaced, on the external lines, by spinor amplitudes of the particles)¹² is:

$$iM_{fi} = \bar{u}_i(p_3) \bar{u}_k(p_4) \left[-i\Gamma_{ik,lm}(p_3, p_4; p_1, p_2) \right] u_l(p_1) u_m(p_2). \quad (\text{XI.80})$$

By virtue of (XI.79) this amplitude automatically possesses the required antisymmetry with respect to permutation of the electrons.

¹²We shall see later (see § 110) that in constructing amplitudes of real processes one need not include self-energy parts on the free ends of the diagrams.

§ 107. Dyson equations

The exact propagators and the vertex part are connected with each other by definite integral relations.

The concept of irreducibility of diagrams extends to any diagrams. Let us consider compact self-energy electron diagrams. Among them only one is irreducible — the second-order diagram. All vertex corrections can be assigned to one of the two vertices.¹³ Therefore the mass operator $\mathcal{M}$ is depicted by a single skeleton diagram:

(XI.81)

Analytically:¹⁴

$$\mathcal{M}(p) = \mathcal{G}^{-1}(p) - G^{-1}(p) = -ie^2 \int \gamma^\nu \mathcal{G}(p+k) \Gamma^\mu(p+k, p; k) \mathcal{D}_{\mu\nu}(k) \frac{d^4 k}{(2\pi)^4}. \quad (\text{XI.82})$$

An analogous expression can be written for the polarisation operator $\mathcal{P}$. Among the compact photon self-energy parts only one is irreducible:

(XI.83)

The corresponding analytic equality is:

$$\frac{\mathcal{P}_{\mu\nu}(k)}{4\pi} = \mathcal{D}_{\mu\nu}^{-1}(k) - D_{\mu\nu}^{-1}(k) = ie^2 \int \text{Sp} \gamma_\mu \mathcal{G}(p+k) \Gamma_\nu(p+k, p; k) \mathcal{G}(p) \frac{d^4 p}{(2\pi)^4} \quad (\text{XI.84})$$

(in (XI.82) and (XI.84) bispinor indices are omitted). Equations (XI.82) and (XI.84) are called the *Dyson equations*.

These equations can also be derived analytically. For the derivation of (XI.82), we consider

$$(\gamma \hat{p} - m)_{il} \mathcal{G}_{lk}(x - x') = -i(\gamma \hat{p} - m)_{il} \langle 0 | T \psi_l(x) \bar{\psi}_k(x') | 0 \rangle \quad (\text{XI.85})$$

($\hat{p} = i\partial$ — the differentiation operator with respect to x). We compute this expression using (XI.5), in the same way as was done in § 75 for the free-particle propagator. The result is:

$$(\gamma \hat{p} - m)_{il} \mathcal{G}_{lk}(x - x') = -ie\gamma_{il}^\nu \langle 0 | T A_\nu(x) \psi_l(x) \bar{\psi}_k(x') | 0 \rangle + \delta_{ik} \delta^{(4)}(x - x'). \quad (\text{XI.86})$$

The δ -function term here is the same as in (75.7), since the equal-time commutation relations for operators in the Heisenberg and interaction representations are the same. The first term on the right is $-ie\gamma^\nu K_{ik}^\nu(x, x, x')$, so that (dropping bispinor indices):

$$(\gamma \hat{p} - m) \mathcal{G}(x - x') = -ie\gamma^\nu K_\nu(x, x, x') + \delta^{(4)}(x - x'). \quad (\text{XI.87})$$

¹³For clarity we emphasise that although we obtain the entire required collection of diagrams by introducing corrections to only one vertex, for each specific diagram the structure of the correction block generally depends on which vertex it is assigned to.

¹⁴If in (XI.81) the exact vertex part is assigned to the left vertex, then in equation (XI.82) the factors γ and Γ are interchanged. Both forms of the equation are, of course, essentially equivalent.

We now pass to Fourier components. If we integrate the definition (XI.66) over $d^4k d^4p_2/(2\pi)^8$, we get:

$$\int K^\mu(p+k, p; k) \frac{d^4k}{(2\pi)^4} = \int K^\mu(x, x, x') e^{ip(x-x')} d^4(x-x'), \quad (\text{XI.88})$$

which shows that the integral on the left-hand side is a Fourier component of the function $K^\mu(x, x, x')$. Taking the Fourier transform of both sides of (XI.87), using the definition (XI.72), and recalling that $\gamma p - m = G^{-1}(p)$, we obtain:

$$G^{-1}(p) \mathcal{G}(p) = 1 - ie^2 \int \gamma^\nu \mathcal{G}(p+k) \Gamma^\mu(p+k, p; k) \mathcal{G}(p) \mathcal{D}_{\mu\nu}(k) \frac{d^4k}{(2\pi)^4}. \quad (\text{XI.89})$$

Multiplying the right-hand side by $\mathcal{G}^{-1}(p)$, we return to equation (XI.82).

§ 108. Ward identity

Yet another relation between the photon propagator and the vertex part, simpler than the Dyson equation, arises as a consequence of gauge invariance.

Let us perform the gauge transformation (102.8), assuming that $\chi(x) \equiv \delta\chi(x)$ is an infinitely small simple (non-operator) function of the 4-coordinates x . Then the electron propagator acquires the addition:

$$\delta\mathcal{G}(x, x') = ie\mathcal{G}(x - x') [\delta\chi(x) - \delta\chi(x')]. \quad (\text{XI.90})$$

Note that this gauge transformation violates spatial-temporal homogeneity and $\delta\mathcal{G}$ depends on x and x' separately, not just on the difference $x - x'$. We decompose $\delta\mathcal{G}$ in a Fourier integral over x and x' separately:

$$\delta\mathcal{G}(p_2, p_1) = \int \int \delta\mathcal{G}(x, x') e^{ip_2x - ip_1x'} d^4x d^4x'.$$

Substituting (XI.90) and integrating over $d^4\xi d^4x'$ (where $\xi = x - x'$), we obtain:

$$\delta\mathcal{G}(p + q, p) = ie \delta\chi(q) [\mathcal{G}(p) - \mathcal{G}(p + q)]. \quad (\text{XI.91})$$

On the other hand, under the same gauge transformation the operator $\hat{A}_\mu(x)$ receives an addition:

$$\delta A_\mu^{(e)}(x) = -\frac{\partial}{\partial x^\mu} \delta\chi, \quad (\text{XI.92})$$

which can be regarded as an infinitely small external field. In the momentum representation:

$$\delta A_\mu^{(e)}(q) = iq_\mu \delta\chi(q). \quad (\text{XI.93})$$

The quantity $\delta\mathcal{G}$ can be computed as the change of the propagator under the influence of this field. In first-order accuracy, it is represented by the single skeleton diagram:

$$(\text{XI.94})$$

Here the bold dashed line represents the effective external field line, associated with the factor (see (103.15)):

$$\delta A_\mu^{(e)}(q) + \delta A_\lambda^{(e)}(q) \frac{\mathcal{P}^{\lambda\nu}(q)}{4\pi} \mathcal{D}_{\mu\nu}(q).$$

But the 4-vector $\delta A_\lambda^{(e)}(q)$ is longitudinal (proportional to q_λ), and the tensor $\mathcal{P}^{\lambda\nu}$ is transverse. Therefore the second term vanishes, and the effective external field reduces to simply the field $\delta A^{(e)}$:

The analytical expression is:

$$\delta \mathcal{G} = e \mathcal{G}(p+q) \Gamma^\mu(p+q, p; q) \mathcal{G}(p) \cdot \delta A_\mu^{(e)}. \quad (\text{XI.95})$$

Substituting (XI.93) and comparing with (XI.91):

$$\mathcal{G}(p+q) - \mathcal{G}(p) = -\mathcal{G}(p+q) \Gamma^\mu(p+q, p; q) \mathcal{G}(p) \cdot q_\mu,$$

or, passing to inverse matrices:

$$\mathcal{G}^{-1}(p+q) - \mathcal{G}^{-1}(p) = q_\mu \Gamma^\mu(p+q, p; q) \quad (\text{XI.96})$$

(H. S. Green, 1953).

Letting $q \rightarrow 0$ and comparing the coefficients of the infinitely small q_μ on both sides, we obtain:

$$\frac{\partial}{\partial p_\mu} \mathcal{G}^{-1}(p) = \Gamma^\mu(p, p; 0). \quad (\text{XI.97})$$

This is the *Ward identity* (J. C. Ward, 1950). The derivative of $\mathcal{G}^{-1}(p)$ with respect to the momentum equals the vertex operator at zero momentum transfer.¹⁵

For the derivative of $\mathcal{G}(p)$ itself:

$$-\frac{\partial}{\partial p_\mu} i\mathcal{G}(p) = i\mathcal{G}(p) [-i\Gamma^\mu(p, p; 0)] i\mathcal{G}(p). \quad (\text{XI.98})$$

One could similarly obtain higher derivatives from higher-order terms of the expansion of $\delta\chi$ in powers of the coordinates. We shall not need such formulae.

Now let us consider the derivative $\partial\mathcal{P}(k)/\partial k_\mu$ of the polarisation operator. Unlike $\mathcal{G}(p)$, the operator $\mathcal{P}(k)$ is gauge-invariant and does not change when the fictitious external field (XI.93) is introduced. Therefore the derivative of $\mathcal{P}$ cannot be computed in the same manner. It is, however, possible to obtain a diagrammatic expression.

Consider the first diagram entering the definition of $\mathcal{P}$ — the second-order diagram:

$$(\text{XI.99})$$

The solid lines correspond to the factors $iG(p)$ and $iG(p+k)$. Differentiating with respect to k replaces the second factor by $\partial G(p+k)/\partial k$, and according to the identity (XI.98) this replacement is equivalent to adding an extra vertex on the electron line:

¹⁵In the zeroth approximation, i.e. for the free-particle propagator, this identity is obvious: $G^{-1}(p) = \gamma p - m$, and therefore $\partial G^{-1}/\partial p_\mu = \gamma^\mu$.

(XI.100)

In the first non-vanishing order the desired derivative is thus expressed through a diagram with three photon ends (“photon three-tail”). Note that this diagram by itself does *not* give the amplitude of splitting of one photon into two. The amplitude of such a process would be the sum of diagram (XI.100) and the same diagram with the reversed direction of traversal of the loop; by Furry’s theorem this sum vanishes. But diagram (XI.100) by itself is non-zero.

Similarly one can differentiate more complex diagrams, adding the vertex with $k' = 0$ to all electron lines that depend on k . There are also diagrams in which the dependence on k resides in internal photon lines, for example:

The derivative of the diagram enclosed in curly brackets is represented by introducing a new graphical element — a *fictitious three-photon vertex* (a point at which three dashed lines meet):

$$4\pi i \frac{\partial D^{-1}}{\partial k^\nu} = 2ik_\mu \equiv v_\mu. \quad (\text{XI.101})$$

Now any diagram depending on k can be differentiated by adding the vertices v_μ or γ_μ along all lines carrying momentum k . Computing further:

$$-\frac{1}{4\pi} \frac{\partial \mathcal{P}_{\mu\nu}}{\partial k^\lambda} = V_{\mu\lambda\nu}, \quad (\text{XI.102})$$

where $ie^2 V_{\mu\lambda\nu}$ is the sum of the internal parts of all “photon three-tails” obtained in this manner.

For the second derivative of the polarisation operator, differentiating (XI.102) once more:

$$\frac{1}{4\pi} \frac{\partial^2 \mathcal{P}_{\mu\nu}}{\partial k^\rho \partial k^\lambda} = \mathcal{G}_{\mu\rho\lambda\nu} + \mathcal{G}_{\mu\lambda\rho\nu}, \quad (\text{XI.103})$$

where $ie^2 \mathcal{G}$ is the sum of the internal parts of all “photon four-tails”:

(XI.104)

(including diagrams with fictitious three-photon vertices (XI.101)).

§ 109. Electron propagator in an external field

If the system is in a given external field $A^{(e)}(x)$, then the exact electron propagator is defined by the same formula (XI.48), but the Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}$ now includes, together with the interaction between particles, also the interaction with the external field:

$$\hat{V} = e \int \hat{A}_\mu \hat{j}^\mu d^3x + e \int A_\mu^{(e)} \hat{j}^\mu d^3x. \quad (\text{XI.105})$$

Since the external field violates the homogeneity of space-time, the propagator $\mathcal{G}(x, x')$ depends on x and x' separately, not merely on the difference $x - x'$.

Passage to the interaction representation leads to the standard diagrammatic technique, with the difference that in the diagrams there appear, alongside the virtual photon lines,

also lines of the external field. Such a technique is inconvenient in those cases where the external field cannot be treated as a small perturbation — when particles may be in bound states in the field. The electron propagator in an external field is needed precisely for the purpose of studying the properties of bound states when radiative corrections are taken into account.

For the construction of such a propagator, it is natural to start from a representation of the operators in which the external field is already taken into account in the zeroth approximation, while only the electron–photon interaction itself is treated as a perturbation (W. H. Furry, 1951).

We shall assume the external field to be stationary (time-independent).

The required representation of the ψ -operators is given by the formulae of second quantisation in the external field:

$$\begin{aligned}\hat{\psi}^{(e)}(t, \mathbf{r}) &= \sum_n \left\{ \hat{a}_n \psi_n^{(+)}(\mathbf{r}) e^{-i\varepsilon_n^{(+)}t} + \hat{b}_n^\dagger \psi_n^{(-)}(\mathbf{r}) e^{i\varepsilon_n^{(-)}t} \right\}, \\ \hat{\bar{\psi}}^{(e)}(t, \mathbf{r}) &= \sum_n \left\{ \hat{a}_n^\dagger \bar{\psi}_n^{(+)}(\mathbf{r}) e^{i\varepsilon_n^{(+)}t} + \hat{b}_n \bar{\psi}_n^{(-)}(\mathbf{r}) e^{-i\varepsilon_n^{(-)}t} \right\},\end{aligned}\quad (\text{XI.106})$$

where $\psi_n^{(\pm)}(\mathbf{r})$ and $\varepsilon_n^{(\pm)}$ are the wave functions and energy levels of the electron and the positron — solutions of the single-particle Dirac equation in the field.

It is easy to see that the operators (XI.106) are the ψ -operators in a certain representation (Furry representation), intermediate between the Heisenberg and interaction representations:

$$\begin{aligned}\hat{\psi}^{(e)}(t, \mathbf{r}) &= e^{i\hat{H}_1 t} \hat{\psi}(\mathbf{r}) e^{-i\hat{H}_1 t}, \\ \hat{\bar{\psi}}^{(e)}(t, \mathbf{r}) &= e^{i\hat{H}_1 t} \hat{\bar{\psi}}(\mathbf{r}) e^{-i\hat{H}_1 t},\end{aligned}\quad (\text{XI.107})$$

where

$$\hat{H}_1 = \hat{H}_0 + e \int A_\mu^{(e)}(x) \hat{j}^\mu(x) d^3x.$$

The electromagnetic field operator $\hat{A}_\mu$ commutes with the second term in $\hat{H}_1$, so that for this operator the Furry representation coincides with the interaction representation.

The zeroth-approximation electron propagator in the new representation is:

$$G_{ik}^{(e)}(x, x') = -i \langle 0 | T \hat{\psi}_i^{(e)}(x) \hat{\bar{\psi}}_k^{(e)}(x') | 0 \rangle. \quad (\text{XI.108})$$

The operator $\hat{\psi}^{(e)}(t, \mathbf{r})$ satisfies the Dirac equation in the external field:

$$[\gamma \hat{p} - e \gamma A^{(e)}(x) - m] \hat{\psi}^{(e)}(t, \mathbf{r}) = 0, \quad (\text{XI.109})$$

and correspondingly for $G^{(e)}$:

$$[\gamma \hat{p} - e \gamma A^{(e)}(x) - m] G^{(e)}(x, x') = \delta^4(x - x') \quad (\text{XI.110})$$

(cf. the derivation of (107.5)).

The diagrammatic technique for the expansion of the exact propagator $\mathcal{G}$ in a series in powers of e^2 is constructed — by passing from the Heisenberg to the Furry representation

and then to the interaction representation — in exactly the same way as the usual technique. It leads to diagrams of the same type, with the sole difference that the solid lines in the diagrams now correspond to the factor $iG^{(e)}$ (instead of iG).

A slight difference arises in the analytic rules corresponding to the diagrams, since $G^{(e)}$ is a function not only of the difference $x - x'$, but of x and x' separately. In a constant external field the homogeneity of time is preserved, so that t and t' enter only through the difference $\tau = t - t'$. Thus $G^{(e)} = G^{(e)}(\tau, \mathbf{r}, \mathbf{r}')$.

Performing the Fourier transform over each of its arguments, we have:

$$G^{(e)}(\tau, \mathbf{r}, \mathbf{r}') = \int e^{i(\varepsilon\tau - \mathbf{p}_2 \cdot \mathbf{r} + \mathbf{p}_1 \cdot \mathbf{r}')} G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) \frac{d\varepsilon}{2\pi} \frac{d^3 p_1}{(2\pi)^3} \frac{d^3 p_2}{(2\pi)^3}. \quad (\text{XI.111})$$

Each solid line now carries the factor $iG^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1)$, with one energy value ε but two momentum values — the initial $\mathbf{p}_1$ and the final $\mathbf{p}_2$:

$$iG^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1). \quad (\text{XI.112})$$

Integration is performed over $d\varepsilon/(2\pi)$ for each independent loop, and over $d^3 p_1/(2\pi)^3$ and $d^3 p_2/(2\pi)^3$ independently, with conservation of momentum at each vertex. For example:

$$= e^2 \int G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}'') \gamma^\mu G^{(e)}(\varepsilon - \omega, \mathbf{p}'' - \mathbf{k}, \mathbf{p}' - \mathbf{k}) \gamma^\nu G^{(e)}(\varepsilon, \mathbf{p}', \mathbf{p}_1) \\ \times D_{\mu\nu}(\omega, \mathbf{k}) \frac{d^4 k}{(2\pi)^4} \frac{d^3 p'}{(2\pi)^3} \frac{d^3 p''}{(2\pi)^3}. \quad (\text{XI.113})$$

It is important to note that in this technique one must also include diagrams with “self-closed” electron lines (loops), which in the usual technique are discarded as corresponding to a “vacuum current.” In the presence of an external field this current need not vanish, owing to the “polarisation of the vacuum.” For example, in the diagram:

$$(\text{XI.114})$$

the upper loop carries the factor

$$i \int G^{(e)}(\omega, \mathbf{p} + \mathbf{k}, \mathbf{p}) \frac{d^3 p}{(2\pi)^3} \frac{d\omega}{2\pi}. \quad (\text{XI.115})$$

Here one must clarify the meaning of the integral over ω . The point is that integrating the Fourier component of $G^{(e)}(\tau)$ over ω amounts to taking the value of this function at $\tau = 0$; but $G^{(e)}(\tau)$ is discontinuous at this point, and one must specify which of its two limiting values should be taken. To clarify this, we note that the integral (XI.115) arises from the contraction of ψ -operators at the same point: $\hat{j}^\mu = \hat{\psi}^{(e)}(t, \mathbf{r}) \gamma^\mu \hat{\psi}^{(e)}(t, \mathbf{r})$, where $\hat{\psi}^{(e)}$ stands to the left. According to the definition of the propagator (XI.108), this ordering of the factors corresponds to $t = t'$ if t' is understood as $t' = t + 0$, i.e. one

must take the limiting value of $G^{(e)}(t - t')$ at $t - t' \rightarrow -0$. In other words, the integral over $d\omega/2\pi$ in (XI.115) should be understood as

$$\int \dots e^{-i\omega\tau} \frac{d\omega}{2\pi}, \quad \tau \rightarrow -0. \quad (\text{XI.116})$$

The mass operator in the external field is defined in the same manner as in § 105: $-i\mathcal{M}$ is the sum of all compact self-energy blocks. It is now a function of the energy ε and of the momenta $\mathbf{p}_1$ and $\mathbf{p}_2$ on the external lines entering and leaving the block:

$$-i\mathcal{M}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1). \quad (\text{XI.117})$$

Proceeding in exactly the same way as in deriving (XI.53):

$$\begin{aligned} \mathcal{G}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) - G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) \\ = \int \int G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}'') \mathcal{M}(\varepsilon, \mathbf{p}'', \mathbf{p}') \mathcal{G}(\varepsilon, \mathbf{p}', \mathbf{p}_1) \frac{d^3 p'}{(2\pi)^3} \frac{d^3 p''}{(2\pi)^3}. \end{aligned} \quad (\text{XI.118})$$

It is more natural to write this equation in the coordinate representation:

$$\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}') = \int \int \mathcal{G}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) e^{i(\mathbf{p}_2 \cdot \mathbf{r} - \mathbf{p}_1 \cdot \mathbf{r}')} \frac{d^3 p_1 d^3 p_2}{(2\pi)^6}, \quad (\text{XI.119})$$

and analogously for the other quantities. Performing the inverse Fourier transform in (XI.118):

$$\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}') - G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}') = \int \int G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}_2) \mathcal{M}(\varepsilon, \mathbf{r}_2, \mathbf{r}_1) \mathcal{G}(\varepsilon, \mathbf{r}_1, \mathbf{r}') d^3 x_1 d^3 x_2.$$

We apply to both sides of this equation the operator $\gamma^0 \varepsilon - \gamma \hat{\mathbf{p}} - e\gamma A^{(e)}(\mathbf{r})$ (where ε is a number, and $\hat{\mathbf{p}} = -i\nabla$ differentiates with respect to $\mathbf{r}$). According to (XI.110):

$$[\gamma^0 \varepsilon - \gamma \hat{\mathbf{p}} - e\gamma A^{(e)}(\mathbf{r})] G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'). \quad (\text{XI.120})$$

The result is:

$$[\gamma^0 \varepsilon - \gamma \hat{\mathbf{p}} - e\gamma A^{(e)}(\mathbf{r})] \mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}') - \int \mathcal{M}(\varepsilon, \mathbf{r}, \mathbf{r}_1) \mathcal{G}(\varepsilon, \mathbf{r}_1, \mathbf{r}') d^3 x_1 = \delta(\mathbf{r} - \mathbf{r}'). \quad (\text{XI.121})$$

The particular value of the function $\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}')$ is that its poles as a function of ε determine the energy levels of the electron in the external field.

Let us show this first for the approximate function $G^{(e)}$. Substituting the operators (XI.106) into the definition of the propagator (XI.108), we get, in analogy with the formulae (75.12) for the free-particle propagator:

$$G_{ik}^{(e)}(t - t', \mathbf{r}, \mathbf{r}') = \begin{cases} -i \sum_n \psi_{ni}^{(+)}(\mathbf{r}) \bar{\psi}_{nk}^{(+)}(\mathbf{r}') e^{-i\varepsilon_n^{(+)}(t-t')}, & t > t', \\ i \sum_n \psi_{ni}^{(-)}(\mathbf{r}) \bar{\psi}_{nk}^{(-)}(\mathbf{r}') e^{i\varepsilon_n^{(-)}(t-t')}, & t < t', \end{cases} \quad (\text{XI.122})$$

and after Fourier transformation:

$$G_{ik}^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}') = \sum_n \left\{ \frac{\psi_{ni}^{(+)}(\mathbf{r}) \bar{\psi}_{nk}^{(+)}(\mathbf{r}')}{\varepsilon - \varepsilon_n^{(+)} + i0} + \frac{\psi_{ni}^{(-)}(\mathbf{r}) \bar{\psi}_{nk}^{(-)}(\mathbf{r}')}{\varepsilon + \varepsilon_n^{(-)} - i0} \right\}. \quad (\text{XI.123})$$

The function $G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}')$, considered as an analytic function of ε , has poles on the positive real half-axis at the electron energy levels, and on the negative half-axis at the positron levels. The values $\varepsilon_n^{(\pm)} > m$ form a continuous spectrum, and the corresponding poles merge into two branch cuts: from $-\infty$ to $-m$ and from m to $+\infty$. On the interval $|\varepsilon| < m$ lie the poles that determine the discrete energy levels.

For the exact propagator $\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}')$ one obtains an analogous expansion by expressing it through matrix elements of the Schrödinger–Heisenberg operators:

$$\langle m | \hat{\psi}(t, \mathbf{r}) | n \rangle = \langle m | \hat{\psi}(\mathbf{r}) | n \rangle e^{-i(E_n - E_m)t}. \quad (\text{XI.124})$$

Here E_n are the exact (with all radiative corrections) energy levels of the system in the external field. The operator $\hat{\psi}$ increases the charge by unity, and $\hat{\bar{\psi}}$ decreases it by unity. In the matrix elements $\langle n | \hat{\psi} | 0 \rangle$ and $\langle 0 | \hat{\bar{\psi}} | n \rangle$, the states $|n\rangle$ must correspond to charge -1 — one positron plus electron–positron pairs and photons; we denote the energies $E_n^{(-)}$. Similarly, in $\langle 0 | \hat{\psi} | n \rangle$ and $\langle n | \hat{\bar{\psi}} | 0 \rangle$, the states have charge $+1$ — one electron plus pairs and photons; energies $E_n^{(+)}$.

We obtain:¹⁶

$$\mathcal{G}_{ik}(t - t', \mathbf{r}, \mathbf{r}') = \begin{cases} -i \sum_n \langle 0 | \hat{\psi}_i(\mathbf{r}) | n \rangle \langle n | \hat{\bar{\psi}}_k(\mathbf{r}') | 0 \rangle e^{-iE_n^{(+)}(t-t')}, & t > t', \\ i \sum_n \langle 0 | \hat{\bar{\psi}}_k(\mathbf{r}') | n \rangle \langle n | \hat{\psi}_i(\mathbf{r}) | 0 \rangle e^{iE_n^{(-)}(t-t')}, & t < t', \end{cases} \quad (\text{XI.125})$$

and hence:

$$\mathcal{G}_{ik}(\varepsilon, \mathbf{r}, \mathbf{r}') = \sum_n \left\{ \frac{\langle 0 | \hat{\psi}_i(\mathbf{r}) | n \rangle \langle n | \hat{\bar{\psi}}_k(\mathbf{r}') | 0 \rangle}{\varepsilon - E_n^{(+)} + i0} + \frac{\langle 0 | \hat{\bar{\psi}}_k(\mathbf{r}') | n \rangle \langle n | \hat{\psi}_i(\mathbf{r}) | 0 \rangle}{\varepsilon + E_n^{(-)} - i0} \right\}. \quad (\text{XI.126})$$

Let ε be close to one of the discrete energy levels $E_n^{(+)}$ (or $E_n^{(-)}$). Then in the sum (XI.126) only the single pole term survives. Substituting into (XI.121), the factors depending on the second argument $\mathbf{r}'$ drop out (when $\mathbf{r} \neq \mathbf{r}'$), giving a homogeneous integro-differential equation for the function $\langle 0 | \hat{\psi}(\mathbf{r}) | n \rangle$ (or $\langle n | \hat{\bar{\psi}}(\mathbf{r}) | 0 \rangle$), which we denote $\Psi_n(\mathbf{r})$ for brevity.¹⁷ Dropping the index n :

$$[\gamma^0 \varepsilon + i\gamma \nabla - e\gamma A^{(e)}(\mathbf{r})]_{ik} \Psi_k(\mathbf{r}) - \int \mathcal{M}_{ik}(\varepsilon, \mathbf{r}, \mathbf{r}_1) \Psi_k(\mathbf{r}_1) d^3x_1 = 0 \quad (\text{XI.127})$$

(J. Schwinger, 1951). The discrete energy levels E_n appear as eigenvalues of this equation, providing the regular procedure for determining these levels.

¹⁶It is assumed that the external field vanishes at spatial infinity.

¹⁷Neglecting radiative corrections, $\Psi_n(\mathbf{r})$ coincides (for states containing one electron or one positron) with the wave functions $\psi_n^{(\pm)}$ — solutions of the Dirac equation.

Let us express, for example, the first-order correction to a discrete electron energy level ε_n , obtained from the Dirac equation:

$$[\gamma^0 \varepsilon_n + i\boldsymbol{\gamma} \nabla - e\gamma A^{(e)}(\mathbf{r})]\psi_n(\mathbf{r}) = 0, \quad (\text{XI.128})$$

with normalisation

$$\int \psi_n^* \psi_n d^3x = 1. \quad (\text{XI.129})$$

We write the eigenfunction in the form

$$\Psi_n(\mathbf{r}) = \psi_n(\mathbf{r}) + \psi_n^{(1)}(\mathbf{r}), \quad (\text{XI.130})$$

where $\psi_n^{(1)}$ is the correction. Substituting (XI.130) into equation (XI.127), multiplying from the left by $\bar{\psi}_n(\mathbf{r})$, and integrating over d^3x ,¹⁸ we obtain:

$$E_n - \varepsilon_n \approx \iint \bar{\psi}_{ni}(\mathbf{r}) \mathcal{M}_{ik}(\varepsilon_n, \mathbf{r}, \mathbf{r}_1) \psi_{nk}(\mathbf{r}_1) d^3x d^3x_1. \quad (\text{XI.131})$$

§ 110. Physical conditions of renormalisation

The theory presented so far in this chapter has been to a significant extent formal in character. We operated with all quantities as if they were finite, intentionally leaving aside the infinities arising in the theory. However, in an actual computation of the functions $\mathcal{D}$, $\mathcal{G}$, Γ by perturbation theory, divergent integrals appear which cannot, without additional considerations, be ascribed any definite value. The emergence of such divergences is a manifestation of the logical incompleteness of quantum electrodynamics. We shall see, however, that in this theory one can establish certain prescriptions that allow an unambiguous “subtraction of infinities” and the obtaining of finite values for all quantities possessing direct physical meaning. At the basis of these prescriptions lie obvious physical requirements: that the photon mass be zero and that the charge and mass of the electron equal their observed values.

Let us begin with the conditions on the photon propagator.

Consider scattering through single-particle intermediate states containing one virtual photon. The amplitude must have a pole at $P^2 = 0$ (the square of the total 4-momentum of the initial particles P coincides with the square of the real photon mass). As we saw in §79, this follows from the unitarity condition. The pole term arises from diagrams of the type (79.1):

$$(\text{XI.132})$$

with radiative corrections, both parts of the diagram being connected by a bold dashed line (exact photon propagator). Therefore $\mathcal{D}(k^2)$ must have a pole at $k^2 = 0$:

$$\mathcal{D} \rightarrow \frac{4\pi Z}{k^2} \quad \text{when} \quad k^2 \rightarrow 0, \quad (\text{XI.133})$$

¹⁸In the integration one must use the self-adjointness of the differential operator of equation (XI.128) for transferring its action from $\psi_n^{(1)}$ to $\bar{\psi}_n$.

where Z is a constant. For the polarisation operator this means, according to (103.21):

$$\mathcal{P}(0) = 0. \quad (\text{XI.134})$$

The coefficient $1/Z = 1 - \mathcal{P}(k^2)/k^2|_{k^2 \rightarrow 0}$.

Further restrictions on $\mathcal{P}(k^2)$ follow from the physical definition of electric charge. Two classical (arbitrarily heavy) particles at rest at a large distance from one another must interact according to the Coulomb law $U = e^2/r$ ($r \gg 1/m$). On the other hand, this interaction is expressed by the diagram:

$$(\text{XI.135})$$

where the upper and lower lines correspond to the classical particles. The photon self-energy corrections are included in the virtual photon line. Other corrections, involving the heavy-particle lines, vanish in the limit $M \rightarrow \infty$: adding internal lines to the diagram (XI.135) (for example, connecting lines a and c , or a and b , by a photon line) leads to virtual heavy-particle propagators containing the mass M in the denominator and vanishing as $M \rightarrow \infty$.

From the diagram (XI.135), the factor $e^2 \mathcal{D}(k^2)$ must represent (up to sign) the Fourier transform of the interaction potential. Stationarity of the particles means the virtual photon frequency $\omega = 0$; large distances mean small wave vectors $\mathbf{k}$. The Fourier transform of the Coulomb potential is $4\pi e^2/\mathbf{k}^2$. Since $\mathcal{D}$ depends only on $k^2 = \omega^2 - \mathbf{k}^2$, we obtain:

$$\mathcal{D} \rightarrow \frac{4\pi}{k^2}, \quad k^2 \rightarrow 0. \quad (\text{XI.136})$$

Thus the coefficient Z in (XI.133) must equal unity: $Z = 1$ (the sign in (XI.136) is obvious: $\mathcal{D}(k^2)$ tends to the free photon propagator $D(k^2)$). For the polarisation operator this means:

$$\frac{\mathcal{P}(k^2)}{k^2} \rightarrow 0, \quad k^2 \rightarrow 0. \quad (\text{XI.137})$$

Besides condition (XI.134), this also gives:

$$\mathcal{P}'(0) = 0. \quad (\text{XI.138})$$

In §103 it was noted that the effective external photon line factor (103.15), with account of (103.16) and (103.20), is:

$$\left[1 + \frac{\mathcal{P}(0) \mathcal{D}(0)}{4\pi} \right] e^\mu.$$

Now, with (XI.136) and (XI.137), the correction term vanishes. In other words, in external photon lines one need not include radiative corrections at all.

Thus natural physical requirements lead to the establishment of definite (zero) values of $\mathcal{P}(0)$ and $\mathcal{P}'(0)$. Computing these quantities from perturbation-theory diagrams would lead to divergent integrals. The method of eliminating infinities consists in ascribing to the divergent expressions the values prescribed by the physical requirements. This procedure is called *renormalisation* of the corresponding quantities.¹⁹

¹⁹The idea of this approach was first expressed by Kramers (H. Kramers, 1947). Systematic application of the renormalisation method in quantum electrodynamics was carried out in the works of Dyson, Tomonaga (S. Tomonaga), Feynman and Schwinger.

The procedure can also be formulated differently. For charge renormalisation one introduces a non-physical “bare” charge e_0 as the parameter in the electromagnetic interaction Hamiltonian in the formal perturbation theory. After this, the renormalisation condition is formulated as: $e_0^2 \mathcal{D}(k^2) \rightarrow 4\pi e^2/k^2$ (at $k^2 \rightarrow 0$), where e is the true physical charge. From this $e_0^2 Z = e^2$, and together with Z the non-physical quantity e_0 is eliminated. By requiring $Z = 1$ from the outset, we perform the renormalisation “on the fly” and avoid introducing fictitious quantities even in intermediate calculations.

Now let us turn to the conditions on the electron propagator. Consider scattering through a single-particle intermediate state containing one virtual electron. The amplitude must have a pole when the square of the total 4-momentum of the initial particles P_i coincides with the square of the real electron mass: $P_i^2 = m^2$. The pole term in the amplitude arises from diagrams of the type

$$(XI.139)$$

where, with radiative corrections taken into account, the bold solid line is the exact electron propagator. This means that the function $\mathcal{G}(p)$ must have a pole at $p^2 = m^2$, i.e. must have the limiting form

$$\mathcal{G}(p) \approx Z_1 \frac{\gamma p + m}{p^2 - m^2 + i0} + g(p), \quad p^2 \rightarrow m^2, \quad (XI.140)$$

where Z_1 is a scalar constant, and $g(p)$ remains finite as $p^2 \rightarrow m^2$. The matrix structure of the pole term in (XI.140) (proportionality to $\gamma p + m$) is a consequence of the same unitarity condition from which the requirement of the existence of the pole itself arose. Let us demonstrate this, simultaneously clarifying an important question about the renormalisation conditions for external electron lines.

If $\mathcal{G}(P)$ has the limiting form (XI.140), the inverse matrix

$$\mathcal{G}^{-1}(p) \approx \frac{1}{Z_1} (\gamma p - m) - (\gamma p - m) g (\gamma p - m), \quad p^2 \rightarrow m^2. \quad (XI.141)$$

The mass operator

$$\mathcal{M} = G^{-1} - \mathcal{G}^{-1} \approx \left(1 - \frac{1}{Z_1}\right) (\gamma p - m) + (\gamma p - m) g (\gamma p - m), \quad p^2 \rightarrow m^2. \quad (XI.142)$$

The effective external (say, incoming) electron line carries the factor (cf. (103.15))

$$\mathcal{U}(p) = u(p) + \mathcal{G}(p) \mathcal{M}(p) u(p), \quad (XI.143)$$

where $u(p)$ is the ordinary spinor amplitude of the electron, satisfying the Dirac equation $(\gamma p - m)u = 0$. By virtue of the requirements of relativistic invariance ($\mathcal{U}$, like u , is a bispinor), the limiting value of $\mathcal{U}(p)$ as $p^2 \rightarrow m^2$ can differ from $u(p)$ only by a constant scalar factor:

$$\mathcal{U}(p) = Z' u(p). \quad (XI.144)$$

This factor Z' is connected in a definite way with Z_1 , but finding this connection by direct substitution of (XI.141), (XI.142) into (XI.143) is not possible, owing to the

indeterminacy that arises: the result depends on the order in which the limit is taken for the various factors in (XI.143).

One can, however, avoid the question about the correct way to take the limit, by appealing instead to the unitarity condition applied to the reaction depicted by diagram (XI.139). The unitarity relation applies, generally speaking, not to individual diagrams but to amplitudes of processes as a whole. But when $p^2 \rightarrow m^2$ the pole diagram (XI.139) gives the dominant contribution to the corresponding amplitude M_{fi} , so that the other diagrams belonging to the same reaction can be disregarded.

By virtue of the unitarity requirements, as was shown in § 79, the single-particle intermediate state leads to the appearance in the imaginary part of the reaction amplitude of a term with a δ -function:

$$i\pi\delta(p^2 - m^2) \sum_{\text{pol}} M_{fn} M_{in}^*, \quad (\text{XI.145})$$

where in the present case the index n refers to a state containing one real electron, and the summation over polarisations is carried out as required by the symmetrisation of both sides of the unitarity relation over the helicities of the initial and final particles (as in § 79; then $M_{fi} = M_{if}$). The amplitude M_{fn} corresponds to the process depicted by the diagram and has the form

$$M_{fn} = (M'_{fn} \mathcal{U}) = Z' (M'_{fn} u),$$

where M'_{fn} is a quantity with one free bispinor index.²⁰ Analogously the amplitude M_{in}^* has the structure

$$M_{in}^* = (\bar{\mathcal{U}} M'^*_{in}) = Z' (\bar{u} M'^*_{in}).$$

The summation over electron polarisations replaces the product $(M'_{fn} u)(\bar{u} M'^*_{in})$ by $M'_{fn}(\gamma p + m) M'^*_{in}$, so that the term (XI.145) in the amplitude M'_{fi} takes the form

$$Z'^2 i\pi\delta(p^2 - m^2) [M'_{fn}(\gamma p + m) M'^*_{in}].$$

On the other hand, computing this same amplitude directly from diagram (XI.139) gives

$$iM_{fi} = iM'_{fn} \cdot i\mathcal{G}(p) \cdot iM'^*_{in}.$$

Comparing the two expressions confirms the limiting form for $\mathcal{G}(p)$ (the first term in (XI.140)), with

$$Z' = \sqrt{Z_1}. \quad (\text{XI.146})$$

Let us now show that once the required limiting form of the electron propagator has been established, there is no need for any further conditions on the vertex operator.

Consider the diagram

²⁰Strictly speaking, a clarification is needed here. An electron as a stable particle cannot actually decay into a different set of real particles. One can, however, formally regard as “final particles” certain fictitious particles with masses that would permit such a decay. The resulting relation should then be understood in the sense of analytic continuation to the real values of the masses.

(XI.147)

describing electron scattering in an external field $A^{(e)}(k)$ (in the first Born approximation), with all radiative corrections taken into account. In the limit $k \rightarrow 0$, $p_2 \rightarrow p_1 \equiv p$, the self-energy corrections to the external field line vanish (recall that these corrections vanish for any k when $k^2 = 0$). The diagram then corresponds to the amplitude

$$M_{fi} = -e \bar{\mathcal{U}}(p) \Gamma(p, p; 0) \mathcal{U}(p) A^{(e)}(k \rightarrow 0) \quad (\text{XI.148})$$

— the product of $A^{(e)}$ with the electron transition current $\bar{\mathcal{U}} \Gamma \mathcal{U}$. But when $k \rightarrow 0$ the potential $A^{(e)}(x)$ reduces to one independent of coordinates and time. Such a potential does not correspond to any physical field (it is a particular case of a gauge transformation), so that it cannot cause any change in the electron current. In other words, in the limit under consideration the transition current $\bar{\mathcal{U}} \Gamma \mathcal{U}$ must simply coincide with the free current $\bar{u} \gamma^\mu u$:

$$\bar{\mathcal{U}}(p) \Gamma^\mu(p, p; 0) \mathcal{U}(p) = Z_1 \bar{u}(p) \Gamma^\mu u(p) = \bar{u}(p) \gamma^\mu u(p). \quad (\text{XI.149})$$

This is, in essence, also an expression of the definition of the physical charge of the electron. It is easy to see that it is automatically satisfied regardless of the value of Z_1 . Indeed, substituting $\mathcal{G}^{-1}(p)$ from (XI.141) into the Ward identity (XI.97), we find

$$\Gamma^\mu(p, p; 0) = Z_1^{-1} \gamma^\mu - \gamma^\mu g(p) (\gamma p - m) - (\gamma p - m) g(p) \gamma^\mu,$$

and equality (XI.149) is satisfied by virtue of the equations $(\gamma p - m)u = 0$, $\bar{u}(\gamma p - m) = 0$.

We see that in constructing the amplitude of a physical process the “renormalisation constant” Z_1 drops out altogether. Moreover, taking advantage of the indeterminacy arising from the divergences in computing Γ , one can simply require that

$$\bar{u}(p) \Gamma^\mu(p, p; 0) u(p) = \bar{u}(p) \gamma^\mu u(p), \quad p^2 = m^2, \quad (\text{XI.150})$$

i.e. set $Z_1 = 1$. The convenience of this definition is that the need for introducing corrections to external electron lines disappears: we simply have

$$\mathcal{U}(p) = u(p).$$

One can also verify this directly: noting that when $Z_1 = 1$ the mass operator (XI.142) becomes

$$\mathcal{M} = (\gamma p - m) g (\gamma p - m) \quad (\text{XI.151})$$

and the second term in (XI.143) obviously vanishes. Thus, external lines — of both photons and electrons — will not require “renormalisation” corrections.²¹

²¹In renormalising the photon propagator the condition $Z = 1$ arose as a necessary physical requirement, and after this the disappearance of corrections to external photon lines occurs automatically. From a formal viewpoint, however, the situation for photon and electron external lines is analogous: if $Z \neq 1$ then the photon wave amplitude e_μ of a real photon would be multiplied by $\sqrt{Z}$.

§ 111. Analytic properties of the photon propagator

The investigation of the analytic properties of the photon propagator is conveniently begun by studying the properties of the function $\Pi(k^2)$. Direct use of the definition (103.1) is complicated by the gauge ambiguity of the operators $\hat{A}^\mu(x)$.

Starting from the expression for the photon self-energy function in terms of matrix elements of the gauge-invariant current operator, the integral representation of $\Pi(k^2)$ (104.11) was obtained in § 104. Denoting k^2 by t ,²² consider the properties of $\Pi(t)$ in the complex t -plane. From the integral representation:

$$\Pi(t) = \int_0^\infty \frac{\rho(t') dt'}{t - t' + i0}. \quad (\text{XI.152})$$

On the negative real half-axis, $\Pi(t)$ is real, and on the entire remaining plane satisfies:

$$\Pi(t^*) = \Pi^*(t). \quad (\text{XI.153})$$

$\Pi(t)$ can have singularities only at points of the function $\rho(t)$. These lie at values $t = k^2$ which are thresholds for production of various collections of real particles by the virtual photon. At these values new types of intermediate states “come into play” in the sum (104.9). The contribution of these states is zero below threshold and nonzero above, leading to singularities at the threshold points. These threshold values are real and non-negative.²³

Therefore the singular points of $\Pi(t)$ lie on the positive real half-axis of t . Making a cut along this half-axis, $\Pi(t)$ is analytic on the entire cut plane.

The $+i0$ in the denominator of (XI.152) shows that the pole $t' = t$ must be bypassed from below. Thus by the “value of $\Pi(t)$ ” on the real axis we mean its value on the upper edge of the cut. Using the rule (75.18):

$$\frac{1}{x \pm i0} = \mathcal{P} \frac{1}{x} \mp i\pi\delta(x), \quad (\text{XI.154})$$

we find for real t :

$$\text{Im } \Pi(t) \equiv \text{Im } \Pi(t + i0) = -\pi\rho(t). \quad (\text{XI.155})$$

On the lower edge of the cut the imaginary part has opposite sign, while $\text{Re } \Pi$ is the same. Hence the jump across the cut:

$$\Pi(t + i0) - \Pi(t - i0) = -2\pi i \rho(t). \quad (\text{XI.156})$$

The integral representation (XI.152) can itself be viewed as a Cauchy formula for the analytic function $\Pi(t)$. Indeed, applying the Cauchy formula:

$$\Pi(t) = \frac{1}{2\pi i} \oint_C \frac{\Pi(t') dt'}{t' - t} \quad (\text{XI.157})$$

to the contour

²²Not to be confused with the time designation!

²³For example, $k^2 = 0$ is the threshold for three (or more odd number of) real photons, $k^2 = 4m^2$ is the threshold for one electron-positron pair, etc.

(XI.158)

wrapping around the cut. Under the assumption of sufficiently fast decrease of $\Pi(t)$ at infinity, the integral over the large circle vanishes, and the integrals along the edges of the cut give the *dispersion relation*:

$$\Pi(t) = \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \Pi(t' + i0)}{t' - t} dt' = \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \Pi(t')}{t' - t - i0} dt'. \quad (\text{XI.159})$$

Substituting (XI.155), we recover (XI.152).²⁴

The analytic properties of $\mathcal{P}(t)$ and $\mathcal{D}(t)$ coincide with those of $\Pi(t)$, through which they are expressed by simple formulas (104.2) and (103.21). For $\mathcal{D}(t)$:

$$\mathcal{D}(t) = \frac{4\pi}{t(1 + \Pi(t)/t)}. \quad (\text{XI.160})$$

On the positive real half-axis ($t > 0$), t should be understood as $t + i0$. The imaginary part of $\mathcal{D}(t)$ can be computed using (XI.154) and (XI.155), taking into account that according to (XI.137) $\Pi(t)/t \rightarrow 0$ as $t \rightarrow 0$. Then:

$$\text{Im } \mathcal{D}(t) = -4\pi^2 \delta(t) + \frac{4\pi}{t^2} \text{Im } \Pi(t) = -4\pi^2 \delta(t) - \frac{4\pi^2}{t^2} \rho(t). \quad (\text{XI.161})$$

Applying the dispersion relation of type (XI.159) to $\mathcal{D}(t)$:

$$\mathcal{D}(t) = \frac{4\pi}{t + i0} + 4\pi \int_0^\infty \frac{\rho(t')}{t'^2} \frac{dt'}{t - t' + i0}. \quad (\text{XI.162})$$

This formula is called the *Källén–Lehmann representation* (G. Källén, 1952; H. Lehmann, 1954).

There is a close connection between the position of the cut for $\mathcal{D}(t)$ (and its imaginary part on the cut) and the unitarity condition for the amplitude of the process $a + b \rightarrow c + d$ depicted by diagram (110.4).

The unitarity condition (71.2):

$$T_{fi} - T_{if}^* = i(2\pi)^4 \sum_n T_{fn} T_{in}^* \delta^{(4)}(P_f - P_i).^{25} \quad (\text{XI.163})$$

The summation on the right is over all physical intermediate states n , which are evidently the states of systems of real pairs and photons — the same states that appear in the matrix elements in the definition of $\rho(k^2)$ (104.9). The amplitudes M_{fi} and M_{if} contain factors $\mathcal{D}(k^2)$ and $\mathcal{D}^*(k^2)$ respectively, and their difference contains the imaginary part $\text{Im } \mathcal{D}(k^2)$. We see that the connection (from (XI.155)) between $\text{Im } \mathcal{D}$ and the existence of the indicated intermediate states is a consequence of the necessary requirements of unitarity.

²⁴Dispersion relations were introduced into quantum field theory by M. Gell-Mann, M. L. Goldberger, and W. E. Thirring (1954).

²⁵Recall that amplitudes T_{fi} differ from M_{fi} only by factors (see (64.10)).

In subsequent work we shall see that in actual perturbation-theory computations of $\mathcal{D}(t)$ (or equivalently $\mathcal{P}(t)$) it is convenient to begin by computing $\text{Im } \mathcal{P}$, which involves no divergent expressions; the real part is then obtained from the dispersion formula of type (XI.159), and the integral may turn out to be divergent, requiring additional subtraction operations to satisfy the conditions $\mathcal{P}(0) = 0$ and $\mathcal{P}'(0) = 0$. This subtraction can be performed without explicit manipulation of divergent integrals. To do this, one applies the dispersion relation (XI.159) not to $\mathcal{P}(t)$ itself but to $\mathcal{P}(t)/t^2$. Then $\mathcal{P}(t)$ takes the form:

$$\mathcal{P}(t) = \frac{t^2}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t'^2(t' - t - i0)} dt'. \quad (\text{XI.164})$$

This integral converges, and the function $\mathcal{P}(t)$ obtained automatically satisfies the required conditions.

Equation (XI.164) is called a dispersion relation “with two subtractions.” The meaning of the passage to $\mathcal{P}(t)/t^2$ becomes particularly clear if one writes (XI.164) in the form:

$$\mathcal{P}(t) = \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t' - t - i0} dt' - \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t'} dt' - \frac{t}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t'^2} dt'. \quad (\text{XI.165})$$

If one denotes the first (“unregularised”) integral as $\tilde{\mathcal{P}}(t)$, then the entire expression on the right equals

$$\tilde{\mathcal{P}}(t) - \tilde{\mathcal{P}}(0) - t \tilde{\mathcal{P}}'(0).$$

§ 112. Regularisation of Feynman integrals

The physical renormalisation conditions considered in § 110 allow, in principle, obtaining unambiguously finite values for the amplitude of any electrodynamic process when computed in any approximation of perturbation theory.

Let us first clarify the character of the divergences in Feynman integrals. Consider an n -th order diagram (n vertices) with N_e external electron lines and N_γ external photon lines.

The number N_e is always even; the electron lines in the diagram form $N_e/2$ continuous sequences, each beginning and ending at an external end. The number of internal electron lines in each such sequence is one less than the number of vertices lying on it; therefore the total number of internal electron lines in the diagram equals $n - N_e/2$.

Each internal photon line connects two vertices; their total number is $(n - N_\gamma)/2$.

Each internal photon line carries a propagator $D(k) \sim k^{-2}$. Each internal electron line carries a propagator $G(p) \sim p^{-1}$. The total power of 4-momenta in the denominator of the integrand is thus

$$2 \frac{n - N_\gamma}{2} + \frac{n - N_e/2}{1} = 2n - \frac{1}{2}N_e - N_\gamma.$$

The number of independent integrations (over d^4p or d^4k) equals the total number of internal lines minus the $n - 1$ additional constraints imposed by 4-momentum conservation

at the vertices (one conservation law relates external momenta and does not constrain internal ones). Thus the total number of integrations over all components of 4-momenta is

$$4 \left[\frac{n - N_\gamma}{2} + n - \frac{N_e}{2} - (n - 1) \right] = 2(n - N_e - N_\gamma + 2).$$

The degree of divergence r of the diagram is the difference between this number and the power of momenta in the denominator:

$$r = 4 - \frac{3}{2} N_e - N_\gamma. \quad (\text{XI.166})$$

Remarkably, this number does not depend on the order n of the diagram.

The condition $r < 0$ for the diagram as a whole is, however, insufficient for convergence of the integral: every internal block of the diagram must likewise have a negative degree of divergence r' . The presence of blocks with $r' > 0$ can increase the degree of divergence of the diagram as a whole.

Since N_e and N_γ are positive integers, only a few pairs of values (N_e, N_γ) give $r \geq 0$. Let us enumerate these cases, immediately excluding $N_e = N_\gamma = 0$ (vacuum closed loops, devoid of physical meaning) and $N_e = 0, N_\gamma = 1$ (a “vacuum current,” to be discarded according to § 103). The remaining cases are:

$$(\text{XI.167})$$

In the first case the divergence is quadratic; in all the others ($r = 0$ or $r = 1$) it is logarithmic.

Diagram d) is the first correction to the vertex operator. It must satisfy condition (XI.150), which we write in the form

$$\bar{u}(p) \Lambda^\mu(p, p; 0) u(p) = 0, \quad p^2 = m^2, \quad (\text{XI.168})$$

where

$$\Lambda^\mu = \Gamma^\mu - \gamma^\mu. \quad (\text{XI.169})$$

Denote the Feynman integral written directly from the diagram rules by $\bar{\Lambda}^\mu(p_2, p_1; k)$. This integral diverges logarithmically and does not by itself satisfy (XI.168). However, a quantity satisfying this condition is obtained by forming the difference

$$\Lambda^\mu(p_2, p_1; k) = \bar{\Lambda}^\mu(p_2, p_1; k) - \bar{\Lambda}^\mu(p_1, p_1; 0) \Big|_{p_1^2 = m^2}. \quad (\text{XI.170})$$

The leading divergent part of the integral $\bar{\Lambda}^\mu$ is obtained by regarding the virtual photon 4-momentum f as arbitrarily large in the integrand. It has the form²⁶

$$-4\pi i e^2 \int \frac{\gamma^\nu(\gamma f) \gamma^\mu(\gamma f) \gamma_\nu}{f^2 \cdot f^2 \cdot f^2} \frac{d^4 f}{(2\pi)^4}$$

and does not depend on the values of the 4-momenta of the external lines. Therefore in the difference (XI.170) the divergence cancels and a finite quantity is obtained. This

²⁶The full expression of the integral is given in § 117 (see (117.2)).

operation of eliminating the divergence by subtraction is called *regularisation* of the integral.

Note that regularisability by a single subtraction is ensured by the fact that in this case the divergence is only logarithmic—the weakest possible. If the integral contained divergences of various orders, one subtraction might be insufficient.

Having determined the first correction Γ^μ (the first term of the expansion of Λ^μ), the first correction to the electron propagator (diagram (XI.167,c)) can be computed using the Ward identity (XI.97), which may also be written as

$$\frac{\partial \mathcal{M}(p)}{\partial p_\mu} = -\Lambda^\mu(p, p; 0), \quad (\text{XI.171})$$

introducing the mass operator $\mathcal{M}$ in place of $\mathcal{G}$ and Λ^μ in place of Γ^μ . This equation must be integrated with the boundary condition

$$\bar{u}(p) \mathcal{M}(p) u(p) = 0, \quad p^2 = m^2, \quad (\text{XI.172})$$

which follows from (110.20).

Finally, for the first term in the expansion of the polarisation operator, we use identity (108.14); after contraction over pairs of indices it gives

$$\frac{3}{4\pi} \frac{\partial^2 \mathcal{P}}{\partial k_\rho \partial k^\sigma} = 2\mathcal{G},$$

relating the scalar functions $\mathcal{P} = \frac{1}{3}\mathcal{P}^\mu{}_\mu$ and $\mathcal{G} = \mathcal{G}^\mu{}_\rho{}^{\rho\mu}$. Both depend only on the scalar k^2 :

$$2k^2 \mathcal{P}''(k^2) + \mathcal{P}'(k^2) = \frac{4}{3} \mathcal{P}(k^2), \quad (\text{XI.173})$$

where primes denote differentiation with respect to k^2 . Together with condition $\mathcal{P}'(0) = 0$, this also gives

$$\mathcal{P}(0) = 0. \quad (\text{XI.174})$$

In the first approximation, $\mathcal{P}(k^2)$ is determined by diagram (XI.167,e) (with 4-momenta of the ends $k, k, 0, 0$). The corresponding Feynman integral (denote it $\bar{\mathcal{G}}(k^2)$) diverges logarithmically, and its regularisation is accomplished by a single subtraction using condition (XI.174):

$$\mathcal{G}(k^2) = \bar{\mathcal{G}}(k^2) - \bar{\mathcal{G}}(0).$$

Then $\mathcal{P}(k^2)$ is determined as the solution of equation (XI.173) with boundary conditions $\mathcal{P}(0) = 0, \mathcal{P}'(0) = 0$.

In the next approximation, the correction to the vertex operator ($\Lambda_\mu^{(2)}$) is determined by diagrams (XI.73,d–i). The irreducible diagram (XI.73,d) is computed by the same regularisation (XI.170) as in the computation of the first-approximation $\Lambda_\mu^{(1)}$. In the reducible diagrams, the internal self-energy and vertex parts of lower order are replaced by the already-known (regularised) first-approximation quantities ($\mathcal{P}^{(1)}, \mathcal{M}^{(1)}, \Lambda_\mu^{(1)}$), after which the integrals are once again regularised using (XI.170).²⁷

²⁷In diagrams of still higher approximations it may also be necessary to replace “four-tails” by the regularised values of $\mathcal{G}$.

The systematic procedure described gives, in principle, finite expressions for $\mathcal{P}$, $\mathcal{M}$, and Λ_μ in any approximation of perturbation theory. Thereby it becomes possible to compute the amplitudes of physical scattering processes, whose diagrams contain the blocks $\mathcal{P}$, $\mathcal{M}$, Λ_μ as constituent parts.

We see that the physical conditions established in §§ 110 and 111 turn out to be sufficient for the unambiguous regularisation of all Feynman integrals. This is by no means a trivial property of quantum electrodynamics, and is called *renormalisability*.²⁸

For practical computation of radiative corrections the described procedure may not be the simplest or most rational. In the next chapter we shall see that it is often more convenient to begin by computing the imaginary parts of the corresponding quantities; these are given by integrals that do not diverge and contain no divergent integrals. The full quantity is then determined by analytic continuation using dispersion relations. This avoids the need for direct regularisation by subtraction.

²⁸Another approach to the theory of renormalisation in quantum electrodynamics is given in: N. N. Bogolyubov, D. V. Shirkov, *Introduction to the Theory of Quantised Fields*, Moscow: Nauka, 1984.

Radiative Corrections

Asymptotic Formulae of Quantum Electrodynamics

Electrodynamics of Hadrons